The Alaoglu–Erdős Integer-Exponent Conjecture

Machine-Checked Partial Results, Exact Conditional Structure, Determinant Methods, and the Remaining Barrier

A unified report on the corpus in

https://github.com/VladimirReshetnikov/ProveIt

Bottom line. The conjecture — that $2^x, 3^x \in \mathbb{Z}$ forces $x \in \mathbb{Z}$ — remains open, and the reduction to the four exponentials conjecture explains why no routine combination of present methods can close it. What the corpus does establish is unusually complete. Kernel-verified in Lean: a zero-free region $2^x < 16384$; transcendence of every hypothetical counterexample; multiplicative rank at most two with a canonical primitive odd core; and — conditional on a single counterexample — the *exact* classification

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}, \qquad \mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\},$$

with unique coordinates, exact dyadic block counts, a sharp quadratic cumulative asymptotic, and vanishing block Fourier modes. These reduce the problem to one narrow statement: no primitive odd $w > 1$ has a positive power of w^ϑ in $\mathbb{Z}[1/3]$.

The remaining mathematics is organized by method rather than provenance. Determinant and interpolation sections develop arbitrary-node repulsion, generalized Vandermonde and Smith profiles, exact boundary residuals, rational Loewner certificates, geometric Newton interpolation, and the precise archimedean/nonarchimedean constants that prevent the known integerizations from becoming subunit. Algebraic sections classify auxiliary bases and output ranks, analyze radical and Kummer fields, and isolate both a prime-log-product sufficient condition and a shifted Newton–Kummer algebraic integer. Dynamical and information-theoretic sections give compact-orbit laws, Sturmian and carry languages, decoder bounds, exact height entropy, and the limits of fixed observables. Analytic sections cover canonical products, natural boundaries, Mellin and special-function transfer, Hausdorff moments, and an explicit critical Stieltjes interpolant.

Three especially sharp interfaces survive. Pointwise-positive rational entire interpolants are flexible, but geometric log-convexity or log-concavity would close the conjecture from $1, 2, 3$, and nonnegative Taylor coefficients would close it from $1, 2, 3, 6$. A balanced Newton exhaustion would need an output-specific denominator *upper* bound despite nested fresh-prime exhaustions and direct Markov–Padé ratios forcing denominator mass in the opposite direction; the finite targets are now a saturated two-generator divisor of the Padé-defect integer or control of the vanishing and p-adic lifts of the transverse bad-cluster factor. And a Kummer operation would have to preserve signed cross-base cancellation through exceptional cross-conjugate ideal overlap: ordinary traces, character projections, ideal gcds, and partial norms have exact deficits, while augmentation-zero signed ratios satisfy an exact unit/divisor obstruction unless such overlap occurs. The fixed-level order-one private-prime theorem supplies no polynomial-factor transfer to the unshifted or adaptive cocycle. None of these bridges follows from the present hypotheses, and the four-exponentials barrier remains intact.

Abstract

Let $\vartheta = \log_2 3$. The Alaoglu–Erdős conjecture asserts that a real x with $2^x, 3^x \in \mathbb{Z}$ must be an integer; equivalently, the only positive integers m with $m^\vartheta \in \mathbb{Z}$ are the powers of two. It is the first unresolved 2×2 instance of the four exponentials problem, while its three-base analogue follows from six exponentials and is formalized here by an interpolation determinant.

This report catalogues the machine-checked corpus with exact Lean identifiers: zero-free regions, the Gelfond–Schneider dichotomy, multiplicative rank two, the canonical primitive odd core, the rational-power index and radical degree, localized-radical equivalences, arbitrary-order arithmetic-progression obstructions, shrinking-target Diophantine bounds, fractional-part gap equivalences, and the exact conditional structure theory — the free monoid $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ with a unique least generator, exact block counts, and the sharp quadratic asymptotic. Second, it develops two new determinant methods. Arbitrary candidate tuples on the lattice curve $y = x^\vartheta$ satisfy

$$\prod_{0 \le i < j \le r} (m_j - m_i) \;\ge\; \frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\, m_0^{r-\vartheta},$$

with no arithmetic-progression hypothesis; and the mixed exponent semigroup supplies positive integer generalized Vandermonde determinants whose repulsion exponent tends to one. The paper proof yields the explicit unconditional bound $\#(\mathcal{C} \cap [X, 2X]) \le 10000 \max\{1, \log X\}^2$ by a route independent of the prime-valuation rank theorem; the corresponding accepted Lean endpoint carries the explicit hypothesis `NewtonFactorization`. On the nonarchimedean side, the accepted corpus now identifies the exact Beatty minimizing fiber and gap, groups every tropical layer by row partitions, truncates the normalized determinant exactly modulo p^K, bounds the number of surviving row partitions by a stars-and-bars coefficient, and extracts finite first-fiber divisors. In the block-scaled consecutive-power model, the all-layer unit gain adds directly to the tropical minimum, giving $p^{\tau+U} \mid \det$ uniformly across the assignment cascade. These gains are lower order; exact synthetic experiments expose genuine cross-layer carrying but do not exhibit a leading-order cascade at the tested sizes. Complementary sections organize the remaining exact results by method: matrix-coefficient and sparse-interpolation equivalences; rational-orbit, factorial, and Smith divisors; compact/profinite and Sturmian laws; polynomial, Padé, and rational-contact rigidity; common-zero divisors and natural boundaries; matrix-power and Loewner certificates; finite-place transversality and prime switching; structural-prime residuals and resultants; information-theoretic decoders and carry languages; special-function and Mahler transfer; spectral and moment representations; rank geometry; and projective defect bounds.

The critical analytic synthesis determines the canonical entire interpolant selected by finite Lagrange interpolation, its exact second-order critical growth, and the corresponding positive Stieltjes denominator target. Its one-edge residuals satisfy sharp quadratic Jensen barriers and a rational-coefficient uncertainty law $4\sigma_2\delta_2 \ge 1$; the canonical entire q-Newton completion realizes the exact boundary defect. In the cross-coprime determinant branch, the former $7/10$ ceiling is sharpened to the identity $7/10 - \mathcal{E}_{\mathrm{val}}/20$, with an explicit arithmetic deficit of order x^{-4}. A four-value covariance theorem closes any nonnegative-coefficient representative, while pointwise-positive rational sum-of-squares representatives always exist. One-signed geometric curvature nevertheless closes the problem from the three values at $1, 2, 3$, and every globally positive nonintegral representative pays the full refined critical growth. Finite positive-scale filters cannot remove the canonical alternating tail; normally convergent power-substitution families have a supercubic coefficient threshold, and the basic four-dilation rank-one defect remains exactly critical after division by the node product. Reciprocal Stieltjes moments give a full Vandermonde-divisible Hankel determinant whose residual alternant dominates the input Vandermonde for every node set and whose exact T^5 asymptotic remains strictly positive. Direct-node Markov–Padé ratios force denominator repulsion through half the interpolation order and isolate a positive Padé-defect integer; arbitrarily deep one-axis collisions show that raw Hankel Smith content cancels from its primitive form. Meanwhile, sparse fresh-prime spikes can force almost the full node-product mass into reduced denominators; both effects point away from closure. Uniform Corvaja–Zannier estimates show that pairwise gap synchronization for one fixed hypothetical output table has only subcubic mass. An exact collision-tree initial form proves transfer for a unique simple affine cluster, but the first complete triangle already exhibits collective cancellation and fixed four-node smooth supports have unbounded collective excess at one prime. The tied, deeper, or non-affine side necessarily carries the full cubic source capacity; the unresolved full-triangle quantity is its output-sensitive transverse cancellation. A shifted higher Newton–Kummer cocycle gives an exponentially small algebraic integer, while character projections, the exact common-ideal norm, and signed-orbit augmentation expose both a one-factor partial-norm deficit and a conditional unit/divisor obstruction. At fixed ℓ and Newton order one, an exact Fermat-like resultant splitting reduces the missing private-prime radical to the standard number-field *abc* conjecture; known unconditional squarefree-factor estimates remain sublinear. The fixed-level argument is nonuniform and has no polynomial-factor descent to the unshifted or adaptive cocycle: the defining polynomials at distinct shifts are pairwise coprime, and the order-one zero curve is not contained in any higher Newton zero curve. Complete synchronized gap products give integer orbit polynomials, exact discriminants, and a universal quadratic surplus $\kappa = 0.7079245703208558\ldots$ between the triadic and dyadic products. The same audit shows that the natural Lambert/Hadamard, fixed-S, reciprocal-Hankel, and pure-2/3-denominator amplifications fail at quantified scales. Overall, the determinant hierarchy misses the conditional population by exactly one factor of $\log X$. The report also records equivalent and sufficient conditions — block-growth dichotomies (including the observation that the conjecture is equivalent to the dyadic block count being 1 for *infinitely many* exponents), a compact synchronized S-integral orbit formulation, and the prime-log-product independence criterion — and closes with a prioritized research program and a formalization blueprint.

Contents

1 The problem and its equivalent forms

1.1 Original statement

Conjecture 1.1 (Alaoglu–Erdős, two-base integer form)**.** *For every $x \in \mathbb{R}$,*

$$2^x \in \mathbb{Z} \quad \textit{and} \quad 3^x \in \mathbb{Z} \qquad \Longrightarrow \qquad x \in \mathbb{Z}.$$

Status: recorded in Lean as the proposition `AlaogluErdosConjecture`, without being asserted.

Since 2^x and 3^x are positive, the hypotheses actually put them in $\mathbb{N}$; and $2^x \in \mathbb{N}$ forces $x \geq 0$, since a negative exponent gives $0 < 2^x < 1$. All solutions therefore live in $[0, \infty)$ (`nonneg_of_two_rpow_integer`), and "integer" may be replaced throughout by "positive natural number."

Historical placement. Questions of this type go back to the 1944 study of colossally abundant numbers by Alaoglu and Erdős [1], where the arithmetic nature of simultaneous rational powers p^x, q^x of distinct primes controls the quotients of consecutive colossally abundant numbers — a theme reaching back to Ramanujan's superior highly composite numbers [2]. Alaoglu and Erdős isolated the two-prime assertion explicitly, regarded it as very likely, could not prove it, and reported a communicated three-prime result of Siegel. In modern language Siegel's case is supplied by the six exponentials theorem, whereas the two-base statement sits at the unresolved 2×2 threshold of the four exponentials conjecture (Schneider's first problem [24]; see Waldschmidt [25, 28]). The elementary-looking integrality question is thus one of the canonical concrete tests of the gap between six and four exponentials, not an isolated recreational puzzle. The four exponentials conjecture remains open.

1.2 Candidate-base normalization

Put

$$\vartheta := \log_2 3 = 1.584962500721156\ldots$$

and define the *candidate set*

$$\mathcal{C} := \{m \in \mathbb{N} : m > 0,\ m^\vartheta \in \mathbb{N}\}.$$

Lemma 1.2 (Exponent–candidate bijection)**.** *The maps $x \mapsto 2^x$ and $m \mapsto \log_2 m$ are mutually inverse bijections between $\mathcal{S} := \{x \in \mathbb{R} : 2^x, 3^x \in \mathbb{Z}\}$ and $\mathcal{C}$. Under the bijection, $3^x = m^\vartheta$.*

Proof. If $x \in \mathcal{S}$, set $m = 2^x \in \mathbb{N}$. Then $m^\vartheta = \exp(\log m \cdot \log_2 3) = \exp(x \log 3) = 3^x \in \mathbb{N}$. Conversely, if $m \in \mathcal{C}$ and $x = \log_2 m$, then $2^x = m$ and the same identity gives $3^x = m^\vartheta \in \mathbb{N}$. □

Status: [kernel-verified Lean] —
`exists_twoBaseNaturalCandidate_of_two_three_rpow_integer`,
`TwoBaseNaturalCandidate.integer_powers`.

Integral solutions correspond exactly to powers of two, by the identifier

`twoBaseCandidateExponent_isInteger_iff_isPowerOfTwo`,

so

$$\boxed{\text{Conjecture 1.1} \iff \mathcal{C} = \{2^n : n \in \mathbb{N}_0\}.} \tag{1}$$

This equivalence is itself formalized, as

`alaogluErdosConjecture_iff_candidates_are_powers_of_two`.

The candidate formulation is well suited to unique factorization, finite certification, finite differences of t^{ϑ}, multiplicative-rank arguments, and the determinant constructions of this report, and it is the formulation in which most of the corpus is stated.

Two closure properties of $\mathcal{S}$ are elementary but essential: $\mathcal{S}$ is closed under addition and under multiplication by positive naturals, so it is an additive submonoid of $[0, \infty)$ containing $\mathbb{N}$ (`rpow_integer_add`, `rpow_integer_nat_mul`).

1.3 The logarithmic determinant

Writing

$$M = 2^x, \qquad A = 3^x,$$

a solution is equivalently a positive integer pair satisfying

$$\log M \, \log 3 - \log A \, \log 2 = 0, \tag{2}$$

and the conjecture says that all positive integer solutions of (2) are

$$(M, A) = (2^n, 3^n), \qquad n \in \mathbb{N}_0.$$

Thus the problem is the vanishing of a 2×2 determinant of logarithms of algebraic numbers,

$$\det \begin{pmatrix} \log M & \log 2 \\ \log A & \log 3 \end{pmatrix} = 0,$$

with the additional information that two of the four entries are logarithms of *unknown integers*. This is not merely an analogy with the four exponentials conjecture; it is its exact local geometry.

1.4 One-parameter subgroup form

Let

$$\Gamma = \{(2^t, 3^t) : t \in \mathbb{R}\} \subset \mathbb{R}^2_{>0}.$$

Then Conjecture 1.1 is the statement

$$\Gamma \cap \mathbb{N}^2 = \{(2^n, 3^n) : n \in \mathbb{N}_0\}.$$

The curve Γ is a real one-parameter subgroup of the multiplicative torus $\mathbb{G}_m^2(\mathbb{R})$, and the assertion is that it meets the integer lattice points of the torus only at the obvious places. The difficulty is that its direction vector $(\log 2, \log 3)$ is transcendental rather than algebraic, so standard algebraic-subgroup intersection theorems do not apply directly.

1.5 The four-exponentials reduction

Suppose x is nonintegral and $2^x, 3^x$ are algebraic. The rational-exponent argument (Lemma 1.3) shows that x cannot be rational, so $1, x$ are $\mathbb{Q}$-linearly independent; and $\log 2, \log 3$ are $\mathbb{Q}$-linearly independent since $2^a 3^b = 1$ forces $a = b = 0$. The four exponentials built from these two pairs are

$$e^{1 \cdot \log 2} = 2, \quad e^{1 \cdot \log 3} = 3, \quad e^{x \log 2} = 2^x, \quad e^{x \log 3} = 3^x,$$

all four algebraic — contrary to the four exponentials conjecture. This reduction is formalized: `alaogluErdosConjecture_of_realFourExponentialsConjecture`. **[kernel-verified Lean]**

The reduction is a diagnostic, not just context: any proposed elementary argument must exploit integrality, positivity, or arithmetic structure beyond bare algebraicity of those four values, or else it would prove a substantial case of the four exponentials conjecture by another name.

1.6 Rational and algebraic exponents

Lemma 1.3 (Rational exponent)**.** *If $x \in \mathbb{Q}$ and $2^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$. Consequently every nonintegral solution is irrational.*

Proof. Write $x = a/b$ in lowest terms with $b > 0$. If $2^{a/b} = m \in \mathbb{N}$, then $2^a = m^b$; comparing the exponent of 2 gives $a = b\, v_2(m)$, so $b \mid a$ and $b = 1$. □

Status: **[kernel-verified Lean]** —
`IntegerExponent.integer_of_rational_of_two_rpow_integer`,
`irrational_of_not_integer_of_two_rpow_integer`.

Gelfond–Schneider upgrades "irrational" to "transcendental" and removes the algebraic irrational case entirely.

Lemma 1.4 (Transcendence dichotomy)**.** *If $2^x \in \mathbb{Z}$, then either $x \in \mathbb{Z}$ or x is transcendental. In particular every counterexample to Conjecture 1.1 is a transcendental number, and $\vartheta = \log_2 3$ is itself transcendental.*

Status: **[kernel-verified Lean]** —
`transcendental_of_not_integer_of_two_three_rpow_integer`,
`transcendental_logThreeDivLogTwo`; the supporting Gelfond–Schneider material is discussed in Section 3.

Together with the finite verification these lemmas already remove several whole regimes. In summary form, the pinned corpus proves the following unconditional restrictions on a hypothetical counterexample x.

- x is irrational, and in fact transcendental. **[kernel-verified Lean]**
- ϑ is transcendental. **[kernel-verified Lean]**
- $x \geq 14$, by the kernel-checked zero-free region $2^x < 16384$. **[kernel-verified Lean]**

- $2^x \geq 2^{27}$, hence $x \geq 27$, by the directed-rounding scan; this raises the barrier only under the stated software trust boundary and is not a kernel certificate. **[software computation]**
- The real four exponentials conjecture implies Conjecture 1.1. **[kernel-verified Lean]**

These facts eliminate the rational, algebraic-irrational, small, and routine three-exponentials cases. They also explain why an elementary manipulation of logarithms is unlikely to finish the proof: the unknown lies exactly where present transcendence theory changes from theorem to conjecture.

1.7 The three-base theorem and its determinant architecture

By contrast with the two-base problem, three multiplicatively independent bases fall to the six exponentials mechanism, and the special case needed here is fully formalized.

Theorem 1.5 (Three-base theorem)**.** *If $2^x, 3^x, 5^x \in \mathbb{Z}$ for a real x, then $x \in \mathbb{Z}$.*

Status: **[kernel-verified Lean]** — modules `IntegerExponent`, `IntegerExponent.DeterminantBound`, `IntegerExponent.ExponentialKernel`, `IntegerExponent.Grid`, following Waldschmidt's square-determinant proof [27].

In outline, for irrational x one forms a large square matrix $\mathcal{A}_{ij} = \exp(a_i b_j)$, where the row arguments a_i encode monomials in $2, 3, 5$ and the column arguments b_j encode affine forms in $1, x$. Irrationality makes the column parameters pairwise distinct and multiplicative independence makes the row parameters pairwise distinct; a total-positivity argument gives $\det \mathcal{A} \neq 0$; integrality of all entries forces $|\det \mathcal{A}| \geq 1$; and an interpolation estimate with explicit size choices gives $|\det \mathcal{A}| < 1$, a contradiction.

The dimension count is exactly where the argument stops short of Conjecture 1.1. With three independent row directions and two column directions the analytic upper bound decays faster than the arithmetic lower bound 1 permits, and the contradiction closes. Shrinking to two bases removes one row direction, and the same balance then lands precisely on the classical boundary where the decay below one fails. Nothing in the proof is wasteful enough to be recovered by sharper constants: a successful two-base determinant of this shape would be a new four-exponentials argument, which is why Section 18 and Section 19 enlarge the *node set* rather than tune the estimate.

A denominator-controlled rational third output. The rational-output extension now formalized is deliberately conditional. Let $a > 1$ and assume the monomial map

$$(i, j, k) \longmapsto 2^i 3^j a^k$$

is injective. If $2^x = m_2$ and $3^x = m_3$ are natural numbers, $a^x = q \in \mathbb{Q}$, and for some $u, v \in \mathbb{N}_0$ the reduced denominator satisfies

$$\mathrm{den}(q) \mid m_2^u m_3^v,$$

then $x \in \mathbb{Q}$, and in fact $x \in \mathbb{Z}$ because 2^x is integral. A concrete corollary obtains the monomial injectivity when a has a prime divisor different from 2 and 3.

On the determinant side, a nonzero $m \times m$ real determinant whose entries become integers after multiplication by a common positive denominator Q satisfies $|\det A| \geq Q^{-m}$. The explicit analytic estimate also retains a denominator margin: for $m = 2r$, $r > 2$, $192C \le r$, and $bs + 2r < r^2$, the verified upper-bound expression remains strictly below 1 even after multiplication by b^s.

Status: **[kernel-verified Lean]** — `one_div_pow_le_abs_det_of_common_denominator`, `exponential_det_numeric_bound_with_denominator`, `rational_of_two_three_a_rpow_rational_of_den_dvd_outputs`, and `integer_of_two_three_a_rpow_rational_of_prime_factor_of_den_dvd_outputs` in `RationalThirdBase`. These are denominator-divisibility and explicit-margin results; the unrestricted determinant grid for an arbitrary rational third-base output has *not* been completed.

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2 Trust and verification convention

The mathematical status of a claim is determined by its displayed hypotheses and by the trust labels below, not by the chronology of repository commits. The theorem-level status matrix and the module-by-module formalization inventory appear in Appendices A and B.

2.1 Evidence convention

The evidence convention of this report is source-tree plus build-level, not rebuild-on-demand. Every claim marked **[kernel-verified Lean]** was checked against the cited Lean source and theorem statement at the repository state above, and the corresponding module is on the aggregate import surface `ExponentialIdentities.lean`, where it is recorded as accepted by the Lean kernel. The label does not assert that a separate fresh build was run while preparing this document; it asserts that the theorem exists, in the stated form, on an import surface the kernel has accepted.

Three weaker labels are used deliberately. **[software computation]** marks a finite interval computation with an explicit software trust boundary and no kernel certificate; **[paper proof]** marks a conventional proof given in this report, written with formalization in mind but not yet translated; and **[Lean draft, not yet accepted]** marks a Lean module that exists in the tree as a draft but has *not* been accepted by the kernel, and therefore carries no more weight than **[paper proof]**. The complete label-by-label inventory is in Appendix A, and the per-module inventory in Appendix B.

3 Verified local and finite obstructions

Throughout this section x denotes a hypothetical nonintegral solution, $m = 2^x \in \mathbb{N}$, $n = 3^x \in \mathbb{N}$. Every statement here is **[kernel-verified Lean]** unless labeled otherwise.

3.1 Fractional-part gaps and shrinking targets

An integral power caught strictly between consecutive integral powers of its base must clear both by at least one full unit; dividing through normalizes this to the fractional part.

Theorem 3.1 (Fractional-part gaps). *Let $x \in (N, N+1)$ with $N \in \mathbb{N}$, $t = x - N$, and suppose $2^x, 3^x \in \mathbb{Z}$. Then*

$$1 + 2^{-N} \le 2^t \le 2 - 2^{-N}, \qquad 1 + 3^{-N} \le 3^t \le 3 - 3^{-N}.$$

Status: [kernel-verified Lean] — `two_three_fractional_part_gaps`, `two_three_rpow_gaps_near_nat`, from `rpow_integer_gap_between_consecutive_powers`.

Near the endpoints the excluded zones have width $\asymp 2^{-N}$: they *shrink* as the integer part grows. Applied to all multiples kx (which are again solutions), this yields explicit Diophantine lower bounds.

Theorem 3.2 (Effective irrationality bound). *Let $x \notin \mathbb{Z}$ satisfy $2^x \in \mathbb{Z}$ and write $m = 2^x$. Then for all integers p and all $k \ge 1$,*

$$\left| x - \frac{p}{k} \right| \;\ge\; \frac{1}{2\,k\,m^k}.$$

Exact (slightly stronger) radii, and two-base versions taking the better of the base-2 and base-3 radii, are formalized as well.

Status: [kernel-verified Lean] — `rational_approximation_exponential_lower_bound`, `two_three_multiple_distance_to_integer_lower_bound`, `exists_natural_base_rational_approximation_lower_bound` in `ShrinkingTarget`.

In shorthand, $\|kx\|_{\mathbb{Z}} \gtrsim 2^{-kx}$. This bound is compatible with ordinary Diophantine approximation: Dirichlet guarantees infinitely many k with $\|kx\|_{\mathbb{Z}} < 1/k$, and the shrinking radius 2^{-kx} is incomparably smaller. The interplay between this exponential gap and equidistribution is discussed in Section 24.

The bound dualizes into a growth constraint on the continued fraction of a solution: if $|x - p/q| < 1/(qQ)$ with $q, Q \ge 1$, then $Q < 2m^q$. Applied to consecutive convergent denominators, which satisfy $|x - p_n/q_n| < 1/(q_n q_{n+1})$, this gives

$$q_{n+1} \;<\; 2\,m^{\,q_n} :$$

the partial quotients of a hypothetical nonintegral solution are capped by a single exponential in the preceding denominator, with base its own output $m = 2^x$.

Status: [kernel-verified Lean] — `approximation_quality_lt_of_noninteger_solution` in `DenominatorGrowth`.

3.2 Verified zero-free regions and certificate architecture

Theorem 3.3 (Zero-free region). *If $2^x, 3^x \in \mathbb{Z}$ and $2^x < 16384$, then $x \in \mathbb{Z}$. Equivalently, every nonintegral solution satisfies $x \ge 14$, and every candidate that is not a power of two is at least 16384.*

Status: [kernel-verified Lean] — `integer_of_two_three_rpow_integer_of_two_rpow_lt_16384`, `fourteen_le_of_not_integer_of_two_three_rpow_integer`, `le_of_twoBaseNonintegerCandidate_16384` in `FiniteCheck16384`, extending `FiniteCheck8192` ($2^x < 8192$, hence $x \geq 13$), `FiniteCheck4096` ($x \geq 12$), `FiniteCheck256` ($x \geq 8$), and `FiniteCheck` ($2^x < 100$). Statements elsewhere in this report that quote an earlier kernel bound remain valid a fortiori; each module is an independent, separately replayable certificate.

The certificate architecture is worth recording, both because it is the repository's model of a trustworthy large computation and because scaling it is a named research direction. For each non-power-of-two m the certificate is a pair of exact integer power comparisons against rational enclosures $L_p/L_q < \vartheta < U_p/U_q$ drawn from continued-fraction convergents of ϑ: from

$$a^{L_q} < m^{L_p} \quad \text{and} \quad m^{U_p} < (a+1)^{U_q}$$

one traps $a < m^\vartheta < a+1$, so $m^\vartheta \notin \mathbb{Z}$. The $2^x < 256$ stage verifies per-value certificates by `norm_num` with three enclosure tiers (exponent sizes ∼500, ∼1500, ∼25000). The extension to 4096 verifies 3836 certificates through a single kernel-replayed `decide` over a table generated by PARI/GP and revalidated independently with exact big-integer arithmetic in Python. Five tiers suffice, with tier populations 78, 488, 3241, 28, 1. The further extension to 8192 (`FiniteCheck8192`) replays 4095 certificates the same way, in nine tiers built from the convergent enclosures below together with one *semiconvergent*, $478245/301739 < \vartheta$, which is needed only for $m = 5143$ and $m = 6714$ (the two values in $[4096, 8192)$ whose m^ϑ lies within $4 \cdot 10^{-4}$ of an integer); the next convergent would have required exponents near $1.7 \cdot 10^7$, while the semiconvergent keeps the largest exponent at 478245 and the whole table replays in under two minutes. The tier populations for $4096 \leq m < 8192$ are 826, 3120, 71, 12, 33, 2, 2, 28, 2. The extension to 16384 (`FiniteCheck16384`) replays 8191 rows in 39 tiers drawn from the semiconvergent ladder; its hardest value, $m = 9329$, needs the enclosure $5914137/3731405$ (exponents near $6 \cdot 10^6$, integers of 78 Mbit), which the kernel's GMP-backed `Nat.pow` compares in about ten seconds, and the whole table replays in under eight minutes. The tables were generated by a single script (`mpmath` floor, exact big-integer re-verification of every row, tiers as distinct enclosure pairs), so each further doubling is mechanical. For the 4096 table:

tier	lower convergent L_p/L_q	upper convergent U_p/U_q
0	569/359	485/306
1	1054/665	1539/971
2	25781/16266	24727/15601
3	50508/31867	125743/79335
4	176251/111202	301994/190537

Only the value $m = 2762$, whose ϑ-power is within $3 \cdot 10^{-9}$ of an integer in logarithmic scale, needs tier 4; its certificate compares integers of roughly 90,000 digits, which the kernel's exact arithmetic replays without difficulty. No axioms beyond the Lean kernel are involved; in particular no `native_decide`.

3.3 Arithmetic descent

Theorem 3.4 (Affine descent)**.** *If the two integral outputs factor simultaneously as $2^x = 2^r u^k$ and $3^x = 3^r v^k$ with $r \geq 0$, $k \geq 1$, then $(x-r)/k$ is again a solution. Combined with the $x \geq 8$ region this gives the unconditional constraint*

$$r + 8k \leq x$$

for nonintegral x; in particular common perfect-power shapes of the two outputs are strongly restricted.

Status: [kernel-verified Lean] — `affine_descent_integral_powers`, `base_depth_add_eight_mul_degree_le`, `min_output_base_factorization_add_eight_le` in `ArithmeticRigidity`. With the 4096 region the constant 8 improves to 12 by bookkeeping; with the interval scan of Section 23 it would improve to 20 at the [software computation] level.

For the *least* counterexample the same descent later becomes an exact primitivity statement in Section 4: no nontrivial matched decomposition exists at all.

3.4 Local progression obstructions and zero density

The function $t \mapsto t^{\vartheta}$ is strictly convex with second differences at moderate steps lying strictly between 0 and 1 — hence never integers.

Theorem 3.5 (Committed AP obstruction)**.** *Let $n, h \geq 1$ with $h^5 \leq n$. Then $n, n+h, n+2h$ are not all candidates. In particular no three consecutive naturals are candidates, and among the positive integers up to $3N$ at most $2N$ are candidates.*

Status: [kernel-verified Lean] — `not_three_arithmetic_progression_twoBaseNaturalCandidates`, `not_three_consecutive_logThreeDivLogTwo_rpow_integers`, `twoBaseNaturalCandidate_Icc_card_le`. The sharper criterion $\vartheta(\vartheta-1)h^2 < n^{2-\vartheta}$ is kernel-verified in `SharperProgressions`, and the analogous four-term criterion is kernel-verified in `ThirdDifference`. The arbitrary-order statement for every $r \geq 2$ is `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` in `HigherDifference`, which formalizes a generic iterated forward-difference mean-value identity and applies uniformly in the order r.

For finite verification the same curvature mechanism is also available in a purely rational effective form: if $h^{400} \leq n^{83}$ — step exponent $83/400 = 0.2075$, against the $1/5$ of Theorem 3.5 — then $n, n+h, n+2h$ are not all candidates.

Status: [kernel-verified Lean] — `not_three_arithmetic_progression_twoBaseNaturalCandidates_sharp` in `SharpProgression`, with the kernel-replayed enclosure $\vartheta < 317/200$ obtained from the exact integer comparison $3^{200} < 2^{317}$.

The hypothesis bounding h is essential: for steps comparable to n the second difference is large and the obstruction vanishes. This locality is a genuine barrier — a *global* progression obstruction would, for example, forbid candidates with odd part $2^s \pm 1$ via progressions such as $(2^i, 2^{i+t-1}, 2^i(2^t-1))$, but no such global statement is available; under a counterexample the exact monoid of Section 4 shows sparse candidates whose additive relations the curvature argument cannot see.

Feeding the local obstruction into the theory of Roth numbers gives quantitative sparsity.

Theorem 3.6 (Zero density). *For every $H \geq 1$ the number of candidates below N is at most $H^5 + \lceil (N - H^5)/H \rceil\, r_3(H)$, where $r_3(H)$ is the maximal size of a 3-AP-free subset of $\{0, \ldots, H-1\}$. Consequently the candidate counting function is $o(N)$: the candidates, and a fortiori the values 2^x of nonintegral solutions, have asymptotic density zero.*

Status: [kernel-verified Lean] — `twoBaseNaturalCandidatesBelow_card_le`, `twoBaseNaturalCandidatesBelow_isLittleO_id`, `tendsto_twoBaseNonintegerCandidatesBelow_density_zero` in `ZeroDensity` and `NonintegerDensity`.

3.5 Transcendence

The corpus contains a complete Lean proof of the Gelfond–Schneider theorem (Hilbert's seventh problem), following Hua's exposition [17], ported from Michail Karatarakis's Mathlib formalization (Mathlib PR #42911, Apache 2.0).

Theorem 3.7 (Gelfond–Schneider). *If α, β are algebraic, $\alpha \neq 0, 1$, and β is irrational, then α^β is transcendental.*

Status: [kernel-verified Lean] — `GelfondSchneider.transcendental_cpow_of_isAlgebraic_of_irrational`.

Corollary 3.8 (Algebraic–integral dichotomy). *If $2^x \in \mathbb{Z}$ then x is either an integer or transcendental; algebraicity of such x is equivalent to integrality. Every nonintegral two-base solution is transcendental, and $\vartheta = \log_2 3$ is transcendental.*

Status: [kernel-verified Lean] — `transcendental_of_not_integer_of_two_rpow_integer`, `isAlgebraic_iff_integer_of_two_rpow_integer`, `transcendental_of_not_integer_of_two_three_rpow_integer`, `transcendental_logThreeDivLogTwo`.

The dichotomy does not resolve the conjecture — the unknown exponent is allowed to be transcendental — but it constrains every construction below: the primitive generator β of Section 4 is transcendental, as is $\log_2 w$ for the odd core.

3.6 Unconditional gap equivalences

Because natural multiples of a solution are solutions, one nonintegral solution generates infinitely many, with irrational generator; Dirichlet approximation and a rotation-stepping argument spread their fractional parts densely.

Theorem 3.9 (Density dichotomy). *If some $x \notin \mathbb{Z}$ satisfies $2^x, 3^x \in \mathbb{Z}$, then for every $t \in [0, 1]$ and $\varepsilon > 0$ there is a nonintegral solution y with $|\{y\} - t| < \varepsilon$; and the set of nonintegral solutions is infinite.*

Theorem 3.10 (Gap equivalences). *Each of the following is equivalent to Conjecture 1.1:*

(i) *some nonempty open interval $(u, v) \subseteq (0, 1)$ contains no fractional part of a nonintegral solution;*

(ii) *the fractional parts of nonintegral solutions are bounded away from 1;*

(iii) *the fractional parts of nonintegral solutions are bounded away from 0.*

Status: [kernel-verified Lean] — `dense_fract_of_noninteger_solution`, `infinite_noninteger_solutions_of_exists`, `alaogluErdosConjecture_iff_fract_avoids_interval`, `alaogluErdosConjecture_iff_fract_bounded_away_one`, `alaogluErdosConjecture_iff_fract_bounded_away_zero` in `FractionalPartDensity`, built on the reusable `exists_pos_nat_fract_close_of_irrational`.

Remark 3.11 (Consistency pincer). Theorem 3.1 bounds $\{x\}$ away from 0 and 1 by margins $\asymp 2^{-\lfloor x \rfloor}$ that decay with the height of the solution, while Theorem 3.9 produces fractional parts arbitrarily close to 0 and 1 along multiples of unbounded height. The two results meet edge-to-edge without contradiction, and Theorem 3.10 shows the decay is essential: *any* height-independent gap, however small and wherever located, is exactly as strong as the full conjecture. The elementary gap bounds are therefore of optimal shape, and the fractional-part axis measures the entire remaining distance to Conjecture 1.1.

These equivalences are *unconditional* biconditionals. Conditional on failure, the exact structure of the next section gives kernel-verified convergence for every finite trigonometric polynomial and the paper-level strengthening to full blockwise Weyl equidistribution (Section 4).

4 Exact conditional structure of all solutions

The most consequential fact in the current corpus is not another exclusion range. It is the complete description of *all* solutions under the hypothesis that even one nonintegral solution exists. Conditional on failure of Conjecture 1.1, the solution set is not merely sparse or contained in some rank-two set: it is a free rank-two additive monoid with an explicit canonical generator, exactly countable in every dyadic block, and equidistributed blockwise on the circle.

Throughout this section we write

$$\mathcal{S} := \{x \in \mathbb{R} : 2^x \in \mathbb{N} \text{ and } 3^x \in \mathbb{N}\}, \qquad \mathcal{C} := \{2^x : x \in \mathcal{S}\}, \qquad \mathcal{P} := \{(2^x, 3^x) : x \in \mathcal{S}\},$$

and we say a counterexample exists when $\mathcal{S} \not\subseteq \mathbb{N}_0$. The canonical primitive odd core, the rational-power index, the power-of-three denominator normalization, the unique least solution, the simultaneous primitive output normalization, the exact free-monoid classification, the exact unit-block and dyadic counts, the cumulative quadratic asymptotic, and the finite-Fourier block averages are all **[kernel-verified Lean]**. Only the extension of the block averages from finite trigonometric polynomials to arbitrary continuous tests or interval indicators (Theorem 4.10) remains **[paper proof]**.

4.1 The primitive common odd core

Theorem 4.1 (Primitive common odd core)**.** *Assume $\mathcal{C}$ contains a non-power of two. Then there is a unique odd integer $w > 1$ that is power-free in the sense*

$$\gcd\{v_p(w) : p \mid w\} = 1,$$

and every candidate $b \in \mathcal{C}$ has a unique representation

$$b = 2^i w^j, \qquad i, j \in \mathbb{N}_0.$$

Proof. Choose a non-power-of-two candidate a. Its odd valuation vector $A = (A_q)_{q \neq 2}$, $A_q = v_q(a)$, is nonzero and finitely supported. Let

$$g = \gcd\{A_q : A_q > 0\}, \qquad c_q = A_q/g, \qquad w = \prod_{q \neq 2} q^{c_q}.$$

Then w is odd, greater than one, and the positive c_q have gcd one.

Let $b \in \mathcal{C}$ and set $B_q = v_q(b)$. Fix an odd prime p with $A_p > 0$. By the cross-valuation identity of Section 3, $A_p B_q = B_p A_q$ for all odd q. If $A_q = 0$, this forces $B_q = 0$. On the support of A it says

$$B_q = \frac{B_p}{A_p} A_q = \Big(\frac{gB_p}{A_p}\Big) c_q.$$

Put $t = gB_p/A_p \in \mathbb{Q}_{\geq 0}$. Every tc_q is an integer. Since the c_q have gcd one, there are integers r_q, finitely many nonzero, with $\sum r_q c_q = 1$; therefore $t = \sum_q r_q(tc_q) \in \mathbb{Z}$. Writing $j = t \in \mathbb{N}_0$, the odd part of b is exactly w^j, so $b = 2^i w^j$ with $i = v_2(b)$.

Uniqueness of (i, j) for fixed w follows from the odd valuations and the 2-adic valuation. If w' is another primitive common core, the odd valuation vectors of w and w' generate the same rational ray and are both primitive integer vectors, hence equal. □

Status: **[kernel-verified Lean]**

`existsUnique_primitive_common_odd_core` in `PrimitiveOddCore`. The module defines `oddPrimeValuationGCD`, proves its equivalence with power-theoretic primitivity for $w > 1$, and kernel-checks both uniqueness of the normalized core and uniqueness of every pair (i, j). The earlier non-normalized existence statement `exists_common_odd_core` recorded in Section 3 is subsumed by it.

The gcd normalization is exactly what makes w canonical: it prevents a hidden perfect power from being absorbed into the exponent j, so that both the core and the coordinates are determined by $\mathcal{C}$ alone.

Set

$$\alpha := \log_2 w, \qquad E := 3^\alpha = w^\vartheta.$$

Every candidate exponent then has the form $x = i + j\alpha$. The number α is irrational (otherwise the rational-exponent lemma applied to $2^\alpha = w$ would make it integral, forcing the odd $w > 1$ to be a power of two), and in fact transcendental by the algebraic–integral dichotomy of Section 3. The candidate condition becomes

$$3^{i+j\alpha} = 3^i E^j \in \mathbb{N},$$

and at this stage the admissible pairs (i, j) still form an unknown subset of $\mathbb{N}_0^2$. The next subsection shows that this subset is completely rigid.

4.2 The rational-power index and the denominator normalization

Because a non-power-of-two candidate exists, some $j > 0$ has $E^j \in \mathbb{Q}$. Let

$$d := \min\{j \in \mathbb{N}_{>0} : E^j \in \mathbb{Q}\}.$$

Lemma 4.2 (All rational powers are multiples of the least one)**.** *For every $j \in \mathbb{Z}$: $E^j \in \mathbb{Q} \iff d \mid j$.*

Proof. The set $J := \{j \in \mathbb{Z} : E^j \in \mathbb{Q}\}$ is an additive subgroup of $\mathbb{Z}$ (it contains 0, products add exponents, and reciprocals negate them), and every nonzero subgroup of $\mathbb{Z}$ equals $d\mathbb{Z}$ for its least positive element d. □

Status: [kernel-verified Lean]

`rationalPowerIndices`, `AddSubgroup.Int.exists_least_positive_generator`, and `exists_rationalPowerIndex` in `RationalPowerIndex`.

Write the positive rational E^d in lowest terms. Its denominator has no prime other than 3: taking a candidate with odd-core exponent $j = dk > 0$ and 2-exponent i, integrality of $3^i(E^d)^k$ shows that any other prime in the reduced denominator would survive in the k-th power. Hence there are unique integers $a \geq 0$ and $A \geq 1$ with

$$E^d = \frac{A}{3^a}, \qquad \text{and} \qquad 3 \nmid A \ \text{ when } a > 0, \tag{3}$$

so that $a = 0$ or $3 \nmid A$ in every case. Define

$$\beta := a + d\alpha = a + d\log_2 w, \qquad M := 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^a E^d. \tag{4}$$

Thus β is itself a solution, and it is irrational because α is. (Where the denominator normalization is written with a named integer numerator c, the identity $E^d = c/3^a$ with $a = 0$ or $3 \nmid c$ is exactly (3) with $c = A = 3^\beta$.)

Status: [kernel-verified Lean]

The denominator-support step is `exists_three_pow_denominator_of_rational_power` in `ThreeDenominatorNormalization`; the normalized integrality criterion is `three_pow_mul_normalized_pow_integer_iff` in `RationalPowerIndex`.

Equation (3) also has a field-theoretic reading: E is a root of the binomial $X^d - A/3^a$, which is the minimal polynomial of E over $\mathbb{Q}$, so $[\mathbb{Q}(E) : \mathbb{Q}] = d$. That computation, together with the equivalent one-number and localized-radical formulations of the conjecture that follow from it, is developed in Section 5; we do not repeat it here.

4.3 The exact primitive-generator theorem

Theorem 4.3 (Exact primitive generator and solution monoid)**.** *Assume Conjecture 1.1 is false. With w, d, a, β, M, A as above:*

(i) *every solution has a unique representation $x = n + k\beta$ with $n, k \in \mathbb{N}_0$, that is,*

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\};$$

(ii) *every candidate has a unique representation $m = 2^n M^k$ with $n, k \in \mathbb{N}_0$, that is,*

$$\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\};$$

(iii) *every output pair has a unique representation $(2^x, 3^x) = (2^n M^k, 3^n A^k)$;*

(iv) *β is the unique least nonintegral solution.*

Proof. Let $x \in \mathcal{S}$. By Theorem 4.1, $2^x = 2^i w^j$ for unique $i, j \in \mathbb{N}_0$, so $x = i + j\alpha$. From $3^x = 3^i E^j \in \mathbb{N}$ we get $E^j \in \mathbb{Q}$, so Lemma 4.2 gives $j = dk$ for some $k \in \mathbb{N}_0$. Using (3),

$$3^x = 3^i (E^d)^k = 3^{\,i-ak} A^k.$$

Because $A/3^a$ is reduced, this is an integer exactly when $i \geq ak$. Put $n = i - ak \in \mathbb{N}_0$; then

$$x = i + dk\alpha = n + k(a + d\alpha) = n + k\beta.$$

Conversely, for any $n, k \in \mathbb{N}_0$,

$$2^{\,n+k\beta} = 2^n M^k \in \mathbb{N}, \qquad 3^{\,n+k\beta} = 3^n A^k \in \mathbb{N},$$

so $n + k\beta \in \mathcal{S}$. If $n + k\beta = n' + k'\beta$ with $k \neq k'$, then $\beta = (n' - n)/(k - k') \in \mathbb{Q}$, a contradiction; hence the representation is unique. This proves (i), and (ii)–(iii) follow by exponentiation. A nonintegral solution has $k \geq 1$; the least such value is attained at $(n, k) = (0, 1)$ and equals β, uniquely so by uniqueness of the representation. ∎

Status: [kernel-verified Lean]

`exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture` in `PrimitiveGenerator`, built from `exists_exact_solution_monoid_of_fixed_common_oddCore` in `RationalPowerIndex`. The Lean statement supplies the canonical odd core w, the rational-power index d, the depth a, the numerator A, and the unique-coordinate clauses in one package. Conditional on failure, $\mathcal{S}$ is not merely contained in a rank-two additive set: every solution has a unique pair of coordinates in the free additive monoid on 1 and the irrational least generator β. The set of occurring odd-core exponents is exactly $d\,\mathbb{N}_0$, with the power of two governed by one linear threshold.

Packaged algebraically, Theorem 4.3 says that

$$(\mathbb{N}_0^2,+) \xrightarrow{\sim} (\mathcal{S},+), \quad (n,k) \mapsto n + k\beta, \qquad (\mathbb{N}_0^2,+) \xrightarrow{\sim} (\mathcal{C},\cdot), \quad (n,k) \mapsto 2^n M^k,$$

are monoid isomorphisms, and that the output-pair monoid $\mathcal{P}$ is freely generated by $(2,3)$ and (M,A). Every multiplicative relation among candidates is the obvious coordinate relation; there is no room for an unexpected coincidence. This exact conditional model is the benchmark against which any future density, progression, or descent argument must be tested.

The phrase "least nonintegral" is materially stronger than choosing an arbitrary counterexample. It forces affine primitivity: no nontrivial common root can be extracted from the two output integers while preserving the same base-adic offset. It is the arithmetic analogue of choosing a primitive lattice direction, and the next subsection turns it into exact exclusions.

4.4 Primitivity of the least output pair

Corollary 4.4 (No matched base depth)**.** *For the primitive pair $(M,A) = (2^\beta, 3^\beta)$ we have $\min\{v_2(M)\,, v_3(A)\} = 0$; that is, M is odd or $3 \nmid A$ (possibly both).*

Proof. If both valuations were positive, then $2^{\beta-1} = M/2$ and $3^{\beta-1} = A/3$ would be integers, so $\beta - 1$ would be a smaller nonintegral solution. ☐

Corollary 4.5 (No common perfect-power degree)**.** *There is no $k \geq 2$ for which both M and A are perfect k-th powers.*

Proof. If $M = U^k$ and $A = V^k$, then $2^{\beta/k} = U$ and $3^{\beta/k} = V$, so β/k is a smaller nonintegral solution. ☐

Corollary 4.6 (Exact affine primitivity)**.** *If $M = 2^r U^k$ and $A = 3^r V^k$ with $r \in \mathbb{N}_0$, $k \in \mathbb{N}_{>0}$, and $U, V \in \mathbb{N}$, then $r = 0$ and $k = 1$.*

Proof. The affine descent theorem of Section 3 produces the solution $(\beta - r)/k$, which is nonintegral since β is irrational; if $r > 0$ or $k > 1$ it is strictly smaller than β, contradicting leastness. ☐

Status: [kernel-verified Lean]

`exists_least_twoBaseNonintegerSolution`, `IsLeastTwoBaseNonintegerSolution.unique`, `IsLeastTwoBaseNonintegerSolution.not_both_base_dvd_outputs`, `IsLeastTwoBaseNonintegerSolution.no_common_perfect_power`, and `IsLeastTwoBaseNonintegerSolution.affine_primitive` in `LeastSolution`.

The odd-core normalization of M has a symmetric counterpart on the other output, and the two can be imposed simultaneously.

Theorem 4.7 (Simultaneous primitive output normalization). *If the conjecture fails, its least nonintegral solution β admits simultaneous factorizations*

$$M = 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^b v^e,$$

where $w > 1$ is odd and power-primitive, $v > 1$ is power-primitive with $3 \nmid v$, and $d, e > 0$. Moreover $a = 0$ or $b = 0$; the exponents a and b are the exact base-adic depths; d and e are the gcds of the prime valuations of the corresponding base-free parts; and

$$\gcd(\gcd(d, e),\, |b - a|) = 1.$$

Since every counterexample satisfies $\beta \geq 12$, these normalizations also obey the size dichotomy

$$w^d \geq 4096 \qquad \text{or} \qquad v^e \geq 3^{12}.$$

Status: [kernel-verified Lean]

`exists_simultaneous_primitive_output_normalization`, `IsLeastTwoBaseNonintegerSolution.simultaneousCoordinateGCD_eq_one`, and `exists_large_primitiveCore_of_not_alaogluErdosConjecture` in `OutputNormalization`. The gcd-one statement simultaneously controls the two outer power degrees and the difference of the two base-adic depths; it is strictly stronger than merely excluding a common perfect-power degree (Corollary 4.5).

These restrictions are strong, but they still admit infinitely many formal integer parameter patterns. The unresolved analytic content is untouched: the pair (M, A) must satisfy

$$\log_2 M = \log_3 A,$$

and no known local valuation argument turns the depth dichotomy $a = 0$ or $b = 0$, or the gcd-one condition, into a contradiction. Every constraint proved so far is compatible with the existence of such a pair; what fails to exist must fail for a transcendence reason, not a valuation-theoretic one.

4.5 Exact counting

The free-monoid description makes the population of a hypothetical counterexample completely explicit, block by block.

Theorem 4.8 (Exact unit-block and dyadic counts)**.** *Assume a counterexample exists and let β be the primitive generator. For every $N \in \mathbb{N}_0$, the nonintegral solutions in $[N, N+1)$ are precisely*

$$x_{N,k} = \lceil N - k\beta \rceil + k\beta, \qquad 1 \le k \le \Big\lfloor \frac{N+1}{\beta} \Big\rfloor,$$

so their number is exactly $\lfloor (N+1)/\beta \rfloor$. Consequently

$$\#(\mathcal{S} \cap [N, N+1)) = \#(\mathcal{C} \cap [2^N, 2^{N+1})) = 1 + \Big\lfloor \frac{N+1}{\beta} \Big\rfloor.$$

Proof. For fixed $k \ge 1$ there is at most one integer n with $N \le n + k\beta < N+1$; such an $n \ge 0$ exists if and only if $k\beta < N+1$, and then $n = \lceil N - k\beta \rceil$. Irrationality of β makes $(N+1)/\beta$ nonintegral, which gives the stated range. The unique integral solution in the block is N itself, whence the additional 1; passing through $m = 2^x$ turns the exponent block $[N, N+1)$ into the magnitude block $[2^N, 2^{N+1})$ bijectively. □

Status: [kernel-verified Lean]

`ExactCounting` proves the explicit ceiling-image parametrization, `twoBaseNonintegerSolutionsInUnitBlock_ncard`, `twoBaseNaturalCandidatesInDyadicBlock_card`, and the exact finite-sum identity `cumulativeTwoBaseCandidateCount_eq`. `QuadraticAsymptotic` identifies that cumulative count with the candidate count below 2^T and proves the sharp bounds and the limit in (5).

In particular a failed conjecture would produce only linearly many candidates in the logarithmic block index,

$$\#(\mathcal{C} \cap [X, 2X]) \asymp \log X,$$

and the kernel-verified zero-free region gives $\beta \ge 12$, hence $\beta > 12$ because β is irrational. The number of *nonintegral* solutions in the block is therefore at most $\lfloor N/12 \rfloor$, while the full solution count — equivalently the dyadic candidate count — is at most $1 + \lfloor N/12 \rfloor$. Under the directed-rounding scan these bounds improve respectively to $\lfloor N/27 \rfloor$ and $1 + \lfloor N/27 \rfloor$, at the weaker **[software computation]** trust level. Summing over blocks, the cumulative candidate count below 2^T is

$$C(T) = T + \sum_{r=1}^{T} \Big\lfloor \frac{r}{\beta} \Big\rfloor = \frac{T^2}{2\beta} + O(T). \tag{5}$$

More precisely, Lean proves the two-sided bound and the limit

$$\frac{T(T+1)}{2\beta} \le C(T) \le \frac{T(T+1)}{2\beta} + T, \qquad \frac{C(T)}{T^2} \longrightarrow \frac{1}{2\beta}.$$

This is conditionally sharper than the unconditional verified bounds ($(c\log_2 N)^2$ and $c\log_2 N \cdot c\log_3 N$ of Section 3) and matches them in leading order: with $\beta > 12$ the leading coefficient $1/(2\beta)$ is below $1/24$.

There is also an unconditional growth reformulation, obtained by comparing the two cases. Under the conjecture the count below 2^T is exactly T; under failure the preceding limit is the strictly positive constant $1/(2\beta)$. Hence

$$\text{Conjecture 1.1 holds} \iff \frac{\#(\mathcal{C} \cap [1, 2^T))}{T^2} \longrightarrow 0, \tag{6}$$

which is `alaogluErdosConjecture_iff_candidate_count_below_two_pow_subquadratic`. [kernel-verified Lean]

This exact population is the correct benchmark for every proposed density or determinant argument. A method that proves only $O((\log X)^2)$ candidates in $[X, 2X]$, for instance, is a valid unconditional theorem but cannot contradict Theorem 4.8: it is weaker than the conditional truth it would need to refute.

4.6 Blockwise irrational rotation and equidistribution

By Theorem 4.8 the fractional part of $x_{N,k}$ is $\{k\beta\}$, so a high unit block contains exactly an initial segment

$$\{\beta\},\ \{2\beta\},\ \ldots,\ \{K_N\beta\}, \qquad K_N = \Big\lfloor \frac{N+1}{\beta} \Big\rfloor,$$

of one irrational rotation orbit, and $K_N \to \infty$.

Theorem 4.9 (Finite Fourier tests on exact blocks)**.** *Assume a counterexample exists and let β be its primitive generator. For every finitely supported coefficient function $c : \mathbb{Z} \to \mathbb{C}$, put*

$$P_c(t) = \sum_{h\in\mathbb{Z}} c(h)e^{2\pi i h t}.$$

Then, on the exact ceiling-parametrized points $x_{N,k} = \lceil N - k\beta \rceil + k\beta$,

$$\frac{1}{K_N}\sum_{k=1}^{K_N} P_c(x_{N,k}) \longrightarrow c(0) \qquad (N \to \infty).$$

Status: [kernel-verified Lean]

`BlockWeyl` proves vanishing of every nonzero normalized Fourier mode on the exact block points, and `BlockEquidistribution` proves `tendsto_primitiveUnitBlockTrigonometricAverage` for every finitely supported coefficient function, with limit `c 0`. This is convergence against finite trigonometric polynomials; it is not yet a theorem for arbitrary continuous tests.

Theorem 4.10 (Blockwise Weyl equidistribution)**.** *Assume a counterexample exists. For every interval $I \subseteq [0, 1]$ whose boundary has measure zero,*

$$\frac{\#\{x \in (\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1) : \{x\} \in I\}}{\#((\mathcal{S} \setminus \mathbb{N}_0) \cap [N, N+1))} \longrightarrow |I| \qquad (N \to \infty).$$

Proof. By Theorem 4.8 the relevant fractional parts are $\{\beta\}, \{2\beta\}, \ldots, \{K_N\beta\}$ with $K_N = \lfloor (N+1)/\beta \rfloor \to \infty$. For every nonzero integer h the Weyl sum is a finite geometric series,

$$\frac{1}{K}\sum_{k=1}^{K} e^{2\pi i h k\beta} = \frac{e^{2\pi i h\beta}\big(1 - e^{2\pi i h K\beta}\big)}{K\big(1 - e^{2\pi i h\beta}\big)} \longrightarrow 0,$$

since β is irrational and hence $e^{2\pi i h\beta} \neq 1$. Weyl's criterion concludes. □

Status: [paper proof]

The finite-trigonometric-polynomial input is kernel-verified by Theorem 4.9, but the Stone–Weierstrass and Weyl-criterion extension to all continuous tests, and then to interval indicators, has not been formalized. No effective discrepancy rate follows without further Diophantine input on β: the geometric-sum proof controls each fixed Fourier mode but not uniformly over many modes.

This regularity is attractive, and it is worth stating plainly why it does not by itself decide anything. The integrality condition being tested is not a fixed positive-measure target in the circle; it is an exponentially shrinking target, since the relevant approximation quality must improve like the reciprocal of the integer sizes involved. Equidistribution at a fixed scale is simply the wrong instrument for a target whose measure tends to zero geometrically, and that mismatch recurs in the analytic approaches examined later in this report.

5 Equivalent one-number and radical formulations

The exact conditional structure of Section 4 compresses a statement quantified over all real x into a statement about a single real number at a time. This section records that compression in its two kernel-verified forms — a localization at the prime 3 and a radical-algebraicity normalization — and then names the exact arithmetic obstruction that remains.

5.1 Localization at the prime three

Theorem 5.1 (Localization reformulation)**.** *Conjecture 1.1 is false if and only if there exists an odd integer $u > 1$ with*

$$u^{\vartheta} \in \mathbb{Z}[1/3],$$

equivalently, an odd $u > 1$ and $a \in \mathbb{N}_0$ with $3^a u^{\vartheta} \in \mathbb{N}$.

Proof. If x is a counterexample, write $2^x = 2^a u$ with u odd; since 2^x is not a power of two, $u > 1$. Using $(2^a)^{\vartheta} = 3^a$,

$$u^{\vartheta} = \frac{(2^x)^{\vartheta}}{(2^a)^{\vartheta}} = \frac{3^x}{3^a} \in \mathbb{Z}[1/3].$$

Conversely, if $u^{\vartheta} = B/3^a$ with $B \in \mathbb{N}$, set $x = a + \log_2 u$; then $2^x = 2^a u \in \mathbb{N}$ and $3^x = 3^a u^{\vartheta} = B \in \mathbb{N}$, and $x \notin \mathbb{Z}$ since otherwise the odd $u > 1$ would be a power of two. ☐

Status: [kernel-verified Lean] —
`alaogluErdosConjecture_iff_odd_rpow_not_localized` in `Localization` and `alaogluErdosConjecture_iff_oddLocalizedRadicalExclusion` in `RadicalReformulation`; the candidate-level variant `alaogluErdosConjecture_iff_no_odd_multiple_candidate`, also in `Localization`, is proved independently and phrases the same content in terms of admissible candidate bases. The equivalence is kernel-verified; the localized-radical exclusion on its right-hand side remains the open conjecture. The allowance of 3-power denominators is exactly the freedom created by shifting the exponent by an integer.

The denominator restriction to powers of 3 is not cosmetic. Only the prime 3 can occur in the denominator, because the exponent shift $x \mapsto x+1$ multiplies 2^x by 2 and 3^x by 3; the odd part u absorbs the 2-adic freedom completely and $\mathbb{Z}[1/3]$ absorbs the remaining 3-adic freedom. Everything else has been eliminated.

5.2 Reduction to power-primitive odd bases

An odd $u > 1$ that is a proper power, $u = w^d$ with w odd and power-primitive, satisfies $u^\vartheta = (w^\vartheta)^d$. Unique primitive-power decomposition therefore lets the quantifier in Theorem 5.1 range over power-primitive bases only, at the cost of an extra exponent $d \geq 1$.

Status: [kernel-verified Lean] — `PrimitiveRadical` proves unique primitive-power decomposition of every integer $u > 1$ together with the exact equivalence `alaogluErdosConjecture_iff_primitiveOddLocalizedRadicalExclusion`. Proper-power choices of u introduce no genuinely new cases.

5.3 Canonical radical algebraicity

In the notation of the least generator of Section 4, a counterexample produces natural numbers w, d, a, A with $w > 1$ odd and power-primitive, $d > 0$, and

$$\beta = a + d\log_2 w, \qquad 2^\beta = 2^a w^d, \qquad 3^\beta = A,$$

normalized so that $a = 0$ or $3 \nmid A$. Setting

$$E := w^\vartheta = 3^{\log_2 w},$$

one has $E^d = A/3^a \in \mathbb{Q}$, and d is the least positive exponent for which E^d is rational. Minimality of d has an exact algebraic consequence.

Proposition 5.2 (Exact radical degree)**.** *The binomial $X^d - A/3^a$ is irreducible over $\mathbb{Q}$, it is the minimal polynomial of E, and E is algebraic of exact degree d.*

Proof. For a positive rational q, the binomial $X^d - q$ can be reducible only if q is a p-th power in $\mathbb{Q}$ for some prime $p \mid d$ (the additional $4 \mid d$ exceptional case concerns -4 times a fourth power and is irrelevant for $q > 0$). If $q = r^p$ with $p \mid d$, the positive real $E^{d/p}$ is the positive p-th root of q, hence rational, contradicting minimality of d. ☐

Status: [kernel-verified Lean] — `irreducible_X_pow_sub_C_of_least_rational_power` in `RadicalDegree` proves the general norm argument, and `oddCoreRpow_exact_radical_degree` specializes it to the least normalized radical index. It identifies the minimal polynomial with $X^d - A/3^a$ and proves that the degree is exactly d.

The same normalization also gives an exact translated-power norm. Fix $k \geq 1$ and $n \geq 0$, and put

$$g = \gcd(k, d), \qquad K = k/g, \qquad D = d/g, \qquad q = A/3^a = E^d.$$

Since the k-th powers of the d conjugates $\zeta_d^j E$ run through every D-th root of unity exactly g times, one obtains

Theorem 5.3 (Translated canonical-radical norm)**.**

$$\left|N_{\mathbb{Q}(E)/\mathbb{Q}}(E^k - 3^n)\right| = |q^K - 3^{nD}|^g = \frac{|A^K - 3^{aK+nD}|^g}{3^{ak}}. \tag{7}$$

The numerator is nonzero, and

$$|E^k - 3^n| = \frac{|A^K - 3^{aK+nD}|}{3^{aK}\sum_{j=0}^{D-1}(E^k)^{D-1-j}(3^n)^j} \geq \frac{3^{-aK}}{D\max(E^k, 3^n)^{D-1}}. \tag{8}$$

Indeed, multiplication over the conjugates gives $\pm((E^k)^D - 3^{nD})^g$ and $Kg = k$. If the numerator vanished, positivity would imply $E^k = 3^n$ and hence $w^k = 2^n$, impossible for odd $w > 1$; the last display is the ordinary factorization of $X^D - Y^D$. The identity compresses the entire conjugate product to one two-logarithm gap. It is therefore a precise calibration of norm-only arguments, not a claim that every possible use of selected or signed conjugates is exhausted. Baker's two-logarithm estimate [3] sharpens the elementary lower bound to polynomial relative separation, but remains compatible with the rational-return scale.

Status: [paper proof] — exact paper-level norm and factorization identities. They extend the degree and norm data above but are not yet formalized; the methodological conclusion is deliberately restricted to arguments that retain conjugates only through the full field norm.

A counterexample therefore produces the very special algebraic number $E = 3^{\log_2 w}$, reached from a transcendental exponent. Taking logarithms of $E^d = A/3^a$:

$$d\,\log w\,\log 3 = (\log A - a\log 3)\log 2. \tag{9}$$

This is a *bilinear* relation among logarithms of integers: each side is a product of two logarithms, not a linear form in logarithms with algebraic coefficients. Baker's theorem [3] bounds linear forms $\sum \alpha_i \log\gamma_i$ with algebraic α_i away from zero, and no substitution turns (9) into such a form, because the coefficient of $\log w$ on the left is $d\log 3$, itself a logarithm rather than an algebraic number. The four exponentials conjecture does forbid the relation. Equation (9) is thus a precise description of the transcendence-theoretic wall, not a disguised elementary factorization problem.

5.4 Where the wall stands in the smallest case

The smallest structural question already meets the open core. A 3-smooth candidate $m = 2^a 3^b$ with $b \geq 1$ would require $3^a E_3^b \in \mathbb{N}$ for

$$E_3 = 3^{\log_2 3} = e^{(\log 3)\log_2 3} = 5.7045224946911176\ldots,$$

and even the irrationality of E_3 is open: it is a "three-logarithms" quantity outside the reach of Gelfond–Schneider, Baker's method, and the six exponentials theorem. No currently known technique excludes the odd core $w = 3$; this is why the corpus emphasizes counting, structure, and verified finite regions rather than infinite families of exclusions, and why Conjecture 5.4 below is the precise name of the obstruction.

Conjecture 5.4 (Radical four-exponentials target)**.** *For every odd integer $u > 1$, $u^{\log_2 3} \notin \mathbb{Z}[1/3]$. Equivalently, for every power-primitive odd $w > 1$ and every $d \geq 1$, $(3^{\log_2 w})^d \notin \mathbb{Z}[1/3]$.*

By Theorem 5.1 and the primitive reduction above, both formulations are exactly equivalent to Conjecture 1.1; neither is proved. A proof of the special case $u = 3$ — even of the weaker assertion that $3^{\log_2 3}$ is irrational — by any genuinely new mechanism would already be a meaningful breakthrough.

6 The exact four-exponentials barrier

6.1 The two-by-two exponential matrix

Suppose x is a nonintegral solution. Consider the row vector of parameters $(1, x)$ and the column vector of parameters $(\log 2, \log 3)$. The four exponentials formed from the products are

$$\exp(1 \cdot \log 2) = 2, \quad \exp(1 \cdot \log 3) = 3, \quad \exp(x \log 2) = 2^x, \quad \exp(x \log 3) = 3^x,$$

all four algebraic — indeed positive integers. The two row parameters $1, x$ are $\mathbb{Q}$-linearly independent because x is irrational: a rational nonintegral exponent is excluded by the rational-exponent argument of Section 1. The two column parameters $\log 2, \log 3$ are $\mathbb{Q}$-linearly independent because $2^a 3^b = 1$ forces $a = b = 0$, that is, no nonzero powers of 2 and 3 agree. The real four exponentials conjecture predicts that at least one of the four displayed values must be transcendental, a contradiction [25].

Status: [kernel-verified Lean] —
`alaogluErdosConjecture_of_realFourExponentialsConjecture` in
`FourExponentialsReduction` formalizes exactly this implication. It does not assume four exponentials as an axiom: the conjectural statement is defined as a proposition and the implication is proved conditionally on it. Four exponentials itself is open, so this is context and diagnosis, not a proof of Conjecture 1.1.

The reduction is a diagnostic and not merely background. Any proposed elementary argument must exploit integrality, positivity, or arithmetic structure beyond bare algebraicity of those four values, or else it would prove a substantial case of the four exponentials conjecture by another name.

6.2 Why six exponentials does not specialize

The six exponentials theorem applies to two independent row parameters together with three independent column parameters, or to three rows and two columns. Here only two canonical logarithmic directions are available. Adding a third base 5 supplies the missing direction and yields the kernel-verified three-base theorem

$$2^x, 3^x, 5^x \in \mathbb{Z} \quad \Longrightarrow \quad x \in \mathbb{Z},$$

proved by an interpolation determinant. In the two-base problem, any additional base built from powers of 2 and 3 has logarithm in the rational span of $\log 2$ and $\log 3$ and therefore does not create an independent column.

The dimension count is what decides the outcome. With three independent row directions and two column directions the analytic size estimate beats the arithmetic lower bound $|\det| \geq 1$; in the 2×2 configuration the same balance lands exactly on the classical boundary, where the required decay below one fails. A successful two-base determinant argument would, by construction, be a new four-exponentials argument.

The classification of auxiliary bases makes the obstruction precise: `ThreeSmooth` and the denominator-controlled results in `RationalThirdBase` show that an auxiliary base with a prime factor outside $\{2, 3\}$ would supply a genuinely new column, but its corresponding exponential value is not forced to be an integer — or even algebraic — by the original hypotheses. Bases supported only on 2 and 3 do not increase the transcendence rank.

6.3 Integrality is stronger than algebraicity, but not by enough

Four exponentials asks only for algebraicity. The values here are integers, so there is extra discrete structure available, and the determinant machinery of Section 18 and Section 19 extracts some of it: the relevant determinants have integer entries, so a nonzero determinant is a nonzero integer and therefore satisfies $|\det| \geq 1$.

That inequality is the whole of the extra archimedean information. A nonzero integer has absolute value at least one, and nothing in the integrality hypotheses improves the constant 1 by itself. To cross the four-exponentials threshold one would need a strictly stronger arithmetic input, for instance:

- a proof that the determinants are divisible by a large integer, replacing $|\det| \geq 1$ by $|\det| \geq D$ with D growing in the size parameters; or
- a proof that a suitable common denominator is anomalously small, improving the arithmetic side rather than the analytic side.

Both routes ask for divisibility rather than size, which is precisely why the p-adic direction is given priority in the research program of Section 25.

Status: **[paper proof]** for the unconditional paper proofs of the determinant hierarchy. **[kernel-verified Lean]** for the accepted algebraic, integrality, nonvanishing, and positivity layers in `ArbitraryNodeDeterminant`, `SemigroupDeterminant`, `MixedExponentSemigroup`, and `OrderedChamberPositivity`. The quantitative Lean endpoints carry their still-missing calculus inputs as explicit theorem hypotheses rather than axioms. The reduction of Conjecture 1.1 to four exponentials is **[kernel-verified Lean]**, but four exponentials is open and the conjecture remains open with it.

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7 Rational-function rigidity and closure properties

The exact primitive-generator theorem of Section 4 determines the additive structure of $\mathcal{S}$ completely under failure of Conjecture 1.1. It has a consequence that deserves to be isolated: a hypothetical solution monoid is not merely additively free of rank two, it is *rigid* against every nonlinear rational operation over the algebraic numbers. Addition and multiplication by naturals are the only operations that stay inside $\mathcal{S}$, and every apparent exception collapses to an identity in one indeterminate. The results below are exact equivalences, several of them unconditional, and they translate directly into closure reformulations of the conjecture itself.

7.1 Evaluation injectivity at the generator

Throughout this section, β denotes the canonical least nonintegral solution supplied by the generator theorem of Section 4, so that

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}$$

with unique coordinates n, k. The transcendence dichotomy of Section 3 makes β transcendental over $\mathbb{Q}$. The rational-function statements need the slightly stronger form over the algebraic closure.

Lemma 7.1 (Evaluation is injective over $\overline{\mathbb{Q}}$)**.** *Let $\gamma \in \mathbb{C}$ be transcendental over $\mathbb{Q}$. Then no nonzero $P \in \overline{\mathbb{Q}}[T]$ satisfies $P(\gamma) = 0$; equivalently, the evaluation homomorphism $\overline{\mathbb{Q}}[T] \to \mathbb{C}$, $P \mapsto P(\gamma)$, is injective.*

Proof. If $P(\gamma) = 0$ for some nonzero $P \in \overline{\mathbb{Q}}[T]$, then γ is algebraic over $\overline{\mathbb{Q}}$. Since $\overline{\mathbb{Q}}/\mathbb{Q}$ is an algebraic extension, the tower property makes γ algebraic over $\mathbb{Q}$, contradicting the hypothesis. Injectivity follows by applying this to differences. □

Status: [paper proof] — the transcendence of β itself is **[kernel-verified Lean]** (Section 3, `Transcendence`); the only new content here is the standard tower argument that upgrades transcendence over $\mathbb{Q}$ to transcendence over $\overline{\mathbb{Q}}$.

7.2 The rational-function intersection theorem

Theorem 7.2 (Rational-function rigidity at the primitive generator)**.** *Assume Conjecture 1.1 is false and let β be the canonical generator. Let $R(T) \in \overline{\mathbb{Q}}(T)$ have no pole at β. Then*

$$\boxed{R(\beta) \in \mathcal{S} \quad \Longleftrightarrow \quad R(T) = n + kT \text{ in } \overline{\mathbb{Q}}(T) \text{ for some } n, k \in \mathbb{N}_0.}$$

Proof. The reverse implication is additive closure of $\mathcal{S}$ together with closure under multiplication by naturals: $R(\beta) = n + k\beta \in \mathcal{S}$.

For the forward implication write $R = P/Q$ with $P, Q \in \overline{\mathbb{Q}}[T]$ and $Q(\beta) \neq 0$. If $R(\beta) \in \mathcal{S}$, the exact coordinates give $R(\beta) = n + k\beta$ for unique $n, k \in \mathbb{N}_0$, hence

$$P(\beta) - (n + k\beta)\, Q(\beta) = 0.$$

The left-hand side is the value at β of the polynomial $P(T) - (n + kT)Q(T) \in \overline{\mathbb{Q}}[T]$. By Lemma 7.1 that polynomial is identically zero, so $P(T) = (n + kT)Q(T)$ and $R(T) = n + kT$ in $\overline{\mathbb{Q}}(T)$. □

Status: [paper proof]

The theorem allows poles away from β, permits arbitrary cancellation, and takes the full algebraic closure as coefficient field. It can be read as the exact intersection

$$\mathcal{S} \cap \overline{\mathbb{Q}}(\beta) = \mathbb{N}_0 + \mathbb{N}_0\beta, \tag{10}$$

with the extra information that every rational expression realizing an element of that intersection is *identically* the corresponding affine expression — there is no accidental representation.

Corollary 7.3 (Rational self-maps of the solution set)**.** *Under failure, the rational functions over $\overline{\mathbb{Q}}$ that are regular on $\mathcal{S}$ and map $\mathcal{S}$ into itself are exactly the maps*

$$T \longmapsto n + kT, \qquad n, k \in \mathbb{N}_0.$$

Proof. Such a function in particular sends β into $\mathcal{S}$, so Theorem 7.2 identifies it as $n + kT$ with $n, k \in \mathbb{N}_0$. Conversely, translation by a natural and dilation by a natural preserve $\mathbb{N}_0 + \mathbb{N}_0\beta$. □

Status: [paper proof]

Proposition 7.4 (Multivariate reduction to one indeterminate)**.** *Assume failure, let $x_i = n_i + k_i\beta \in \mathcal{S}$ for $i = 1, \ldots, r$, and let $F \in \overline{\mathbb{Q}}(X_1, \ldots, X_r)$ be regular at $(x_1, \ldots, x_r)$. Then*

$$F(x_1, \ldots, x_r) \in \mathcal{S}$$

if and only if the one-variable specialization

$$G(T) := F(n_1 + k_1T, \ldots, n_r + k_rT)$$

is identically $p + qT$ in $\overline{\mathbb{Q}}(T)$ for some $p, q \in \mathbb{N}_0$.

Proof. Write $F = P/Q$ with $P, Q \in \overline{\mathbb{Q}}[X_1, \ldots, X_r]$ and $Q(x_1, \ldots, x_r) \neq 0$. The specialized denominator $Q(n_1 + k_1T, \ldots, n_r + k_rT)$ is a polynomial in T that does not vanish at $T = \beta$, hence is not the zero polynomial. Therefore G is a well-defined element of $\overline{\mathbb{Q}}(T)$ with no pole at β, and $G(\beta) = F(x_1, \ldots, x_r)$. Now apply Theorem 7.2. □

Status: [paper proof]

Every rational operation on finitely many hypothetical solutions is thus reduced to an elementary identity between polynomials in a single indeterminate, with coordinates in $\mathbb{N}_0$. No analytic input is required beyond the transcendence already verified by the kernel.

7.3 Exact multiplication criterion

The next statement needs no assumption at all: the branch in which the conjecture holds is trivial, and the branch in which it fails is governed by the generator.

Theorem 7.5 (Products of two solutions)**.** *For all $x, y \in \mathcal{S}$, unconditionally,*

$$\boxed{xy \in \mathcal{S} \iff x \in \mathbb{N}_0 \text{ or } y \in \mathbb{N}_0.}$$

Proof. If one factor, say x, lies in $\mathbb{N}_0$, then xy is a natural multiple of a solution and therefore a solution; this direction is unconditional.

For the converse, split on Conjecture 1.1. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$ and the right-hand side is automatically true, so there is nothing to prove. Under failure, write the unique coordinates

$$x = n + k\beta, \qquad y = m + \ell\beta, \qquad n, k, m, \ell \in \mathbb{N}_0.$$

If neither factor is integral then $k, \ell > 0$ and

$$xy = nm + (n\ell + mk)\beta + k\ell\,\beta^2,$$

with $k\ell > 0$. Membership $xy \in \mathcal{S}$ would force $xy = r + s\beta$ for some $r, s \in \mathbb{N}_0$, producing a nonzero polynomial of degree exactly two over $\mathbb{Q}$ vanishing at the transcendental number β. Equivalently, apply Theorem 7.2 to $R(T) = (n + kT)(m + \ell T)$, which is not affine. Both routes are impossible. ☐

Status: [paper proof] — note that only transcendence over $\mathbb{Q}$ is used, which is [kernel-verified Lean]; the appeal to $\overline{\mathbb{Q}}$ in Theorem 7.2 is a convenience, not a necessity.

Corollary 7.6 (Finite products)**.** *Let $x_1, \dots, x_r \in \mathcal{S}$ be positive. Then*

$$\prod_{i=1}^{r} x_i \in \mathcal{S} \quad \Longleftrightarrow \quad \textit{at most one } x_i \textit{ is nonintegral.}$$

If a zero factor is allowed, the product is $0 \in \mathcal{S}$ and the criterion is vacuous.

Proof. If the conjecture holds, every x_i is integral and both sides are true. Under failure, the positive integral factors multiply to a positive natural, so after removing them the product is a natural multiple of a product of nonintegral solutions. Writing each remaining factor as $n_i + k_i\beta$ with $k_i > 0$ and expanding, the result is a polynomial in β whose degree equals the number of nonintegral factors and whose leading coefficient is positive. Degree at least two is excluded by Theorem 7.2, while degree zero or one is preserved by natural dilation and translation. ☐

Status: [paper proof]

Corollary 7.7 (Internal power test)**.** *For every $x \in \mathcal{S}$ and every integer $r \geq 2$,*

$$x^r \in \mathcal{S} \quad \Longleftrightarrow \quad x \in \mathbb{N}_0.$$

In particular $x \in \mathbb{N}_0$ holds exactly when $x^2 \in \mathcal{S}$: a single squaring test detects integrality.

Proof. Apply Theorem 7.5 with $y = x$ and induct, or apply Corollary 7.6 to r equal factors, noting that $x = 0$ satisfies both sides. ☐

Status: [paper proof]

7.4 Exact quotient criterion

The same evaluation argument classifies division. Division is more delicate than multiplication because even the true solution set $\mathbb{N}_0$ is not closed under arbitrary quotients, so the criterion cannot be a clean dichotomy; it is instead an exact divisibility condition on coordinates.

Theorem 7.8 (Quotients of solutions)**.** *Assume Conjecture 1.1 is false and write the unique coordinates*

$$x = n + k\beta, \qquad y = m + \ell\beta > 0.$$

Exactly one of the two cases $\ell = 0$ and $\ell > 0$ applies, and in that case $x/y \in \mathcal{S}$ if and only if the corresponding condition holds:

(a) *$\ell = 0$, so that $y = m$ is a positive natural, and $m \mid n$ and $m \mid k$;*

(b) *$\ell > 0$, and $x = qy$ for some $q \in \mathbb{N}_0$, equivalently $n = qm$ and $k = q\ell$.*

In particular, if x and y are both nonintegral, then

$$\frac{x}{y} \in \mathcal{S} \quad \Longleftrightarrow \quad \frac{x}{y} \in \mathbb{N}_0.$$

Proof. Suppose $x/y = r + s\beta$ with $r, s \in \mathbb{N}_0$. Clearing the denominator gives

$$n + k\beta = mr + (ms + \ell r)\beta + \ell s\,\beta^2.$$

The difference of the two sides is a polynomial in β with rational coefficients, so transcendence of β forces every coefficient to vanish:

$$\ell s = 0, \qquad k = ms + \ell r, \qquad n = mr.$$

If $\ell = 0$ then $y = m$ and $y > 0$ gives $m \geq 1$; the equations read $n = mr$ and $k = ms$, which is precisely $m \mid n$ and $m \mid k$. If $\ell > 0$ then $s = 0$, so $n = mr$ and $k = \ell r$, that is $x = ry$ with $r \in \mathbb{N}_0$. The converses follow by direct substitution.

For the final claim, both x and y nonintegral means $k, \ell > 0$, so case (b) applies and $x/y = r \in \mathbb{N}_0$. ☐

Status: [paper proof]

Corollary 7.9 (Reciprocals)**.** *Unconditionally, for $x \in \mathcal{S}$ with $x > 0$,*

$$\frac{1}{x} \in \mathcal{S} \quad \Longleftrightarrow \quad x = 1.$$

Proof. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$ and the claim is the elementary fact that a positive natural with natural reciprocal equals 1. Under failure apply Theorem 7.8 with numerator $1 = 1 + 0 \cdot \beta$ and denominator $y = x$. If $x = m$ is integral, case (a) gives $m \mid 1$, so $m = 1$. If x is nonintegral, case (b) gives $1 = qx$ for some $q \in \mathbb{N}_0$; here $q = 0$ is impossible, and $q \geq 1$ would make $x = 1/q$ rational, contradicting the irrationality of every nonintegral solution. ☐

Status: [paper proof]

A hypothetical solution monoid is therefore additively rich — a free additive monoid of rank two — and almost maximally hostile to every multiplicative operation: products, powers, reciprocals and quotients all escape $\mathcal{S}$ except in the degenerate cases forced by the additive structure itself.

7.5 Closure formulations of the conjecture

Because the multiplication criterion is unconditional, it converts directly into equivalent statements of the conjecture in terms of closure.

Theorem 7.10 (Multiplicative closure equivalences)**.** *The following are equivalent.*

(i) *Conjecture 1.1 is true.*

(ii) *$\mathcal{S}$ is closed under multiplication.*

(iii) *$\mathcal{S}$ is closed under squaring.*

(iv) *For every real x,*

$$2^x, 3^x \in \mathbb{Z} \quad \Longrightarrow \quad 2^{x^2}, 3^{x^2} \in \mathbb{Z}.$$

Proof. If the conjecture holds then $\mathcal{S} = \mathbb{N}_0$, which is closed under multiplication, so (i) implies (ii); and (ii) trivially implies (iii). Statements (iii) and (iv) are the same assertion, written once for $\mathcal{S}$ and once for the defining integrality conditions. Finally, assume (iii). For any $x \in \mathcal{S}$ we get $x^2 \in \mathcal{S}$, so Corollary 7.7 forces $x \in \mathbb{N}_0$. Hence $\mathcal{S} = \mathbb{N}_0$, which is Conjecture 1.1. □

Status: **[paper proof]**

The candidate formulation of Section 1 carries the same content in multiplicative language. For positive reals define the commutative operation

$$m \star n := 2^{(\log_2 m)(\log_2 n)} = m^{\log_2 n} = n^{\log_2 m}. \tag{11}$$

If $m = 2^x$ and $n = 2^y$, then $m \star n = 2^{xy}$, so $\star$ is exactly the transport of multiplication of exponents through the exponent–candidate bijection of Section 1.

Corollary 7.11 (Candidate star-product)**.** *For $m, n \in \mathcal{C}$,*

$$m \star n \in \mathcal{C} \quad \Longleftrightarrow \quad m \text{ is a power of } 2 \text{ or } n \text{ is a power of } 2.$$

Consequently Conjecture 1.1 is equivalent both to closure of $\mathcal{C}$ under $\star$ and to the apparently weaker diagonal condition

$$m \in \mathcal{C} \quad \Longrightarrow \quad m \star m \in \mathcal{C}.$$

Proof. Put $x = \log_2 m$ and $y = \log_2 n$, so that $x, y \in \mathcal{S}$ and $m \star n = 2^{xy}$. Membership $m \star n \in \mathcal{C}$ is equivalent to $xy \in \mathcal{S}$, and m is a power of two exactly when $x \in \mathbb{N}_0$; now apply Theorem 7.5. The two closure equivalences are the image of Theorem 7.10(ii) and (iii) under the same bijection. □

Status: **[paper proof]**

These are genuine equivalences and not a proof. There is at present no independent mechanism forcing nonlinear closure: nothing in the archimedean, valuation-theoretic, or determinant toolkits of Sections 3, 18 and 19 supplies a reason for $\mathcal{C}$ to be closed under $\star$. Their value is diagnostic — they identify the missing property as a single crisp failure of semiring-like structure, rather than as a quantitative deficit.

7.6 Formalization cost

The multiplication, power and quotient criteria are inexpensive Lean targets, and a route exists that avoids building a general rational-function library. For a new module — `RationalRigidity` would be a natural name — the plan is:

1. case split on `AlaogluErdosConjecture`, discharging the true branch by $\mathcal{S} = \mathbb{N}_0$;
2. in the failure branch obtain the canonical generator and the unique coordinates from the packaged statement

   `exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture`

   in the module `PrimitiveGenerator`;
3. expand the relevant equality as a polynomial of degree at most two in β with rational coefficients;
4. invoke the kernel-verified transcendence in `Transcendence` to force every coefficient to vanish;
5. discharge the remaining natural-number divisibility and coordinate conditions with `omega`.

Only the full rational-function theorem, Theorem 7.2, needs evaluation injectivity over $\overline{\mathbb{Q}}$ and hence Lemma 7.1; the multiplication theorem, the power test, the quotient criterion and the closure equivalences need nothing beyond transcendence over $\mathbb{Q}$, which is already accepted by the kernel.

Status: Planned; no draft yet. The mathematics is **[paper proof]** throughout this section, and the prerequisites are **[kernel-verified Lean]**.

7.7 What the rigidity statements buy

The practical payoff is that they sharply restrict how a counterexample could be manufactured from a smaller one. A natural strategy for producing new solutions from old ones is to apply some algebraic operation — square a solution, multiply two of them, take a ratio, invert, or substitute solutions into a rational function — and hope to land back in $\mathcal{S}$, either to run a descent or to build an infinite family with better arithmetic properties than the canonical generator. Equation (10) closes that door completely: the only rational functions over $\overline{\mathbb{Q}}$ that map solutions to solutions are the affine maps $T \mapsto n + kT$ with natural coefficients, which reproduce nothing that the additive monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$ did not already contain. Any construction of a new counterexample from an old one must therefore be transcendental in an essential way, or must leave the solution set and return by a route that is not rational over the algebraic numbers. In the other direction, the closure equivalences of Theorem 7.10 and Corollary 7.11 convert the conjecture into a purely structural question — is $\mathcal{C}$ closed under $\star$? — which is attractive precisely because it has no quantitative parameter to optimize, and which is a warning that no counting or equidistribution bound alone will produce the answer. Finally, all of these statements are short, elementary consequences of results already accepted by the kernel, which makes them the cheapest genuine additions to the verified corpus that this report proposes.

8 Three-smooth outputs and transcendental product exponents

Call a positive integer *three-smooth* when it has no prime factor outside $\{2, 3\}$, that is, when it equals $2^u 3^v$ for some $u, v \in \mathbb{N}_0$. The kernel-verified classification of auxiliary bases recorded in Section 3 settles which bases B satisfy $B^x \in \mathbb{Z}$ for a *fixed* nonintegral solution x: exactly the three-smooth ones. This section turns that classification onto the outputs $2^x, 3^x$ themselves, and then onto second-level exponents xy formed from two solutions.

Two distinct things come out, and they have different provenance. The first is an unconditional integrality test: a solution is integral precisely when its own two outputs are three-smooth. It uses only the kernel-verified transcendence of ϑ and unique factorization, so it is cheap to add to the Lean development. The second is an upgrade of "not an integer" to "transcendental", obtained from the six exponentials theorem. That upgrade converts several apparently weaker *algebraicity* statements into exact reformulations of Conjecture 1.1, at the cost of a classical input that the repository has not formalized in general form.

8.1 The two outputs cannot both be three-smooth

Theorem 8.1 (Three-smooth output test)**.** *For every $x \in \mathcal{S}$ the following are equivalent:*

(i) $x \in \mathbb{N}_0$*;*

(ii) *both* 2^x *and* 3^x *are three-smooth;*

(iii) 6^x *is three-smooth.*

Proof. Condition (i) gives (ii) immediately, since 2^x and 3^x are then powers of 2 and 3 respectively. Conditions (ii) and (iii) are equivalent because 2^x and 3^x are positive integers with product 6^x, and a product of positive integers is three-smooth exactly when both factors are.

For (ii)⇒(i), write

$$2^x = 2^p 3^q, \qquad 3^x = 2^r 3^s$$

with $p, q, r, s \in \mathbb{N}_0$. Taking logarithms and dividing by $\log 2$ and $\log 3$ respectively gives

$$x = p + q\vartheta, \qquad x = s + \frac{r}{\vartheta},$$

where $\vartheta = \log_2 3$. Eliminating x and clearing the denominator yields the quadratic relation

$$q\,\vartheta^2 + (p - s)\,\vartheta - r = 0 \tag{12}$$

with rational — indeed integer — coefficients. The repository proves ϑ transcendental, so every coefficient in (12) vanishes: $q = 0$, $p = s$, and $r = 0$. Hence $2^x = 2^p$ and $x = p \in \mathbb{N}_0$. $\square$

Status: [paper proof]

The statement is unconditional and does not use the conditional generator, the exact solution monoid, or any six-exponentials input. Its only nonelementary ingredient is the kernel-verified transcendence of ϑ,

`transcendental_logThreeDivLogTwo`

in `Transcendence`, together with unique factorization. A Lean proof is expected to be short; see Section B and the skeleton at the end of this section.

The test is sharp in a useful direction: it rules out one of the four a priori smoothness patterns of the least output pair.

Corollary 8.2 (Primitive-core trichotomy). *Assume Conjecture 1.1 fails, and use the simultaneous least-output normalization of Section 4,*

$$M = 2^{\beta} = 2^{a} w^{d}, \qquad A = 3^{\beta} = 3^{b} v^{e},$$

with $w > 1$ odd and power-primitive, $v > 1$ power-primitive and coprime to 3, and $d, e > 0$. Then at most one of $w = 3$ and $v = 2$ holds, and exactly one of the following three patterns occurs:

M	*A*	*primitive cores*
three-smooth	*not three-smooth*	$w = 3,\ v \neq 2$
not three-smooth	*three-smooth*	$w \neq 3,\ v = 2$
not three-smooth	*not three-smooth*	$w \neq 3,\ v \neq 2$

The missing fourth pattern, both outputs three-smooth, is impossible.

Proof. An odd, three-smooth, power-primitive integer greater than one is a power of 3, and primitivity forces the exponent to be 1; hence M is three-smooth exactly when $w = 3$. Symmetrically a three-smooth power-primitive integer greater than one that is coprime to 3 is exactly 2, so A is three-smooth exactly when $v = 2$. The least nonintegral solution β lies in $\mathcal{S} \setminus \mathbb{N}_0$, so Theorem 8.1 forbids both outputs from being three-smooth, which is the exclusion $\neg(w = 3 \wedge v = 2)$. ☐

Status: [paper proof]

The normalization itself, including power-primitivity of w and v, the depth dichotomy $a = 0$ or $b = 0$, and the gcd-one condition, is **[kernel-verified Lean]** in `OutputNormalization` (Section 4). Only the exclusion of the fourth pattern is new here.

The two exceptional rows are precisely the cases the corpus already identifies as hardest: $w = 3$ makes 3^{ϑ} the sharp radical target of Section 5, and $v = 2$ is its base-swapped mirror with $2^{1/\vartheta} = 2^{\log_3 2}$. The third row is generic and carries two independent outside-prime supports.

8.2 Third-base transcendence from six exponentials

The remaining statements of this section rest on one published transcendence theorem, which we state in the form we use.

Theorem 8.3 (Six exponentials theorem; Siegel, Lang, Ramachandra)**.** *Let u_1, u_2 be complex numbers linearly independent over $\mathbb{Q}$, and let v_1, v_2, v_3 be complex numbers linearly independent over $\mathbb{Q}$. Then at least one of the six numbers $e^{u_i v_j}$, $1 \le i \le 2$, $1 \le j \le 3$, is transcendental.*

Status: [**classical/open background**] — classical, not formalized in this repository in general form.

See Waldschmidt [25, 26]. Every consequence below is tagged [**classical/open background**] for the same reason: the repository formalizes only the integer-entry specialization of the underlying interpolation determinant, which yields the 3-smooth classification of Section 3 but not the transcendence conclusion.

Lemma 8.4 (Third-base transcendence)**.** *Let $y \in \mathcal{S} \setminus \mathbb{N}_0$ and let $B \in \mathbb{N}_{>0}$ be not three-smooth. Then B^y is transcendental.*

Proof. The numbers $\log 2, \log 3, \log B$ are linearly independent over $\mathbb{Q}$: a nontrivial rational relation among them would give a nontrivial multiplicative relation among 2, 3, and B, which unique factorization forbids because B has a prime factor outside $\{2, 3\}$. Also 1 and y are linearly independent over $\mathbb{Q}$, since y is irrational — a rational solution is integral by the kernel-verified rational-exponent lemma of Section 3, and $y \notin \mathbb{N}_0$.

Apply Theorem 8.3 with $u_1 = 1$, $u_2 = y$ and $v_1 = \log 2$, $v_2 = \log 3$, $v_3 = \log B$. The six values are

$$2, \quad 3, \quad B, \quad 2^y, \quad 3^y, \quad B^y,$$

and the first five are integers, hence algebraic: $2, 3, B$ by hypothesis and $2^y, 3^y$ because $y \in \mathcal{S}$. Therefore the sixth, B^y, is transcendental. □

Status: [**classical/open background**] — consequence of the six exponentials theorem (Theorem 8.3).

Combining Lemma 8.4 with the kernel-verified 3-smooth classification of Section 3 gives an exact algebraicity trichotomy for integer bases: for every $y \in \mathcal{S} \setminus \mathbb{N}_0$ and every $B \in \mathbb{N}_{>0}$,

$$B^y \in \overline{\mathbb{Q}} \iff B^y \in \mathbb{Z} \iff B = 2^u 3^v \text{ for some } u, v \in \mathbb{N}_0. \tag{13}$$

There is no algebraic-irrational middle case. Any base that is not three-smooth produces a transcendental value outright.

8.3 Products of noninteger solutions

The set $\mathcal{S}$ is closed under addition and under multiplication by natural numbers, but it is not known to be closed under multiplication of two of its own nonintegral elements. The next theorem says exactly what happens at the product exponent xy.

Theorem 8.5 (Product-exponent dichotomy)**.** *Let $x, y \in \mathcal{S}$.*

(a) *If $x \in \mathbb{N}_0$ or $y \in \mathbb{N}_0$, then $2^{xy} \in \mathbb{N}$ and $3^{xy} \in \mathbb{N}$.*

(b) *If $x \notin \mathbb{N}_0$ and $y \notin \mathbb{N}_0$, then each of 2^{xy} and 3^{xy} is either an integer or transcendental, at least one of the two is transcendental, and*

$$6^{xy} \text{ is transcendental.}$$

Proof. Part (a) is natural-number dilation: if $y = n \in \mathbb{N}_0$ then $xy = nx$ is a natural multiple of a solution and therefore a solution, and symmetrically in x.

For part (b) put $M_x = 2^x$ and $A_x = 3^x$, both positive integers, and note

$$2^{xy} = M_x^y, \qquad 3^{xy} = A_x^y.$$

If M_x is three-smooth, the 3-smooth classification of Section 3 makes M_x^y an integer; if M_x is not three-smooth, Lemma 8.4 makes it transcendental. The same alternative holds for A_x^y. By Theorem 8.1 applied to the nonintegral solution x, the two outputs M_x and A_x are not both three-smooth, so at least one of $2^{xy}, 3^{xy}$ is transcendental.

Finally $6^x = M_x A_x$ is a positive integer, and it is not three-smooth, since a prime outside $\{2, 3\}$ dividing one factor divides the product. Applying Lemma 8.4 with base 6^x and exponent y gives $(6^x)^y = 6^{xy}$ transcendental. □

Status: [**classical/open background**] for part (b) — via Theorem 8.3.

Part (a) is elementary and its underlying closure property, that natural multiples of a solution are solutions, is [**kernel-verified Lean**].

Corollary 8.6 (Diagonal transcendence)**.** *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$. Then 6^{x^2} is transcendental and at least one of 2^{x^2}, 3^{x^2} is transcendental. More precisely,*

$$2^{x^2} \in \mathbb{N} \iff 2^x \text{ is three-smooth,}$$
$$3^{x^2} \in \mathbb{N} \iff 3^x \text{ is three-smooth,}$$

and failure of either equivalence's right-hand side makes the corresponding value transcendental.

Proof. Take $y = x$ in Theorem 8.5(b). The two displayed equivalences are the alternative used in that proof, applied to the bases $M_x = 2^x$ and $A_x = 3^x$: a three-smooth base gives an integer by the 3-smooth classification of Section 3, and a base that is not three-smooth gives a transcendental value by Lemma 8.4. □

Corollary 8.7 (Mixed three-smooth bases)**.** *Let $x, y \in \mathcal{S} \setminus \mathbb{N}_0$ and let $u, v \geq 1$ be integers. Then $(2^u 3^v)^{xy}$ is transcendental.*

Proof. The positive integer

$$(2^u 3^v)^x = (2^x)^u (3^x)^v$$

is divisible by every prime occurring in either output, because $u, v \geq 1$. By Theorem 8.1 at least one output has a prime factor outside $\{2, 3\}$, so this base is not three-smooth. Apply Lemma 8.4 with exponent y. □

Status: [**classical/open background**] for both corollaries — via Theorem 8.3.

8.4 A global second-level trichotomy

The exact generator theorem of Section 4 makes the preceding dichotomy uniform: which of the three patterns occurs is a property of the counterexample as a whole, not of the particular pair (x, y).

Theorem 8.8 (Uniform second-level pattern)**.** *Assume Conjecture 1.1 fails and let $M = 2^\beta$, $A = 3^\beta$ for the primitive generator β. For every $x, y \in \mathcal{S} \setminus \mathbb{N}_0$, exactly one of the following three patterns holds, and the pattern does not depend on x or y:*

condition on (M, A)	2^{xy}	3^{xy}
M three-smooth, A not	*integer*	*transcendental*
A three-smooth, M not	*transcendental*	*integer*
neither three-smooth	*transcendental*	*transcendental*

Proof. By the exact primitive-generator theorem of Section 4, write $x = n + k\beta$ with $n, k \in \mathbb{N}_0$ and $k \geq 1$, so that

$$2^x = 2^n M^k, \qquad 3^x = 3^n A^k.$$

Multiplying by a power of the distinguished smooth prime cannot remove a prime factor outside $\{2, 3\}$, and $k \geq 1$ keeps every such prime of M present in M^k; hence 2^x is three-smooth exactly when M is, and likewise 3^x is three-smooth exactly when A is. Now apply Theorem 8.5(b) together with the alternative used in its proof. The fourth pattern, both M and A three-smooth, is excluded by Theorem 8.1 applied to β. $\square$

Status: [classical/open background] — via Theorem 8.3.

The structural input, namely $\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\}$ with unique coordinates, is **[kernel-verified Lean]** in `PrimitiveGenerator` (Section 4).

This suggests a clean three-case research split, matching Corollary 8.2. The two one-smooth-output cases reduce to the exceptional primitive cores $w = 3$ and $v = 2$, where the target radicals are 3^ϑ and $2^{1/\vartheta}$ respectively and where Section 5 already locates the wall. In the generic case both second iterates are transcendental and two independent outside-prime supports are available, which is the configuration a mixed archimedean and non-archimedean determinant would exploit; see Section 25.

8.5 Algebraicity formulations equivalent to the conjecture

Theorem 8.9 (Algebraicity tests)**.** *Each of the following is equivalent to Conjecture 1.1.*

(i) *For every $x \in \mathcal{S}$, $6^{x^2} \in \overline{\mathbb{Q}}$.*

(ii) *For every $x \in \mathcal{S}$, both $2^{x^2} \in \overline{\mathbb{Q}}$ and $3^{x^2} \in \overline{\mathbb{Q}}$.*

(iii) *For every $x, y \in \mathcal{S}$, $6^{xy} \in \overline{\mathbb{Q}}$.*

(iv) *For one fixed pair of integers $u, v \geq 1$, every $x \in \mathcal{S}$ satisfies $(2^u 3^v)^{x^2} \in \overline{\mathbb{Q}}$.*

(v) *For every $x \in \mathcal{S}$, the integer 6^x is three-smooth.*

Proof. If Conjecture 1.1 holds, then $\mathcal{S} = \mathbb{N}_0$, every exponent appearing in (i)–(iv) is a nonnegative integer, and all the listed values are positive integers, hence algebraic; condition (v) is immediate because $6^x = 2^x 3^x$ is then a power of 6.

Conversely, suppose the conjecture fails and fix $x \in \mathcal{S} \setminus \mathbb{N}_0$. Then (i) and (ii) fail by Corollary 8.6, (iii) fails by taking $y = x$ in Theorem 8.5(b), (iv) fails by Corollary 8.7 with $y = x$, and (v) fails by Theorem 8.1. □

Status: [**classical/open background**] for (i)–(iv) — via Theorem 8.3. [**paper proof**] for (v).

Equivalence (v) needs only Theorem 8.1, hence only the kernel-verified transcendence of ϑ; it is therefore the one item in this list that is within immediate reach of the Lean development.

Items (i)–(iv) replace integrality by the apparently much weaker target of algebraicity, and the six exponentials theorem makes the replacement lossless. That sharpens the statement of what is missing: one would need an independent reason why some nonlinear iterate such as 6^{x^2} is algebraic, derived from the two original integer values 2^x and 3^x alone. No mechanism in the present corpus produces algebraicity at the second level, which is consistent with the four-exponentials diagnosis of Section 6: the two-base configuration supplies only two independent logarithmic directions, and squaring the exponent does not create a third.

8.6 Formalization boundary

The division of labour in this section is unusually clean, and it marks exactly where the repository can advance without new transcendence theory.

Theorem 8.1, Corollary 8.2, part (a) of Theorem 8.5, and equivalence (v) of Theorem 8.9 use only kernel-verified ingredients: transcendence of ϑ, unique factorization, the closure of $\mathcal{S}$ under natural multiples, the simultaneous output normalization, and the 3-smooth classification in `ThreeSmooth`. A proposed module `OutputSmoothness` would host them. With the predicate

$$\mathrm{ThreeSmooth}(m) :\Longleftrightarrow \exists\, u, v \in \mathbb{N}_0,\ m = 2^u 3^v,$$

the headline statement is a biconditional between integrality of the exponent and joint smoothness of the two outputs:

```
theorem integer_iff_outputs_threeSmooth
    {x : R} (hx : TwoBaseIntegralSolution x) :
    x in Set.range ((^) : Z -> R) <->
      ThreeSmooth (natOutputTwo hx) /\
      ThreeSmooth (natOutputThree hx)
```

Listing 1: Proposed output-smoothness signature.

The reverse direction is the interesting one, and it follows the proof of Theorem 8.1: take logarithms of the two smooth factorizations, assemble (12) as a polynomial of degree at most two in ϑ with integer coefficients, apply

`transcendental_logThreeDivLogTwo`

to force every coefficient to vanish, and discharge the remaining natural-number arithmetic with `omega`. Mathlib's polynomial evaluation interface is the natural way to build $qX^2 + (p - s)X - r$ and invoke the definition of transcendence.

Everything else in this section — Lemma 8.4, part (b) of Theorem 8.5, Corollaries 8.6 and 8.7, Theorem 8.8, and items (i)–(iv) of Theorem 8.9 — needs the six exponentials theorem in general form. The repository's interpolation determinant is currently specialized to integer entries: a candidate matrix has integer entries, a nonzero integer determinant satisfies $|D| \geq 1$, and that inequality is what drives the 3-smooth classification. The natural formal route to Lemma 8.4 is to replace the bound $|D| \geq 1$ by a number-field norm bound for a nonzero *algebraic* determinant, retaining the same analytic upper bound from the interpolation estimate. Packaging that as an "algebraic entries of bounded height" interface would recover the transcendence conclusion while reusing the existing determinant infrastructure, and would place the general six exponentials theorem within the same framework. This is a substantial addition, not a refactor, and it is listed among the transcendence-theoretic priorities of Section 25.

9 Perfect-power content, affine descent, and primitive-output statistics

Section 4 closes the conditional structure question: if Conjecture 1.1 fails, then $\mathcal{S}$ is the free rank-two additive monoid generated by 1 and the canonical least nonintegral solution β, and the least output pair $(M, A) = (2^\beta, 3^\beta)$ is affinely primitive. Both facts are **[kernel-verified Lean]**. What that corpus does not yet record is the *exact* perfect-power content of the output pair at an arbitrary point of the monoid: the kernel-verified corollaries exclude matched decompositions at β, but say nothing about which matched decompositions actually occur elsewhere.

This section supplies that classification. It determines exactly which simultaneous factorizations

$$2^x = 2^r U^d, \qquad 3^x = 3^r V^d$$

can occur for $x \in \mathcal{S}$, identifies the maximal common perfect-power degree of the output pair with a gcd of coordinates, and then converts the classification into exact limiting statistics for primitive output pairs. Everything here is conditional on failure and rests on the kernel-verified monoid classification; the classification and the statistics themselves are **[paper proof]**.

Throughout, assume failure and fix the canonical generator β of Section 4, with

$$M = 2^\beta, \qquad A = 3^\beta.$$

For $x = n + k\beta$ with $n, k \in \mathbb{N}_0$, exact coordinates give

$$2^x = 2^n M^k, \qquad 3^x = 3^n A^k, \tag{14}$$

and the pair (n, k) is uniquely determined by x. The coordinate gcd $\gcd(n, k)$ turns out to be an intrinsic arithmetic invariant of the output pair, not merely of its coordinates.

9.1 Simultaneous perfect powers

Theorem 9.1 (Exact common-power criterion)**.** *Let $x = n + k\beta \in \mathcal{S}$ and let $r \geq 1$. The following are equivalent:*

(i) *both* 2^x *and* 3^x *are perfect* r*-th powers in* $\mathbb{N}$*;*

(ii) $x/r \in \mathcal{S}$*;*

(iii) $r \mid n$ *and* $r \mid k$*.*

Proof. If $2^x = U^r$ and $3^x = V^r$ with $U, V > 0$, uniqueness of positive real r-th roots gives

$$2^{x/r} = U, \qquad 3^{x/r} = V,$$

so $x/r \in \mathcal{S}$. The converse follows by raising the two integer values at x/r to the r-th power.

For the coordinate form, write $x/r = p + q\beta$ in the unique coordinates supplied by the exact primitive-generator theorem of Section 4. Then $n + k\beta = rp + rq\beta$, and uniqueness of coordinates forces $n = rp$ and $k = rq$, which is exactly the pair of divisibility conditions. Conversely, if $r \mid n$ and $r \mid k$, then $(n/r) + (k/r)\beta \in \mathcal{S}$ and equals x/r. ☐

For $x > 0$ define the *common output power degree*

$$g(x) := \max\{r \geq 1 : 2^x \text{ and } 3^x \text{ are both perfect } r\text{-th powers}\}.$$

Corollary 9.2 (Coordinate gcd)**.** *If* $x = n + k\beta > 0$*, then*

$$g(x) = \gcd(n, k),$$

with the convention $\gcd(n, 0) = n$*. The output pair is simultaneously power-primitive exactly when* $\gcd(n, k) = 1$*.*

Proof. By Theorem 9.1 the admissible degrees r are precisely the common divisors of n and k, and the set of common divisors of a pair that is not $(0, 0)$ has a largest element, namely $\gcd(n, k)$. ☐

Consequently every nonzero output pair has a unique decomposition

$$(2^x, 3^x) = (U^g, V^g), \qquad g = \gcd(n, k),$$

where $(U, V) = (2^{x/g}, 3^{x/g})$ is a simultaneously power-primitive output pair belonging to the solution x/g. The repository proves power-primitivity for the least pair only; this corollary computes the intrinsic power degree at every point of the solution monoid. For integral $x = n$ it returns $g(n) = n$, which is the familiar statement that 2^n and 3^n are perfect n-th powers and no better; for $x = \beta$ it returns $g(\beta) = \gcd(0, 1) = 1$, which is exactly the kernel-verified exclusion of a common perfect-power degree at the least pair.

Status: [paper proof] — Theorem 9.1 and Corollary 9.2 are paper results of this report. Their inputs are kernel-verified: the unique-coordinate clause of the exact primitive-generator theorem (`PrimitiveGenerator`) and the least-solution corollaries of `LeastSolution`. The implication (i)⇒(ii) is the special case $r = 0$ of the Lean affine descent theorem `affine_descent_integral_powers` in `ArithmeticRigidity`; the converse and the coordinate criterion (iii) are new here.

9.2 Matched affine decompositions

A descent that strips the same distinguished-base depth and the same outer power from both outputs is the natural common generalization of the two operations above. The repository's affine descent theorem shows that such a stripping produces a solution; the following criterion says exactly when one exists.

Theorem 9.3 (Matched affine decomposition criterion)**.** *Let $x = n + k\beta \in \mathcal{S}$, let $d \geq 1$, and let $r \in \mathbb{N}_0$. The following are equivalent:*

(i) *there exist $U, V \in \mathbb{N}_{>0}$ with*
$$2^x = 2^r U^d, \qquad 3^x = 3^r V^d;$$

(ii) $(x - r)/d \in \mathcal{S}$*;*

(iii) $0 \leq r \leq n$*,* $d \mid k$*, and* $d \mid (n - r)$*.*

When these conditions hold,

$$\frac{x-r}{d} = \frac{n-r}{d} + \frac{k}{d}\,\beta, \qquad U = 2^{(x-r)/d}, \qquad V = 3^{(x-r)/d}.$$

Proof. Conditions (i) and (ii) are equivalent by uniqueness of positive real roots, applied after removing the common factor 2^r from the first output and 3^r from the second. Expanding (ii) in unique coordinates gives
$$n + k\beta = r + d(p + q\beta)$$
for some $p, q \in \mathbb{N}_0$, so $n = r + dp$ and $k = dq$; this is exactly (iii), and it also yields the displayed formulas. Conversely, (iii) lets one define $p = (n-r)/d \in \mathbb{N}_0$ and $q = k/d \in \mathbb{N}_0$, and then $p + q\beta \in \mathcal{S}$ equals $(x - r)/d$. □

At the least pair the criterion collapses. For $x = \beta$ we have $n = 0$ and $k = 1$, so (iii) forces $r = 0$ and $d = 1$: no nontrivial matched decomposition of (M, A) exists. That is the kernel-verified exact affine primitivity statement of Section 4, recovered here as one instance of a criterion valid at every point.

For a nonintegral solution $x = n + k\beta$ with $k > 0$, Euclidean division $n = qk + r$ with $0 \leq r < k$ gives a canonical reduced affine presentation

$$x = r + k(q + \beta), \tag{15}$$

and hence
$$2^x = 2^r\bigl(2^q M\bigr)^k, \qquad 3^x = 3^r\bigl(3^q A\bigr)^k.$$

The normalization $0 \leq r < k$ makes the presentation unique. It separates the outer affine degree k — an invariant of the point, being its β-coordinate — from the reduced common base depth r.

Status: [paper proof] — Theorem 9.3 and the reduced presentation (15) are paper results. The implication (i)⇒(ii) is the kernel-verified affine descent theorem; the equivalence with the coordinate condition (iii), and hence the classification of *all* matched decompositions, is new.

9.3 What is already formal, and what is new here

The table below separates the two layers. The middle column is the kernel-verified statement about the least pair (M, A); the right column is what this section adds for a general point $x = n + k\beta$ of the monoid.

Statement	Kernel-verified at β	Extension in this section
Matched base depth	$\min\{v_2(M), v_3(A)\} = 0$, in `LeastSolution`	Admissible depths at x are exactly $0 \le r \le n$ with $d \mid (n-r)$
Common perfect-power degree	no $k \ge 2$ with M, A both k-th powers, in `LeastSolution`	the exact degree is $g(x) = \gcd(n, k)$ (Corollary 9.2)
Affine primitivity	$M = 2^r U^k$, $A = 3^r V^k$ force $r = 0$, $k = 1$, in `LeastSolution`	full criterion for every x, with the least pair as the case $(n, k) = (0, 1)$ (Theorem 9.3)
Affine descent	$(x - r)/k \in \mathcal{S}$ from a matched factorization, in `ArithmeticRigidity`	the descent condition is an equivalence, not merely an implication
Output normalization	$M = 2^a w^d$, $A = 3^b v^e$ with $a = 0$ or $b = 0$ and $\gcd(\gcd(d, e), \lvert b - a\rvert) = 1$, in `OutputNormalization`	unchanged; that invariant constrains the two *separate* degrees, whereas g measures only the common one

The kernel-verified identifiers behind the middle column are

`IsLeastTwoBaseNonintegerSolution.not_both_base_dvd_outputs`

`IsLeastTwoBaseNonintegerSolution.no_common_perfect_power`

`IsLeastTwoBaseNonintegerSolution.affine_primitive`

in `LeastSolution`, together with `exists_simultaneous_primitive_output_normalization` in `OutputNormalization` and `affine_descent_integral_powers` in `ArithmeticRigidity`. None of the right-hand column is machine-checked. A formalization would be light: the coordinate criteria follow from the already accepted unique-representation clause, so the missing work is bookkeeping over $\mathbb{N}_0^2$ rather than new arithmetic.

9.4 Why this does not yet descend below β

If n and k have a common divisor, Theorem 9.1 descends to $x/\gcd(n, k)$. If a matched decomposition exists, Theorem 9.3 descends to $(x - r)/d$. But exact coordinates guarantee that the descended point is again $p + q\beta$ with $p, q \in \mathbb{N}_0$, and it lies below β only when $q = 0$, in which case it is integral and carries no information. The least nonintegral point β is already primitive for every descent of this shape, which is precisely why the kernel-verified corollaries at β are exclusions rather than contradictions.

A successful descent argument therefore needs a transformation not captured by common perfect powers and common distinguished-base depths. Both operations act inside the monoid $\mathbb{N}_0^2$; anything that could close the problem must leave it.

9.5 Total and primitive counts

Corollary 9.2 converts ordinary coprimality statistics of lattice points into exact statistics of hypothetical output pairs. For $T > 0$ put

$$N_\beta(T) = \#\{(n, k) \in \mathbb{N}_0^2 \setminus \{(0, 0)\} : n + k\beta < T\}$$

and

$$P_\beta(T) = \#\{(n, k) \in \mathbb{N}_0^2 : n + k\beta < T,\ \gcd(n, k) = 1\}.$$

The first counts nonzero solutions below T; by Corollary 9.2 the second counts exactly those output pairs with no nontrivial common perfect-power degree.

Counting each column separately, for every real $U > 0$,

$$N_\beta(U) = \sum_{0 \le k < U/\beta} \lceil U - k\beta \rceil - 1 = \frac{U^2}{2\beta} + O_\beta(U + 1), \tag{16}$$

uniformly in U. For integer T this is the kernel-verified cumulative count $C(T)$ of Section 4 minus the point $x = 0$, so (16) agrees in leading order with `QuadraticAsymptotic`; the uniform real-variable form used below is elementary.

Theorem 9.4 (Primitive output pairs)**.** *As $T \to \infty$,*

$$P_\beta(T) = \frac{T^2}{2\beta\zeta(2)} + O_\beta(T \log T) = \frac{3T^2}{\pi^2\beta} + O_\beta(T \log T).$$

Consequently

$$\frac{P_\beta(T)}{N_\beta(T)} \longrightarrow \frac{1}{\zeta(2)} = \frac{6}{\pi^2}.$$

Proof. Every nonzero coordinate pair has the unique factorization

$$(n, k) = d\,(n_0, k_0), \qquad d = \gcd(n, k), \quad \gcd(n_0, k_0) = 1,$$

and $n + k\beta < T$ holds if and only if $n_0 + k_0\beta < T/d$. Hence

$$N_\beta(T) = \sum_{d \ge 1} P_\beta(T/d),$$

a finite sum for each T, and Möbius inversion gives

$$P_\beta(T) = \sum_{d \le T} \mu(d)\, N_\beta(T/d).$$

Inserting the uniform estimate (16),

$$\begin{aligned} P_\beta(T) &= \frac{T^2}{2\beta} \sum_{d \le T} \frac{\mu(d)}{d^2} + O_\beta\left(\sum_{d \le T} \Big(\frac{T}{d} + 1\Big)\right) \\ &= \frac{T^2}{2\beta\zeta(2)} + O_\beta(T \log T), \end{aligned}$$

where the tail $\sum_{d>T} \mu(d)/d^2 = O(1/T)$ contributes $O_\beta(T)$. Dividing by (16) gives the stated limit. $\square$

The constant $6/\pi^2$ is universal: it does not depend on the hypothetical generator at all. The geometry of the monoid contributes the factor $1/(2\beta)$, and the primitive-lattice density contributes $1/\zeta(2)$. In particular no choice of counterexample can make the primitive proportion small.

9.6 Distribution of the maximal common power degree

For fixed $g \geq 1$ let

$$P_{\beta,g}(T) := \#\{x = n + k\beta : 0 < x < T,\ \gcd(n,k) = g\}$$

be the number of solutions below T whose output pair has common power degree exactly g.

Corollary 9.5 (Exact-degree distribution)**.** *For fixed $g \geq 1$,*

$$P_{\beta,g}(T) = P_\beta(T/g) = \frac{3T^2}{\pi^2 \beta g^2} + O_\beta\left(\frac{T}{g} \log \frac{2T}{g}\right).$$

The limiting probability that an output pair below T has maximal common power degree exactly g is

$$\frac{6}{\pi^2 g^2}.$$

Proof. The map $(n_0, k_0) \mapsto g\,(n_0, k_0)$ is a bijection from the primitive pairs below T/g onto the pairs of gcd exactly g below T, so $P_{\beta,g}(T) = P_\beta(T/g)$; now apply Theorem 9.4 and divide by (16). $\square$

The probabilities sum to one, since $\sum_{g\geq 1} 6/(\pi^2 g^2) = 1$. Likewise, by Theorem 9.1 both outputs are perfect r-th powers exactly when $r \mid n$ and $r \mid k$, so

$$\#\{0 < x < T : 2^x \text{ and } 3^x \text{ are both perfect } r\text{-th powers}\} = N_\beta(T/r), \tag{17}$$

and the limiting proportion is $1/r^2$.

9.7 Exact generating series

The coordinate monoid has an equally explicit Laplace series. For $s > 0$,

$$\begin{aligned} Z_\beta(s) &:= \sum_{x \in \mathcal{S}\setminus\{0\}} e^{-sx} = \sum_{n,k\geq 0} e^{-s(n+k\beta)} - 1 \\ &= \frac{1}{(1-e^{-s})(1-e^{-\beta s})} - 1, \end{aligned} \tag{18}$$

the two factors being the two free generators 1 and β. Möbius inversion in the degree gives the primitive-output series

$$Z_\beta^{\text{prim}}(s) = \sum_{d\geq 1} \mu(d)\, Z_\beta(ds).$$

At the candidate level the same factorization appears multiplicatively: since $\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}$ with unique coordinates,

$$\sum_{m\in\mathcal{C}} m^{-s} = \frac{1}{(1-2^{-s})(1-M^{-s})}, \qquad \Re s > 0. \tag{19}$$

These identities make the two free generators visible in analytic form and give clean starting points for weighted Tauberian refinements of Theorem 9.4.

Status: [paper proof] — Theorem 9.4, Corollary 9.5, and the series (18)–(19) are paper results of this report. Their only nonelementary input is the kernel-verified exact monoid classification of Section 4, whose leading-order count is itself **[kernel-verified Lean]** through `ExactCounting` and `QuadraticAsymptotic`. The finite scans of Section 23 play no role here; the exponent bound in force remains the kernel-verified $x \geq 12$.

9.8 Interpretation

Roughly 60.79% of hypothetical output pairs would be simultaneously power-primitive, and the remaining ones split across degrees $g \geq 2$ with the exact frequencies $6/(\pi^2 g^2)$. This is not pressure toward a contradiction. It is precisely the primitive proportion expected of a free rank-two monoid, and it would be the same for any generator: the statistic is diagnostic, not restrictive.

Its diagnostic value is a warning about method. The kernel-verified primitivity corollaries at β, and the classification of this section at a general point, both say that perfect-power and matched-depth structure is the exception among hypothetical solutions. Any descent argument built on extracting a common root or a common base depth therefore acts on a minority of the conditional population — asymptotically the proportion $1 - 6/\pi^2 \approx 0.3921$ of pairs that admit any nontrivial common degree at all — and leaves the primitive majority untouched. A method that closes the problem must engage the power-primitive pairs directly, which is exactly where the transcendence content of Section 6 sits.

10 Candidate divisibility and ambient root saturation

The exact classification of Section 4 describes the conditional candidate set by its *generators*: under failure, $\mathcal{C} = \{2^n M^k : n, k \in \mathbb{N}_0\}$ with unique coordinates. Generation alone says nothing about how candidates sit inside $\mathbb{N}$ under the two operations that descent arguments and searches actually use — divisibility, and extraction of roots. This section answers both exactly. The candidate monoid has a simple valuation geometry: divisibility is coordinatewise order on a two-dimensional lattice cone, $\mathcal{C}$ is closed under gcd and lcm, and the set of integers some power of which is a candidate is again an explicit cone — one that contains $\mathcal{C}$ with an exact finite-index defect measured by the rational-power index d of Section 4, which is also the exact radical degree of Section 5.

Every statement below is **[paper proof]**. Its inputs — the canonical primitive odd core, the rational-power index, and the exact primitive-generator theorem — are **[kernel-verified Lean]**; what is added here is elementary lattice combinatorics on top of them. None of it is a step toward a contradiction: as the closing discussion makes explicit, the closure properties proved here are compatible with failure by construction, and the kernel-verified exponent bound remains $x \geq 12$.

10.1 A two-dimensional valuation cone

Lemma 10.1 (Valuation coordinates and divisibility)**.** *Assume Conjecture 1.1 is false and let w, d, a, β, M be the canonical data of Section 4, so that $w > 1$ is odd and power-primitive, $d > 0$, and*

$$M = 2^\beta = 2^a w^d.$$

Put $U := w^d$, the odd part of M, and for $n, k \in \mathbb{N}_0$ write

$$m(n, k) := 2^n M^k = 2^{n+ak} U^k, \qquad s := v_2(m(n, k)) = n + ak.$$

Then:

(i) *$U > 1$;*

(ii) *the map $m(n, k) \mapsto (s, k)$ is a bijection from $\mathcal{C}$ onto the lattice cone*

$$\Lambda_a := \{(s, k) \in \mathbb{N}_0^2 : s \geq ak\};$$

(iii) *writing $m_i = m(n_i, k_i)$ and $s_i = n_i + ak_i$,*

$$m_1 \mid m_2 \quad \Longleftrightarrow \quad s_1 \leq s_2 \ \ \text{and} \ \ k_1 \leq k_2. \tag{20}$$

Proof. (i) If $U = 1$ then $M = 2^a$ and $\beta = a \in \mathbb{Z}$, contradicting the irrationality of the least generator.

(ii) By the exact primitive-generator theorem of Section 4, every candidate is $m(n, k)$ for a unique pair $(n, k) \in \mathbb{N}_0^2$. The reparameterization $(n, k) \mapsto (n + ak, k)$ is affine with inverse $(s, k) \mapsto (s - ak, k)$, and $n \geq 0$ holds exactly when $s \geq ak$.

(iii) Since U is odd, $v_2(m(n, k)) = s$, and for an odd prime p we have $v_p(m(n, k)) = k\, v_p(U)$, which vanishes unless $p \mid U$. Divisibility in $\mathbb{N}$ is comparison of all prime valuations. If $m_1 \mid m_2$ then comparison at 2 gives $s_1 \leq s_2$, and comparison at any prime $p \mid U$ — one exists by (i) — gives $k_1 v_p(U) \leq k_2 v_p(U)$ with $v_p(U) > 0$, hence $k_1 \leq k_2$. Conversely those two inequalities force every prime valuation of m_1 to be at most the corresponding valuation of m_2. □

Thus candidate divisibility is nothing but the coordinatewise order on Λ_a. The inequality $s \geq ak$ is the only constraint, and it records the base-adic depth a of the least generator: a candidate cannot carry the odd part U^k without carrying at least ak factors of two along with it.

10.2 The candidate gcd/lcm lattice

Theorem 10.2 (Candidate gcd/lcm lattice)**.** *Unconditionally, $\mathcal{C}$ is closed under* gcd *and* lcm*. If Conjecture 1.1 fails, then in the notation of Lemma 10.1,*

$$\begin{aligned}
\gcd(m_1, m_2) &= m(\min(s_1, s_2) - a\min(k_1, k_2),\ \min(k_1, k_2))\,,\\
\operatorname{lcm}(m_1, m_2) &= m(\max(s_1, s_2) - a\max(k_1, k_2),\ \max(k_1, k_2))\,;
\end{aligned}$$

equivalently, in cone coordinates gcd *and* lcm *are coordinatewise minimum and maximum on Λ_a. Candidate divisibility is therefore a distributive lattice, and Λ_a is a sublattice of $\mathbb{N}_0^2$.*

Proof. If the conjecture holds then $\mathcal{C} = \{2^n : n \in \mathbb{N}_0\}$, whose gcds and lcms are again powers of two; this gives the unconditional half.

Assume failure. By Lemma 10.1 the candidate $m(n,k)$ has valuation vector s at the prime 2 and $k\,v_p(U)$ at each odd p. Gcd and lcm take coordinatewise minima and maxima of valuation vectors, so

$$\gcd(m_1, m_2) = 2^{\min(s_1,s_2)} U^{\min(k_1,k_2)}, \qquad \operatorname{lcm}(m_1, m_2) = 2^{\max(s_1,s_2)} U^{\max(k_1,k_2)}.$$

It remains to check that both exponent pairs lie in Λ_a. Suppose $s_1 \le s_2$; then $\min(s_1, s_2) = s_1 \ge ak_1 \ge a\min(k_1, k_2)$ because $a \ge 0$. For the maximum, suppose $k_1 \le k_2$; then $\max(s_1, s_2) \ge s_2 \ge ak_2 = a\max(k_1, k_2)$. Hence Λ_a is closed under coordinatewise min and max, that is, it is a sublattice of $\mathbb{N}_0^2$, and the translation back to (n, k) coordinates is the stated substitution $n = s - ak$. A sublattice of a product of chains is distributive. □

Status: [kernel-verified Lean] — `twoBaseNaturalCandidate_gcd` and `twoBaseNaturalCandidate_lcm` in `CandidateLattice`, resting on the coordinatewise arithmetic lemmas `gcd_two_pow_mul_odd_pow` and `lcm_two_pow_mul_odd_pow` of the same module and on the [kernel-verified Lean] exact primitive-generator theorem of Section 4. The formal statements are unconditional: the Lean proof splits on the conjecture and disposes of the affirmative branch through the powers of two. No transcendence input and no analytic estimate enters.

Corollary 10.3 (Counting candidate divisors)**.** *Under failure, the number of candidates dividing $m(n,k)$, including 1 and $m(n,k)$ itself, is*

$$(k+1)(n+1) + \frac{ak(k+1)}{2}.$$

Proof. By (20) a candidate divisor corresponds to a point $(s', j) \in \Lambda_a$ with $0 \le j \le k$ and $aj \le s' \le n + ak$. For fixed j there are $n + ak - aj + 1$ admissible values of s'. Summing the arithmetic progression over $j = 0, 1, \ldots, k$ gives $(k + 1)(n + 1) + a\sum_{i=0}^{k} i$, which is the stated value. □

Remark 10.4 (Candidates are not divisor-closed). The full divisor count of $m(n,k) = 2^s U^k$ is $(s + 1)\prod_{p|U}(k\,v_p(U) + 1)$, which for $k \ge 1$ and U with more than one prime factor grows much faster than the count of Corollary 10.3. So $\mathcal{C}$ is emphatically not closed under taking divisors; it is closed only under the two lattice operations. The gap between the two counts is a concrete measure of how thin the candidate sublattice is inside the divisor lattice of a single candidate.

10.3 Fixed prime support

The lattice picture has an immediate consequence for the shape of a hypothetical counterexample branch: the supports do not vary at all.

Corollary 10.5 (Prime-support rigidity)**.** *Assume failure and use the simultaneous primitive output normalization of Section 4,*

$$M = 2^\beta = 2^a w^d, \qquad A = 3^\beta = 3^b v^e.$$

Then for $x = n + k\beta$,

$$2^x = 2^{n+ak} w^{dk}, \qquad 3^x = 3^{n+bk} v^{ek},$$

so every nonintegral solution — that is, every solution with $k \geq 1$ — has exactly the same odd prime support $\operatorname{supp}(w)$ *in its base-2 output, and exactly the same prime support away from 3, namely* $\operatorname{supp}(v)$*, in its base-3 output. The only variation across the entire nonintegral branch is whether the distinguished primes 2 and 3 occur at all.*

Proof. Substituting M and A into $(2^x, 3^x) = (2^n M^k, 3^n A^k)$ gives the displayed factorizations. For $k \geq 1$ the odd part of 2^x is w^{dk}, and $\operatorname{supp}(w^{dk}) = \operatorname{supp}(w)$ because $dk \geq 1$; similarly for v^{ek}. The exponents $n + ak$ and $n + bk$ may vanish, which is exactly the stated freedom at 2 and 3. $\square$

A counterexample therefore locks the whole nonintegral branch onto a single fixed pair of finite prime sets. Computational reconnaissance should exploit this directly, enumerating primitive cores and supports rather than raw integers scanned independently; the finite verifications reported in Section 23 are **[software computation]** and this observation does not change their status, only the shape of the search that would extend them.

10.4 Ambient root saturation and the index d

Reparameterizing Lemma 10.1 by the exponent of the primitive core w rather than of $U = w^d$ gives the description that governs roots. With $s = v_2(m)$ and t the exponent of w in the odd part of m,

$$\mathcal{C} = \{2^s w^t : s, t \in \mathbb{N}_0,\ d \mid t,\ ds \geq at\}. \tag{21}$$

Indeed $m = 2^n M^k$ has $s = n + ak$ and $t = dk$, so $d \mid t$, and $n \geq 0$ is equivalent to $s \geq at/d$, that is, to $ds \geq at$.

Definition 10.6 (Ambient root saturation)**.** For a multiplicative submonoid $H \subseteq \mathbb{N}_{>0}$ put

$$\operatorname{Sat}(H) := \{u \in \mathbb{N}_{>0} : u^r \in H \text{ for some } r \geq 1\},$$

the set of positive integers admitting some positive power inside H.

Theorem 10.7 (Exact ambient root saturation)**.** *Assume Conjecture 1.1 is false. Then*

$$\operatorname{Sat}(\mathcal{C}) = \{2^s w^t : s, t \in \mathbb{N}_0,\ ds \geq at\},$$

and for $u = 2^s w^t$ in this set the least positive r with $u^r \in \mathcal{C}$ is

$$r_{\min}(u) = \frac{d}{\gcd(d,t)}.$$

Proof. First let $u = 2^s w^t$ with $s, t \in \mathbb{N}_0$ and $ds \geq at$, and put $r = d/\gcd(d,t)$. Then $rt = d \cdot t/\gcd(d,t)$ is divisible by d, and multiplying $ds \geq at$ by $r > 0$ gives $d(rs) \geq a(rt)$. By (21), $u^r = 2^{rs} w^{rt} \in \mathcal{C}$, so $u \in \operatorname{Sat}(\mathcal{C})$. Conversely, if $u^{r'} \in \mathcal{C}$ for some $r' \geq 1$ then (21) forces $d \mid r't$, since w is power-primitive and hence the exponent of w in the odd part of $u^{r'}$ is exactly $r't$. Now $d \mid r't$ is equivalent to $\big(d/\gcd(d,t)\big) \mid r'$, so $r' \geq r$ and r is least.

Now suppose merely that $u \in \mathbb{N}_{>0}$ satisfies $u^r \in \mathcal{C}$ for some $r \geq 1$, and write $u^r = 2^S w^{dk}$ with $dS \geq a\,dk$. A prime divides u if and only if it divides u^r, so no prime outside $\{2\} \cup \operatorname{supp}(w)$ divides u. Put $c_p := v_p(w)$ for $p \mid w$; comparing valuations in $u^r = 2^S w^{dk}$ gives

$$r\, v_p(u) = dk\, c_p \qquad \text{for every } p \mid w.$$

Because w is power-primitive, $\gcd\{c_p : p \mid w\} = 1$, so there are integers r_p, almost all zero, with $\sum_p r_p c_p = 1$; then

$$\frac{dk}{r} = \sum_p r_p \cdot \frac{dk\, c_p}{r} = \sum_p r_p\, v_p(u) \in \mathbb{Z}.$$

Call this integer $t \geq 0$. Then $v_p(u) = tc_p$ for every $p \mid w$, so the odd part of u is exactly w^t, while $S = rs$ for $s = v_2(u)$. Substituting $S = rs$ and $dk = rt$ into $dS \geq a\,dk$ and dividing by $r > 0$ yields $ds \geq at$. Hence $u = 2^s w^t$ lies in the stated cone. □

Status: [paper proof] — the only nonformal ingredient is the Bézout step, which is the same primitivity mechanism already kernel-verified in `PrimitiveOddCore` through `oddPrimeValuationGCD`. Everything else is unique factorization and integer arithmetic.

Corollary 10.8 (Ambient saturation index)**.** *Under failure, $\mathcal{C} = \operatorname{Sat}(\mathcal{C})$ if and only if $d = 1$, and in general*

$$d = \min\{r \geq 1 : \exists\, s \in \mathbb{N}_0,\ (2^s w)^r \in \mathcal{C}\}.$$

Proof. Take $t = 1$ and any $s \geq a$, so that $ds \geq a = at$; by Theorem 10.7 the integer $2^s w$ lies in $\operatorname{Sat}(\mathcal{C})$ with $r_{\min} = d/\gcd(d, 1) = d$. This proves the displayed formula, and it shows that $2^s w \in \mathcal{C}$ forces $d = 1$. Conversely, if $d = 1$ then the divisibility condition $d \mid t$ in (21) is vacuous and the two sets coincide. □

Corollary 10.9 (Which integer roots of a candidate are again candidates)**.** *Assume failure and let $m = 2^n M^k \in \mathcal{C}$ and $r \geq 1$.*

(a) *There is an integer u with $u^r = m$ if and only if $r \mid n + ak$ and $r \mid dk$; such a u always lies in $\operatorname{Sat}(\mathcal{C})$.*

(b) *That integer root is itself a candidate if and only if $r \mid n$ and $r \mid k$, in which case $u = 2^{n/r} M^{k/r}$.*

Proof. (a) Write $m = 2^s w^t$ with $s = n + ak$ and $t = dk$. If $u^r = m$ then, exactly as in the proof of Theorem 10.7, u is supported in $\{2\} \cup \operatorname{supp}(w)$, $r \mid s$, and $t/r \in \mathbb{Z}$ by the Bézout step; conversely $u = 2^{s/r} w^{t/r}$ is an integer r-th root when $r \mid s$ and $r \mid t$. Membership in $\operatorname{Sat}(\mathcal{C})$ is immediate from Definition 10.6.

(b) If $u \in \mathcal{C}$ then $u = 2^{n'} M^{k'}$ for unique $n', k' \in \mathbb{N}_0$, and $u^r = 2^{rn'} M^{rk'}$ together with uniqueness of coordinates gives $n = rn'$ and $k = rk'$. The converse is the direct computation $\left(2^{n/r} M^{k/r}\right)^r = m$. □

Observation 10.10 (Normal intrinsically, saturated only up to index d)**.** Let $L := \{2^s w^t : s, t \in \mathbb{Z}\}$ be the ambient multiplicative group generated by 2 and w inside $\mathbb{Q}^{\times}_{>0}$, and let $L_{\mathcal{C}}$ be the group generated by $\mathcal{C}$. Then

$$L_{\mathcal{C}} = \{2^s w^t : s \in \mathbb{Z},\ d \mid t\}, \qquad [L : L_{\mathcal{C}}] = d, \qquad \mathcal{C} = \operatorname{Sat}(\mathcal{C}) \cap L_{\mathcal{C}}.$$

In particular $\operatorname{Sat}(\mathcal{C})$ is itself a submonoid of $\mathbb{N}_{>0}$, namely the full lattice points of the rational cone $ds \geq at$, $s, t \geq 0$.

Proof. $L_{\mathcal{C}}$ is generated by 2 and $M = 2^a w^d$, so it consists of the elements $2^{i+aj} w^{dj}$ with $i, j \in \mathbb{Z}$, which is the stated set; the quotient $L/L_{\mathcal{C}}$ is isomorphic to $\mathbb{Z}/d\mathbb{Z}$ through the exponent of w. Intersecting the cone description of Theorem 10.7 with the congruence $d \mid t$ returns (21). Closure of the cone under multiplication is additivity of the inequality $ds \geq at$ in (s, t). □

A terminology caution is worth stating explicitly, because the word "normal" is used differently in affine-semigroup theory. There, normality of a semigroup is measured relative to the group that the semigroup itself generates. By Observation 10.10 that group is $L_{\mathcal{C}}$, in which the congruence $t \equiv 0 \pmod{d}$ already holds identically, and $\mathcal{C}$ is the full set of cone points of $L_{\mathcal{C}}$ for every value of d: intrinsically, $\mathcal{C}$ is normal. Theorem 10.7 measures something different, namely saturation inside the larger ambient lattice L containing every $2^s w^t$. So d is an over-lattice saturation defect of finite index, not a failure of intrinsic normality. This is precisely parallel to its two other exact meanings established elsewhere in this report: d is the least rational-power index of Section 4 and the exact radical degree $[\mathbb{Q}(w^{\vartheta}) : \mathbb{Q}]$ of Section 5. One integer invariant, three characterizations — algebraic degree, least rational power, and lattice saturation defect.

10.5 Why lattice saturation is a good Lean target

Two points must be kept apart. The gcd/lcm closure of Theorem 10.2 and the saturation formula of Theorem 10.7 are compatible with failure by construction: they describe the counterexample world rather than obstruct it, and no contradiction can be extracted from them alone. They are structural information and search normalization. With that said, they are unusually attractive as the next formalization increment.

The mathematical inputs are already kernel-verified. The canonical primitive odd core, the rational-power index, the three-power denominator normalization, and the exact primitive generator with unique coordinates are all **[kernel-verified Lean]** and live in `ExponentialIdentities/TwoBaseIntegerExponent/`. Everything added in this section is finite combinatorics over those results: unique factorization in $\mathbb{N}$, one Bézout step that already exists in `PrimitiveOddCore` as `oddPrimeValuationGCD`, and elementary min/max arithmetic on $\mathbb{N}_0^2$. There is no analysis, no transcendence input, and no numerical certificate anywhere in the chain — a rare profile in this corpus, and the reason the work should be cheap relative to its yield. A single module, sitting beside `PrimitiveGenerator` and `RationalPowerIndex`, would carry the coordinate bijection of Lemma 10.1, the divisibility criterion (20), the lattice operations, the divisor count, the cone form (21), the saturation set, the exact value of $r_{\min}$, and the index characterization of d. The natural name is

```
CandidateLattice
```

and the verification conventions of Section B apply unchanged.

The contribution to the search is threefold. First, a kernel-checked normal form: candidate enumeration can be phrased over lattice coordinates rather than over raw integers, so rejection tests become integer inequalities of the form $ds \geq at$ and $d \mid t$ instead of factorizations recomputed for every scanned value. This is exactly the discipline recommended in the computational priority of Section 25, and it applies equally to the certificate architecture that any extension of the **[software computation]** finite verifications of Section 23 would need; the kernel-verified region itself would still stop at $x \geq 12$ until new certificates are replayed. Second, a single invariant with three independently proved meanings: once Corollary 10.8 is formal alongside the radical-degree proposition of Section 5, any future argument that bounds d — from Kummer theory, from ramification, or from a determinant estimate — bounds all three at once, and the Lean statements make that transfer mechanical rather than a matter of prose. Third, and most concretely, root extraction is the step where descent arguments live. Every descent so far — no matched base depth, no common perfect-power degree, exact affine primitivity in Section 4 — proceeds by extracting a root of the output pair and contradicting leastness of β. Corollary 10.9 says exactly which integer roots of a candidate survive as candidates and which merely land in the saturation, so a formalized version supplies the reusable side condition that such arguments currently re-prove by hand. That is a modest but genuine improvement in the leverage of the conditional structure, and it costs no new mathematics.

11 A sufficient condition: independence of prime-log products

The reductions of Section 5 isolate the obstruction as a statement about one radical at a time. This section isolates it differently, as a single statement about $\mathbb{Q}$-linear independence of a natural family of real numbers. The resulting criterion is weaker in appearance than the four exponentials conjecture and is of a different kind: it does not mention exponentials at all.

11.1 Failure as a vanishing linear form in log-products

Suppose the conjecture fails, and let x be a nonintegral solution with outputs

$$M = 2^x \in \mathbb{N}, \qquad A = 3^x \in \mathbb{N}.$$

Taking logarithms of $M = 2^x$ and $A = 3^x$ and eliminating x gives the defining relation already recorded in (2):

$$\log M \, \log 3 - \log A \, \log 2 = 0. \tag{22}$$

Now expand both integers over their prime factorizations,

$$M = \prod_p p^{\alpha_p}, \qquad A = \prod_p p^{\gamma_p}, \qquad \alpha_p, \gamma_p \in \mathbb{N}_0,$$

so that $\log M = \sum_p \alpha_p \log p$ and $\log A = \sum_p \gamma_p \log p$. Substituting into (22),

$$\sum_p \alpha_p (\log p \, \log 3) - \sum_p \gamma_p (\log p \, \log 2) = 0. \tag{23}$$

Every term of (23) is an integer multiple of a product of exactly two logarithms of primes. Collecting coefficients in the family

$$\{\log p \, \log q \ : \ p \le q \text{ prime}\},$$

the unordered pair $\{p, 3\}$ with $p \notin \{2, 3\}$ receives the coefficient α_p; the pair $\{p, 2\}$ with $p \notin \{2, 3\}$ receives $-\gamma_p$; the diagonal pair $\{3, 3\}$ receives α_3; the diagonal pair $\{2, 2\}$ receives $-\gamma_2$; and the mixed pair $\{2, 3\}$ receives $\alpha_2 - \gamma_3$.

Thus *failure of the conjecture is exactly the existence of a nontrivial vanishing integer-coefficient linear form in products of two prime logarithms*, with the additional positivity constraint that the coefficients arise from prime factorizations.

11.2 The independence criterion

Definition 11.1 (Prime-log-product independence)**.** Say that *prime-log products are independent* if the real numbers $\log p \, \log q$, indexed by pairs of primes $p \le q$, are linearly independent over $\mathbb{Q}$.

Equivalently, in the integer-coefficient form convenient for formalization: for every finite set S of primes and every $c : S \times S \to \mathbb{Z}$,

$$\sum_{p\in S}\sum_{q\in S} c_{pq} \log p \, \log q = 0 \qquad \Longrightarrow \qquad c_{pq} + c_{qp} = 0 \ \text{ for all } p, q \in S.$$

(The diagonal case $p = q$ reads $2c_{pp} = 0$, hence $c_{pp} = 0$.) Clearing denominators shows the two formulations agree.

Theorem 11.2 (Prime-log-product independence implies the conjecture)**.** *If prime-log products are independent, then Conjecture 1.1 holds.*

Proof. Let x satisfy $2^x, 3^x \in \mathbb{Z}$. Both powers are positive, so $M := 2^x$ and $A := 3^x$ are positive integers, and $x \ge 0$. Relation (22) holds, hence so does (23).

Apply the independence hypothesis to the finite set $S = \mathrm{primes}(M) \cup \mathrm{primes}(A) \cup \{2, 3\}$ and to the coefficient family

$$c_{pq} = \begin{cases} \alpha_p, & q = 3, \\ -\gamma_p, & q = 2, \\ 0, & \text{otherwise}, \end{cases}$$

which reproduces the left-hand side of (23). The conclusion $c_{pq} + c_{qp} = 0$ yields, coefficient by coefficient:

- taking $q = 3$ and $p \in S \setminus \{2, 3\}$: $\alpha_p + 0 = 0$, so $\alpha_p = 0$;
- taking $p = q = 3$: $2\alpha_3 = 0$, so $\alpha_3 = 0$;
- taking $q = 2$ and $p \in S \setminus \{2, 3\}$: $-\gamma_p + 0 = 0$, so $\gamma_p = 0$;
- taking $p = q = 2$: $-2\gamma_2 = 0$, so $\gamma_2 = 0$;
- taking $p = 2$, $q = 3$: $\alpha_2 - \gamma_3 = 0$.

Primes outside S divide neither M nor A, so their exponents vanish as well. Hence $\alpha_p = 0$ for every $p \ne 2$, that is

$$M = 2^{\alpha_2},$$

and symmetrically $A = 3^{\gamma_3} = 3^{\alpha_2}$. Writing $n = \alpha_2 \in \mathbb{N}_0$ we get $2^x = M = 2^n$, and strict monotonicity of $t \mapsto 2^t$ gives $x = n \in \mathbb{Z}$. ☐

Status: [kernel-verified Lean] — `LogProductIndependence` implements this argument, with the independence hypothesis recorded as an explicit `Prop` and never asserted; the accepted conditional theorem is `alaogluErdosConjecture_of_primeLogProductIndependence`.

11.3 What the criterion is, and is not

Three remarks place Theorem 11.2 correctly.

It is a consequence of Schanuel's conjecture. Schanuel's conjecture implies that logarithms of multiplicatively independent positive rationals are algebraically independent over $\mathbb{Q}$; in particular the numbers $\log p$, over all primes p, would be algebraically independent, and then the products $\log p \log q$ are $\mathbb{Q}$-linearly independent because distinct monomials of degree two in algebraically independent elements are linearly independent. So Definition 11.1 is strictly weaker than Schanuel and lies well inside the standard expectations.

Its strength beyond the conjecture lies in pairs that never occur. This calibration matters and is easy to miss. Collecting the relation (23) back up,

$$\sum_p \alpha_p(\log p \log 3) - \sum_p \gamma_p(\log p \log 2) = \log 3\, \log M - \log 2\, \log A,$$

so the only products ever exercised are those of the form $\log 2 \log p$ and $\log 3 \log p$. Write F for that sub-family. A vanishing rational relation over F factors as $\log 2\, \log u + \log 3\, \log v = 0$ for positive rationals u, v, i.e. says precisely that some positive rational $\neq 1$ has a rational ϑ-th power. Equivalently, $\mathbb{Q}$-linear independence of F is the *rational-output* form of the problem:

$$2^x \in \mathbb{Q} \;\text{ and }\; 3^x \in \mathbb{Q} \qquad \Longrightarrow \qquad x \in \mathbb{Z}. \tag{24}$$

This is worth separating from Conjecture 1.1. It implies the conjecture, since integers are rational, and it is *a priori strictly stronger*: a rational counterexample such as $2^x = 3/2$ is not an integer counterexample, and nothing in the corpus reduces one to the other. Both follow from four exponentials, because rationals are algebraic.

The proof of Theorem 11.2 establishes (24) verbatim: for rational outputs the prime exponents α_p, γ_p are integers of either sign, and the argument never used their positivity. So the criterion is not a restatement of the conjecture; it delivers the stronger rational-output form. What the calibration does show is where its surplus lies: independence of pairs $\log p \log q$ with $p, q \geq 5$ is never exercised, so one should not expect the full criterion to be easier than (24). Its unconditional dividend is the two-prime case of Section 11.5.

Status: [kernel-verified Lean] for the integer-output form (`alaogluErdosConjecture_of_primeLogProductIndependence`); the rational-output strengthening (24) is **[paper proof]** and awaits a Lean version stated over `Rat` valuations.

It is nevertheless of a different kind from four exponentials. The four exponentials conjecture is a statement about 2×2 matrices of logarithms and their exponentials; the present criterion mentions no exponentials and is purely a linear-independence assertion about a countable family of real numbers. Neither is known, and neither is known to imply the other in general; what Theorem 11.2 shows is that *either* suffices for the Alaoglu–Erdős conjecture. This is worth recording because the two hypotheses invite different techniques: four exponentials is attacked by auxiliary functions and zero estimates, while linear independence of log-products is a question about the arithmetic of the multiplicative group of $\mathbb{Q}_{>0}$ and its second symmetric power.

The smallest instances are already open. Independence of the family includes, as its very first nontrivial case, the assertion that $(\log 3)^2$, $(\log 2)(\log 3)$ and $(\log 2)^2$ are $\mathbb{Q}$-linearly independent, equivalently that $\vartheta = \log_2 3$ is not a root of a rational quadratic — a statement about the irrationality of ϑ^2 modulo $\mathbb{Q}(\vartheta)$ that is not decided by present transcendence theory. Indeed the relation exhibited in Section 5 for a 3-smooth counterexample would already contradict the single independence

$$\alpha_2(\log 2)(\log 3) + \alpha_3(\log 3)^2 - \gamma_2(\log 2)^2 - \gamma_3(\log 2)(\log 3) \neq 0$$

for the relevant nonzero integer vector. Restricted to the two primes 2 and 3, the criterion needed is thus exactly: 1*,* ϑ *and* ϑ^2 *are linearly independent over* $\mathbb{Q}$, which is the 3-smooth shadow of Conjecture 5.4.

11.4 A finite form of the criterion

Because a counterexample has a fixed finite prime support, the full strength of Definition 11.1 is never needed at once.

Corollary 11.3 (Finite prime-support form)**.** *Let S be a finite set of primes containing* 2 *and* 3*. If the products* $\log p \log q$ *with* $p \leq q$ *in* S *are* $\mathbb{Q}$*-linearly independent, then no counterexample to Conjecture 1.1 has both outputs supported on* S*.*

Proof. The proof of Theorem 11.2 used only the pairs drawn from the prime support of MA together with 2 and 3. □

11.5 The two-prime case is a theorem, and it is sharp

The finite form is more than a bookkeeping device, because its smallest instance is *provable*.

Proposition 11.4 (Two-prime log-product independence)**.** *Let* $p \neq q$ *be primes. Then* $(\log p)^2$*,* $\log p\, \log q$ *and* $(\log q)^2$ *are linearly independent over* $\mathbb{Q}$*.*

Proof. A vanishing relation $a(\log p)^2 + b \log p \log q + c(\log q)^2 = 0$ with rational a, b, c, divided by $(\log p)^2$, reads $a + bt + ct^2 = 0$ where $t = \log_p q$. If $(a, b, c) \neq 0$ this makes t a root of a nonzero rational polynomial, hence algebraic. But t is transcendental: it is irrational by unique factorization, and were it algebraic irrational then Gelfond–Schneider applied to $p^t = q$ would make the integer q transcendental. □

Consequently the hypothesis of Corollary 11.3 is *unconditionally true* for $S = \{2, 3\}$, and the corollary becomes an unconditional exclusion. Stated directly, and proved in Lean without any appeal to Definition 11.1:

Theorem 11.5 (Three-smooth outputs force an integral exponent)**.** *If* $2^x \in \mathbb{Z}$ *and* $3^x \in \mathbb{Z}$ *are both* 3*-smooth — divisible by no prime outside* $\{2, 3\}$ *— then* $x \in \mathbb{Z}$*.*

Proof. Write $2^x = 2^a 3^b$ and $3^x = 2^c 3^d$. Taking base-two logarithms gives the two relations $x = a + b\vartheta$ and $x\vartheta = c + d\vartheta$. Substituting the first into the second eliminates x and leaves

$$b\,\vartheta^2 + (a - d)\,\vartheta - c = 0.$$

If $b \neq 0$ this makes ϑ a root of a nonzero rational quadratic, contradicting its transcendence. Hence $b = 0$ and $x = a$. □

Corollary 11.6 (A smooth candidate has a rough partner)**.** *If x is a nonintegral solution and 2^x is 3-smooth, then 3^x has a prime factor at least 5. The two outputs of a counterexample can never both be built from $\{2,3\}$.*

Status: [kernel-verified Lean] — `integer_of_threeSmooth_outputs` and `exists_prime_five_le_dvd_of_threeSmooth_candidate` in `SmoothOutputs`. The proofs use only transcendence of ϑ, itself kernel-verified in this corpus.

Two remarks fix the position of this result precisely.

The two-sided hypothesis is essential. Assuming only that 2^x is 3-smooth gives $x = a + b\vartheta$ and then requires $\left(3^{\vartheta}\right)^b \in \mathbb{Q}$, which is exactly the open localized-radical condition of Section 5. What the second smoothness hypothesis supplies is a *second* linear relation between x and ϑ; two such relations eliminate x and leave an algebraic equation in ϑ alone. This degree drop is the whole mechanism, and it is worth isolating as a template: *each independent multiplicative constraint on an output buys one linear relation, and two relations force algebraicity of ϑ.* The difficulty of the general problem is precisely that one never gets a second constraint for free.

Three primes is where the criterion becomes conjectural. Proposition 11.4 exhausts what transcendence of a single logarithmic ratio can give. For $S = \{2,3,5\}$ a vanishing relation among the six products, divided by $(\log 2)^2$, reads

$$a + b\vartheta + c\vartheta_5 + d\vartheta^2 + e\vartheta\vartheta_5 + f\vartheta_5^2 = 0, \qquad \vartheta_5 = \log_2 5,$$

that is, the point (ϑ, ϑ_5) lies on a rational conic. Both coordinates are transcendental and $1, \vartheta, \vartheta_5$ are $\mathbb{Q}$-independent by unique factorization, but no present method excludes a quadratic relation between them. So Definition 11.1 is a theorem for two primes, open from three onwards, and the Alaoglu–Erdős conjecture sits exactly on that boundary.

12 The algebraic auxiliary-base spectrum

Section 8 classified the *natural* auxiliary bases whose x-th powers remain algebraic, and had to tag the transcendence half of that classification as classical background: the repository's interpolation determinant carried integer entries, so it could prove $B^x \notin \mathbb{Q}$ but not $B^x \notin \overline{\mathbb{Q}}$. That boundary has moved. The same square determinant now accepts algebraic entries of bounded height inside a number field, and the resulting classification is exact over *all* positive real algebraic bases. A hypothetical nonintegral solution turns out to have one fixed, explicitly described set of good bases — a dense but very thin subgroup of $\mathbb{R}_{>0}$ — and every positive algebraic base outside it produces a transcendental value.

12.1 The saturated group of positive two-three units

The multiplicative group $\overline{\mathbb{Q}}_{>0}$ of positive real algebraic numbers is a vector space over $\mathbb{Q}$ once rational scalars act through the unique positive real root:

$$q \cdot \alpha = \alpha^q, \qquad q \in \mathbb{Q}, \quad \alpha \in \overline{\mathbb{Q}}_{>0}.$$

For multiplicatively independent $a, b \in \overline{\mathbb{Q}}_{>0}$ define the *saturated group*

$$\Gamma(a,b) = \{a^r b^s : r, s \in \mathbb{Q}\}.$$

The map $\mathbb{Q}^2 \to \Gamma(a,b)$, $(r,s) \mapsto a^r b^s$, is an isomorphism of $\mathbb{Q}$-vector spaces; injectivity is exactly multiplicative independence after clearing denominators. For the original problem write

$$\Gamma = \Gamma(2,3) = \{2^r 3^s : r, s \in \mathbb{Q}\}.$$

This group has a second description that is the one the Lean development actually uses. Let $\langle 2,3\rangle_{\mathbb{Z}} = \{2^i 3^j/(2^k 3^\ell) : i,j,k,\ell \in \mathbb{N}_0\}$ be the group of positive rational $2,3$-units, and let

$$\sqrt[\infty]{\langle 2,3\rangle_{\mathbb{Z}}} := \{\gamma > 0 : \exists\, n \geq 1,\ \gamma^n \in \langle 2,3\rangle_{\mathbb{Z}}\}$$

be its divisible hull inside $\mathbb{R}_{>0}$.

Observation 12.1. $\sqrt[\infty]{\langle 2,3\rangle_{\mathbb{Z}}} = \Gamma$. In particular every element of the divisible hull is automatically algebraic.

Indeed, if $\gamma^n = 2^i 3^j/(2^k 3^\ell)$ then γ is the unique positive real n-th root of a positive rational, hence $\gamma = 2^{(i-k)/n} 3^{(j-\ell)/n} \in \Gamma$; conversely $2^r 3^s$ raised to a common denominator of r, s lands in $\langle 2,3\rangle_{\mathbb{Z}}$. The Lean predicate `HasPositiveTwoThreeUnitPower` is the literal transcription of membership in the hull, so it and membership in Γ may be used interchangeably below.

12.2 The exact spectrum theorem

Three positive algebraic bases with injective nonnegative monomials cannot all have algebraic x-th powers when x is irrational. The next theorem is the coordinate-free synthesis for an arbitrary independent pair; its algebraic relation API and six-exponentials reduction are kernel-checked in `AlgebraicBaseSpectrum`.

Theorem 12.2 (Exact algebraic-base spectrum)**.** *Let $a, b \in \overline{\mathbb{Q}}_{>0}$ be multiplicatively independent, let $x \in \mathbb{R} \setminus \mathbb{Q}$, and assume $a^x, b^x \in \overline{\mathbb{Q}}$. Then, for every $\alpha \in \overline{\mathbb{Q}}_{>0}$,*

$$\alpha^x \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \alpha \in \Gamma(a,b).$$

[kernel-verified Lean]

Proof. If $\alpha = a^r b^s$ with $r, s \in \mathbb{Q}$, positivity gives $\alpha^x = (a^x)^r (b^x)^s$, which is algebraic.

Conversely, suppose $\alpha \notin \Gamma(a,b)$. Then $\log a$, $\log b$, $\log \alpha$ are linearly independent over $\mathbb{Q}$: a rational relation clears to an integer multiplicative relation, and if the coefficient of $\log \alpha$ vanishes it contradicts the multiplicative independence of a, b, while otherwise it places α in the saturated group. The pair $1, x$ is linearly independent over $\mathbb{Q}$ because x is irrational. Apply the six exponentials theorem (Theorem 8.3) to the three rows $\log a, \log b, \log \alpha$ and the two columns $1, x$. Five of the six exponential values,

$$a, \quad b, \quad \alpha, \quad a^x, \quad b^x,$$

are algebraic. Therefore the sixth, α^x, is transcendental. $\square$

Conceptually this says more than any list of good auxiliary bases: two algebraic outputs have already exhausted the maximum algebraic multiplicative rank that six exponentials permits.

Specializing to $a = 2$, $b = 3$ gives the statement the repository proves outright.

Corollary 12.3 (Complete algebraic spectrum of a counterexample)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$. For every $\alpha \in \overline{\mathbb{Q}}_{>0}$,*

$$\alpha^x \in \overline{\mathbb{Q}} \iff \alpha \in \Gamma.$$

In particular the spectrum does not depend on which hypothetical nonintegral solution is chosen. ***[kernel-verified Lean]***

A rational solution is integral, so x is irrational and Theorem 12.2 applies; the formal route is the divisible-hull equivalence of the next subsection together with Observation 12.1.

Status: **[kernel-verified Lean]** — the divisible-hull equivalence lives in `AlgebraicBaseUnitPower`.

Its two forms are

`real_rpow_isAlgebraic_iff_integer_or_hasPositiveTwoThreeUnitPower`

together with its counterexample-relative form

`real_rpow_isAlgebraic_iff_hasPositiveTwoThreeUnitPower`

in the `TwoBaseNonintegerSolution` namespace, both resting on the injectivity criterion `real_two_three_gamma_monomial_injective_iff`. The natural-base and rational-base specializations are `rpow_isAlgebraic_iff_integer_or_eq_two_pow_mul_three_pow` in `AlgebraicThirdBase` and `rat_rpow_isAlgebraic_iff_isTwoThreeUnit_of_not_integer` in `AlgebraicRationalBase`; the half-plane, mixed-iterate and rank-one ratio consequences are in `AlgebraicOutputLocus`; and the simultaneous exclusion of both canonical radicals from the entire hull is in `CanonicalRadicalDivisibleHull`.

The group Γ is countable, divisible, of rational rank two, and dense in $\mathbb{R}_{>0}$ — density already follows from the subgroup $\{2^r : r \in \mathbb{Q}\}$. A counterexample would therefore possess a dense set of algebraic bases with algebraic output, while every algebraic base outside that exact dense subgroup would give a transcendental one.

Corollary 12.4 (Rank bound for any family of algebraic outputs)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$ and let $\alpha_1, \dots, \alpha_t \in \overline{\mathbb{Q}}_{>0}$ satisfy $\alpha_i^x \in \overline{\mathbb{Q}}$. Then every α_i lies in Γ and the multiplicative rank of $\alpha_1, \dots, \alpha_t$ is at most two.* ***[paper proof]***

Both ingredients are kernel-verified — membership in the hull by `AlgebraicBaseUnitPower` and the rank ceiling by `AlgebraicRankThreeEquivalence` — but the combined family statement is recorded here at paper level.

12.3 The divisible-hull formulation

Stated without reference to a fixed counterexample, the kernel-verified form is a single biconditional valid for every positive real algebraic base. For $\gamma \in \overline{\mathbb{Q}}_{>0}$, under $2^x, 3^x \in \mathbb{Z}$,

$$\gamma^x \in \overline{\mathbb{Q}} \iff x \in \mathbb{Z} \quad \text{or} \quad \gamma \in \sqrt[\infty]{\langle 2, 3 \rangle_{\mathbb{Z}}}. \tag{25}$$

The arithmetic core is an exact combinatorial equivalence: γ has *no* positive natural power in $\langle 2,3\rangle_{\mathbb{Z}}$ precisely when the monomial map $(i,j,k) \mapsto 2^i 3^j \gamma^k$ is injective on $\mathbb{N}_0^3$. The forward implication of (25) then feeds that injectivity into the generic algebraic determinant, and the reverse implication descends algebraicity from a positive integral power. Specializing γ to a natural number recovers the 3-smooth classification of Section 8 with "rational" upgraded to "algebraic", so for a hypothetical nonintegral solution the three properties $a^x \in \mathbb{Z}$, $a^x \in \mathbb{Q}$ and $a^x \in \overline{\mathbb{Q}}$ coincide for $a \in \mathbb{N}_{>0}$, and hold exactly at $2,3$-smooth a. Specializing to a positive rational q gives $q^x \in \overline{\mathbb{Q}} \iff q \in \langle 2,3\rangle_{\mathbb{Z}}$; there is no algebraic-irrational middle case anywhere in the picture.

12.4 The determinant engines behind the spectrum

Two generic determinant theorems, neither of which mentions the bases $2,3$, carry the whole classification. The rational one states that if $a,b,c \in \mathbb{Q}_{>0}$ have injective nonnegative monomials $(i,j,k) \mapsto a^i b^j c^k$ and $a^x, b^x, c^x \in \mathbb{Q}$, then $x \in \mathbb{Q}$. The algebraic one removes both restrictions: if a,b,c are positive real algebraic numbers with injective nonnegative monomials and a^x, b^x, c^x are all algebraic, then $x \in \mathbb{Q}$. Both allow bases below 1 and negative exponents, and both use $1+|x|$ in the size estimate.

Status: [kernel-verified Lean] — `rational_of_three_rat_rpows_rational_of_monomial_injective` in `RationalSixExponentials`, and `rational_of_three_real_rpows_isAlgebraic_of_monomial_injective` in `AlgebraicSixExponentials`.

The step from rational to algebraic outputs is exactly the replacement of the trivial bound $|D| \geq 1$ for a nonzero integer determinant by a field-norm bound for a nonzero algebraic integer determinant, which is the extension anticipated in Section 8. The six input and output values are placed in one number field K with one common nonzero natural scale s making every entry lie in $\mathcal{O}_K$. For side parameter n the square matrix has size n^6 and each entry acquires the factor s^{n^5}, so the determinant acquires $s^{n^{11}}$; at every embedding other than the distinguished one the six factor groups in an entry are bounded by T^{6n^5}, and the norm of the nonzero algebraic-integer determinant gives

$$\frac{1}{\left((n^6)!\, T^{6n^{11}}\right)^{[K:\mathbb{Q}]-1}} \;\leq\; |\sigma(\det A)|.$$

The verified estimates $(n^6)! \leq 2^{n^{11}}$ and $\left((n^6)!\, T^{6n^{11}}\right)^{[K:\mathbb{Q}]-1} \leq \left((2T^6)^{[K:\mathbb{Q}]-1}\right)^{n^{11}}$ put the entire conjugate and degree loss on the same n^{11} scale as the old rational denominator loss, where the existing explicit analytic margin already absorbs it. No new analytic estimate was required — only careful bookkeeping.

Status: [kernel-verified Lean] — `one_div_conjugate_entry_bound_le_norm_det` and `one_div_entry_bound_le_scaled_norm_det` in `AlgebraicIntegerDeterminant`, with the number-field realizations `AlgebraicOutputBridge.ofRealIsAlgebraic` and `FiniteAlgebraicOutputBridge.ofFinSixIsAlgebraic`.

12.5 The strong spectrum via Baker and Roy

Let $\mathcal{L}$ denote the $\overline{\mathbb{Q}}$-vector space generated by 1 and by all logarithms of nonzero algebraic numbers, with arbitrary complex branches allowed; for positive real algebraic numbers we use the real logarithm, which is one of those branches. Roy's strong six exponentials theorem [23, 26] states that if three complex numbers are linearly independent over $\overline{\mathbb{Q}}$ and two more are linearly independent over $\overline{\mathbb{Q}}$, then at least one of the six pairwise products lies outside $\mathcal{L}$.

Theorem 12.5 (Strong algebraic-base spectrum)**.** *Under the hypotheses of Theorem 12.2, if $\alpha \notin \Gamma(a, b)$, then*

$$x \log \alpha \notin \mathcal{L}.$$

In particular α^x is transcendental. ***[paper proof]***

Proof. As in Theorem 12.2, the logarithms $\log a, \log b, \log \alpha$ are $\mathbb{Q}$-linearly independent, and Baker's homogeneous theorem [3] upgrades that to linear independence over $\overline{\mathbb{Q}}$. The number x is transcendental: were it algebraic, its irrationality together with $a \neq 0, 1$ would make a^x transcendental by Gelfond–Schneider. Hence $1, x$ are linearly independent over $\overline{\mathbb{Q}}$. Apply Roy's theorem to the rows $\log a, \log b, \log \alpha$ and the columns $1, x$. Five products lie in $\mathcal{L}$, namely $\log a$, $\log b$, $\log \alpha$, $x \log a = \log(a^x)$ and $x \log b = \log(b^x)$. Therefore the sixth product $x \log \alpha$ lies outside $\mathcal{L}$. □

Definition 12.6. Call a positive exponential value e^z *log-strongly transcendental* when $z \notin \mathcal{L}$. This implies ordinary transcendence of e^z, since the logarithm of a positive algebraic number belongs to $\mathcal{L}$.

So outside Γ the failure is not merely that α^x is transcendental: the number $x \log \alpha$ is not even a $\overline{\mathbb{Q}}$-linear combination of 1 and logarithms of algebraic numbers. This is a genuinely stronger exclusion than Corollary 12.3, and it is the form in which the spectrum interacts with linear forms in logarithms.

12.6 One base is enough: detectors and rank equivalences

Because the spectrum is the same for every hypothetical nonintegral solution, a *single* auxiliary base outside Γ already decides the whole conjecture.

Theorem 12.7 (Algebraic detector equivalence)**.** *Fix $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$. Then the Alaoglu–Erdős conjecture is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \Longrightarrow \alpha^x \in \overline{\mathbb{Q}},$$

and equally to the assertion that every such x satisfies $x \log \alpha \in \mathcal{L}$. ***[paper proof]***

Proof. If the conjecture holds, every solution is a nonnegative integer and α^x is algebraic, with $x \log \alpha \in \mathcal{L}$. Conversely a nonintegral solution would make α^x transcendental by Corollary 12.3 and would put $x \log \alpha$ outside $\mathcal{L}$ by Theorem 12.5. □

This is a paper-level universalization of a pointwise locus that the kernel already knows. Valid detectors include, for instance,

$$5, \qquad \sqrt{5}, \qquad 1 + \sqrt{2}, \qquad \frac{7 + 3\sqrt{5}}{2},$$

provided the chosen number does not accidentally lie in Γ; for an algebraic integer with a prime divisor outside $\{2, 3\}$ that exclusion is immediate, and in general it is a multiplicative independence check. Nothing arithmetically special about 5 is used.

Four instances of this phenomenon are themselves kernel-verified, and they are the most quotable consequences of the whole algebraic-output layer.

One-base equivalents of the conjecture. Each of the following is equivalent to the Alaoglu–Erdős conjecture, and each equivalence is accepted by the Lean kernel.

(i) Every x with $2^x, 3^x \in \mathbb{Z}$ has 5^x algebraic.

(ii) Every x with $2^x, 3^x \in \mathbb{Z}$ has 6^{x^2} algebraic.

(iii) For every x with $2^x, 3^x \in \mathbb{Z}$, the positive-algebraic output locus

$$\{\alpha \in \overline{\mathbb{Q}}_{>0} : \alpha^x \in \overline{\mathbb{Q}}\}$$

has multiplicative rank three, i.e. contains a, b, c with injective nonnegative monomials.

(iv) For every x with $2^x, 3^x \in \mathbb{Z}$, the rational squared-output ratio locus has rank two, i.e. contains algebraic members $(2^i/3^j)^{x^2}$ and $(2^k/3^\ell)^{x^2}$ with $i\ell \neq kj$.

A counterexample would collapse (iii) to rank at most two and confine the entire locus in (iv) to a single rational ray in the exponent lattice. **[kernel-verified Lean]**

Status: **[kernel-verified Lean]** — all four equivalences are accepted by the kernel; the Lean names follow.

Items (i) and (ii) are `alaogluErdosConjecture_iff_five_rpow_isAlgebraic` and `alaogluErdosConjecture_iff_six_squared_exponent_isAlgebraic` in `AlgebraicConsequences`, which also carries the two-iterate variant `alaogluErdosConjecture_iff_both_iterated_outputs_isAlgebraic`. Item (iii) is

`alaogluErdosConjecture_iff_algebraicRpowOutputLocusHasRankThree`

in `AlgebraicRankThreeEquivalence`, and item (iv) is

`alaogluErdosConjecture_iff_ratioSquaredAlgebraicLocusHasRankTwo`

in `AlgebraicOutputLocus`. The same modules also record the pointwise forms `ratioSquaredAlgebraicLocusHasRankTwo_iff_integer` and `algebraicRpowOutputLocusHasRankThree_iff_integer`, and the explicit p-adic line `exists_external_prime_algebraic_output_ratio_padic_line` on which a counterexample would force every algebraic ratio output to sit.

12.7 Exact exponent, rational-function, and tower loci

Fix a hypothetical nonintegral solution x and put

$$\mathcal{E}_x = \{y \in \mathbb{R} : 2^y, 3^y \in \overline{\mathbb{Q}}\}.$$

The transposed six-exponentials determinant determines this set exactly:

$$\mathcal{E}_x = \mathbb{Q} + \mathbb{Q}x. \tag{26}$$

Thus every exponent outside the affine plane makes at least one of 2^y and 3^y transcendental. In particular $y = x^n$ for every $n \geq 2$, and also $y = x^{-1}$, lie outside the plane because x is transcendental. For every fixed $n \geq 2$, requiring both 2^{x^n} and 3^{x^n} to be algebraic at every original two-base solution is therefore equivalent to Conjecture 1.1. The theorem says "at least one": it neither selects which base is transcendental nor supplies the missing algebraicity needed to prove the conjecture.

The exact plane makes every genuinely nonaffine rational function an equally sharp detector. For $P, Q \in \mathbb{Q}[X]$ with $Q(x) \neq 0$,

$$2^{P(x)/Q(x)}, 3^{P(x)/Q(x)} \in \overline{\mathbb{Q}} \quad \iff \quad P = (q + rX)Q \quad \text{for some } q, r \in \mathbb{Q}. \tag{27}$$

The reverse direction is literal affine membership in (26); the forward direction uses transcendence of x to turn equality after evaluation into a polynomial identity. Consequently any fixed nonaffine rational function represented as P/Q with the displayed denominator Q having no real zero gives a global formulation equivalent to Conjecture 1.1, and every $P \in \mathbb{Q}[X]$ of degree at least two does so with $Q = 1$. These are detector equivalences, not new algebraicity results: they relocate the open conclusion to another exponent without proving it.

Under failure, the same module chooses the canonical generator β so that the two loci can be compared on one rational-function chart. For every P/Q defined at β, both $2^{P(\beta)/Q(\beta)}$ and $3^{P(\beta)/Q(\beta)}$ are algebraic exactly when $P = (q + rX)Q$ with $q, r \in \mathbb{Q}$, whereas $P(\beta)/Q(\beta)$ is itself an original integral solution exactly when $P = (n + kX)Q$ with $n, k \in \mathbb{N}_0$. Thus the exact solution monoid is the nonnegative integral cone inside the rational affine algebraic-output plane; the comparison is exact, but it does not collapse the larger plane to that cone.

There is also an exact two-dimensional picture for the squared monomials. Under a counterexample, the set

$$\mathcal{F}_x = \{(i, j) \in \mathbb{N}_0^2 : (2^i 3^j)^{x^2} \in \overline{\mathbb{Q}}\}$$

is exactly one of

$$\mathbb{N}_0 \times \{0\}, \qquad \{0\} \times \mathbb{N}_0, \qquad \{(0, 0)\}.$$

Writing $M = 2^x$ and $B = 3^x$, the three cases say respectively that M is $2,3$-smooth and B is not, that B is $2,3$-smooth and M is not, or that neither is; both cannot be $2,3$-smooth for a nonintegral solution. Under the original two integrality hypotheses this coordinate-face property is pointwise equivalent to nonintegrality, and the conjecture is equivalent to excluding it for every solution. All three faces remain compatible with the present theory, so the trichotomy does not itself yield a contradiction.

A related transition theorem has a different, solution-independent plane. With $\vartheta = \log_2 3$, define

$$\mathcal{T}_x = \{y \in \mathbb{R} : 2^y \in \overline{\mathbb{Q}} \text{ and } 2^{xy} \in \overline{\mathbb{Q}}\}.$$

For every hypothetical nonintegral solution,

$$\mathcal{T}_x = \mathbb{Q} + \mathbb{Q}\vartheta. \tag{28}$$

Two consecutive algebraic base-two tower levels therefore correspond exactly to one exponent in this fixed plane. More strongly, at most one positive depth n can satisfy $x^n \in \mathcal{T}_x$. Indeed, transcendence of x makes the ϑ-coefficients in the two affine expressions nonzero; raising the two expressions to cross-powers and using transcendence of ϑ promotes equality at ϑ to a polynomial identity, whose degree comparison forces the two depths to coincide. Hence every three consecutive positive-depth levels

$$2^{x^n}, \qquad 2^{x^{n+1}}, \qquad 2^{x^{n+2}}$$

contain a transcendental value, and over any finite set the collection of depths starting an adjacent algebraic pair has cardinality at most one. Conversely, for any fixed distinct positive depths $n \neq m$, the assertion that every original two-base solution has algebraic adjacent pairs at both depths n and m is equivalent to Conjecture 1.1. The restriction still permits isolated algebraic levels and a single adjacent algebraic pair; it does not prove the full conjecture.

The symmetric transition picture is stronger than merely repeating the same argument with base three. Put

$$\mathcal{S}_x = \{y \in \mathbb{R} : 3^y \in \overline{\mathbb{Q}} \text{ and } 3^{xy} \in \overline{\mathbb{Q}}\}.$$

Under a nonintegral solution,

$$y \in \mathcal{S}_x \quad \Longleftrightarrow \quad \vartheta y \in \mathbb{Q} + \mathbb{Q}\vartheta. \tag{29}$$

No positive depth in the base-two tower can start an algebraic adjacent pair simultaneously with any positive depth in the base-three tower, even when the two depths are unrelated. The base-three tower itself has at most one such positive depth. Consequently, across the two principal towers together there is at most one positive transition depth, and at that depth at most one of the two bases can occur. There is also a direction-uniform version in the multiplicatively dependent-output branch: once its kernel relation produces a nonzero rational $2,3$-unit direction with a depth-one transition, every nonzero rational $2,3$-unit direction is excluded at every depth $n \geq 2$. These are strict structural obstructions, but they still allow the single depth-one transition forced by the kernel relation and therefore do not close that branch.

Status: [kernel-verified Lean] — all statements in this subsection are accepted by the kernel.

The exact exponent plane and its natural-power and reciprocal consequences are `TwoBaseNonintegerSolution.twoThreeRpowAlgebraic_iff_affine`, `transcendental_two_or_three_rpow_nat_pow`, `transcendental_two_or_three_rpow_inv`, and `alaogluErdosConjecture_iff_twoThreeRpowAlgebraic_nat_pow` in `AlgebraicExponentLocus`. Equation (27) and its global and polynomial specializations are `TwoBaseNonintegerSolution.twoThreeRpowAlgebraic_rationalFunction_iff`, `alaogluErdosConjecture_iff_twoThreeRpowAlgebraic_rationalFunction`, and `alaogluErdosConjecture_iff_twoThreeRpowAlgebraic_polynomial_of_two_le_natDegree` in `RationalFunctionAlgebraicLocus`. Its simultaneous two-locus package is `exists_generator_with_exact_rationalFunction_algebraic_and_integral_loci`. The exact squared-coordinate-face statements are `TwoBaseNonintegerSolution.exists_exact_algebraic_output_monomial_locus`, `algebraic_two_three_squared_monomial_locus_trichotomy`, `squaredTwoThreeMonomialAlgebraicLocusIsCoordinateFace_iff_not_integer`, and `alaogluErdosConjecture_iff_no_squaredTwoThreeMonomial_coordinateFace` in `AlgebraicOutputLocus`. Finally, (28), global uniqueness of a transition depth, and the fixed-distinct-depth equivalence are `TwoBaseNonintegerSolution.twoRpowAlgebraicTransition_iff_affine_logRatio`, `transcendental_one_of_three_consecutive_two_rpow_tower`, `not_two_distinct_algebraic_tower_transitions`, `card_algebraicTowerTransitionDepths_le_one`, and `alaogluErdosConjecture_iff_two_distinct_tower_transitions_algebraic` in `AlgebraicTowerTransition`. The symmetric locus (29), the cross-base exclusion, the combined uniqueness theorem, and the rational-unit-direction exclusions are respectively `TwoBaseNonintegerSolution.threeRpowAlgebraicTransition_iff_scaled_affine_logRatio`, `not_baseTwo_and_baseThree_tower_transitions`, `eq_of_twoOrThree_tower_transitions`, and `not_higher_unitDirectionTowerTransition_of_kernelRelation` in `CrossBaseTowerTransition`. Their axiom audit uses only `propext`, `Classical.choice`, and `Quot.sound`. None asserts Conjecture 1.1.

There is a corresponding zero-one law: assuming the conjecture fails and fixing $\alpha \in \overline{\mathbb{Q}}_{>0}$, exactly one of two things happens. Either $\alpha \in \Gamma$ and α^x is algebraic for *every* nonintegral solution x, or $\alpha \notin \Gamma$ and α^x is transcendental for every nonintegral solution. No base can behave one way for one counterexample and the other way for another.

12.8 Why the exact spectrum does not settle the conjecture

It is worth being precise about what has and has not been gained. The spectrum is an exact answer to the question "which algebraic bases stay algebraic?", and its answer is $\Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\}$ — but that is precisely the set of bases whose logarithms lie in the rational span of $\log 2$ and $\log 3$. Every base the spectrum certifies is, logarithmically, a rational recombination of the two columns already present in the four-exponentials matrix of Section 6. Six exponentials needs a third row genuinely independent of $\log 2, \log 3$ together with algebraicity of its output; the spectrum theorem shows that a counterexample supplies algebraic outputs at exactly the bases that fail this requirement, and transcendental outputs at exactly the bases that would meet it. The classification is therefore self-consistent with a counterexample rather than obstructive to one: it converts every attempt to manufacture a third independent base into the very transcendence statement one was trying to prove. The kernel-verified radical work makes the same wall concrete: the simultaneous normalization of the two canonical radicals produces an explicit multiplicative relation among 2, the primitive odd core, and the swapped radical, so that the most tempting three-base application of the algebraic six-exponentials theorem fails its monomial-injectivity hypothesis for a structural, kernel-verified reason (`swappedRadical_relation_blocks_monomial_injectivity` in `CanonicalRadicalFieldNorm`). Closing the gap needs a new source of size — nonarchimedean, or from a sharper five/six-exponentials interface — not a wider supply of algebraic bases. The conjecture remains open.

13 Rational and integral outputs inside the spectrum

The algebraic spectrum identifies the divisible group

$$\Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\}$$

as the exact set of positive algebraic bases whose powers at a hypothetical nonintegral solution remain algebraic: outside Γ every such power is transcendental, and inside Γ every such power is algebraic. A finer arithmetic question survives inside the spectrum. If $\alpha = 2^r 3^s \in \Gamma$, when is α^x rational, and when is it a positive integer?

For a fixed solution the answer is completely encoded by prime valuations of the output pair. Rationality becomes a congruence condition, integrality adds a cone condition, and the gap between the obvious rational bases $2^u 3^v$ with $u, v \in \mathbb{Z}$ and the full rational locus is one explicit finite abelian group whose order is a single gcd of 2×2 minors. This is a strict refinement of the divisible-hull description of the algebraic locus, and it is exact rather than asymptotic.

13.1 The valuation matrix of an output pair

Let $x \in \mathcal{S} \setminus \mathbb{Z}$ be a nonintegral solution and write

$$M = 2^x \in \mathbb{N}, \qquad A = 3^x \in \mathbb{N}.$$

Let P be the finite union of the prime supports of M and A, and form the column vectors

$$\mathbf{m} = (v_p(M))_{p \in P}, \qquad \mathbf{a} = (v_p(A))_{p \in P}, \qquad V_x = \begin{pmatrix} \mathbf{m} & \mathbf{a} \end{pmatrix} \in M_{P \times 2}(\mathbb{Z}).$$

For a column $q = (r, s)^T \in \mathbb{Q}^2$ put $\alpha(q) = 2^r 3^s$, so that Γ is exactly the image of $q \mapsto \alpha(q)$. Taking the unique positive real interpretation of the rational powers,

$$\alpha(q)^x = \left(2^x\right)^r \left(3^x\right)^s = M^r A^s = \prod_{p \in P} p^{r v_p(M) + s v_p(A)}. \tag{30}$$

The exponent vector appearing in (30) is precisely $V_x q$.

Lemma 13.1 (The valuation matrix has rank two). *The two columns of V_x are linearly independent over $\mathbb{Q}$.*

Proof. Suppose $r\mathbf{m}+s\mathbf{a}=0$ with $r,s\in\mathbb{Q}$. Then every prime exponent in (30) vanishes, so $M^rA^s=1$. Taking real logarithms and using $\log M = x\log 2$, $\log A = x\log 3$ gives $x(r\log 2+s\log 3)=0$. Since a solution with $M=2^x\in\mathbb{N}$, $M>1$ has $x>0$, and since $\vartheta=\log_2 3$ is irrational, $r=s=0$. □

Proposition 13.2 (Exact rational and integral auxiliary-output spectrum). *Let $x\in\mathcal{S}\setminus\mathbb{Z}$, let V_x be as above, and let $q\in\mathbb{Q}^2$. Then*

$$\alpha(q)^x\in\mathbb{Q}_{>0} \iff V_xq\in\mathbb{Z}^P, \qquad\qquad \alpha(q)^x\in\mathbb{N} \iff V_xq\in\mathbb{N}_0^P.$$

*Together with the algebraic spectrum, this classifies algebraic, rational and integral outputs for every positive algebraic auxiliary base. **[paper proof]***

Proof. Equation (30) exhibits $\alpha(q)^x$ as a finite product of primes with rational exponents. Choose a common denominator d for the entries of V_xq. If $\alpha(q)^x$ is rational, then raising to the d-th power gives a rational number whose ordinary p-adic valuation equals d times the p-th entry of V_xq; comparing valuations of numerator and denominator shows every entry of V_xq is an integer. Conversely, integer exponents produce a positive rational number. Such a rational number is a positive integer exactly when all of its prime valuations are nonnegative, which is the second equivalence. □

Three nested spectra therefore occur, all determined by V_x:

$$\begin{aligned}\mathcal{B}_{\mathrm{alg}}(x) &= \{\alpha(q): q\in\mathbb{Q}^2\}=\Gamma,\\ \mathcal{B}_{\mathrm{rat}}(x) &= \{\alpha(q): q\in\Lambda_x\},\\ \mathcal{B}_{\mathrm{int}}(x) &= \{\alpha(q): q\in\Lambda_x^+\},\end{aligned} \qquad \begin{aligned}\Lambda_x &= \{q\in\mathbb{Q}^2 : V_xq\in\mathbb{Z}^P\},\\ \Lambda_x^+ &= \{q\in\Lambda_x : V_xq\ge 0\}.\end{aligned}$$

The outer spectrum does not depend on x; the two inner ones may.

13.2 Smith normal form and the finite rational-output defect

By Lemma 13.1 the matrix V_x has rank two, so Λ_x is a rank-two lattice in $\mathbb{Q}^2$ containing $\mathbb{Z}^2$ with finite index. Write

$$\delta_1(V_x)=\gcd\{\text{all entries of } V_x\}, \qquad \delta_2(V_x)=\gcd\left\{\left|\det\begin{pmatrix} v_p(M) & v_p(A)\\ v_{p'}(M) & v_{p'}(A)\end{pmatrix}\right| : p,p'\in P\right\}$$

for the two determinantal divisors. The second is positive precisely because the rank is two.

Proposition 13.3 (Smith classification of the rational locus). *Let $d_1\mid d_2$ be the Smith invariant factors of V_x, so that $d_1=\delta_1(V_x)$ and $d_1d_2=\delta_2(V_x)$. After a unimodular change of coordinates in $\mathbb{Q}^2$,*

$$\Lambda_x=\tfrac{1}{d_1}\mathbb{Z}\oplus\tfrac{1}{d_2}\mathbb{Z},$$

and consequently

$$\Lambda_x/\mathbb{Z}^2\cong\mathbb{Z}/d_1\mathbb{Z}\oplus\mathbb{Z}/d_2\mathbb{Z}, \qquad \left|\Lambda_x/\mathbb{Z}^2\right|=\delta_2(V_x).$$ ***[paper proof]***

Proof. Choose unimodular matrices $U \in \mathrm{GL}_P(\mathbb{Z})$ and $W \in \mathrm{GL}_2(\mathbb{Z})$ with

$$UV_xW = \begin{pmatrix} d_1 & 0 \\ 0 & d_2 \\ 0 & 0 \\ \vdots & \vdots \end{pmatrix}.$$

Since U preserves $\mathbb{Z}^P$, the condition $V_xq \in \mathbb{Z}^P$ is equivalent to $UV_xq \in \mathbb{Z}^P$, and substituting $q = Wq'$ turns it into $d_1q'_1 \in \mathbb{Z}$ and $d_2q'_2 \in \mathbb{Z}$. Since W preserves $\mathbb{Z}^2$, it identifies $\Lambda_x/\mathbb{Z}^2$ with the quotient computed in the primed coordinates, which is $\mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z}$. The identifications $d_1 = \delta_1(V_x)$ and $d_1d_2 = \delta_2(V_x)$ are the standard determinantal-divisor formulae for the Smith normal form. □

The first invariant factor has a direct output interpretation: d_1 is the largest integer d such that M and A are both perfect d-th powers, because $d \mid \delta_1(V_x)$ says exactly that d divides every prime valuation of M and of A. For the canonical least nonintegral solution β, primitivity of the least output pair (Section 4) gives $d_1 = 1$, and then

$$\Lambda_\beta/\mathbb{Z}^2 \cong \mathbb{Z}/\delta_2(V_\beta)\mathbb{Z}.$$

Thus a single integer — the gcd of all 2×2 valuation minors of the least output pair — measures the entire finite defect between the rational β-outputs at algebraic $2,3$-radical bases and the obvious rational $2,3$-unit bases. We call $\delta_2(V_\beta)$ the *rational-output defect*. It is a finite, in principle computable, invariant of a hypothetical counterexample, and it equals 1 exactly when no auxiliary base outside $\{2^u3^v : u, v \in \mathbb{Z}\}$ produces a rational output.

The integral spectrum is the intersection of this lattice with the rational polyhedral cone $V_xq \geq 0$. By Gordan's lemma Λ_x^+ is a finitely generated normal affine semigroup, so $\mathcal{B}_{\mathrm{int}}(x)$ has a finite generating set of auxiliary bases. This is an exact structural replacement for attempting to enumerate algebraic auxiliary bases one at a time.

Status: [**kernel-verified Lean**] and [**paper proof**]

The lattice statements themselves — Lemma 13.1, Proposition 13.2, Proposition 13.3 and the Gordan consequence — are [**paper proof**]: they are proved on paper in this report and are not yet formalized in that form. What *is* kernel-verified is the arithmetic core on which they rest, in three modules.

`CanonicalRadicalValuation` proves the divisibility dictionary between perfect powers and valuations that underlies the meaning of d_1: a divisor of the odd-prime valuation gcd of a positive integer is a genuine exponent of an exact power (`exists_pow_eq_of_dvd_oddPrimeValuationGCD`), the least rational-power index is coprime to the joint gcd of the denominator exponent and the numerator's valuation gcd (`rationalPowerIndex_gcd_denominator_valuationGCD_eq_one`), and every prime dividing that index leaves some prime valuation of the normalized numerator undivided, giving the exact radical degree together with the valuation restriction (`exists_exact_radical_degree_and_valuation_restriction`).

`SimultaneousRadicalDegrees` kernel-verifies the exactness statements that the $d_1 = 1$ normalization buys: under the simultaneous primitive normalization the displayed exponents are the least rational-power indices and hence the exact degrees over $\mathbb{Q}$ (`oddCoreRpow_exact_rationalPowerIndex_of_simultaneous_normalization` and `threeFreeCoreRpow_exact_radical_degree_of_simultaneous_normalization`), and when both base-adic depths vanish the two indices are coprime and the mixed product has exact degree equal to their product (`mixed_coreRpow_exact_radical_degree_of_zero_baseDepths`). The descent used there is precisely the one behind $d_1 = 1$: a proper index divisor would make both outputs of the least nonintegral solution common perfect powers, which is excluded by `IsLeastTwoBaseNonintegerSolution.no_common_perfect_power` in `LeastSolution` (Section 4). That exclusion is exactly the assertion $d_1 = 1$ for V_β.

`CanonicalRadicalMixedOutput` kernel-verifies the transcendence side of the mixed case: a genuinely mixed positive monomial in the two canonical radicals retains an external prime in every rational power, so all of its outputs at the least nonintegral solution are transcendental, and an algebraic mixed output is therefore forced onto a coordinate face. The theorem is

`TwoBaseNonintegerSolution.transcendental_mixed_canonical_radical_output`,

packaged under failure of the conjecture as

`exists_simultaneous_canonical_radical_mixed_output_transcendence`.

These are statements about the canonical radicals rather than about Λ_x itself; they confirm the geometry of the lattice picture without yet formalizing it.

Exact simultaneous radical degrees and mixed-output exclusion. Under failure of the conjecture, let β be the least nonintegral solution and choose its simultaneous primitive normalization

$$2^\beta = 2^a w^d, \qquad 3^\beta = 3^b v^e, \qquad a = 0 \text{ or } b = 0.$$

For

$$E = w^\vartheta, \qquad F = v^{1/\vartheta},$$

the simultaneous power identities are

$$E^d = \frac{3^b v^e}{3^a}, \qquad F^e = \frac{2^a w^d}{2^b}.$$

The least positive rational-power indices of E and F are exactly d and e. Consequently their minimal polynomials are the displayed pure binomials

$$X^d - \frac{3^b v^e}{3^a}, \qquad X^e - \frac{2^a w^d}{2^b},$$

and their degrees over $\mathbb{Q}$ are exactly d and e. In the zero-depth case $a = b = 0$ one has $\gcd(d, e) = 1$, and for $G = EF$,

$$\operatorname{minpoly}_{\mathbb{Q}}(G) = X^{de} - v^{e^2} w^{d^2}, \qquad [\mathbb{Q}(G) : \mathbb{Q}] = de.$$

Moreover, the canonical data supplied by failure contain an external prime, and for every $i, j > 0$ the mixed output $(E^i F^j)^\beta$ is transcendental; if a nonnegative mixed output is algebraic, its exponent pair lies on one of the two coordinate axes. **[kernel-verified Lean]**

These conclusions are mutually compatible, not contradictory. The degree statements concern the algebraic numbers E, F, and EF, whereas the real irrational-power outputs at β are transcendental. A field embedding of $\mathbb{Q}(E, F)$ need not commute with the chosen real operation $z \mapsto z^\beta$, so taking conjugates or norms does not turn the mixed-output transcendence into a forbidden algebraic relation. In the zero-depth case the product degree de is the maximal degree allowed by the two coprime pure-radical degrees, not an impossible dependence. Thus the simultaneous theory sharpens the exact algebraic data and excludes the interior of the mixed-output quadrant, but leaves both coordinate axes — and the full Alaoglu–Erdős conjecture — open.

13.3 Two translated solutions remove the fractional defect

The algebraic spectrum is solution-independent, but the rational locus Λ_x need not be. Translation of the exponent by one gives a sharp intersection identity, and it shows that the defect cannot persist across the whole solution monoid.

Proposition 13.4 (Translation intersection)**.** *Let $x \in \mathcal{S} \setminus \mathbb{Z}$. Then*

$$\Lambda_x \cap \Lambda_{x+1} = \mathbb{Z}^2.$$

*Equivalently, for $\alpha \in \overline{\mathbb{Q}}_{>0}$, if α^x and α^{x+1} are both rational, then $\alpha = 2^u 3^v$ with $u, v \in \mathbb{Z}$. **[paper proof]***

Proof. One inclusion is immediate: for $q \in \mathbb{Z}^2$ the base $\alpha(q) = 2^u 3^v$ has $\alpha(q)^y = M^u A^v \in \mathbb{Q}_{>0}$ for every solution y, in particular for $y = x$ and $y = x + 1$. Conversely, suppose α^x and α^{x+1} are rational. Both are algebraic, so the algebraic spectrum gives $\alpha = 2^r 3^s$ with $r, s \in \mathbb{Q}$, and

$$\alpha = \frac{\alpha^{x+1}}{\alpha^x} \in \mathbb{Q}_{>0}.$$

A positive rational of the form $2^r 3^s$ with $r, s \in \mathbb{Q}$ has $r, s \in \mathbb{Z}$, by comparing ordinary 2-adic and 3-adic valuations after clearing a common denominator. $\square$

Corollary 13.5 (Universal rational and integral spectra under failure)**.** *Assume Conjecture 1.1 fails and let $\alpha \in \overline{\mathbb{Q}}_{>0}$.* ***[paper proof]***

1. *α^y is rational for every nonintegral solution y if and only if $\alpha = 2^u 3^v$ with $u, v \in \mathbb{Z}$.*
2. *α^y is a positive integer for every solution y if and only if $\alpha = 2^u 3^v$ with $u, v \in \mathbb{N}_0$.*

Proof. For the first part, apply Proposition 13.4 to a nonintegral solution x and to $x + 1$, which is again a nonintegral solution. Conversely, rational $2, 3$-units have rational outputs $M^u A^v$ at every solution.

For the second part, assume $\alpha^y \in \mathbb{N}$ for every $y \in \mathcal{S}$. Taking $y = x$ and $y = x+1$ makes $\alpha = \alpha^{x+1}/\alpha^x$ a positive rational, say a/b in lowest terms. Put $z = \alpha^x \in \mathbb{N}$. Every $x + n$ with $n \geq 0$ is again a solution, so the number $z(a/b)^n = \alpha^{x+n}$ is an integer for every n, whence b^n divides the fixed integer z for every n, forcing $b = 1$ and $\alpha \in \mathbb{N}$. Since α^x is an integer, the algebraic spectrum gives $\alpha \in \Gamma$, and an integer in Γ is exactly $2^u 3^v$ with $u, v \in \mathbb{N}_0$. The converse is immediate. $\square$

Corollary 13.5 says that the rational-output defect is a genuinely per-solution phenomenon. A base can enjoy a rational output at one nonintegral solution only by failing at the next one, unless it is a rational $2, 3$-unit already.

13.4 Smooth-output feedback on the valuation lattice

Theorem 11.5 and Corollary 11.6 already give the complete two-sided smooth-output exclusion, including its Lean implementation and the essential role of having two independent multiplicative constraints. In the present notation they say that the valuation matrix V_x of a counterexample always has a row indexed by a prime $p \geq 5$.

This exclusion feeds back into the lattice. Because $P \not\subseteq \{2, 3\}$, the valuation matrix of a counterexample is never square, so $\delta_2(V_x)$ is a genuine gcd over at least three rows rather than a single determinant, and the rational locus Λ_x is cut out by at least three congruences. In particular the rational-output defect is bounded by the absolute value of any one 2×2 minor of V_x, and any two minors that are coprime force $\delta_2(V_x) = 1$, in which case no auxiliary base outside $\{2^u 3^v : u, v \in \mathbb{Z}\}$ has a rational output at x at all.

14 The second-iterate kernel and multiplicative dependence

The algebraic spectrum of Section 12 controls one application of the exponent x. Asking which bases of Γ *return* to Γ after that application turns second iterates into finite-dimensional linear algebra, and — as this section records — into an entirely elementary statement about four integers.

14.1 The kernel and its invariance

For a nonintegral solution x with outputs $M = 2^x$ and $A = 3^x$, set

$$K_x = \{2^r 3^s : (r,s) \in \mathbb{Q}^2,\ M^r A^s \in \Gamma\}, \qquad \Gamma = \{2^u 3^v : u, v \in \mathbb{Q}\}. \tag{31}$$

We freely identify K_x with its coordinate subspace $\{(r,s) \in \mathbb{Q}^2 : M^r A^s \in \Gamma\}$ under $\Gamma \cong \mathbb{Q}^2$; thus its identity is written 1 multiplicatively and $(0,0)$ or 0 additively. Writing $\mathbf{m}_{\text{ext}} = (v_p(M))_{p \neq 2,3}$ and $\mathbf{a}_{\text{ext}} = (v_p(A))_{p \neq 2,3}$ for the external valuation vectors, membership of $M^r A^s$ in Γ is exactly the vanishing of every prime valuation outside $\{2,3\}$, so

$$K_x \cong \ker\big((r,s) \mapsto r\,\mathbf{m}_{\text{ext}} + s\,\mathbf{a}_{\text{ext}}\big). \tag{32}$$

Proposition 14.1 (Kernel dimension)**.** *The coordinate subspace of K_x has dimension at most one; it contains no vector with $rs > 0$; it equals $\mathbb{Q}(1,0)$ exactly when M is 3-smooth and $\mathbb{Q}(0,1)$ exactly when A is 3-smooth.*

Proof. At least one external vector is nonzero: otherwise both M and A would be 3-smooth, which Theorem 11.5 excludes. Hence the map in (32) has rank at least one and its kernel has dimension at most one. If $r, s > 0$ then any external prime dividing M or A has positive valuation in $M^r A^s$, and if $r, s < 0$ it has negative valuation, so no same-sign vector lies in the kernel. The two smooth cases are immediate from (32) by testing $(1,0)$ and $(0,1)$. □

Status: [kernel-verified Lean] — `SecondIterateKernel` carries the kernel as an integral subgroup `kernelAddSubgroup` of $\mathbb{Z}^2$, with `kernelPairs_mul_nonpos` for the absence of same-sign vectors, `kernelPairs_det_eq_zero` and `kernelPairs_prop_dim` for the rank bound, `mem_kernelPairs_one_zero_iff` for the two smooth cases, and `saturatedKernel_invariant` for $K_x = K_y$. The input that M and A cannot both be 3-smooth is `integer_of_threeSmooth_outputs` in `SmoothOutputs`. Clearing denominators lets the formal statement use integer rather than rational pairs, which is equivalent and lighter.

The kernel is not an artifact of the chosen counterexample. If x, y are two nonintegral solutions and $\alpha = 2^r 3^s$, then $B = \alpha^x$ is algebraic, and applying the spectrum theorem to y gives $\alpha^{xy} = B^y \in \overline{\mathbb{Q}}$ if and only if $B \in \Gamma$, that is, if and only if $(r,s) \in K_x$. The left-hand condition is symmetric in x and y, so $K_x = K_y$: the kernel is an invariant of the problem, not of the solution.

14.2 Multiplicative dependence of four integers

The kernel condition has a formulation with no logarithms, no projective linear group, and no transcendence vocabulary at all. It is the observation of this section.

Theorem 14.2 (Multiplicative form of the kernel dichotomy)**.** *Let x be a nonintegral solution with outputs $M = 2^x$ and $A = 3^x$. The following are equivalent:*

(i) $K_x \neq \{0\}$*;*

(ii) *the four integers M, A, 2, 3 are multiplicatively dependent, that is,*

$$M^a A^b 2^c 3^d = 1 \qquad \textit{for some } (a,b,c,d) \in \mathbb{Z}^4 \setminus \{0\};$$

(iii) 1, ϑ, x, $x\vartheta$ *are linearly dependent over* $\mathbb{Q}$;

(iv) x *lies in the* $\mathrm{PGL}_2(\mathbb{Q})$*-orbit of* ϑ, *that is,* $x = (u + v\vartheta)/(r + s\vartheta)$ *with rational* u, v, r, s *and* $us - vr \neq 0$.

Proof. (i) implies (ii). If $(r, s) \in K_x$ is nonzero then $M^r A^s = 2^u 3^v$ with $u, v \in \mathbb{Q}$; clearing a common denominator of r, s, u, v produces integers with $M^a A^b 2^c 3^d = 1$ and $(a, b) \neq (0, 0)$.

(ii) implies (i). Given such a relation, note first that $(a, b) \neq (0, 0)$: otherwise $2^c 3^d = 1$, and unique factorization forces $c = d = 0$, contradicting nontriviality. Then $M^a A^b = 2^{-c} 3^{-d} \in \Gamma$, so $(a, b) \in K_x \setminus \{0\}$.

(ii) is equivalent to (iii). Taking logarithms of $M^a A^b 2^c 3^d = 1$ and using $\log M = x \log 2$ and $\log A = x \log 3$ gives

$$(ax + c) \log 2 + (bx + d) \log 3 = 0,$$

and division by $\log 2$ turns this into

$$x\,(a + b\vartheta) + (c + d\vartheta) = 0, \tag{33}$$

which is precisely a nontrivial rational linear relation among $1, \vartheta, x, x\vartheta$. Each step reverses.

(iii) is equivalent to (iv). In (33) the coefficient $a + b\vartheta$ cannot vanish, since ϑ is irrational and $(a, b) \neq (0, 0)$; hence $x = -(c + d\vartheta)/(a + b\vartheta)$. If the determinant $ad - bc$ vanished, numerator and denominator would be rationally proportional and x would be rational, hence integral by Lemma 1.3, contrary to hypothesis. Conversely, any such expression cross-multiplies to (33). □

Status: [kernel-verified Lean] — `KernelDichotomy` formalizes the equivalence of (ii), (iii) and (iv), as `multiplicativelyDependentOutputs_iff_secondIterateKernelRelation` together with `exists_moebius_of_multiplicativelyDependentOutputs` and `multiplicativelyDependentOutputs_of_moebius`; the packaged branching is `twoBaseNonintegerSolution_kernel_dichotomy`. Clause (i) is `SecondIterateKernel`. The formalization is slightly stronger than the statement above: (ii) $\iff$ (iii) needs only irrationality of ϑ, and nonintegrality of x enters just for the nonvanishing of the Möbius determinant.

14.3 Why the multiplicative form is worth having

Two consequences follow from stating the dichotomy multiplicatively.

It removes the transcendence vocabulary from the branching. Every counterexample falls into exactly one of two cases, and the case distinction is a question about four ordinary integers: either they satisfy a multiplicative relation or they do not. In the dependent branch Proposition 14.1 says the relation is essentially unique up to scaling, and its exponents on M and A have opposite signs unless one of the outputs is 3-smooth.

It makes Baker's theorem applicable to the independent branch. This is the point that the Diophantine formulations obscure. In the independent branch $M, A, 2, 3$ are multiplicatively independent, so $\log M, \log A, \log 2, \log 3$ are $\mathbb{Q}$-linearly independent *logarithms of algebraic numbers* — all four arguments are positive integers. Baker's homogeneous theorem then gives more than $\mathbb{Q}$-linear independence: these four logarithms are linearly independent over $\overline{\mathbb{Q}}$.

Corollary 14.3 (Algebraic-coefficient rigidity in the independent branch)**.** *Suppose x is a nonintegral solution for which $M, A, 2, 3$ are multiplicatively independent. Then no relation*

$$\gamma_1 \log M + \gamma_2 \log A + \gamma_3 \log 2 + \gamma_4 \log 3 = 0$$

holds with algebraic $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ not all zero.

Status: [**classical/open background**] — immediate from Baker's homogeneous theorem on $\overline{\mathbb{Q}}$-linear independence of logarithms of algebraic numbers, applied to the four positive integers $M, A, 2, 3$.

It must be said plainly what this does and does not do. It does *not* close the independent branch: the defining relation $\log M \log 3 = \log A \log 2$ has coefficients $\log 3$ and $-\log 2$, which are transcendental by Lindemann, so it is not a $\overline{\mathbb{Q}}$-linear relation and Corollary 14.3 does not apply to it. That is exactly the "Baker is linear, not bilinear" obstruction recorded in Appendix L. What changes is that the independent branch is no longer structureless: it carries a genuine unconditional rigidity statement, and any future argument that manages to produce an algebraic-coefficient relation among these four logarithms — for instance by clearing the transcendental coefficients against a second, independent relation — would close it immediately.

15 Depth-three tower rigidity

A hypothetical counterexample makes the first level of the tower $2 \mapsto 2^x$ integral by assumption. This section answers the exact question that the corpus keeps circling: starting from an arbitrary positive algebraic base, how many times can one apply the same nonintegral exponent x before transcendence is forced? The answer is at most three, the bound is attained, and the depth at which the first transcendental value appears is determined by a single rational-linear invariant.

For positive real bases there is no branch ambiguity between iterated powers and powers of powers, so the tower is unambiguous:

$$((\gamma^x)^x)^x = \gamma^{x^3}.$$

Throughout this section x denotes a hypothetical nonintegral solution, that is $x \in \mathcal{S} \setminus \mathbb{N}_0$, and we write

$$m = 2^x \in \mathbb{N}, \qquad n = 3^x \in \mathbb{N}.$$

The letter γ is reserved for a positive algebraic base, matching the Lean naming in `AlgebraicBaseUnitPower`.

15.1 Spectrum and return-kernel input

The tower uses two earlier classifications without repeating them. Corollary 12.3 identifies the algebraic spectrum with $\Gamma = \{2^r 3^s : r, s \in \mathbb{Q}\}$, while Theorem 12.5 says that a base outside Γ is already log-strongly transcendental after one application of x. Inside the spectrum, the return kernel K_x is the subgroup defined in (31). Proposition 14.1 gives its valuation formula, dimension at most one, absence of same-sign directions, and the two smooth coordinate-axis cases. The only additional dynamical fact needed here is that this kernel admits no nontrivial internal orbit of length two.

Lemma 15.1 (No two-step internal orbit). *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$ and let $\gamma \in K_x \setminus \{1\}$. Then $\gamma^x \notin K_x$; equivalently,*

$$\gamma \in \Gamma \text{ and } \gamma^x \in \Gamma \quad \Longrightarrow \quad \gamma^{x^2} \notin \Gamma.$$

Proof. Suppose γ and $\gamma_1 = \gamma^x$ both lie in $K_x \setminus \{1\}$. By Proposition 14.1 the space K_x is at most one-dimensional over $\mathbb{Q}$, so two of its nonzero elements are rational scalar multiples of one another in the additive notation: $\gamma_1 = \gamma^q$ for some $q \in \mathbb{Q}$. Since also $\gamma_1 = \gamma^x$ and $\gamma \neq 1$, comparing logarithms gives $x = q$, contradicting the irrationality of x. Hence $\gamma_1 \notin K_x$, which by (31) says exactly that $\gamma_1^x = \gamma^{x^2} \notin \Gamma$. □

There are therefore no cycles, no fixed rays, and no length-two paths lying entirely inside Γ: a rational direction may survive one return to the algebraic spectrum, but the returned direction cannot itself survive.

15.2 Exact survival depth

Theorem 15.2 (Depth-three algebraic-tower theorem). *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$ and let $\gamma \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$. Then at least one of*

$$\gamma^x, \qquad \gamma^{x^2}, \qquad \gamma^{x^3}$$

is transcendental. More precisely, the first transcendental term occurs at exactly one of the following three depths.

(1) ***Depth one.*** *If $\gamma \notin \Gamma$, then γ^x is log-strongly transcendental.*

(2) ***Depth two.*** *If $\gamma \in \Gamma \setminus K_x$, then $\gamma^x \in \overline{\mathbb{Q}} \setminus \Gamma$ and γ^{x^2} is log-strongly transcendental.*

(3) ***Depth three.*** *If $\gamma \in K_x \setminus \{1\}$, then $\gamma^x \in \Gamma$, $\gamma^{x^2} \in \overline{\mathbb{Q}} \setminus \Gamma$, and γ^{x^3} is log-strongly transcendental.*

Equivalently, for some $j \in \{1, 2, 3\}$ one has $x^j \log \gamma \notin \mathcal{L}$.

Proof. Depth one is Theorem 12.5.

Assume $\gamma \in \Gamma$, so that $\gamma_1 = \gamma^x$ is algebraic by Corollary 12.3. If $\gamma \notin K_x$, then $\gamma_1 \notin \Gamma$ by the definition of K_x in (31), and γ_1 is a positive algebraic base outside Γ; the strong spectrum applied to γ_1 and the same exponent x makes $\gamma_1^x = \gamma^{x^2}$ log-strongly transcendental.

If instead $\gamma \in K_x \setminus \{1\}$, then $\gamma_1 \in \Gamma$, so $\gamma_2 = \gamma_1^x = \gamma^{x^2}$ is algebraic. By Lemma 15.1 we have $\gamma_1 \notin K_x$, hence $\gamma_2 \notin \Gamma$. One last application of the strong spectrum, now to the base γ_2, gives $\gamma_2^x = \gamma^{x^3}$ log-strongly transcendental.

The three cases are mutually exclusive and exhaustive, and in each case all earlier terms of the tower are algebraic, so the stated depth is exact rather than an upper bound. □

The classification is compact enough to tabulate. For a nonintegral solution x and a positive algebraic base $\gamma \neq 1$, the first level of the tower whose logarithm escapes $\mathcal{L}$ is:

Location of γ	First log outside $\mathcal{L}$	Structural reason
$\gamma \notin \Gamma$	$x \log \gamma$	exact algebraic spectrum
$\gamma \in \Gamma \setminus K_x$	$x^2 \log \gamma$	the first image leaves Γ
$\gamma \in K_x \setminus \{1\}$	$x^3 \log \gamma$	no two-step internal orbit

Status: [paper proof] — paper proof in this report, resting on Theorem 12.5 and hence on Baker's theorem and Roy's strong six exponentials theorem. The ordinary-transcendence version of depth one is **[kernel-verified Lean]**; the ordinary-transcendence version of depth two for the mixed integer bases $2^i 3^j$ with $i, j > 0$ is also **[kernel-verified Lean]**, as recorded below.

The theorem places a strict ceiling on how far an algebraic power tower can imitate the integral first level of a counterexample. A counterexample buys two algebraic outputs; it never buys a third independent one, and the tower makes the deficit visible after at most three steps regardless of the base chosen.

15.3 Exact behaviour of the bases two, three, and six

Corollary 15.3 (Base-two tower dichotomy)**.** *Let $x \in \mathcal{S} \setminus \mathbb{N}_0$ and $m = 2^x$.*

(a) *If m is not three-smooth, then 2^{x^2} is log-strongly transcendental.*

(b) *If m is three-smooth, say $m = 2^a 3^b$ with $a, b \in \mathbb{N}_0$, then*

$$2^{x^2} = m^x = (2^x)^a (3^x)^b = m^a n^b \in \mathbb{N}, \tag{34}$$

and 2^{x^3} is log-strongly transcendental.

The mirror statement holds with 2 replaced by 3 and m by n.

Proof. The base 2 lies in Γ, and its coordinate vector is $(1, 0)$, which lies in K_x exactly when $m \in \Gamma$, that is, by Proposition 14.1, exactly when the integer m is three-smooth. Now apply Theorem 15.2: the non-smooth case is depth two and the smooth case is depth three. In the smooth case the intermediate value is computed by (34), where the middle equality is the identity $(2^a 3^b)^x = (2^x)^a (3^x)^b$ for positive real bases and the last membership holds because $m, n \in \mathbb{N}$. □

Equation (34) is worth pausing on. In the smooth branch the second level of the tower is not merely algebraic but an explicit positive integer, expressed in the two original outputs with the same exponents that describe the first output. The tower does not become transcendental because the arithmetic runs out immediately; it becomes transcendental because the third level has nowhere left to land.

Corollary 15.4 (Base six always fails at depth two)**.** *For every $x \in \mathcal{S} \setminus \mathbb{N}_0$ and all $r, s \in \mathbb{Q}$ of the same nonzero sign,*

$$x^2 (r \log 2 + s \log 3) \notin \mathcal{L}.$$

In particular $x^2 \log 6 \notin \mathcal{L}$, so 6^{x^2} is log-strongly transcendental.

Proof. The base $\gamma = 2^r 3^s$ lies in Γ, and its coordinate vector (r, s) has positive coordinate product, so $\gamma \notin K_x$ by Proposition 14.1. Its exact survival depth is therefore two, by Theorem 15.2(2). The case $r = s = 1$ is $\gamma = 6$. □

This recovers and strengthens a kernel-verified family. The Lean theorem

```
TwoBaseNonintegerSolution.transcendental_mixed_two_three_squared_exponents
```

in `AlgebraicOutputLocus` proves that $(2^i 3^j)^{x^2}$ is transcendental for all natural $i, j > 0$, which is exactly the statement that no genuinely mixed three-smooth integer base survives to the second level. Corollary 15.4 extends that to all rational same-sign directions, including negative ones, and upgrades the conclusion from transcendence to escape from $\mathcal{L}$; more importantly it explains *why* the mixed bases fail, namely that the exceptional directions of K_x are confined to the two coordinate axes, which the mixed monomials avoid by construction. The kernel-verified case is the positive-integer corner of a statement that is really about a line in $\mathbb{Q}^2$.

Status: **[paper proof]** for Corollaries 15.3 and 15.4 as stated. **[kernel-verified Lean]** for the ordinary transcendence of $(2^i 3^j)^{x^2}$ with integer $i, j > 0$, in `AlgebraicOutputLocus`. The gap between the two is exactly the general six exponentials input discussed in Section 8.

15.4 One-base tower equivalences

The depth theorem converts into a reformulation of the conjecture that is striking in its indifference to the base.

Universal three-level detector (Theorem 15.5). Fix *any* positive algebraic $\gamma \neq 1$ — $\gamma = 5$, $\gamma = \sqrt{2}$, $\gamma = 2$, $\gamma = 3$, $\gamma = \frac{1}{2}(7 + 3\sqrt{5})$, anything at all. Conjecture 1.1 is equivalent to

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \implies \gamma^x,\ \gamma^{x^2},\ \gamma^{x^3} \in \overline{\mathbb{Q}},$$

and equally to the assertion that all three numbers $x \log \gamma$, $x^2 \log \gamma$, $x^3 \log \gamma$ lie in $\mathcal{L}$. No arithmetic feature of the base is used: three levels of the tower detect a counterexample for every positive algebraic base simultaneously. **[paper proof]**

Theorem 15.5 (Universal three-level detector). *Fix $\gamma \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$. Then Conjecture 1.1 is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \implies \gamma^x, \gamma^{x^2}, \gamma^{x^3} \in \overline{\mathbb{Q}}, \tag{35}$$

and also to the same implication with $\overline{\mathbb{Q}}$-algebraicity replaced by the three membership conditions $x^j \log \gamma \in \mathcal{L}$, $j = 1, 2, 3$.

Proof. If the conjecture holds, then $\mathcal{S} = \mathbb{N}_0$, every exponent x, x^2, x^3 appearing in (35) is a nonnegative integer, and all three powers of the algebraic number γ are algebraic, with logarithms in $\mathcal{L}$. If the conjecture fails, apply Theorem 15.2 to a nonintegral solution: one of the three values is log-strongly transcendental, so (35) fails and so does its $\mathcal{L}$-version. □

Bases already inside Γ have their first level algebraic for free, which removes one of the three conditions and yields the shortest form of the equivalence.

Corollary 15.6 (Base-two cubic-tower equivalence)**.** *Conjecture 1.1 is equivalent to*

$$\forall x \in \mathbb{R}, \quad 2^x, 3^x \in \mathbb{Z} \implies 2^{x^2} \in \overline{\mathbb{Q}} \ \textit{ and } \ 2^{x^3} \in \overline{\mathbb{Q}}. \tag{36}$$

The same statement holds with the base 3 *in place of the base* 2*.*

Proof. Take $\gamma = 2$ in Theorem 15.5 and drop the first condition, which is automatic because 2^x is an integer under the hypothesis. □

This is a genuinely different equivalence from the second-level algebraicity tests of Section 8, which require *both* 2^{x^2} and 3^{x^2} to be algebraic. One original base suffices if one is willing to climb a single extra level. The trade is informative: a second base and a third tower level carry exactly the same amount of information, which is another way of seeing that the two-base configuration supplies only two independent logarithmic directions no matter how the exponent is iterated.

Status: **[paper proof]** — paper proof in this report. Both equivalences inherit the classical inputs of Theorem 12.5; the ordinary-algebraicity forms (35) and (36) need only the six exponentials theorem, since Corollary 12.3 is **[kernel-verified Lean]** and only the depth-two and depth-three steps require the general form. Neither is a proof of Conjecture 1.1, which remains open: the tower reformulations relocate the difficulty to the second and third levels without supplying a mechanism that produces algebraicity there.

16 Symmetric tower constants

The exponent $\vartheta = \log_2 3$ and its reciprocal

$$\eta = \frac{1}{\vartheta} = \log_3 2$$

generate two symmetric "next outputs",

$$E_3 = 3^{\vartheta} = 5.704522494691117635\ldots, \qquad E_2 = 2^{\eta} = 1.548562652630242907\ldots$$

The first is the canonical radical of Section 5 in the special branch where the primitive odd core equals 3; the second is its base-swapped mirror. Neither constant is known to be transcendental, and for E_3 even irrationality is open. What is available unconditionally is a statement about the *pair*, and what a counterexample would add is an exact classification of which of the two becomes algebraic.

16.1 An unconditional strong dichotomy

Theorem 16.1 (At least one symmetric tower constant is strongly transcendental)**.** *At least one of the two numbers*

$$\vartheta \log 3 = \log E_3, \qquad \eta \log 2 = \log E_2$$

lies outside $\mathcal{L}$. In particular, at least one of E_3 and E_2 is transcendental.

Proof. The three numbers $1, \vartheta, \eta$ are linearly independent over $\overline{\mathbb{Q}}$: a relation $a + b\vartheta + c\eta = 0$ with algebraic coefficients, multiplied by ϑ, becomes $b\vartheta^2 + a\vartheta + c = 0$, which is impossible for a transcendental ϑ unless $a = b = c = 0$. The two numbers $\log 2, \log 3$ are linearly independent over $\mathbb{Q}$ by unique factorization, hence over $\overline{\mathbb{Q}}$ by Baker's theorem [3].

Apply Roy's strong six exponentials theorem [23, 26] to the three rows $1, \vartheta, \eta$ and the two columns $\log 2, \log 3$. Four of the six pairwise products are already logarithms of algebraic numbers,

$$\log 2, \qquad \log 3, \qquad \vartheta \log 2 = \log 3, \qquad \eta \log 3 = \log 2,$$

and therefore lie in $\mathcal{L}$. Hence at least one of the two remaining products, $\vartheta \log 3$ and $\eta \log 2$, lies outside $\mathcal{L}$. ☐

Status: [paper proof] — paper proof in this report, using Baker's theorem and Roy's strong six exponentials theorem. The ordinary six exponentials theorem gives the weaker dichotomy that one of E_3, E_2 is transcendental. The strong form is worth the extra input because it survives verbatim into the conditional branches below.

Note what the theorem does not say. It does not identify which constant is transcendental, and no known method identifies either one individually; that is precisely the wall described in Section 5. The value of the dichotomy is that a counterexample would resolve it, and would resolve it in a completely explicit way.

16.2 Exact conditional classification

Throughout this subsection $x \in \mathcal{S} \setminus \mathbb{N}_0$, $m = 2^x$, and $n = 3^x$.

Theorem 16.2 (The E_3 equivalences)**.** *The following are equivalent.*

(i) $E_3 = 3^\vartheta$ *is algebraic;*

(ii) $\vartheta \log 3 \in \mathcal{L}$*;*

(iii) m *is three-smooth;*

(iv) *there are* $a, b \in \mathbb{N}_0$ *with* $b > 0$ *such that* $m = 2^a 3^b$ *and* $x = a + b\vartheta$*;*

(v) $x \in \mathbb{Q} + \mathbb{Q}\vartheta$*.*

If these conditions fail, then $\vartheta \log 3 \notin \mathcal{L}$ *and* E_3 *is log-strongly transcendental. If they hold, then*

$$E_3^b = \frac{n}{3^a} \in \mathbb{Q}_{>0}, \tag{37}$$

so E_3 *is an explicit algebraic radical.*

Proof. (iv)⇒(iii)⇒(v): the first is trivial, and taking logarithms of $m = 2^a 3^b$ gives $x = a + b\vartheta$, which lies in $\mathbb{Q} + \mathbb{Q}\vartheta$.

(v)⇒(iv): write $x = u + v\vartheta$ with $u, v \in \mathbb{Q}$ and clear a common positive denominator d, so that $m^d = 2^p 3^q$ with $p, q \in \mathbb{Z}$. The left side is a positive integer, so prime valuations force $p, q \geq 0$ and $d \mid p$, $d \mid q$. Hence $u = a$ and $v = b$ with $a, b \in \mathbb{N}_0$, and $b > 0$ because $x \notin \mathbb{N}_0$.

(i)⇒(iv): apply the six exponentials theorem to the rows $1, x, \vartheta$ and the columns $\log 2, \log 3$. The six exponential values are

$$2, \quad 3, \quad m, \quad n, \quad 3, \quad E_3,$$

all algebraic by hypothesis. The theorem therefore forbids both independence conditions simultaneously; the columns $\log 2, \log 3$ are $\mathbb{Q}$-independent, so the rows are $\mathbb{Q}$-dependent and $x = u + v\vartheta$ for some $u, v \in \mathbb{Q}$. Now (v)⇒(iv) applies.

(iv)⇒(i): from $x = a + b\vartheta$ we get $n = 3^x = 3^a E_3^b$, which is (37) and exhibits E_3 as a positive real radical of a positive rational, hence algebraic.

Finally the strong statement. Suppose m is not three-smooth. Then $\log 2, \log m, \log 3$ are linearly independent over $\mathbb{Q}$: a rational relation with nonzero coefficient on $\log m$ would place m in Γ, and an integer in Γ is three-smooth, while a relation with zero coefficient on $\log m$ contradicts the independence of $\log 2, \log 3$. Baker's theorem upgrades this to independence over $\overline{\mathbb{Q}}$, so after division by $\log 2$ the three numbers $1, x, \vartheta$ are $\overline{\mathbb{Q}}$-linearly independent. Apply Roy's theorem to the rows $1, x, \vartheta$ and the columns $\log 2, \log 3$: five of the six products, namely $\log 2, \log 3, \log m, \log n$, and $\vartheta \log 2 = \log 3$, lie in $\mathcal{L}$, so $\vartheta \log 3 \notin \mathcal{L}$. Thus failure of (iii) implies failure of (ii); and (i) implies (ii) because $\log E_3 = \vartheta \log 3$ is then the logarithm of an algebraic number. ☐

Theorem 16.3 (The E_2 equivalences)**.** *The following are equivalent.*

(i) $E_2 = 2^\eta$ *is algebraic;*

(ii) $\eta \log 2 \in \mathcal{L}$*;*

(iii) n *is three-smooth;*

(iv) *there are* $c, d \in \mathbb{N}_0$ *with* $c > 0$ *such that* $n = 2^c 3^d$ *and* $x = d + c\eta$*;*

(v) $x \in \mathbb{Q} + \mathbb{Q}\eta$*.*

If these conditions fail, then $\eta \log 2 \notin \mathcal{L}$ *and* E_2 *is log-strongly transcendental. If they hold, then*

$$E_2^c = \frac{m}{2^d} \in \mathbb{Q}_{>0}. \tag{38}$$

Proof. Interchange the roles of 2 and 3, of m and n, and of ϑ and η in the proof of Theorem 16.2. For the strong step, use the $\mathbb{Q}$-linear independence of $\log 3, \log n, \log 2$ when n is not three-smooth. ☐

Corollary 16.4 (The two branches are mutually exclusive)**.** *For a nonintegral solution,* E_3 *and* E_2 *cannot both be algebraic. Equivalently, at least one of* $\vartheta \log 3$ *and* $\eta \log 2$ *lies outside* $\mathcal{L}$ *under a counterexample as well as unconditionally.*

Proof. By Theorems 16.2 and 16.3, algebraicity of E_3 is equivalent to three-smoothness of m and algebraicity of E_2 to three-smoothness of n. Theorem 11.5 forbids both: eliminating x from $x = a + b\vartheta$ and $x = d + c\eta$ produces the same nonzero rational quadratic in the transcendental number ϑ used in its proof. ☐

So under a counterexample the unconditional dichotomy of Theorem 16.1 becomes a genuine classification rather than an alternative: at most one of the two constants is algebraic, and each is algebraic exactly in its own smooth-output branch, with an explicit radical description.

Smooth branch	Return kernel	Symmetric constants
$m = 2^a 3^b$, $b > 0$	$\{2^r : r \in \mathbb{Q}\}$	$E_3^b = n/3^a \in \mathbb{Q}_{>0}$; E_2 log-strongly transcendental
$n = 2^c 3^d$, $c > 0$	$\{3^s : s \in \mathbb{Q}\}$	$E_2^c = m/2^d \in \mathbb{Q}_{>0}$; E_3 log-strongly transcendental
neither output smooth	no coordinate axis	both E_3 and E_2 log-strongly transcendental

The three rows of the table are exactly the three surviving rows of the primitive-core trichotomy of Section 8: the first is the case $w = 3$, in which the canonical radical $E = w^\vartheta$ of Section 5 degenerates to E_3 itself with index $d = b$; the second is its mirror $v = 2$; and the third is the generic case with two independent external prime supports. The missing fourth pattern, both outputs three-smooth, is the one excluded by Theorem 11.5.

Two remarks are worth making about the direction of the implications. First, nothing here proves that E_3 or E_2 is transcendental: a counterexample would make one of them an explicit algebraic radical, and Corollary 16.4 only says that it cannot make both algebraic. The conjecture remains open, and the smallest instance — irrationality of $E_3 = 3^{\log_2 3}$ — is exactly the wall named in Section 5. Second, the implication that is usable in the other direction is the strong one: proving $\vartheta \log 3 \notin \mathcal{L}$ would kill the entire $w = 3$ branch, and proving $\eta \log 2 \notin \mathcal{L}$ would kill the $v = 2$ branch, leaving only the generic case, in which both outputs carry prime support outside $\{2, 3\}$ and the local determinant methods have two independent external primes to work with.

Status: [paper proof] — paper proofs in this report. Theorems 16.2 and 16.3 use the six exponentials theorem for the algebraicity equivalences and Baker's theorem together with Roy's strong six exponentials theorem for the $\mathcal{L}$-statements; none of those classical inputs is formalized in the repository. Corollary 16.4 adds only Theorem 11.5, whose sole nonelementary ingredient is the kernel-verified transcendence of ϑ, so the mutual exclusion of the two smooth branches — as opposed to the transcendence conclusions attached to them — is within immediate reach of the Lean development.

17 Finite-difference obstructions

The first local obstruction recorded in the corpus forbids three candidates in arithmetic progression under the convenient hypothesis $h^5 \le n$. That hypothesis is an artifact of the estimate, not of the mechanism: the sharp arbitrary-order curvature theorem is now kernel-verified for every $r \ge 2$, and it contains the three- and four-term cases as specializations.

For a function f and a step $h > 0$ write $\Delta_h f(t) = f(t+h) - f(t)$ and $\Delta_h^r = \Delta_h \circ \cdots \circ \Delta_h$ (r factors). Iterating the fundamental theorem of calculus gives the mean-value identity in integral form,

$$\Delta_h^r f(n) = \int_{[0,h]^r} f^{(r)}(n + t_1 + \cdots + t_r)\, dt_1 \cdots dt_r. \tag{39}$$

For $f(t) = t^\vartheta$ and $r \ge 2$ the derivative is

$$f^{(r)}(t) = (\vartheta)_{\underline{r}}\, t^{\vartheta - r}, \qquad (\vartheta)_{\underline{r}} := \prod_{j=0}^{r-1} (\vartheta - j) = \vartheta(\vartheta - 1) \cdots (\vartheta - r + 1).$$

Since $1 < \vartheta < 2$ the falling factorial never vanishes, so $f^{(r)}$ has a fixed nonzero sign on $(0, \infty)$, and $|f^{(r)}|$ decreases in t. These two facts — nonvanishing and decay — are the whole content of the obstruction.

Theorem 17.1 (Higher-difference progression obstruction)**.** *Let $r \geq 2$, let $n, h \in \mathbb{N}$ be positive, and set $c_r = |\,(\vartheta)_{\underline{r}}\,|$. If all of $n, n+h, \ldots, n+rh$ are candidates, then*

$$1 \leq c_r\, h^r n^{\vartheta - r}.$$

Such a progression is therefore impossible whenever

$$c_r\, h^r < n^{\,r-\vartheta}. \tag{40}$$

Proof. If all $n + jh$ are candidates, every $(n + jh)^\vartheta$ is an integer, so the integer combination $\Delta_h^r f(n) = \sum_{j=0}^{r} (-1)^{r-j} \binom{r}{j} f(n + jh)$ is an integer. By (39) and the fixed sign of $f^{(r)}$ it is nonzero, hence of absolute value at least one. Bounding $|f^{(r)}|$ by its value at the left endpoint n,

$$1 \leq |\Delta_h^r f(n)| \leq c_r\, h^r n^{\vartheta - r}.$$ □

Status: [kernel-verified Lean] — `HigherDifference` proves the generic iterated mean-value identity `exists_iterated_fwdDiff_eq_pow_mul`, its real-power specialization `exists_iterated_fwdDiff_rpow_point`, integrality of the binomial forward difference, and the obstruction `not_arithmetic_progression_twoBaseNaturalCandidates_of_curvature` uniformly for every $r \geq 2$. The displayed iterated-integral identity (39) is not used by the Lean proof; the formal argument iterates the mean-value theorem instead, which avoids any integration theory.

17.1 The sharp three-term criterion

For $r = 2$ the criterion (40) reads

$$\vartheta(\vartheta - 1)h^2 < n^{\,2-\vartheta}, \qquad \text{i.e.} \qquad h < 1.038547802\ldots \cdot n^{0.2075187\ldots}, \tag{41}$$

strictly weaker as a hypothesis than $h \leq n^{1/5}$, whose exponent is only 0.2. The real form is `second_difference_logThreeDivLogTwo_ap_of_curvature` in `SharperProgressions`, built on the curvature estimate `second_difference_logThreeDivLogTwo_ap_bound`.

Almost all of the gain survives in a purely rational hypothesis, which is what a kernel can check without any real-number decision procedure. The step exponent $83/400 = 0.2075$ is admissible:

Theorem 17.2 (Sharp rational three-term criterion)**.** *Let $n, h \geq 1$ be integers with $h^{400} \leq n^{83}$. Then n, $n + h$, $n + 2h$ are not all candidates.*

Status: [kernel-verified Lean] — `SharpProgression` proves `second_difference_logThreeDivLogTwo_ap_sharp` and `not_three_arithmetic_progression_twoBaseNaturalCandidates_sharp`. The two numerical ingredients are the enclosure $\vartheta < 317/200$, replayed by the kernel from the exact power inequality $3^{200} < 2^{317}$ through `logThreeDivLogTwo_lt_317_div_200`, and the consequence $\vartheta(\vartheta - 1) < 37089/40000 < 1$. For $r = 3$, `ThirdDifference` kernel-checks the exact third-difference identity and the corresponding four-candidate obstruction.

17.2 Thresholds for higher order

Writing (40) as a bound on the step, a progression of length $r+1$ starting at n is impossible once

$$h < C_r\, n^{\,1-\vartheta/r}, \qquad C_r = c_r^{-1/r}.$$

Table 1 lists the first thresholds.

r	length $r+1$	exponent $1-\vartheta/r$	C_r
2	3	0.207518750	1.038547802
3	4	0.471679166	1.374849673
4	5	0.603759375	1.164124485
5	6	0.683007500	0.946705202
6	7	0.735839583	0.778537835
7	8	0.773576786	0.652646038
8	9	0.801879687	0.557368999

Table 1: Step thresholds for the higher-difference obstruction. A progression of length $r+1$ with step $h < C_r n^{1-\vartheta/r}$ cannot consist of candidates.

17.3 Why longer progressions do not close the problem

As r grows the admissible step exponent $1-\vartheta/r$ approaches one, which looks tantalizing: for large r the theorem forbids progressions with steps almost as large as the starting point itself. It does not contradict the exact candidate monoid, and it cannot be bootstrapped into a proof, for a purely combinatorial reason.

Observation 17.3 (Combinatorial limitation)**.** Conditionally on the structure described in Section 4, a dyadic block near X contains $O(\log X)$ candidates, so the average additive gap inside the block is of order $X/\log X$. For each fixed r this exceeds $X^{\,1-\vartheta/r}$ once X is large. A rank-two multiplicative monoid carries no reason whatsoever to contain long *additive* progressions; the hypothesis of Theorem 17.1 is therefore vacuous for the configurations that actually occur.

Extracting leverage from long progressions would require r growing with the height, control of the rapidly growing constant $c_r = |\,(\vartheta)_{\underline{r}}\,|$, and — decisively — a combinatorial source guaranteeing such progressions exist. None of the three is available. The productive response is to drop the additive-structure hypothesis altogether, which is what Section 18 does.

17.4 The optimized variable-order scale

Theorem 17.1 is uniform in r, so nothing forces the difference order to stay fixed while the starting point grows. Table 1 records the first fixed-order thresholds, but each row is only a snapshot of a one-parameter family that may be re-optimized at every height. Carrying out that optimization strengthens every fixed-order corollary at once and, more valuably, replaces the vague complaint that the candidate density "feels too small" by a single explicit constant. Throughout this subsection log denotes the natural logarithm.

The exact derivative coefficient. Fix $r \geq 2$ and recall the constant $c_r = |\,(\vartheta)_{\underline{r}}\,|$ of Theorem 17.1. Since $1 < \vartheta < 2$, the first two factors ϑ and $\vartheta - 1$ are positive while every later factor $\vartheta - j$ with $j \geq 2$ is negative, so taking absolute values termwise gives

$$c_r = \vartheta(\vartheta - 1) \prod_{j=2}^{r-1} (j - \vartheta),$$

and the recursion $\Gamma(z+1) = z\Gamma(z)$ telescopes the product into closed form:

$$c_r = \frac{\vartheta(\vartheta - 1)}{\Gamma(2 - \vartheta)}\, \Gamma(r - \vartheta). \tag{42}$$

So c_r grows at factorial speed, which is exactly why no fixed order can be optimal for all n: the constant eventually defeats the improving exponent. Stirling's formula applied to (42) gives the root asymptotic that the optimization needs,

$$c_r^{1/r} = \frac{r}{e}\left(1 + O\!\left(\frac{\log r}{r}\right)\right). \tag{43}$$

Optimizing the step threshold over the order. The formalized condition (40) forbids a progression of length $r+1$ as soon as $h < c_r^{-1/r} n^{\,1-\vartheta/r}$, and substituting (43) rewrites that threshold as

$$c_r^{-1/r} n^{\,1-\vartheta/r} = \frac{en}{r} \exp\!\left(-\frac{\vartheta \log n}{r}\right)\left(1 + O\!\left(\frac{\log r}{r}\right)\right). \tag{44}$$

The two r-dependent factors pull in opposite directions: $c_r^{-1/r} \sim e/r$ decays in r, while $n^{-\vartheta/r}$ increases to 1. Discarding the lower-order factor and differentiating the logarithm of the leading term,

$$\frac{d}{dr}\left(1 + \log n - \log r - \frac{\vartheta \log n}{r}\right) = -\frac{1}{r} + \frac{\vartheta \log n}{r^2},$$

which is positive for $r < \vartheta \log n$, vanishes at $r = \vartheta \log n$, and is negative afterwards. The leading term is therefore unimodal in real $r > 0$ with a unique maximum at $r = \vartheta \log n$, and at that point the explicit factor e cancels against $\exp(-\vartheta \log n / r) = e^{-1}$, leaving the clean scale $n/(\vartheta \log n)$.

Theorem 17.4 (Optimized variable-order obstruction)**.** *Let $0 < \varepsilon < 1$. For all sufficiently large n choose an integer*

$$r_n = \vartheta \log n + O(1), \qquad r_n \geq 2.$$

If

$$1 \leq h \leq (1 - \varepsilon)\,\frac{n}{\vartheta \log n},$$

then $n, n+h, \ldots, n + r_n h$ are not all candidates. Moreover, the largest leading-order step scale obtainable by optimizing the curvature condition (40) over r is

$$h_{\max}(n) \sim \frac{n}{\vartheta \log n} = \log_3 2 \cdot \frac{n}{\log n} = 0.6309297535\ldots \cdot \frac{n}{\log n}.$$

Proof. Take r_n to be either nearest integer to $\vartheta \log n$. Then $r_n \to \infty$, so (43) and hence (44) apply, and the quotient of $c_{r_n}^{-1/r_n} n^{\,1-\vartheta/r_n}$ by $n/(\vartheta \log n)$ tends to 1. The fixed margin $1 - \varepsilon$ absorbs both the Stirling error $1 + O(\log r / r)$ and the $O(1)$ rounding of $\vartheta \log n$ to an integer, so (40) holds for every h in the stated range, and Theorem 17.1 excludes the progression.

For the optimality assertion, the leading approximation in (44) is unimodal in real $r > 0$ with its unique maximum at $r = \vartheta \log n$, where its value is $n/(\vartheta \log n)$. Any regime in which r stays bounded, or grows outside a constant multiple of $\log n$, returns a strictly smaller order or a strictly smaller leading constant. □

Status: [paper proof] — the obstruction being optimized is **[kernel-verified Lean]** and holds uniformly for every $r \geq 2$ with no upper bound on the order, so substituting $r = r_n$ inside the formalized statement is legitimate; the relevant module is `HigherDifference`. The optimization itself, the Gamma evaluation (42), and the Stirling asymptotic (43) are paper arguments and have not been submitted to the kernel.

Comparison with the conditional candidate spacing. A dyadic block $[X, 2X)$ is an interval of length one in exponent space, so the exact block count of Section 4 applies verbatim: conditional on a counterexample with least nonintegral generator β,

$$\#(\mathcal{C} \cap [X, 2X)) = \frac{\log_2 X}{\beta} + O(1) = \frac{\log X}{\beta \log 2} + O_\beta(1). \tag{45}$$

The average additive spacing between consecutive candidates inside that block is therefore

$$\sim \beta \log 2 \cdot \frac{X}{\log X},$$

whereas Theorem 17.4 taken at $n \asymp X$ reaches only

$$h_{\max} \sim \frac{1}{\vartheta} \cdot \frac{X}{\log X}.$$

Both are of order $X/\log X$, so the comparison is a pure constant, and in it the transcendental ϑ cancels against the $\log 2$:

$$\frac{\beta \log 2}{1/\vartheta} = \beta\, \vartheta \log 2 = \beta \log 3. \tag{46}$$

The kernel-verified zero-free region $2^x < 4096$ forces $\beta > 12$ (Section 4), which already makes the ratio exceed $12 \log 3 = 13.1833\ldots$; the interval scan of Section 23 raises it past $20 \log 3 = 21.9722\ldots$, and the extended directed-rounding scan through 2^{27} raises it past $27 \log 3 = 29.6625\ldots$. The last two improvements are **[software computation]** and do not enter any kernel-verified statement; the exponent bound the kernel actually certifies remains $x \geq 12$.

The same constant governs a second, purely cardinal comparison, which is the more telling of the two. The configuration forbidden by Theorem 17.4 consists of

$$r_n + 1 \sim \vartheta \log X$$

candidates arranged in one structured pattern, while the whole dyadic block contains only $\sim \log X/(\beta \log 2)$ candidates by (45). The ratio is again exactly $\beta \log 3$: the optimized theorem asks a block to supply about $\beta \log 3$ times as many candidates as it can possibly hold. Concretely, at $X = 2^{100}$ the block $[X, 2X)$ holds $100/\beta + O(1)$ candidates — about eight if β were as small as its kernel-certified floor, and about three under the 2^{27} scan — while the shortest progression the theorem forbids at that height already needs $r_n + 1 \approx 111$ terms.

Remark 17.5 (Quantified barrier). Optimizing over r materially strengthens the local theorem: the admissible step scale rises from $n^{1-\vartheta/r}$ at fixed r to $n^{1-o(1)}$, missing n itself only by the single logarithmic factor $\vartheta \log n$. It still cannot contradict the conditional monoid, because the conditional candidate density in a dyadic block is too small by the fixed factor $\beta \log 3 > 13.18$. No argument using only the number of candidates in a dyadic block can force the progression that the current finite-difference functional requires. A successful continuation must exploit additional arithmetic correlation — for example valuation congruences that force a progression of a special shape — or replace Δ_h^r by a different integer-valued functional with a better curvature-to-spacing constant.

This is a limitation of the method, not evidence against Conjecture 1.1. Its value is to replace "the density seems too weak" by a sharp asymptotic constant and a concrete numerical target that any improved functional must beat.

Agreement with the one-logarithm deficit. Section 21 audits the semigroup determinant in the same currency and identifies the same bottleneck. In both analyses the binding quantity is the conditional dyadic population $\asymp \log X$ of (45), and in both the archimedean inequality turns out to be comfortably satisfied by the exact monoid rather than violated by it. To that extent the two diagnoses agree. They differ, instructively, on the size of the shortfall. The semigroup determinant needs $r \asymp (\log X)^2$ nodes and is supplied $\asymp \log X$: a deficit of a full power of $\log X$. The optimized finite difference needs $\vartheta \log X$ nodes and is supplied $\log X/(\beta \log 2)$: a deficit of the bounded factor $\beta \log 3$. The finite difference is thus quantitatively much closer — but only because it buys that closeness with a hypothesis it cannot discharge, namely that the candidates lie in exact arithmetic progression, which Observation 17.3 says a rank-two multiplicative monoid has no reason to provide. Discarding that hypothesis is exactly what the determinants of Sections 18 and 19 do, and the one-logarithm deficit is the precise price charged for it. Read together, the two sections say that the present archimedean toolkit is short by one factor of $\log X$ in generality or by a factor of $\beta \log 3$ in strength, and that tuning constants converts neither shortfall into the other.

18 Arbitrary-node lattice repulsion

The higher-difference theorem of Section 17 uses equally spaced nodes, because the forward-difference operator is built from an arithmetic progression. The same integer-versus-small-real principle extends to *arbitrary* nodes through interpolation determinants. The resulting statement constrains every configuration of candidates, not only the additively structured ones, which is exactly the gap identified in Observation 17.3.

18.1 The determinant identity

For distinct real numbers $x_0 < \cdots < x_r$ and a function f, write $f[x_0, \ldots, x_r]$ for the divided difference. The standard determinant identity is

$$\det \begin{pmatrix} 1 & x_0 & \cdots & x_0^{r-1} & f(x_0) \\ 1 & x_1 & \cdots & x_1^{r-1} & f(x_1) \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & x_r & \cdots & x_r^{r-1} & f(x_r) \end{pmatrix} = V(x_0, \ldots, x_r)\, f[x_0, \ldots, x_r], \tag{47}$$

where

$$V(x_0,\dots,x_r) = \prod_{0\le i<j\le r} (x_j - x_i).$$

If $f \in C^r$ on the interval spanned by the nodes, the generalized mean-value theorem for divided differences supplies some $\xi \in (x_0, x_r)$ with

$$f[x_0,\dots,x_r] = \frac{f^{(r)}(\xi)}{r!}. \tag{48}$$

For $f(t) = t^\vartheta$ this is again $f^{(r)}(t) = (\vartheta)_{\underline{r}}\, t^{\vartheta-r}$, and since ϑ is irrational — in particular not an integer — the coefficient $(\vartheta)_{\underline{r}}$ never vanishes.

18.2 The arbitrary-node theorem

Theorem 18.1 (Vandermonde repulsion for candidates)**.** *Let $r \ge 2$ and let*

$$1 \le m_0 < m_1 < \cdots < m_r$$

be natural candidates, so that $m_i^\vartheta \in \mathbb{N}$ for every i. Then

$$\prod_{0\le i<j\le r} (m_j - m_i) \ge \frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\, m_0^{r-\vartheta}. \tag{49}$$

Proof. Apply (47) to $f(t) = t^\vartheta$ with the nodes m_i. Every entry of the matrix is an integer: the ordinary powers m_i^k are integral by construction, and the last column is integral precisely because the m_i are candidates. By (48) the determinant equals

$$D = V(m_0,\dots,m_r)\,\frac{(\vartheta)_{\underline{r}}}{r!}\,\xi^{\vartheta-r} \qquad (m_0 < \xi < m_r).$$

Each factor is nonzero, so $D \in \mathbb{Z}\setminus\{0\}$ and therefore $|D| \ge 1$. Because $r \ge 2 > \vartheta$ the exponent $\vartheta - r$ is negative, so $\xi^{\vartheta-r} \le m_0^{\vartheta-r}$. Therefore

$$1 \le |D| \le V(m_0,\dots,m_r)\,\frac{|\,(\vartheta)_{\underline{r}}\,|}{r!}\, m_0^{\vartheta-r},$$

which rearranges to (49). ☐

Status: [paper proof] for the unconditional paper theorem; [kernel-verified Lean] for the formal determinant and repulsion chain. `ArbitraryNodeDeterminant` proves integrality, the Vandermonde identities, and the hypothesis-carrying repulsion step, while `ArbitraryNodeRepulsion` packages the final bound under the explicit hypothesis `RpowDividedDifferenceMeanValue`. The corpus does not assert that analytic hypothesis.

Remark 18.2 (Formalization state)*.* The accepted module `ArbitraryNodeDeterminant` supplies the algebraic half deliberately free of analysis — the node matrix, an explicit integral model, integrality of the determinant with the resulting bound $1 \le |\det|$, the Vandermonde product and its positivity on strictly increasing nodes. The later endpoint theorem is also accepted, but its divided-difference mean-value statement for $t \mapsto t^\vartheta$ remains a visible hypothesis rather than a proved ingredient. Thus the conditional Lean implication may be cited as machine-checked; the unconditional analytic discharge is supplied only by the paper proof here.

18.3 Content-minimal barycentric certificates

The determinant normalization above can be sharpened for any fixed smooth node set. Let $T = (t_0 < \cdots < t_k)$ and put

$$d_j(T) = \prod_{i\neq j}(t_j - t_i), \qquad L(T) = \mathrm{lcm}_j\, |d_j(T)|, \qquad \mathcal{D}_T(s) = L(T)\sum_{j=0}^{k} \frac{t_j^s}{d_j(T)}.$$

The coefficients $L(T)/d_j(T)$ are primitive integers and annihilate every monomial of degree below k. If all t_j are 3-smooth, then $\mathcal{D}_T(x)$ is an integer under the conjectural hypothesis. The divided-difference mean-value theorem gives

$$\mathcal{D}_T(x) = \frac{L(T)}{k!}(x)_k \xi^{x-k} \quad (t_0 < \xi < t_k), \tag{50}$$

where $(x)_k = x(x-1)\cdots(x-k+1)$. Thus, unless $x \in \{0, \ldots, k-1\}$,

$$1 \le \frac{L(T)}{k!}|(x)_k| \max(t_0^{x-k}, t_k^{x-k}). \tag{51}$$

This is the content-minimal version of the arbitrary-node repulsion certificate: it removes the common content of the Vandermonde cofactors before applying the integer lower bound.

When $t_0 = 1$, it excludes the strict endpoint windows

$$k - 1 - \frac{k}{L(T)} < x < k - 1, \qquad k - 1 < x < k - 1 + \frac{1}{L(T)}. \tag{52}$$

At either outer endpoint one may substitute directly only after checking that the functional is nonzero and has absolute value below one; at the central point $x = k - 1$ it vanishes. For $T = (1, 2, 3, 4)$, $L(T) = 6$ and

$$\mathcal{D}_T(x) = 4^x - 3 \cdot 3^x + 3 \cdot 2^x - 1.$$

For $1 < x < 2$ this is a nonzero negative integer, while $|\mathcal{D}_T(x)| < x(x-1)(2-x) < 1$, excluding the entire strip without enumerating M.

Exact search among subsets of the first eighteen smooth nodes that contain 1 gives the following compact calibration; the two window widths are $1/L$ and k/L.

k	minimizing node set found	$L(T)$
2	$1, 2, 3$	2
3	$1, 2, 3, 4$	6
4	$1, 2, 3, 4, 6$	120
5	$1, 2, 3, 4, 6, 8$	1680
6	$1, 2, 3, 4, 6, 8, 9$	5040
7	$1, 2, 3, 4, 6, 8, 9, 12$	3991680
8	$1, 2, 3, 4, 6, 8, 9, 12, 18$	4071513600
9	$1, 2, 3, 4, 6, 8, 9, 12, 16, 18$	741015475200
10	$1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24$	818081084620800

The rapid growth of $L(T)$ is the obstruction, not a rounding artifact. A broader exact target chooses primitive integer coefficients c_j with $\sum_j c_j t_j^q = 0$ for $q < k$ and a nonzero one-signed Peano kernel

$$K_c(u) = \sum_j c_j (t_j - u)_+^{k-1}.$$

If additionally $\sup_{m<s<m+1} |\sum_j c_j t_j^s| < 1$, the corresponding strip is excluded. This is a finite integer-optimization problem that genuinely extends pure divided differences. The denominator clearing, moment annihilation, and low-node certificates are **[kernel-verified Lean]** in `ClearedDividedDifference`; the mean-value and Peano-kernel steps are the paper layer.

18.4 Three points: a lattice-triangle inequality

For $r = 2$ the determinant is

$$D = \det \begin{pmatrix} 1 & m_0 & m_0^\vartheta \\ 1 & m_1 & m_1^\vartheta \\ 1 & m_2 & m_2^\vartheta \end{pmatrix},$$

and $|D|$ is twice the area of the lattice triangle with vertices (m_i, m_i^ϑ). Strict convexity of $t \mapsto t^\vartheta$ makes the area positive; lattice integrality makes it at least $1/2$. The bound below is the quantitative form of that picture.

Corollary 18.3 (Three-candidate span)**.** *If three candidates lie in $[N, N+H]$ with $N \geq 1$, then*

$$H \geq \left(\frac{8}{\vartheta(\vartheta-1)}\right)^{1/3} N^{(2-\vartheta)/3}. \tag{53}$$

Rounded to ten decimal places the constant is 2.0510723957 and the exponent 0.1383458331; in particular $H \geq 2.0510723956\, N^{0.1383458330}$.

Proof. Write $a = m_1 - m_0$ and $b = m_2 - m_1$. Then $a, b > 0$ and $a + b \leq H$, so the Vandermonde product satisfies

$$V = ab(a+b) \leq \frac{H^3}{4},$$

the continuous maximum being attained at $a = b = H/2$. Theorem 18.1 at $r = 2$ gives

$$V \geq \frac{2}{\vartheta(\vartheta-1)} N^{2-\vartheta}.$$

Combining the two inequalities yields (53). □

18.5 General local packing bound

Put

$$R_r := \binom{r+1}{2} = \frac{r(r+1)}{2}, \qquad C_r := \left(\frac{r!}{|(\vartheta)_{\underline{r}}|}\right)^{1/R_r}, \qquad \alpha_r := \frac{r-\vartheta}{R_r} = \frac{2(r-\vartheta)}{r(r+1)}.$$

Every pairwise gap among points of $[N, N+H]$ is at most H, so the Vandermonde product is at most H^{R_r}, and Theorem 18.1 immediately gives the following.

Corollary 18.4 (At most r points below the repulsion scale)**.** *If $r+1$ candidates lie in $[N, N+H]$, then*

$$H \geq C_r N^{\alpha_r}. \tag{54}$$

Equivalently, every interval of length strictly less than $C_r N^{\alpha_r}$ beginning at or after N contains at most r candidates.

The exponent can be optimized in r. As a function of a real variable $r > \vartheta$,

$$\alpha(r) = \frac{2(r-\vartheta)}{r(r+1)}$$

attains its maximum at

$$r_* = \vartheta + \sqrt{\vartheta^2 + \vartheta} = 3.6090841941\ldots,$$

so the best integer order in this ordinary-power family is $r = 4$ — not the three-point case that geometry makes most vivid. Table 2 lists the constants.

r	R_r	$\lvert (\vartheta)_{\underline{r}} \rvert$	α_r	C_r
2	3	0.9271436280	0.1383458331	1.2920946431
3	6	0.3847993728	0.2358395832	1.5805909191
4	10	0.5445055422	0.2415037499	1.4602287461
5	15	1.3150013031	0.2276691666	1.3510881062
6	21	4.4907787617	0.2102398809	1.2735045942
7	28	19.8269566337	0.1933941964	1.2187063909
8	36	107.3637136680	0.1781954861	1.1790125192

Table 2: Ordinary-power arbitrary-node repulsion constants. For $r = 2$ the sharper three-point constant of Corollary 18.3 uses the exact maximum of the three-gap product rather than the crude bound H^{R_2}.

Corollary 18.5 (Best ordinary-power local statement)**.** *Every interval*

$$[N, N+H], \qquad H < 1.4602287460\, N^{0.2415037499},$$

contains at most four natural candidates. The exact constant is $C_4 = (4!/\lvert (\vartheta)_{\underline{4}} \rvert)^{1/10} = 1.4602287461$ *to ten decimal places, and the exact exponent is* $\alpha_4 = (4-\vartheta)/10 = 0.2415037499\ldots$

18.6 A bound for any longer interval

The span estimate converts into a counting inequality. Let $m_1 < \cdots < m_s$ be all candidates in $[N, N+H]$. For each $i \le s - r$ the consecutive window $m_i, \ldots, m_{i+r}$ has span at least $C_r N^{\alpha_r}$ by Corollary 18.4. Summing these spans, each adjacent gap $m_{j+1} - m_j$ is counted at most r times, so

$$(s-r) C_r N^{\alpha_r} \le r(m_s - m_1) \le rH.$$

Corollary 18.6 (Arbitrary-node counting inequality)**.** *For every* $r \ge 2$*,*

$$\#(\mathcal{C} \cap [N, N+H]) \le r + \frac{r}{C_r}\, H N^{-\alpha_r}. \tag{55}$$

This is a genuine strengthening of progression avoidance: it constrains every possible configuration of candidates in an interval, with no additive hypothesis at all. It remains far from the conditional benchmark of Section 4. At the optimal $r = 4$ and $H = N$, the right-hand side of (55) is $O(N^{0.7585\ldots})$, whereas the exact population of a failed conjecture would be only $O(\log N)$. The gap is enormous, and closing it by ordinary powers alone is hopeless — which motivates enlarging the column space, the subject of Section 19.

18.7 Relation to higher differences

For equally spaced nodes $m_i = n + ih$ the Vandermonde factor is explicit:

$$V = h^{R_r} \prod_{j=1}^{r} j!.$$

Substituting in (47) recovers an integer interpolation remainder closely related to the rth forward difference of Section 17. The committed repository theorem is sharper for this special configuration, because it uses the exact binomial combination rather than bounding all pairwise gaps by a common span. Conversely, Theorem 18.1 applies when the nodes carry no additive structure whatsoever. The two results are complementary, and only the arbitrary-node form escapes the vacuity diagnosed in Observation 17.3.

19 Semigroup generalized Vandermonde determinants

The determinant of Section 18 used only the integral columns $1, m, m^2, \dots$ and one extra integral column m^ϑ. But a candidate carries much more integral data. If $m^\vartheta = n \in \mathbb{N}$, then for every $a, b \in \mathbb{N}_0$,

$$m^{a+b\vartheta} = m^a n^b \in \mathbb{N}. \tag{56}$$

The full set of available exponents is therefore

$$\Lambda_\vartheta := \{a + b\vartheta : a, b \in \mathbb{N}_0\}.$$

Since ϑ is irrational, all pairs (a, b) give distinct exponents.

19.1 Positivity of generalized Vandermonde determinants

Let

$$0 \le \lambda_0 < \lambda_1 < \cdots < \lambda_r \qquad \text{and} \qquad 0 < x_0 < x_1 < \cdots < x_r.$$

The functions x^{λ_j} form an extended complete Chebyshev system on $(0, \infty)$. Their Wronskian is classical in the Karlin–Studden theory of Tchebycheff systems:

$$W(x) = x^{\sum_j \lambda_j - R_r} \prod_{0 \le i < j \le r} (\lambda_j - \lambda_i) > 0. \tag{57}$$

Indeed, after factoring x^{λ_j} from column j and x^{-i} from derivative row i, the remaining determinant is the Vandermonde determinant of the λ_j, expressed in the falling-factorial basis. Positivity of all initial Wronskians implies strict positivity of every ordered evaluation determinant.

For completeness, there is also an elementary zero-count proof. Put $x = e^t$. A nonzero linear combination of x^{λ_j} becomes an exponential polynomial $\sum c_j e^{\lambda_j t}$. After factoring the lowest exponential, differentiation removes the constant term and leaves one fewer exponential. Induction with Rolle's theorem shows that a combination of $r + 1$ such functions has at most r real zeros counting multiplicity. Consequently an evaluation determinant at $r + 1$ ordered nodes cannot vanish. Its sign is constant on the connected chamber $x_0 < \cdots < x_r$ and is positive in the coalescing-node limit by (57).

Lemma 19.1 (Generalized Vandermonde positivity). *For strictly increasing nonnegative exponents λ_j and strictly increasing positive nodes x_i,*

$$\det(x_i^{\lambda_j})_{0\le i,j\le r} > 0. \tag{58}$$

The two arguments just sketched are carried out in full, with all induction bookkeeping made explicit, in Appendix D.

19.2 Integer determinants from candidates

Theorem 19.2 (Semigroup integer determinant). *Let $m_0 < \cdots < m_r$ be natural candidates. Choose distinct pairs $(a_j, b_j) \in \mathbb{N}_0^2$ and order*

$$\lambda_j = a_j + b_j\vartheta$$

so that $\lambda_0 < \cdots < \lambda_r$. Then

$$D_{\boldsymbol{\lambda}}(\boldsymbol{m}) := \det(m_i^{\lambda_j})_{0\le i,j\le r} \tag{59}$$

is a positive integer. In particular, $D_{\boldsymbol{\lambda}}(\boldsymbol{m}) \ge 1$.

Proof. By (56), each entry equals

$$m_i^{a_j}(m_i^{\vartheta})^{b_j} \in \mathbb{N},$$

so the determinant is an integer. It is strictly positive by Lemma 19.1. □

Status: [paper proof]

This theorem packages all mixed integrality consequences into one reusable object. It is not the same as the repository's exponential interpolation determinant in `MultiplicativeRank.lean`: here the rows are arbitrary natural candidates and the columns are chosen from the two-dimensional exponent semigroup.

19.3 Divided-difference factorization

Let $f_j(x) = x^{\lambda_j}$ and let

$$V(\boldsymbol{m}) = \prod_{i<j}(m_j - m_i).$$

Applying the Newton divided-difference row transformation gives

$$D_{\boldsymbol{\lambda}}(\boldsymbol{m}) = V(\boldsymbol{m})\det\big(f_j[m_0,\ldots,m_i]\big)_{0\le i,j\le r}. \tag{60}$$

The transformation is exact; its determinant is $V(\boldsymbol{m})^{-1}$.

Suppose

$$N \le m_0 < \cdots < m_r \le N + H \le 2N.$$

For each divided difference, the mean-value theorem gives a point in $[N, 2N]$ and

$$|f_j[m_0,\ldots,m_i]| \le \frac{|\,(\lambda_j)_{\underline{i}}\,|}{i!} \sup_{N\le x\le 2N} x^{\lambda_j - i}. \tag{61}$$

We need a uniform elementary coefficient estimate.

Lemma 19.3 (Falling-factorial coefficient bound)**.** *For $\lambda \geq 0$ and $i \in \mathbb{N}_0$,*

$$\frac{|\,(\lambda)_{\underline{i}}\,|}{i!} \leq 2^{\lambda+i}. \tag{62}$$

Proof. If $0 \leq \lambda \leq 1$, each factor satisfies $|\lambda - k| \leq k+1$, so the quotient is at most 1. If $\lambda \geq 1$, let $n = \lceil \lambda \rceil$. Then

$$\frac{|\,(\lambda)_{\underline{i}}\,|}{i!} \leq \prod_{k=0}^{i-1} \frac{\lambda + k}{k+1} \leq \prod_{k=0}^{i-1} \frac{n+k}{k+1} = \binom{n+i-1}{i}.$$

The last binomial coefficient is at most the sum of its row, 2^{n+i-1}, and $n-1 \leq \lambda$. Hence it is at most $2^{\lambda+i}$. □

For $x \in [N, 2N]$,

$$x^{\lambda-i} \leq 2^{\lambda} N^{\lambda-i}. \tag{63}$$

If $\lambda - i \geq 0$, use $x \leq 2N$; if it is negative, use $x \geq N$. Combining (61), (62), and (63),

$$|f_j[m_0, \ldots, m_i]| \leq 2^{2\lambda_j+i} N^{\lambda_j - i}. \tag{64}$$

19.4 Quantitative semigroup gap theorem

Set

$$R := R_r = \frac{r(r+1)}{2}, \qquad \Sigma := \sum_{j=0}^{r} \lambda_j.$$

By the Leibniz formula and (64), every permutation term in the small determinant in (60) is at most

$$2^{2\Sigma+R} N^{\Sigma-R},$$

and there are $(r+1)!$ terms. Since every pairwise node difference is at most H, $V(\boldsymbol{m}) \leq H^R$. The positive integer lower bound from Theorem 19.2 now gives the following.

Theorem 19.4 (Semigroup determinant gap)**.** *Let $r \geq 1$. Let $\lambda_0 < \cdots < \lambda_r$ be any $r+1$ exponents in Λ_ϑ, and put $R = r(r+1)/2$ and $\Sigma = \sum_j \lambda_j$. If $r+1$ natural candidates lie in*

$$[N, N+H] \subseteq [N, 2N], \qquad N \geq 1,$$

then

$$H \geq ((r+1)!)^{-1/R} 2^{-1-2\Sigma/R} N^{1-\Sigma/R}. \tag{65}$$

Proof. Equations (60) and (64) imply

$$1 \leq D_{\boldsymbol{\lambda}}(\boldsymbol{m}) \leq H^R (r+1)! 2^{R+2\Sigma} N^{\Sigma-R}.$$

Taking Rth roots and rearranging proves (65). □

The exponent $1 - \Sigma/R$ is the key. We should choose the $r+1$ smallest available semigroup exponents to minimize Σ.

19.5 The ordered exponent semigroup

Write the elements of Λ_ϑ in increasing order:

$$0 = \lambda_0 < \lambda_1 < \lambda_2 < \cdots .$$

The beginning is

$$0,\ 1,\ \vartheta,\ 2,\ 1+\vartheta,\ 3,\ 2\vartheta,\ 2+\vartheta,\ 4,\ 1+2\vartheta,\ 3+\vartheta,\ 3\vartheta,\ 5,\ldots$$

The counting function is elementary:

$$\begin{aligned} L(T) &:= \#\{\lambda \in \Lambda_\vartheta : \lambda \le T\} \\ &= \sum_{0\le b\le T/\vartheta} (\lfloor T - b\vartheta \rfloor + 1) = \frac{T^2}{2\vartheta} + O(T). \end{aligned} \tag{66}$$

Irrationality of ϑ ensures that different pairs (a, b) are counted distinctly. Inverting (66) and summing the order statistics gives

$$\lambda_r = \sqrt{2\vartheta r} + O(1), \tag{67}$$

$$\Sigma_r := \sum_{j=0}^{r} \lambda_j = \frac{2}{3}\sqrt{2\vartheta}\, r^{3/2} + O(r), \tag{68}$$

$$\frac{\Sigma_r}{R_r} = \frac{4}{3}\sqrt{\frac{2\vartheta}{r}} + O(r^{-1}). \tag{69}$$

Consequently the power of N in (65) tends to 1:

$$1 - \frac{\Sigma_r}{R_r} = 1 - \frac{4}{3}\sqrt{\frac{2\vartheta}{r}} + O(r^{-1}). \tag{70}$$

r	Σ_r	$1 - \Sigma_r/R_r$
2	2.5849625007	0.1383458331
3	4.5849625007	0.2358395832
4	7.1699250014	0.2830074999
5	10.1699250014	0.3220049999
6	13.3398500029	0.3647690475
7	16.9248125036	0.3955424106
8	20.9248125036	0.4187552082
9	25.0947375050	0.4423391666
10	29.6797000058	0.4603690908
12	39.4345875079	0.4944283653
20	87.7939875195	0.5819333928

Table 3: Exponent gain from using the first $r+1$ mixed exponents in $\mathbb{N}_0 + \mathbb{N}_0\vartheta$. The ordinary-power family peaks and then decays; the semigroup family improves monotonically in these data and tends to exponent 1.

19.6 An explicit coarse form

For the global counting corollary we need no delicate lattice-point error term. A simple bound suffices.

Lemma 19.5 (Crude semigroup weight bound)**.** *For $r \geq 2$, the first $r+1$ semigroup exponents satisfy*

$$\lambda_r \leq 6\sqrt{r}, \qquad \frac{\Sigma_r}{R_r} \leq \frac{12}{\sqrt{r}}. \tag{71}$$

Proof. Let $q = \lceil \sqrt{r+1} \rceil$. The $(q+1)^2$ pairs $0 \leq a, b \leq q$ give distinct exponents and each is at most $q(1+\vartheta) < 3q$. Since $(q+1)^2 > r+1$, the rth ordered exponent is below $3q$. For $r \geq 2$, $q \leq \sqrt{r+1} + 1 \leq 2\sqrt{r}$, hence $\lambda_r \leq 6\sqrt{r}$. Then

$$\Sigma_r \leq (r+1)\lambda_r \leq 6(r+1)\sqrt{r},$$

and division by $R_r = r(r+1)/2$ gives the second inequality. □

For $r \geq 144$, (71) gives $\Sigma_r/R_r \leq 1$. Also

$$((r+1)!)^{1/R_r} \leq (r+1)^{(r+1)/R_r} = (r+1)^{2/r} \leq 3$$

for $r \geq 2$. Therefore Theorem 19.4 yields a particularly clean form.

Corollary 19.6 (Explicit near-linear span)**.** *If $r \geq 144$ and $r+1$ natural candidates lie in $[N, N+H] \subseteq [N, 2N]$, then*

$$H \geq \frac{1}{24} N^{1-12/\sqrt{r}}. \tag{72}$$

Proof. In (65), use $((r+1)!)^{-1/R} \geq 1/3$, $2^{-1-2\Sigma/R} \geq 2^{-3} = 1/8$, and, since $N \geq 1$, $N^{1-\Sigma/R} \geq N^{1-12/\sqrt{r}}$. □

19.7 Counting candidates in a general interval

As in the ordinary-power count of Section 18, sum the spans of consecutive windows of $r+1$ candidates. Every adjacent gap is counted at most r times.

Corollary 19.7 (Semigroup interval count)**.** *For $r \geq 144$ and $1 \leq H \leq N$,*

$$\#(\mathcal{C} \cap [N, N+H]) \leq r + 24rHN^{-1+12/\sqrt{r}}. \tag{73}$$

Proof. If s is the number of candidates and $s \leq r$, there is nothing to prove. Otherwise, Corollary 19.6 gives a lower bound $L = N^{1-12/\sqrt{r}}/24$ for each span $m_{i+r} - m_i$. Thus

$$(s-r)L \leq rH,$$

which is equivalent to (73). □

19.8 An independent dyadic logarithmic-square bound

Corollary 19.8 (Explicit $O((\log X)^2)$ dyadic bound)**.** *For every real* $X \geq 1$*,*

$$\#(\mathcal{C} \cap [X, 2X]) \leq 10000 \max\{1, \log X\}^2. \tag{74}$$

Proof. Put $L = \max\{1, \log X\}$ and

$$r = \left\lceil 144L^2 \right\rceil.$$

Then $r \geq 144$, $\sqrt{r} \geq 12L$, and

$$X^{12/\sqrt{r}} = \exp\left(\frac{12 \log X}{\sqrt{r}}\right) \leq e.$$

Apply (73) with $N = H = X$:

$$\#(\mathcal{C} \cap [X, 2X]) \leq r + 24rX^{12/\sqrt{r}} \leq (1 + 24e)r.$$

Since $L \geq 1$, $r \leq 145L^2$, and $(1 + 24e)145 < 10000$. □

Status: [paper proof]

The bound is numerically coarse by design. Its value is conceptual independence: it comes from positivity and integrality of generalized Vandermonde determinants, not from the prime-factor valuation rank of candidates. Refining constants is much less important than finding an additional source of determinant lower bound.

Status: [kernel-verified Lean] and **[paper proof]**

The hierarchy is now kernel-accepted in `MixedExponentSemigroup`, `SemigroupDeterminant`, `FallingFactorialBound`, and `SemigroupWeightBound`. The final span and $10^4(\log X)^2$ deductions are accepted in `SemigroupGapBound` conditional on its explicit `NewtonFactorization` argument. That predicate is a visible input, not an axiom; the unconditional analytic proof remains the paper proof in this section.

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20 The min-plus ceiling: why term-wise p-adic amplification cannot close a determinant argument

Every interpolation-determinant argument in this report squeezes a nonzero integer determinant D between a lower and an upper bound. A recurring proposal — registered in the deficit analysis of Section 21 as the most natural way to supply the missing logarithm — is to manufacture the lower bound from p-adic content: choose the nodes and the column exponents so that every monomial in the Leibniz expansion of D is divisible by a high power of 2 and of 3, and conclude

$$v_p(D) \;\geq\; V_p^* := \min_{\sigma} \sum_i v_p\Big(E_{i\sigma(i)}\Big), \qquad |D| \;\geq\; 2^{V_2^*} 3^{V_3^*}. \tag{75}$$

This section records why the proposal cannot work, and — as importantly — corrects the quantitative form in which the obstruction has been stated.

20.1 The threshold

Fix a column set $S = \{(a_j, b_j)\}_{j=0}^{r}$ of distinct mixed exponents, put $\lambda_j = a_j + b_j\vartheta$ and $W = \sum_j \lambda_j$, and let D be the semigroup determinant $\det\big(c_i^{a_j}(c_i^{\vartheta})^{b_j}\big)$ built on the candidates $c_0 < \cdots < c_r$ of a single dyadic block $[2^T, 2^{T+1})$. Every entry satisfies $\log_2 E_{ij} \le (T+1)\lambda_j$, so the Leibniz bound gives

$$\log_2 |D| \ \le\ (T+1)W + \log_2(r+1)! \ =\ TW\,(1+o(1)). \tag{76}$$

An amplification scheme yielding a guaranteed lower bound $|D| \ge 2^V$ therefore needs $V > (T+1)W + \log_2(r+1)!$ for a contradiction. We call the right-hand side of (76) the *bar*.

20.2 The ceiling

The obstruction is not quantitative at all. It is the observation that the $\{2,3\}$-part of an integer divides the integer.

Theorem 20.1 (Min-plus ceiling)**.** *Let $n \ge 1$ be a natural number, let $V_2, V_3 \ge 0$ satisfy $2^{V_2} \mid n$ and $3^{V_3} \mid n$, and suppose $n \le 2^B$. Then*

$$V_2 + V_3\,\vartheta \ \le\ B.$$

More generally, for any finite set S of primes, $\sum_{p\in S} v_p(n)\log p \le \log n$.

Proof. 2^{V_2} and 3^{V_3} are coprime and both divide n, so $2^{V_2}3^{V_3} \mid n$ and hence $2^{V_2}3^{V_3} \le n \le 2^B$. Taking $\log_2$ gives the claim. For general S the same argument applies to $\prod_{p\in S} p^{v_p(n)}$, which divides n by pairwise coprimality. □

Status: [kernel-verified Lean] — `minPlusContent_le_of_le` and `minPlusContent_le_of_natAbs_le` in `MinPlusCeiling`, with the general prime-set form `primeContent_le_log` and the divisibility input `prod_primePow_dvd`.

Corollary 20.2 (Term-wise amplification is inert)**.** *Let $D \ne 0$ be the determinant of an interpolation system, let V_2^*, V_3^* be any certified min-plus lower bounds as in (75), and let B be any correct archimedean upper bound $\log_2 |D| \le B$. Then $V_2^* + V_3^*\vartheta \le B$.*

Consequently no choice of node system, no choice of column set, and no enlargement of the prime set can make a term-wise divisibility bound contradict a correct archimedean bound.

The corollary is unconditional and needs no equidistribution input, no averaging, and no property of the candidates beyond $D \in \mathbb{Z}\setminus\{0\}$. Its content is that term-wise divisibility contributes nothing beyond what $|D| \ge 1$ already gives: it is a lower bound for $|D|$ derived from $|D|$ itself.

20.3 A correction to the averaging form of the obstruction

The obstruction has previously been argued by averaging: the minimum over permutations is at most the mean over permutations, the mean of $\sum_i v_p(E_{i\sigma(i)})$ is $\sum_j\big(a_j\bar\nu + b_j\bar\mu\big)$ with $\bar\nu, \bar\mu$ the row-averaged valuations, and the depths n_k equidistribute over $[0, T]$, so $\bar\nu = T/2$. That computation is correct, and its conclusion — that the ceiling is governed by the *same* block equidistribution that defeats the shrinking-target route of Section 22 — is a genuine structural insight worth keeping: at the structural places the two walls of this problem are one wall.

The numerical conclusion drawn from it, however, is too strong. Bounding the 2-adic place and the ϑ-weighted 3-adic place separately gives $TW/2$ *each*; their sum is TW, which is the bar, not half of it. With $M = 2^\beta$ and $A = 3^\beta$ and the normalization $s_M = 2^{v_2(M)}3^{v_3(M)}$, $s_A = 2^{v_2(A)}3^{v_3(A)}$ for the 3-smooth parts, the mean bound reads

$$V^* \;\le\; \frac{T}{2}\left[W + \frac{1}{\beta}\sum_j \Big(a_j \log_2 s_M + b_j \log_2 s_A\Big)\right], \tag{77}$$

and the bracket is at most $2W$, with equality exactly when M and A are both 3-smooth. Theorem 11.5 excludes that case, so the mean bound is strictly below the bar; but the deficiency it provides is proportional to the roughness of the outputs and degenerates as the outputs approach smoothness.

The factor $1/2$ requires $3 \nmid M$ *and* $2 \nmid A$. The matched-depth theorem of Section 4 excludes only the pair $2 \mid M$ and $3 \mid A$; the crossed pair $3 \mid M$ and $2 \mid A$ is not excluded, and it is exactly the configuration in which the ceiling climbs.

20.4 Calibration: the exact ceilings

The mean bound is not the true ceiling, because the minimum is genuinely lower than the mean. The exact constants follow from the optimal-transport computation below and replace the heuristic figures that arise from mean-value estimates.

Write $\mathfrak{G}$ for the Gini norm of a normalized prime-valuation slope vector, and c_p for the slope vector at p. The termwise all-prime divisor fraction of the balanced mixed-semigroup determinant is exactly $1 - \frac{3}{8}\sum_p \mathfrak{G}(c_p) + o(1)$, and the primitive condition $\min\big(v_2(M)\,,v_3(A)\big) = 0$ forces $\sum_p \mathfrak{G}(c_p) \ge 8/15$.

System	Termwise-divisor ceiling
Ordinary input/output Vandermonde product	< 0.8710491
Mixed single block, arbitrary branch	$4/5$
Mixed single block, cross-coprime	$7/10$
Mixed single block, places 2 and 3 only, cross-coprime	$3/10$
Complete triangular row grid	$\pi/4 = 0.7853982\ldots$
First tied fibers at 2 and 3	$o(1)$ of the block budget

Two figures that have been proposed for this obstruction are not valid in the stated generality. A structural ceiling of $1/2$ requires dropping the depths $k\,u\,v_2(M)$ and $k\,v\,v_3(A)$ simultaneously, which assumes $v_2(M) = v_3(A) = 0$; the primitive theorem of Section 4 supplies only $\min\big(v_2(M)\,,v_3(A)\big) = 0$. An all-prime ceiling of $2/3$ does not follow from "minimum $\le$ average": for an external positive weight proportional to $X_1 + X_2$ on the standard triangle one has $\mathbb{E}Y = 2/3$ and $\Phi(1,1) = 4/15$, so the minimum assignment is $4/5$ of half the mean, not $2/3$.

An independent numerical check, computing exact assignment minima for the block node system over the places 2 and 3 only, in the cross-coprime branch and with a square-box column family, gives a ratio settling near 0.33 across block heights $T = 400, 1200, 3600, 10000$, consistent with the exact continuum value $3/10$ for those places. The same computation shows what happens outside the balanced primitive normalization: with a degenerate column family such as the pure-b columns $(0, j)$, and with A within $\log_2 3$ of being a power of three, the ratio rises to 0.9964. That configuration violates none of the kernel-verified constraints, but it saturates *tautologically* — the entries are then nearly powers of three, their 3-adic content is nearly their size, and (75) degenerates. The same degeneration occurs on the a-side at $b = 0$, where $D = \prod_{i<i'}(c_{i'} - c_i)$ and $v_2(D) = \sum_{i<i'} \min(n_i, n_{i'})$ exactly. This is why the balanced and primitive normalizations are not cosmetic: they are what make the 4/5 and 7/10 ceilings meaningful.

Status: [paper proof] for the exact ceilings — they follow from the closed-form Gini identity, not from experiment. [software computation] for the independent 3/10 cross-check and the degenerate-family observation: exact rational node data, exact Jonker–Volgenant assignment minima, double-precision ϑ. Neither is needed for Corollary 20.2, which is [kernel-verified Lean] and unconditional.

20.5 Where cancellation can live

The ceilings bound only the *tropical* divisor. They say nothing about the true valuation $v_p(D)$, which can be larger when several permutations attain the minimum. The following observation localizes that possibility exactly.

Proposition 20.3 (A unique minimizer forbids cancellation)**.** *Let $D = \det(E_{ij})$ be a nonzero integer determinant and let p be a prime. If a unique permutation σ_0 minimizes $\sum_i v_p(E_{i\sigma(i)})$, then $v_p(D) = \sum_i v_p(E_{i\sigma_0(i)})$.*

Proof. Divide the Leibniz expansion by p^{τ_p} with τ_p the minimum. The term indexed by σ_0 becomes a p-adic unit and every other term stays divisible by p, so the sum is a unit modulo p. □

Status: [kernel-verified Lean] — `det_dvd_and_not_dvd_of_unique_min` in `MinPlusCeiling`, stated in the divisibility form ‘$q^\tau \mid \det$ and $q^{\tau+1} \nmid \det$’ so that no p-adic valuation machinery is needed.

The corresponding tied-minimum statement is now also exact at the algebraic level. Suppose each Leibniz product has been written

$$\prod_i E_{i,\sigma(i)} = p^\tau q_\sigma$$

with $q_\sigma \in \mathbb{Z}$. Then the determinant factors as

$$\det E = p^\tau \sum_\sigma \operatorname{sgn}(\sigma) q_\sigma, \tag{78}$$

and, for every $k \geq 0$,

$$p^{\tau+k} \mid \det E \iff p^k \mid \sum_\sigma \operatorname{sgn}(\sigma) q_\sigma. \tag{79}$$

If p is prime and S is the tied minimum fiber, while the quotients outside S are divisible by p, the first extra factor is characterized exactly by the initial form

$$p^{\tau+1} \mid \det E \iff \sum_{\sigma \in S} \operatorname{sgn}(\sigma) q_\sigma = 0 \quad \text{in } \mathbb{Z}/p\mathbb{Z}. \tag{80}$$

Thus higher-cost terms disappear modulo p and all first-layer cancellation is concentrated in the tied fiber.

Status: [kernel-verified Lean] — `TropicalInitialForm` proves the factorization and all-depth divisibility criterion as `det_eq_pow_mul_signedLeibnizQuotientSum` and `det_pow_add_dvd_iff_signedLeibnizQuotientSum_dvd`; its prime first-layer forms are `det_pow_succ_dvd_iff_signedLeibnizQuotientSum_eq_zero`, `signedLeibnizQuotientSum_mod_eq_tiedFiberSum`, and `det_pow_succ_dvd_iff_tiedFiberSum_eq_zero`.

This module is a purely algebraic prerequisite and diagnostic. It proves neither that the tied sum vanishes nor that it has a large higher p-adic valuation; in particular it supplies no cancellation lower bound and does not close Conjecture 1.1.

The abstract block factorization needed for the report's first-fiber calculation is also now kernel-verified. If a block map b describes the surviving permutations and A_k is the square submatrix on the fiber $b^{-1}(k)$, then

$$\sum_{\sigma \text{ preserving } b} \operatorname{sgn}(\sigma) \prod_i A_{\sigma(i),i} = \prod_k \det A_k. \tag{81}$$

The formula allows varying block sizes and a fixed preliminary row permutation, whose sign is the only extra factor. When, after reindexing, each A_k consists of consecutive powers of nodes $x_{k,0}, \ldots, x_{k,s_k-1}$, it becomes the exact Vandermonde product

$$\prod_k \prod_{0 \le i < j < s_k} (x_{k,j} - x_{k,i}). \tag{82}$$

Composing (81) with the tropical quotient criterion gives an exact first-extra-factor test: under the explicit hypotheses that every quotient outside the block-preserving set vanishes modulo p and that each block-preserving quotient equals the corresponding full Leibniz product of A, $p^{\tau+1} \mid \det E$ if and only if $\prod_k \det A_k$ vanishes modulo p; in the consecutive-power specialization the right side is precisely (82) modulo p.

Status: [kernel-verified Lean] — `BlockFiberFactorization` proves `sum_blockPreserving_eq_prod_blockDet`, `sum_basePerm_mul_blockPreserving_eq_sign_mul_prod_blockDet`, `sum_blockPreserving_eq_prod_vandermonde_of_reindex`, and `sum_blockPreserving_consecutivePowers_eq_prod_vandermonde`. `TropicalBlockApplication` packages the composition as `det_pow_succ_dvd_iff_prod_blockDet_eq_zero` and `det_pow_succ_dvd_iff_prod_vandermonde_eq_zero`.

The rank-one Beatty scaffolding is now kernel-verified separately. For matrices

$$E_{ij} = p^{n_i c(b_j)} A_{ij}, \qquad n_i = T - \lfloor i\beta \rfloor,$$

with strictly decreasing Beatty depths, sorted block labels b_j, and strictly increasing block values c, `BeattyTropicalFiber` proves that the minimum assignments are exactly the block-preserving permutations. Every other assignment gains at least $\lfloor \beta \rfloor$, its normalized quotient is divisible by $p^{\lfloor \beta \rfloor}$, and the determinant has the exact expansion

$$\det E = p^{\tau} \left(\prod_k \det A_k + p^{\lfloor \beta \rfloor} Z \right).$$

The headline declarations are `rankOneAssignmentCost_eq_iff_blockPreserving`, `beattyAssignmentCost_add_floor_le_of_not_blockPreserving`, `beattyPower_pow_floor_dvd_quotient_of_not_blockPreserving`, and `det_beattyPowerMatrix_eq_pow_mul_blockDet_add_pow_floor_mul`. Thus the minimizing fiber, the tropical gap, and the full-product quotient identity no longer remain paper-only hypotheses. The generic scaled consecutive-power wiring is now formalized below, and `GlobalMixedPrefixCascade` closes the formerly paper-level finite specialization: its sorted box prefix is the genuine global first-N mixed-semigroup prefix, its coordinate fibers are lower-closed, and it constructs the lexicographic and colexicographic fiber equivalences needed for the dyadic and triadic determinant identities. What remains globally is the valuation and cancellation of the residual low-excess sums. No leading-order upper bound follows from the exact expansion.

The complete assignment cascade can now be grouped more economically than the raw Leibniz sum. A *row-block assignment* records only which row set is sent to each repeated column weight, with the prescribed fiber cardinalities. For the rank-one power matrix, the whole Leibniz fiber over such an assignment a factors as one sign times a product m_a of square unit minors. If $c(a)$ is its structural assignment cost and τ is the minimum, then

$$\det E = p^{\tau} \sum_a p^{c(a)-\tau} m_a. \tag{83}$$

More sharply, for every cutoff K this is the exact finite-depth decomposition

$$\det E = p^{\tau} \left(\sum_{c(a)-\tau<K} p^{c(a)-\tau} m_a + p^K \sum_{c(a)-\tau \geq K} p^{c(a)-\tau-K} m_a \right). \tag{84}$$

Thus modulo p^K every high-excess row partition disappears, and the surviving coefficients are explicit products of unit minors rather than sums over individual permutations. In the Beatty specialization, every noncanonical row partition has excess at least $\lfloor \beta \rfloor$.

Status: [kernel-verified Lean] — `RowBlockLaplaceExpansion` proves the fiber factorization `sum_permutationRowBlockAssignmentFiber_eq_sign_mul_prod_blockDet`, the normalized all-layer identity `det_rankOnePowerMatrix_eq_minPow_mul_sum_assignmentExcess`, the exact cutoff formula `det_rankOnePowerMatrix_eq_minPow_mul_truncatedAssignmentSum`, and the Beatty gap `beattyRowBlockAssignment_excess_ge_floor`.

This is an exact support theorem for every finite cascade depth. It gives no upper bound for the number of surviving row partitions by itself, no upper bound for the valuations of the residual minor-product quotients, and no noncancellation statement for their signed sum. It therefore reorganizes the remaining wall but does not bound $v_p(\det E) - \tau$.

There is, however, now an exact entropy bound for the first of those quantities. For a row partition a, let $P_k(a)$ be the excess of the total block value assigned to the first $k+1$ rows over the oppositely sorted assignment, and define its crossing energy by

$$\mathcal{E}(a) = \sum_{k=0}^{N-1} P_k(a).$$

Every $P_k(a)$ is nonnegative, the vector $(P_k(a))_k$ determines a when the block values are distinct, and summation by parts identifies $\mathcal{E}(a)$ with the excess for the unit depth staircase. For Beatty depths,

$$c(a) - \tau \geq \lfloor \beta \rfloor \, \mathcal{E}(a). \tag{85}$$

Consequently, for $N, K \geq 1$, if $q = \lfloor (K-1)/\lfloor \beta \rfloor \rfloor$, stars and bars gives the explicit finite support upper bound

$$\#\{a : c(a) - \tau < K\} \leq \binom{N+q-1}{q}. \tag{86}$$

For fixed normalized depth q this is polynomial in N; for growing q it is an explicit entropy bound, not a cancellation estimate.

Status: [kernel-verified Lean] — `AssignmentExcessGeometry` proves the exact identity `rowBlockCrossingEnergy_eq_reverseFinRank_excess`, the Beatty lower bound `beattyFloor_mul_rowBlockCrossingEnergy_le_excess`, the stars-and-bars injection `card_rowBlockAssignment_crossingEnergy_lt_choose_of_pos`, and the specialized bound `card_beattyRowBlockAssignment_excess_lt_of_pos`.

This controls how many row partitions can occur in the cutoff sum. It supplies no upper bound for their minor-product valuations and no noncancellation theorem for their signed sum, so (86) is cascade-support control rather than a proof of the conjecture.

The fixed-divisor contribution can be threaded through this entire expansion, rather than applied only to the canonical block. For a block-scaled consecutive-power unit matrix, let s_k be the fixed size of block k, define

$$U_2(s) = \sum_{0 \leq j < s} (j + v_2(j!)), \qquad U_3(s) = \sum_{0 \leq j < s} \left(\left\lfloor \frac{j}{2} \right\rfloor + v_3\left(\left\lfloor \frac{j}{2} \right\rfloor ! \right) \right),$$

and put

$$\mathcal{U}_2 = \sum_k U_2(s_k), \qquad \mathcal{U}_3 = \sum_k U_3(s_k).$$

Every row-block assignment has the same column-fiber sizes, so its minor product is divisible by $2^{\mathcal{U}_2}$, respectively $3^{\mathcal{U}_3}$. Factoring this common divisor from the exact row-partition expansion gives

$$2^{\tau_2+\mathcal{U}_2} \mid \det E_2, \qquad 3^{\tau_3+\mathcal{U}_3} \mid \det E_3. \tag{87}$$

The row scale is allowed to depend on both the target block and the row; this includes the mixed factors M^{ku} in the dyadic blocks and A^{kv} in the triadic blocks. Thus the gain is genuinely uniform across every tropical layer, not a statement only about the first initial form.

Status: [kernel-verified Lean] — `RowBlockUnitDivisibility` defines `fiberScaledConsecutivePowerMatrix`, proves the common-minor divisors `two_pow_rowBlockOrderingExponent_dvd_minorProduct_fiberScaledConsecutivePowers` and `three_pow_rowBlockOrderingExponent_dvd_minorProduct_fiberScaledConsecutivePowers`, and packages (87) as `two_pow_minCost_add_rowBlockOrderingExponent_dvd_det_rankOnePowerMatrix` and `three_pow_minCost_add_rowBlockOrderingExponent_dvd_det_rankOnePowerMatrix`.

The exponent is exactly the sum of the block gains from `UnitOrderingGain` and is independent of assignment cost. Consequently it factors every cutoff sum in (84), but it cannot distinguish layers or upper-bound the valuation of the residual signed sum.

There is also a cascade-independent fixed divisor of the unit determinants. A monic triangular change of columns from $1, X, X^2, \ldots$ to an integer polynomial basis preserves the full evaluation determinant. On positive odd nodes with consecutive local degrees, the dyadic p-ordering basis

$$\prod_{0\le t<j} (X-(2t+1))$$

extracts the column weight

$$j + v_2(j!),$$

while on positive nodes prime to three with consecutive local degrees, the ordering $1, 2, 4, 5, 7, 8, \ldots$ extracts

$$\left\lfloor \frac{j}{2} \right\rfloor + v_3\left(\left\lfloor \frac{j}{2} \right\rfloor !\right).$$

The resulting prime powers divide the *whole* univariate Vandermonde determinant, independently of which tropical layer survives. At the univariate block level this extracts a universal lower-order divisor rather than estimating successive initial forms.

Status: [kernel-verified Lean] — `AllLayerUnitBasis` proves the generic column-content and determinant-one column-change lemmas `prod_columnDivisor_dvd_det`, `pow_sum_columnWeight_dvd_det_of_unimodular_mul`, and `pow_sum_dvd_det_vandermonde_of_monicBasis`. Its strengthened unit specializations are `two_pow_sum_degree_add_factorialVal_dvd_det_vandermonde` and `three_pow_sum_halfDegree_add_factorialVal_dvd_det_vandermonde`.

The companion module `UnitOrderingGain` proves exact digit-sum identities for the block weights and the finite aggregate bounds

$$\sum_k U_2(s_k) \le S^3, \qquad 2\sum_k U_3(s_k) \le S^3$$

whenever there are at most S blocks and every block has size at most S (`sum_twoUnitBlockGain_le_cube` and `two_mul_sum_threeUnitBlockGain_le_cube`). For the report's $O(L)$ fibers of size $O(L)$, the remaining paper-level block count therefore gives $O(L^3)$, equivalently $O(N^{3/2})$, lower order than $H_T \asymp T^{5/2}$. In fact, for the triangular cutoff $u + \vartheta v \le L$ and $N \sim L^2/(2\vartheta)$, elementary summation gives the paper-level leading terms for the total gains

$$\mathcal{U}_2 \sim \frac{2^{3/2}}{3\sqrt{\vartheta}} N^{3/2} = 0.7488834673\ldots N^{3/2}, \qquad \mathcal{U}_3 \sim \sqrt{\frac{\vartheta}{8}} N^{3/2} = 0.4451070799\ldots N^{3/2}.$$

The finite divisors and cubic aggregation are kernel-verified; the triangular fiber asymptotics and displayed constants are not Lean theorems. The determinant-one fiberwise change and its additive combination with the tropical minimum are kernel-verified for the generic scaled block model by `RowBlockUnitDivisibility`; the concrete triangular index bridge, including column permutation and transfer back to the raw mixed integer matrix, is kernel-verified under explicit finite fiber certificates in `MixedDeterminantUnitBridge`. The module `GlobalMixedPrefixCascade` now constructs those certificates for the genuine global first-N mixed-semigroup enumeration and proves the exact all-layer and cutoff-truncated dyadic and triadic formulas directly. The resulting all-layer gain neither supplies nor upper-bounds the remaining leading-order cancellation excess.

Consequently the missing fifth, or more, can arise only at primes where the assignment problem ties, and the tie geometry is explicit: for $p \ne 2, 3$ columns tie along the rational lines orthogonal to the external valuation vector, and in the cross-coprime branch the 2-adic cost depends only on u and the 3-adic cost only on v, so every vertical or horizontal column fiber creates equal-cost permutations. The next calculation evaluates the first such tied layer. The kernel-verified finite cube budget, combined with the elementary paper-level count of the Beatty fiber sizes, gives $O(T^2)$ against $H_T \asymp T^{5/2}$, hence $o(1)$ of the budget; the sharper quoted two-logarithm estimate improves this to $O\big(T^{3/2}(\log T)^2\big)$.

20.6 What this leaves

Three statements survive every normalization.

First, term-wise amplification is retired unconditionally against the crude entry bound. Corollary 20.2 is a structural identity, not a numerical near-miss, and it holds for every prime set, every node system and every column set.

Second, the quantitative ceilings above say how much of the archimedean budget automatic divisibility can supply in each concrete construction, and none reaches 1. Adding every prime in the fixed support does not close any determinant; it improves the diagnosis from a heuristic "about one half" to a proved 4/5.

Third — and this is the redirection — a successful p-adic proof has to be a *cancellation* theorem, not a divisibility-by-every-term theorem. Proposition 20.3 says exactly where such a theorem could live: inside the equal-cost fibers, and it would have to repeat through many tropical layers or be combined with a sharper archimedean free energy than the crude bar. Quantifying the gain of the refined divided-difference bounds of Sections 18 and 19 over the crude bar is the concrete companion question, and it concerns the archimedean side alone.

Status: [paper proof] for the reading of the obstruction; **[kernel-verified Lean]** for Theorem 20.1, Corollary 20.2 and Proposition 20.3. The structural observation that the ceiling and the shrinking-target obstruction are governed by the same block equidistribution is recorded as an interpretive claim, not a theorem.

---

21 Why the new hierarchy still stops short

The semigroup determinant is much stronger locally than the ordinary-power determinant: its repulsion exponent approaches 1. It is therefore important to identify exactly why it does not prove the conjecture.

21.1 How many nodes are needed to approach scale X?

The asymptotic exponent in (70) is

$$\delta_r := 1 - \frac{\Sigma_r}{R_r} = 1 - \frac{c_\vartheta}{\sqrt{r}} + O(r^{-1}), \qquad c_\vartheta = \frac{4}{3}\sqrt{2\vartheta} = 2.3739044262\ldots.$$

For nodes near X, the lower span has the shape

$$X^{\delta_r} = X \exp\left(-\frac{c_\vartheta \log X}{\sqrt{r}} + O\left(\frac{\log X}{r}\right)\right). \tag{88}$$

To make the exponential loss in (88) bounded independently of X, one needs

$$\sqrt{r} \asymp \log X, \qquad \text{that is,} \qquad r \asymp (\log X)^2. \tag{89}$$

This is precisely why Corollary 19.8 has a logarithmic square.

21.2 How many nodes a counterexample supplies

Conditional on a counterexample, the exact dyadic block count established in Section 4 gives

$$\#(\mathcal{C} \cap [X, 2X]) = \frac{\log_2 X}{\beta} + O(1) \asymp \log X. \tag{90}$$

Thus the actual conditional population is smaller than the determinant threshold (89) by one factor of $\log X$.

Taking $r \asymp \log X$ in (88) yields only

$$X \exp(-C\sqrt{\log X}),$$

which is much smaller than the average spacing $X/\log X$ among the $\asymp \log X$ conditional candidates. The inequality is therefore compatible with the exact monoid even at the level of expected spacing.

21.3 Broader multiplicative intervals do not help

The cumulative conditional count below X is $\asymp (\log X)^2$, so one might try to use all those points at once. But their nodes range from 1 to X, and the gap theorem depends on the smallest node. Restricting to $[X^\gamma, X]$ with fixed $0 < \gamma < 1$ does retain $\asymp (\log X)^2$ points, but its additive length is $\asymp X$, while the lower scale is at most $X^{\gamma+o(1)}$. Again there is no contradiction.

This is not a mere defect of coarse constants. It reflects a mismatch between additive node geometry and the logarithmic triangular lattice of exponents (n, k) in $m = 2^n M^k$.

21.4 The determinant needs one more source of size

The archimedean argument uses only

$$D \in \mathbb{Z}_{>0} \quad \Longrightarrow \quad D \geq 1.$$

If the exact monoid coordinates forced

$$D \equiv 0 \pmod{Q} \qquad \text{with} \quad \log Q \gg R \log X/\sqrt{r},$$

then the lower bound $D \geq Q$ could remove the one-logarithm deficit. Alternatively, one could seek a determinant with total exponent weight $\Sigma = O(r)$ rather than $\Sigma \asymp r^{3/2}$, but the two-dimensional semigroup counting law $L(T) \asymp T^2$ makes that impossible using only distinct nonnegative mixed monomials.

This gives a concrete design specification:

> A successful determinant proof must add nonarchimedean divisibility, introduce lower-weight functions outside the mixed-monomial semigroup, or exploit the exact two-dimensional row grid so efficiently that $O(\log X)$ nodes suffice.

Status: [paper proof]

The deficit analysis of this section is a paper argument about the limits of the method, not a formalized theorem. It does not weaken any statement proved above; it explains why the explicit $O((\log X)^2)$ bound of Corollary 19.8 cannot be pushed to the conditional $\asymp \log X$ population by tuning constants alone. The conjecture remains open.

22 Further equivalent reformulations

Several equivalent forms are already formalized in the repository and were catalogued in Section 3. The exact conditional structure of Section 4 also yields useful paper-level reformulations that expose different possible proof technologies. Each restatement below is an exact equivalence, not a heuristic analogy: a proof of any one of them is a proof of Conjecture 1.1.

22.1 Integer points on a transcendental lattice curve

The candidate formulation of Section 1 says that

$$\{(m,n) \in \mathbb{N}^2 : n = m^{\vartheta}\} = \{(2^k, 3^k) : k \in \mathbb{N}_0\}. \tag{91}$$

The curve $n = m^{\vartheta}$ is strictly convex and transcendental, and (91) asserts that its only lattice points are the trivial exponential ones. The arbitrary-node repulsion theorem of Section 18 can be read as a lower bound for the volumes of lattice simplices cut from this curve; the deficit analysis of Section 21 measures exactly how far the current bound is from excluding the nontrivial points.

22.2 Vanishing determinant of four logarithms

Writing $M = 2^x$ and $A = 3^x$, a solution is a positive integer pair with

$$\log M \, \log 3 - \log A \, \log 2 = 0, \quad M, A \in \mathbb{N}, \quad \Longrightarrow \quad (M, A) = (2^k, 3^k). \tag{92}$$

The hypothesis of (92) is a vanishing 2×2 determinant of logarithms of positive integers. This is the most direct bridge to the four-exponentials circle of ideas and to algebraic independence of logarithms, and it is the form in which the kernel-verified four-exponentials reduction is stated.

22.3 Primitive localized-radical exclusion

The kernel-verified radical form of Section 5 is

$$\forall\, w > 1 \text{ odd and power-primitive}, \quad \forall\, d > 0,\ a \geq 0, \qquad 3^a \left(w^{\vartheta}\right)^d \notin \mathbb{N}. \tag{93}$$

If (93) fails with least rational-power index d, then $X^d - c/3^a$ is the exact minimal polynomial of w^{ϑ} over $\mathbb{Q}$ for the corresponding rational c.

This formulation is well suited to Kummer-theoretic, norm, and ramification investigations, because all redundant proper powers have been removed by primitivity and the algebraic degree of the hypothetical radical is known exactly rather than merely bounded.

22.4 Exact growth dichotomy

Write

$$B(T) := \#\left(\mathcal{C} \cap [2^T, 2^{T+1})\right)$$

for the Tth dyadic block count. The repository formalizes the cumulative subquadratic equivalence of Section 4; the exact block formula there, namely

$$B(T) = 1 + \left\lfloor \frac{T+1}{\beta} \right\rfloor \qquad \text{under failure with least generator } \beta, \tag{94}$$

gives several immediate variants.

Proposition 22.1 (Block-growth reformulations)**.** *Conjecture 1.1 is equivalent to each of the following:*

1. *the dyadic block counts $B(T)$ are bounded;*
2. *$B(T) = o(T)$;*
3. *$\liminf_{T\to\infty} B(T)/T = 0$;*
4. *$B(T) = 1$ for every $T \in \mathbb{N}_0$.*

Proof. If the conjecture holds, every block contains only 2^T, so all four properties hold. If it fails, (94) gives $B(T) = 1 + \lfloor (T+1)/\beta \rfloor$, so $B(T)/T \to 1/\beta > 0$ and none of the first three properties holds. The fourth is visibly equivalent to the candidate statement. □

This is useful because a future analytic theorem need not classify candidates individually. Any uniform $o(T)$ bound for the number of candidates in the Tth dyadic block would prove the conjecture.

The exactness of (94) permits a sharpening of item (4) that is, at first sight, dramatically weaker than the conjecture and yet still equivalent to it.

Proposition 22.2 (Infinitely many singleton blocks suffice)**.** *Conjecture 1.1 is equivalent to the statement*

$$B(T) = 1 \quad \text{for infinitely many } T \in \mathbb{N}_0.$$

Proof. If the conjecture holds then $B(T) = 1$ for every T, so certainly for infinitely many. Conversely, suppose the conjecture fails, and let $\beta > 0$ be the irrational least generator of the exact solution monoid. By (94),

$$B(T) = 1 \iff \left\lfloor \frac{T+1}{\beta} \right\rfloor = 0 \iff T + 1 < \beta,$$

which holds for at most the finitely many indices $T < \beta - 1$. Hence failure forces $B(T) = 1$ for only finitely many T, and the contrapositive gives the equivalence. □

Remark 22.3. Proposition 22.2 cannot be weakened further to "$B(T) = 1$ for some T". That statement is already known unconditionally and carries no information: the kernel-verified zero-free region $2^x < 4096$ gives $\beta \geq 12$, hence $B(T) = 1$ for $T = 0, \ldots, 10$, and the interval computation of Section 23 pushes the same conclusion to $T \leq 19$ at the weaker **[software computation]** trust level. The gap between "for some T" and "for infinitely many T" is exactly the gap between a finite check and the conjecture, because none of the present results bounds β from above: a counterexample with a very large least generator would have $B(T) = 1$ for every $T < \beta - 1$, that is, on any range a fixed finite computation could ever reach. Only the infinitude of singleton blocks is decisive.

Status: [kernel-verified Lean] for Proposition 22.1: `BlockGrowthReformulations` proves the bounded, little-o, liminf-zero, and identically one equivalences. [paper proof] for Proposition 22.2, whose short paper proof rests on the same kernel-verified exact dyadic block formula.

22.5 Compact synchronized S-integral orbit

The S-integral orbit of Section 4 yields another equivalence, this time one that isolates the synchronization of the two denominators.

Proposition 22.4 (Compact-orbit reformulation). *Conjecture 1.1 fails if and only if there exist integers $M, A > 1$ and an irrational $\beta > 0$ such that, for every $k \geq 1$, the point*

$$\left(\frac{M^k}{2^{\lfloor k\beta \rfloor}}, \frac{A^k}{3^{\lfloor k\beta \rfloor}}\right) \tag{95}$$

lies on the compact arc

$$\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}.$$

Under these conditions the orbit is dense in $\mathcal{A}$.

Proof. Failure supplies the least generator β, with $M = 2^\beta$ and $A = 3^\beta$; then (95) is exactly $(2^{\{k\beta\}}, 3^{\{k\beta\}})$, and density follows from irrational rotation.

Conversely, arc membership gives $t_k \in [0, 1)$ such that

$$k \log_2 M - \lfloor k\beta \rfloor = t_k = k \log_3 A - \lfloor k\beta \rfloor.$$

Thus $\log_2 M = \log_3 A =: \gamma$, and $0 \leq k\gamma - \lfloor k\beta \rfloor < 1$ for every k. Hence $\lfloor k\gamma \rfloor = \lfloor k\beta \rfloor$, so $|k(\gamma - \beta)| < 1$ for every k; therefore $\gamma = \beta$. Consequently $M = 2^\beta$ and $A = 3^\beta$ are integers while β is nonintegral, so the conjecture fails. The same identity now shows that the orbit equals the irrational-rotation orbit and is dense. □

Status: [kernel-verified Lean] — `not_alaogluErdosConjecture_iff_exists_compactOrbit` and `alaogluErdosConjecture_iff_no_compactOrbit` in `CompactOrbit`.

This formulation isolates the synchronized-denominator feature that a specialized S-integral counting theorem would need to exploit: the two coordinates are S-integral for disjoint sets of primes, yet their denominator exponents are forced to agree exactly at every height.

22.6 Two-dimensional exponential grid

Under failure, every solution coordinate (n, k) gives

$$m_{n,k} = 2^n M^k, \qquad m^\vartheta_{n,k} = 3^n A^k,$$

and for every mixed exponent $a + b\vartheta$ with $a, b \in \mathbb{N}_0$,

$$m_{n,k}^{a+b\vartheta} = (2^a 3^b)^n (M^a A^b)^k. \tag{96}$$

Thus the candidate evaluation matrix factors through a two-dimensional row lattice and a two-dimensional column lattice. The conjecture can be viewed as the assertion that this integer exponential grid cannot exist unless (M, A) is a common integral power of $(2, 3)$. This is the most natural starting point for a bespoke 2×2 interpolation determinant that uses more than the arbitrary-node geometry of Section 18, and it is the structure the semigroup hierarchy of Section 19 exploits only partially.

22.7 An operator formulation on the prime-logarithm space

All the reformulations above are Diophantine. There is also a purely linear-algebraic one, which recasts the problem as a statement about multiplication by ϑ acting on a single $\mathbb{Q}$-vector space.

Let

$$W := \operatorname{span}_{\mathbb{Q}}\{\log p \;:\; p \text{ prime}\} \subset \mathbb{R},$$

a $\mathbb{Q}$-vector space of countably infinite dimension, with the $\log p$ as a basis by unique factorization. Multiplication by ϑ does not preserve W, so consider the largest part of W that it does return to W:

$$U := \{\, w \in W \;:\; \vartheta\, w \in W \,\}.$$

This is a $\mathbb{Q}$-subspace, and it is never zero: $\vartheta \log 2 = \log 3 \in W$, so $\mathbb{Q} \log 2 \subseteq U$.

Proposition 22.5 (Operator form)**.** *$\dim_{\mathbb{Q}} U = \operatorname{rank} K$, where $K = \{v \in \mathbb{Q}_{>0} : v^{\vartheta} \in \mathbb{Q}\}$ is the multiplicative group of Section 11. Consequently the rational-output form (24) — and hence Conjecture 1.1 — is equivalent to*

$$\boxed{\dim_{\mathbb{Q}}\{\, w \in W : \vartheta\, w \in W \,\} = 1.}$$

Proof. An element of W is $w = \sum_p c_p \log p$ with rational c_p, almost all zero. Clearing a common denominator N gives $Nw = \log v$ for a positive rational v. Now $\vartheta w \in W$ says $\vartheta \log v = \sum_p d_p \log p$ with rational d_p; clearing a denominator M turns this into $\vartheta \log(v^M) = \log u$ with $u \in \mathbb{Q}_{>0}$, that is $\left(v^M\right)^{\vartheta} = u$, i.e. $v^M \in K$. So U is exactly the $\mathbb{Q}$-span of $\log K$ inside W, whence $\dim_{\mathbb{Q}} U = \operatorname{rank} K$. The conjecture in its rational-output form says $K = \langle 2 \rangle$, of rank one. □

Status: [paper proof] — an elementary translation, but a genuinely different framing: it replaces a Diophantine assertion by the statement that the operator "multiply by ϑ" carries only a single line of W back into W.

Two remarks on what this framing does and does not give.

It explains why finite-dimensional arguments cannot apply directly. If some nonzero finite-dimensional subspace $V \subseteq W$ were *ϑ-stable*, then ϑ would be an eigenvalue of the rational matrix representing multiplication by ϑ on V, hence algebraic, contradicting transcendence. So a proof cannot proceed by exhibiting an invariant finite-dimensional subspace: none exists, unconditionally. The subspace

U above is not ϑ-stable — $\vartheta U \subseteq W$ but generally $\vartheta U \not\subseteq U$ — and this failure of stability is precisely what blocks the eigenvalue argument.

The natural stable candidate is empty for the same reason. Setting $U_\infty = \{w \in W : \vartheta^k w \in W \text{ for all } k \geq 0\}$ does produce a ϑ-stable space, but it is either zero or infinite-dimensional, and infinite dimension is automatic whenever it is nonzero: if $0 \neq w \in U_\infty$, the powers $\vartheta^k w$ are $\mathbb{Q}$-linearly independent, since a relation $\sum_k c_k \vartheta^k w = 0$ with $w \neq 0$ would make ϑ algebraic. So the operator picture reproduces the transcendence barrier exactly, rather than circumventing it. Its value is diagnostic: it shows the obstruction is not a missing estimate but the absence of any finite-dimensional invariant structure to work with.

23 Independent finite verification

Exactly one zero-free region in this report is a kernel certificate. Lean proves that a solution with $2^x < 4096$ forces $x \in \mathbb{Z}$, so that a nonintegral solution satisfies $x \geq 12$; the statement lives in `FiniteCheck4096` as

`twelve_le_of_not_integer_of_two_three_rpow_integer`

and it is the only finite exclusion that any other statement here may use without adding trust assumptions. **[kernel-verified Lean]**

One further finite computation pushes the excluded range far past 4096: a directed-rounding exclusion through 2^{27}, described in Section 34. It supersedes the earlier interval scans, which are not restated here. It is a software result: it does not replay in the Lean kernel, it is not interchangeable with `FiniteCheck4096`, and it does not raise the kernel-verified exponent bound above $x \geq 12$. They are recorded here because they strengthen the explicit conditional picture, test the conditional models of Section 4 against data, and supply the extremal cases that a future generated certificate must clear.

23.1 What each scan must certify

Write $\vartheta = \log_2 3$, and let m be a positive integer that is not a power of two. Excluding m means proving $m^\vartheta \notin \mathbb{N}$, and it suffices to exhibit an integer n with $n < m^\vartheta < n+1$. Exponentiation is never performed: the pair of strict inequalities is equivalent, after taking logarithms and clearing the positive factor $\log 2$, to

$$\log m \, \log 3 - \log n \, \log 2 > 0, \tag{97}$$

$$\log(n+1) \, \log 2 - \log m \, \log 3 > 0. \tag{98}$$

Every logarithm and every product occurring in the scanned ranges is positive, so the required rounding directions are monotone and unambiguous.

In both scans an ordinary floating-point evaluation merely *proposes* $n \approx \lfloor m^\vartheta \rfloor$. Once certified lower bounds for (97) and (98) are both strictly positive, the floating proposal has no remaining logical role: it is a witness generator, not a step of the argument. A locator error can therefore create a spurious failure report, but it cannot create a spurious exclusion. Powers of two are skipped rather than tested, since 2^q has the integral value $(2^q)^\vartheta = 3^q$; an exactly integral value at a non-power of two would satisfy neither strict inequality for any n and would be reported as a failure.

23.2 Superseded scans

Two earlier software exclusions — an interval-arithmetic scan through 2^{20} and a directed-rounding scan through 2^{26} — have been absorbed by the extension through 2^{27} of Section 34, which uses the same directed-rounding method, the same margin conventions and the same manifest discipline over the wider range. Only the strongest exclusion is stated in this report; the superseded ranges and their chunk tables are not reproduced.

23.3 The trust boundary, stated plainly

Kernel. $2^x < 16384$ and $2^x, 3^x \in \mathbb{Z}$ imply $x \in \mathbb{Z}$; a nonintegral solution has $x \geq 14$. **[kernel-verified Lean]** `FiniteCheck16384` (and, independently, $2^x < 8192$ and $2^x < 4096$ in `FiniteCheck8192`, `FiniteCheck4096`).
Interval scan. Under CPython and `mpmath` interval arithmetic, a nonintegral solution has $x > 20$. **[software computation]**
Directed-rounding scan. Under the C source, GCC code generation, OpenMP scheduling, the MPFR and GMP shared libraries, the x86-64 arithmetic environment, and the operating system, a nonintegral solution has $x > 27$. **[software computation]**

MPFR's correct-rounding contract makes the numerical reasoning crisp, and the interval scan is reproducible from a short script, but neither removes the software assumptions listed above. Nothing in Section 4 or Section 25 that is tagged **[kernel-verified Lean]** depends on either scan; wherever a numeric threshold is used inside a kernel-verified statement, that threshold is 12, 13, or 14.

The 8192 and 16384 extensions. The kernel region was raised from $2^x < 4096$ to $2^x < 8192$ and then to $2^x < 16384$ by two further unsplit tables (`FiniteCheck8192`, `FiniteCheck16384`, Section 3.2); the only new ingredient is the use of semiconvergent enclosures for the hardest values. An earlier chunked draft of the 16384 region had failed on the per-chunk length lemmas; the unsplit table avoids the problem entirely, and the 8191-row list literal and its length lemma are accepted with `maxRecDepth` 200000. The next doubling, $[16384, 32768)$, will need the 10^7-denominator convergent or its semiconvergents for a handful of rows; the $m = 9329$ timing shows this is minutes, not hours.

From scan to kernel certificate. The repository's own certificate architecture (Section 3) shows the stronger trust model: generate rational certificates externally, revalidate them independently, and replay only exact integer comparisons in Lean by `decide`. The next step is certificate extraction, not a larger blind scan. For each block one records rational lower and upper approximants to ϑ together with integer inequalities sufficient to replay every exclusion, exploiting monotonicity and convexity so that a single certificate covers a run of consecutive m rather than storing one record per input. A scalable formal certificate should share convergent enclosures across ranges, chunk the data modules to bound elaboration memory — the failure mode of the 16384 draft is precisely that the chunk bookkeeping, not the arithmetic, is the expensive part — and rely on verified binary powering rather than giant literals. The extremal pairs identified by both scans predict the required tier depth. A practical intermediate target is a generated, kernel-replayed certificate beyond 4096 (with the existing 16384 draft as an engineering test), followed by a hierarchical certificate that approaches the current software frontier 2^{27}.

24 Attempts at a full proof and their exact failure points

This section records the principal avenues that were pushed until they broke, together with the exact place where each one breaks. The purpose is not merely negative. Every failure point names a quantitative or structural feature that a successful proof must improve, and several of them turn out to be the same feature seen from different sides; the last subsection gives that common feature its exact name.

24.1 Collapsing rank two to rank one

The multiplicative-rank bound is kernel-verified: the exponent vectors of the candidate set span a subgroup of rank at most two. Under failure the rank-two possibility is not vague but known exactly, $\mathcal{C} = \langle 2, M \rangle$, with M the primitive generator of Section 4. Nothing in the rank machinery distinguishes this free rank-two monoid from the allowed monoid of powers of two.

Failure point. A rank bound is an upper bound on dimension; it supplies no mechanism for forcing the dimension down to one. Excluding the second generator *is* the original transcendence problem, restated.

24.2 Rational approximation to a least generator

Assume failure and let β be the least nonintegral solution, with $M = 2^\beta$ and $A = 3^\beta$. Let p/q approximate β and put $\varepsilon = q\beta - p$. Then

$$M^q - 2^p = 2^p\big(e^{\varepsilon \log 2} - 1\big), \tag{99}$$

$$A^q - 3^p = 3^p\big(e^{\varepsilon \log 3} - 1\big). \tag{100}$$

Both left-hand sides are integers, so a nonzero value has absolute value at least 1, which for small ε gives bounds of the shape

$$|\varepsilon| \gtrsim 2^{-p} \qquad \text{and} \qquad |\varepsilon| \gtrsim 3^{-p}.$$

These are exponentially tiny in q. Continued fractions only guarantee infinitely many approximants with $|\varepsilon| \ll q^{-1}$, and Baker-type lower bounds for ordinary linear forms in logarithms are polynomial in the relevant heights [3, 25]. Neither scale conflicts with a lower bound as weak as M^{-q} or A^{-q}. Multiplying (99) by (100), or eliminating ε between them, only recovers $\log_2 3$ to first order; the higher terms are far too small to create an integer contradiction.

Failure point. The two integer gaps are correlated rather than independent, so the two exponential inequalities carry essentially one piece of information, not two.

Observation 24.1. Any rational-approximation proof needs more than "a nonzero integer has absolute value at least one." It needs a divisibility lower bound growing exponentially with q, or an approximation to β exponentially better than continued fractions supply in general.

24.3 Equidistribution against exponentially shrinking targets

At paper level the orbit $\{k\beta\}$ is equidistributed modulo 1 (Section 4); the finite-Fourier convergence needed for Weyl's criterion, covering every fixed mode and every finite trigonometric polynomial, is kernel-verified. Meanwhile, integrality of 2^{n+t} between adjacent integral powers pushes the fractional part t away from both endpoints by an amount comparable with 2^{-n}, and the base-3 condition imposes a comparable 3^{-n} restriction; at height k the admissible target width is of exponential scale M^{-k}, a bound also kernel-verified in the shrinking-target module.

Equidistribution controls visits to fixed intervals and, with effective discrepancy, to polynomially shrinking ones along subsequences. It does not force visits to a summable family of targets of length e^{-ck}: the Borel–Cantelli heuristic predicts only finitely many such visits for a typical rotation, and numbers with bounded partial quotients show that equidistributed multiples can avoid exponentially small targets forever. The exact conditional monoid is perfectly compatible with that behaviour.

Failure point. The blockwise Weyl theorems are structurally correct but point the wrong way: they describe how already existing solutions distribute inside each block, not how likely arbitrary exponents are to hit the integer lattice. Closing the route needs Diophantine information specific to the transcendental β with $2^\beta, 3^\beta \in \mathbb{N}$, not generic metric theory.

24.4 Finite differences and additive progressions

For $f(t) = t^\vartheta$ the repository proves the exact mean-value formula for every forward difference order r,

$$\Delta_h^r f(n) = (\vartheta)_{\underline{r}}\, h^r \xi^{\vartheta - r} \qquad (n < \xi < n + rh),$$

so that if all $r+1$ values $f(n), f(n+h), \ldots, f(n+rh)$ are integers, the left side is an integer; under $|\,(\vartheta)_{\underline{r}}\,|h^r < n^{r-\vartheta}$ its absolute value lies strictly between 0 and 1. This is a sharp uniform obstruction to candidate arithmetic progressions, and its rational effective three-term shadow $h^{400} \le n^{83}$ is kernel-verified together with the arbitrary-order form.

The difficulty is combinatorial rather than analytic. Conditional failure produces only $O(\log X)$ candidates per multiplicative block, far below the positive-density regime of Roth-type theorems, and a rank-two multiplicative monoid need not contain long additive progressions at all: many points in a short interval need not be equally spaced. A three-term progression of candidates would solve the S-unit equation

$$2^{n_1} M^{k_1} + 2^{n_3} M^{k_3} = 2^{n_2+1} M^{k_2},$$

and general S-unit theory yields finiteness statements, not the required nonexistence.

Failure point. Sparse exact structure and sharp local curvature are mutually consistent. The arbitrary-node determinant of Section 18 removes the equal-spacing hypothesis, but its packing exponent is still too weak to contradict the exact conditional count.

24.5 Radical fields and conjugates

The normalized radical $E = w^\vartheta$ is algebraic with exact binomial minimal polynomial $X^d - c/3^a$, and its conjugates are $\zeta_d^j E$; degree, norm, and denominator support are all kernel-verified (Section 5). What this does not supply is a way to transport the real analytic identity $\log E = \vartheta \log w$ to the other conjugates.

Failure point. The chosen positive real logarithm is not Galois-equivariant: the remaining roots differ by complex logarithmic periods, and taking norms merely reproduces $E^d = c/3^a$. Ramification is equally inconclusive, since the equation relates the constrained primes through a quotient of logarithms rather than through a field embedding; standard Kummer theory sees the radical equation but not the special value $\vartheta = \log_2 3$.

Observation 24.2. A radical-field proof needs a genuinely mixed invariant: something simultaneously sensitive to the algebraic conjugates of E and to the logarithmic relation among $2, 3, w, E$. At present the algebraic and the real-analytic arguments meet only at the four-logarithm determinant.

24.6 The compact S-integral orbit

From the least generator define

$$U_k := \frac{M^k}{2^{\lfloor k\beta \rfloor}} = 2^{\{k\beta\}} \in \mathbb{Z}[1/2], \qquad V_k := \frac{A^k}{3^{\lfloor k\beta \rfloor}} = 3^{\{k\beta\}} \in \mathbb{Z}[1/3].$$

The points (U_k, V_k) lie on the compact analytic arc $\mathcal{A} = \{(2^t, 3^t) : 0 \leq t < 1\}$ and are dense there because β is irrational. This is the compact-orbit reformulation of Section 22, and it invites o-minimal and S-integral point counting; Butler relates special cases of Wilkie-type counting to real three- and four-exponentials statements [18].

Failure point. The counting is not strong enough by an enormous margin. The logarithmic heights of the orbit points grow linearly in k, so the arc carries only about $\log H$ orbit points of height at most H, far below the subpolynomial count H^ε that Pila–Wilkie style theorems already tolerate outside algebraic parts. What would suffice is a denominator-sensitive refinement counting only points whose coordinate denominators are supported exactly on $\{2\}$ and on $\{3\}$ with synchronized exponents; no theorem of that strength appears in the inspected corpus.

24.7 Forcing an auxiliary base

If one could exhibit an integer $b > 1$ multiplicatively independent of 2 and 3 with b^x algebraic — or rational with suitably controlled denominator — then the six exponentials theorem, or the repository's rational-third-base theorem, would force $x \in \mathbb{Q}$ and hence $x \in \mathbb{Z}$. The five exponentials theorem [25, 23] similarly makes $e^{\gamma x}$ transcendental for every nonzero algebraic γ, given the four algebraic corner exponentials, and the kernel-verified 3-smooth classification is the formalized integer-level shadow of the six exponentials theorem.

Failure point. The hypotheses provide no such b. The natural auxiliary quantities $p^x = e^{x \log p}$ have transcendental coefficient $\log p$, so the five exponentials theorem does not convert a divisor of M or A into a contradiction; and the six exponentials theorem controls b^x, not the distinct question of whether b divides the integer 2^x. The obvious constructions $b = 2^u 3^v$, and any construction from M and A, collapse to the same rank-two logarithmic span. An auxiliary base must come from an arithmetic operation whose exponential behaviour stays controlled while its logarithm leaves that span — an algebraic independence problem in disguise.

24.8 Exploiting the primitive output pair

The primitive pair (M, A) is canonical and rigid: kernel-verified theorems show that it admits no matched depth, no common perfect power, and no nontrivial affine decomposition, and that the odd core and its coordinates are unique. Every structural handle one might hope to grip has therefore been shown to be absent by design rather than by accident.

Failure point. After all of that normalization, what remains is exactly the single relation $\log M\ \log 3 = \log A\ \log 2$, a product of logarithms lying outside the scope of linear-forms machinery.

24.9 The linear-form-in-log-products formulation of the wall

The obstruction can be given one further exact name. A counterexample is a pair of integers $m, A \geq 2$ with $\log m\ \log 3 = \log A\ \log 2$. Expanding both integers over their prime factorizations $m = \prod_p p^{m_p}$ and $A = \prod_p p^{a_p}$, failure of Conjecture 1.1 is equivalent to the nontrivial vanishing of

$$D\ =\ \sum_p (m_p \log 3 - a_p \log 2) \log p, \tag{101}$$

an *integer-coefficient linear form in the products of two prime logarithms* $\{\log p \cdot \log 2,\ \log p \cdot \log 3\}$. Baker's theory bounds linear forms in logarithms with *algebraic* coefficients [3]; here every coefficient of $\log p$ is itself a transcendental logarithm, and no lower-bound technology exists for such forms. Indeed even the $\mathbb{Q}$-linear independence of the family $\{\log p \cdot \log q : p < q \text{ prime}\}$ is open — it follows from, and is essentially a fragment of, the four exponentials circle of conjectures. Section 11 turns exactly that independence statement into a sufficient condition for the conjecture.

Failure point. Every route examined above — determinants, product formulas in the radical field, auxiliary exponentials, equidistribution against shrinking targets, orbit counting — terminates at a special case of this one independence statement. A genuine proof therefore requires either a fundamentally new lower bound for linear forms in log-products, or a mechanism exploiting the positivity and integrality of the coefficients (m_p, a_p) in (101) that has no analogue in the general four-exponentials setting.

25 Consolidated research program

The two source programs overlapped substantially; the exact structure theory of Section 4 settles some directions outright and re-prioritizes the rest. The list below is ordered by how directly each item attacks the quantitative deficit of Section 21 — the one missing logarithm between what the determinant hierarchy delivers and what a contradiction requires.

25.1 Priority A: p-adic amplification of semigroup determinants

The semigroup determinant $D = \det\big(m_i^{a_j}(m_i^{\vartheta})^{b_j}\big)$ is a positive integer, and under the exact monoid every entry has explicit valuations obtained from $m_i = 2^{n_i} M^{k_i}$ and $m_i^{\vartheta} = 3^{n_i} A^{k_i}$:

$$\begin{aligned} v_2\Big(\mathrm{entry}_{ij}\Big) &= a_j\big(n_i + v_2(M)\,k_i\big) + b_j v_2(A)\,k_i, \\ v_3\Big(\mathrm{entry}_{ij}\Big) &= a_j v_3(M)\,k_i + b_j\big(n_i + v_3(A)\,k_i\big). \end{aligned}$$

What this would buy is decisive: archimedean reasoning offers only the universal lower bound $|D| \geq 1$, whereas a systematic p-adic factor supplies $|D| \geq p^L$ and could erase the missing logarithm outright.

Question 25.1. *Can one choose $O(\log X)$ candidate rows and mixed-monomial columns so that the resulting nonzero determinant satisfies $\log |D|_p^{-1} \gg (\log X)^2$ for $p = 2$ or $p = 3$, while the archimedean estimate remains $\log |D|_\infty = o\big((\log X)^2\big)$?*

The difficulty is that a naive row or column factor is lost because the exponent set contains $(0,0)$; only more subtle min-plus determinant estimates, applied after suitable row and column finite-difference operations, have a chance. The exact affine primitivity and the gcd condition on the normalized degrees should be fed into the estimate rather than discarded.

25.2 Priority B: exploit the full (n,k) grid, not sorted magnitudes

The arbitrary-node argument forgets the canonical coordinates and retains only the sorted integers m_i. Restoring them exposes rectangular, often Kronecker-like, bivariate exponential matrices, which is exactly the balanced 2×2 setting where ordinary six exponentials estimates stop; integrality, positivity, and the one-depth-zero normalization may create a margin absent from the general four-exponentials problem. The right experiment is not another large generic determinant but an optimized minor selected from triangular coordinate regions $n+k\beta\le T$ and $a+b\vartheta\le L$, with exact accounting of archimedean size, zero estimates, and 2- and 3-adic valuations. The hard part is that the optimization is genuinely two-parameter and the useful shapes are not predicted by any current heuristic, so a search must precede a theorem.

Section 48 turns this priority into an exact fewnomial target. On R distinct counterexample points, every vanishing auxiliary polynomial has at least $R+1$ monomials; therefore one must either construct a relation with one fewer monomial than ordinary interpolation would use, or exploit additional arithmetic structure not measured by support. On the $2N$ grid, the BK-degree calibration shows a factor-four gap between the extremal interpolant and the general Matrix Coefficient target. The boundary-dilation defects are the most concrete finite compatibility equations currently available for attacking that gap.

25.3 Priority C: denominator-sensitive counting on the compact arc

The compact-orbit reformulation asks for infinitely many points on $y=x^{\vartheta}$ in $[1,2)\times[1,3)$ whose first denominator is a power of 2, whose second denominator is a power of 3, and whose denominator exponents are synchronized by the same integer $\lfloor k\beta\rfloor$. A theorem asserting that a compact transcendental arc with nonzero curvature carries $o(H)$ points of logarithmic height at most H with denominators supported on two prescribed disjoint prime sets and a common exponent parameter would close the case, since the failed-conjecture orbit has $\asymp H$ such points. What makes it hard is that standard rational-point counting is sensitive to denominator *size* but not to denominator *support* or synchronization; Butler's connection between Wilkie-type statements and real exponential conjectures [18] suggests this direction sits on the transcendence frontier rather than being a routine application.

25.4 Priority D: mixed archimedean–radical invariants

The normalized radical $E=w^{\vartheta}$ has exact degree d and binomial minimal polynomial, so the Kummer field $\mathbb{Q}(E,\zeta_d)$ is completely explicit. The questions worth pursuing are whether the logarithmic relation $\log E\log 2-\log w\log 3=0$ forces an impossible regulator or height relation there; whether simultaneous normalization at both outputs constrains the two normalized degrees through field intersections or discriminants beyond the known gcd condition; whether all complex branches $\log(\zeta_d^jE)=\log E+2\pi ij/d$ can populate a multi-logarithm determinant with algebraic entries of controlled height; and whether a p-adic logarithm branch is compatible with the real identity after raising to the normalized degree. A warning is essential, and it is the same one that closed Section 24.5: merely taking norms or conjugates of $X^d - c/3^a$ reproduces known relations, so the new invariant must mix conjugation with the special logarithmic direction.

25.5 Priority E: discharge and sharpen the determinant hypotheses

The algebraic and combinatorial determinant hierarchy has now been formalized and accepted:

`ArbitraryNodeDeterminant`, `ExponentialPolynomialZeros`,
`OrderedChamberPositivity`, `MixedExponentSemigroup`,
`SemigroupDeterminant`, `FallingFactorialBound`,
`SemigroupWeightBound`, `ArbitraryNodeRepulsion`, `SemigroupGapBound`.

They are all on the aggregate import surface. **[kernel-verified Lean]** The two quantitative endpoints retain their analytic inputs as explicit hypotheses, `RpowDividedDifferenceMeanValue` and `NewtonFactorization`; those propositions are not asserted by the corpus. The priority is therefore no longer to translate the algebra, but to prove these named hypotheses in the required uniform form, or to replace them by a stronger factorization whose archimedean loss is small enough to interact with only $O(\log X)$ nodes. The exact tropical initial-form criterion of `TropicalInitialForm` is a useful prerequisite on the nonarchimedean side, but it does not supply the cancellation lower bound that this program needs.

25.6 Priority F: force a third algebraic exponential

The denominator-controlled theorem in `RationalThirdBase` is a precise gateway: for an auxiliary integer q containing a prime outside $\{2, 3\}$ it suffices, in several forms, to prove that q^x is rational with denominator supported by the known outputs, after which monomial injectivity and the three-base determinant force $x \in \mathbb{Q}$ and hence $x \in \mathbb{Z}$. What this would buy is a complete proof. The obstacle is the one identified in Section 24.7: straight monomials cannot introduce a new prime direction, so candidates must come from norms, resultants, or algebraic combinations arising in the primitive radical field, whose xth powers would then have to simplify.

25.7 Priority G: five and sharp six exponentials

Formalizing the five exponentials theorem, or Roy's sharp six exponentials theorem [23], would add genuinely new restrictions — for instance transcendence of $e^{\gamma x}$ for algebraic $\gamma \neq 0$ — and would extend the rational-third-base result from a controlled denominator to an unrestricted rational output. The effort is comparable to the completed Gelfond–Schneider port, which is encouraging; the risk is that it be undertaken without a specific downstream lemma in mind, since as Section 24.7 shows these theorems constrain b^x and not the divisibility facts the conjecture actually needs.

25.8 Priority H: larger kernel-replayed regions and effective measures

The zero-free region $2^x < 16384$ is kernel-verified, while the current software frontier is 2^{27} (**[software computation]**) in Section 34. Each further doubling is one generated, kernel-replayed table; the semiconvergent device of `FiniteCheck8192` and `FiniteCheck16384` keeps the exponents manageable. Every descent constant improves by bookkeeping once the region grows, so the payoff is broad but shallow. The bottleneck is a compact kernel-replayable proof object: adaptive convergent enclosures, chunked certificate modules, and tiers shared along ranges. In the same vein, effective irrationality measures for ϑ from Páde or hypergeometric methods would replace per-value certificate tiers with a uniform polynomial criterion and likely extend verified regions by orders of magnitude. Their limitation is sharp: they say nothing directly about the generator $\beta = \log_2 M$ for unknown M, which would require an effective measure derived from the log-product relation (101) and therefore lies beyond ordinary linear forms in logarithms.

25.9 Priority I: computational reconnaissance with exact certificates

Finite scanning is evidence, not a path to infinity, but computation remains valuable in three focused roles: searching candidate determinant shapes for large systematic 2- or 3-adic valuations, feeding Priority A and Priority B; enumerating small primitive radical patterns (w, d, a, c) and recording local obstructions with exact rational inequalities rather than floating-point rejection; and rejecting candidates early using the primitive-pair conditions — no matched depth, no common perfect power, no matched affine decomposition — while recording near misses keyed by odd core, denominator, and rational-power index. The goal must be pattern discovery and theorem generation; the difficulty is the discipline not to spend the budget on extending a decimal barrier that proves nothing new.

25.10 Priority J: the colossally abundant interface

Formalizing the original application — that the rational-power form of the conjecture implies that quotients of consecutive colossally abundant numbers are prime [1] — would connect the corpus to its historical source and give the whole development a visible consequence. It is logically downstream of everything above, and the work is combinatorial and definitional rather than transcendental, so it should follow the core structure modules rather than compete with them.

26 Comparison of methods

26.1 Prime-valuation rank versus interpolation geometry

The module `MultiplicativeRank` proves that any three candidates have linearly dependent prime-exponent vectors, of rank at most two. It does so by a three-base, six-exponentials determinant: if three candidate bases were multiplicatively independent, integrality of all three ϑth powers would force ϑ rational, contradicting its irrationality. A finite candidate set therefore lies in a rational plane inside the free vector space on primes. Choosing two prime coordinates injectively encodes the set and yields an explicit logarithmic-square count.

The semigroup determinant of Section 19 proves a logarithmic-square dyadic bound by an entirely different mechanism.

Prime-valuation method	Semigroup determinant method
Uses prime factorization of candidate bases	Uses only integer values on $y = x^{\vartheta}$
Global multiplicative rank at most two	Local total positivity and interpolation
Invokes a six-exponentials consequence	Uses elementary determinant integrality once ϑ is fixed
Produces coordinates in two prime directions	Produces repulsion for arbitrary ordered nodes
Naturally global in magnitude	Naturally local in additive intervals
Kernel-verified **[kernel-verified Lean]**	Paper theorem **[paper proof]**; hypothesis-carrying Lean endpoint **[kernel-verified Lean]**

Their agreement at logarithmic-square scale is suggestive rather than coincidental. Both methods see a two-dimensional structure, but they see different two-dimensional structures: prime-exponent rank two on one side, mixed-exponent lattice dimension two on the other. Neither refinement alone has closed the gap, and a future proof may well need to combine the two dual lattices rather than push either one further in isolation.

26.2 Exact conditional rank one over two generators

Conditional on failure of the conjecture, the candidate monoid is not merely of rank at most two — it is exactly free on 2 and M. In prime-valuation space its points form the nonnegative lattice generated by the prime-exponent vectors of 2 and M. In mixed-exponent evaluation space, the values factor across a two-dimensional grid indexed by the pair (n, k) with value $2^n M^k$. This exact grid is far stronger than the unconditional rank theorem, and yet every presently known consequence remains compatible with it. That compatibility is the correct measure of how much room is left.

The strongest likely synthesis is a determinant whose rows use primitive-generator coordinates and whose columns use mixed exponents. Such a matrix would be simultaneously:

- integral, hence subject to the $|\det| \geq 1$ lower bound;
- totally positive after ordering by magnitude, hence nonvanishing for free;
- explicitly factored into powers of $2, 3, M, A$;
- constrained by affine primitivity and by simultaneous output normalization.

No current module combines all four properties. `MultiplicativeRank` has the first and third; the semigroup modules have the first and second; `PrimitiveGenerator` and `OutputNormalization` have the fourth. Building the module that has all four is the most concrete open formalization task suggested by this report.

26.3 Finite differences as confluent determinants

Forward differences are the equally spaced specialization of interpolation, and Wronskians are its confluent limit as nodes coalesce. The determinant hierarchy of Sections 18 and 19 therefore provides a single conceptual envelope for four things that the corpus currently treats separately:

- second-, third-, and arbitrary-order finite differences;
- arbitrary-node ordinary-power determinants;
- mixed-monomial generalized Vandermonde determinants;
- confluent determinants using derivatives or repeated coordinate directions.

The corresponding formalization strategy is to build a general interpolation-determinant layer once and recover `HigherDifference` as a specialization of it, rather than to keep the two developments parallel. That would remove duplicated real analysis and, more usefully, make experimental determinant choices cheap to certify: a new choice of nodes and exponents would need only a divided-difference bound, with integrality and nonvanishing supplied by the shared layer.

27 A synchronized Sturmian model of a counterexample

27.1 Fractional-part normalization

Write

$$\beta = B + \alpha, \qquad B = \lfloor \beta \rfloor \in \mathbb{N}_0, \qquad 0 < \alpha < 1.$$

Since a nonintegral solution is transcendental, α is irrational. Define the rational localized multipliers

$$U = \frac{M}{2^B} = 2^{\alpha} \in \mathbb{Q} \cap (1,2), \qquad V = \frac{A}{3^B} = 3^{\alpha} \in \mathbb{Q} \cap (1,3). \tag{102}$$

For $k \geq 0$, put

$$e_k = \lfloor k\alpha \rfloor, \qquad s_k = e_{k+1} - e_k \in \{0,1\}.$$

The binary sequence $s = s_0 s_1 s_2 \cdots$ is the lower mechanical word of slope α. Define compact normalized orbits

$$R_k = \frac{U^k}{2^{e_k}} = 2^{\{k\alpha\}} \in [1,2), \qquad S_k = \frac{V^k}{3^{e_k}} = 3^{\{k\alpha\}} \in [1,3). \tag{103}$$

Both orbits use the *same* fractional part and hence the same symbolic itinerary.

Theorem 27.1 (Synchronized dyadic–triadic Sturmian recurrence)**.** *Assume a primitive nonintegral solution β exists. Then the rational sequences R_k, S_k in (103) satisfy*

$$R_0 = S_0 = 1, \qquad R_{k+1} = \frac{UR_k}{2^{s_k}}, \qquad S_{k+1} = \frac{VS_k}{3^{s_k}}, \tag{104}$$

where

$$s_k = 1 \iff \{k\alpha\} \geq 1 - \alpha \iff UR_k \geq 2 \iff VS_k \geq 3. \tag{105}$$

Moreover,

$$\sum_{j=0}^{N-1} s_j = \lfloor N\alpha \rfloor \qquad (N \geq 1). \tag{106}$$

Proof. The recurrence follows by subtracting consecutive floor exponents:

$$R_{k+1} = \frac{U^{k+1}}{2^{e_{k+1}}} = \frac{UU^k}{2^{e_k+s_k}} = \frac{UR_k}{2^{s_k}},$$

and the same calculation with base 3 gives the formula for S_k. Since $e_{k+1} = e_k + 1$ exactly when $\{k\alpha\}+\alpha \geq 1$, the first equivalence in (105) holds. The identities $UR_k = 2^{\alpha+\{k\alpha\}}$ and $VS_k = 3^{\alpha+\{k\alpha\}}$ give the other two. Finally, (106) telescopes: $\sum_{j<N} s_j = e_N - e_0 = \lfloor N\alpha \rfloor$. □

27.2 Sturmian consequences

The following facts are classical consequences of irrational mechanical coding [47, 48].

Corollary 27.2 (Aperiodicity, balance, and minimal complexity)**.** *The word s in Theorem 27.1 is aperiodic. Any two factors of equal length contain numbers of* 1*'s differing by at most one, and the number of distinct factors of length n is exactly $n+1$.*

Aperiodicity has an elementary proof in this setting. If s were eventually periodic, then its limiting density of 1's would be rational. Equation (106) shows that this density is α, contradicting irrationality. Balance and factor complexity are the standard Sturmian characterization.

The point is not merely terminological. A hypothetical counterexample is no longer described only by two rational numbers U, V satisfying a logarithmic equality. It determines one minimal-complexity aperiodic word, and that same word must be realized by two rational piecewise-multiplicative systems with multiplicatively independent reset bases 2 and 3. This is a much more rigid synchronized object.

27.3 Exact reduced denominators

The primitive factor depths determine how much cancellation occurs in (103). Write

$$M = 2^a W, \quad W \text{ odd}, \qquad A = 3^b Z, \quad 3 \nmid Z.$$

Because M is not a power of 2 and A is not a power of 3, one has $W, Z > 1$.

Proposition 27.3 (Denominator exponents and common jump word)**.** *In lowest terms,*

$$R_k = \frac{W^k}{2^{q_k^{(2)}}}, \qquad q_k^{(2)} = \lfloor k\beta \rfloor - ak = k(B-a) + \lfloor k\alpha \rfloor, \tag{107}$$

$$S_k = \frac{Z^k}{3^{q_k^{(3)}}}, \qquad q_k^{(3)} = \lfloor k\beta \rfloor - bk = k(B-b) + \lfloor k\alpha \rfloor. \tag{108}$$

Consequently

$$q_{k+1}^{(2)} - q_k^{(2)} = B - a + s_k, \qquad q_{k+1}^{(3)} - q_k^{(3)} = B - b + s_k.$$

At least one of the two sequences has baseline jump B, because $\min(a,b) = 0$*.*

Proof. Using $\lfloor k\beta \rfloor = kB + \lfloor k\alpha \rfloor$,

$$R_k = \frac{M^k}{2^{\lfloor k\beta \rfloor}} = \frac{2^{ak}W^k}{2^{\lfloor k\beta \rfloor}} = \frac{W^k}{2^{\lfloor k\beta \rfloor - ak}}.$$

The numerator is odd, so this is reduced. The base-3 calculation is identical. Taking successive differences gives the jump formulas, and the last claim is the inherited primitive condition (4.4). □

Thus the same Sturmian bit s_k records whether *both* reduced denominators take their large or small permitted jump at time k. One denominator can have a shallower constant baseline, but the aperiodic fluctuation is synchronized exactly.

27.4 Exact synchronized return gaps

The same word controls an integral recurrence before normalization. Put

$$m_n = \lfloor n\beta \rfloor, \qquad \delta_n = \{n\alpha\} = \{n\beta\}, \qquad g_{2,n} = M^n - 2^{m_n}, \qquad g_{3,n} = A^n - 3^{m_n}.$$

For a nonintegral solution these gaps are positive and

$$\frac{g_{2,n}}{2^{m_n}} = 2^{\delta_n} - 1, \qquad \frac{g_{3,n}}{3^{m_n}} = 3^{\delta_n} - 1. \tag{109}$$

Thus

$$\rho_n := \frac{g_{3,n}2^{m_n}}{g_{2,n}3^{m_n}} = \frac{3^{\delta_n} - 1}{2^{\delta_n} - 1} \in \mathbb{Q}, \qquad \log_2 3 < \rho_n < 2. \tag{110}$$

The ratio on the right is strictly increasing, and at zero has the expansion

$$\frac{3^t - 1}{2^t - 1} = \log_2 3 \left(1 + \frac{\log 3 - \log 2}{2} t + \frac{(\log 3 - \log 2)(2\log 3 - \log 2)}{12} t^2 + O(t^3)\right).$$

Each return therefore gives a rational approximation to $\log_2 3$ from above, but its native prime height is already exponential. If $\lambda = v_2(M)$ and $\mu = v_3(A)$, then for all large n

$$v_2(g_{2,n}) = \lambda n, \qquad v_3(g_{3,n}) = \mu n; \tag{111}$$

after reducing (110), its numerator retains at least $2^{m_n - \lambda n}$ and its denominator at least $3^{m_n - \mu n}$. This is why the linear Taylor gain alone does not give an irrationality-measure contradiction.

More structure survives in the unnormalized gaps. With $c_{n,r} = \lfloor \delta_n + \delta_r \rfloor \in \{0, 1\}$,

$$g_{2,n+r} = 2^{m_n} g_{2,r} + 2^{m_r} g_{2,n} + g_{2,n} g_{2,r} - (2^{c_{n,r}} - 1) 2^{m_n + m_r}, \tag{112}$$

$$g_{3,n+r} = 3^{m_n} g_{3,r} + 3^{m_r} g_{3,n} + g_{3,n} g_{3,r} - (3^{c_{n,r}} - 1) 3^{m_n + m_r}. \tag{113}$$

The same Sturmian carry drives both equations, with different correction weights. These exact recurrences retain gcd and divisibility data destroyed by normalization; exploiting that shared data remains open.

27.5 Complete synchronized gap products and discriminant energy

The preceding one-lag recurrence extends to the complete product over all pairs. For $N \geq 2$ and $1 \leq d < N$, set

$$L_d = N - d, \qquad h_d = m_d, \qquad \eta_d = \delta_d, \qquad \varepsilon_{i,d} = m_{i+d} - m_i - h_d \in \{0, 1\},$$

and let $C_{d,N} = \sum_{i=0}^{L_d - 1} \varepsilon_{i,d}$. Thus $\delta_{i+d} = \delta_i + \eta_d - \varepsilon_{i,d}$, and the carry is one exactly when $\delta_i \geq 1 - \eta_d$. For $b > 1$ define the unsquared orbit discriminant

$$\Delta_{b,N} = \prod_{0 \leq i < j < N} |b^{\delta_j} - b^{\delta_i}|. \tag{114}$$

Splitting the lag-d factors according to the carry gives the exact identity

$$\prod_{i=0}^{L_d-1} |b^{\delta_{i+d}} - b^{\delta_i}| = b^{\sum_{i<L_d} \delta_i + C_{d,N}(\eta_d - 1)} (b^{\eta_d} - 1)^{L_d - C_{d,N}} (b^{1-\eta_d} - 1)^{C_{d,N}}. \tag{115}$$

Multiplication over $d = 1, \ldots, N-1$ recovers $\Delta_{b,N}$ and retains the entire carry word rather than replacing it by its mean.

For an arithmetic pair $(b, B) = (2, M)$ or $(3, A)$, put

$$\begin{aligned} E_N &= \sum_{j=1}^{N-1} j m_j, \\ P_{b,B}(N) &= \prod_{0 \le i < j < N} |B^j - B^i b^{m_j - m_i}|, \\ \mathcal{Q}_{b,B}(N) &= \prod_{d=1}^{N-1} |B^d - b^{h_d}|^{L_d - C_{d,N}} |B^d - b^{h_d+1}|^{C_{d,N}}. \end{aligned} \tag{116}$$

Directly clearing the pairwise denominators gives the first equality below; factoring B^i from every lag numerator gives the second:

$$P_{b,B}(N) = b^{E_N} \Delta_{b,N} = B^{\binom{N}{3}} \mathcal{Q}_{b,B}(N), \qquad E_N = \frac{\beta}{3} N^3 + O_\beta(N^2). \tag{117}$$

When $b \nmid B$, every pairwise numerator is a b-adic unit, so the structural denominator in the first equality is exact and $P_{b,B}(N)$ is coprime to b. This qualification is essential: the formula always holds, but exact reduced denominators are asserted only in the primitive branch $b \nmid B$.

The same data form one integer polynomial,

$$F_{b,B,N}(X) = \prod_{k=0}^{N-1} (b^{m_k} X - B^k) \in \mathbb{Z}[X]. \tag{118}$$

If $R_N = \sum_{k<N} m_k$ and $S_N = \sum_{i=0}^{N-2} (N-1-i) m_i$, then

$$\operatorname{lc}(F_{b,B,N}) = b^{R_N}, \qquad |F_{b,B,N}(0)| = B^{\binom{N}{2}}, \qquad |\operatorname{Disc} F_{b,B,N}| = b^{2S_N} P_{b,B}(N)^2. \tag{119}$$

Thus the dyadic and triadic orbit polynomials have exactly matched normalized endpoints; their nontrivial separation lies in the discriminants.

The logarithmic energy of the rotation determines that separation. Baker's fixed-M estimate [3] makes β finite type, so discrepancy plus the Diophantine spacing bound gives uniform integrability of the logarithmic kernel. More explicitly, truncating at height T controls the excess $(-\log y - T)_+$ by $O(e^{-T})$ after $N \to \infty$, while the omitted T times the close-pair proportion is $O(Te^{-T})$. Hence

$$\frac{2}{N^2} \log \Delta_{b,N} \longrightarrow I_b, \qquad I_b = \int_0^1 \int_0^1 \log |b^s - b^t| \, ds \, dt, \tag{120}$$

with the closed form

$$I_b = \frac{2 \log b}{3} - \frac{\pi^2}{3 \log b} + \frac{2(\zeta(3) - \operatorname{Li}_3(1/b))}{(\log b)^2}. \tag{121}$$

There is also a pointwise comparison, stronger than mere asymptotic separation. For $0 < u < 1$ define

$$\Phi(u) = \log_3(3^u - 1) - \log_2(2^u - 1).$$

Because $\frac{d}{du}\log_b(b^u - 1) = (1 - b^{-u})^{-1}$ decreases strictly with b, the function Φ is strictly decreasing and

$$\Phi(u) > \log_3 2. \tag{122}$$

Applying this both to $u = \delta_d$ and to the ceiling branch $u = 1 - \delta_d$ yields, for $h \in \{\lfloor d\beta \rfloor, \lceil d\beta \rceil\}$,

$$|A^d - 3^h| > 2\,|M^d - 2^h|^{\vartheta}. \tag{123}$$

Summing over every factor, define

$$\mathcal{G}_N = \log_3 \mathcal{Q}_{3,A}(N) - \log_2 \mathcal{Q}_{2,M}(N). \tag{124}$$

The monomial factors in (117) and the structural factors in the discriminants cancel exactly, leaving the equivalent formulas

$$\begin{aligned} \mathcal{G}_N &= \log_3 P_{3,A}(N) - \log_2 P_{2,M}(N) \\ &= \log_3 \Delta_{3,N} - \log_2 \Delta_{2,N} = \sum_{0 \le i < j < N} \Phi(|\delta_j - \delta_i|). \end{aligned} \tag{125}$$

Consequently

$$\mathcal{G}_N > \binom{N}{2} \log_3 2, \qquad \frac{\mathcal{G}_N}{N^2} \longrightarrow \kappa, \tag{126}$$

where

$$\begin{aligned} \kappa &= \frac{1}{2}\left(\frac{I_3}{\log 3} - \frac{I_2}{\log 2}\right) \\ &= \frac{\pi^2}{6}\left(\frac{1}{(\log 2)^2} - \frac{1}{(\log 3)^2}\right) + \frac{\zeta(3) - \mathrm{Li}_3(1/3)}{(\log 3)^3} - \frac{\zeta(3) - \mathrm{Li}_3(1/2)}{(\log 2)^3} \\ &= 0.70792457032085580787 44\ldots. \end{aligned} \tag{127}$$

This analytic surplus forces ordinary arithmetic mass. If $H_N = \gcd(\mathcal{Q}_{2,M}(N), \mathcal{Q}_{3,A}(N))$ and $Y_N = \mathcal{Q}_{3,A}(N)/H_N$, then

$$Y_N > 2^{\binom{N}{2}}, \qquad \liminf_{N \to \infty} \frac{\log Y_N}{N^2} \ge \kappa \log 3. \tag{128}$$

Prime by prime, this means positive excess valuations after subtracting the dyadic multiplicities; it does not mean that those primes are absent from the dyadic product. If $3 \nmid A$, the triadic product is coprime to $3A$, so this excess lies outside that fixed support. If $A = 3^e V$ with $e > 0$, $3 \nmid V$, and $\gamma = \beta - e$, its exact structural part is

$$v_3(\mathcal{Q}_{3,A}(N)) = e\binom{N+1}{3} + \sum_{\substack{1 \le d < N \\ d\gamma < 1}} (N - d - C_{d,N}) v_3(V^d - 1) = e\binom{N+1}{3} + O_A(N). \tag{129}$$

The symmetric dyadic formula holds when $2 \mid M$. The primitive-depth condition prevents both cubic structural terms from occurring simultaneously.

Equations (124) and (127) also calibrate the remaining local target: any proposed upper bound for $N^{-2}\mathcal{G}_N$ with limsup strictly below κ would contradict the exact limit, so it must add genuinely new arithmetic input rather than restate unique factorization. The quadratic unmatched quotient is substantial, but by itself it provides no upper bound and hence no contradiction.

Status: **[paper proof]** — exact finite product identities, a paper proof of the logarithmic-energy limit with the close-pair term included, and the resulting arithmetic surplus. The strict one-gap inequality and its finite-product amplification are **[kernel-verified Lean]** in `MatchedGapAmplification`; the orbit-product identities, energy limit, κ, and unmatched quotient remain paper-level. The numerical value of κ is ancillary; the closed formula and strict finite inequality are the substantive statements. No computational table or uncertified floor calculation is used.

27.6 An exact equivalent exclusion principle

The localization reformulations above include conditions such as $2^\alpha \in \mathbb{Z}[1/2]$ and $3^\alpha \in \mathbb{Z}[1/3]$. The recurrence packages them into a symbolic statement.

Theorem 27.4 (Synchronized Sturmian exclusion is equivalent to Alaoglu–Erdős)**.** *The Alaoglu–Erdős conjecture is equivalent to the nonexistence of data*

$$\alpha \in (0,1) \setminus \mathbb{Q}, \qquad U \in \mathbb{Q} \cap (1,2), \qquad V \in \mathbb{Q} \cap (1,3)$$

such that, for the lower mechanical word $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$, the recurrences

$$R_0 = S_0 = 1, \qquad R_{k+1} = UR_k/2^{s_k}, \qquad S_{k+1} = VS_k/3^{s_k}$$

remain in $[1,2) \cap \mathbb{Q}$ and $[1,3) \cap \mathbb{Q}$, respectively, for every k, with $U = 2^\alpha$ and $V = 3^\alpha$. Equivalently, it suffices to rule out irrational α for which

$$2^\alpha \in \mathbb{Z}[1/2], \qquad 3^\alpha \in \mathbb{Z}[1/3].$$

Proof. A counterexample $\beta = B + \alpha$ gives the data by Theorem 27.1. Conversely, if such α, U, V exist, choose an integer B at least as large as the powers of 2 and 3 occurring in the respective reduced denominators of U and V. Then

$$2^{B+\alpha} = 2^B U \in \mathbb{N}, \qquad 3^{B+\alpha} = 3^B V \in \mathbb{N},$$

while $B + \alpha \notin \mathbb{Z}$. The recurrence identities are equivalent to $R_k = 2^{\{k\alpha\}}$ and $S_k = 3^{\{k\alpha\}}$, so they add an exact dynamic coding but no hidden hypothesis. □

27.7 Cobham-type and Mahler-type diagnostics

Cobham's theorem says that a sequence automatic in two multiplicatively independent integer bases is ultimately periodic [49, 50]. Sturmian words of irrational slope are not automatic in any integer base. Theorem 27.4 therefore suggests a tempting strategy: show that the dyadic rational recurrence makes s 2-automatic and that the triadic recurrence makes it 3-automatic.

That implication is *not* presently available. Rationality of the multiplier does not turn the compact rational state R_k or S_k into a finite-state object; the exact reduced denominators grow without bound. Cobham's theorem becomes applicable only after proving a finite-kernel or finite-substitution property, and that is essentially a new arithmetic statement. The synchronized formulation is useful because it identifies this missing lemma cleanly rather than treating "automaticity" as a metaphor.

Research target 27.5 (Finite-kernel extraction)**.** Show that simultaneous rational realizability of the same irrational mechanical word by (104) forces a finite 2-kernel and a finite 3-kernel, or derive a weaker pair of incompatible substitutive structures.

A Mahler-function variant is also conceivable: encode the jump word by $F(z) = \sum_{k\geq 0} s_k z^k$. Aperiodicity and the finite coefficient set already imply, by classical results of Fatou type, that F is transcendental over $\mathbb{Q}(z)$. A proof would have to derive incompatible functional equations from the two rational recurrences. No such equations were found here; the recurrence is multiplicative in the state, not self-similar in the index.

28 Effective logarithmic separation

28.1 From linear forms in two logarithms to an irrationality measure

The shrinking-target theorem gives an elementary denominator-growth statement: a very good rational approximation to a hypothetical solution β would force a correspondingly large output, and consecutive continued-fraction denominators obey a single-exponential recurrence. Since β is itself a ratio of logarithms of fixed integers, Baker's theory gives a substantially stronger conclusion.

Theorem 28.1 (Effective finite irrationality measure for the primitive generator)**.** *Assume a primitive counterexample β exists, and put $M = 2^\beta$. There are effectively computable constants $c_M > 0$ and $\mu_M < \infty$ such that, for every $p \in \mathbb{Z}$ and $q \geq 2$,*

$$\left|\beta - \frac{p}{q}\right| \geq c_M q^{-\mu_M}. \tag{130}$$

One may choose μ_M bounded by an effectively computable absolute constant times $1 + \log M$, hence by an absolute constant times $1 + \beta$. The same conclusion follows from $\beta = \log_3 A$.

Proof. Consider the linear form

$$\Lambda = q \log M - p \log 2.$$

It is nonzero: otherwise $M^q = 2^p$, which would make $\beta = p/q$ rational; a rational solution is integral by the elementary rational-exponent classification already formalized in the source corpus. The Baker–Wüstholz theorem, or Matveev's explicit lower bound for a homogeneous linear form in logarithms of algebraic numbers, gives

$$\log|\Lambda| \geq -C\,(1 + \log M)\log H - C'(1 + \log M),$$

where $H \geq \max(3, |p|, q)$ and C, C' are effective absolute constants after the fixed number 2 is absorbed. For approximations with $|p| \leq (\beta + 1)q$, one has $H \ll_M q$; for all other p the desired inequality is trivial after decreasing c_M. Since

$$\left|\beta - \frac{p}{q}\right| = \frac{|\Lambda|}{q \log 2},$$

absorbing fixed factors and one additional power of q gives (130). Matveev's dependence on the logarithmic heights is linear in $\log M = \beta \log 2$ here, which gives the final scale assertion. □

Status: [classical/open background] plus [paper proof]

The lower-bound theorem for logarithmic forms is classical [3, 33, 42]. Its application to the primitive generator and the continued-fraction consequences below are paper-level deductions. No useful small numerical value of μ_M is claimed; standard explicit constants are very large.

28.2 Polynomial recurrence for convergent denominators

Let p_n/q_n be the continued-fraction convergents of β. The standard upper bound

$$\left|\beta - \frac{p_n}{q_n}\right| < \frac{1}{q_n q_{n+1}}$$

combined with (130) gives the following.

Corollary 28.2 (Polynomial denominator growth). *There is an effective constant $C_M > 0$ such that*

$$q_{n+1} \le C_M q_n^{\mu_M - 1} \tag{131}$$

for every n with $q_n \ge 2$. Consequently the next partial quotient satisfies $a_{n+1} = O_M(q_n^{\mu_M-2})$.

This is qualitatively stronger than the inherited elementary estimate $q_{n+1} < 2M^{q_n}$. It rules out Liouville behavior and says that the exceptionally long near-repetitions of the Sturmian word are polynomially controlled in the preceding standard word length. The exponent depends on the unknown integer M, however, and may be enormous.

28.3 Polynomial spacing in conditional blocks

The irrationality measure also upgrades the separation of the block nodes. From (130), for $d \ge 1$ and the nearest integer r to $d\beta$,

$$\|d\beta\|_{\mathbb{Z}} = |d\beta - r| \ge c_M d^{1-\mu_M}. \tag{132}$$

For an integer $T \ge 0$, put

$$K_T = \left\lfloor \frac{T+1}{\beta} \right\rfloor, \qquad x_{T,k} = T - \lfloor k\beta \rfloor + k\beta = T + \{k\beta\}, \quad 0 \le k \le K_T.$$

The associated candidate and output nodes are

$$m_{T,k} = 2^{x_{T,k}} = 2^{T-\lfloor k\beta \rfloor} M^k, \qquad a_{T,k} = 3^{x_{T,k}} = 3^{T-\lfloor k\beta \rfloor} A^k. \tag{133}$$

Corollary 28.3 (Effective block separation). *For distinct $i, j \in \{0, \dots, K_T\}$,*

$$|x_{T,j} - x_{T,i}| \ge c_M K_T^{1-\mu_M}, \tag{134}$$

$$|m_{T,j} - m_{T,i}| \ge (\log 2) c_M 2^T K_T^{1-\mu_M}, \tag{135}$$

$$|a_{T,j} - a_{T,i}| \ge (\log 3) c_M 3^T K_T^{1-\mu_M}. \tag{136}$$

Proof. The ordinary distance between two fractional parts is at least their circular distance, so

$$|\{j\beta\}-\{i\beta\}|\geq \|(j-i)\beta\|_{\mathbb{Z}}\geq c_M|j-i|^{1-\mu_M}\geq c_M K_T^{1-\mu_M},$$

because $1-\mu_M<0$. This proves (134). The mean value theorem applied to 2^x and 3^x on $[T,T+1]$ gives the remaining inequalities. □

28.4 Separation on the full (n,k) lattice

The same estimate applies to the two-dimensional conditional exponent grid.

Corollary 28.4 (Polynomial separation of conditional exponents)**.** *There are effective $c_M,\mu_M>0$ such that, for distinct integer pairs $(n,k),(n',k')$ with $0\leq k,k'\leq H$,*

$$|(n+k\beta)-(n'+k'\beta)|\geq \min\{1,c_M H^{1-\mu_M}\}.$$

Proof. If $k=k'$, the difference is a nonzero integer. Otherwise divide the linear form $(n-n')+(k-k')\beta$ by $|k-k'|$ and apply (130). □

The column exponent lattice $u+\vartheta v$ has an analogous effective polynomial spacing with absolute constants, because $\vartheta=\log_2 3$ is a fixed ratio of two logarithms. More refined explicit estimates are known for linear forms in $\log 2$ and $\log 3$; for example Wu's simultaneous linear-independence measure leads, in the usual normalization, to an irrationality exponent for ϑ below approximately 8.62 [46]. The exact convention-dependent numerical conversion is not needed below.

28.5 Why the Baker upgrade does not close the conjecture

The effective spacing is useful, but its scale is still compatible with the exact conditional population. A dyadic exponent block contains only $K_T+1\asymp T$ nodes. A polynomial lower bound on their minimum spacing imposes no contradiction on T points in a fixed interval. Likewise, the full triangular grid has $\asymp T^2$ points in an interval of exponent length T, and polynomially small gaps are expected.

Baker's theorem therefore strengthens compactness and determinant estimates but does not supply the missing logarithm by itself. Its most promising roles are indirect:

- controlling the smallest row and column gaps in an exact Harish–Chandra–Itzykson–Zuber factorization;
- bounding how often a p-adic assignment problem can be nearly degenerate;
- limiting the lengths of near-periodic Sturmian blocks that a congruence argument must handle;
- making computational searches over continued-fraction branches provably finite once a candidate output M is fixed.

29 Exact all-support valuations of ordinary block Vandermondes

29.1 The block determinants

Fix an integer T and abbreviate $K = K_T$. Let $m_k = m_{T,k}$ and $a_k = a_{T,k}$ be the $K+1$ input–output pairs in (133). Their ordinary Vandermonde determinants are

$$V_T^{(2)} = \det(m_i^j)_{0\le i,j\le K} = \prod_{0\le i<j\le K}(m_j - m_i), \tag{137}$$

$$V_T^{(3)} = \det(a_i^j)_{0\le i,j\le K} = \prod_{0\le i<j\le K}(a_j - a_i). \tag{138}$$

They are nonzero integers. Since $m_k \in [2^T, 2^{T+1})$ and $a_k \in [3^T, 3^{T+1})$,

$$\log|V_T^{(2)}| \le R_T T\log 2, \tag{139}$$

$$\log|V_T^{(3)}| \le R_T(T\log 3 + \log 2), \tag{140}$$

where

$$R_T = \binom{K+1}{2} = \frac{K(K+1)}{2}.$$

The additive $R_T \log 2$ in (140) is lower order.

29.2 The input determinant: an exact formula at every support prime

Factor

$$M = 2^a W, \qquad W \text{ odd}, \qquad \delta_2 = \beta - a = \log_2 W.$$

As noted above, $W > 1$, so $W \ge 3$ and

$$\delta_2 \ge \log_2 3 > 1. \tag{141}$$

Because a is an integer,

$$m_k = 2^{T-\lfloor k\delta_2\rfloor}W^k. \tag{142}$$

The 2-adic valuations in (142) strictly decrease with k, while every odd support-prime valuation strictly increases.

Proposition 29.1 (Exact support valuations for $V_T^{(2)}$)**.** *For $0 \le i < j \le K$,*

$$v_2(m_j - m_i) = T - \lfloor j\delta_2\rfloor, \tag{143}$$

$$v_p(m_j - m_i) = i\, v_p(W) \qquad (p \mid W). \tag{144}$$

Consequently

$$v_2(V_T^{(2)}) = \sum_{j=1}^{K} j(T - \lfloor j\delta_2\rfloor), \tag{145}$$

$$v_p(V_T^{(2)}) = v_p(W)\sum_{i=0}^{K-1} i(K-i) \qquad (p \mid W). \tag{146}$$

Proof. Put $\rho_k = T - \lfloor k\delta_2 \rfloor$. By (141), $\rho_i > \rho_j$ when $i < j$. From (142),

$$m_j - m_i = 2^{\rho_j} W^i \left(W^{j-i} - 2^{\rho_i - \rho_j} \right).$$

The parenthesis is odd, proving (143). For an odd prime $p \mid W$, the two terms m_i and m_j have distinct valuations $iv_p(W) < jv_p(W)$, so the ultrametric equality gives (144). Summing over pairs gives (145)–(146). $\square$

Let $Q_T^{(2)}$ be the part of $V_T^{(2)}$ supported on the primes dividing $2M$. Then Proposition 29.1 gives the exact identity

$$\log Q_T^{(2)} = \log 2 \sum_{j=1}^{K} j(T - \lfloor j\delta_2 \rfloor) + \log W \sum_{i=0}^{K-1} i(K-i). \tag{147}$$

The two elementary sums have asymptotics

$$\sum_{j=1}^{K} j(T - \lfloor j\delta_2 \rfloor) = \frac{TK^2}{2} - \frac{\delta_2 K^3}{3} + O_\beta(T^2),$$
$$\sum_{i=0}^{K-1} i(K-i) = \frac{K^3}{6} + O(K^2).$$

Using $\log W = \delta_2 \log 2$ and $K/T \to 1/\beta$ yields

$$\frac{\log Q_T^{(2)}}{R_T T \log 2} \longrightarrow 1 - \frac{\delta_2}{3\beta}. \tag{148}$$

This includes every prime appearing in the entries. Any further prime divisor of the Vandermonde comes from subtraction and hence from genuine cancellation beyond termwise entry valuations.

29.3 The output determinant and bounded tie corrections

Factor

$$A = 3^b Z, \qquad 3 \nmid Z, \qquad \delta_3 = \beta - b = \log_3 Z > 0.$$

Then

$$a_k = 3^{T - \lfloor k\delta_3 \rfloor} Z^k. \tag{149}$$

Unlike δ_2, the number δ_3 can be smaller than 1; for example the least possible base-free factor is 2. Equal consecutive 3-adic valuations can therefore occur. They contribute only a lower-order correction.

Proposition 29.2 (Output support valuations)**.** *For every prime $p \mid Z$ and $0 \le i < j \le K$,*

$$v_p(a_j - a_i) = i\, v_p(Z).$$

At $p = 3$,

$$v_3(a_j - a_i) = T - \lfloor j\delta_3 \rfloor$$

after adding a nonnegative correction that is zero whenever $\lfloor i\delta_3 \rfloor < \lfloor j\delta_3 \rfloor$. The total correction over all pairs is $O_{A,\beta}(K^2)$. Consequently, if $Q_T^{(3)}$ denotes the part of $V_T^{(3)}$ supported on primes dividing $3A$, then

$$\frac{\log Q_T^{(3)}}{R_T T \log 3} \longrightarrow 1 - \frac{\delta_3}{3\beta}. \tag{150}$$

Proof. The assertion for $p \mid Z$ is the same distinct-valuation argument as before. Put $\rho_k = T - \lfloor k\delta_3 \rfloor$. If $\rho_i > \rho_j$, then

$$a_j - a_i = 3^{\rho_j} Z^i \left(Z^{j-i} - 3^{\rho_i - \rho_j} \right)$$

and the parenthesis is not divisible by 3, so the valuation is exactly ρ_j. If $\rho_i = \rho_j$, then

$$v_3(a_j - a_i) = \rho_j + v_3(Z^{j-i} - 1).$$

Equality of the floors implies $(j - i)\delta_3 < 1$, so $j - i$ belongs to a fixed finite set depending only on δ_3. Hence the extra valuation is bounded by a constant depending on A and β. There are $O(K^2)$ pairs. The asymptotic calculation is then identical to (147). □

29.4 The depth-sensitive product ceiling

Combining (139), (140), (148), and (150) gives the principal conclusion of the ordinary Vandermonde audit.

Theorem 29.3 (All-support ordinary Vandermonde fraction)**.** *For a hypothetical primitive counterexample,*

$$\frac{\log(Q_T^{(2)} Q_T^{(3)})}{R_T T \log 6} \longrightarrow 1 - \frac{\delta_2 \log 2 + \delta_3 \log 3}{3\beta \log 6}, \qquad \delta_2 = \beta - a, \quad \delta_3 = \beta - b. \tag{151}$$

Because $\min(a, b) = 0$*, the limit is at most*

$$1 - \frac{\log 2}{3 \log 6} = 0.871049\,064\ldots. \tag{152}$$

If $b = 0$*, the sharper ceiling is*

$$1 - \frac{\log 3}{3 \log 6} = 0.795599\,921\ldots.$$

If $a = b = 0$*, the limit is exactly* $2/3$*.*

Proof. Only the numerical ceiling remains to be shown. If $a = 0$, then $\delta_2 = \beta$, so the deficit in (151) is at least $\log 2/(3 \log 6)$. If $b = 0$, it is at least $\log 3/(3 \log 6)$. Taking the larger possible fraction gives (152). If both vanish, the numerator of the deficit is $\beta \log 6$. □

Table 4: Leading termwise-divisor ceilings for ordinary block Vandermondes.

Branch information	Exact/upper fraction	Decimal
$a = b = 0$	$2/3$	$0.666666\ldots$
$b = 0$	$\le 1 - \log 3/(3 \log 6)$	$0.795599\ldots$
No matched depth only	$\le 1 - \log 2/(3 \log 6)$	$0.871049\ldots$

29.5 Why even 87% is not enough

The deficit in Theorem 29.3 is a fixed positive fraction once the hypothetical primitive pair is fixed. The archimedean height is $\asymp T^3$, whereas all factorial, spacing, and tie corrections are $O_\beta(T^2 \log T)$. Thus no lower-order refinement of the ordinary Vandermonde can close the remaining gap.

This conclusion is stronger than the earlier observation that the 2-adic valuation alone captures one third in the base-free case. It includes every prime already present in the entries and every possible base depth. A successful ordinary-Vandermonde proof would need a new leading-order source of divisibility in the prime factors of the *differences*; the fixed-support valuations are completely accounted for by Theorem 29.3.

30 The mixed-semigroup determinant and its Gini norm

30.1 Balanced low-weight columns

Let

$$\Lambda = \{u + \vartheta v : (u, v) \in \mathbb{N}_0^2\}.$$

Because ϑ is irrational, every element has a unique pair (u, v). List the pairs in increasing order of $u + \vartheta v$:

$$(u_0, v_0), (u_1, v_1), \dots .$$

For $N \geq 1$, let

$$L_N = \max_{0 \leq j < N} (u_j + \vartheta v_j).$$

The elementary lattice count

$$\#\{(u, v) \in \mathbb{N}_0^2 : u + \vartheta v \leq L\} = \frac{L^2}{2\vartheta} + O_\vartheta(L) \tag{153}$$

shows that

$$L_N \sim \sqrt{2\vartheta N}. \tag{154}$$

Moreover, the empirical distribution of

$$X_j = \left(\frac{u_j}{L_N}, \frac{\vartheta v_j}{L_N} \right) \tag{155}$$

converges to uniform area measure on the standard triangle

$$\mathcal{T} = \{(x, z) \in \mathbb{R}^2 : x \geq 0, z \geq 0, x + z \leq 1\}. \tag{156}$$

Here and below "uniform" means probability measure with density 2 on this triangle.

For the block at height T, set $K = K_T$ and $N = K + 1$. Define

$$D_T = \det \left(m_{T,k}^{u_j} a_{T,k}^{v_j} \right)_{0 \leq k, j < N}. \tag{157}$$

The generalized-Vandermonde theorem of Section 19 implies that D_T is a nonzero integer. Put

$$B_j = 2^{u_j} 3^{v_j}.$$

Since $m_{T,k} = 2^{x_{T,k}}$ and $a_{T,k} = 3^{x_{T,k}}$,

$$m_{T,k}^{u_j} a_{T,k}^{v_j} = B_j^{x_{T,k}} = B_j^{T+\{k\beta\}}. \tag{158}$$

Thus the Leibniz expansion gives the clean archimedean bound

$$\log |D_T| \le (T+1) \sum_{j<N} \log B_j + \log N!. \tag{159}$$

The leading height is

$$H_T = T \sum_{j<N} \log B_j. \tag{160}$$

By (154) and the mean $\mathbb{E}(X_1 + X_2) = 2/3$ on $\mathcal{T}$,

$$H_T \sim \frac{2}{3} T N L_N \log 2 \asymp_\beta T^{5/2}. \tag{161}$$

Both $\sum_j \log B_j$ and $\log N!$ in the error of (159) are $o(H_T)$.

30.2 The all-prime tropical divisor

For a prime p, define the assignment minimum

$$\tau_{p,T} = \min_{\sigma \in S_N} \sum_{k=0}^{N-1} v_p\left(m_{T,k}^{u_{\sigma(k)}} a_{T,k}^{v_{\sigma(k)}} \right). \tag{162}$$

Every Leibniz term is divisible by $p^{\tau_{p,T}}$, so

$$Q_T^{\mathrm{trop}} = \prod_p p^{\tau_{p,T}} \quad \text{divides} \quad D_T. \tag{163}$$

Only primes dividing $6MA$ occur in the product.

The problem is now to compute the leading asymptotic of $\log Q_T^{\mathrm{trop}}/H_T$. A crude estimate "minimum $\le$ average" loses the precise branch dependence. The exact answer is controlled by a simple probability norm.

30.3 Antimonotone assignment and the Gini norm

Let $X = (X_1, X_2)$ be uniform on $\mathcal{T}$, and let X' be an independent copy. For $c = (r, s) \in \mathbb{R}^2$, define

$$\mathfrak{G}(r,s) = \mathbb{E} \left| r(X_1 - X_1') + s(X_2 - X_2') \right|. \tag{164}$$

This is a norm on $\mathbb{R}^2$: homogeneity and the triangle inequality are immediate, and the support of $X - X'$ spans $\mathbb{R}^2$, so only $(r,s) = (0,0)$ has norm zero.

Let $Y = rX_1 + sX_2$, with ascending quantile $Q_Y(t)$. Pairing an increasing uniform row parameter with the decreasing ordering of Y gives the continuum assignment functional

$$\Phi(r,s) = \int_0^1 t\, Q_Y(1-t)\, dt. \tag{165}$$

Lemma 30.1 (Gini formula for the minimum assignment)**.** *For every* $(r,s) \in \mathbb{R}^2$*,*

$$\Phi(r,s) = \frac{1}{2}\,\mathbb{E}Y - \frac{1}{4}\,\mathfrak{G}(r,s) = \frac{r+s}{6} - \frac{1}{4}\,\mathfrak{G}(r,s). \tag{166}$$

Proof. The rearrangement inequality says that the minimum scalar product pairs the increasing row parameter with the decreasing list of column values. Changing variables $u = 1-t$ in (165) gives

$$\Phi = \int_0^1 (1-u) Q_Y(u)\,du.$$

If Y' is an independent copy, the minimum of Y, Y' has quantile density $2(1-u)$, so

$$\Phi = \frac{1}{2}\mathbb{E}\min(Y,Y').$$

Since $\min(Y,Y') = (Y+Y'-|Y-Y'|)/2$, this becomes $\frac{1}{2}\mathbb{E}Y - \frac{1}{4}\mathbb{E}|Y-Y'|$. Finally $\mathbb{E}X_1 = \mathbb{E}X_2 = 1/3$. $\square$

30.4 Closed form of the Gini norm

The norm (164) is elementary. Sort the three vertex values

$$0, \qquad r, \qquad s$$

as $\xi_0 \le \xi_1 \le \xi_2$, and put

$$A_0 = \xi_1 - \xi_0, \qquad B_0 = \xi_2 - \xi_1.$$

Proposition 30.2 (Explicit triangular Gini norm)**.** *If* $A_0 + B_0 > 0$*, then*

$$\mathfrak{G}(r,s) = \frac{2\left(2A_0^2 + 3A_0B_0 + 2B_0^2\right)}{15(A_0+B_0)}. \tag{167}$$

At $(r,s) = (0,0)$ *the value is* 0*. In particular,*

$$\mathfrak{G}(1,0) = \mathfrak{G}(0,1) = \mathfrak{G}(1,1) = \frac{4}{15}, \qquad \mathfrak{G}(1,-1) = \frac{7}{15}. \tag{168}$$

Proof. Translation of the three vertex values does not change a difference $Y - Y'$, so take $\xi_0 = 0$, $\xi_1 = A_0$, and $\xi_2 = A_0 + B_0 = L$. Projection of uniform area measure on a triangle by a linear function has density

$$f(y) = \begin{cases} \dfrac{2y}{A_0 L}, & 0 \le y \le A_0, \\ \dfrac{2(L-y)}{B_0 L}, & A_0 \le y \le L. \end{cases}$$

Thus its distribution function is $F(y) = y^2/(A_0L)$ on the first interval and $F(y) = 1-(L-y)^2/(B_0L)$ on the second. The standard identity

$$\mathbb{E}|Y-Y'| = 2\int_{-\infty}^{\infty} F(y)(1-F(y))\,dy$$

gives (167) after two polynomial integrations. Cases with one gap zero follow by continuity. $\square$

For later use, if $\gamma \ge 0$, then

$$\mathfrak{G}(-1,\gamma) = \frac{2(2+3\gamma+2\gamma^2)}{15(1+\gamma)} = \frac{4}{15} + \frac{2\gamma(1+2\gamma)}{15(1+\gamma)} \ge \frac{4}{15}. \tag{169}$$

30.5 Normalized prime-valuation slope vectors

Recall

$$a = v_2(M), \quad b = v_3(A), \quad c = v_2(A), \quad d = v_3(M).$$

For each prime, define a normalized slope vector $c_p \in \mathbb{R}^2$ by

$$c_2 = \left(-1 + \frac{a}{\beta},\ \frac{c}{\beta\vartheta}\right), \tag{170}$$

$$c_3 = \left(\frac{\vartheta d}{\beta},\ -1 + \frac{b}{\beta}\right), \tag{171}$$

$$c_p = \left(\frac{v_p(M)\log p}{\beta\log 2}, \frac{v_p(A)\log p}{\beta\log 3}\right), \qquad p \neq 2, 3. \tag{172}$$

The logarithmic factorizations of M and A give the conservation law

$$\sum_p c_p = (0, 0). \tag{173}$$

Indeed, the first coordinate sum is

$$-1 + \frac{a\log 2 + d\log 3 + \sum_{p\neq 2,3} v_p(M)\log p}{\beta\log 2} = 0,$$

and similarly for the second coordinate.

The meaning of these vectors can be read directly from the valuation matrices. Put $s_k = \beta k/T$. Up to $o(L_N)$ after division by T, the logarithmic p-adic cost of the entry indexed by a scaled column (x, z) is

Prime	Cost divided by $TL_N \log 2$
2	$x + s_k\, c_2 \cdot (x, z)$
3	$z + s_k\, c_3 \cdot (x, z)$
$p \neq 2, 3$	$s_k\, c_p \cdot (x, z)$

For example,

$$v_2(m_k^u a_k^v) = u(T - \lfloor k\beta\rfloor + ak) + vck,$$

and substituting $u = L_N x$, $v = L_N z/\vartheta$, $k \sim Ts_k/\beta$ gives the first row. The $O(u)$ errors from the floors sum to $O(NL_N) = o(H_T)$.

30.6 The exact continuum limit

Theorem 30.3 (All-prime tropical fraction for the mixed block determinant)**.** *Under a hypothetical primitive counterexample, the determinant* (157) *satisfies*

$$\frac{\log Q_T^{\mathrm{trop}}}{H_T} \longrightarrow 1 - \frac{3}{8}\sum_p \mathfrak{G}(c_p), \tag{174}$$

where the finitely many slope vectors c_p are given by (170)*–*(172)*.*

Proof. The row parameters $s_k = \beta k/T$ become uniformly distributed on $[0, 1]$, because $K_T \sim T/\beta$. The scaled columns become uniformly distributed on $\mathcal{T}$ by (153). The finite rearrangement inequality and Riemann-sum convergence therefore show that the assignment minimum at prime p, after normalization by $TL_N N \log 2$, tends to its constant-column mean plus $\Phi(c_p)$.

Only $p = 2$ has constant part x, and only $p = 3$ has constant part z. Their combined mean is $2/3$. By Lemma 30.1, the sum of the slope contributions is

$$\sum_p \Phi(c_p) = \frac{1}{6}\sum_p (c_{p,1} + c_{p,2}) - \frac{1}{4}\sum_p \mathfrak{G}(c_p).$$

The first sum vanishes by (173). Hence

$$\frac{\log Q_T^{\mathrm{trop}}}{TL_N N \log 2} \longrightarrow \frac{2}{3} - \frac{1}{4}\sum_p \mathfrak{G}(c_p).$$

On the other hand, (161) gives $H_T/(TL_N N \log 2) \to 2/3$. Dividing proves (174). ☐

This theorem is exact at the level of minimum Leibniz-term valuations. It replaces a family of separate p-adic estimates by one finite norm sum satisfying a conservation law.

The four base and cross depths already give a useful branch-sensitive bound without knowing any external prime factorization.

Corollary 30.4 (Depth-sensitive three-vector bound)**.** *For every hypothetical primitive counterexample,*

$$\limsup_{T\to\infty} \frac{\log Q_T^{\mathrm{trop}}}{H_T} \le 1 - \frac{3}{8}\Big(\mathfrak{G}(c_2) + \mathfrak{G}(c_3) + \mathfrak{G}(-c_2 - c_3)\Big). \tag{175}$$

Equality in this upper bound can occur only when all nonzero external slope vectors are positively collinear.

Proof. The external vectors satisfy $\sum_{p\neq 2,3} c_p = -c_2 - c_3$. Hence the triangle inequality gives

$$\sum_{p\neq 2,3} \mathfrak{G}(c_p) \ge \mathfrak{G}(-c_2 - c_3).$$

Substitute this in Theorem 30.3. Equality in a norm triangle inequality requires all summands to lie on one nonnegative ray. ☐

The equality condition detects exactly the second-iterate kernel branch reviewed in the source report. Indeed the external vectors c_p are rescaled versions of $(v_p(M), v_p(A))$. They are all collinear if and only if the two external valuation vectors of M and A are rationally dependent, equivalently if the kernel K_β is nonzero. Thus:

Corollary 30.5 (The tropical norm sees the kernel dichotomy)**.** *In the multiplicatively independent branch $K_\beta = 0$, the inequality in (175) is strict. In the dependent branch $K_\beta \neq 0$, the external triangle inequality is an equality (including the axis cases in which one output is $\{2, 3\}$-smooth). For fixed M, A, the strict gap in the independent branch is explicit:*

$$\Delta_{\mathrm{ext}} = \sum_{p\neq 2,3} \mathfrak{G}(c_p) - \mathfrak{G}\left(\sum_{p\neq 2,3} c_p\right) > 0. \tag{176}$$

There is no uniform positive lower bound for Δ_{ext}: distinct rational rays can approach one another as the valuations grow. Nevertheless, the quantity is a computable branch invariant and gives a sharper determinant threshold for every fixed hypothetical primitive pair.

30.7 The universal $4/5$ ceiling

Theorem 30.6 (Sharp universal min-plus ceiling)**.** *For every hypothetical primitive counterexample,*

$$\sum_p \mathfrak{G}(c_p) \geq \frac{8}{15}. \tag{177}$$

Consequently

$$\limsup_{T\to\infty} \frac{\log Q_T^{\text{trop}}}{H_T} \leq \frac{4}{5}. \tag{178}$$

The constant $4/5$ is the sharp supremum allowed by the normalized valuation constraints, although equality would require a degenerate boundary incompatible with a genuine nonintegral solution.

Proof. By the primitive theorem, either $a = 0$ or $b = 0$. Suppose first that $a = 0$. Then $c_2 = (-1, \gamma)$ with $\gamma = c/(\beta\vartheta) \geq 0$. By (169), $\mathfrak{G}(c_2) \geq 4/15$. Conservation and the triangle inequality for the norm give

$$\sum_{p\neq 2} \mathfrak{G}(c_p) \geq \mathfrak{G}\left(\sum_{p\neq 2} c_p\right) = \mathfrak{G}(-c_2) = \mathfrak{G}(c_2).$$

Thus the total is at least $8/15$. If $b = 0$, apply the same argument to $c_3 = (\eta, -1)$, where $\eta = \vartheta d/\beta \geq 0$. Substitution in (174) gives $1 - (3/8)(8/15) = 4/5$. □

The proof is short because the difficult arithmetic has been compressed into two facts: the slope vectors sum to zero, and one structural vector reaches from -1 to a nonnegative vertex. No averaging estimate alone reveals this geometry.

30.8 Cross-coprime constants: $7/10$ and $3/10$

Assume the stronger branch condition

$$\gcd(MA, 6) = 1. \tag{179}$$

Then $a = b = c = d = 0$,

$$c_2 = (-1, 0), \qquad c_3 = (0, -1), \qquad \sum_{p\neq 2,3} c_p = (1, 1).$$

Corollary 30.7 (All-prime cross-coprime ceiling)**.** *Under* (179),

$$\limsup_{T\to\infty} \frac{\log Q_T^{\text{trop}}}{H_T} \leq \frac{7}{10}. \tag{180}$$

If the external normalized valuation vectors are not all positively collinear, the inequality is strict by a branch-dependent positive amount.

Proof. By (168), the structural vectors each contribute $4/15$. The triangle inequality gives

$$\sum_{p\neq 2,3} \mathfrak{G}(c_p) \geq \mathfrak{G}(1,1) = \frac{4}{15}.$$

Thus the total norm is at least $4/5$, and $1-(3/8)(4/5) = 7/10$. Equality in the triangle inequality requires positive collinearity. ☐

30.9 The exact valuation-slope energy below seven tenths

In the cross-coprime branch the external vectors

$$(u_p, v_p) := c_p = \left(\frac{v_p(M)\log p}{\beta \log 2}, \frac{v_p(A)\log p}{\beta\log 3}\right), \qquad p \geq 5,$$

are two finitely supported probability distributions: $\sum_p u_p = \sum_p v_p = 1$. The first-quadrant specialization of Proposition 30.2 is

$$\mathfrak{G}(u,v) = \frac{2}{15}\left(u+v+\frac{(u-v)^2}{\max(u,v)}\right), \tag{181}$$

where the final quotient is zero at $(u,v) = (0,0)$. Define the valuation-slope energy

$$\mathcal{E}_{\mathrm{val}}(M,A) = \sum_{p\geq 5} \frac{(u_p - v_p)^2}{\max(u_p,v_p)} = \frac{1}{\beta\log 3}\sum_{p\geq 5} \log p\, \frac{(v_p(M)\vartheta - v_p(A))^2}{\max(v_p(M)\vartheta, v_p(A))}, \tag{182}$$

with absent primes omitted. Summing (181) and adding the two structural vectors gives the exact refinement

$$\boxed{\frac{\log Q_T^{\mathrm{trop}}}{H_T} \longrightarrow \frac{7}{10} - \frac{1}{20}\mathcal{E}_{\mathrm{val}}(M,A).} \tag{183}$$

The energy is strictly positive. Equality in a nonzero summand would give $v_p(M)\log 3 = v_p(A)\log 2$, impossible by unique factorization; the external support is nonempty. Thus the qualitative equality case in Corollary 30.7 never occurs for an arithmetic pair.

There is also a uniform, albeit small, arithmetic separation. Put

$$\delta_p = v_p(M)\vartheta - v_p(A).$$

The identity $\sum_{p\geq 5}(\log p)\delta_p = 0$ and the irrationality of ϑ produce distinct primes $p, q \geq 5$ with $\delta_p > 0 > \delta_q$. Their integral valuation minor

$$\omega_{pq} = v_p(M)v_q(A) - v_q(M)v_p(A) \tag{184}$$

is positive, hence at least one. The normalized wedge is therefore

$$u_pv_q - u_qv_p = \frac{\log p \log q}{\beta^2 \log 2\log 3}\,\omega_{pq} \geq \frac{\kappa_\wedge}{\beta^2}, \qquad \kappa_\wedge := (\log_2 5)(\log_3 7). \tag{185}$$

For two probability vectors define $\mathrm{TV}(u,v) = \frac{1}{2}\sum_p |u_p - v_p|$. Cauchy–Schwarz and $\sum_p \max(u_p,v_p) = 1+\mathrm{TV}(u,v)$ give

$$\frac{4\,\mathrm{TV}(u,v)^2}{1+\mathrm{TV}(u,v)} \leq \mathcal{E}_{\mathrm{val}}(M,A) \leq 2\,\mathrm{TV}(u,v). \tag{186}$$

For the straddling pair, $0 < u_p v_q - u_q v_p \le \mathrm{TV}(u,v)$. Combining this with (185) yields

$$\boxed{\mathcal{E}_{\mathrm{val}}(M,A) \ge \frac{4\kappa_\wedge^2}{\beta^4+\kappa_\wedge\beta^2}, \qquad \frac{\log Q_T^{\mathrm{trop}}}{H_T} \longrightarrow \frac{7}{10} - \frac{\mathcal{E}_{\mathrm{val}}}{20} \le \frac{7}{10} - \frac{\kappa_\wedge^2}{5(\beta^4+\kappa_\wedge\beta^2)}.} \tag{187}$$

Since the finite exclusion gives $\beta > 12$, one may simplify the final deficit to $\kappa_\wedge^2/(10\beta^4) = 1.6914306423\ldots/\beta^4$.

This correction moves the guaranteed divisor in the unfavorable direction: it lies strictly *below* $7/10$. It therefore does not prove the conjecture. Its value is to remove the last equality case and to quantify the additional cancellation or divisibility that a successful rank-four determinant would have to recover.

Status: **[paper proof]** — the arithmetic specialization, straddling minor, exact energy correction, and β^{-4} separation are paper results. The abstract finite energy–total-variation and wedge-to-energy inequalities are **[kernel-verified Lean]** in `ValuationSlopeEnergy`; the prime-valuation adapter remains paper-level.

The places 2 and 3 alone have an even smaller exact continuum contribution.

Corollary 30.8 (The structural-place 3/10 constant)**.** *Under* (179)*, let* $Q_T^{(2,3)} = 2^{\tau_{2,T}} 3^{\tau_{3,T}}$*. Then*

$$\frac{\log Q_T^{(2,3)}}{H_T} \longrightarrow \frac{3}{10}. \tag{188}$$

Proof. For $c = (-1, 0)$,

$$\Phi(c) = \frac{-1}{6} - \frac{1}{4}\cdot\frac{4}{15} = -\frac{7}{30},$$

and the same holds for $(0,-1)$. Adding the constant mean $2/3$ gives $2/3 - 7/15 = 1/5$. Dividing by the height mean $2/3$ gives $3/10$. □

Equivalently, one can compute this constant from quantiles. The descending quantile of X_1 on $\mathcal{T}$ is $1-\sqrt{t}$, so

$$\int_0^1 t(1-\sqrt{t})\,dt = \frac{1}{10}, \qquad \mathbb{E}X_1 = \frac{1}{3}.$$

The ratio is $(1/10)/(1/3) = 3/10$ in each structural coordinate.

30.10 Where extra divisibility can occur

Proposition 20.3 shows that the tropical minimum equals the true determinant valuation whenever its minimizing permutation is unique.

Consequently, the missing 20% or more can arise only from primes at which the assignment problem has several minimizers. The tie geometry is explicit.

- For $p \ne 2, 3$, columns tie when $v_p(M)(u-u') + v_p(A)(v-v') = 0$, so ties lie on rational lines orthogonal to the external valuation vector.

- In the cross-coprime branch, the 2-adic cost depends only on u and the 3-adic cost only on v. Every vertical or horizontal column fiber creates many equal-cost permutations. Any leading extra valuation must be produced inside these fibers.
- Cross valuations $v_2(A)$ and $v_3(M)$ tilt the fibers. The floor term in the row profile perturbs the exact affine ordering but does not change the continuum slope vector.

This turns "look for p-adic cancellation" into a concrete algebra problem: factor or control the residual determinants inside equal-slope column fibers, and show that their valuation supplies at least the missing Gini deficit.

31 The complete triangular grid and a $\pi/4$ transport ceiling

The one-block determinant deliberately uses the exact $O(T)$ population in a unit interval of exponent space. The full two-dimensional coordinate grid instead gives, conditional on a counterexample, $\asymp T^2$ genuine integer rows below exponent height T, exactly the population that the mixed-semigroup interpolation argument seemed to be missing. It is therefore important to determine whether entrywise p-adic divisibility becomes strong enough on this larger node set.

The answer is again negative, but for a different and pleasantly universal reason. Once all places are summed, the prime-by-prime assignment minima are bounded by one antimonotone transport problem for the logarithmic entry matrix. Both the row heights and the balanced column weights have the same triangular distribution, and the ratio between antimonotone and monotone transport is exactly $\pi/4$.

31.1 Rows and columns

For $T > 0$ define the complete conditional row set

$$\mathcal{R}_T = \{(n,k) \in \mathbb{N}_0^2 : n + k\beta \le T\}, \qquad N_T = \#\mathcal{R}_T. \tag{189}$$

For $i = (n_i, k_i) \in \mathcal{R}_T$ put

$$x_i = n_i + k_i\beta, \qquad m_i = 2^{n_i} M^{k_i} = 2^{x_i}, \qquad a_i = 3^{n_i} A^{k_i} = 3^{x_i}. \tag{190}$$

Because β is irrational, the x_i are pairwise distinct.

Let $\mathcal{C}_T$ be the first N_T points $(u,v) \in \mathbb{N}_0^2$ in increasing order of

$$\lambda = u + \vartheta v, \qquad y = \log(2^u 3^v) = \lambda \log 2.$$

Write L_T for the largest selected λ. Standard lattice-point estimates for the two triangles give

$$N_T = \frac{T^2}{2\beta} + O_\beta(T), \qquad L_T = \sqrt{2\vartheta N_T} + O_\vartheta(1) = T\sqrt{\frac{\vartheta}{\beta}} + O_\beta(1). \tag{191}$$

The associated square determinant is

$$D_T^{\triangle} = \det\Big(m_i^{u_j} a_i^{v_j}\Big)_{i,j=1}^{N_T} = \det(e^{x_i y_j})_{i,j=1}^{N_T}. \tag{192}$$

Every entry is an integer. Ordering the x_i and y_j increasingly, strict total positivity of the exponential kernel makes $D_T^{\triangle} > 0$; equivalently, this is the same exponential-polynomial zero theorem used by the generalized-Vandermonde part of the source report.

Status: [paper proof] for the asymptotic transport statements below. Integrality and nonvanishing are inherited from the mixed-semigroup determinant architecture [kernel-verified Lean].

31.2 The common triangular distribution

The normalized row heights $h_i = x_i/T$ and column weights $\ell_j = y_j/(L_T \log 2)$ both live in $[0,1]$. Their empirical measures have the same limit.

Lemma 31.1 (Triangular height law)**.** *For every continuous $f : [0,1] \to \mathbb{R}$,*

$$\frac{1}{N_T}\sum_{i=1}^{N_T} f(h_i) \longrightarrow \int_0^1 2t f(t)\,dt, \tag{193}$$

$$\frac{1}{N_T}\sum_{j=1}^{N_T} f(\ell_j) \longrightarrow \int_0^1 2t f(t)\,dt. \tag{194}$$

Thus the common limiting distribution has cumulative distribution function $F(t) = t^2$ and quantile function

$$Q(s) = \sqrt{s}, \qquad 0 \le s \le 1. \tag{195}$$

Proof. After scaling (n,k) to $(n/T, \beta k/T)$, the row lattice becomes equidistributed in the unit right triangle

$$\{(r,s) : r, s \ge 0,\ r+s \le 1\}.$$

The subtriangle on which $r+s \le t$ has area $t^2/2$, while the whole triangle has area $1/2$. Hence the height $r+s$ has distribution function t^2. Boundary discrepancies are $O(T)$ against $N_T \asymp T^2$, proving (193).

For the columns, scale (u,v) to $(u/L_T, \vartheta v/L_T)$. The same unit triangle appears, and $\ell = (u+\vartheta v)/L_T$ is again the sum of the scaled coordinates. The boundary shell created by taking the first exactly N_T points has $O(L_T)$ points against $N_T \asymp L_T^2$, which proves (194). □

31.3 Summing all primewise assignment minima

For each prime p, let

$$\tau_{p,T} = \min_{\sigma \in S_{N_T}} \sum_{i=1}^{N_T} v_p\Big(m_i^{u_{\sigma(i)}} a_i^{v_{\sigma(i)}}\Big), \qquad Q_T^{\triangle} = \prod_p p^{\tau_{p,T}}. \tag{196}$$

As before, $Q_T^{\triangle}$ divides every Leibniz term and hence divides $D_T^{\triangle}$. The key observation is that taking a separate minimizing permutation at each prime can only decrease the sum relative to using one permutation at all primes.

Lemma 31.2 (All-place transport domination)**.** *For every T,*

$$\log Q_T^{\triangle} \le \min_{\sigma \in S_{N_T}} \sum_{i=1}^{N_T} x_i y_{\sigma(i)}. \tag{197}$$

Proof. For a permutation σ, put

$$C_p(\sigma) = \sum_i v_p\Big(m_i^{u_{\sigma(i)}} a_i^{v_{\sigma(i)}}\Big) \log p.$$

Then

$$\sum_p \min_\sigma C_p(\sigma) \le \min_\sigma \sum_p C_p(\sigma).$$

Unique factorization and (190) give, entry by entry,

$$\sum_p v_p(m_i^{u_j} a_i^{v_j}) \log p = \log(m_i^{u_j} a_i^{v_j}) = x_i y_j.$$

Substitution proves (197). □

Notice how little arithmetic remains in the last inequality: it is valid in every valuation branch, includes all primes, and does not use the primitive-depth normalization. The price is that it is only an upper bound for the common termwise divisor. That is exactly what is needed for a no-go theorem.

31.4 Optimal transport and the $\pi/4$ ratio

Let

$$E_T^- = \min_{\sigma \in S_{N_T}} \sum_i x_i y_{\sigma(i)}, \tag{198}$$

$$E_T^+ = \max_{\sigma \in S_{N_T}} \sum_i x_i y_{\sigma(i)}. \tag{199}$$

The rearrangement inequality pairs the two ordered lists oppositely for E_T^- and in the same order for E_T^+.

Theorem 31.3 (Triangular-grid transport constants)**.** *As $T \to \infty$,*

$$\frac{E_T^-}{N_T T L_T \log 2} \longrightarrow \int_0^1 \sqrt{t(1-t)}\,dt = \frac{\pi}{8}, \tag{200}$$

$$\frac{E_T^+}{N_T T L_T \log 2} \longrightarrow \int_0^1 t\,dt = \frac{1}{2}. \tag{201}$$

Consequently

$$\frac{E_T^-}{E_T^+} \longrightarrow \frac{\pi}{4} = 0.7853981633\ldots. \tag{202}$$

Proof. By Lemma 31.1, the increasing empirical quantile functions of the two normalized lists converge in L^1 to $Q(t) = \sqrt{t}$. The discrete rearrangement sums are Riemann sums for the monotone and antimonotone couplings. Thus

$$\frac{E_T^-}{N_T T L_T \log 2} \longrightarrow \int_0^1 Q(t)Q(1-t)\,dt, \qquad \frac{E_T^+}{N_T T L_T \log 2} \longrightarrow \int_0^1 Q(t)^2\,dt.$$

The first integral is the beta integral $B(3/2, 3/2) = \Gamma(3/2)^2/\Gamma(3) = \pi/8$; the second is $1/2$. ☐

The determinant has $N_T!$ Leibniz terms, each at most $e^{E_T^+}$. Since $N_T \asymp T^2$ while $T L_T N_T \asymp T^4$,

$$\log(N_T!) = O(T^2 \log T) = o(N_T T L_T). \tag{203}$$

Combining the preceding statements gives the promised ceiling.

Corollary 31.4 (Full-grid termwise-divisor ceiling)**.** *With the notation above,*

$$\limsup_{T\to\infty} \frac{\log Q_T^{\triangle}}{E_T^+ + \log(N_T!)} \le \frac{\pi}{4}. \tag{204}$$

Thus the complete $\asymp T^2$ row grid does not make the all-prime common divisor exceed the sharp largest-Leibniz-term archimedean bound. Entrywise valuation alone leaves at least the fraction $1 - \pi/4 = 0.2146018\ldots$ uncovered at leading order.

Proof. Lemma 31.2 gives $\log Q_T^{\triangle} \le E_T^-$. Divide by the standard determinant upper bound $E_T^+ + \log(N_T!)$ and apply Theorem 31.3 and (203). ☐

Caution 31.5 (What the $\pi/4$ theorem does not exclude)**.** The denominator in (204) is the largest-term upper bound, not the true size of $D_T^{\triangle}$. Generalized Vandermonde determinants exhibit substantial alternating cancellation. A sufficiently sharp archimedean estimate could lower the relevant budget below E_T^-, in which case the termwise divisor might become decisive. The theorem excludes the naive "more rows plus the same Hadamard/Leibniz estimate" route; it does not exclude a joint p-adic/free-energy argument.

31.5 The Harish–Chandra–Itzykson–Zuber identity

The exact analytic object behind that warning is classical. For diagonal Hermitian matrices $X = \operatorname{diag}(x_1, \ldots, x_N)$ and $Y = \operatorname{diag}(y_1, \ldots, y_N)$ with distinct increasing entries, normalized Haar measure on the unitary group satisfies

$$\int_{U(N)} \exp\bigl(\operatorname{Tr}(XUYU^*)\bigr)\,dU = \frac{\prod_{r=1}^{N-1} r!}{\Delta(x)\Delta(y)} \det(e^{x_i y_j})_{i,j=1}^N, \tag{205}$$

where $\Delta(x) = \prod_{i<j}(x_j - x_i)$ and similarly for $\Delta(y)$ [51, 52, 53]. Hence

$$\log D_T^{\triangle} = \log \Delta(x) + \log \Delta(y) - \sum_{r=1}^{N_T - 1} \log(r!) + \log \mathcal{I}_T, \tag{206}$$

with $\mathcal{I}_T$ the unitary integral in (205).

The large powers of T hidden separately in the two Vandermonde factors and the factorial product cancel at leading logarithmic scale. What remains is a genuine free-energy problem for two triangular empirical measures. The Baker separation theorem of Section 28 gives a polynomial lower bound for the spacings in $\Delta(x)$, so the row Vandermonde cannot collapse through anomalously close conditional exponents. What is not known is a bound on the full combination (206) strong enough to beat the primewise divisor.

Research target 31.6 (HCIZ–valuation comparison)**.** Determine the T^4-scale limit, or a sufficiently sharp upper bound, of

$$\log D_T^{\triangle} - \log Q_T^{\triangle}$$

for the conditional triangular row lattice and balanced mixed columns. It would suffice to prove that this quantity is negative for all sufficiently large T. Corollary 31.4 shows that such a result must use either the HCIZ/Vandermonde cancellation in (206), extra p-adic cancellation beyond the assignment minima, or both.

This formulation has a useful advantage over a generic interpolation determinant: both limiting measures and all lattice densities are explicit. It is therefore suitable for numerical free-energy reconnaissance before attempting a proof.

32 The first tied p-adic layer is lower order

Theorem 30.6 says that extra divisibility must come from ties among minimum-valuation Leibniz terms. Ties are especially abundant in the cross-coprime branch (179): at $p = 2$ all columns with the same u have the same valuation profile, and at $p = 3$ all columns with the same v do. It is natural to hope that cancellation inside those large fibers supplies the missing leading-order fraction.

The present section computes that first cancellation layer exactly. It factors into ordinary Vandermondes of the odd row parts. There are two useful bounds. Standard upper bounds for p-adic linear forms in two logarithms give the sharper $O(T^{3/2}(\log T)^C)$ estimate. More importantly for formalization, the elementary fact $p^{v_p(z)} \le |z|$ already gives an unconditional $O(T^2)$ bound. Either is lower order than the mixed determinant height $\asymp T^{5/2}$. This does not bound the final determinant valuation—higher tropical layers may cancel again—but it rules out a one-step fiber explanation of the missing fraction without requiring deep p-adic approximation theory.

32.1 Tropical initial form at two

Assume throughout this section that $\gcd(MA, 6) = 1$. For the block rows, write

$$q_k = a_{T,k} = 3^{n_k} A^k, \qquad n_k = T - \lfloor k\beta \rfloor. \tag{207}$$

Both M and every q_k are odd, and

$$m_{T,k}^u a_{T,k}^v = 2^{un_k} M^{ku} q_k^v. \tag{208}$$

For each integer $u \ge 0$, let

$$V_u = \{v : (u, v) \text{ is one of the selected } N \text{ columns}\}, \qquad s_u = \#V_u.$$

Because columns are selected by the inequality $u + \vartheta v \le L_N$ up to one boundary shell, each nonempty V_u is an initial interval of integers. Sort the distinct u values in increasing order. Since $n_0 > n_1 > \cdots > n_{N-1}$, every minimizing assignment for the cost un_k sends one consecutive row block R_u of size s_u to the columns in the fiber $\{u\} \times V_u$.

Let $\tau_{2,T}$ be the minimum from (162). The sum of all Leibniz terms having exactly this valuation is the 2-adic tropical initial form. Up to an overall sign and an odd factor, it is

$$I_{2,T} = \prod_u \det(q_k^v)_{k\in R_u,\ v\in V_u} = \prod_u \prod_{\substack{i<j \\ i,j\in R_u}} (q_j - q_i). \tag{209}$$

The first equality follows by grouping the optimal permutations by fibers and factoring the odd number M^{ku} from each row in a fixed u block; the second is the ordinary Vandermonde identity because V_u is consecutive.

Proposition 32.1 (Exact first 2-adic initial form)**.** *There is an integer $Z_{2,T}$ such that*

$$2^{-\tau_{2,T}} D_T = \varepsilon_{2,T} I_{2,T} + 2^{\lfloor\beta\rfloor} Z_{2,T}, \tag{210}$$

where $\varepsilon_{2,T}$ is odd. Consequently

$$v_2(D_T) - \tau_{2,T} \ge \min\{v_2(I_{2,T}), \lfloor\beta\rfloor\}. \tag{211}$$

No equality assertion follows without controlling the valuation of the full outside-fiber remainder $Z_{2,T}$.

Proof. The rearrangement inequality with repeated column weights identifies the minimizing permutations exactly as above, and summing their signed unit parts gives (209). Any nonminimizing assignment must move at least one column across two adjacent u fibers. An exchange across the corresponding boundary changes the cost by

$$(u' - u)(n_i - n_j).$$

Here $u' - u \ge 1$, while consecutive row depths differ by $\lfloor(k+1)\beta\rfloor - \lfloor k\beta\rfloor \ge \lfloor\beta\rfloor$. Hence every nonminimal Leibniz term has valuation at least $\tau_{2,T} + \lfloor\beta\rfloor$, proving the expansion. The valuation statement follows from the ultrametric inequality. □

32.2 The symmetric initial form at three

Put

$$r_k = m_{T,k} = 2^{n_k} M^k. \tag{212}$$

At $p = 3$ the entry factors as

$$m_{T,k}^u a_{T,k}^v = 3^{vn_k}\, r_k^u A^{kv}.$$

For a selected value of v, let U_v be the consecutive set of selected u values and let R'_v be the corresponding consecutive block of rows in the minimum assignment. The same argument gives

$$I_{3,T} = \prod_v \prod_{\substack{i<j \\ i,j\in R'_v}} (r_j - r_i) \tag{213}$$

and

$$3^{-\tau_{3,T}} D_T = \varepsilon_{3,T} I_{3,T} + 3^{\lfloor\beta\rfloor} Z_{3,T}, \tag{214}$$

with $\varepsilon_{3,T}$ a 3-adic unit.

32.3 Two-logarithm bounds inside the fibers

For $i < j$, set $d = j - i$ and $h = n_i - n_j = \lfloor j\beta \rfloor - \lfloor i\beta \rfloor$. The common factors in the row values are units at the relevant place, so

$$v_2(q_j - q_i) = v_2(A^d - 3^h), \tag{215}$$

$$v_3(r_j - r_i) = v_3(M^d - 2^h). \tag{216}$$

Neither difference vanishes: $A^d = 3^h$ or $M^d = 2^h$ would give $d\beta = h$, contrary to the irrationality of β.

A standard consequence of p-adic linear-form theory is that for fixed multiplicatively independent positive rationals α_1, α_2 and a fixed prime p, there are effectively computable constants C_0, C_1 such that

$$v_p(\alpha_1^d - \alpha_2^h) \le C_0 + C_1\big(\log(2 + \max\{d, h\})\big)^2. \tag{217}$$

Much sharper versions, with linear dependence on $\log B$, are known in important two-logarithm regimes; the square is more than sufficient here [43, 44, 45]. Apply this to $(A, 3, p = 2)$ and $(M, 2, p = 3)$. Since $d, h = O_\beta(N)$, each factor in (209)–(213) has valuation $O_{M,A}((\log N)^2)$.

For the qualitative scale separation, however, no linear-form theorem is needed. If p is prime and $a^d \ne b^h$, the divisor-size inequality gives the explicit finite bound

$$v_p(a^d - b^h) \le 1 + d\,\lceil \log_p a \rceil + h\,\lceil \log_p b \rceil. \tag{218}$$

Indeed, $p^{v_p(a^d - b^h)}$ divides the nonzero difference, while $|a^d - b^h| \le a^d + b^h$ and each power is bounded by the corresponding power of p. Inside a Beatty row block, $h \le B_\beta d$ for a fixed integer B_β, so the right side is $O_{M,A}(d)$. A block of size s then contributes $O_{M,A}(s^3)$, because $\sum_{0 \le i < j < s}(j - i) = O(s^3)$. There are $O(L_N)$ fibers of size $O(L_N)$, hence the total elementary budget is

$$O_{M,A}(L_N^4) = O_{M,A}(N^2) = O_{M,A}(T^2). \tag{219}$$

The number of pairs within the vertical fibers is

$$\sum_u \binom{s_u}{2} = \frac{L_N^3}{6\vartheta^2} + O_\vartheta(L_N^2) = O_\vartheta(N^{3/2}), \tag{220}$$

while the number within the horizontal fibers is

$$\sum_v \binom{\#U_v}{2} = \frac{L_N^3}{6\vartheta} + O_\vartheta(L_N^2) = O_\vartheta(N^{3/2}). \tag{221}$$

We obtain the following rigorous scale separation.

Theorem 32.2 (First tied layer is lower order)**.** *In the cross-coprime branch,*

$$v_2(I_{2,T}) + v_3(I_{3,T}) = O_{M,A}(N^2) = O_{M,A}(T^2). \tag{222}$$

If one invokes the quoted two-logarithm estimate (217)*, the stronger bound $O_{M,A}(N^{3/2}(\log N)^2)$ also holds. In particular,*

$$\frac{v_2(I_{2,T})\log 2 + v_3(I_{3,T})\log 3}{H_T} \longrightarrow 0. \tag{223}$$

Proof. For the elementary assertion, sum (218) over the pairs in each fiber and use (219). For the sharper assertion, sum (217) over the pair counts (220) and (221), using (215)–(216). Finally use $N \asymp_\beta T$ and $H_T \asymp_\beta T^{5/2}$. □

Status: **[kernel-verified Lean]** for the exact finite valuation and cube-budget skeleton; [**classical/open background**] only for the optional sharper two-logarithm estimate.

The module `BlockVandermondeValuationBudget` proves the exact collision-depth identity `padicValNat_natVandermondeProduct_eq_sum_collisionCount`, the elementary power-difference bound `padicValNat_natAbs_pow_sub_pow_le`, and the variable-block and power-difference cube bounds `padicValNat_blockVandermondeNatProduct_le_cubeBudget` and `padicValNat_blockVandermondeNatProduct_le_powDifferenceCubeBudget`. The displayed $O(T^2)$ specialization still uses the elementary paper-level count of Beatty fiber sizes; the finite inequality on which it rests is kernel-verified. The sharper $O(T^{3/2}(\log T)^2)$ statement uses the cited classical estimate and is not claimed as a Lean theorem.

32.4 What a leading excess would have to do

Theorem 32.2 does not prove $v_2(D_T) - \tau_{2,T} = o(H_T)$ or its 3-adic analogue. If the initial form is divisible by a power exceeding the first tropical gap, terms from the next valuation layer enter; those can cancel, exposing another layer, and so on. A leading-order excess is therefore possible only through control of the valuations and cancellations in the truncated sums at cutoffs growing with T. It could arise from unusually divisible low-excess minor products, cancellation within one layer, or carrying across several layers. Cancellation confined to the minimum-permutation fibers is quantitatively negligible. Formula (84) makes this statement finite and exact: at depth K one need only control the signed products of unit minors attached to row partitions of excess less than K. The unit-adapted bases above extract a universal divisor from the relevant univariate minors, but its paper-level blockwise scale is only $O(N^{3/2})$. What is missing is an upper bound on the valuation of the residual truncated sum, uniform for cutoffs growing with T.

This yields a more precise version of the p-adic research target.

Research target 32.3 (No shallow-layer proof)**.** Either prove that the tropical cancellation cascade has total depth $o(H_T)$—which would retire the single-block p-adic strategy—or identify a structural identity forcing $\gg H_T$ cumulative cancellation across successively nonminimal assignments. Pairwise congruences such as (215) and (216) cannot by themselves supply the latter, because p-adic logarithm bounds make each congruence only polylogarithmic in the row separation.

32.5 Exact finite cascade reconnaissance

Exact integer experiments help rule out two tempting but false simplifications of Target 32.3. For synthetic cross-coprime parameters M, A, put

$$\beta = \log_2 M, \qquad n_k = T - \lfloor k\beta \rfloor,$$

and take the first $N = 1 + \lfloor (T+1)/\beta \rfloor$ exponent pairs (u, v) in increasing order of $u + \vartheta v$. The tested determinant is exactly

$$D_T = \det\left((2^{n_k} M^k)^u (3^{n_k} A^k)^v\right)_{0 \le k < N, (u,v)}.$$

Let $e_p = v_p(D_T) - \tau_p$, let I_p denote the first-fiber Vandermonde product, and let U_p be the common row-block ordering exponent determined by the column-fiber sizes. The weights come from `AllLayerUnitBasis`, their finite sums from `UnitOrderingGain`, and their uniform contribution to every row-block term and the full determinant from `RowBlockUnitDivisibility`. For $(M, A) = (5, 7)$, exact DVR elimination gives:

T	N	e_2	U_2	$v_2(I_2)$	e_3	U_3	$v_3(I_3)$	$(e_2 \log 2 + e_3 \log 3)/H_T$
30	14	26	17	29	14	9	14	0.0360124
100	44	157	136	157	87	74	87	0.0101461
300	130	868	796	870	517	471	530	0.00353027

The data are compatible with an $N^{3/2}$ excess and incompatible with no proved asymptotic; their purpose is diagnostic.

Two exact small cases are more informative. At $(M, A, T, p) = (5, 7, 30, 2)$ one has

$$\tau_2 = 177, \qquad v_2(D_T) = 203, \qquad e_2 = 26 < 29 = v_2(I_2).$$

Writing $2^{-\tau_2} D_T = \sum_\delta 2^\delta C_\delta$, the first layer has $v_2(C_0) = 29$, whereas the layer $\delta = 6$ has $v_2(C_6) = 20$ and therefore smaller effective valuation 26. A nonminimal layer can genuinely beat the initial form. Conversely, at $(M, A, T, p) = (11, 5, 20, 2)$ the layers $\delta = 0$ and $\delta = 4$ both have effective valuation 9, but their sum is divisible by 2^{10}; all other layers have effective valuation at least 12, and the final excess is 10. Thus even "take the minimum effective layer" is false because different layers can carry into one another.

Status: [exact computation] — all displayed determinant, assignment, layer, and valuation integers use exact integer arithmetic; the logarithmic ratios are ordinary floating approximations used only diagnostically. The no-dependency scripts `scripts/cascade_experiments.py`, `scripts/cascade_layer_report.py`, and `scripts/cascade_verify.py` reproduce the computations; the last directly Bareiss-checks the same twenty small determinants and agrees on their valuations.

The parameters are deliberately synthetic: they satisfy $\gcd(6, MA) = 1$, but do not assert the unknown identity $A = 3^{\log_2 M}$. The decreasing ratios and the apparent $N^{3/2}$ scale are evidence only. The two finite counterexamples to naive layer formulas are exact for the displayed mixed determinant family.

33 Exact correction of heuristic tropical ceilings

Two natural averaging heuristics propose a structural 1/2 ceiling and an all-prime 2/3 ceiling for termwise min-plus divisibility. Both preserve the correct distinction between tropical divisibility and cancellation beyond the minimum, but neither constant is valid in the stated generality.

The structural averaging argument omitted the base and cross depths that appear in (170)–(171). Even under $v_2(M) = 0$, the terms $v_2(A)kv$ and $v_3(M)ku + v_3(A)kv$ do not disappear. The exact structural contribution is obtained by retaining c_2 and c_3 in Theorem 30.3; it can exceed one half of H_T.

The all-prime step used the assertion that an oppositely sorted nonnegative triangular weight has assignment minimum at most two thirds of its mean. For the weight $Y = X + Z$ on the standard triangle, the exact values from Lemma 30.2 are

$$\mathbb{E}Y = \frac{2}{3}, \qquad \Phi(1,1) = \frac{4}{15}, \qquad \frac{\Phi(1,1)}{\mathbb{E}Y/2} = \frac{4}{5}.$$

Thus 4/5, not 2/3, is already attained by the relaxed external rank-one grading. The Gini-norm identity explains the general case and leads to the rigorous ceiling in Theorem 30.6.

Status: [paper proof] — the correction follows from the exact transport formula, not from numerical experimentation.

34 Directed-rounding finite verification through 2^{27}

The reproducible MPFR scan excludes every non-power-of-two candidate below 2^{27}. The interval $[2^{26}, 2^{27})$ was divided into sixteen chunks of width 2^{22} and checked with 192-bit outward-rounded endpoints; all 67,108,863 non-power inputs passed. The verifier and invocation details are in Appendix I.

The smallest new lower and upper logarithmic margins are respectively

$$5.9082259992879571 \times 10^{-21} \quad \text{at } (132{,}833{,}765, 7{,}501{,}348{,}219{,}569),$$

and

$$8.3073231820333117 \times 10^{-22} \quad \text{at } (129{,}799{,}345, 7{,}231{,}571{,}292{,}851).$$

Both inputs were replayed independently at 320 and 512 bits.

Theorem 34.1 (Software-level finite exclusion through 2^{27})**.** *At the stated MPFR trust level, for every integer $1 \le m < 2^{27}$,*

$$m^{\vartheta} \in \mathbb{N} \quad \Longrightarrow \quad m \textit{ is a power of two.}$$

Consequently a nonintegral solution satisfies $2^x > 2^{27}$ and hence

$$x > 27. \tag{224}$$

Status: [software computation] — reproducible directed-rounding software, not a kernel proof.

The formal finite floor remains the Lean-certified bound; this scan is a separately labelled computational extension.

35 Additional proof routes pushed to their failure points

The determinant calculations were the main new line of attack, but they were not the only one. This section records four other routes in enough detail to prevent the same hidden loss from being rediscovered under different notation.

35.1 Transferring convergents of β to ϑ

Let p/q be a continued-fraction convergent of a hypothetical primitive generator β and put

$$\varepsilon = q\beta - p.$$

For all sufficiently large q, $|\varepsilon| < 1/2$. Define the two nonzero integer gaps

$$D_2 = M^q - 2^p = 2^p(2^\varepsilon - 1), \qquad D_3 = A^q - 3^p = 3^p(3^\varepsilon - 1). \tag{225}$$

They manufacture an exact rational approximation to ϑ:

$$R_{p,q} := \frac{2^p D_3}{3^p D_2} = \frac{3^\varepsilon - 1}{2^\varepsilon - 1} \in \mathbb{Q}. \tag{226}$$

Taylor expansion at zero gives

$$R_{p,q} = \vartheta + \tfrac{1}{2}\log_2 3\,(\log 3 - \log 2)\,\varepsilon + O(\varepsilon^2), \tag{227}$$

so $|R_{p,q} - \vartheta| \ll |\varepsilon|$.

This looks promising because ϑ has an explicit irrationality measure. Wu's simultaneous linear-independence estimate for $1, \log 2, \log 3, \log 5$ gives, after setting the unused coefficients to zero, a conservative exponent

$$\left|\vartheta - \frac{r}{s}\right| \geq c_\vartheta s^{-\mu_\vartheta}, \qquad \mu_\vartheta = 8.616 \tag{228}$$

for all sufficiently large denominators s [46]. The obstacle is the denominator created by (226). Before reduction it is at most

$$3^p|D_2| \ll 3^p 2^p |\varepsilon| = 6^p|\varepsilon|. \tag{229}$$

Combining (227)–(229) with (228) gives only

$$|\varepsilon| \gg 6^{-p\mu_\vartheta/(1+\mu_\vartheta)}. \tag{230}$$

Since

$$6^{-\mu_\vartheta/(1+\mu_\vartheta)} = 2^{-2.316\ldots},$$

this is substantially weaker than the elementary bound $|\varepsilon| \gg 2^{-p}$ obtained directly from $D_2 \neq 0$. Sharper real irrationality measures cannot repair the architecture: every irrationality exponent is at least two, and the unreduced denominator already contains both 2^p and 3^p.

Proposition 35.1 (Exact denominator-loss diagnosis)**.** *The convergent-transfer construction can improve the elementary gap only if one proves an exponential lower bound for*

$$G_{p,q} = \gcd(2^p D_3, 3^p D_2) \tag{231}$$

that survives reduction of $R_{p,q}$. In the absence of such a bound, every known irrationality measure for ϑ is defeated by the factor 6^p in (229).

The two gaps share the same small real parameter ε, but real closeness alone does not impose the exponentially large common prime divisor required here.

35.2 Two-row determinants and fixed-support S-units

For two block rows $i < j$, put $d = j - i$ and $h = n_i - n_j = \lfloor j\beta \rfloor - \lfloor i\beta \rfloor$. Their 2×2 determinant has the exact factorization

$$m_{T,i}a_{T,j} - m_{T,j}a_{T,i} = 6^{n_j}(MA)^i(2^hA^d - 3^hM^d). \tag{232}$$

Moreover

$$\frac{2^hA^d}{3^hM^d} = \left(\frac{3}{2}\right)^{\beta d - h}. \tag{233}$$

Thus a good rational approximation h/d to β makes the bracket in (232) a small *relative* difference of two integers supported on the fixed prime set dividing $6MA$.

Let

$$U = 2^hA^d, \qquad V = 3^hM^d, \qquad G = \gcd(U, V).$$

The reduced nonzero integer $(U - V)/G$ gives

$$|\beta d - h| \gg_{M,A} \frac{G}{\max\{U, V\}}. \tag{234}$$

The exponent of the right-hand side can be written explicitly as the positive part of a valuation-vector difference:

$$\log\frac{U}{G} = \sum_p \max\{h\mathbf{1}_{p=2} + dv_p(A) - h\mathbf{1}_{p=3} - dv_p(M), 0\}\log p. \tag{235}$$

As $h/d \to \beta$, this is $\kappa_{M,A}d + o(d)$ for a positive constant $\kappa_{M,A}$. Hence (234) is merely exponential in d, much weaker than the Baker–Matveev polynomial separation of Section 28.

The *abc* conjecture applied to the reduced equation $U/G - V/G = (U - V)/G$ would make the difference nearly as large as a fixed positive power of U/G, but it would still allow a relative gap $e^{-\epsilon d}$. Continued-fraction gaps of order $1/d$ are much larger. Thus this direct S-unit equation does not become a proof even after inserting *abc*; one would need an additional theorem controlling the prime support or radical of the difference.

35.3 Automaticity and the synchronized Sturmian word

The coding $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$ from Section 27 is aperiodic Sturmian. A standard theorem in automatic-sequence theory says that no aperiodic Sturmian word is automatic in any integer base [49, 50]. One quick obstruction is frequency: the frequency of 1 in s is the irrational number α, whereas an automatic sequence having a letter frequency has rational frequency.

This observation blocks a tempting but invalid shortcut. The rationality of $U = 2^\alpha$ does *not* make the threshold itinerary of $R \mapsto UR/2^{\mathbf{1}_{UR\geq 2}}$ finite-state; its exact rational denominators grow with time. The same is true for V in base three. Therefore Cobham's theorem cannot be invoked merely because the same word is produced by a dyadic and a triadic rational system. A viable automata proof would first have to manufacture genuine finite kernels in two multiplicatively independent bases, and doing so is already a new arithmetic theorem about U and V.

35.4 Real and p-adic logarithms

At the real place the counterexample is exactly

$$\log M\ \log 3 - \log A\ \log 2 = 0. \tag{236}$$

At $p = 2$ or 3, after stripping valuation and torsion parts, the units among $M, A, 2, 3$ admit p-adic logarithms. The determinant entries then have unit parts whose p-adic logarithms are bilinear forms in the row and column coordinates. This is the natural language for the cancellation cascade isolated in Section 32.

What is missing is a transfer principle from (236) to a congruence or rank drop among those p-adic logarithms. Real exponentiation by the transcendental number β does not define a compatible p-adic exponent, and the equality $M = 2^\beta$ has no p-adic meaning. Known p-adic linear-form theorems control a nonzero form once its integer coefficients and algebraic bases are specified; they do not create a p-adic form from a real one.

Research target 35.2 (Real/p-adic compatibility)**.** Find an algebraic construction, valid for the integer pair (M, A) alone, that converts (236) into either

1. a nontrivial congruence among p-adic unit logarithms with precision $\gg T$, repeated over $\gg T^{3/2}$ fiber pairs; or
2. a rank drop in the tropical initial matrices persisting through $\gg T$ valuation layers.

Without such a construction, ordinary p-adic logarithm bounds point in the opposite direction: they show that individual congruences are only polylogarithmically deep.

35.5 Transcendental-curve point counting

The graph $y = x^\vartheta$ is definable in the real exponential field, so Pila–Wilkie type results [54] give subpolynomial bounds for rational points outside algebraic pieces. Those generic bounds are far too weak here. The curve already contains the infinite sequence $(2^n, 3^n)$, giving $\asymp \log H$ integer points of height at most H, while a hypothetical second generator produces $\asymp (\log H)^2$ points globally. A proof by point counting would therefore need a sharp $O(\log H)$ theorem with a structural classification of the extremal sequence, not merely $O(H^\epsilon)$. Such a theorem would essentially be another form of the rank-one conclusion sought by the conjecture.

36 Exact sufficient conditions extracted from the new analysis

A useful continuation should not stop at saying that the present determinant fails. The calculations above isolate numerical thresholds which, if crossed by a future theorem, would settle the conjecture. This section records those thresholds as precise implications.

36.1 The mixed-block cancellation-excess criterion

Let

$$\mathcal{P} = \{p : p \mid 6MA\}$$

be the fixed support of all determinant entries, and retain the mixed block determinant D_T, its tropical minima $\tau_{p,T}$, and the archimedean scale H_T. Define the support-prime cancellation excess

$$\mathcal{E}_T = \sum_{p \in \mathcal{P}} \left(v_p(D_T) - \tau_{p,T}\right) \log p \geq 0 \tag{237}$$

and the exact tropical deficit

$$\delta(M, A) = \frac{3}{8} \sum_{p \in \mathcal{P}} \mathfrak{G}(c_p). \tag{238}$$

Theorem 30.3 says $\log Q_T^{\mathrm{trop}} = (1 - \delta(M, A) + o(1))H_T$, while the elementary Leibniz upper bound is $\log D_T \leq H_T + o(H_T)$.

Theorem 36.1 (Cancellation-excess criterion)**.** *If a hypothetical primitive counterexample satisfies*

$$\liminf_{T \to \infty} \frac{\mathcal{E}_T}{H_T} > \delta(M, A), \tag{239}$$

then no such counterexample exists. In particular it is enough to prove an excess greater than $H_T/5$ uniformly in all primitive branches; in the cross-coprime branch the exact threshold is at least $3H_T/10$.

Proof. The selected support primes alone give

$$\log D_T \geq \sum_{p \in \mathcal{P}} v_p(D_T) \log p = \log Q_T^{\mathrm{trop}} + \mathcal{E}_T.$$

Under (239), the right-hand side exceeds H_T by a fixed positive multiple for all large T, contradicting the Leibniz upper bound. The numerical thresholds follow from Theorem 30.6 and Corollary 30.7. □

This criterion is not tautological: it involves only valuations at the fixed finite set of primes already present in $6MA$, whereas D_T can acquire arbitrary new prime factors. It states exactly how much same-support cancellation is required.

36.2 A denominator-cancellation criterion

For the transferred approximant $R_{p,q}$ of Section 35.1, let $s_{p,q}$ be its reduced denominator. If one could prove along infinitely many convergents that

$$s_{p,q} \ll q^{\eta} \qquad \text{for some } \eta < \frac{1 + \mu_\vartheta}{\mu_\vartheta}, \tag{240}$$

then Wu's measure would give $|R_{p,q} - \vartheta| \gg q^{-\eta\mu_\vartheta}$, whereas (227) and the convergent bound give $|R_{p,q} - \vartheta| \ll q^{-1}$. Since $\eta\mu_\vartheta < 1 + \mu_\vartheta$ is not by itself enough, one tracks $|\varepsilon| < 1/q_+$ more precisely; a contradiction follows whenever $q_+ \gg q^{\eta\mu_\vartheta + \epsilon}$ along the same subsequence. A simpler sufficient condition is

$$s_{p,q} = q^{o(1)} \tag{241}$$

for infinitely many convergents with $q_+ \geq q^{1+\epsilon}$.

Equivalently, the gcd $G_{p,q}$ in (231) would have to cancel essentially the entire exponential factor $6^p|\varepsilon|$, leaving a subpolynomial denominator. This is a sharp formulation of the otherwise vague request for "correlation between the two integer gaps."

36.3 An automaticity criterion

The synchronized coding gives another exact, if presently inaccessible, implication.

Proposition 36.2 (Finite-kernel criterion)**.** *If under a hypothetical counterexample the common word $s_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$ were automatic in any integer base $q \geq 2$, then the Alaoglu–Erdős conjecture would follow.*

Proof. An automatic Sturmian word is ultimately periodic. Its slope, equal to the frequency of the letter 1, is therefore rational. But $\alpha = \beta - \lfloor \beta \rfloor$ would then be rational, and the rational-exponent theorem would make β integral. □

The missing step is not symbolic dynamics; it is proving automaticity from the two rational multipliers U and V. The growing denominators in (103) show exactly why the obvious state space is infinite.

37 Power-saving counting from logarithmic separation

The exact cumulative count in Section 4.5 has an $O(T)$ remainder without using Diophantine approximation. The effective finite irrationality measure from Theorem 28.1 sharpens that remainder by controlling the discrepancy of the rotation sequence $\{k\beta\}$.

Proposition 37.1 (Quadratic counting with a power saving)**.** *Assume a primitive counterexample β exists. There is an effective $\delta > 0$ such that*

$$\#(\mathcal{S} \cap [0,T]) = \frac{T^2}{2\beta} + \frac{\beta+1}{2\beta}T + O_\beta(T^{1-\delta}).$$

Proof. Put $K = \lfloor T/\beta \rfloor$. The exact monoid gives

$$\#(\mathcal{S} \cap [0,T]) = \sum_{k=0}^{K} \big(\lfloor T - k\beta \rfloor + 1\big).$$

Theorem 28.1, the Erdős–Turán inequality, and the geometric-series bound for $\sum_{k \leq K} e^{2\pi i h k\beta}$ give

$$\sum_{k=0}^{K} \{T - k\beta\} = \frac{K+1}{2} + O(K^{1-\delta})$$

for an effective $\delta > 0$. Substitution and $K = T/\beta - \{T/\beta\}$ cancel the apparently oscillatory linear terms and yield the stated coefficient. □

The same discrepancy estimate quantifies equidistribution of the rational points

$$\left(2^{\{k\beta\}}, 3^{\{k\beta\}}\right) = \left(\frac{M^k}{2^{\lfloor k\beta \rfloor}}, \frac{A^k}{3^{\lfloor k\beta \rfloor}}\right)$$

on the compact exponential arc. Their heights still grow exponentially in k, so this power saving does not reach a rational-point contradiction.

38 Valuation-lattice saturation and an exact duality

The denominator argument has a useful invariant formulation. It also has a symmetric consequence that is easy to miss if one works only with the least counterexample β.

Write

$$a_p = v_p(A), \qquad b_p = v_p(B),$$

and form the two-column integer matrix whose rows are

$$(\mathbf{1}_{p=2}, a_p) \quad \text{and} \quad (\mathbf{1}_{p=3}, b_p) \qquad (p \text{ prime}).$$

Only finitely many entries in the second column are nonzero. Let $\mathcal{L}$ be the rank-two lattice generated by the columns in the direct sum of all row copies of $\mathbb{Z}$.

Theorem 38.1 (Saturated valuation lattice)**.** *The lattice $\mathcal{L}$ is primitive (saturated):*

$$(\mathcal{L} \otimes_{\mathbb{Z}} \mathbb{Q}) \cap \bigoplus_p (\mathbb{Z} \oplus \mathbb{Z}) = \mathcal{L}.$$

Equivalently, the greatest common divisor of all maximal minors of the matrix is 1*. In the present sparse matrix this is exactly*

$$\gcd(\{a_p : p \neq 2\} \cup \{b_p : p \neq 3\} \cup \{a_2 - b_3\}) = 1.$$

Proof. The nonzero maximal minors are, up to sign, exactly the displayed a_p, b_p, and $a_2 - b_3$: the first column is nonzero only in the row belonging to $p = 2$ in the first valuation block and the row belonging to $p = 3$ in the second. It remains to prove that their gcd is 1.

If a number $d > 1$ divided all these minors, choose $t \in \{0, \dots, d-1\}$ with $t \equiv -a_2 \pmod{d}$. The congruence $a_2 \equiv b_3 \pmod{d}$ then gives $t \equiv -b_3 \pmod{d}$, while every other valuation of A or B is already divisible by d. Consequently both $2^t A$ and $3^t B$ are perfect dth powers in $\mathbb{N}$. Hence

$$\frac{\beta + t}{d} \in S.$$

This number is nonintegral, and it is strictly between 0 and β because $\beta > 1$ and $0 \leq t \leq d - 1$, contradicting the choice of β. Thus the gcd of the maximal minors is 1. The rank-two Smith normal form criterion then says exactly that the column lattice is saturated. □

A particularly concrete form of the theorem records exactly how extraction of common roots behaves throughout the solution monoid.

Corollary 38.2 (Perfect-power content is preserved)**.** *For $u, v \in \mathbb{N}_0$, not both zero, define*

$$X_{u,v} = 2^u A^v, \qquad Y_{u,v} = 3^u B^v.$$

Then

$$\gcd(\{v_p(X_{u,v}) : p \text{ prime}\} \cup \{v_p(Y_{u,v}) : p \text{ prime}\}) = \gcd(u, v).$$

Equivalently, for every $d \geq 1$, the two integers $X_{u,v}$ and $Y_{u,v}$ are simultaneously perfect dth powers if and only if $d \mid u$ and $d \mid v$.

Proof. Every divisor of $\gcd(u,v)$ plainly divides all displayed valuations. In the other direction, if d divides all of them, then $X_{u,v} = C^d$ and $Y_{u,v} = D^d$ for positive integers C, D. Thus $(u + v\beta)/d \in S$. Uniqueness of the representation in the irrational basis $(1, \beta)$ gives $u/d, v/d \in \mathbb{N}_0$. □

For example, A and B cannot themselves be simultaneous proper powers; and for every $t \geq 0$, neither pair

$$(2^t A, 3^t B), \qquad (2A^t, 3B^t)$$

can consist of simultaneous proper powers of the same degree. These are not independent ad hoc restrictions: all of them are specializations of one saturated rank-two valuation lattice.

There is also an exact reciprocal description. Put

$$S_{A,B} = \{y \in \mathbb{R} : A^y \in \mathbb{Z},\ B^y \in \mathbb{Z}\}.$$

Theorem 38.3 (Reciprocal solution monoid)**.** *Under the counterexample hypothesis,*

$$S_{A,B} = \mathbb{N}_0 + \frac{1}{\beta}\mathbb{N}_0.$$

In particular $1/\beta$ *is the least positive nonintegral element of* $S_{A,B}$*.*

Proof. If $y \in S_{A,B}$, then $x = \beta y$ satisfies

$$2^x = A^y \in \mathbb{Z}, \qquad 3^x = B^y \in \mathbb{Z}.$$

Hence $x = n + k\beta$ with $n, k \in \mathbb{N}_0$, and $y = k + n/\beta$. Conversely,

$$A^{k+n/\beta} = A^k 2^n, \qquad B^{k+n/\beta} = B^k 3^n$$

are integers for all $n, k \geq 0$. □

This reciprocal monoid shows that the two columns of the valuation matrix may be interchanged without losing arithmetic information: the original and dual root-extraction problems are the same saturated lattice viewed in opposite bases.

39 Exact-grid interpolation determinants

The free monoid of the exact primitive-generator theorem (Section 4) supplies two exact rank-two grids: one in the logarithms of the bases and one in the solution exponents. Retaining both coordinates is the central step of this continuation.

For finite sets $\mathcal{R}, \mathcal{C} \subset \mathbb{N}_0^2$ with $|\mathcal{R}| = |\mathcal{C}| = N$, define

$$\begin{aligned} u_{a,b} &:= a\alpha + b\gamma, \qquad (a,b) \in \mathcal{R}, \\ v_{n,k} &:= n + k\beta, \qquad (n,k) \in \mathcal{C}, \end{aligned}$$

and the matrix

$$M_{(a,b),(n,k)} := \exp(u_{a,b} v_{n,k}) = 2^{an} 3^{bn} A^{ak} B^{bk}. \tag{242}$$

Every entry is a positive integer.

39.1 Nonvanishing and total positivity

Lemma 39.1 (Strict positivity)**.** *Order the u-values and the v-values increasingly. Then*

$$D(\mathcal{R},\mathcal{C}) := \det\bigl(\exp(u_i v_j)\bigr)_{i,j=1}^{N} > 0.$$

In particular it is a positive integer and hence at least 1*.*

Proof. The u-values are distinct because 2 and 3 are multiplicatively independent, and the v-values are distinct because β is irrational. The kernel $K(u,v) = e^{uv}$ is strictly totally positive on $\mathbb{R} \times \mathbb{R}$. Equivalently, the functions $e^{v_1 u}, \dots, e^{v_N u}$ with $v_1 < \cdots < v_N$ form an extended complete Chebyshev system: their Wronskian is

$$e^{u(v_1+\cdots+v_N)} \prod_{i<j} (v_j - v_i) > 0.$$

The generalized mean-value theorem for Chebyshev systems gives the desired sign. □

39.2 Archimedean upper bound

Write $\Delta(u) = \prod_{i<j}(u_j - u_i)$ and similarly for v, and let $G(N+1) = \prod_{j=1}^{N-1} j!$ be the relevant Barnes product.

Lemma 39.2 (HCIZ determinant inequality)**.** *For increasingly ordered real vectors u, v,*

$$0 < \det(e^{u_i v_j}) \le \frac{\Delta(u)\,\Delta(v)}{G(N+1)} \exp\left(\sum_{i=1}^{N} u_i v_i\right). \tag{243}$$

Proof. The Harish–Chandra–Itzykson–Zuber formula with normalized Haar measure is

$$\int_{U(N)} e^{\mathrm{tr}(\mathrm{diag}(u) U\, \mathrm{diag}(v) U^*)}\, dU = G(N+1) \frac{\det(e^{u_i v_j})}{\Delta(u)\,\Delta(v)}.$$

Von Neumann's trace inequality bounds the exponent in the integral above by $\sum_i u_i v_i$. □

The factor $G(N+1)$ is crucial. When both node sets come from two-dimensional lattice regions, the leading $N^2 \log N$ contributions of the two Vandermonde products cancel the $N^2 \log N$ contribution of $G(N+1)$ exactly. The problem therefore lives at the next, N^2-constant, scale. This is the determinant manifestation of the critical nature of four exponentials.

39.3 Finite-prime lower bound as an assignment problem

For a prime p, the valuation of an entry in (242) is

$$w_p((a,b),(n,k)) = \mathbf{1}_{p=2} an + \mathbf{1}_{p=3} bn + a_p ak + b_p bk. \tag{244}$$

Lemma 39.3 (Tropical determinant bound)**.** *For every prime p,*

$$v_p(D(\mathcal{R},\mathcal{C})) \ge \min_{\sigma \in S_N} \sum_{i=1}^{N} w_p(i, \sigma(i)).$$

Proof. Each Leibniz term has the displayed summed valuation. The nonarchimedean triangle inequality says that the valuation of their signed sum is at least the minimum of their valuations. $\square$

This converts divisibility into a finite optimal-transport problem. Even a coarse componentwise use of it already produces an N^2-scale divisor.

40 Square grids: a universal divisor and the critical cancellation

Take $\mathcal{R} = \mathcal{C} = I_m := \{0, \ldots, m-1\}^2$ and $N = m^2$. Denote the corresponding determinant by D_m. Let L_m be the minimum of $\sum_i a_i n_{\sigma(i)}$ over bijections between the two copies of I_m. The multiset of first coordinates consists of each integer $0, \ldots, m-1$ repeated m times. The rearrangement inequality gives

$$L_m = m \sum_{j=0}^{m-1} j(m-1-j) = \frac{m^2(m-1)(m-2)}{6}. \tag{245}$$

The same value applies to each of the four pairings an, bn, ak, bk.

Theorem 40.1 (Universal square-grid divisor)**.** *For every $m \geq 1$,*

$$\log D_m \geq L_m(\log 2 + \log 3 + \log A + \log B) = L_m(1+\beta)\log 6. \tag{246}$$

Equivalently, the integer D_m is at least $6^{(1+\beta)L_m}$ in logarithmic size.

Proof. For a fixed prime, the minimum of a sum of nonnegative cost components is at least the sum of their separate minima. Apply Lemma 39.3 to (244); each present component has minimum L_m. Sum the resulting lower bounds with weights $\log p$ and use $\sum_p a_p \log p = \log A$ and $\sum_p b_p \log p = \log B$. $\square$

The estimate is much stronger than the bare fact $D_m \geq 1$. Nevertheless, a balanced rectangular asymptotic does not yet contradict it. Let $z = \sqrt{\alpha\gamma\beta}$. After choosing rectangle aspect ratios so that the two contributions to each projected logarithm are equal, the normalized node distribution is that of $Z = (X+Y)/2$ for independent $X, Y \sim \mathrm{Unif}[0,1]$. One has

$$\mathbb{E}Z^2 = \frac{7}{24}, \qquad \iint \log|s-t| \, d\,\mathrm{Law}(Z)(s) d\,\mathrm{Law}(Z)(t) = \frac{1}{3}\log 2 - \frac{25}{12}.$$

Combining these with Lemma 39.2 and Theorem 40.1 yields the necessary rectangular inequality

$$h(z) := \frac{1}{2}\log(4z) + \frac{1}{3}\log 2 - \frac{4}{3} + \frac{z}{2} \geq 0. \tag{247}$$

For the range allowed even by $A \geq 3$, this inequality is already positive. The square-grid divisor is therefore real progress in the arithmetic estimate, but rectangular geometry spends too much archimedean size.

41 Triangular grids and a new exclusion inequality

The natural way to reduce the archimedean cost is to take the first N nodes rather than a rectangular block.

Let $\mathcal{R}_N$ consist of the N pairs $(a,b) \in \mathbb{N}_0^2$ with the smallest values of $u = a\alpha + b\gamma$, and let $\mathcal{C}_N$ consist of the N pairs $(n,k) \in \mathbb{N}_0^2$ with the smallest values of $v = n + k\beta$. Ties do not occur. Let D_N be the positive integer determinant from these sets.

Define

$$U := \sqrt{2\alpha\gamma}, \qquad V := \sqrt{2\beta}, \qquad z := \sqrt{\alpha\gamma\beta},$$

so $UV = 2z$.

41.1 Limiting distributions

Elementary lattice-point counting gives

$$\#\{(a,b) \in \mathbb{N}_0^2 : a\alpha + b\gamma \le T\} = \frac{T^2}{2\alpha\gamma} + O(T),$$

and similarly for $n + k\beta$. Thus the Nth thresholds are asymptotic to $U\sqrt{N}$ and $V\sqrt{N}$. After division by $\sqrt{N}$, the projected nodes converge to the distributions

$$d\mu_U(t) = \frac{2t}{U^2}\,dt \quad (0 \le t \le U), \qquad d\nu_V(t) = \frac{2t}{V^2}\,dt \quad (0 \le t \le V).$$

Their quantile functions are

$$Q_{\mu_U}(s) = U\sqrt{s}, \qquad Q_{\nu_V}(s) = V\sqrt{s}.$$

The logarithmic energies are

$$\mathcal{E}(\mu_U) = \log U - \frac{7}{4}, \qquad \mathcal{E}(\nu_V) = \log V - \frac{7}{4}. \tag{248}$$

The finite irrationality type supplied by linear forms in logarithms ensures that exceptionally close projected lattice points do not spoil convergence of the logarithmic energies; details are given in Appendix M.

41.2 The archimedean constant

The Barnes asymptotic is

$$\log G(N+1) = \frac{N^2}{2}\log N - \frac{3N^2}{4} + o(N^2).$$

Using (248), the two Vandermonde terms in Lemma 39.2, and comonotone coupling for the trace term, gives

$$\limsup_{N\to\infty} \frac{\log D_N}{N^2} \le \frac{1}{2}\log(UV) - 1 + \frac{UV}{2}. \tag{249}$$

Indeed,

$$\int_0^1 Q_{\mu_U}(s)Q_{\nu_V}(s)\,ds = UV\int_0^1 s\,ds = \frac{UV}{2}.$$

41.3 The finite-prime constant

Uniform measure on the row triangle can be written as

$$a = \frac{U}{\alpha}X, \qquad b = \frac{U}{\gamma}Y, \qquad X, Y \geq 0, \quad X + Y \leq 1,$$

where (X, Y) is uniform on the standard simplex. Each coordinate has density $2(1-x)$ and quantile

$$q(s) = 1 - \sqrt{1-s}.$$

The column coordinates have the corresponding scales $n = VX'$ and $k = (V/\beta)Y'$. The counter-monotone assignment constant is

$$J := \int_0^1 q(s)q(1-s)\,ds = \frac{\pi}{8} - \frac{1}{3}. \tag{250}$$

Consequently each of the four logarithmically weighted components $an \log 2$, $bn \log 3$, $ak \log A$, and $bk \log B$ contributes asymptotically $UVJN^2$ to the componentwise valuation lower bound. Therefore

$$\liminf_{N\to\infty} \frac{\log D_N}{N^2} \geq 4UVJ. \tag{251}$$

Theorem 41.1 (Triangular determinant inequality)**.** *Every nonintegral counterexample β satisfies*

$$\boxed{g\Big(\sqrt{\beta \log 2 \log 3}\Big) \geq 0, \qquad g(z) = \frac{1}{2}\log(2z) - 1 + \left(\frac{11}{3} - \pi\right) z.} \tag{252}$$

Proof. Subtract the lower bound (251) from the upper bound (249). Since $UV = 2z$ and $4J = \pi/2 - 4/3$, the resulting necessary inequality is exactly (252). ☐

The derivative $g'(z) = 1/(2z) + 11/3 - \pi$ is positive for $z > 0$, so g has a unique positive zero

$$z_0 = 1.12895010749713, \qquad \beta_\triangle := \frac{z_0^2}{\log 2 \log 3} = 1.67370758736671.$$

At the first arithmetically possible nonintegral value $\beta = \log_2 3$, one has $z = \log 3$ and

$$g(\log 3) = -0.029549732685145 < 0.$$

The numerical consequence of Theorem 41.1 is weaker than what is already known: it gives $\beta \geq \beta_\triangle = 1.6737\ldots$, whereas the kernel-verified finite check of Section 23 gives $\beta \geq 12$ and the directed-rounding scan reaches 2^{27}. The inequality is recorded for its method, not for its bound: it is the first exclusion obtained from a uniform asymptotic determinant estimate rather than from a per-value certificate, and it exposes the arithmetic constant that would have to improve in order to reach all A at once.

41.4 Numerical anatomy of the two grid geometries

The table compares the triangular gap g with the balanced-rectangle gap h from (247). Negative values would contradict a counterexample at that β.

β	$z = \sqrt{\alpha\gamma\beta}$	$g(z)$	$h(z)$
1	0.8726396796	-0.2633422644	-0.04093352603
$\log_2 3$	1.098612289	-0.02954973269	0.1871929656
$\beta_\triangle$	1.128950107	-1.054219794e-81	0.2159820074
2	1.23409887	0.09973735836	0.3130828643
$\log_2 5$	1.329717364	0.1872568287	0.3982047952
3	1.511456262	0.3467367942	0.5531278372
5	1.951281643	0.7053840783	0.900746934
10	2.759528964	1.303060538	1.47815739

The triangular region improves the constant substantially, but g is increasing and becomes comfortably positive in the range $\beta \geq \log_2 5$. To continue, one must either improve the finite-prime lower bound, find a better variational region, or enrich the determinant.

42 Rational changes of basis and the product formula

A natural reaction to the positive gap is to replace 3 by a ratio such as 3/2, whose logarithm is smaller. Ignoring denominators makes this look like an immediate proof. The product formula shows exactly why that argument is false and, more usefully, turns it into quantitative denominator constraints.

42.1 A signed arithmetic functional

For a rational $q > 1$, write it in lowest terms as $q = \operatorname{num}(q)/\operatorname{den}(q)$ and define

$$\rho(q) := \frac{J \log \operatorname{num}(q) - K \log \operatorname{den}(q)}{\log q}, \qquad J = \frac{\pi}{8} - \frac{1}{3}, \quad K = \frac{1}{6}. \tag{253}$$

The constant J is the countermonotone coordinate cost in a simplex, while K is the comonotone cost, since a simplex coordinate has second moment 1/6. If q is an integer, then $\rho(q) = J$. A denominator contributes with K, not J, because a negative valuation reverses the relevant assignment bound.

Theorem 42.1 (Adelic triangular inequality)**.** *Let $r_1, r_2 > 1$ be multiplicatively independent positive rationals and suppose $R_i = r_i^\beta \in \mathbb{Q}_{>1}$ for $i = 1, 2$. Put*

$$z_r := \sqrt{\beta \log r_1 \log r_2}.$$

Then the existence of these four rational exponentials implies

$$\frac{1}{2}\log(2z_r) - 1 + z_r - 2z_r(\rho(r_1) + \rho(r_2) + \rho(R_1) + \rho(R_2)) \geq 0. \tag{254}$$

Equivalently, if

$$\Delta_{\mathrm{den}} := \sum_{q\in\{r_1,r_2,R_1,R_2\}} \log_q \mathrm{den}(q),$$

then

$$g(z_r) + 2z_r\left(\frac{1}{2} - \frac{\pi}{8}\right)\Delta_{\mathrm{den}} \geq 0. \tag{255}$$

Proof. Use the same first-N triangular sets for the positive logarithms $\log r_1, \log r_2$ and for $1, \beta$. The determinant entries are now rational. For a fixed prime, split each signed valuation coefficient into its positive and negative parts. In the tropical determinant bound, a positive coefficient is multiplied by the countermonotone minimum J, while a negative coefficient is multiplied by the comonotone maximum K. Summing over primes and using the product formula for the nonzero rational determinant gives the lower constant

$$2z_r\big(\rho(r_1) + \rho(r_2) + \rho(R_1) + \rho(R_2)\big).$$

The HCIZ upper constant remains $\frac{1}{2}\log(2z_r) - 1 + z_r$. This proves (254). Finally,

$$\rho(q) = J - (K - J)\log_q \mathrm{den}(q),$$

and substitution gives (255). □

42.2 All unimodular bases of $\langle 2, 3\rangle$

Let

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z})$$

and suppose

$$\lambda_1 = a\alpha + b\gamma > 0, \qquad \lambda_2 = c\alpha + d\gamma > 0.$$

Set

$$r_1 = 2^a 3^b, \quad r_2 = 2^c 3^d, \qquad R_1 = A^a B^b, \quad R_2 = A^c B^d.$$

Then Theorem 42.1 applies. This supplies an infinite family of necessary inequalities indexed by unimodular changes of basis. Continued-fraction vectors make one λ_i small, but the corresponding rational base has a large denominator; Equation (255) measures the precise price.

Proposition 42.2 (Optimality of the integral basis)**.** *Among unimodular bases for which both transformed bases are integers, the product $\lambda_1\lambda_2$ is at least $\alpha\gamma$. Equality occurs only for the original basis $(2, 3)$ up to permutation. Likewise, among unimodular bases of the solution lattice whose two transformed exponents remain in $\mathcal{S}$, the product of the positive exponent generators is at least β.*

Proof. If $2^a 3^b$ is an integer, then $a, b \geq 0$. Thus an integral unimodular basis has nonnegative entries. Expanding,

$$(a\alpha + b\gamma)(c\alpha + d\gamma) \geq (ad + bc)\alpha\gamma \geq \alpha\gamma,$$

because $|ad - bc| = 1$. Equality forces the standard basis up to permutation. For the exponent lattice, the exact primitive-generator theorem says that a transformed exponent is simultaneously integral at bases 2 and 3 exactly when its coefficients in $(1, \beta)$ are nonnegative. The same argument applies. □

This proposition explains why a denominator-free lattice reduction cannot improve the triangular inequality: the exact monoid has already selected the smallest integral basis.

42.3 The basis $(2, 3/2)$ and a gcd constraint

Take

$$r_1 = 2, \qquad r_2 = \frac{3}{2}, \qquad R_1 = A, \qquad R_2 = \frac{B}{A}.$$

Let $d = A/\gcd(A, B)$ be the denominator of B/A in lowest terms and put $\delta = \log(3/2)$. Then

$$\Delta_{\mathrm{den}} = \frac{\alpha}{\delta} + \frac{\log d}{\beta\delta}. \tag{256}$$

Theorem 42.1 therefore gives a rigorous lower bound on d, whenever the right side obtained by solving (255) is positive. In words: if $3/2$ substantially improves the archimedean configuration, then B/A must carry enough denominator to pay for that improvement. A counterexample cannot have excessive divisibility $A \mid B$ in the range where the reduced-basis gap is negative.

For orientation, the next table substitutes the hypothetical value $\beta = \log_2 A$. The column $d_{\min}$ is the least integer denominator not excluded by the inequality alone; it is not a claim that the corresponding B exists.

A	$\log_2 A$	z_r	gap before $\log d$	$d_{\min}$
3	1.5849625	0.667419621	-0.260297461	4
5	2.32192809	0.807818616	-0.0396109308	2
6	2.5849625	0.852347316	0.0269341919	1
7	2.80735492	0.88825597	0.0795954278	1
9	3.169925	0.94387388	0.159569469	1
10	3.32192809	0.966239056	0.191227186	1
11	3.45943162	0.986033907	0.219022705	1
12	3.5849625	1.00376439	0.243748126	1
13	3.70043972	1.01980266	0.265979188	1
14	3.80735492	1.03443012	0.286146736	1
15	3.9068906	1.04786443	0.304581094	1
17	4.08746284	1.07180649	0.337231592	1
18	4.169925	1.08256404	0.351820061	1
19	4.24792751	1.09264233	0.36544257	1
20	4.32192809	1.10211837	0.378212214	1
21	4.39231742	1.11105698	0.39022374	1
22	4.45943162	1.11951323	0.401557272	1
23	4.52356196	1.12753426	0.412281139	1

A second useful basis is

$$r_1 = \frac{3}{2}, \qquad r_2 = \frac{4}{3}, \qquad R_1 = \frac{B}{A}, \qquad R_2 = \frac{A^2}{B}.$$

Its two logarithms are smaller, but Equation (255) now charges the denominators of all four displayed ratios. It therefore constrains the possibility that B/A and A^2/B are simultaneously close to integers.

More generally, continued-fraction bases produce a network of constraints on

$$\frac{B^q}{A^p} \quad \text{when} \quad q\log 3 - p\log 2 \text{ is small.}$$

This is a concrete future route: prove that the denominator requirements for a carefully chosen finite family of bases are incompatible with the prime valuation vectors of A and B.

43 Joint prime transport and shared-factor gains

The componentwise lower bounds used above are intentionally conservative. At a fixed prime, a single permutation must handle all valuation components at once. If a prime occurs in more than one generator, the separate countermonotone assignments can conflict, making the true minimum strictly larger.

For the original triangular determinant, define

$$\mathcal{T}_p(N) := \frac{1}{N^2}\min_{\sigma\in S_N}\sum_i w_p(i,\sigma(i)).$$

Then the strongest direct product-formula lower bound is

$$\liminf_{N\to\infty}\frac{\log D_N}{N^2} \geq \sum_p (\log p)\liminf_{N\to\infty}\mathcal{T}_p(N).$$

The baseline $4UVJ$ replaces each $\mathcal{T}_p$ by the sum of four independent one-coordinate minima.

A simple rigorous gain estimate is available for a prime $p \notin \{2,3\}$ that divides both A and B. Put $r = v_p(A)$, $s = v_p(B)$ and

$$R = \frac{r}{\alpha}, \qquad S = \frac{s}{\gamma}.$$

On the normalized row simplex, the relevant scalar is $RX + SY$, while the column coordinate k has the marginal law of a simplex coordinate. The countermonotone cost is at least the product of the means minus the product of the standard deviations. Since

$$\mathbb{E}X = \mathbb{E}Y = \frac{1}{3}, \quad \operatorname{Var}X = \operatorname{Var}Y = \frac{1}{18}, \quad \operatorname{Cov}(X,Y) = -\frac{1}{36},$$

one obtains

$$\liminf \mathcal{T}_p(N) \geq \frac{UV}{\beta}\left(\frac{R+S}{9} - \frac{\sqrt{R^2+S^2-RS}}{18}\right). \tag{257}$$

The componentwise baseline for the same two terms is

$$\frac{UV}{\beta}J(R+S).$$

Taking the larger of these two lower bounds gives a fully explicit shared-prime improvement. Summed over common prime divisors of A and B, it yields a sufficient contradiction criterion whenever the gain exceeds the positive gap $g(z)$.

The estimate is strongest when r/α and s/γ are comparable and vanishes continuously as one valuation goes to zero. It therefore does not supply a universal proof: the worst arithmetic configuration may have nearly disjoint prime support. But it turns "a large gcd should help" into a quantitative theorem and identifies exact valuation patterns worth targeting.

44 Universal congruence divisibility is lower order

There is also divisibility at primes not dividing $6AB$. Modulo such a prime p, a row of the matrix depends on (a, b) only modulo $p-1$, by Fermat's little theorem. Hence there are at most $(p-1)^2$ distinct rows modulo p. For an $N \times N$ determinant,

$$v_p(D_N) \geq N - \operatorname{rank}_{\mathbb{F}_p}(M) \geq N - (p-1)^2$$

whenever the right side is positive. Summing over primes $p \leq \sqrt{N}$ and using the prime number theorem gives the universal extra estimate

$$\sum_{\substack{p \leq \sqrt{N} \\ p \nmid 6AB}} (N - (p-1)^2) \log p = \left(\frac{2}{3} + o(1)\right) N^{3/2}. \tag{258}$$

Higher prime powers and column congruences can improve the lower-order term. They do not alter the N^2 constant in Theorem 41.1. This provides a useful barrier result: congruence collisions that depend only on finite-group periodicity are too small by a factor of $\sqrt{N}$. A complete determinant proof needs counterexample-specific valuation structure or a different determinant geometry.

45 A variational reformulation of the remaining determinant problem

The preceding calculations extend to scaled lattice regions $R, C \subset \mathbb{R}^2_{\geq 0}$ of equal area. Push forward uniform measure on R by $(a, b) \mapsto a\alpha + b\gamma$, and uniform measure on C by $(n, k) \mapsto n + k\beta$. The HCIZ side is determined by their logarithmic energies and comonotone transport. The finite-prime side is a sum of countermonotone or, more accurately, joint optimal-transport costs for the valuation bilinear forms (244).

This suggests the following concrete variational program.

1. Choose regions R, C and compute the archimedean functional

$$\mathcal{A}(R, C) = \frac{1}{2}\mathcal{E}(\mu_R) + \frac{1}{2}\mathcal{E}(\nu_C) + \frac{3}{4} + \int_0^1 Q_{\mu_R}(t) Q_{\nu_C}(t)\, dt,$$

   with the scale normalization determined by $\operatorname{area}(R) = \operatorname{area}(C)$.

2. Compute or bound the exact prime transport functional

$$\mathcal{P}(R, C; A, B) = \sum_p \log p \inf_\pi \int w_p(r, \pi(r))\, dr.$$

3. Any counterexample must satisfy $\mathcal{A}(R, C) \geq \mathcal{P}(R, C; A, B)$ for every admissible pair of regions. A single strict reverse inequality proves a contradiction.

Rectangles and origin triangles are analytically soluble test cases. The next step is not merely to scan aspect ratios: it is to optimize jointly over region shape and valuation configurations allowed by the two-three unit corollary and the primitive-gcd lemma of Section 4. The latter constraints are discrete and may prevent the valuation directions that make the transport cost artificially small.

46 Confluent and integer-valued determinant extensions

Ordinary evaluations use one row per lattice point. A potentially stronger construction uses integer jets. View (242) as evaluating the monomials X^nY^k at the integer points

$$(X,Y) = (2^a3^b, A^aB^b).$$

The Euler operators

$$D_X = X\frac{\partial}{\partial X}, \qquad D_Y = Y\frac{\partial}{\partial Y}$$

act by

$$D_X^rD_Y^s(X^nY^k) = n^rk^sX^nY^k,$$

which remains integral at the lattice points. Hasse derivatives give the alternative integer values

$$\binom{n}{r}\binom{k}{s}X^{n-r}Y^{k-s}.$$

Thus one can build confluent evaluation determinants without introducing the irrational factors $n + k\beta$ that ordinary derivatives along the real curve would produce.

The unresolved issues are nonvanishing and optimal scaling. A useful target is an integer-jet determinant whose archimedean limit behaves like a confluent HCIZ determinant while its p-adic assignment problem acquires an additional N^2 contribution. This is more promising than generic factorial normalization: the elementary divisibility obtained from binomial bases alone is only of order $N^{3/2}\log N$ for balanced two-dimensional grids.

47 A proof-producing finite verification

The structural and determinant arguments above are uniform, but they leave an infinite range of possible values of $A = 2^\beta$. A complementary finite calculation can remove an initial segment without assuming any unproved transcendence statement. The following computation is deliberately based on outward-rounded intervals and an explicit positive-series remainder, rather than on a test such as "the floating-point value is not visually close to an integer."

Theorem 47.1 (Certified finite range)**.** *For every integer A with*

$$2 \le A \le 100{,}000,$$

the number $A^{\log_2 3}$ is an integer if and only if A is a power of 2. Consequently, a nonintegral Alaoglu–Erdős counterexample must satisfy

$$2^\beta > 100{,}000, \qquad \beta > \log_2(100{,}000) \approx 16.609640474437.$$

Certificate method. For fixed A, put

$$F_A(b) = (\log b)(\log 2) - (\log A)(\log 3), \qquad b > 0.$$

The function F_A is strictly increasing, and $A^{\log_2 3} = b$ is equivalent to $F_A(b) = 0$. Every logarithm is enclosed without calling a transcendental floating-point routine in the decisive comparison. For $0 \le z \le 1/3$ we use

$$\log \frac{1+z}{1-z} = 2 \sum_{j=0}^{M-1} \frac{z^{2j+1}}{2j+1} + R_M(z), \qquad 0 < R_M(z) < \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

For an integer $n \ge 1$, choose $k = \lfloor \log_2 n \rfloor$ and write

$$\log n = k \log 2 + \log \frac{1+z_n}{1-z_n}, \qquad z_n = \frac{n-2^k}{n+2^k} \in [0, 1/3).$$

Thus all terms are positive rational operations, and directed rounding gives certified lower and upper endpoints. The run used 70 decimal digits and $M = 28$ terms. A binary64 value was used only as an initial guess for the integer immediately below the root; the program then moved monotonically until it had certified, for consecutive integers $b, b+1$,

$$F_A(b) < 0 < F_A(b+1).$$

Hence no integer can be the root. Powers of 2 are separated at the start, and for $A = 2^m$ the corresponding value is exactly 3^m.

The completed run checked 99,983 nonpowers of 2. The largest number of integer steps needed to repair an initial binary64 guess was 0. The smallest absolute interval endpoint encountered was `6.263278510736485942331545732E-13`, far outside the interval uncertainty. Runtime on the working container was approximately 15.238 seconds. The exact verifier used for the run is reproduced in Appendix Q. □

Remark 47.2. The numerical bound is not presented as evidence that near misses cannot occur later. Its role is narrower and rigorous: it turns all A in the displayed finite interval into closed cases, while the determinant inequalities address whole infinite parameter ranges and arithmetic configurations at once.

48 Matrix coefficients, sparse interpolation, and boundary defects

This section uses the current module statuses, corrected candidate count, and correct first-layer valuation inequality throughout. Its mathematical core is a matrix-coefficient reformulation, an exact sparse-interpolation threshold, a Newton-polygon degree calibration, and a boundary-defect identity for the extremal interpolant. The matrix coefficient, formal prime-symbol, and sparse-interpolation components now have dedicated kernel-checked Lean modules.

Status: The full conjecture remains open. The restricted matrix-coefficient equivalence, formal prime-symbol zero classification, Structural Rank equivalence, finite sparse power-curve lemmas, and boundary-dilation algebra below are **[kernel-verified Lean]**. Newton-polygon geometry, the packaged BK-degree threshold, the counting reinterpretations, and the complex monodromy consequence remain conventional mathematics or consequences of named classical inputs.

48.1 The restricted matrix-coefficient principle

For positive integers M, A, put

$$\mathsf{L}(M,A) = \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log A \end{pmatrix}.$$

The determinant vanishes exactly when there is a real β with

$$M = 2^\beta, \qquad A = 3^\beta.$$

In that case

$$\mathsf{L}(M,A) = \begin{pmatrix} \log 2 \\ \log 3 \end{pmatrix} \begin{pmatrix} 1 & \beta \end{pmatrix},$$

and for rational vectors $w = (r,s)^T$ and $v = (u,z)^T$ one has the exact factorization

$$w^T \mathsf{L}(M,A) v = (r \log 2 + s \log 3)(u + z\beta). \tag{259}$$

Define $\mathrm{MCC}_{2,3}^{\mathbb{Z}}$ to assert that every singular matrix of this restricted form has nonzero rational w, v with $w^T \mathsf{L}(M,A)v = 0$.

Theorem 48.1 (Restricted Matrix Coefficient equivalence)**.** *The Alaoglu–Erdős conjecture is equivalent to* $\mathrm{MCC}_{2,3}^{\mathbb{Z}}$.

Proof. If the conjecture holds, singularity supplies a common exponent β, which must be an integer m; the rational right vector $(-m,1)^T$ lies in the kernel. Conversely, suppose β is a nonintegral common exponent. Integrality of 2^β implies that β is irrational. In (259) the left factor cannot vanish for nonzero rational w, by multiplicative independence of 2 and 3, and the right factor cannot vanish for nonzero rational v, by irrationality of β. Thus the required zero matrix coefficient does not exist. □

The new module `RestrictedMatrixCoefficient` formalizes the concrete 2×2 matrix, its determinant, the common-exponent equivalence, the rank-one identity `restrictedLogMatrix_eq_rankOne`, the coefficient factorization and anisotropy, and the exact endpoint

`alaogluErdosConjecture_iff_restrictedMatrixCoefficientPrinciple`.

This is **[kernel-verified Lean]**. It is a strict specialization of the Matrix Coefficient Conjecture of Dasgupta–Kakde [59]; in the real 2×2 case their fewnomial condition implies the desired matrix coefficient, while constructing that fewnomial from singularity remains the open direction.

48.2 The formal prime-symbol determinant

Let $a_p = v_p(M)$ and $b_p = v_p(A)$, and introduce one formal variable X_p for each prime in $\{2,3\} \cup \mathrm{supp}(MA)$. The formal determinant is

$$\Phi_{M,A}(\mathbf{X}) = X_2 \sum_p b_p X_p - X_3 \sum_p a_p X_p. \tag{260}$$

Substitution $X_p = \log p$ gives $\det \mathsf{L}(M, A)$. Coefficient comparison gives

$$\Phi_{M,A} \equiv 0 \quad \Longleftrightarrow \quad \exists m \in \mathbb{N}_0,\ M = 2^m \text{ and } A = 3^m. \tag{261}$$

Indeed, all b_p away from 3 and all a_p away from 2 must vanish, while the X_2X_3 coefficient forces $b_3 = a_2$. Hence the conjecture is equivalently the assertion that specialization of this signed sparse quadratic at the prime-logarithm point cannot vanish unless the polynomial is formally zero.

For a hypothetical counterexample $\Phi_{M,A}$ is nonzero and irreducible over $\mathbb{Q}$. A factorization of a homogeneous quadratic would yield two rational linear factors; one of them would vanish after $X_p = \log p$, contradicting rational linear independence of logarithms of distinct primes. The new `FormalPrimeDeterminant` module defines the genuine `MvPolynomial formalPrimeDeterminant`, proves its specialization `formalPrimeDeterminant_evalâĈĈ_log`, the unconditional classification `formalPrimeDeterminant_eq_zero_iff`, and the exact equivalence `alaogluErdosConjecture_iff_sparsePrimeLogStructuralRankPrinciple`. These claims are **[kernel-verified Lean]**. The homogeneous-factor irreducibility bookkeeping is intentionally kept separate and remains paper-level.

48.3 Exact sparse interpolation on an irrational power curve

For irrational β, a monomial X^rY^s restricted to $(X, Y) = (e^t, e^{\beta t})$ becomes

$$e^{(r+s\beta)t}.$$

Distinct pairs $(r, s) \in \mathbb{N}_0^2$ give distinct frequencies. The classical Chebyshev zero theorem for real exponential polynomials, already formalized in `ExponentialPolynomialZeros`, therefore gives the following sharp lower bound.

Theorem 48.2 (Sparse power-curve zero bound)**.** *An m-term nonzero polynomial restricted to $Y = X^\beta$, with β irrational, has at most $m - 1$ distinct positive zeros. Equivalently, vanishing at R distinct positive curve points requires at least $R + 1$ nonzero monomials.*

The `SparsePowerCurve` module proves arbitrary-order and finite-zero-set forms of this theorem, as well as nonsingularity of the square evaluation matrix. It also proves the corresponding integral statement: if $y_i = x_i^\beta$ are natural numbers, then

$$\det\big(x_i^{r_j} y_i^{s_j}\big) \neq 0$$

for distinct monomial pairs and ordered positive x_i. Thus the determinant is a nonzero integer and has absolute value at least one. These statements are **[kernel-verified Lean]**.

The matching upper construction is the node polynomial

$$P_0(T) = \prod_{i=1}^{R} (T - x_i).$$

Because all x_i are positive, every elementary symmetric coefficient is nonzero, so P_0 has exactly $R + 1$ monomials. This coefficient assertion and the vanishing at every node are also formalized as `positiveNodeInterpolant_support_card` and `positiveNodeInterpolant_eval`. Consequently the following packaged minimum is a paper-level corollary whose two substantive halves are kernel checked:

$$\min\left\{\#\operatorname{supp} P : \substack{0\neq P\in\mathbb{C}[X,Y]\\ P(x_i,x_i^{\beta})=0\ (1\le i\le R)}\right\} = R+1. \tag{262}$$

Here the paper-level minimum allows complex coefficients; the kernel-checked zero-bound and matching positive-node construction are stated over the real core needed in the application.

In the restricted integer-output setting this gives an exact dichotomy. If the common exponent is an integer m, the minimum support on any nonempty family is 2, attained by $Y - X^m$; a single nonzero monomial cannot vanish at a positive point. If the exponent is nonintegral, it is irrational and the minimum on R distinct points is $R+1$ by (262). The irrational branch is the kernel-checked sparse theorem; the integral branch is the displayed elementary two-term witness and positivity argument.

For a counterexample (M, A, β) and a finite $\Omega \subset \mathbb{N}_0^2$, the points

$$(2^a 3^b, M^a A^b) = (e^{z_{a,b}}, e^{\beta z_{a,b}}), \qquad z_{a,b} = a\log 2 + b\log 3,$$

have distinct parameters. Hence $R = \#\Omega$ gives the exact minimum $R+1$. On the $2N\times 2N$ box, any polynomial with at most $4N^2$ monomials would therefore save one term over trivial interpolation and rule out a counterexample. Dasgupta–Kakde ask for the stronger support bound $< N^2$ on the same $4N^2$ points; for Alaoglu–Erdős, one saved monomial already suffices.

48.4 BK degree and the exact factor-four threshold

For $0 \neq P \in \mathbb{Q}[X,Y]$, first divide out the greatest common monomial of its terms, as in the Dasgupta–Kakde translation-invariant normalization. Let P^{red} denote the resulting polynomial, let $\Delta(P)$ be its Newton polygon, and set

$$\operatorname{bkd}(P) = \max\{\deg P^{\mathrm{red}},\ 2\operatorname{Area}\Delta(P)\}.$$

Pick's theorem gives

$$\#\operatorname{supp} P \le \operatorname{bkd}(P) + 2, \tag{263}$$

with $+1$ when the Newton polygon is one-dimensional. For irrational β and $R \ge 2$ distinct positive rational nodes whose ordinates x_i^{β} are rational, let $L \in \mathbb{Q}[X]$ be the Lagrange interpolation polynomial of degree at most $R-1$ with $L(x_i) = x_i^{\beta}$. Combining (262) with the thin interpolant $Y - L(X)$ forces all $R+1$ available coefficients to occur and yields the exact paper-level threshold

$$\min\left\{\operatorname{bkd}(P) : \substack{0\neq P\in\mathbb{Q}[X,Y]\\ P(x_i,x_i^{\beta})=0}\right\} = R-1. \tag{264}$$

For the $K\times K$ semigroup grid with $K \ge 2$ this is K^2-1; on the Dasgupta–Kakde $2N$ grid it is $4N^2-1$. Their target $< N^2$ must therefore cross a genuine asymptotic factor-four gap. No Pick-theorem or normalized lattice-area library is currently present in the pinned Mathlib, so (263)–(264) remain **[paper proof]**.

48.5 Boundary defects of the extremal interpolant

Assume $K \ge 2$ and a counterexample normalization $M = 2^{\beta}$, $A = 3^{\beta}$ with β irrational. Let $Q_K \in \mathbb{Q}[X]$ be the degree-K^2-1 polynomial interpolating the second coordinate on the $K\times K$

grid, and let

$$\Pi_{r,s}(X) = \prod_{\substack{0\le i<r\\0\le j<s}} (X - 2^i 3^j).$$

The interior grid is stable under the two dilations. Therefore

$$Q_K(2X) - MQ_K(X) = \Pi_{K-1,K}(X)U_K(X), \tag{265}$$
$$Q_K(3X) - AQ_K(X) = \Pi_{K,K-1}(X)V_K(X), \tag{266}$$

where U_K, V_K are unique of degree $K-1$; their leading coefficients are respectively $q_K(2^{K^2-1} - M)$ and $q_K(3^{K^2-1} - A)$ when q_K is the leading coefficient of Q_K. The irrationality of β makes both displayed factors nonzero. More generally, the same degree conclusion requires the explicit noncancellation hypotheses $2^{K^2-1} \neq M$ and $3^{K^2-1} \neq A$.

The dilation operators $D_2 f = f(2X) - Mf(X)$ and $D_3 f = f(3X) - Af(X)$ commute. Substituting (265)–(266) and cancelling the common interior factor $\Pi_{K-1,K-1}$ produces an identity of degree at most $2K-2$ involving the top and right boundary pieces together with their bottom and left dilation preimages. This boundary compression is a concrete finite target: it turns K^2 interpolation constraints into a low-degree compatibility equation. It does not by itself produce the missing fewnomial, and no such claim is made.

The module `BoundaryDilationDefect` proves the residual polynomial algebra over a general field and the commutator identity over an infinite field. Its theorem `existsUnique_squareGridResiduals` gives both unique residuals, their exact degree and leading coefficients; `boundaryDilation_commutator` proves the displayed commuting boundary identity; and `squareBoundaryExpressions_natDegree_le` proves the $2K-2$ bounds. The interpolation values, node injectivity, and noncancellation of the top coefficient remain visible hypotheses. These statements are **[kernel-verified Lean]**.

48.6 Counting thresholds and rational monodromy

A counterexample gives the rational points

$$P_{n,k} = (2^n M^k, 3^n A^k) = (2^{n+k\beta}, 3^{n+k\beta}), \qquad n, k \in \mathbb{N}_0,$$

on the fixed curve $Y = X^\vartheta$. Since the second coordinate controls the height, a triangular lattice count gives

$$\#\{(n,k) \in \mathbb{N}_0^2 : n + k\beta \le \log_3 H\} = \frac{(\log H)^2}{2\beta(\log 3)^2} + O_\beta(\log H).$$

Thus any uniform $o((\log H)^2)$ upper bound for rational points on the unbounded power curve would prove the conjecture. The synchronized compact orbit from `CompactOrbit` gives a sharper arithmetic target: its first k points have height at most A^k, so an $o(\log H)$ bound for points retaining the common floor level would also suffice. These are height reinterpretations of the accepted exact and quadratic conditional counts, not a new point-counting theorem.

Finally, classical six exponentials forces a complex consequence which the current real determinant formalization does not cover. If β were a nonintegral solution and $q \in \mathbb{Q}^\times$, then

$$e^{2\pi i q\beta} \quad \text{is transcendental,} \tag{267}$$

and so are its sine and cosine. Apply six exponentials to the independent rows $1, \beta$ and the independent columns $\log 2, \log 3, 2\pi i q$; five of the six exponentials are algebraic, forcing the sixth to be transcendental. This is a valid consequence of the named classical theorem, not a new kernel theorem.

48.7 The documented 2^{29} scan and its trust boundary

Bundled documentation records an MPFR directed-rounding extension through 2^{29}, including chunk tables, extremal records, high-precision replays, the literal C source, and a run summary. The logical consequence would be that a nonintegral solution has $x > 29$. However, the repository contains no standalone scanner or canonical run logs, and the long run was not independently repeated during this audit. It is therefore retained as a documented **[software computation]** result, while the repository-reproducible software frontier in Section 34 remains 2^{27}.

48.8 Scope and remaining gap

These exact finite reformulations do not give an unconditional proof of the conjecture. The resulting hierarchy is:

1. the conjecture is exactly the restricted matrix-coefficient principle (**[kernel-verified Lean]**);
2. the formal prime-symbol determinant specializes to the logarithmic determinant, and its sparse Structural Rank principle is again exactly the conjecture (**[kernel-verified Lean]**);
3. a counterexample has maximal sparse interpolation complexity $R + 1$, with the lower bound, matching positive interpolant, and evaluation determinant kernel checked;
4. the corresponding BK-degree threshold is exactly $R-1$ (**[paper proof]**), exposing a factor-four gap to the current Matrix Coefficient auxiliary-polynomial target;
5. dilation defects compress the extremal grid interpolant to boundary polynomials of degree $K - 1$ (**[kernel-verified Lean]**), but an arithmetic construction saving one monomial remains open;
6. generic polylogarithmic point counting is insufficient unless its exponent is below two globally, or below one on the synchronized compact orbit.

The Massold v1 parameter substitution [60] does not cover the 2×2 case: its displayed choice gives $t = 1$, $\mu = \nu = 1$, making the required strict inequality $1/2 > 1$. This is a narrow version-specific warning, not a claim about every possible correction.

49 Support spectra and arithmetic determinant audits

Three themes are developed here: the exact support spectrum of a finite orbit, integer-valued-polynomial divisors and primitive Smith normalization, and an adelic audit of exact-grid determinants including a signed-progression no-go theorem. We do not import a stated "quadratic divisor criterion": an inequality containing an unspecified $o(N^2)$ term is not a sufficient eventual strict inequality. Likewise, the proposed cusp lattice realization is kept conditional on the uniform logarithmic-energy convergence that its current proof does not establish.

49.1 The exact orbit-collision support spectrum

For a finite nonempty $\Omega \subset \mathbb{Z}^2$, define

$$T_x(\Omega) = \{i + jx : (i,j) \in \Omega\}, \qquad D_x(\Omega) = \#T_x(\Omega),$$

and let $\mu_x(\Omega)$ be the least number of monomials in a nonzero Laurent polynomial in $\mathbb{C}[X^{\pm 1}, Y^{\pm 1}]$ vanishing at all points

$$X_x(\Omega) = \{(2^{i+jx}, 3^{i+jx}) : (i,j) \in \Omega\}.$$

Multiplication by a monomial shows that the same minimum is obtained with ordinary polynomials. Restriction to the orbit turns every monomial into an exponential with frequency $a \log 2 + b \log 3$; these frequencies are distinct by unique factorization. Consequently the exponential-polynomial zero bound and the positive-node polynomial give the exact formula

$$\boxed{\mu_x(\Omega) = D_x(\Omega) + 1.} \tag{268}$$

More generally, every evaluation matrix over $\mathbb{C}$ has rank $\min(D_x(\Omega), \#S)$ for any finite monomial support S. Repeated zeros counted with multiplicity obey the same rule: prescribing total multiplicity R costs at least $R+1$ monomials.

For $L, K \geq 1$, on the rectangle $\Omega_{L,K} = \{0, \dots, L-1\} \times \{0, \dots, K-1\}$ the collision count is explicit. If $x \notin \mathbb{Q}$, then $D_x(L,K) = LK$. If $x = r/s \in \mathbb{Q}_{\geq 0}$ is in lowest terms, then

$$D_{r/s}(L,K) = LK - (L-r)_+(K-s)_+, \qquad \mu_{r/s}(L,K) = D_{r/s}(L,K) + 1. \tag{269}$$

Indeed two vertices collide exactly along the primitive direction $(r, -s)$; the collision classes are paths, and vertices minus edges gives (269). Thus rationality is equivalent to saving one monomial on some square grid, while an irrational exponent has $\mu_x(L,L) = L^2 + 1$ for every L. For integer outputs, a complex dependence may be replaced by a rational one and denominators cleared, giving an integer certificate. This is the exact finite-support version of the restricted Matrix Coefficient target.

The module `RationalOrbitGrid` kernel-checks the exact rectangle formula, including the zero-numerator edge case, the irrational full-cardinality statement, the irrational square-grid sparse lower bound, and the canonical one-monomial-saving rational-orbit certificate once the grid exceeds the reduced numerator and denominator. Together with `SparsePowerCurve`, this certifies the real, nonnegative-exponent core of the spectrum above. The arbitrary finite Laurent support/rank formulation and the confluent multiplicity packaging remain paper-level. **[kernel-verified Lean]**

49.2 Coefficient height and the generic-modulus no-go

The exact support count does not control coefficient height. For $M \geq 1$, under an irrational exponent x with $A = 2^x \in \mathbb{Z}_{>0}$, the canonical annihilator of the M^2 first-coordinate nodes is

$$Q_M(T) = \prod_{0 \leq i,j < M} (T - 2^i A^j) \in \mathbb{Z}[T].$$

It has exactly $M^2 + 1$ nonzero coefficients. Write $H(Q) = \max_k |q_k|$ for $Q(T) = \sum_k q_k T^k$. Writing $C_M = (2A)^{M^2(M-1)/2}$, positivity of the roots gives the exact finite bounds

$$C_M \leq \|Q_M\|_1 \leq 2^{M^2} C_M, \qquad \frac{C_M}{M^2+1} \leq H(Q_M) \leq 2^{M^2} C_M, \tag{270}$$

and hence

$$\log H(Q_M) = \frac{(1+x)\log 2}{2} M^2(M-1) + O(M^2 + \log M).$$

Thus the canonical support is sharp but carries an $\exp(\Theta_x(M^3))$ archimedean cost. Adding confluent multiplicities does not lower the support cost: the exponential Chebyshev bound charges one extra monomial for each unit of prescribed total multiplicity.

The corresponding adelic criterion is elementary and exact. If $P \in \mathbb{Z}[X, Y]$ and, at every integer grid point z, a finite set of primes S satisfies

$$|P(z)|_\infty \prod_{p \in S} |P(z)|_p < 1,$$

then the product formula forces $P(z) = 0$. Equivalently, it is enough to find a divisor $q_z \mid P(z)$ with $|P(z)| < q_z$ at every row. This identifies the right target but does not construct the coefficients.

A cardinality-only modular pigeonhole does not suffice. Let $E \in \mathbb{Z}^{R \times K}$ with $1 \le K \le R$, let $Q \ge 2$ and $H \ge 0$, and assume $|E_{rj}| \le M_r$ with $M_r \ge 1$. Using coefficient vectors in $\{0, \dots, H\}^K$ and one modulus Q requires $(H+1)^K > Q^R$, which forces $H \ge Q$; the generic bound $|(Ed)_r| \le KHM_r$ can then never prove $|(Ed)_r| < Q$. Even row-dependent moduli do not repair this architecture: for $H \ge 1$ and $q_r \ge 2$, the two sufficient conditions

$$(H+1)^K > \prod_{r=1}^{R} q_r, \qquad q_r > KHM_r \quad (1 \le r \le R)$$

are incompatible. Finally, if one uses the full adelic product—or a finite prime set containing every prime divisor of the row content g_r—then dividing that row by g_r changes neither the archimedean–p-adic product nor the meaningful cancellation: the product formula for g_r exactly cancels the apparent common divisibility. The useful arithmetic object is therefore the Smith profile of the row-primitive evaluation matrix, not a raw common modulus.

These coefficient-height, adelic, pigeonhole, and row-content statements are paper-level. The same support proof is base-independent for every multiplicatively independent positive base tuple: distinct monomials restrict to exponentials with distinct real frequencies.

49.3 Universal factorial content and primitive interpolation

For $r \ge 1$, write

$$\mathfrak{F}_r = \prod_{j=0}^{r-1} j!.$$

For arbitrary integers $m_0, \dots, m_r, y_0, \dots, y_r$, the interpolation determinant with columns $1, m, \dots, m^{r-1}, y$ is divisible by $\mathfrak{F}_r$. Replacing the power columns by $\binom{m}{0}, \dots, \binom{m}{r-1}$ proves the claim by a triangular Stirling change of basis, and the nodes $0, \dots, r$ with ordinate vector $(0, \dots, 0, 1)$ show that the universal divisor is best possible. Applied to candidate ordinates, this strengthens the arbitrary-node repulsion constant by the factor $\mathfrak{F}_r^{1/\binom{r+1}{2}}$ but leaves its exponent unchanged.

The focused module `SuperfactorialDeterminant` kernel-checks the nonnegative-integer-node instance of this universal divisor, its exact consecutive-node sharpness witness, the resulting lower bound for a nonzero candidate interpolation determinant, and the strengthened divided-difference repulsion and packing inequalities. As in `ArbitraryNodeRepulsion`, the final two endpoints retain `RpowDividedDifferenceMeanValue` as an explicit hypothesis; the superfactorial divisibility itself is unconditional in that formalized scope. The arbitrary signed-integer node formulation above remains a paper consequence. **[kernel-verified Lean]**

The bivariate version is equally exact. If $\mathcal{S} \subset \mathbb{N}_0^2$ is a finite coordinate lower set, then for exactly $|\mathcal{S}|$ integer points (X_i, Y_i),

$$\prod_{(a,b)\in\mathcal{S}} a!b! \ \Big|\ \det\big(X_i^a Y_i^b\big)_{i,(a,b)\in\mathcal{S}}. \tag{271}$$

The proof uses the integer-valued basis $\binom{X}{a}\binom{Y}{b}$. On a weighted triangular lower set of cardinality N, the logarithm of this divisor is $O(N^{3/2}\log N) = o(N^2)$. It is a genuine source of primes absent from the matrix entries, but it cannot by itself repair a positive quadratic-scale determinant screen.

There is a sharper invariant for the unique $(M+1)$-term interpolating relation. Let $M \geq 2$ and let $u_1 < \cdots < u_M$ be distinct integers, with $v_1, \ldots, v_M \in \mathbb{Z}$. For the $M \times (M+1)$ matrix whose ith row is $(v_i, 1, u_i, \ldots, u_i^{M-1})$, let Δ_M be the gcd of all signed maximal minors and let

$$G_M = \gcd_i \prod_{a<b,\ a,b\neq i} (u_b - u_a), \qquad V_M = \prod_{a<b}(u_b - u_a), \qquad R_i = \prod_{j\neq i} |u_i - u_j|.$$

Then a primitive cofactor generator, determined only up to global sign, has Y-coefficient satisfying $|c_Y| = V_M/\Delta_M$, and

$$\mathfrak{F}_{M-1} \mid G_M \mid \Delta_M, \qquad G_M = \frac{V_M}{\operatorname{lcm}_i R_i}. \tag{272}$$

Moreover G_M is the greatest divisor forced uniformly by the nodes as the integer ordinate vector varies. The remaining quotient Δ_M/G_M is therefore the precise output-dependent Smith mass that an arithmetic proof must enlarge. These Smith statements and their 2^i3^j-grid asymptotics remain paper-level.

49.4 The adelic–HCIZ accounting identity

Let $N \geq 1$. For an exact-grid matrix $E_{ij} = \exp(u_i v_j) \in \mathbb{N}$ with strictly increasing $u_1 < \cdots < u_N$ and $v_1 < \cdots < v_N$, let $D_N > 0$ be its determinant. Put

$$C_N^- = \sum_i u_i v_{N+1-i}, \quad M_N = \frac{(\sum_i u_i)(\sum_j v_j)}{N}, \quad C_N^+ = \sum_i u_i v_i.$$

For each prime let τ_p be the minimum valuation of a Leibniz term, and set $T_N = \sum_p \tau_p \log p$ and $Q_N = \prod_p p^{\tau_p}$. The HCIZ identity writes

$$\log D_N = V_N + \log I_N, \qquad V_N = \log\Delta(u) + \log\Delta(v) - \log\prod_{j=1}^{N-1} j!.$$

Here I_N is the Harish–Chandra–Itzykson–Zuber integral $\int_{U(N)} \exp(\operatorname{tr}(\operatorname{diag}(u)U\operatorname{diag}(v)U^*))\,dU$ for normalized Haar probability measure. Rearrangement, averaging, Jensen, and von Neumann's trace inequality give the exact sandwich

$$T_N \leq C_N^- \leq M_N \leq \log I_N \leq C_N^+. \tag{273}$$

Defining

$$\Xi_N = \log(D_N/Q_N), \quad S_N = V_N + M_N - C_N^-, \quad H_N = \log I_N - M_N, \quad A_N = C_N^- - T_N$$

therefore gives

$$\boxed{\Xi_N = S_N + H_N + A_N}, \qquad H_N, A_N, \Xi_N \geq 0. \tag{274}$$

This separates geometric screen, random-matrix entropy, and incompatible primewise assignments instead of hiding them in one determinant estimate.

For the triangular spectra above, the best possible joint screen has normalized limit

$$f_{\text{joint}}(c) = \tfrac{1}{2}\log c - 1 + \left(\tfrac{4}{9} - \tfrac{\pi}{8}\right)c, \qquad c = 2\sqrt{\beta \log 2 \log 3}.$$

Its unique zero corresponds to $\beta = 6.9269442924\ldots$. Hence throughout the kernel-verified range $\beta \geq 12$, no combination of an exact HCIZ evaluation and termwise prime valuations can close this triangular determinant family. This is a no-go theorem for that geometry, not for all determinant methods.

A power-cusp proposal has a formal continuum screen tending to $-\infty$. The continuum formula itself is valid under convergence of logarithmic energies, but weak lattice convergence does not supply that hypothesis near the diagonal. A complete lattice realization needs an explicit spacing/truncation estimate. Even after such a repair, the separated-prime-support branch makes the adelic alignment defect grow linearly in the cusp parameter, so cusp concentration alone cannot be a universal proof.

49.5 Signed progressions and the denominator tax

Assume $N \geq 1$, $p, q, r, s, M, A \in \mathbb{N}_{>0}$, and the counterexample relations $M = 2^\beta$, $A = 3^\beta$ with β irrational. For $0 \leq i, j < N$, use the nonnegative exponent vectors

$$(a_i, b_i) = (iq, (N-1-i)p), \qquad (n_j, k_j) = (jr, (N-1-j)s).$$

Set

$$X = 2^{qr}, \quad Y = 3^{pr}, \quad Z = M^{qs}, \quad W = A^{ps}, \qquad P = XW, \quad Q = YZ.$$

Define the original integer matrix and its determinant by

$$E_{ij} = 2^{a_i n_j} 3^{b_i n_j} M^{a_i k_j} A^{b_i k_j}, \qquad D_N = \det(E_{ij})_{0 \leq i,j < N},$$

and let Q_N^{trop} be the termwise tropical divisor of this matrix. Then

$$\log(P/Q) = (q \log 2 - p \log 3)(r - s\beta),$$

so $P \neq Q$: unique factorization makes the first factor nonzero, while irrationality makes the second nonzero. A direct reduction to a geometric Vandermonde gives

$$|D_N| = (PQ)^{\binom{N}{3}} \prod_{d=1}^{N-1} |P^d - Q^d|^{N-d}. \tag{275}$$

After writing $g = \gcd(P, Q)$, $\widehat{P} = P/g$, $\widehat{Q} = Q/g$, every row, column, and common gauge factor cancels from the tropical quotient:

$$\frac{|D_N|}{Q_N^{\text{trop}}} = \prod_{d=1}^{N-1} |\widehat{P}^{\,d} - \widehat{Q}^{\,d}|^{N-d}. \tag{276}$$

Let $B = \min(\widehat{P}, \widehat{Q})$ and $\eta = |\log(\widehat{P}/\widehat{Q})|$. With $G(N+1) = \prod_{d=1}^{N-1} d^{N-d}$, the exact expansion is

$$\log \frac{|D_N|}{Q_N^{\mathrm{trop}}} = \binom{N+1}{3} \log B + \binom{N}{2} \log \eta + \log G(N+1) + R_N, \tag{277}$$

$$0 \le R_N \le \binom{N+1}{3} \eta. \tag{278}$$

The integer gap $|\widehat{P} - \widehat{Q}| \ge 1$ forces $\log B + \log \eta \ge -\eta$. Thus the product of two continued-fraction errors is not free: the denominator height repays the apparent quadratic smallness at cubic scale. This exact algebra rules out the naive double-continued-fraction progression, while leaving open more structured cyclotomic cancellation inside $\widehat{P}^d - \widehat{Q}^d$.

The module `SignedProgressionDeterminant` kernel-checks the geometric Vandermonde factorization, the primitive invariant quotient and its coprimality with both primitive bases, the exact finite logarithmic decomposition, the nonnegative remainder bound, and the integer-gap inequality. The original four-parameter matrix-to-tropical-quotient reduction, the continued-fraction asymptotics, and any general impossibility theorem for nontropical determinant cancellation remain paper-level. **[kernel-verified Lean]**

Status: The universal univariate superfactorial divisor and its strengthened repulsion/packing consequences are **[kernel-verified Lean]** in `SuperfactorialDeterminant`; the finite signed-progression and denominator-tax algebra is **[kernel-verified Lean]** in `SignedProgressionDeterminant`. The general support-spectrum, bivariate factorial/Smith, and HCIZ statements have paper proofs. For signed progressions, the original four-parameter/tropical identification and continued-fraction conclusion remain paper-level. No cusp lattice limit or full conjecture is asserted.

50 Phase randomization and positive-kernel certificates

The matrix-coefficient and sparse-grid preliminaries are already recorded in Sections 48.1–48.3. The distinct contribution here is a probability-theoretic control experiment that separates a generic random phase from the arithmetically distinguished phase zero. The experiment does not declare ϑ generic and does not prove the conjecture; it identifies one exact finite Fourier estimate that would suffice.

50.1 A uniform metric estimate and random-phase extinction

Fix $0 < a < b$ and, for $m \ge 2$ and $0 < \varepsilon \le 1/4$, set

$$E_m(\varepsilon) = \{\alpha \in [a, b] : \|m^\alpha\|_{\mathbb{Z}} < \varepsilon\}.$$

The monotone substitution $y = m^\alpha$ and a harmonic-sum estimate give

$$\mathrm{Leb}(E_m(\varepsilon)) \le \frac{8}{3}\left(b + \frac{1 + \log 3}{\log 2}\right)\varepsilon. \tag{279}$$

The constant is independent of m. Consequently every nonnegative summable sequence $\sum_m \varepsilon_m < \infty$ satisfies

$$\|m^\alpha\|_{\mathbb{Z}} < \varepsilon_m \quad \text{for only finitely many } m$$

for almost every $\alpha > 0$, by the first Borel–Cantelli lemma. This is a metric theorem, not a specialization theorem for the named exponent ϑ.

For the synchronized model, fix $N \geq 1$ and put

$$D_N = \{a \in \mathbb{Z} : 2^N < a < 2^{N+1}\}, \qquad Z_N(u;\eta) = \#\{a \in D_N : \left\|a^\vartheta - u\right\|_{\mathbb{Z}} < \eta\},$$

where u is a phase on $\mathbb{R}/\mathbb{Z}$. A uniform phase makes each radius-η arc have Haar measure 2η, so at the natural resolution $\eta = 3^{-N}$,

$$\mathbb{E}_u Z_N(u;3^{-N}) = 2(2^N - 1)3^{-N} < 2(2/3)^N. \tag{280}$$

Summability and Borel–Cantelli show that almost every single phase has no sufficiently late near hits. No independence between blocks is used.

Phase zero is qualitatively different. Let

$$\mathcal{H}_N = \#\{a \in D_N : a^\vartheta \in \mathbb{Z}\}.$$

If the conjecture holds then $\mathcal{H}_N = 0$. Under failure, the least nonintegral generator β and `ExactCounting` give the exact nonintegral count

$$\mathcal{H}_N = \left\lfloor \frac{N+1}{\beta} \right\rfloor, \qquad \frac{\mathcal{H}_N}{N} \longrightarrow \frac{1}{\beta}, \qquad Z_N(0;3^{-N}) \geq \mathcal{H}_N. \tag{281}$$

This is deliberately the nonintegral count: the full candidate count in the half-open dyadic block convention used elsewhere in this document has the additional integral point and equals $1 + \lfloor (N+1)/\beta \rfloor$.

50.2 The Fejér certificate

Write $e(t) = e^{2\pi i t}$ and, for an integer $H \geq 2$, define

$$F_H(t) = \frac{1}{H}\left|\sum_{j=0}^{H-1} e(jt)\right|^2, \qquad \Phi_H(t) = \frac{F_H(t)}{H}.$$

Then $\Phi_H \geq 0$, $\Phi_H(0) = 1$, and

$$\Phi_H(t) = \frac{1}{H}\sum_{|h|<H}\left(1 - \frac{|h|}{H}\right) e(ht).$$

With

$$S_N(h) = \sum_{a \in D_N} e(ha^\vartheta), \qquad Q_N(H) = \sum_{a \in D_N} \Phi_H(a^\vartheta),$$

positivity and the finite Fourier expansion give the exact certificate

$$\mathcal{H}_N \leq Q_N(H), \tag{282}$$

$$Q_N(H) = \frac{|D_N|}{H} + \frac{2}{H}\sum_{h=1}^{H-1}\left(1 - \frac{h}{H}\right)\Re S_N(h). \tag{283}$$

Thus the conjecture follows if, along an unbounded sequence, one can choose $H = H_N^*$ with $|D_N|/H_N^* = o(N)$ and make the weighted Fourier term in (283) equal to $o(N)$. The natural bandwidth $H = 3^N$ has exponentially small random-phase mean, but it demands information at exponentially many frequencies; fixed-mode equidistribution from `BlockWeyl` is therefore far too coarse.

The same diagnosis appears in the exact second moment. If $d_{a,a'} = \left\|\{a^\vartheta\} - \{a'^\vartheta\}\right\|_{\mathbb{Z}}$ and $0 < \eta \le 1/4$, then

$$\mathbb{E}_u Z_N(u;\eta)^2 = 2\eta|D_N| + 2\sum_{a<a'}(2\eta - d_{a,a'})_+.$$

The second term is the microscopic pair-correlation energy that a counterexample must force to be large at phase zero.

50.3 Why naive random sparse coefficients do not bridge the gap

The integrality amplifier behind the sparse certificate is exact: if $D \in \mathbb{N}_{>0}$, $0 \ne P \in \mathbb{Q}[X,Y]$, $DP \in \mathbb{Z}[X,Y]$, and $|P(g)| < D^{-1}$ at every conditional integer-grid point g, then $P(g) = 0$ everywhere. Sections 48.3 and 49.1 then turn a subcritical support into rationality of the common exponent.

Unconditioned random signs do not naturally reach this threshold. For distinct monomials M_ν, real c_ν, and independent signs ϵ_ν, put $P_\epsilon = \sum_\nu \epsilon_\nu c_\nu M_\nu$. On every finite grid $G \subset \mathbb{R}^2_{>0}$,

$$\mathbb{E}\sum_{g\in G}|P_\epsilon(g)|^2 = \sum_\nu |c_\nu|^2 \sum_{g\in G}|M_\nu(g)|^2. \tag{284}$$

At a fixed g, if $\sigma_g^2 = \sum_\nu |c_\nu M_\nu(g)|^2$, the fourth-moment bound and Paley–Zygmund give

$$\Pr\{|P_\epsilon(g)| \ge \sigma_g/\sqrt{2}\} \ge \frac{1}{12}.$$

When $\sigma_g = 0$ this is immediate; otherwise it is the usual Paley–Zygmund application to $|P_\epsilon(g)|^2$. This gives an RMS-scale baseline at each fixed point. It does not exclude rare simultaneous small-ball sign assignments, but any improvement must exploit more structure than this second-moment calculation records.

Status: The metric, Haar-phase, second-moment, and random-sign statements above have self-contained paper proofs. The phase-zero count is a repackaging of `ExactCounting`. The finite positive-kernel bridge, exact weighted Fourier score, and integer-phase hit bound are **[kernel-verified Lean]** in `FejerKernelCertificate`; Haar integration and Borel–Cantelli are not claimed there. Neither almost-everywhere extinction nor fixed-mode equidistribution is specialized to ϑ.

51 Compact orbit laws and approximation rigidity

Two complementary questions govern this section: what fixed measure-theoretic, profinite, and adelic observables can see under a hypothetical counterexample, and whether polynomial, rational, or Hermite–Padé approximation can exploit the exact integer lattice. Neither direction closes the conjecture. Every fixed compact factor is already maximally distributed, while the most natural approximation scales lose their gain when denominators are cleared.

51.1 Difference groups, triangle–Haar laws, and profinite Beatty factors

Assume in this subsection that the conjecture fails and let $\beta = B + \alpha$ be its least nonintegral solution, where $B = \lfloor \beta \rfloor$ and $0 < \alpha < 1$. Write

$$M = 2^\beta = 2^a W, \qquad A = 3^\beta = 3^b Z, \qquad W > 1 \text{ odd}, \quad Z > 1, \quad 3 \nmid Z.$$

The exact structure theorem gives $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ and unique coordinates. Taking differences produces an exact measure-theoretic dichotomy:

$$\mathcal{S} - \mathcal{S} = \begin{cases} \mathbb{Z}, & \text{if the conjecture holds,} \\ \mathbb{Z} + \mathbb{Z}\beta, & \text{if it fails.} \end{cases} \tag{285}$$

Modulo $\mathbb{Z}$, the failure branch is the countable dense Haar-null subgroup $\{k\alpha : k \in \mathbb{Z}\} \subset \mathbb{T}$. Thus closure, Haar measure, and the arithmetic set itself give three sharply different pictures of the same orbit. This is the first reason that a fixed weak-limit argument cannot settle the problem.

The macroscopic and microscopic coordinates can nevertheless be combined exactly. Put

$$L_T = \{(n, k) \in \mathbb{N}_0^2 : n + \beta k \le T\}, \qquad N_T = \#L_T = \frac{T^2}{2\beta} + O_\beta(T),$$

and $\Delta = \{(u, v) \in \mathbb{R}^2_{\ge 0} : u + v \le 1\}$. For every continuous $F : \Delta \times \mathbb{T} \to \mathbb{C}$,

$$\frac{1}{N_T} \sum_{(n,k) \in L_T} F\left(\frac{n}{T}, \frac{\beta k}{T}, \{k\beta\}\right) \longrightarrow 2 \int_\Delta \int_\mathbb{T} F(u, v, t)\, dt\, du\, dv. \tag{286}$$

For a nonzero circle Fourier mode, Abel summation reduces the assertion to bounded geometric partial sums of $e(hk\beta)$; the zero mode is an anisotropic Riemann sum. In the variables $H = u + v$ and $C = v/(u + v)$, the limiting density is $2H\, dH\, dC$, independently of the Haar phase. This is the precise independence statement; the separate Dirichlet description of $(u, v, 1 - u - v)$ is merely another description of uniform triangle area, not a replacement for the (H, C) factorization.

The finite congruence refinement is stronger. For every irrational α, every $d \ge 1$, residues $r, j \pmod d$, and interval $I \subset \mathbb{T}$,

$$\frac{1}{K} \#\{0 \le k < K : k \equiv r \pmod d,\ \lfloor k\alpha \rfloor \equiv j \pmod d,\ \{k\alpha\} \in I\} \longrightarrow \frac{|I|}{d^2}. \tag{287}$$

Indeed, after writing $k = d\ell + r$, the map $x \mapsto (\lfloor dx \rfloor, \{dx\})$ sends Lebesgue measure to uniform measure on $\mathbb{Z}/d\mathbb{Z}$ times Lebesgue measure. Equivalently,

$$k \longmapsto (k, \lfloor k\alpha \rfloor, \{k\alpha\})$$

is Haar distributed in $\widehat{\mathbb{Z}}^2 \times \mathbb{T}$. Every fixed family of congruence conditions is therefore asymptotically independent of the real phase. With the finite irrationality measure $\|h\beta\|_\mathbb{Z} \ge c_M h^{1-\mu_M}$, where we harmlessly enlarge μ_M so that $\mu_M > 1$, the same argument controls growing moduli only in the range $d^{\mu_M + 1} = o(K)$. The true denominator moduli grow exponentially in k, so this effective range remains far too small.

51.2 Exact denominator drift and noncompact escape

The compact arc coordinates

$$R_k = \frac{M^k}{2^{\lfloor k\beta \rfloor}}, \qquad S_k = \frac{A^k}{3^{\lfloor k\beta \rfloor}}$$

separate into primitive numerators and denominator exponents

$$q_k^{(2)} = \lfloor k\beta \rfloor - ka, \qquad q_k^{(3)} = \lfloor k\beta \rfloor - kb.$$

Writing $t_k = \{k\beta\}$ gives the exact identities

$$q_k^{(2)} = k(\beta - a) - t_k, \qquad q_k^{(3)} = k(\beta - b) - t_k, \tag{288}$$

$$q_k^{(2)} - q_k^{(3)} = k(b - a). \tag{289}$$

Both centered valuation defects are therefore the same bounded quantity $-t_k$, rather than independent noise. Since $\beta - a = \log_2 W > 0$ and $\beta - b = \log_3 Z > 0$,

$$|R_k|_2 = 2^{q_k^{(2)}} \longrightarrow \infty, \qquad |S_k|_3 = 3^{q_k^{(3)}} \longrightarrow \infty. \tag{290}$$

Thus the empirical measures escape every compact subset of $\mathbb{Q}_2 \times \mathbb{Q}_3$. Removing the denominator powers leaves (W^k, Z^k) in the compact group $\mathcal{H} = \overline{\langle (W, Z) \rangle} \subset \mathbb{Z}_2^\times \times \mathbb{Z}_3^\times$; the pair (t_k, W^k, Z^k) is distributed according to Lebesgue measure times Haar measure on $\mathcal{H}$. In a moving frame the limiting measure is the pushforward of $dt\, dm_{\mathcal{H}}$ under

$$(t, u, v) \longmapsto (t, -t, -t, u, v).$$

This retains exactly the phase information that one-point compactification would erase.

The integer definition, the cast identity, the common centered defect, the cross-base difference, the natural-exponent layer, and the resulting 2- and 3-adic escape are **[kernel-verified Lean]** in `AdelicDrift`. The module deliberately defines the raw exponent in $\mathbb{Z}$ before proving nonnegativity, so no hidden natural-subtraction assumption enters the statement.

51.3 Sturmian rigidity and fixed-observable blindness

Choose a representative $t \in [0, 1)$ of an initial circle point and set

$$\xi_j(t) = \mathbf{1}_{[1-\alpha,1)}(\{t + j\alpha\}).$$

Each bit is the carry in addition by α, so the exact telescoping identity is

$$\sum_{j=0}^{N-1} \xi_j(t) = \lfloor t + N\alpha \rfloor - \lfloor t \rfloor. \tag{291}$$

For uniform t, this sum takes only $\lfloor N\alpha \rfloor$ and $\lceil N\alpha \rceil$, with mean $N\alpha$ and variance $\{N\alpha\}\,(1-\{N\alpha\}) \leq 1/4$. If $\delta_\ell = \{\ell\alpha\}$, then

$$\operatorname{Cov}(\xi_0, \xi_\ell) = \max(0, \alpha - \delta_\ell) + \max(0, \alpha + \delta_\ell - 1) - \alpha^2. \tag{292}$$

The correlations do not decay. The process is a zero-entropy, pure-point factor of an irrational rotation; Bernoulli, Poisson, and central-limit heuristics are therefore the wrong model for its carries.

More generally, let G be a profinite group and let $\pi : \widehat{\mathbb{Z}}^2 \to G$ be continuous. Combining (286) and (287) shows that for continuous observables the scaled triangle coordinate, $\pi(k, \lfloor k\alpha \rfloor)$, and the circle phase have the product limit. The same holds for indicators whose discontinuity set has limiting measure zero; this qualification is needed for Sturmian bits, which are not continuous functions. Consequently every fixed compact congruence or unit factor is blind to the counterexample. A successful observable must move with height, retain an unbounded valuation, or resolve an exact shrinking target.

Two further paper-level calibrations reinforce the point. First,

$$\frac{1}{\log X}\int_1^X \delta_{(\{u\},\{u^\vartheta\})}\,\frac{du}{u} \xrightarrow{\;*\;} m_{\mathbb{T}^2};$$

nonzero Fourier modes follow from a first-derivative estimate. Second, for $E_m(\psi) = \{c \in [a,b] : \|m^c\|_{\mathbb{Z}} < \psi_m\}$ one has $\lambda(E_m(\psi)) \ll_{a,b} \psi_m$, and for $\psi_m = m^{-\tau}$,

$$\dim_{\mathrm{H}} \limsup_m E_m(m^{-\tau}) \le \min\left(1, \frac{b+1}{b+\tau}\right). \tag{293}$$

This can place ϑ in a zero-dimensional exceptional set without excluding that named number. The local mass estimate near exact hits above is only a scale diagnostic unless overlaps and uniform Taylor remainders are controlled, and is not promoted here to a theorem. The near-Sidon statement for conditional truncations is likewise retained only as a conditional consequence of the **[classical/open background]** Evertse–Schlickewei–Schmidt finite-rank unit-equation theorem.

51.4 Arithmetic resolution of normalized polynomial approximation

For the approximation problem, start from a candidate $m \in [2^T, 2^{T+1})$, put $u = m/2^T \in [1,2)$ and $n = m^\vartheta = 3^T u^\vartheta \in \mathbb{N}$, and keep careful track of the denominators introduced by normalization. If $p \in \mathbb{Q}[t]$, $\deg p \le d$, and $Hp \in \mathbb{Z}[t]$, then

$$H2^{Td}(n - 3^T p(u)) \in \mathbb{Z}. \tag{294}$$

Hence an error smaller than $(H3^T 2^{Td})^{-1}$ forces exact equality at every candidate in the block. For integral A, B of degrees at most d, the correct rational-approximation object is the numerator residual, not the quotient error:

$$2^{Td}(B(u)n - 3^T A(u)) = 3^T 2^{Td}(B(u)u^\vartheta - A(u)) \in \mathbb{Z}. \tag{295}$$

A small quotient can therefore be arithmetically useless when B is large or has costly coefficients.

The decisive analytic comparison is elementary. For every $d \ge 1$ and every real polynomial p of degree at most d,

$$\|t^\vartheta - p(t)\|_{[1,2]} \ge c_d := \frac{|\,(\vartheta)_{\underline{d+1}}\,|\,2^{\vartheta-2d-2}}{(d+1)^{d+1}}. \tag{296}$$

Indeed, with $h = (d+1)^{-1}$, the $(d+1)$-st forward difference of p vanishes, the triangle inequality costs 2^{d+1}, and the iterated finite-difference mean-value theorem evaluates the same difference of t^ϑ at an intermediate point. For $d = 0$, endpoint separation gives the lower bound 1.

Combining (294) and (296) yields the all-degree obstruction:

Theorem 51.1 (Normalization-tax obstruction)**.** *If $T \geq 4$, $d \geq 0$, $H \geq 1$, $\deg p \leq d$, and $Hp \in \mathbb{Z}[t]$, then*

$$H3^T 2^{Td} \|t^\vartheta - p(t)\|_{[1,2]} \geq 1. \tag{297}$$

Thus no normalized rational polynomial, in any degree, crosses the arithmetic forcing threshold.

The proof reduces to $T = 4, H = 1$. Writing $L_d = 3^4 2^{4d} c_d$, the rational bounds $3/2 < \vartheta < 8/5$ show $L_1 > \cdots > L_5 > 4$, $L_6 > 1$, and $L_{d+1}/L_d > 1$ for $d \geq 6$; in particular the exact calculation includes $L_5 > 13328/3125$. No finite search bound is used. The forward-difference estimate, polynomial degree floor, exact denominator clearing, and the all-degree tax theorem are **[kernel-verified Lean]** in `PolynomialApproximationTax`; the sharp Bernstein-ellipse rate and its degree constant remain classical paper inputs.

Working in the physical variable avoids the factor 2^{Td}, but then the same lower bound requires degree

$$d \geq \log_{4e} 3\, T - O(\log T) = 0.460384\ldots T - O(\log T).$$

Classical Bernstein approximation improves the slope to $\log_{3+2\sqrt{2}} 3 = 0.623238\ldots$. Either slope exceeds the conditional block population slope by a wide margin; this is a structural mismatch, not a numerical failure at the beginning of the search range.

51.5 Rational contact rigidity and the area-to-diameter collapse

The local approximation audit has a sharper exact conclusion. For a polynomial P and a rational point (a, b), let $\operatorname{mult}_{(a,b)} P$ be the least total degree in the Taylor expansion of $P(a + U, b + V)$.

Theorem 51.2 (Rational contact equals algebraic multiplicity)**.** *Let τ be transcendental, $0 \neq P \in \mathbb{Q}[X, Y]$, and $(a, b) \in \mathbb{Q}^2_{>0}$. If $s = \operatorname{mult}_{(a,b)} P$, then*

$$\operatorname{ord}_{z=0} P(ae^z, be^{\tau z}) = s. \tag{298}$$

If $H_s(U, V) = \sum_{j=0}^{s} h_j U^{s-j} V^j$ is the first nonzero homogeneous Taylor layer, the coefficient of z^s is $H_s(a, b\tau) = \sum_j h_j a^{s-j} b^j \tau^j$. This is a nonzero rational polynomial evaluated at a transcendental number, so it cannot vanish. In particular, for bidegree at most (d, e) the contact order is at most $d + e$; equality is attained only by a rational scalar multiple of $(X - a)^d (Y - b)^e$. The square of $(d + 1)(e + 1)$ coefficient slots thus collapses to the diameter $d + e$ over $\mathbb{Q}$. Over $\mathbb{Q}(\tau)$ the ordinary distinct-frequency Vandermonde instead permits order $(d + 1)(e + 1) - 1$.

For not-both-zero $A, B \in \mathbb{Q}[t]$ of degree at most d this gives

$$\operatorname{ord}_{t=1}(B(t)t^\tau - A(t)) \leq d + 1.$$

If $A(1) = B(1)$ and $B(1) \neq 0$, the intersection is already transverse because

$$B(1)\tau + B'(1) - A'(1) \neq 0. \tag{299}$$

The rational-polynomial transcendence lemma, nonvanishing of the homogeneous leading coefficient, the derivative identity, and the Padé transversality endpoint are **[kernel-verified Lean]** in `RationalContactRigidity`. The general Taylor-order packaging in Theorem 51.2 remains a self-contained paper proof.

Finally, for a finite support $S \subset \mathbb{N}_0^2$ and nonzero integer coefficients,

$$F(x) = \sum_{(i,j)\in S} c_{ij} x^{i+j\vartheta}$$

is integral at every candidate. If $\|F\|_I < 1$, it vanishes at all candidates in I, while the accepted exponential-polynomial zero bound gives at most $|S| - 1$ positive zeros. Hence

$$\#(\mathcal{C} \cap I) \le |S| - 1. \tag{300}$$

This arithmetic Müntz certificate is a direct corollary of the existing `ExponentialPolynomialZeros` machinery, not a new approximation theorem. The genuinely open target is to find an integral coefficient vector with norm below one after all archimedean, denominator, and p-adic height costs are charged. The schematic adelic height functional proposed here records this target but is not itself a proved estimate.

Status: The exact drift/escape package is **[kernel-verified Lean]** in `AdelicDrift`; the finite-difference and normalization-tax package is **[kernel-verified Lean]** in `PolynomialApproximationTax`; the exact carry, prefix, mean, variance, joint-probability, and covariance package is **[kernel-verified Lean]** in `SturmianProbability`; and the homogeneous contact obstruction plus rational Padé transversality are **[kernel-verified Lean]** in `RationalContactRigidity`. The triangle–Haar, profinite Beatty, logarithmic Haar, Hausdorff-dimension, full contact-order, and area-to-diameter statements above have self-contained paper proofs. The effective discrepancy uses the Baker–Matveev interface, the near-Sidon law uses the external finite-rank unit-equation theorem, and the sharp Bernstein rate is classical. The documented 2^{29} computation is not promoted: the reproducible trust frontier remains 2^{27}.

52 Analytic common divisors and natural boundaries

The approximation material is already consolidated in Section 51. The distinct contribution here is a three-part analytic reformulation: cancel the automatic integer zeros of the two sine compositions, identify the resulting conditional zero lattice and its canonical divisor, and encode the synchronized denominator exponents by integer power series with a natural boundary. A final compact-height count sharpens the earlier qualitative compact-orbit equivalence.

52.1 Removing the automatic zeros and counting the conditional divisor

Put

$$F_a(z) = \sin(\pi a^z), \qquad a \in \{2, 3\}.$$

The accepted complex-localization theorem already shows that every common zero of F_2 and F_3 is real and simple. The nonnegative integers are automatic common zeros. They can be removed without introducing new ones by filling the removable singularities in

$$G_a(z) = \Gamma(-z)\sin(\pi a^z). \tag{301}$$

Indeed, at $n \in \mathbb{N}_0$ one has

$$G_a(n) = \frac{\pi(\log a)a^n}{n!}(-1)^{n+1+a^n} \neq 0, \tag{302}$$

so G_2 and G_3 have a common zero exactly when the conjecture has a nonintegral solution.

Under failure, let β be the least nonintegral solution. The exact conditional monoid classification identifies the de-integralized common zero set as

$$\Lambda_\beta = \{n + k\beta : n \in \mathbb{N}_0,\ k \in \mathbb{N}\}.$$

Its representation is unique, and elementary lattice counting gives

$$N_\beta(R) := \#(\Lambda_\beta \cap [0, R]) = \frac{R^2}{2\beta} + O_\beta(R). \tag{303}$$

Consequently its exponent of convergence is exactly two. With $E_2(w) = (1-w)\exp(w + w^2/2)$, the genus-two product

$$P_\beta(z) = \prod_{\lambda \in \Lambda_\beta} E_2(z/\lambda) \tag{304}$$

converges locally uniformly, has precisely those simple zeros, and divides both G_2 and G_3 in the ring of entire functions. Conversely, any common nonunit entire divisor has a zero and hence gives a nonintegral solution. Thus the conjecture is equivalently the absence of a common nonunit entire divisor.

This is not an ordinary finite-order zero-counting problem: along suitable horizontal lines, $\log M(r, G_a) \geq c_a a^r$ for infinitely many r. The conditional divisor has only quadratic zero density inside functions of infinite order. It is also too large for a bounded-characteristic function on the right half-plane, since partial summation gives

$$\sum_{\lambda \in \Lambda_\beta} \frac{\lambda}{1+\lambda^2} = \infty. \tag{305}$$

This supplies a precise but still open analytic target: a growth-controlled common divisor or Bézout theorem for the pair G_2, G_3. Canonical-product convergence and the bounded-characteristic/Blaschke implication are classical analytic interfaces, not new Lean claims.

52.2 The synchronized word has a natural boundary

Write $\beta = B + \alpha$ with $0 < \alpha < 1$, and define

$$e_k = \lfloor k\alpha \rfloor, \qquad s_k = e_{k+1} - e_k \in \{0, 1\}.$$

For the formal power series

$$\mathcal{S}(X) = \sum_{k \geq 0} s_k X^k, \qquad \mathcal{E}(X) = \sum_{k \geq 1} e_k X^k,$$

telescoping gives the exact identity

$$X\mathcal{S}(X) = (1 - X)\mathcal{E}(X). \tag{306}$$

Both analytic series have radius one. The binary Sturmian word (s_k) is not eventually periodic because α is irrational. A rational finite-alphabet power series has an eventually periodic coefficient sequence; hence $\mathcal{S}$ is not rational. The classical Pólya–Carlson dichotomy now makes the unit circle a natural boundary for $\mathcal{S}$, and (306) transfers that boundary to $\mathcal{E}$.

The exact reduced denominator exponents have the form

$$q_{r,k} = rk + e_k, \qquad \mathcal{Q}_r(X) = \frac{rX}{(1-X)^2} + \mathcal{E}(X).$$

Thus every $\mathcal{Q}_r$ has the same natural boundary, and

$$\mathcal{Q}_r - \mathcal{Q}_s = \frac{(r-s)X}{(1-X)^2}. \tag{307}$$

The carry telescope, formal-series identity, denominator decomposition, twice-cleared defect, and cross-base cancellation are **[kernel-verified Lean]** in `SturmianGeneratingSeries`. The natural-boundary conclusion itself invokes the classical Pólya–Carlson theorem. It is a one-slope diagnostic, not a contradiction: every irrational rotation already has this boundary, so a successful theorem must use both prime supports together.

52.3 A logarithmic compact-height criterion

Let

$$\mathcal{C}_{2,3} = \{(2^t, 3^t) : 0 \le t < 1\} \cap (\mathbb{Z}[1/2] \times \mathbb{Z}[1/3]),$$

and for a reduced rational pair define $H(a/b, c/d) = \max(|a|, b, |c|, d)$. Write $N_{2,3}(H)$ for the number of points of $\mathcal{C}_{2,3}$ of height at most H. The accepted compact-orbit theorem gives the qualitative equivalence $\mathcal{C}_{2,3} = \{(1,1)\}$ if and only if the conjecture holds. This gives the exact growth dichotomy

$$N_{2,3}(H) = \begin{cases} 1, & \text{if the conjecture holds,} \\ \asymp_\beta \log H, & \text{if it fails.} \end{cases} \tag{308}$$

In the failure branch, the distinct points $(2^{\{k\alpha\}}, 3^{\{k\alpha\}})$ exhaust the localized arc, and their reduced numerator and denominator heights grow between two fixed exponentials in k. Therefore each of

$$N_{2,3}(H) = O(1), \qquad N_{2,3}(H) = o(\log H)$$

is equivalent to the conjecture. Compact normalization has removed the automatic integer direction from the established $\asymp (\log H)^2$ unbounded count and leaves precisely one conditional generator.

Status: The formal generating-series algebra is **[kernel-verified Lean]** in `SturmianGeneratingSeries`. The Gamma cancellation, quadratic lattice count, compact height dichotomy, and Blaschke calculation have self-contained paper proofs; canonical-product and Pólya–Carlson steps are classical interfaces. The duplicated Bernstein approximation material is consolidated in Section 51. The documented 2^{29} scan is not promoted; the reproducible trust frontier remains 2^{27} computation.

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53 Matrix powers, Loewner certificates, and the one-half barrier

This matrix-theoretic approach produces one exact reformulation and a family of genuine determinant certificates, but it also proves that polynomial representations of the original matrix carry no information beyond the two assumed integer outputs. The optimized determinant family reaches a limiting one-half height exponent and therefore remains on the same side of the four-exponentials wall.

53.1 One fixed integral matrix

Consider

$$A_0 = \begin{pmatrix} 0 & -6 \\ 1 & 5 \end{pmatrix}. \tag{309}$$

Its eigenvalues are $2, 3$, and the complementary integral spectral projectors are $P_2 = 3I - A_0$ and $P_3 = A_0 - 2I$. Real functional calculus therefore gives

$$A_0^t = 2^t P_2 + 3^t P_3 = (3\,2^t - 2\,3^t)I + (3^t - 2^t)A_0. \tag{310}$$

Consequently

$$A_0^t \in M_2(\mathbb{Z}) \iff 2^t, 3^t \in \mathbb{Z}. \tag{311}$$

For the nontrivial direction, if $B = A_0^t$ is integral then $c = B_{21} = 3^t - 2^t$ and $d = B_{11} = 3\,2^t - 2\,3^t$ are integers, while $2^t = d + 2c$ and $3^t = d + 3c$. Equivalently,

$$\text{Alaoglu–Erdős} \iff \{t \in \mathbb{R} : A_0^t \in M_2(\mathbb{Z})\} = \mathbb{N}_0.$$

The concrete matrix, all four entry formulas, the exact entry-integrality equivalence, and this conjectural equivalence are **[kernel-verified Lean]** in `IntegralPowerMatrix`.

This reformulation is exact but intentionally not advertised as extra arithmetic. The unimodular matrix $\left(\begin{smallmatrix} -3 & -2 \\ 1 & 1 \end{smallmatrix}\right)$ diagonalizes A_0 over $\mathbb{Z}$, so the lattice already splits into its 2- and 3-eigensummands. Every polynomial, tensor, symmetric, or exterior-power construction from (A_0, A_0^t) is a polynomial in the two integers $(2^t, 3^t)$ and their monomials. It cannot recover the common real time unless it introduces analytic dependence, a denominator-sensitive inequality, or a new logarithmic direction.

53.2 Cross-Loewner and generalized-alternant certificates

For distinct positive nodes $z_1, \ldots, z_N$ and distinct real exponents $\gamma_1, \ldots, \gamma_N$, strict total positivity of e^{uv} gives

$$\det(z_i^{\gamma_j}) \neq 0,$$

with positive sign when both lists are increasing. If t is a nonintegral solution, the two-block exponent list

$$0, 1, \ldots, r-1, \quad t, t+1, \ldots, t+r-1 \tag{312}$$

is distinct. At $2r$ distinct nodes in the $2, 3$-smooth semigroup every entry is integral, so the alternant is a nonzero integer.

The same certificate admits an exact divided-difference form. For disjoint r-tuples $X = (x_i)$ and $Y = (y_j)$ and a function f on their union, put

$$L_f(X,Y)_{ij} = \frac{f(x_i) - f(y_j)}{x_i - y_j}.$$

If $M_f(X,Y)$ is the $2r$-row alternant with columns $1, z, \dots, z^{r-1}, f(z), zf(z), \dots, z^{r-1}f(z)$, then

$$\det M_f(X,Y) = (-1)^{r(r+1)/2} \left(\prod_{i,j} (x_i - y_j) \right) \det L_f(X,Y). \tag{313}$$

For $f(u) = u^t$ and smooth integer nodes, the right-hand side is therefore a nonzero integer. Writing $t = k + \alpha$ with $0 < \alpha < 1$, the fractional-power kernel

$$K_\alpha(u,v) = \frac{u^\alpha - v^\alpha}{u - v}$$

has the positive Stieltjes representation

$$K_\alpha(u,v) = \frac{\sin(\pi\alpha)}{\pi} \int_0^\infty \frac{\lambda^\alpha}{(u+\lambda)(v+\lambda)}\, d\lambda.$$

Thus its ordered cross minors are strictly positive. Combining positivity with (313) gives the explicit positive integer

$$\left(\prod_i x_i^k y_i^k \right) \left| \prod_{i,j} (x_i - y_j) \right| \det(K_\alpha(x_i, y_j)) \in \mathbb{Z}_{>0}. \tag{314}$$

The positivity holds for every $0 < \alpha < 1$; all arithmetic content is in the displayed normalization. These finite determinant identities have paper proofs here. They are not silently conflated with the already formal generalized-Vandermonde positivity theorem.

53.3 Quantitative repulsion and the optimized barrier

Let $z_i = e^{u_i}$ with pairwise distinct $u_i \in [T, T+\delta]$, $0 < \delta \le 1$, and pairwise distinct exponents $0 \le \gamma_j \le B$. (With repeated nodes or exponents the determinant vanishes and the upper bound is trivial.) Put $\Sigma = \sum_j \gamma_j$, $D = N(N-1)/2$, and $B_* = \max(1, B)$. Newton divided differences followed by Hadamard's inequality give the explicit upper bound

$$\left| \det(z_i^{\gamma_j}) \right| \le e^{T\Sigma} \delta^D \frac{N^{N/2} e^{NB} B_*^D}{\prod_{\ell=0}^{N-1} \ell!}. \tag{315}$$

Together with the nonzero-integer lower bound, this excludes a sufficiently tight cluster of $2r$ smooth nodes. For the equal two-block list (312) it gives an explicit bound of the shape

$$\delta \ge C_{r,t}^{-1/[r(2r-1)]} R^{-(t+r-1)/(2r-1)}.$$

The exponent tends to $1/2$ as $r \to \infty$, much smaller than the logarithmic spacings available from counting smooth numbers.

A return-column specialization makes the remaining denominator target especially explicit. Choose distinct times n_j whose fractional parts $0 < \delta_0 < \cdots < \delta_{N-1} \le \varepsilon$ are ordered, put $m_j = \lfloor n_j \beta \rfloor$, and order the distinct monomials $\lambda_0 < \cdots < \lambda_{N-1}$, where $\lambda_i = a_i \log 2 + b_i \log 3$ with $0 \le a_i \le D_1$, $0 \le b_i \le D_2$. Then

$$\mathcal{A} = \det(e^{\lambda_i \delta_j}), \qquad \left(\prod_j 2^{D_1 m_j} 3^{D_2 m_j}\right) \mathcal{A} \in \mathbb{Z}_{>0}, \tag{316}$$

and the same divided-difference argument gives

$$0 < \mathcal{A} \le e^{NL\varepsilon} N^{N/2} \frac{(L\varepsilon)^{N(N-1)/2}}{\prod_{r=0}^{N-1} r!}, \qquad L = \max_i \lambda_i. \tag{317}$$

For the square box $0 \le a, b \le D$, $N = (D+1)^2$ and $L \le D \log 6$. For fixed $\varepsilon > 0$, taking the first N return times to $(0, \varepsilon]$ gives $\sum_j m_j = O_{\beta,\varepsilon}(N^2)$, so generic column-by-column clearing costs $O_{\beta,\varepsilon}(N^{5/2})$. Meanwhile

$$\log |\mathcal{A}| \le -\frac{1}{4} N^2 \log N + O_{\beta,\varepsilon}(N^2),$$

because the superfactorial contributes $\frac{1}{2} N^2 \log N + O(N^2)$ and the $L^{N(N-1)/2}$ term pays back only one quarter. Thus the archimedean gain is quadratic, but the universal integerizer loses one extra factor $D \asymp \sqrt{N}$. A shared-denominator normalization of cost $o(N^2 \log N)$, or of that order with a sufficiently small leading constant, would cross this particular barrier. No such divisibility theorem is known. Pigeonhole clustering alone supplies only an upper bound on a generic clearing cost and does not prove that every clustered construction fails.

Allowing unequal block sizes does not improve the limiting ratio. With $p + q = N$ and exponent set $0, \ldots, p-1, t, \ldots, t+q-1$, the total exponent is

$$\Sigma(p, q; t) = \frac{p(p-1)}{2} + qt + \frac{q(q-1)}{2} = D + q(q + t - N), \tag{318}$$

so

$$\min_{p+q=N} \frac{\Sigma(p, q; t)}{D} = \frac{1}{2} + O_t(N^{-1}). \tag{319}$$

Initial consecutive shifts already minimize the height inside each coset. Translating signed exponents into the first quadrant restores exactly the denominator height that cancellation appeared to save. This is the matrix one-half barrier.

There is a dual exact identity for candidate exponents. Let $d \ge 1$, let $p, q \in \mathbb{N}$ with $q > 0$, and put $\varepsilon = qt - p$ and $\tau_i = (d-1-i)p + iqt$ for $0 \le i < d$. Then at distinct smooth bases s_j, $0 \le j < d$,

$$\det(s_j^{\tau_i}) = \left(\prod_j s_j^{(d-1)p}\right) \prod_{j<k} (s_k^{\varepsilon} - s_j^{\varepsilon}). \tag{320}$$

Its nonzero-integral lower bound yields an explicit but exponentially weak irrationality estimate for $|qt - p|$. Higher dimension organizes the elementary integer $2^{qt} - 2^p$ but does not change its exponential scale. The affine-exponent determinant, its real-power specialization, the exact semigroup exponent identity, and (320) are **[kernel-verified Lean]** in `CandidateAlternant`; the mean-value upper bound and resulting irrationality estimate remain the paper layer.

Status: The exact fixed matrix reformulation is [kernel-verified Lean] in `IntegralPowerMatrix`. Generalized-Vandermonde nonvanishing and ordered-chamber positivity are separately [kernel-verified Lean] in the established determinant modules, and the exact rational-approximation alternant is [kernel-verified Lean] in `CandidateAlternant`. The Sylvester–Loewner factorization, positive normalized determinant, explicit alternant bound, and optimized one-half barrier are retained as paper theorems. No claim is made that formal representation theory manufactures a third exponential, or that matrix positivity by itself selects integral exponents.

54 Finite-place transversality and local rank-drop certificates

The real prime-symbol polynomial, rational matrix-coefficient obstruction, and sharp support bound are already represented by `FormalPrimeDeterminant`, `RestrictedMatrixCoefficient`, and `SparsePowerCurve`. The genuinely distinct layer compares the singular real logarithm matrix with its finite-place analogues, isolates the first Fermat-quotient obstruction, and shows why ordinary real rational approximation does not propagate to a local rank drop.

54.1 The structural primes are locally transverse

For the Iwasawa logarithm $\log_p : \mathbb{Q}_p^* \to \mathbb{Q}_p$, normalized by $\log_p p = 0$, define

$$L_p(M, A) = \begin{pmatrix} \log_p 2 & \log_p M \\ \log_p 3 & \log_p A \end{pmatrix}, \qquad D_p(M, A) = \log_p M \log_p 3 - \log_p A \log_p 2. \tag{321}$$

On positive rationals the kernel is exactly

$$\log_p u = 0 \quad \iff \quad u = p^k \text{ for some } k \in \mathbb{Z}. \tag{322}$$

Hence a nonintegral counterexample pair $(M, A) = (2^\beta, 3^\beta)$ satisfies

$$D_2(M, A) = \log_2 M \log_2 3 \neq 0, \tag{323}$$

$$D_3(M, A) = -\log_3 A \log_3 2 \neq 0. \tag{324}$$

The real matrix has rank one, but both finite-place matrices at the primes defining the problem have rank two. If, for a prime p, the pair (M, A) belonged to the closure of $\{(2^n, 3^n) : n \in \mathbb{Z}\}$ in $(\mathbb{Q}_p^*)^2$, continuity of $\log_p$ would instead force $D_p(M, A) = 0$. Thus a real-to-local closure principle at either structural prime would prove the conjecture, but no such transfer theorem is known.

This is not merely a change of language. It reverses the naive local expectation: a hypothetical counterexample is singular only at the real place among the three places that are directly visible from its definition. Any adelic proof must compare a real zero with proven local nonzeros, not copy the real relation into $\mathbb{Q}_2$ or $\mathbb{Q}_3$.

54.2 The Fermat-quotient first layer

For an odd prime $p \nmid u$, set

$$q_p(u) = \frac{u^{p-1} - 1}{p} \pmod{p}.$$

The first Iwasawa-logarithm digit is

$$\frac{\log_p u}{p} \equiv -q_p(u) \pmod{p}. \tag{325}$$

Therefore, for $p \geq 5$ and $p \nmid 6MA$, the determinant

$$\Delta_p(M, A) = q_p(M)q_p(3) - q_p(A)q_p(2) \pmod{p} \tag{326}$$

satisfies

$$\frac{D_p(M, A)}{p^2} \equiv \Delta_p(M, A) \pmod{p}. \tag{327}$$

Thus $\Delta_p \neq 0$ certifies the exact first possible valuation $v_p(D_p) = 2$. Conversely, for every integer-exponent pair $(M, A) = (2^n, 3^n)$,

$$q_p(u^n) \equiv nq_p(u) \pmod{p}$$

makes $\Delta_p(M, A) = 0$. A local rank drop must therefore pass a generalized Wieferich congruence and then an infinite tower of higher congruences. The finite near-miss scan is retained only as a software diagnostic; it is not evidence that every primitive pair has a bounded detecting prime.

54.3 Real approximants and the exact adelic wedge

Let r/s be a rational approximation to β, $s > 0$, and put

$$R_2(r, s) = \frac{M^s}{2^r}, \qquad R_3(r, s) = \frac{A^s}{3^r}.$$

At the real place the errors are perfectly synchronized:

$$\log R_2 = (s\beta - r)\log 2, \qquad \log R_3 = (s\beta - r)\log 3.$$

At a finite place set

$$E_p(r, s) = (s\log_p M - r\log_p 2,\ s\log_p A - r\log_p 3).$$

Direct expansion gives the exact wedge identity

$$E_{p,1}(r, s)\log_p 3 - E_{p,2}(r, s)\log_p 2 = sD_p(M, A). \tag{328}$$

At the structural primes the numerator r disappears from one coordinate:

$$\log_2 R_2(r, s) = s\log_2 M, \qquad v_2(\log_2 R_2) = v_2(s) + v_2(\log_2 M), \tag{329}$$

$$\log_3 R_3(r, s) = s\log_3 A, \qquad v_3(\log_3 R_3) = v_3(s) + v_3(\log_3 A). \tag{330}$$

Choosing a better continued-fraction numerator improves the real approximation but has no effect on these two local depths; only divisibility of the denominator s helps. This exact anti-transfer explains why an archimedean convergent does not automatically produce a local near-singularity.

54.4 Cascade depth and the Kummer norm diagnostic

There are also two useful no-go estimates. First, suppose each of the first $J(T)$ active tropical layers contributes at most $CT^{3/2}$ fixed-base logarithmic factors, each of valuation at most $C'(\log T)^2$. Their total exposed valuation is then

$$O\Big(J(T)T^{3/2}(\log T)^2\Big). \tag{331}$$

Against $H_T \asymp T^{5/2}$ this is lower order whenever $J(T) = o(T/(\log T)^2)$. This is a conditional bookkeeping proposition for a factorized layer model, not an upper bound on arbitrary determinant cancellation; a global higher initial form may behave differently.

Second, for positive real Nth roots define the algebraic integer

$$\Xi_N = (M^{1/N} - 1)(3^{1/N} - 1) - (A^{1/N} - 1)(2^{1/N} - 1).$$

The determinant relation cancels the quadratic term and leaves

$$\Xi_N = \frac{\beta \log 2 \log 3\,(\beta - 1)(\log 2 - \log 3)}{2N^3} + O_{M,A}(N^{-4}). \tag{332}$$

Thus the distinguished embedding is polynomially small and nonzero for large N. The field degree and varying, generally uncontrolled root-of-unity twists (subject to the algebraic relations among the radicands) grow with N, however, so a crude product-formula bound becomes worse rather than better. A viable algebraic-integer construction would need simultaneous smallness in many conjugates or exponential smallness at one place.

Status: The real prime-symbol and matrix-coefficient layers are already **[kernel-verified Lean]** in `FormalPrimeDeterminant` and `RestrictedMatrixCoefficient`; its sparse-grid core is **[kernel-verified Lean]** in `SparsePowerCurve`. The exact Fermat-quotient divisibility identity, power law modulo p, determinant interface, and its vanishing on $(2^n, 3^n)$ are **[kernel-verified Lean]** in `FermatQuotientDeterminant`. The passage from that residue determinant to the Iwasawa logarithm uses the classical congruence (325); the structural-prime theorem uses the classical Iwasawa-logarithm kernel. The exact wedge, conditional cascade-depth, and Kummer expansion remain elementary paper results. No p-adic Matrix Coefficient or real-to-local transfer theorem is asserted.

55 Prime switches and microscopic resonance

This section returns to the historical origin of the problem: simultaneous changes in the prime exponents of colossally abundant numbers. This is a valuable structured test family, but its logical scope is narrower than the integer-output conjecture. A switch produces rational values p^ε and q^ε; excluding all switches would prove only a special case of the stronger rational-output statement. No reduction from arbitrary rational outputs, or from this switch family, to Conjecture 1.1 is known.

55.1 Exact switch heights and logarithmic means

For a prime p and an integer $a \geq 1$, set

$$H_{p,a} = p + p^2 + \cdots + p^a, \qquad \varepsilon_{p,a} = \log_p(1 + H_{p,a}^{-1}). \tag{333}$$

Comparing the two consecutive local factors in $\sigma(n)/n^{1+\varepsilon}$ gives the exact tie identity

$$p^{\varepsilon_{p,a}} = \frac{p^{a+1}-1}{p^{a+1}-p} = 1 + \frac{1}{H_{p,a}}. \tag{334}$$

For fixed a this parameter decreases with p, and for fixed p it decreases with a. If $p < q$ and $\varepsilon_{p,a} = \varepsilon_{q,b}$, then

$$H_{p,a} > H_{q,b}, \qquad a > b. \tag{335}$$

Thus simultaneous switches form an antichain in prime size and local exponent.

The analytic center of the calculation is the consecutive logarithmic mean

$$R(H) = \frac{1}{\log(1+1/H)}, \qquad J(H) = H + \frac{1}{2} - \frac{1}{12H}.$$

For every real $H \geq 1$ one has the globally valid two-sided enclosure

$$J(H) < R(H) < J(H) + \frac{1}{24H^2}. \tag{336}$$

The proof differentiates two rational majorants for $\log(1+x)$ on $0 < x \leq 1$; it is an inequality with an exact remainder, not merely an asymptotic expansion. The definitions and the complete enclosure (336) are **[kernel-verified Lean]** in `ConsecutiveLogMeanBounds`.

At a collision, $R(H_{p,a})\log p = R(H_{q,b})\log q$. Writing $H = H_{p,a}$ and $K = H_{q,b}$, (336) therefore gives

$$|J(H)\log p - J(K)\log q| < \frac{\log p}{24H^2} + \frac{\log q}{24K^2}. \tag{337}$$

This is a highly structured Dirichlet-scale approximation: the coefficients are geometric sums, and exploiting that structure rather than merely invoking a general irrationality measure is essential.

55.2 A microscopic prescribed-prime window

For fixed $b \geq 1$, put

$$H_b(t) = t + t^2 + \cdots + t^b, \qquad \Phi_b(t) = J(H_b(t))\log t.$$

Since $\Phi_b'(t) > bt^{b-1}\log t$ on $t > 1$, there is a unique real $Q_{p,a,b} > 1$ satisfying

$$\Phi_b(Q_{p,a,b}) = J(H_{p,a})\log p.$$

If primes $p < q$ switch simultaneously, the mean-value theorem and (337) imply

$$|q - Q_{p,a,b}| < \frac{\frac{\log p}{24H_{p,a}^2} + \frac{\log q}{24H_{q,b}^2}}{b(q-\frac{1}{2})^{b-1}\log(q-\frac{1}{2})} = O_b(q^{1-3b}). \tag{338}$$

The interval has length less than one. Already for $b = 1$ its radius is $O(q^{-2})$. Consequently modern prime-in-short-interval theorems, whose intervals still have positive power length, cannot decide the switch: before primality enters, the nearest integer has already been prescribed. A suggested modernization of the 1944 prime-block argument is retained only as a schematic transference of known prime-distribution input; its required uniform error estimates are not reconstructed here. Likewise, the first-order formula for the location of $Q_{p,a,b}$ is a useful calibration, but its displayed proof in the displayed derivation omits a bootstrap and is not used as a theorem in this synthesis.

55.3 The exact 2–3 convergent sieve

For a possible switch between 2 at level a and 3 at level b, define

$$P = 2^{a+2} - 3, \qquad Q = 3^{b+1} - 2. \tag{339}$$

The second-order enclosure, together with the elementary bounds $3/2 < \vartheta < 8/5$, yields the exact necessary condition

$$0 < \vartheta - \frac{P}{Q} < \frac{1}{2Q^2}. \tag{340}$$

After reduction, P/Q must therefore be a continued-fraction convergent of ϑ. Equivalently, a collision must hit an exponentially shrinking shifted-power target

$$0 < \vartheta(3^{b+1} - 2) - (2^{a+2} - 3) < \frac{1}{2(3^{b+1} - 2)}.$$

This converts the transcendental equality into a discrete S-unit sieve, but no uniform argument over all convergents is currently known.

The literal exact-integer verifier checks this necessary condition for $1 \le b \le 10000$. It certifies 10000 unambiguous floors, finds the sole power-shape hit $(a, b, P, Q) = (5, 3, 125, 79)$, and rejects that hit by the stronger Legendre window; hence there are no survivors in the tested range. An independent rerun reproduced every mathematical field. This remains **[software computation]**, not a kernel proof and not a finite test of the original integer-output conjecture.

55.4 What prime distribution can and cannot force

For fixed integers $U, V \ge 2$, define the universal Fermat-synchronized divisor

$$F_{U,V}(n) = \prod_{\substack{\ell \text{ prime},\ \ell \nmid UV \\ \ell - 1 \mid n}} \ell.$$

Fermat's theorem gives

$$F_{U,V}(n) \mid \gcd(U^n - 1, V^n - 1), \qquad \log F_{U,V}(n) \le \tau(n) \log(n+1) = n^{o(1)}. \tag{341}$$

Along $n_y = \operatorname{lcm}(1, \ldots, \lfloor y \rfloor)$, the prime number theorem and the standard divisor estimate still leave $\log F_{U,V}(n_y) = n_y^{o(1)} = o(n_y)$. This is far below the exponential common-divisor growth that would contradict the Bugeaud–Corvaja–Zannier bound for multiplicatively independent U, V. Generic prime distribution therefore cannot manufacture the missing real-to-finite-place correlation. It becomes relevant only after another argument has produced a rigid prescribed integer, such as the nearest integer in (338) or a convergent satisfying (340).

Status: The exact logarithmic-mean enclosure is **[kernel-verified Lean]** in `ConsecutiveLogMeanBounds`. The switch formula, antichain, resonance, microscopic window, convergent sieve, and Fermat-synchronization barrier are self-contained paper arguments. The low-level switch exclusion is **[software computation]** with the scope and software qualification stated above. Modern short-interval results are scale calibrations only, and the claimed transference to the complete 1944 block argument is not promoted to a theorem. Nothing in this section reduces the rational switch problem to the original integer-output conjecture.

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56 Exterior powers and boundary realizations

This matrix-analysis program is not another reformulation of the Matrix Coefficient principle. It asks what finite-dimensional matrix phenomena a hypothetical counterexample can realize, whether proper exterior powers improve the determinant budget, and whether a K^2-point multiplicative grid can be compressed to its $O(K)$ boundary data.

Throughout this section assume hypothetically that

$$M = 2^\beta \in \mathbb{Z}_{>0}, \qquad A = 3^\beta \in \mathbb{Z}_{>0}, \qquad \beta \notin \mathbb{Q}, \tag{342}$$

and retain $\vartheta = \log_2 3$. Then

$$G = \{2^r M^s : r, s \in \mathbb{Z}\} \subset \mathbb{Q}_{>0} \tag{343}$$

is dense in $\mathbb{R}_{>0}$ and every $g \in G$ has rational ϑ-th power.

56.1 Rational conditioning does not create rational rank

The singular logarithm matrix factors as

$$L = \begin{pmatrix} \log 2 & \log M \\ \log 3 & \log A \end{pmatrix} = \begin{pmatrix} \log 2 \\ \log 3 \end{pmatrix} (1, \beta).$$

For a nonzero $c \in \mathbb{R}^2$ and $B \geq 1$, set

$$\mu_c(B) = \min_{0 \neq z \in \mathbb{Z}^2,\ \|z\|_\infty \leq B} |z^\mathsf{T} c|, \qquad \eta_L(B) = \min_{\substack{0 \neq u, v \in \mathbb{Z}^2 \\ \|u\|_\infty, \|v\|_\infty \leq B}} |u^\mathsf{T} L v|.$$

The rank-one factorization gives the exact identity

$$\eta_L(B) = \mu_{(\log 2, \log 3)}(B)\, \mu_{(1,\beta)}(B). \tag{344}$$

Dirichlet approximation supplies $\eta_L(B) \ll_M B^{-2}$, while effective linear forms in logarithms give only a computable polynomial lower bound. Approximate zero coefficients after bounded rational basis changes are therefore plentiful and entirely compatible with exact nonvanishing. The missing conclusion remains rational orthogonality in one of the two factors, exactly as in `RestrictedMatrixCoefficient`.

56.2 Projective arithmetic density and a Hadamard-power no-go

The density in (343) has a simultaneous finite-matrix strengthening. Given $n \geq 1$, a positive-entry real matrix B, and $\epsilon > 0$, choose each entry from G within a sufficiently small tolerance and clear all negative exponents by one common $\lambda = 2^R M^S$ with $R, S \geq 0$. Then $\lambda, \lambda^\vartheta \in \mathbb{Z}_{>0}$, and this produces

$$C \in M_n(\mathbb{Z}_{>0}), \qquad C^{\circ\vartheta} \in M_n(\mathbb{Z}_{>0}), \tag{345}$$

such that

$$\|\lambda^{-1} C - B\|_{\max} < \epsilon, \qquad \|\lambda^{-\vartheta} C^{\circ\vartheta} - B^{\circ\vartheta}\|_{\max} < \epsilon. \tag{346}$$

The choices may be made symmetrically. More precisely, if $\mathcal{U}$ and $\mathcal{V}$ are separately open and invariant under positive scalar multiplication, then $B \in \mathcal{U}$ and $B^{\circ\vartheta} \in \mathcal{V}$ imply that the integral pair may be chosen with $C \in \mathcal{U}$ and $C^{\circ\vartheta} \in \mathcal{V}$.

This retires a natural positivity route. For distinct positive $r_1, \dots, r_4$ and small $t > 0$, Cauchy–Binet gives

$$\det\big((1 + tr_ir_j)^\alpha\big)_{i,j=1}^4 = t^6 \left(\prod_{k=0}^{3} \binom{\alpha}{k}\right) \prod_{i<j}(r_j - r_i)^2 + O(t^7). \tag{347}$$

For $1 < \alpha < 2$ the leading coefficient is negative. After a small positive diagonal perturbation, the unpowered matrix is positive definite while its entrywise α-power is indefinite. Applying (346) with $\alpha = \vartheta$ yields a positive-definite integer matrix C for which $C^{\circ\vartheta}$ is both integral and indefinite. Thus critical-exponent failure is compatible with a counterexample; it cannot serve as an integrality contradiction.

Supports placed on a rational near-kernel direction give no hidden collective gain either. Their alternants factor into an ordinary Vandermonde times a Schur polynomial; for consecutive support the determinant is exactly a product of pairwise integer binomials. This is the homogeneous counterpart of the affine factorization already **[kernel-verified Lean]** in `CandidateAlternant`, so it is cross-referenced rather than counted as a separate formal advance.

56.3 Determinantal divisors and the compound ceiling

Let $E \in M_N(\mathbb{Z})$ be nonsingular and $1 \le k \le N$. Write $d_k(E)$ for the gcd of its k-minors, let $\sigma_1(E) \ge \cdots \ge \sigma_N(E) > 0$ be its singular values, adopt $v_p(0) = +\infty$, and define

$$\tau_{p,k}(E) = \min_{\substack{|I|=|J|=k \\ \sigma: I \xrightarrow{\sim} J}} \sum_{i\in I} v_p(E_{i,\sigma(i)}), \qquad Q_k(E) = \prod_p p^{\tau_{p,k}(E)}. \tag{348}$$

Only finitely many primes contribute to $Q_k(E)$. The exact adelic compound sandwich is

$$Q_k(E) \mid d_k(E), \qquad d_k(E) \le \|\textstyle\bigwedge^k E\|_2 = \prod_{j=1}^k \sigma_j(E). \tag{349}$$

The divisibility uses only termwise prime divisibility of every minor; the inequality views the minors as the entries of $\bigwedge^k E$. It is the entire Smith/exterior-power analogue of the established full-determinant comparison.

On the complete triangular evaluation grid E_T, put $Q_{T,k} = Q_k(E_T)$ and let $E^-_{T,k}$ and $E^+_{T,k}$ denote, respectively, the minimum reverse-pairing and maximum same-order k-point assignment energies, and take $k/N_T \to \rho \in (0, 1]$. The limiting quantile is $\sqrt{t}$, so

$$\frac{E^-_{T,k}}{N_T T L_T \log 2} \longrightarrow \int_0^\rho \sqrt{t(\rho - t)}\, dt = \frac{\pi}{8}\rho^2, \qquad \frac{E^+_{T,k}}{N_T T L_T \log 2} \longrightarrow \int_{1-\rho}^1 t\, dt = \rho - \frac{\rho^2}{2}. \tag{350}$$

The combinatorial factor $\binom{N_T}{k}k!$ is lower order on the T^4 scale. Under these specific termwise-divisor and largest-minor/Frobenius bounds one obtains the ceiling

$$\limsup_{T\to\infty} \frac{\log Q_{T,k}}{E^+_{T,k} + \log\big(\binom{N_T}{k}k!\big)} \le R(\rho) = \frac{\pi\rho}{8 - 4\rho}, \qquad R(\rho) < R(1) = \frac{\pi}{4} \quad (0 < \rho < 1). \tag{351}$$

Thus every proper compound is worse than the full determinant under the established estimates. This is not a no-go theorem for exterior powers themselves: a sharper singular-value free energy, cancellation among minors, or extra Smith divisibility could still change the comparison.

The same analysis applies a two-variable finite-difference transform to the square-grid evaluation matrix. It extracts diagonal weights $a!b!$. If, for each prime p, the multiset $\{v_p(a!b!) : 0 \le a, b < K\}$ is arranged as $\omega_{p,1} \le \cdots \le \omega_{p,K^2}$ and

$$\mathfrak{F}_{K,k} = \prod_p p^{\sum_{\ell=1}^{k} \omega_{p,\ell}},$$

then the every-compound divisor is

$$\mathfrak{F}_{K,k} \mid d_k(\mathcal{E}_K) \qquad (1 \le k \le K^2). \tag{352}$$

At top degree this becomes

$$\left(\prod_{r=0}^{K-1} r!\right)^{2K} \mid \det \mathcal{E}_K, \qquad \log\left(\prod_{r=0}^{K-1} r!\right)^{2K} = K^3 \log K - \frac{3}{2}K^3 + O(K^2 \log K).$$

The mechanism is the two-variable/all-compound extension of `SuperfactorialDeterminant`, `AllLayerUnitBasis`, and `RowBlockUnitDivisibility`; its contribution is lower order than the K^4 determinant budget and is retained as a paper specialization, not advertised as a new leading divisor.

56.4 Exact boundary displacement and transfer determinants

Let $K > 0$, $\mathcal{I}_K = \{0, \dots, K-1\}^2$, and set

$$\mathcal{E}_K[(i,j),(u,v)] = (2^u M^v)^i (3^u A^v)^j. \tag{353}$$

Put $d_{2;u,v} = 2^u M^v$ and $d_{3;u,v} = 3^u A^v$. Let D_2, D_3 be the diagonal matrices with these respective entries, let S_2, S_3 be the two truncated forward row shifts, and let P_2, P_3 project onto their terminal row boundaries. Directly shifting a row index gives

$$S_2\mathcal{E}_K - \mathcal{E}_K D_2 = -P_2\mathcal{E}_K D_2, \qquad S_3\mathcal{E}_K - \mathcal{E}_K D_3 = -P_3\mathcal{E}_K D_3. \tag{354}$$

After sorting, irrationality of ϑ and β, respectively, makes the row parameters and column frequencies pairwise distinct; strict total positivity of e^{xy} therefore gives $\det \mathcal{E}_K \neq 0$. Since the coordinate multipliers are nonzero, each right side in (354) has rank exactly K. The commuting multiplication operators

$$T_r = \mathcal{E}_K D_r \mathcal{E}_K^{-1} \qquad (r = 2, 3)$$

have the boundary-feedback form $T_r = S_r + R_r$ with $\operatorname{rank} R_r = K$. Their commutativity gives the exact boundary relation

$$[S_2, R_3] + [R_2, S_3] + [R_2, R_3] = 0. \tag{355}$$

If $J_r : \mathbb{R}^K \to \mathbb{R}^{K^2}$ injects the corresponding boundary and $R_r = J_r C_r$, the matrix determinant lemma compresses the full spectrum, for $z \neq 0$, to

$$\prod_{0 \le u,v < K} (z - d_{r;u,v}) = z^{K^2} \det\Big(I_K - C_r(zI - S_r)^{-1} J_r\Big), \qquad (zI - S_r)^{-1} = \sum_{h=0}^{K-1} z^{-h-1} S_r^h. \tag{356}$$

This is an exact $K \times K$ encoding of a K^2-point spectrum. It does not by itself provide an arithmetic gain: the primitive denominators and local Smith factors of the boundary feedback remain unknown.

Because the $d_{2;u,v}$ are distinct, there is also a unique $\widehat{Q}_K \in \mathbb{Q}[X]$ of degree below K^2 with $\widehat{Q}_K(d_{2;u,v}) = d_{3;u,v}$. Simultaneous diagonalization gives the exact paper identity

$$T_3 = \widehat{Q}_K(T_2). \tag{357}$$

Together with (355), this concentrates the unresolved arithmetic in two rank-K boundary feedbacks.

56.5 Kraus–Loewner positivity and its denominator tax

For $f(t) = t^\vartheta$ and $a, x, y > 0$, let $K_a(x, y) = f[x, a, y]$ be the second divided difference. The operator-convex integral representation gives

$$K_a(x, y) = c_\vartheta \int_0^\infty \frac{s^\vartheta}{(x+s)(a+s)(y+s)}\, ds, \qquad c_\vartheta = \frac{\sin(\pi(\vartheta - 1))}{\pi} > 0. \tag{358}$$

Let $0 < x_1 < \cdots < x_n$ and $0 < y_1 < \cdots < y_n$. Andréief's identity and the two Cauchy determinants give a strictly positive multiple integral for $\det(K_a(x_i, y_j))$. Thus the cross-kernel is strictly totally positive.

When a, x_i, y_j are pairwise distinct positive integers and all their ϑ-th powers are integers, each entry is rational and

$$\left(\prod_i (x_i - a)\right)\left(\prod_j (a - y_j)\right)\left(\prod_{i,j} (x_i - y_j)\right) \det(K_a(x_i, y_j)) \in \mathbb{Z} \setminus \{0\} \tag{359}$$

But scaling all nodes by $R > 0$ multiplies this cleared expression by $R^{n^2+n\vartheta}$, a positive power. (It remains an integer after scaling whenever the scaled nodes satisfy the same integrality hypotheses.) The derivative decay is therefore overwhelmed by the naive denominator; a useful argument would need primitive denominator cancellation or new numerator divisibility.

Status: The projective density and Hadamard-power no-go, compound sandwich and triangular ceiling, every-compound finite-difference specialization, transfer determinant, rational-conditioning factorization, and Kraus–Loewner package have self-contained paper proofs. The exact shift/displacement identities are **[kernel-verified Lean]** in `BoundaryDisplacement`; its rank-K endpoints explicitly assume $K > 0$, nonsingularity, and nonzero coordinate multipliers, all supplied for the displayed grid by the paper argument above. The transfer determinant and spectral interpolation remain the paper layer. The rational-direction alternant and factorial-basis mechanisms overlap existing accepted modules and are cited rather than repeated. The compound ceiling applies only to the stated termwise-divisor/Frobenius estimates; the boundary compression and Kraus positivity are exact interfaces, not claimed contradictions.

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57 Structural-prime residuals, dilation resultants, and Smith profiles

The initial finite-place observations recall the established transversality package: the real logarithm matrix has rank one, whereas the Iwasawa logarithm matrices at the structural primes 2 and 3 have rank two; the first good-prime shadow is the Fermat-quotient determinant already isolated in `FermatQuotientDeterminant`. The new layer begins when the square-grid interpolant is placed in its integral coefficient lattice and its two boundary residuals are normalized at a structural prime.

57.1 Local rank is thick, but it is not a product-formula input

Under a hypothetical nonintegral solution write

$$M = 2^{\beta} \in \mathbb{N}, \qquad A = 3^{\beta} \in \mathbb{N}, \qquad \beta \notin \mathbb{Q}.$$

For Iwasawa logarithms the positive-rational kernel is

$$\log_p r = 0 \iff r \in p^{\mathbb{Z}} \qquad (r \in \mathbb{Q}_{>0}).$$

Consequently

$$\det \begin{pmatrix} \log_2 2 & \log_2 M \\ \log_2 3 & \log_2 A \end{pmatrix} = -\log_2 M \log_2 3 \neq 0, \quad \det \begin{pmatrix} \log_3 2 & \log_3 M \\ \log_3 3 & \log_3 A \end{pmatrix} = \log_3 2 \log_3 A \neq 0.$$

After principal-unit normalization, the two-parameter orbit therefore has open closure in a two-dimensional p-adic torus at both structural primes. For a full-rank logarithm matrix B_p there is a constant c_p such that, for $z \neq 0$,

$$\left|\min_i v_p((B_p z)_i) - \min_i v_p(z_i)\right| \leq c_p.$$

Thus for $K \geq 2$, one fixed pair of distinct parameters in $\{0, \ldots, K-1\}^2$ cannot have depths r_p at a finite set S of full-rank primes unless

$$\sum_{p \in S} r_p \log p \leq \log(K-1) + \sum_{p \in S} c_p \log p. \tag{360}$$

This is a genuine local packing ceiling, not a contradiction. The determinants of local logarithms are different transcendental periods at different places; they are not absolute values of one algebraic number and cannot be inserted directly into a product formula.

At a good prime $\ell \nmid 6MA$, after clearing denominators an integer relation $a \log_\ell 2 + b \log_\ell M = 0$ first implies $2^a M^b \in \ell^{\mathbb{Z}}$. Its ℓ-valuation is zero, hence it equals one; multiplicative independence then gives $a = b = 0$. This kernel step is essential before applying Baker–Brumer to a putative local shadow exponent. It is explicitly included here because omitting it would turn multiplicative independence into an invalid assertion about p-adic logarithms.

57.2 The interpolation denominator is a cokernel order

Let $K \geq 2$, $R = K^2$, $n = K - 1$, and enumerate the nodes and values

$$x_{ij} = 2^i 3^j, \qquad y_{ij} = M^i A^j \qquad (0 \leq i, j < K).$$

Let $E_K = (x_{ij}^r)_{(i,j),\,0\le r<R}$ and let $Q_K \in \mathbb{Q}[X]$ be the unique interpolant with $Q_K(x_{ij}) = y_{ij}$. Define

$$\delta_K = \min\{d > 0 : dQ_K \in \mathbb{Z}[X]\}, \qquad P_K = \delta_K Q_K,$$

and put

$$d_{ij}^{(K)} = \prod_{(a,b)\neq(i,j)} (x_{ij} - x_{ab}), \qquad D_K^* = \mathrm{lcm}_{i,j}\, |d_{ij}^{(K)}|.$$

The Lagrange basis and Smith normal form give three exact descriptions:

$$D_K^* = \min\{D > 0 : DE_K^{-1} \in M_R(\mathbb{Z})\} = s_R, \qquad \delta_K = \mathrm{ord}_{\mathbb{Z}^R/E_K\mathbb{Z}^R}[y], \tag{361}$$

where s_R is the largest invariant factor of E_K. If $UE_KV = \mathrm{diag}(s_1, \dots, s_R)$, then more explicitly

$$\delta_K = \mathrm{lcm}_{1\le r\le R} \frac{s_r}{\gcd(s_r, (Uy)_r)}.$$

Thus all node-only denominator growth is separated from the order of the specialized value class. Any unexpectedly small denominator must be an arithmetic statement about that class, not merely a small Vandermonde determinant.

The exact endpoint valuation calculation finds the maximal dyadic barycentric denominator only in the top row and a maximizer at $(n, 0)$; triadically it lies only in the rightmost column with a maximizer at $(0, n)$. This yields two favorable, nonexhaustive branches:

$$M \text{ odd},\ A \text{ even}: \quad v_2([X^{R-1}]P_K) = 0, \tag{362}$$

$$3 \mid M,\ 3 \nmid A: \quad v_3([X^{R-1}]P_K) = 0. \tag{363}$$

The restrictions in (363) are part of the theorem. They do not cover every hypothetical counterexample.

57.3 Exact scaling, roots, and a genuine algebraic resultant

Let $F_K, G_K \in \mathbb{Z}[X]$ be the two degree-n residuals obtained from the dyadic and triadic boundary-dilation factorizations after multiplication by δ_K. In the first branch of (363), one obtains

$$\mathcal{F}_K(Z) = 2^{-n^2} F_K(2^n Z) \in \mathbb{Z}_2[Z], \qquad \overline{\mathcal{F}_K}(Z) \doteq 1 - (Z-1)^n, \tag{364}$$

with unit leading coefficient and $v_2(\mathcal{F}_K(0)) = v_2(M-1)$. In the second branch,

$$\mathcal{G}_K(Z) = 3^{-n^2} G_K(3^n Z) \in \mathbb{Z}_3[Z], \qquad \overline{\mathcal{G}_K}(Z) \doteq R_n(0) - \overline{A}R_n(Z), \quad R_n(Z) = \prod_{i=1}^{n}(Z - 2^i). \tag{365}$$

Consequently

$$v_2([X^r]F_K) \ge n(n-r), \qquad v_3([X^r]G_K) \ge n(n-r),$$

and all roots have structural-prime depth at least n. The total excess depth is exactly $v_2(M-1)$ or $v_3(A-1)$, respectively. When n is odd the reductions are sufficiently separable to determine the exceptional root classes.

The categorical payoff is a single algebraic integer rather than unrelated local periods:

$$\mathcal{R}_{2,K} = \mathrm{Res}(F_K(X), F_K(3X)), \tag{366}$$

$$\mathcal{R}_{3,K} = \mathrm{Res}(G_K(X), G_K(2X)). \tag{367}$$

If the relevant resultant is nonzero, then

$$v_2(\mathcal{R}_{2,K}) \geq n^3 + n + v_2(M-1), \tag{368}$$
$$v_3(\mathcal{R}_{3,K}) \geq n^3 + v_3(A-1). \tag{369}$$

For odd n it is nonzero and equality holds. The finite algebraic hinge is the identity

$$P(3X) - P(X) = 2X\,\mathcal{D}_3(P), \qquad \overline{\mathcal{D}_3(P)} = \overline{P}' \quad \text{in } \mathbb{F}_2[X], \tag{370}$$

where $\mathcal{D}_3(P) \in \mathbb{Z}[X]$ is an explicit geometric-sum quotient. Equation (370) is **[kernel-verified Lean]** in `DilationDifferenceQuotient`.

The resultants are the right kind of adelic objects, but their known archimedean heights are far larger than the local divisor in (369). No product-formula contradiction follows. The next missing theorem would have to control primitive resultant height or force compatibility between the two favorable branches; neither is supplied by local rank or by the Smith denominator alone.

57.4 Paired residuals and the full structural Smith profile

There is nevertheless more arithmetic in the case where the two favorable branches occur simultaneously:

$$M \text{ odd}, \qquad A \text{ even}, \qquad 3 \mid M, \qquad 3 \nmid A. \tag{371}$$

This is a genuine restriction and is not known to follow from a hypothetical counterexample. Nor can a common change of exponent-grid directions manufacture it. Put

$$a = v_2(M), \quad c = v_2(A), \quad d = v_3(M), \quad b = v_3(A), \qquad B_2 = \begin{pmatrix} 1 & 0 \\ a & c \end{pmatrix}, \quad B_3 = \begin{pmatrix} 0 & 1 \\ d & b \end{pmatrix}.$$

If a nonsingular common direction matrix C makes both B_2C and B_3C monomial—one nonzero in each row and column—then, and only then,

$$a = b = 0, \qquad c, d \neq 0, \qquad C \text{ itself is monomial}. \tag{372}$$

This is **[kernel-verified Lean]** as `simultaneous_cross_support_iff` in `GridTransformNoGo`. Translations do not change the direction matrices, and the theorem does not assume unimodularity; hence signed axis swaps, reflections, rectangular finite-index sublattices, and general $GL_2(\mathbb{Z})$ changes do not remove the branch restriction. A local shear may repair one structural prime but necessarily spoils the other. Dividing residuals by their contents is a different operation: it does not repair every fixed depth, and exact examples fail when the boundary degree equals one of the own-prime depths. There is, however, an exact eventual theorem after full content normalization; it is stated below with its strict threshold.

Write

$$F_K(X) = \sum_{r=0}^{n} f_r X^r, \qquad G_K(X) = \sum_{r=0}^{n} g_r X^r, \qquad \tau_r = \binom{n-r}{2}, \qquad e_n = \binom{n}{2}.$$

The boundary commutator does not merely transfer the total structural content. Descending coefficient comparison gives the exact opposite-residual profiles

$$v_2(g_r) = \tau_r, \qquad v_3(f_r) = \tau_r \qquad (0 \leq r \leq n). \tag{373}$$

These complement the direct floors

$$v_2(f_r) \geq n(n-r), \qquad v_3(g_r) \geq n(n-r)$$

from (364)–(365).

Here is the finite mechanism. In the notation of the boundary commutator, the coefficients of both L_3^- and L_3^+ have dyadic valuation τ_r: in the relevant elementary symmetric sum the unique minimum subset is $\{0, \ldots, n-r-1\}$. The L_2^+ terms have the deep direct profile, while the cleared L_2^- term is strictly deeper. Compare the coefficient of X^{n+r} and descend from $r = n$ to $r = 0$. If an already known coefficient is shifted by $d \geq 1$, the doubled valuation gap is exactly

$$2(\tau_{r+d} + nd - \tau_r) = d(2r + d + 1) > 0. \tag{374}$$

The factor A raises the competing half by at least one, while M is a dyadic unit. Thus the coefficient equation forces $v_2(g_r) = \tau_r$. Interchanging $2, F, A, L_3$ with $3, G, M, L_2$ proves the triadic equality.

Let $S_K = \mathrm{Sylv}(F_K, G_K)$ be the $2n \times 2n$ Sylvester matrix. Over the localization at either structural prime, its Smith-factor valuations are exactly

$$\underbrace{0, \ldots, 0}_{n+1 \text{ times}}, \qquad a_k = \frac{k(3k+1)}{2} \quad (1 \leq k < n). \tag{375}$$

Equivalently, if $\Delta_\ell(S_K)$ is the gcd of the ℓ-minors, then for $1 \leq m \leq n$

$$v_2(\Delta_{n+m}(S_K)) = v_3(\Delta_{n+m}(S_K)) = \mu_m := m\binom{m}{2} = \frac{m^2(m-1)}{2}. \tag{376}$$

In particular every $(n+m)$-minor is divisible by 6^{μ_m}, and at the endpoint

$$\mathrm{Res}(F_K, G_K) \neq 0, \qquad v_2 \,\mathrm{Res}(F_K, G_K) = v_3 \,\mathrm{Res}(F_K, G_K) = \frac{n^2(n-1)}{2}. \tag{377}$$

For completeness, the Smith calculation is local and finite. Unit-normalize the direct residual to be monic and pivot on its shifted copies. This gives a unimodular equivalence

$$\mathrm{Sylv}(f, g) \sim I_n \oplus T,$$

where T is multiplication by g mod f on the monomial basis of $R[X]/(f)$. The direct coefficient floor ensures that reduction modulo f does not disturb the shallow coefficients. Below the diagonal the entry costs are $\binom{n-i+j}{2}$; above it the wrap terms are strictly deeper. For any m-matching of selected rows and columns, convexity and the injective-index bounds give total cost at least $m\binom{m}{2}$. The bottom-m by first-m minor has a unique term of exactly this cost. Successive differences

$$\mu_m - \mu_{m-1} = \frac{(m-1)(3m-2)}{2}$$

give (375). The convex matching inequality and the universal Toeplitz-minor divisibility are **[kernel-verified Lean]** in `SmithProfileCore`, whose endpoint is `pow_smithMu_dvd_weightedLowerToeplitz_minor`. The residual-specific commutator induction, wrap estimate, and exact-minor unit remain paper proofs.

The same profile in fact extends beyond the simultaneous favorable branch after dividing by the full integer contents. Assume only that M, A are positive integers and that the two cross ranks are nonzero,

$$a = v_2(M), \qquad b = v_3(A), \qquad c = v_2(A) > 0, \qquad d = v_3(M) > 0, \qquad n > \max(a, b).$$

For a nonzero $H \in \mathbb{Z}[X]$, put $\operatorname{cont}(H) = \gcd_r |[X^r]H|$ and $H^{\mathrm{pr}} = H/\operatorname{cont}(H)$. If $c_K = [X^{K^2-1}]P_K$, the local contents are exactly

$$\begin{aligned} v_2(\operatorname{cont} G_K) &= v_2(c_K), & v_2(\operatorname{cont} F_K) &= v_2(c_K) + a, \\ v_3(\operatorname{cont} F_K) &= v_3(c_K), & v_3(\operatorname{cont} G_K) &= v_3(c_K) + b. \end{aligned} \tag{378}$$

Writing $F_K^{\mathrm{pr}} = \sum f_r X^r$ and $G_K^{\mathrm{pr}} = \sum g_r X^r$, one has

$$v_2(g_r) = v_3(f_r) = \binom{n-r}{2} \qquad (0 \le r \le n), \tag{379}$$

$$v_2(f_{n-e}) \ge e(n-1) + 1, \qquad v_3(g_{n-e}) \ge e(n-1) + 1 \qquad (1 \le e \le n). \tag{380}$$

Consequently the primitive cross-Sylvester matrix has, at both 2 and 3, the full Smith profile (375) and determinantal-divisor orders (376). Thus content normalization enlarges the exact profile from $a = b = 0$ to every cross-rank pair once $n > \max(a, b)$; it does not force the cross-rank hypotheses $c, d > 0$ for a hypothetical counterexample.

The proof is coefficientwise. In the Lagrange expansion the relevant endpoint is uniquely minimal once $a < n$ (and symmetrically $b < n$). Quotient and divided-difference costs give (379), while the direct residual has the stronger descending floor (380). Pivoting the Sylvester matrix again gives $I_n \oplus T$. With row coordinate x and column coordinate j, a nonwrapping entry has doubled cost $(x+j+1)(x+j)$; every wrapping entry has a strictly larger lower bound. A compressed-permutation argument then gives $2m\binom{m}{2}$ for every m-matching, with a unique antidiagonal minimizer. The endpoint gap, direct floor, wrap inequalities, and finite assignment bound are **[kernel-verified Lean]** in `ContentNormalizedProfileFinite`; the Lagrange identities, valuation transfer, Sylvester reduction, and application to the actual residuals are the **[paper proof]** paper layer. The strict threshold is necessary: equality cases $(K, M, A) = (4, 24, 2)$ with $n = a = 3$ and $(3, 3, 18)$ with $n = b = 2$ have noncanonical primitive Smith profiles. Exact finite diagnostics found no failure above the threshold, but those calculations are unbundled evidence rather than part of the theorem.

The exact prime-power cancellation, node-divisor normalizations, change-of-variable formula for resultants, and the implication from the two normalized residue identities to (377) are **[kernel-verified Lean]** in `CrossResidualScaling`, `ResultantScaling`, `NodeDivisorScaling`, and `CrossResultantBridge`. The strongest kernel endpoint, `actual_divisors_favorable_cross_resultant`, keeps the primary interpolant scalings and the two nonzero residue reductions as explicit hypotheses; the barycentric derivation of those hypotheses from (371) remains the paper layer.

The full resultant is not the only exact local object. For $1 \le s \le n$, let $T_s(F_K, G_K)$ be the leading $2s \times 2s$ Sylvester block and put

$$C_s(F_K, G_K) = \det T_s(F_K, G_K) = \operatorname{psc}_{n-s}(F_K, G_K)$$

up to the conventional sign. The direct and opposite coefficient profiles give the exact principal-subresultant chain

$$C_s(F_K, G_K) \ne 0, \qquad v_2(C_s(F_K, G_K)) = v_3(C_s(F_K, G_K)) = \mu_s = \frac{s^2(s-1)}{2}. \tag{381}$$

Indeed every nonzero determinant matching has total displacement s^2. If E units of displacement are assigned to the direct rows, its doubled structural-prime cost is at least

$$2nE + \sum_i d_i(d_i - 1) \geq 2\mu_s,$$

with equality only when $E = 0$ and all s shallow-row displacements equal s. Thus the unique minimal term is $\pm f_n^s g_{n-s}^s$ at 2 and $\pm g_n^s f_{n-s}^s$ at 3. The sharp cost inequality, strict case, and equality classification are **[kernel-verified Lean]** in `DeepSubresultantCost`. Their application to the residual coefficient profiles is the paper layer.

Writing $c = [X^{K^2-1}]P_K$, the first column of T_s is divisible by c; in the simultaneous favorable branch c is a unit at 2 and 3. Hence

$$\Theta_s = C_s/c \in \mathbb{Z} \setminus \{0\}, \qquad v_2(\Theta_s) = v_3(\Theta_s) = \mu_s.$$

For $s = n - j$ with any fixed $j \geq 1$, this proper subresultant retains the full leading $n^3/2$ local mass. It still does not solve the height problem. Define the truncated top-coefficient norms by

$$H_{F,s}^2 = \sum_{d=0}^{\min(n,2s-1)} |f_{n-d}|^2, \qquad H_{G,s}^2 = \sum_{d=0}^{\min(n,2s-1)} |g_{n-d}|^2.$$

The natural Hadamard bound satisfies

$$0 < |\Theta_s| \leq \frac{H_{F,s}^s H_{G,s}^s}{|c|}, \qquad \frac{H_{F,s}^s H_{G,s}^s}{|c|} \geq 6^{s(K^2-2)},$$

so at $s \sim \rho n$ its deficit from the forced divisor is already

$$\left(\rho - \frac{\rho^3}{2}\right)(\log 6)n^3 + O(n^2) > 0.$$

Exact synthetic calculations likewise find a large primitive core: at $K = 4$, $(M, A) = (9, 2)$, and $s = 2$, removing $c6^{\mu_2}$ leaves a primitive leading coefficient with 93 decimal digits. This is an unbundled exact-computation diagnostic, not a common-exponent theorem. The deep chain is a genuine local strengthening, but a cancellation-sensitive primitive-height estimate would still be new.

The all-minor profile still does not close the argument. Structural zero minors prevent a literal application of the all-nonzero-minor inequality of Section 60.1. A support-aware Cauchy–Binet bound can be written as follows. Put $h_F = \|F_K\|_2$, $h_G = \|G_K\|_2$, $k = n + m$. Row normalization and the fact that every k-row subset of an invertible matrix has some nonzero k-minor give the necessary inequality

$$6^{2\mu_m} \sum_{r=m}^{n} \binom{n}{r}\binom{n}{n+m-r} h_F^{-2r} h_G^{-2(n+m-r)} \leq \binom{2n}{n+m}. \tag{382}$$

Even in the optimistic normalization where only the leading-coefficient costs remain, the order-n^3 logarithmic surplus at $m \sim \rho n$ is bounded by

$$\frac{\rho^3}{2}\log 6 - \log 2 - \rho \log 3 \leq -\log 2 < 0 \qquad (0 \leq \rho \leq 1). \tag{383}$$

Thus no intermediate exterior level repairs the endpoint constant using coefficient norms and Hadamard/Cauchy–Binet alone. The displayed rate inequality is **[kernel-verified Lean]** in `CompoundEndpointRate`. A proof still needs a new primitive-height saving, a support-aware cancellation theorem stronger than these norm bounds, or a way to reach the simultaneous residue branch from every hypothetical counterexample.

The full interpolation denominator does not supply the missing saving either. There is an exact smallest-grid counterexample to any universal denominator-survival claim. For the synthetic favorable pair $(M, A) = (9, 2)$ and $n = 1$, the primitive interpolation numerator

$$P = 127X^3 - 1212X^2 + 3227X - 2082$$

takes the values $(60, 120, 540, 1080)$ at $(1, 3, 2, 6)$, so its interpolation denominator is $\delta = 60$. The two boundary residuals are

$$F = 5552 - 127X, \qquad G = 3175X + 1041, \qquad \operatorname{Res}(F, G) = -17759807 = -13 \cdot 31 \cdot 127 \cdot 347.$$

Thus $\gcd(\delta, \operatorname{Res}(F, G)) = 1$, and the rational residual resultant is the reduced fraction $-17759807/60^2$. The entire scalar factor introduced by clearing the two residual denominators can cancel. This certificate is **[kernel-verified Lean]** in `DenominatorCancellationCounterexample`. The pair $(9, 2)$ is test data, not a common-exponent pair; accordingly the example rules out a grid-denominator-only theorem but does not rule out a bound using the still-missing common-β arithmetic.

The obstruction persists asymptotically. For every $n \geq 1$, apply the square-grid construction of side $K = n + 1$ to the same fixed synthetic pair $(M, A) = (9, 2)$; write δ_n for the least interpolation denominator and F_n, G_n for its two boundary residuals. Put

$$A_2(n) = \sum_{j=1}^{n} v_2(3^j - 1), \qquad A_3(n) = \sum_{i=1}^{n} v_3(2^i - 1).$$

The exact square-grid denominator formula and structural resultant order are

$$v_p(\delta_n) = \frac{n^3 + 2n^2 - n}{2} + A_p(n), \qquad v_p \operatorname{Res}(F_n, G_n) = \frac{n^2(n-1)}{2} \quad (p = 2, 3).$$

Since $A_p(n) \geq 0$, subtraction gives the **[paper proof]** all-n inequality

$$2n\, v_p(\delta_n) - v_p \operatorname{Res}(F_n, G_n) \geq n^4 + \frac{3}{2}n^3 - \frac{1}{2}n^2.$$

Thus grid vanishing, collisions, and the local Smith profile alone require an $\Omega(n^4)$ cancellation tax before they can even reach the primitive resultant. This all-n comparison is a paper consequence of the displayed exact formulas; exact-integer checks through $n = 5$ serve only as diagnostics. Its synthetic scope is essential.

There is also an exact endpoint factorization, but it removes no certified structural-prime mass. Let $D = K^2 - 1$, let $Q_K \in \mathbb{Q}[M, A][X]$ be the generic rational interpolant on the square grid, and write $\lambda_K = [X^D]Q_K$, $\kappa_K = Q_K(0)$ in $\mathbb{Q}[M, A]$. Node inversion gives

$$\kappa_K(M, A) = (-1)^D \left(\prod_{i,j<K} 2^i 3^j \right) \lambda_K(M/2, A/3).$$

For the two generic rational boundary residuals U_K, V_K, every monomial of the padded Sylvester determinant contains at least one of their two top coefficients and at least one of their two constant coefficients: killing both tops lowers both degrees, while killing both constants gives the common root zero. Their endpoint formulas therefore give the [paper proof] all-K divisibility identity

$$\mathrm{Res}_X(U_K, V_K) = \lambda_K \kappa_K H_K^{\mathrm{rat}}, \qquad H_K^{\mathrm{rat}} \in \mathbb{Q}[M, A].$$

Thus the primitive numerator has the two universal Cramer endpoint factors, one annihilating the integer exponent curve at $0, \dots, D-1$ and the other at $1, \dots, D$. Exact symbolic factorization for $K = 2, 3, 4$ finds both factors with multiplicity one and a remaining core irreducible over $\mathbb{Q}[M, A]$ of bidegree $(1, 1), (6, 6), (15, 15)$, respectively; the three specialized cores at $(9, 2)$ are pairwise coprime and $2, 3$-adic units. These finite calculations are retained as an unbundled exact-CAS diagnostic, not as [software computation] and not as an all-K irreducibility theorem. They give no stable primitive factor with which to offset the $\Theta(n^4)$ height scale; a deeper clustered subresultant would be a genuinely different object.

Status: Structural-prime rank and Fermat-quotient statements are inherited from the finite-place transversality package. The integral dilation quotient and its mod-2 derivative reduction are [kernel-verified Lean] in `DilationDifferenceQuotient`. The paired-residual cancellation, resultant scaling, actual node-divisor normalization, conditional exact cross-resultant, and Toeplitz Smith-minor core are also [kernel-verified Lean] in the five modules named above. The exact denominator-cancellation certificate is [kernel-verified Lean] in `DenominatorCancellationCounterexample`; the common direction-change obstruction is [kernel-verified Lean] in `GridTransformNoGo`, and the sharp deep-subresultant cost core is [kernel-verified Lean] in `DeepSubresultantCost`. The finite inequalities behind the eventual content-normalized profile are [kernel-verified Lean] in `ContentNormalizedProfileFinite`, while their Lagrange, valuation, and residual application is the paper layer. The denominator-lattice, endpoint, scaled-root, opposite-coefficient, residual-specific full Smith, and self-dilation-resultant theorems are self-contained paper arguments, explicitly restricted to M odd and A even or to $3 \mid M$ and $3 \nmid A$ where stated. The larger symbolic audit is retained only as an unbundled diagnostic: the displayed source does not reconstruct every claimed interpolation, Newton polygon, and resultant. No conclusion here proves the conjecture.

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58 Synchronized information, decoder complexity, and Fourier separation

This section treats a hypothetical semigroup as a finite information source. Its dynamical inputs come from the established Sturmian probability and natural-boundary packages. The new contribution is a sharp fixed-rank sum-channel calculation, an exact two-view decoder, quantitative degree obstructions for algebraic decoders, and a finite Fourier formulation of tied determinant cancellation.

58.1 One magnitude channel and a rank-two valuation profile

For $0 \le n, k < L$ set

$$X_{n,k} = 2^n M^k = 2^{n+\beta k}, \qquad Y_{n,k} = 3^n A^k = 3^{n+\beta k}.$$

The binary digit length of $X_{n,k}$ and ternary digit length of $Y_{n,k}$ are the same integer, $n + \lfloor \beta k \rfloor + 1$. Thus the two apparent size observations are one channel, not two independent constraints. Its entropy is $\log L + O_\beta(1)$, while the source has entropy $2 \log L$. Prefix-code redundancies are bounded quasiperiodic functions of the same rotation phase, so their rate vanishes.

The synchronized carry word has zero entropy rate and logarithmic block entropy; its exact prefix mean, variance, and two-shift covariance are already **[kernel-verified Lean]** in `SturmianProbability`, while the carry/floor generating-series identities are **[kernel-verified Lean]** in `SturmianGeneratingSeries`. Pólya–Carlson supplies a natural boundary only after the classical radius-one and non-eventual-periodicity hypotheses. That one-variable obstruction does not create a two-prime functional equation.

Fixed integer valuation observations have entropy growth equal to the rank of the displayed linear forms times $\log L + O(1)$. For the two output boxes the normalized profile is a rank-two representable polymatroid. Consequently every universal homogeneous linear rank or Shannon inequality is automatically satisfied at leading order. Any contradiction must see heights, growing moduli, or a subleading arithmetic term.

58.2 Fixed-rank sums are asymptotically unordered-pair codes

Let E be an N-element subset of a fixed finitely generated multiplicative group of positive algebraic numbers, of rank r. The Evertse–Schlickewei–Schmidt unit-equation theorem [61] implies that the nondegenerate solutions of $x_1 + x_2 = x_3 + x_4$ contribute only $O_r(N)$ beyond the two trivial permutations. Hence

$$\mathcal{E}_+(E) = 2N^2 - N + O_r(N), \tag{384}$$

$$|E + E| = \frac{N^2}{2} + O_r(N), \tag{385}$$

$$H_2(Z + Z') = 2 \log N - \log 2 + O_r(N^{-1}), \tag{386}$$

$$H(Z + Z') = 2 \log N - \log 2 + O_r((\log N)/N), \tag{387}$$

for independent uniform $Z, Z' \in E$. Thus the sum determines the unordered input pair with error $O_r(N^{-1})$. This is a strong exact-shape theorem, but it is compatible with the hypothetical rank-two boxes: their additive channel is already almost maximally informative.

58.3 Exact synchronized decoding and degree growth

For any real $p > 1$, strict convexity gives a finite injectivity statement with no number theory in its proof:

$$a + b = c + d, \quad a^p + b^p = c^p + d^p, \quad a, b, c, d > 0 \quad \Longrightarrow \quad \{a, b\} = \{c, d\}. \tag{388}$$

Indeed, with $s = a + b$ the function $x \mapsto x^p + (s - x)^p$ is strictly decreasing on $[0, s/2]$. This complete unordered-pair decoder is **[kernel-verified Lean]** in `SynchronizedSumDecoder`; the formal theorem is `eq_or_swap_of_sum_eq_rpow_sum_eq`.

On the smooth semigroup the same convexity has a quantitative integral form. If $p, p + h, q, q + h$ are 3-smooth and $0 < p < p + h \le q < q + h$, then for $x > 1$

$$\Delta_x(p, q; h) = p^x + (q + h)^x - (p + h)^x - q^x = x(x - 1) \int_0^h \int_p^q (u + v)^{x-2}\, du\, dv \in \mathbb{Z}_{>0}. \tag{389}$$

For $1 < x \le 2$ this gives

$$x(x-1)h(q-p)(q+h)^{x-2} \le \Delta_x(p,q;h) \le x(x-1)h(q-p)p^{x-2},$$

with the endpoint powers reversed for $x \ge 2$. In particular a microscopic balanced collision with the upper bound below one is impossible. This strict local-Sidon property is stronger than decoding equality, but projective S-unit finiteness says that fixed four-term collision shapes have only finitely many nondegenerate normalized instances; scaled copies merely multiply the positive integer defect. Any closing additive argument must therefore grow the support, vary weights, or introduce new divisibility rather than repeat a fixed collision.

Exact decodability does not mean bounded algebraic complexity. Suppose $L \ge 2$ and $R = P/Q \in \mathbb{R}(z)$ has $\deg P, \deg Q \le d$, no pole at $2^n + 1$, and

$$R(2^n+1) = 3^n + 1 \qquad (0 \le n < L). \tag{390}$$

After cross multiplication and translation by one, the left side is a nonzero exponential polynomial whose bases lie in

$$\{2^j : 0 \le j \le d\} \cup \{3 \cdot 2^j : 0 \le j \le d\}.$$

These $2d + 2$ bases are distinct. The Chebyshev zero bound therefore gives

$$L \le 2d + 1, \qquad d \ge \left\lceil \frac{L-1}{2} \right\rceil. \tag{391}$$

The cross-multiplied inequality $L \le 2d+1$ is [kernel-verified Lean] as `rational_sum_decoder_card_le` in `RationalSumDecoder`; the formal hypotheses are $\deg P, \deg Q \le d$, $Q \ne 0$, and the cross-multiplied identities $P(2^n+1) = (3^n+1)Q(2^n+1)$ for all $0 \le n < L$. The rational-function formulation follows from its no-pole premise. More generally, a nonzero $F \in \mathbb{R}[z,w]$ of total degree at most d satisfying $F(2^n+1, 3^n+1) = 0$ for $0 \le n < L$ gives

$$L \le \binom{d+2}{2} - 1 = \frac{d(d+3)}{2}. \tag{392}$$

The scope is the displayed exponential slice. A completion criterion may use these bounds only when its synchronized family contains $(2^n, 1)$ for arbitrarily long initial ranges; they do not prove degree growth on every unbounded collection of semigroup pairs.

58.4 Fourier separation identifies, but does not solve, tied cancellation

Let $B \in \{\pm 1\}$ be a balanced parity bit and Z a random element of a finite abelian group G. The signed measure

$$\nu(g) = \mathbb{E}[B\mathbf{1}_{Z=g}]$$

has Fourier coefficients $\widehat{\nu}(\chi) = \mathbb{E}[B\overline{\chi(Z)}]$. Finite Fourier inversion and Parseval give

$$B \text{ independent of } Z \iff \widehat{\nu}(\chi) = 0 \quad (\chi \in \widehat{G}), \tag{393}$$

$$\mathrm{TV}(\mathcal{L}(B,Z), \mathcal{L}(B) \otimes \mathcal{L}(Z)) \le \frac{1}{2}\left(\sum_{\chi \in \widehat{G}} |\widehat{\nu}(\chi)|^2\right)^{1/2}. \tag{394}$$

If B is determined by Z, at least one character correlation has magnitude $|G|^{-1/2}$. Applied to tied determinant terms, this says exactly what would force noncancellation: a legal twist family spanning all residue characters while preserving the archimedean estimate and the tied fiber. No current determinant supplies that family, so (394) is a precise target rather than a completed argument.

Status: The exact ordinary/power-sum decoder is **[kernel-verified Lean]** in `SynchronizedSumDecoder`. The rational decoder degree bound is **[kernel-verified Lean]** in `RationalSumDecoder`. The algebraic decoder degree bound and finite Fourier lemma are unconditional paper proofs; the fixed-rank entropy estimates use the classical unit-equation theorem. The Sturmian and Pólya–Carlson inputs are not restated here. The uniform-decoder criterion is stated only for families containing the exponential slice $(2^n, 1)$ through arbitrarily large initial ranges. Character-twisted determinant separation remains an open construction. None of the entropy identities by itself proves the conjecture.

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59 Transfer spectra, special functions, and mixed Mahler resonance

This section asks whether classical special functions can amplify the multiplicative rank of the two known outputs. The exact spectrum theorem of Section 12 already rules out an independent positive algebraic base; the gamma, Hurwitz, polylogarithmic, Barnes, double-sine, and q-digamma packages below diagnose exact but rank-preserving reformulations.

59.1 Entire cancellation and rank-preserving transfer formulas

For an integer $b \geq 2$, the apparent meromorphic quotient

$$Q_b(z) := \frac{\Gamma(-z)}{\Gamma(1 - b^z)} \tag{395}$$

extends to an entire function: at every $k \in \mathbb{N}_0$ the two simple poles cancel with a nonzero ratio. Its other zeros are the nonintegral points for which b^z is a positive integer. Since the two vertical phase lattices for bases 2 and 3 meet only on the real axis,

$$\text{Alaoglu–Erdős} \iff Z(Q_2) \cap Z(Q_3) = \varnothing. \tag{396}$$

At a nonintegral real zero with $b^x = B \in \mathbb{N}$ one has

$$Q_b'(x) = \Gamma(-x)(-1)^B B! \log b. \tag{397}$$

These paper identities remove all known integral solutions exactly, but they do not create a common nonreal phase lattice: the vertical replicas for bases 2 and 3 remain incommensurable. On the real axis, however, a least nonintegral solution β would generate the genuine rank-two common-zero semigroup $\{n + k\beta : n \in \mathbb{N}_0,\ k \in \mathbb{N}\}$ from Section 52.1, and the associated Barnes product is an exact common divisor. Its anisotropic quadratic-logarithmic indicator is compensated by the two residual quotients, which remain of infinite order; the divisor therefore supplies no contradiction by itself.

Gauss multiplication, Hurwitz distribution, and the Cartier identity for $F_x(z) = \mathrm{Li}_{-x}(z)$ all yield the same multiplier q^x. Under a counterexample their normalized values are algebraic exactly for 3-smooth q, by Theorem 12.2. The dichotomy

$$\mathrm{Li}_{-x}(z) \text{ is D-finite over } \mathbb{C}(z) \iff x \in \mathbb{N}_0 \tag{398}$$

is classical: nonintegral order produces infinitely many independent fractional-power monodromy germs. Likewise, after defining $a_0 = 0$, the sequence $a_n = n^x$ is q-regular exactly for integral x. Integer Cartier eigenvalues at 2 and 3, however, do not imply either finite holonomic rank or finite q-kernel rank; that missing implication is another form of the original wall.

59.2 A finite Mahler identity and the corrected resonance theorem

For a commutative ring and finite cutoffs define

$$H_{A,B}(z) := \sum_{0\le a<A}\sum_{0\le b<B} M^a N^b z^{2^a 3^b}. \tag{399}$$

The exact rectangular inclusion–exclusion identity

$$H_{A+1,B+1}(z) = z + MH_{A,B+1}(z^2) + NH_{A+1,B}(z^3) \tag{400}$$
$$- MNH_{A,B}(z^6) \tag{401}$$

is **[kernel-verified Lean]** as `mixedMahlerTrunc_succ_succ` in `MixedMahlerFinite`. It is a finite algebra theorem only; convergence, natural boundaries, Mellin transforms, and asymptotics are not hidden in its statement.

For positive integers M, N, the limiting integer-coefficient series

$$H_{M,N}(z) = \sum_{a,b\ge 0} M^a N^b z^{2^a 3^b} \tag{402}$$

has radius one and a non-eventually-periodic zero pattern, so the classical Pólya–Carlson/Skolem–Mahler–Lech argument [62, 63] makes the unit circle a natural boundary. This happens for both integral and hypothetical nonintegral exponents and is therefore only a diagnostic.

For $M, N \ge 2$, put

$$\alpha = \log_2 M, \qquad \beta = \log_3 N. \tag{403}$$

Termwise Mellin integration initially gives

$$\int_0^\infty H_{M,N}(e^{-t}) t^{s-1}\, dt = \frac{\Gamma(s)}{(1-M2^{-s})(1-N3^{-s})} \qquad (\Re s > \max\{\alpha, \beta\}). \tag{404}$$

The two geometric pole lattices meet exactly when $\alpha = \beta$, and then only at the real point $s = x$. Consequently resonance gives

$$H_{M,N}(e^{-t}) = t^{-x}\left[\frac{\Gamma(x)}{\log 2 \log 3}\log\frac{1}{t} + B + P_2(\log(1/t)) + P_3(\log(1/t))\right] + O(t^{-x+\eta}), \tag{405}$$

where P_2, P_3 are absolutely convergent analytic periodic functions with periods $\log 2, \log 3$. The restriction $M, N \ge 2$ is essential: $M = 1$ or $N = 1$ introduces an unrelated Gamma-pole collision at $s = 0$. The Barnes kernel

$$\frac{1}{(1-e^{-v\log 2})(1-e^{-v\log 3})} = \sum_{a,b\ge 0} e^{-v(a\log 2 + b\log 3)} \tag{406}$$

is valid as a geometric series for $\Re v > 0$; the corresponding Barnes double-zeta integral starts in $\Re w > 2$, $\Re a > 0$ and is then continued. The double pole detects equality of the two exponents, not their integrality.

59.3 Double-sine, q-digamma, and the surviving target

The reciprocal double-sine normalization with periods $\log 2, \log 3$ packages the two outputs as adjacent shift quotients at $z_x = ix \log 2 \log 3/(2\pi)$. The shift compatibility is built into the function and supplies no contradiction. Separately, Borwein's Padé theorem gives the unconditional irrationality of

$$L_b(B) = \sum_{k\geq 0} \frac{1}{Bb^k - 1} = -\frac{\psi_{1/b}(x) + \log(1 - b^{-1})}{\log b} \qquad (B = b^x \in \mathbb{N},\ b \geq 2). \tag{407}$$

This one-base irrationality [64] also holds for every genuine integral exponent. Finally, ${}_1F_0(-x; ; -z) = (1 + z)^x$ (continued analytically at the displayed arguments) makes the rank-amplification target explicit: the known values at $1 + z = 2, 3$ do not algebraize the value at $1 + z = 5$. A cross-base value theorem doing so would contradict Theorem 12.2, but no such theorem is presently available.

Status: The finite mixed Mahler identity is **[kernel-verified Lean]** in `MixedMahlerFinite`. The entire quotient, corrected $M, N \geq 2$ Mellin resonance, and transfer identities are self-contained paper arguments. The holonomic dichotomy, Pólya–Carlson boundary, Barnes continuation, and Borwein irrationality use named classical inputs. Double-sine rigidity, Cartier-to-holonomy, and all proposed rank-amplification statements remain open. Nothing in the section proves the conjecture.

60 Weighted Smith profiles and rational Loewner clusters

This section replaces the single determinant by the full sequence of determinantal divisors and compares that arithmetic profile with every exterior singular-value level. It also develops a rational cross-Loewner ensemble and proves that ordinary probability and entropy are already saturated by the hypothetical semigroup. The new certificates are exact sufficient criteria, not verified contradictions. The basic exterior/Frobenius identity, fractional Loewner integral, rational positivity, and source-recovery entropy come from the established determinant, Loewner, and information packages. The new refinements are the weighted all-minor inequality, the p^t-level modular divisor, the full geometric-Vandermonde Smith form, and the level-by-level Loewner singular hierarchy.

60.1 From one determinant to every exterior level

For a full-rank $A \in \mathbb{Z}^{n\times n}$ let $\delta_k(A)$ be the gcd of its k-minors and let $\sigma_1 \geq \cdots \geq \sigma_n$ be its singular values. Cauchy–Binet in exterior form gives

$$\sum_{|I|=|J|=k} \det(A_{I,J})^2 = e_k(\sigma_1^2, \ldots, \sigma_n^2). \tag{408}$$

If all k-minors are nonzero, then

$$e_k(\sigma(A)^2) \geq \binom{n}{k}^2 \delta_k(A)^2. \tag{409}$$

Thus, if $A = B + R$, $\operatorname{rank} B \leq r < k$, $\|R\|_2 \leq \varepsilon$, and $\|A\|_2 \leq S$, then

$$\boxed{\sqrt{\binom{n}{k}}\,\delta_k(A) \leq S^r \varepsilon^{k-r}.} \tag{410}$$

Every intermediate k is a possible contradiction level.

For a diagonal gauge $K = D^{-1}AE^{-1}$ with positive diagonal entries ρ_i, κ_j, the exact weighted identity is

$$e_k(\sigma(K)^2) = \sum_{I,J} \frac{\det(A_{I,J})^2}{\rho_I^2 \kappa_J^2} \geq \delta_k(A)^2 e_k(\rho^{-2}) e_k(\kappa^{-2}). \tag{411}$$

This retains Smith mass after removing analytically irrelevant row and column scales, unlike a determinant-only normalization.

60.2 Multilevel residue collapse and a geometric Smith model

For a d-variable integer monomial-evaluation matrix, let $0 \leq \nu_{p,1} \leq \cdots \leq \nu_{p,n}$ be the valuations of its Smith factors and put $r_{p,t} = \#\{i : \nu_{p,i} < t\}$. The exact Ferrers-level identity and residue-class bound are

$$v_p(\delta_k) = \sum_{t \geq 1} (k - r_{p,t})_+, \qquad r_{p,t} \leq p^{td}. \tag{412}$$

Consequently every such matrix satisfies

$$\boxed{D_{k,d} := \prod_p p^{\sum_{t\geq 1}(k-p^{td})_+} \mid \delta_k.} \tag{413}$$

For fixed d,

$$\log D_{k,d} = \left(\frac{d}{d+1} + o(1)\right) k^{1+1/d}; \tag{414}$$

the bivariate constant is $2/3$. This paper theorem applies to arbitrary supports, but its universal size is lower order in the natural global determinant budgets.

The one-variable geometric Vandermonde

$$V_n(q) = (q^{ij})_{0 \leq i,j < n}, \qquad q \geq 2, \tag{415}$$

has the exact Smith factors

$$s_i(q) = q^{\binom{i}{2}} \prod_{r=1}^{i} (q^r - 1), \qquad 0 \leq i < n. \tag{416}$$

The proof changes from monomials to the Newton basis $\prod_{r<i}(X - q^r)$, leaving a diagonal matrix times an upper unitriangular Gaussian-binomial matrix. Hence

$$\log \delta_k(V_n(q)) = \frac{\log q}{3} k^3 + O_q(k^2). \tag{417}$$

This large one-base Smith mass occurs for every ordinary integer q and is therefore a calibration and no-go, not evidence of exceptional two-base structure. The complete determinant-one two-sided reduction, the integral q-Pascal identity, the divisibility chain, and positivity for $q \geq 2$ are **[kernel-verified Lean]** in `GeometricVandermondeSmith`; the formal endpoint is `geometricVandermonde_isUnimodularEquivalent_qSmithDiagonal`.

The factorization also locates the mass exactly. For the nodes $1, q, \ldots, q^d$, put

$$\mathcal{V}_d(q) = \prod_{0 \leq i < j \leq d} (q^j - q^i).$$

Then

$$\mathcal{V}_d(q) = q^{\binom{d+1}{3}} \prod_{r=1}^{d} (q^r - 1)^{d+1-r}. \tag{418}$$

Consequently

$$\begin{aligned} \log \mathcal{V}_d(q) &= \frac{d(d+1)(2d+1)}{6} \log q + O_q(d), \\ \sum_{p \mid q} v_p(\mathcal{V}_d(q)) \log p &= \binom{d+1}{3} \log q, \\ \sum_{r=1}^{d} (d+1-r) \log(q^r - 1) &= \binom{d+2}{3} \log q + O_q(d). \end{aligned} \tag{419}$$

Thus the fixed-base and cyclotomic terms each carry asymptotically one half of the logarithmic determinant. This is not recoverable from any fixed prime set. If S is a fixed finite set of primes, LTE gives only $O_{q,S}(d^2)$ cyclotomic mass at primes in S, and hence

$$\sum_{\substack{p \notin S \\ p \nmid q}} v_p(\mathcal{V}_d(q)) \log p = \binom{d+2}{3} \log q + O_{q,S}(d^2). \tag{420}$$

Equivalently, asymptotically all of the cyclotomic half escapes beyond every fixed residue-order cutoff. Structural primes can account for their half exactly, but a closing Smith argument must capture moving cyclotomic primes rather than enlarge a fixed local audit.

60.3 Pair-specific finite-place tomography

The universal residue bound in (412) forgets the actual pair (M, A). That information can be retained at every Smith layer. For a good prime $p \nmid 6MA$, $t \geq 1$, and finite row and column supports $\Omega, \Sigma \subset \mathbb{Z}^2$, define

$$\begin{aligned} \iota_{p^t}(a, b) &= (2^a M^b,\, 3^a A^b) \pmod{p^t}, & c^{\mathrm{row}}_{p,t}(\Omega) &= \#\iota_{p^t}(\Omega), \\ \iota^*_{p^t}(r, s) &= (2^r 3^s,\, M^r A^s) \pmod{p^t}, & c^{\mathrm{col}}_{p,t}(\Sigma) &= \#\iota^*_{p^t}(\Sigma). \end{aligned} \tag{421}$$

Let $E_{\Omega,\Sigma}$ be the corresponding square monomial-evaluation matrix, assume $\det E_{\Omega,\Sigma} \neq 0$, and write $\nu_1 \leq \cdots \leq \nu_m$ for its Smith valuations at p. With $r_{p,t} = \#\{i : \nu_i < t\}$, the row and column modules over $\mathbb{Z}/p^t\mathbb{Z}$ need at most one generator per orbit or signature class. Nakayama's lemma therefore gives the exact pair-specific refinement

$$r_{p,t} \leq \min\{c^{\mathrm{row}}_{p,t}(\Omega), c^{\mathrm{col}}_{p,t}(\Sigma)\}, \qquad v_p(\det E_{\Omega,\Sigma}) \geq \sum_{t \geq 1} \left(m - \min\{c^{\mathrm{row}}_{p,t}, c^{\mathrm{col}}_{p,t}\} \right). \tag{422}$$

This is stronger than rank loss modulo p: a collision persisting to p^t is counted at all t Smith thresholds. It is also genuinely pair-specific; replacing either image count by p^{2t} recovers only the universal $2/3$ law.

A single short relation is geometrically amplified. For the row box $\Omega_{U,V} = [0, U) \times [0, V)$, suppose

$$2^h M^k \equiv 3^h A^k \equiv 1 \pmod{p^t}, \qquad (h, k) \neq (0, 0), \quad |h| < U, \quad |k| < V.$$

The overlap with its translate by (h, k) has $(U - |h|)(V - |k|)$ edges and is a forest of translation paths. Subtracting every parent row from its child, or applying (422) at levels $1, \ldots, t$, gives

$$p^{\,t(U-|h|)(V-|k|)} \mid \det E_{\Omega_{U,V},\Sigma}. \tag{423}$$

The transpose statement holds for short relations in the column-signature lattice. For several relations the safe aggregate is the least common multiple of their weighted prime-power divisors, not their product; independent relations may nevertheless yield more through the full orbit counts. The generic eliminated-row determinant endpoint of (423) is **[kernel-verified Lean]** in `FinitePlaceShortRelationCore`; the rectangle forest and the full tomography theorem are **[paper proof]** paper arguments.

At level one the duality is completely explicit. For an odd good prime, put $n = p - 1$, choose a primitive root, and form the discrete-log matrix

$$B_p = \begin{pmatrix} \ell_p(2) & \ell_p(M) \\ \ell_p(3) & \ell_p(A) \end{pmatrix} \pmod{n}. \tag{424}$$

The row map is multiplication by B_p and the column map by B_p^{T}. For any integer lifts b_{ij},

$$|\operatorname{im} B_p| = |\operatorname{im} B_p^{\mathsf{T}}| = \frac{n^2}{\gcd(n^2, nb_{11}, nb_{12}, nb_{21}, nb_{22}, \det B_p)}. \tag{425}$$

This is the determinantal-divisor formula for the lattice generated by $[\, nI_2 \mid B_p\,]$. In particular $\det B_p \equiv 0 \pmod{n}$ implies $|\operatorname{im} B_p| \le n$; over the composite ring $\mathbb{Z}/n\mathbb{Z}$ the converse and a free rank-one interpretation are not asserted.

The tomography bound gives a finite proof certificate. For boxes $\Omega_{U,V}$ and $\Sigma_{R,S}$ with $UV = RS = m$, put

$$T_{U,V} = (U - 1) + x(V - 1), \qquad L_{R,S} = (R - 1)\log 2 + (S - 1)\log 3,$$
$$\mathcal{A}_{U,V;R,S} = \sum_{p \nmid 6MA} \sum_{t \ge 1} \Big(m - \min\{c^{\mathrm{row}}_{p,t}(U, V), c^{\mathrm{col}}_{p,t}(R, S)\} \Big) \log p. \tag{426}$$

For fixed boxes this sum is finite: collisions can occur only at primes dividing finitely many nonzero pairwise coordinate differences, and they disappear beyond a finite depth. Total positivity and Hadamard give

$$0 \le \log\det E_{\Omega_{U,V},\Sigma_{R,S}} \le mT_{U,V}L_{R,S} + \frac{m}{2}\log m. \tag{427}$$

Consequently

$$\mathcal{A}_{U,V;R,S} > mT_{U,V}L_{R,S} + \frac{m}{2}\log m \tag{428}$$

is an exact finite contradiction certificate for the proposed pair (M, A).

Along admissible factorizations one can approach the optimal leading budget $4\sqrt{x\log 2\log 3}\,m^2$; for example take $U = ac, V = bd, R = ad, S = bc$ and choose rational ratios $a/b \to \sqrt{x\log_2 3}$ and $c/d \to \sqrt{x\log_3 2}$. If $V \geq 2$ and a set $\mathcal{P}_m$ of good primes admits moving slopes

$$(M, A) \equiv (2, 3)^{c_p} \pmod p, \qquad |c_p| \leq \eta U, \qquad 0 < \eta < 1,$$

then (423) yields

$$\mathcal{A}_{U,V;R,S} \geq (1-\eta)U(V-1)\sum_{p\in\mathcal{P}_m}\log p.$$

Thus logarithmic prime weight exceeding

$$\left(\frac{4\sqrt{x\log 2\log 3}}{1-\eta} + o(1)\right)m$$

would close the determinant comparison. Fixed slopes cannot provide this mass by the fixed-direction gcd bounds in Section 71.12; the needed slopes or p-adic depths must move with the box. No numerical scouting claim is used here.

60.4 Rational cross-Loewner clusters

Write a hypothetical exponent as $x = m + \alpha$ with $0 < \alpha < 1$. For u, v in the dense group $\{2^a3^b : a, b \in \mathbb{Z}\}$, both u^α, v^α are rational, so the cross-Loewner kernel

$$L_\alpha(u, v) = \frac{u^\alpha - v^\alpha}{u - v} = \frac{\sin(\pi\alpha)}{\pi}\int_0^\infty \frac{t^\alpha}{(u+t)(v+t)}\,dt \tag{429}$$

is rational at distinct group nodes. Andreief's identity and the Cauchy determinant give, for disjoint increasing node sets,

$$\det(L_\alpha(u_i, v_j)) = \frac{c_\alpha^n}{n!}\,\Delta(u)\Delta(v) \tag{430}$$

$$\times\int_{(0,\infty)^n}\frac{\prod t_\ell^\alpha\,\Delta(t)^2}{\prod_{i,\ell}(u_i+t_\ell)(v_i+t_\ell)}\,dt, \tag{431}$$

which is strictly positive. For affine clusters $u_i = 1 + \varepsilon\xi_i$, $v_j = 1 + \varepsilon\eta_j$, Selberg's integral yields

$$\det(L_\alpha(u_i, v_j)) = \kappa_{n,\alpha}\Delta(\xi)\Delta(\eta)\varepsilon^{n(n-1)}(1 + O(\varepsilon)), \tag{432}$$

with an explicit positive gamma-product constant. Orthogonal projection in the Stieltjes feature space gives the sharper individual hierarchy

$$\sigma_j = \Theta(\varepsilon^{2(j-1)}) \qquad (1 \leq j \leq n) \tag{433}$$

for fixed nondegenerate affine clusters; the determinant constant is the corresponding Selberg integral [68].

If $A = DKE$ clears a rational cluster to integers, combining (411) and (433) forces

$$\delta_k(A)\left(\frac{e_k(D^{-2})e_k(E^{-2})}{\binom{n}{k}}\right)^{1/2} \leq C_{n,k,\alpha}\varepsilon^{k(k-1)}. \tag{434}$$

A finite violation would prove the conjecture. Naive numerator/denominator clearing is exponentially more expensive than this polynomial target; reduced Smith content or genuine row–column cancellation is still missing.

60.5 Positivity and entropy no-go theorems

The most natural 3-smooth Gram matrices are infinitely divisible: if

$$K_{ij} = 2^{\langle a_i, a_j \rangle} 3^{\langle b_i, b_j \rangle}, \tag{435}$$

then $K^{\circ t} \succeq 0$ for every $t \geq 0$, and under a counterexample $K^{\circ x}$ is integral. Thus positivity of Hadamard powers cannot distinguish integral from nonintegral x without an additional arithmetic positivity principle.

Probability is equally tight. If N, K are independent uniform variables on $\{0, \dots, H\}$ and

$$X_H = 2^N M^K, \qquad Y_H = 3^N A^K, \tag{436}$$

then one external-prime valuation and the structural valuation recover (N, K) from either output separately. Hence

$$\mathsf{H}(X_H) = \mathsf{H}(Y_H) = \mathsf{H}(X_H, Y_H) = 2\log(H+1), \qquad I(X_H; Y_H) = 2\log(H+1). \tag{437}$$

The hypothetical semigroup saturates, rather than violates, ordinary anti-concentration and Shannon inequalities. Random minors and modular scouting remain useful discovery tools only when followed by an exact Smith or interval certificate.

Status: The exact geometric-Vandermonde Smith form is **[kernel-verified Lean]** in `GeometricVandermondeSmith`. The exterior-minor identities, weighted Smith inequalities, multilevel modular divisor, Loewner–Andreief/Selberg formulas, singular hierarchy, and entropy saturation remain self-contained paper arguments with the stated classical linear-algebra and integral inputs. The universal modular divisor is a baseline, not a closing estimate; the geometric Smith form is a one-base no-go; and the Loewner certificate still requires reduced denominator or Smith information not supplied by height clearing. None of these results proves the conjecture.

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61 Logarithmic spectral rank and approximation capacity

This section gives an asymptotic reading of the exact conditional structure. It regards the candidate set as the spectrum of a diagonal operator, resolves the associated heat trace by Mellin inversion, records a second mixed support equation, and compares the best polynomial approximation rate on a fixed multiplicative window with the number of conditional integer nodes. These constructions distinguish a formal rank-one semigroup from a formal rank-two semigroup; by themselves they do not use the additional arithmetic fact that the synchronized ϑth powers are also integers.

61.1 Spectral zeta and logarithmic Weyl rank

On $\ell^2(\mathcal{S})$ define $De_x = 2^x e_x$ and $Ee_x = 3^x e_x$. Positive functional calculus gives $E = D^\vartheta$. For $\Re s > 0$, the exact conditional structure gives

$$Z_D(s) = \mathrm{Tr}(D^{-s}) = \begin{cases} (1-2^{-s})^{-1}, & \mathcal{S} = \mathbb{N}_0, \\ \left((1-2^{-s})(1-M^{-s})\right)^{-1}, & \mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta, \end{cases} \tag{438}$$

and $Z_E(s) = Z_D(\vartheta s)$. Thus the origin is a simple pole in rank one and a double pole in rank two. Equivalently, for

$$N_D(\Lambda) = \#\{x \in \mathcal{S} : 2^x \le \Lambda\}, \tag{439}$$

the exact conditional count, together with the paper-level Baker–discrepancy refinement, gives

$$N_D(\Lambda) = \log_2 \Lambda + O(1) \qquad \text{in rank one,} \tag{440}$$

$$N_D(\Lambda) = \frac{(\log \Lambda)^2}{2\log 2 \log M} + \left(\frac{1}{2\log 2} + \frac{1}{2\log M}\right)\log \Lambda + O_M((\log \Lambda)^{1-\delta}) \qquad \text{in rank two.} \tag{441}$$

Consequently the limit

$$\dim_{\log} D := \lim_{\Lambda\to\infty} \log_{\log \Lambda} N_D(\Lambda) \tag{442}$$

is respectively 1 or 2. This is an exact reformulation of the monoid dichotomy, not an independent method for deciding which case holds.

61.2 Complete heat-trace expansion

For a formal independent pair $(2, M)$ put $a = \log 2$, $b = \log M$ and

$$H_{a,b}(t) = \sum_{n,k\ge 0} \exp(-te^{an+bk}). \tag{443}$$

Mellin transformation gives

$$\int_0^\infty H_{a,b}(t)t^{s-1}\,dt = \frac{\Gamma(s)}{(1-e^{-as})(1-e^{-bs})} \qquad (\Re s > 0). \tag{444}$$

Since 2 and M are multiplicatively independent, Baker's theorem supplies polynomial lower bounds for the near-collisions of the two imaginary pole lattices. The exponential vertical decay of Γ then makes the residue series absolutely convergent. With $L = \log(1/t)$, contour shifting yields, for every $R \ge 0$,

$$H_{a,b}(t) = Q_{a,b}(L) + P_a(L) + P_b(L) \tag{445}$$

$$+ \sum_{r=1}^{R} \frac{(-1)^r t^r}{r!(1-e^{ar})(1-e^{br})} + O_{a,b,R}(t^{R+1/2}), \tag{446}$$

where

$$Q_{a,b}(L) = \frac{L^2}{2ab} + \left(\frac{1}{2a} + \frac{1}{2b} - \frac{\gamma}{ab}\right)L + C_{a,b}, \tag{447}$$

$$P_a(L) = \sum_{\ell\ne 0} \frac{\Gamma(2\pi i\ell/a)}{a(1-e^{-2\pi i\ell b/a})} e^{2\pi i\ell L/a}, \tag{448}$$

and P_b is the symmetric series. The two nonzero frequency sets are disjoint; hence the bounded remainder contains a genuine quasiperiodic component. In rank one the leading term is L/a with one periodic frequency lattice. These paper formulas are mathematically sharper than the rough count, but the output heat trace is only the rescaling $Z_E(s) = Z_D(\vartheta s)$ and supplies no second independent constraint.

61.3 The finite two-scale support equation

For natural scales p, q and a commutative ring, set

$$F_{A,B}(z) = \sum_{0 \le a < A} \sum_{0 \le b < B} z^{p^a q^b}. \tag{449}$$

The exact finite identity

$$F_{A+1,B+1}(z) = z + F_{A,B+1}(z^p) + F_{A+1,B}(z^q) - F_{A,B}(z^{pq}) \tag{450}$$

is [kernel-verified Lean] as `twoScaleSupportTrunc_succ_succ` in `TwoScaleSupportFinite`. It is coefficientwise finite and has no analytic hypotheses.

For multiplicatively independent integers $2, M$, passage to the convergent support series

$$F_{2,M}(z) = \sum_{n,k \ge 0} z^{2^n M^k}, \qquad |z| < 1, \tag{451}$$

gives

$$F_{2,M}(z) - F_{2,M}(z^2) - F_{2,M}(z^M) + F_{2,M}(z^{2M}) = z. \tag{452}$$

Its 0–1 coefficient sequence has density-zero infinite support and is not eventually periodic, so Pólya–Carlson gives the unit circle as a natural boundary. Rank-one lacunary series already have the same boundary; only the radial singularity order sees the rank. Existing Mahler theorems for two separate independent functional equations do not apply to this one mixed inclusion–exclusion equation.

61.4 Spectral interpretation of the fixed-window barrier

The arithmetic optimization in Section 62.1 has a direct spectral reading. On $[1, e^h]$ the nearest singularity of u^ϑ corresponds to the Bernstein parameter $\rho(h) = \coth(h/4)$, while the conditional spectral count in $[X, e^h X]$ is

$$\frac{h}{\beta(\log 2)^2} \log X + o(\log X).$$

The identity

$$\sup_{h>0} h \log \coth(h/4) = 2 \operatorname{arsinh}(1)^2$$

is the logarithmic form of the optimum $c_* = 3 + 2\sqrt{2}$ in (456). With $\beta > 12$, the ratio of available nodes to the necessary approximation degree is below 0.170020. Thus the spectral and integer-valued-polynomial calculations are one barrier in two coordinate systems; the arithmetic section gives the canonical proof and exact unit-locking consequence.

Status: The finite two-scale support identity is [kernel-verified Lean] in `TwoScaleSupportFinite`. The zeta/Weyl reformulations use the kernel-verified exact monoid and leading counting asymptotic, with the displayed power-saving remainder kept at paper level. The complete heat expansion and sharp fixed-window capacity theorem are paper arguments using, respectively, classical Mellin–Baker input and classical best-polynomial-approximation asymptotics. The natural boundary is classical after the finite support algebra. None of the spectral, heat, Mahler, approximation, or additive signatures eliminates a formal rank-two integer semigroup; the missing step is still the arithmetic coupling $M^\vartheta \in \mathbb{N}$.

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62 Arithmetic approximation and geometric Newton interpolation

The fixed-window optimization here converges on the same capacity constant as the spectral calculation, but starts from the integer-valued-polynomial lattice on the 3-smooth nodes rather than from a spectral count. Its complementary finite content is an exact interpolation certificate, the cofactor formula for discrete minimax error, and a geometric Newton expansion with an explicit first extrapolation error. These results isolate the arithmetic unit barrier; they do not manufacture the missing integer-valued approximant.

62.1 Subunit locking and the optimal multiplicative block

Put $\mathcal{B} = \{2^a 3^b : a, b \in \mathbb{N}_0\}$. Under a hypothetical solution β, $q^\beta \in \mathbb{Z}$ for every $q \in \mathcal{B}$. If $E \subset \mathcal{B}$ is finite and a degree-d polynomial P is integer-valued on E, then

$$\max_{q\in E} |q^\beta - P(q)| < 1 \quad \Longrightarrow \quad q^\beta = P(q) \quad (q \in E). \tag{453}$$

The generalized power system $1, t, \ldots, t^d, t^\beta$ is Chebyshev on $(0, \infty)$, so a nonintegral β makes (453) impossible as soon as $|E| \geq d + 2$. The same argument gives an exact unit floor for every integer-coefficient Müntz polynomial with distinct exponents in $\mathbb{N}_0 + \mathbb{N}_0\beta$.

For $c > 1$, the 3-smooth nodes in $[X, cX]$ have cardinality

$$(\log_2 c)(\log_3 X) + o(\log X), \tag{454}$$

whereas Bernstein approximation to t^β on that block requires degree at least $\beta \log_{\rho(c)} X + o(\log X)$, with

$$\rho(c) = \frac{\sqrt{c} + 1}{\sqrt{c} - 1}. \tag{455}$$

The exact optimization is

$$\max_{c>1} \log c \log \rho(c) = 2 \log^2(1 + \sqrt{2}), \qquad c_* = 3 + 2\sqrt{2}. \tag{456}$$

Consequently this particular method could have a positive degree-versus-node margin only for

$$M = 2^\beta < \exp\left(\frac{2 \log^2(1 + \sqrt{2})}{\log 3}\right) = 4.113124586\ldots. \tag{457}$$

Among nonpowers of two only $M = 3$ lies in that formal range, and the accepted finite corpus already excludes it (indeed it proves $M \geq 4096$ for a counterexample). Thus the exact constant is a sharp no-go for ordinary fixed-block polynomial approximation, consistent with the spectral capacity calculation.

62.2 Dyadic certificates and the discrete minimax unit barrier

Let $0 \leq n_0 < \cdots < n_d$, and let $I_{\mathbf{n}} \in \mathbb{Q}[T]$ be the degree-at-most-d interpolant with $I_{\mathbf{n}}(2^{n_j}) = 3^{n_j}$. For an integer M outside the node set, write $I_{\mathbf{n}}(M) = a/b$ in lowest terms, $b > 0$. The exact certificate is

$$\left| M^\vartheta - \frac{a}{b} \right| < \frac{1}{b} \quad \Longrightarrow \quad M^\vartheta \notin \mathbb{Z}. \tag{458}$$

If the target were integral, its nonzero difference from a/b would have size at least $1/b$; equality with the interpolant is excluded by the $d+2$-zero Chebyshev bound. The quadratic interpolants

$$I_{0,1,2}(T) = \frac{T^2+3T-1}{3}, \qquad I_{1,2,3}(T) = \frac{T^2+6T-4}{4} \tag{459}$$

give short exact exclusions for $M = 3, 5, 6, 7$. They are conceptually transparent certificates, not an improvement on the much larger kernel-verified finite range.

More generally, let A be an $(N+1)\times N$ generalized evaluation matrix, f the target column, and C_i the signed maximal cofactors of A. With

$$D = \det[A\ f], \qquad S = \sum_{i=0}^{N} |C_i|, \tag{460}$$

strict total positivity gives the exact real minimax value

$$\min_{c\in\mathbb{R}^N} \|f - Ac\|_\infty = \frac{|D|}{S}. \tag{461}$$

When A and f are integral, clearing this canonical denominator produces $z \in \mathbb{Z}^N$ with

$$Sf - Az = D\,(\operatorname{sgn} C_i)_{i=0}^{N}. \tag{462}$$

Thus the small real residual D/S becomes the nonzero integer residual D. Smith content or a different integer-valued basis must save part of S; ordinary coefficient clearing cannot cross the unit threshold.

62.3 The geometric Newton shadow

For integers $q \ge 2$, $r \ge 1$, define

$$B_k^{(q)}(X) = \prod_{i=0}^{k-1} \frac{X - q^i}{q^k - q^i}, \qquad a_k(q,r) = \prod_{i=0}^{k-1} (r - q^i). \tag{463}$$

At $X = q^n$, the first factor is the Gaussian binomial $\begin{bmatrix} n \\ k \end{bmatrix}_q \in \mathbb{Z}$, and the exact Newton identity is

$$r^n = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix}_q \prod_{i=0}^{k-1} (r - q^i). \tag{464}$$

Hence the truncation through $k = d$ is integer-valued at every geometric node, interpolates r^n for $n \le d$, and has first omitted error

$$r^{d+1} - P_d^{q,r}(q^{d+1}) = \prod_{i=0}^{d} (r - q^i). \tag{465}$$

Unless r is a nonnegative integral power of q, the Newton series never terminates and this extrapolation product grows rather than shrinks. This complements the exact geometric Smith calculation: one-axis integer-valued interpolation is abundant, but it supplies no cross-axis smallness.

The finite expansion has a canonical entire completion. Put

$$\phi_n^{(q)}(z)=\prod_{j=0}^{n-1}(z-q^j),\qquad d_n^{(q)}=\prod_{j=0}^{n-1}(q^n-q^j),\qquad c_n(q,r)=\frac{\prod_{j=0}^{n-1}(r-q^j)}{d_n^{(q)}}$$

and $N_N^{(q,r)}(z)=\sum_{n=0}^{N}c_n(q,r)\phi_n^{(q)}(z)$. Its exact endpoint defect is

$$N_N^{(q,r)}(qz)-rN_N^{(q,r)}(z)=(q^N-r)c_N(q,r)\phi_N^{(q)}(z). \tag{466}$$

Both sides have degree at most N and vanish at $1,q,\dots,q^{N-1}$; their leading coefficients agree. With $Q=q^{-1}$, the locally uniformly convergent limit is

$$\mathcal{N}_{q,r}(z)=\sum_{n\ge 0}\frac{(r;Q)_n(z;Q)_n}{(Q;Q)_n}Q^n={}_2\phi_1\left(\begin{matrix}r,z\\0\end{matrix};Q,Q\right), \tag{467}$$

where $(a;Q)_n=\prod_{j=0}^{n-1}(1-aQ^j)$. Passing to the limit in (466) gives

$$\boxed{\mathcal{N}_{q,r}(qz)-r\mathcal{N}_{q,r}(z)=\frac{(r;Q)_\infty}{(Q;Q)_\infty}(z;Q)_\infty.} \tag{468}$$

For positive r, the constant vanishes exactly when $r=q^m$ for some $m\ge 0$; then the series terminates and $\mathcal{N}_{q,r}(z)=z^m$. Thus one geometric edge is completely solved: its defect quotient is a constant, and the common two-edge cocycle—not either one-axis interpolation problem—is the remaining obstruction. The notation and standard basic-hypergeometric identity are as in [21].

Mixed geometric blocks expose the output-dependent cancellation that a single edge hides. For $q_i>1$ and $n_i\ge 1$, let

$$W=((q_1,n_1),\dots,(q_h,n_h)),\qquad S_i=n_1+\cdots+n_i,\qquad S_0=0,$$

and let $\mathfrak{D}_W=\mathfrak{D}_{q_h,n_h}\circ\cdots\circ\mathfrak{D}_{q_1,n_1}$, where $\mathfrak{D}_{q,n}f(z)=[z,qz,\dots,q^nz]f$. Iterating the one-block formula gives

$$(\mathfrak{D}_Wt^x)(1)=\prod_{i=1}^{h}\frac{\prod_{j=0}^{n_i-1}(q_i^{x-S_{i-1}}-q_i^j)}{D_{q_i,n_i}},\qquad D_{q,n}=q^{\binom{n}{2}}\prod_{j=1}^{n}(q^j-1). \tag{469}$$

If every $q_i\in\{2,3\}$, put $B_2=M$, $B_3=A$. Under a hypothetical solution this is the rational product

$$(\mathfrak{D}_Wt^x)(1)=\prod_{i=1}^{h}q_i^{-S_{i-1}n_i}\frac{\prod_{j=0}^{n_i-1}(B_{q_i}-q_i^{S_{i-1}+j})}{D_{q_i,n_i}}. \tag{470}$$

The exponent intervals $[S_{i-1},S_i)$ partition $\{0,\dots,S_h-1\}$, so the numerator is a word of explicit dyadic or triadic output gaps. If $\overline{W}$ swaps 2 and 3 while preserving the block lengths and $N=S_h$, then

$$(\mathfrak{D}_Wt^x)(1)(\mathfrak{D}_{\overline{W}}t^x)(1)=\frac{\prod_{e=0}^{N-1}(M-2^e)(A-3^e)}{6^{\sum_i S_{i-1}n_i}\prod_i D_{2,n_i}D_{3,n_i}}. \tag{471}$$

The right side depends on the ordered block lengths but not on which blocks were assigned to which base. At a structural prime q, writing $B=q^au$ with $q\nmid u$ gives the exact block valuation

$$v_q\left(\prod_{j=0}^{n-1}(B-q^{S+j})\right)=\sum_{j=0}^{n-1}\min(a,S+j)+\mathbf{1}_{\{S\le a<S+n\}}v_q(u-1). \tag{472}$$

Thus word order is a finite, output-sensitive valuation optimization rather than an opaque determinant numerator. The complementary product also shows its limitation: rearranging blocks cannot change the paired synchronized numerator.

There is a useful sign refinement on the base-two edge. For the data $2^j \mapsto 3^j$, the Newton coefficients have sign $(-1)^{n-2}$ for $n \geq 2$. If $k \geq 1$ and $2^k < A < 2^{k+1}$, then every omitted summand

$$c_n \prod_{j=0}^{n-1} (A - 2^j), \qquad n \geq k+1, \tag{473}$$

has the single sign $(-1)^{k-1}$: the $n-k-1$ negative basis factors cancel the alternating coefficient sign. Tail cancellation therefore cannot be the source of a subunit error at an integer between consecutive dyadic nodes.

Status: The unit-locking, optimal-block, dyadic-certificate, minimax-cofactor, and arithmetic-alternant statements are self-contained paper arguments using the stated classical Chebyshev/Bernstein inputs. The fixed-window constant agrees with the spectral calculation and is already incompatible with the kernel-verified counterexample range. The geometric Newton identity and its exact truncation/remainder are **[kernel-verified Lean]** in `GeometricNewtonInterpolation`, with endpoints `qNewtonExpansion_eq_pow` and `qNewtonTruncValue_first_omitted`. The endpoint dilation defect is also finite algebra and is formalized separately; local-uniform convergence, the q-hypergeometric limit, the infinite defect formula, mixed-word and complementary-word factorizations, exact block valuation, and one-signed tail are **[paper proof]** paper results. All broader Bhargava-ordering, Hermite–Padé, and integer-Chebyshev proposals remain research interfaces, and none of these statements proves the conjecture.

62.4 Connection coefficients, lateral branches, and off-axis transport

The entire completion in (467) has a useful connection theory that is invisible in any fixed truncation. Continue with $Q = q^{-1}$ and put

$$\mathcal{E}_q(z) = (z; Q)_\infty, \qquad \varkappa_q(r) = \frac{(r; Q)_\infty}{(Q; Q)_\infty}. \tag{474}$$

Thus (468) reads

$$\mathcal{N}_{q,r}(qz) - r\mathcal{N}_{q,r}(z) = \varkappa_q(r)\mathcal{E}_q(z).$$

Besides detecting termination exactly, this identity gives the sharp growth dichotomy

$$r \notin q^{\mathbb{N}_0} \quad \Longrightarrow \quad \log M_{\mathcal{N}_{q,r}}(R) = \frac{(\log R)^2}{2 \log q} + O_{q,r}(\log R). \tag{475}$$

The upper bound follows from the defining series, and evaluation of the defect on the negative axis supplies the matching lower bound.

When $q, r \geq 2$ are integers, $N \geq 1$, and $q^N < r < q^{N+1}$,

$$\operatorname{sgn} \varkappa_q(r) = (-1)^{N+1}, \qquad (q-1)^N q^{(N^2-5N-2)/2} < |\varkappa_q(r)| < \frac{q^{(N+1)(N+2)/2}}{(q^{-1}; q^{-1})_\infty}. \tag{476}$$

The sign counts the negative factors of $(r;q^{-1})_\infty$. For the lower bound, the integer gaps in the first N factors and the two boundary factors are at least one; the remaining tail dominates and cancels $(q^{-1};q^{-1})_\infty$. In particular $\log|\varkappa_q(r)| = \frac{1}{2}N^2\log q + O_q(N)$. The dependence on the common fractional phase is also exact: for $0<t<1$,

$$\varkappa_q(q^{N+t}) = (-1)^{N+1}q^{N(N+1)/2+(N+1)t}\frac{(q^{-t};q^{-1})_{N+1}(q^{t-1};q^{-1})_\infty}{(q^{-1};q^{-1})_\infty}. \tag{477}$$

Hence a hypothetical nonintegral solution gives dyadic and triadic coefficients with the same sign and quadratic-exponential size. Their magnitudes need not agree, so this is a constraint rather than a contradiction.

The meromorphic quotient

$$\mathcal{C}_{q,r}(z) = \frac{\mathcal{N}_{q,r}(z)}{\mathcal{E}_q(z)} \tag{478}$$

satisfies

$$(1-qz)\mathcal{C}_{q,r}(qz) - r\mathcal{C}_{q,r}(z) = \varkappa_q(r). \tag{479}$$

Its Taylor coefficients and normally convergent Mittag–Leffler expansion are

$$[z^n]\mathcal{C}_{q,r} = \frac{(rQ^{n+1};Q)_\infty}{(Q;Q)_\infty}, \qquad (1-rQ^n)[z^n]\mathcal{C}_{q,r} = [z^{n-1}]\mathcal{C}_{q,r}, \tag{480}$$

$$\mathcal{C}_{q,r}(z) = \frac{1}{(Q;Q)_\infty}\sum_{n\ge 0}\frac{(-1)^n r^n Q^{n(n+1)/2}}{(Q;Q)_n(1-zQ^n)}. \tag{481}$$

Here the recurrence is for $n\ge 1$, while $(1-r)[z^0]\mathcal{C}_{q,r} = \varkappa_q(r)$. It is deliberately undivided, so it remains valid at the resonances $r=q^m$. Euler's q-binomial identity proves the partial-fraction formula; in particular its poles are at most simple and its residue at q^n is

$$\frac{(-1)^{n+1}r^n Q^{n(n-1)/2}}{(Q;Q)_n(Q;Q)_\infty}.$$

On the negative axis the connection coefficient is intrinsic:

$$\lim_{R\to\infty} R\mathcal{C}_{q,r}(-R) = \varkappa_q(r). \tag{482}$$

More precisely, for every fixed L,

$$\mathcal{C}_{q,r}(-R) = \sum_{\ell=1}^{L}\frac{a_\ell}{R^\ell} + O_{q,r,L}(R^{-L-1}), \quad a_1 = \varkappa_q(r), \quad a_{\ell+1} = (rq^\ell - 1)a_\ell. \tag{483}$$

This follows by subtracting the displayed finite series from (479); the remainder obeys a contracting recurrence on successive logarithmic annuli. For the nonterminating integral multipliers relevant here, a_ℓ has quadratic q-Gevrey growth.

If $r=q^x$ and $x\in\mathbb{R}\setminus\mathbb{Z}$, the principal branch on $\mathbb{C}\setminus(-\infty,0]$ has the exact decomposition

$$z^x = \mathcal{N}_{q,r}(z) + \mathcal{E}_q(z)\mathcal{G}_{q,x}(z), \qquad \mathcal{G}_{q,x} = -\mathcal{C}_{q,r} + \frac{z^x}{\mathcal{E}_q(z)}. \tag{484}$$

The positive geometric poles are removable. Across the negative axis,

$$\mathcal{G}_{q,x,+}(-R) - \mathcal{G}_{q,x,-}(-R) = \frac{2i\sin(\pi x)R^x}{\mathcal{E}_q(-R)}, \tag{485}$$

$$\log|\mathcal{G}_{q,x,+}(-R) - \mathcal{G}_{q,x,-}(-R)| = -\frac{(\log R)^2}{2\log q} + O_x(\log R). \tag{486}$$

Thus both lateral values share the expansion (483) to every algebraic order. This does not hide termination—$a_1 = \varkappa_q(r)$ detects it—but it explains why a finite inverse-power truncation cannot distinguish the lateral branches.

For $M = 2^x$ and $A = 3^x$, the two corrections satisfy the compact system

$$(1-2z)\mathcal{G}_{2,x}(2z) - M\mathcal{G}_{2,x}(z) = -\varkappa_2(M), \tag{487}$$

$$(1-3z)\mathcal{G}_{3,x}(3z) - A\mathcal{G}_{3,x}(z) = -\varkappa_3(A), \tag{488}$$

$$\mathcal{E}_2(z)\mathcal{G}_{2,x}(z) - \mathcal{E}_3(z)\mathcal{G}_{3,x}(z) = \mathcal{N}_{3,A}(z) - \mathcal{N}_{2,M}(z). \tag{489}$$

Both sides represent the same monodromy, but the different quadratic edge types allow two independent leading constants. The system is therefore a precise compatibility interface, not a proof that the constants coincide.

There is also an exact finite transport law away from the dyadic axis. For $c \geq 1$, put

$$g_{a,c} = \mathcal{G}_{2,x}(2^a 3^c), \qquad D_{a,c} = \prod_{j=1}^{a}(1 - 2^j 3^c), \qquad D_{0,c} = 1.$$

Then

$$(1 - 2^{a+1}3^c)g_{a+1,c} - Mg_{a,c} = -\varkappa_2(M), \tag{490}$$

$$D_{a,c}g_{a,c} = M^a g_{0,c} - \varkappa_2(M)\sum_{j=0}^{a-1} M^{a-1-j}D_{j,c}. \tag{491}$$

The symmetric column recurrence holds at base 3. The denominators have the exact scale

$$D_{a,c} = (-1)^a 2^{a(a+1)/2} 3^{ac} \prod_{j=1}^{a}\left(1 - \frac{1}{2^j 3^c}\right),$$

matching the q-Gevrey scale at infinity.

Status: The finite endpoint defect preceding this subsection is already **[kernel-verified Lean]** in `QNewtonEndpointDefect`. The growth dichotomy, connection-coefficient sign/phase/amplification, quotient and Mittag–Leffler formulas, negative-axis expansion, pole–monodromy exchange, and two-axis system are **[paper proof]** paper results using classical normally convergent q-products. The finite denominator transport and closed recurrence are **[kernel-verified Lean]** in `QNewtonOffAxisTransfer`; the formal theorem treats the connection constant as an arbitrary ring element and makes no analytic-existence claim. Neither the synchronized sign nor the common lateral jump forces the two connection constants to agree.

63 Operator coherence and local-to-global arithmetic

This section asks which exact operator extensions would add arithmetic information beyond the two scalar values 2^x and 3^x. Bilateral shifts, two-mode thermodynamics, semisimple functional calculus, and bounded Sturmian cocycles are proved blind; global or finite valuation lattices, one affine coherence, one rational first jet, or one exact finite realization would close the problem. These are sufficient criteria and no-go theorems, not constructions supplied by the original hypotheses.

63.1 Universal two-mode dynamics

On $\ell^2(\mathbb{Q}_{>0})$ let $U_a e_q = e_{aq}$. If a, b are multiplicatively independent, the pair (U_a, U_b) is a countable multiple of the regular representation of $\mathbb{Z}^2$: decompose $\mathbb{Q}_{>0}$ into cosets of $\langle a, b\rangle$. For the substantive case $x > 0$, the pairs attached to $(2, 3)$ and $(2^x, 3^x)$ are therefore unitarily equivalent. On the two-mode Fock space,

$$H_{2^x,3^x} = xH_{2,3}, \tag{492}$$

so heat traces, local KMS data, and Weyl laws are merely reparametrized in time. This matches the spectral calculation: logarithmic spectral rank detects one versus two generators, but cannot quantize a positive rescaling inside the rank-two sector.

The same blindness holds for every rational semisimple matrix T with $(T-2I)(T-3I) = 0$. Writing $M = 2^x$ and $A = 3^x$,

$$T^x = (A - M)T + (3M - 2A)I, \tag{493}$$

which is rational, or integral when the displayed data are integral. Coupling the distinct spectral points probes only their already-known secant.

63.2 Global and finite arithmetic covariance

On $\ell^2(\mathbb{N})$ let $He_n = (\log n)e_n$. For $c > 0$ the following are equivalent:

$$\exists \text{ an isometry } T \text{ with } HT = cTH; \quad n^c \in \mathbb{N}\ (n \in \mathbb{N}); \quad c \in \mathbb{N}. \tag{494}$$

Spectral simplicity gives the middle condition, while the forward difference of order $\lfloor c\rfloor + 1$ is a nonzero integer tending to zero unless c is integral. Conversely $Te_n = e_{n^c}$ works for positive integral c. This is much stronger than the original hypotheses, whose canonical partial intertwiner is supported only on the 3-smooth sector.

A finite prime lattice already suffices. Let $P = \{p_1, \ldots, p_d\}$ contain 2, assume $p_j^c \in \langle P\rangle$ for every j, and suppose 2^c is algebraic. The valuation columns form $V \in M_d(\mathbb{Z})$ with

$$V^{\mathsf{T}}(\log p_i)_i = c(\log p_i)_i. \tag{495}$$

Thus c is an algebraic integer; Gelfond–Schneider and algebraicity of 2^c make it rational, hence integral. If the lattice map is onto, V is unimodular and positive c must equal 1. Finiteness and exact valuation support are the arithmetic input; an abstract Hilbert-space extension does not provide them.

63.3 Affine rulers and the first-jet criterion

On $\ell^2(\mathbb{N}_0)$ let $Se_k = e_{k+1}$ and $V_a e_k = e_{ak}$. For positive integers a, b, d,

$$D_{a,b,d} = V_a S^d - S^{bd} V_a \tag{496}$$

vanishes exactly when $a = b$; otherwise every basis vector is sent to the difference of two distinct basis vectors, so $D_{a,b,d}$ is noncompact and $\|D_{a,b,d}\|_{\mathrm{ess}} \geq \sqrt{2}$. Hence preserving the single relation $V_2 S = S^2 V_2$ after $V_2 \mapsto V_{2^x}$, even modulo compacts, forces $2^x = 2$ and $x = 1$. Translation acts as an external integral ruler absent from the multiplicative subsystem.

At one positive repeated spectral point the terminating functional calculus gives, for $b > 0$ and $N^2 = 0$,

$$(bI + N)^x = b^x I + x b^{x-1} N. \tag{497}$$

If b^x and one nonzero first-jet coefficient xb^{x-1} are rational, then $x = b\,(xb^{x-1})/b^x$ is rational. Under the original integral-output hypotheses this forces $x \in \mathbb{Z}$. Distinct-node Loewner entries are secants and automatic; a self-extension is the first finite-dimensional datum that sees the missing derivative.

The exact false-positive locus can be classified for every rational semisimple base. Let $T \in M_n(\mathbb{Q})$ be semisimple with distinct positive rational eigenvalues $\lambda_1, \ldots, \lambda_s$ and rational spectral projectors $P_1, \ldots, P_s$, assume $\lambda_i^x \in \mathbb{Q}$, put $f_x(z) = z^x$, and write $\mathcal{Z}(T) = \{Y \in M_n(\mathbb{Q}) : YT = TY\}$. For $H \in M_n(\mathbb{Q})$ set

$$\mathcal{E}_T(H) = \begin{pmatrix} T & H \\ 0 & T \end{pmatrix}.$$

The rational matrix space splits as

$$M_n(\mathbb{Q}) = \mathcal{Z}(T) \oplus [T, M_n(\mathbb{Q})], \qquad H_{\mathrm{c}} = \sum_i P_i H P_i, \qquad H_{\mathrm{o}} = \sum_{i \neq j} P_i H P_j. \tag{498}$$

For irrational x, the divided-difference formula gives the exact equivalences

$$Df_{x,T}[H] \in M_n(\mathbb{Q}) \iff H \in [T, M_n(\mathbb{Q})] \iff \mathcal{E}_T(H)^x \in M_{2n}(\mathbb{Q}). \tag{499}$$

Indeed the off-diagonal spectral blocks have rational coefficient $(\lambda_i^x - \lambda_j^x)/(\lambda_i - \lambda_j)$, while the central part is $x \sum_i \lambda_i^{x-1} P_i H P_i$ and can be rational only when every diagonal block vanishes. Conversely, $H = [T, Y]$ gives $Df_{x,T}[H] = [T^x, Y]$ and makes $\mathcal{E}_T(H)$ rationally similar to $T \oplus T$. Thus all rational first-order false positives are precisely the split extensions. Producing one rational powered extension with a nonzero centralizer component would force $x \in \mathbb{Q}$ and hence $x \in \mathbb{Z}$; the original scalar values do not supply such a component.

63.4 Sturmian similarity and exact finite realization

Writing $x = n + \alpha$, define the rotation jump $w(t) = \lfloor t + \alpha \rfloor$ on $[0, 1)$. The rational two-valued cocycles

$$c_b(t) = b^{w(t) - \alpha} \qquad (b = 2, 3) \tag{500}$$

are bounded coboundaries because $w(t) - \alpha = t - \{t + \alpha\}$. Their weighted Koopman operators are therefore similar to the rotation itself; spectra, entropy, Lyapunov growth, and determinant-like invariants are blind. The surviving projection $P = M_w$ has trace $\tau(P) = \alpha$. An exact finite-dimensional trace model would make α rational. Likewise, finite Hankel rank of $w_k = \lfloor (k+1)\alpha \rfloor - \lfloor k\alpha \rfloor$

would make this binary sequence eventually periodic and its mean rational. Either exact realization closes the conjecture; ordinary finite-dimensional approximation does not. The deterministic carry identities overlap the accepted `SturmianProbability` and `SturmianGeneratingSeries` modules.

Status: The bilateral universality, time-homothety, global and finite covariance criteria, affine essential-norm rigidity, semisimple blindness, exact rational self-extension locus, Jordan first-jet criterion, and finite trace/Hankel criteria are self-contained paper arguments with the named standard operator inputs. The maximal algebraic support is already covered more generally by `AlgebraicBaseSpectrum`, and the Sturmian/heat cores overlap earlier accepted modules. The rational scalar and matrix-entry first-jet recovery is [kernel-verified Lean] in `JordanFirstJet`; the theorem `integer_of_two_rpow_integer_of_jordanFirstJet_rational` formalizes the arithmetic closing step once the square-zero holomorphic functional-calculus formula is supplied. No global intertwiner, finite prime completion, affine coherence, or finite realization is derived from the two original outputs. None of these criteria proves the conjecture.

64 Varying the antecedent, the domain, and the codomain

The conjecture fixes three choices that look incidental: the values are asked to be *integers*, the exponent is asked to be *real*, and the codomain is $\mathbb{Z}$ rather than a larger ring. This section records what each choice is worth. The short answer is that the conjecture predicts the same thing in every reading; what changes is only which wall stands guard.

64.1 The antecedent ladder

Write the three antecedents

$$(\mathbb{A}_2)\ 2^x, 3^x \in \overline{\mathbb{Q}} \Rightarrow x \in \mathbb{Q}, \qquad (\mathbb{Q}_2)\ 2^x, 3^x \in \mathbb{Q} \Rightarrow x \in \mathbb{Z}, \qquad (\mathbb{Z}_2)\ 2^x, 3^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}.$$

The three-base theorem decomposes into two fully decoupled mechanisms, and only one of them ever sees the difference between $\mathbb{Z}$, $\mathbb{Q}$ and $\overline{\mathbb{Q}}$.

Mechanism one: transcendence. If $2^x, 3^x, 5^x \in \overline{\mathbb{Q}}$ and $x \notin \mathbb{Q}$ then $\{1, x\}$ is $\mathbb{Q}$-linearly independent, $\{\log 2, \log 3, \log 5\}$ is $\mathbb{Q}$-linearly independent by unique factorization, and the six products exponentiate to $2, 3, 5, 2^x, 3^x, 5^x$, all algebraic, contradicting six exponentials. This step is blind to integrality: it uses only algebraicity. It is the entire transcendence content of the three-base theorem.

Mechanism two: descent from $\mathbb{Q}$ to $\mathbb{Z}$. If the values are rational, write $x = p/q$ in lowest terms. Then $2^{p/q} \in \mathbb{Q}$ is a rational root of the monic $X^q - 2^p$, hence an integer m, and comparing 2-adic valuations in $2^p = m^q$ gives $p = q\, v_2(m)$, so $q \mid p$ and $q = 1$. One rational value at one base kills the denominator, and the same works for values in any subring of $\mathbb{Q}$. The dividing line is simply whether the value ring sits inside $\mathbb{Q}$ or contains irrational algebraic numbers.

Status: [classical/open background] for mechanism one; [kernel-verified Lean] for mechanism two, which is `integer_of_rational_of_two_rpow_integer` in `IntegerExponent`. The implications $(\mathbb{A}_2) \Rightarrow (\mathbb{Q}_2) \Rightarrow (\mathbb{Z}_2)$ follow, the first by mechanism two and the second trivially.

The $\overline{\mathbb{Q}}$-conclusion is sharp. Every rational $x = p/q$ genuinely occurs, since $2^{p/q}, 3^{p/q}, 5^{p/q}$ are algebraic of degree exactly q: the binomial $X^q - 2^p$ is irreducible by Capelli's criterion for $\gcd(p,q) = 1$, because 2^p is positive and is not an ℓ-th power for any $\ell \mid q$. So the $\overline{\mathbb{Q}}$-antecedent solution set is exactly $\mathbb{Q}$, and no strengthening of six exponentials can improve the conclusion. The same degree computation gives a graded refinement: if $2^x, 3^x, 5^x$ are algebraic of degree at most d, then $x \in \mathbb{Q}$ with denominator at most d. The kernel-verified exact radical degree of `RadicalDegree` is the two-base conditional shadow of exactly this statement.

Remark 64.1 (Where primality is used). Mechanism one needs only that the bases be multiplicatively independent, but mechanism two needs a base that is not a perfect power. The bases $4, 9, 25$ are multiplicatively independent, yet $x = 1/2$ gives $4^x = 2$, $9^x = 3$, $25^x = 5$, all integers. For general multiplicatively independent integer bases the correct statement is: rational values force $x \in \mathbb{Q}$ with a denominator q such that every base is a perfect q-th power. This is exactly the phenomenon that the power-primitive normalization of Section 4 exists to quotient away.

The rational form ($\mathbb{Q}_2$) deserves separate billing: it is the conjecture as Alaoglu and Erdős posed it, and it is the form the colossally abundant application consumes. Its solution set is exactly $\mathbb{Z}$, negative exponents included, which restores the $x \leftrightarrow -x$ symmetry that the $\mathbb{Z}$ form breaks by truncating to $\mathbb{N}_0$.

Proposition 64.2 (The exact cost of the rational form)**.** *($\mathbb{Q}_2$) is equivalent to ($\mathbb{Z}_2$) together with a denominator-support lemma: if $2^x = a/b$ and $3^x = c/d$ in lowest terms, then b is a power of 2 and d is a power of 3.*

Proof. Given the support lemma, $y = x + s$ for large s has $2^y = 2^s a/b \in \mathbb{Z}$ and $3^y = 3^s c/d \in \mathbb{Z}$, with y nonintegral exactly when x is, so ($\mathbb{Z}_2$) applies. Conversely ($\mathbb{Q}_2$) makes the support lemma vacuous, every solution being integral. □

Status: [paper proof] — absent the support lemma no reduction is known in either direction. It is the same kind of statement as the kernel-verified rational gateway of `RationalThirdBase`, and the modules `ThreeDenominatorNormalization` and `RationalPowerIndex` operate inside exactly this gap without stating the equivalence.

64.2 Complexifying the exponent is free

On the principal branch the restriction to real exponents is cosmetic.

Theorem 64.3 (Every complex solution is real)**.** *Let $x \in \mathbb{C}$ and let $2^x = \exp(x \log 2)$, $3^x = \exp(x \log 3)$ be the principal determinations. If $2^x, 3^x \in \mathbb{Z}$ then $x \in \mathbb{R}$. The same holds with $\mathbb{Z}$ replaced by $\mathbb{Q}$.*

Proof. Write $x = s + it$. The modulus of 2^x is 2^s, which reproduces the real hypothesis, and its argument is $t \log 2$; an integer value is real, so $t \log 2 \in \pi\mathbb{Z}$, and likewise $t \log 3 \in \pi\mathbb{Z}$. Write $t \log 2 = \pi a$ and $t \log 3 = \pi b$ with $a, b \in \mathbb{Z}$. If $t \neq 0$ then $a \neq 0$, and dividing gives $\vartheta = b/a \in \mathbb{Q}$, contradicting irrationality of ϑ. □

Status: [kernel-verified Lean] for the integer-codomain version — `im_eq_zero_of_complexTwoBaseIntegralSolution` and `twoBaseIntegralSolution_re_of_complex` in `ComplexSolutions`. The rational-codomain realness statement follows by the same paper argument but is not exported there. Only irrationality of ϑ enters; no transcendence input is used.

Geometrically the sets $\{x : 2^x \in \mathbb{Z}\}$ and $\{x : 3^x \in \mathbb{Z}\}$ lie on horizontal gratings with spacings $\pi/\log 2$ and $\pi/\log 3$, and the two spacings are incommensurable *precisely because* ϑ is irrational, so the gratings meet only on the real axis. The conjecture may therefore be stated over $\mathbb{C}$ at no mathematical cost.

64.3 The algebraic antecedent over the complex numbers

Here the reduction to $\mathbb{R}$ genuinely fails — algebraic values need not have argument in $\{0, \pi\}$ — but something cleaner is true. Since $\overline{\mathbb{Q}}$ is closed under conjugation and division, $z \in \overline{\mathbb{Q}} \setminus \{0\}$ has $|z| = \sqrt{z\bar{z}} \in \overline{\mathbb{Q}}$. So the complex solution set splits as a direct sum

$$\{x \in \mathbb{C} : 2^x, 3^x \in \overline{\mathbb{Q}}\} = \{s \in \mathbb{R} : 2^s, 3^s \in \overline{\mathbb{Q}}\} + i\,\{t \in \mathbb{R} : 2^{it}, 3^{it} \in \overline{\mathbb{Q}}\}.$$

The real slice is $(\mathbb{A}_2)$; the imaginary slice is the unit-circle problem. Both are four-exponentials instances — genuinely different instances, with rows $\{1, s\}$ and $\{1, it\}$ against the same columns $\{\log 2, \log 3\}$ — so the complex two-base algebraic problem is exactly two decoupled instances of the same conjecture, and under it the conclusion is still $x \in \mathbb{Q}$, still sharp.

For three bases everything closes unconditionally: six exponentials is a statement over $\mathbb{C}$, the real slice gives $s \in \mathbb{Q}$, and the imaginary slice is the standard corollary that $2^{it}, 3^{it}, 5^{it}$ cannot all be algebraic for real $t \neq 0$. The three-base algebraic solution set over $\mathbb{C}$ is therefore exactly $\mathbb{Q}$; complexification adds nothing. One unconditional crumb survives for two bases: nonreal *algebraic* x are excluded, because Gelfond–Schneider holds for complex algebraic irrational exponents, so the transcendence theorem of Section 3 extends to $\mathbb{C}$ verbatim.

64.4 Multivalued branches widen the wall by one prime

Demand only that *some* determination of each power be an integer: $e^{xL_2} = m$ and $e^{xL_3} = m'$ with $L_2 = \log 2 + 2\pi i J$ and $L_3 = \log 3 + 2\pi i K$. Writing $xL_2 = \log|m| + i\pi a$ and $xL_3 = \log|m'| + i\pi b$ and eliminating x, the imaginary part gives a $\mathbb{Z}$-linear relation among logarithms, hence by unique factorization the multiplicative constraint $|m|^{2K}3^a = |m'|^{2J}2^b$, while the real part gives

$$\log|m|\log 3 - \log|m'|\log 2 = 2\pi^2\,(aK - bJ). \tag{501}$$

If $aK - bJ = 0$ the left side vanishes, which is the log-product wall of Section 11: either $|m| = |m'| = 1$, forcing $x = 0$, or $(|m|, |m'|)$ is a real counterexample pair. If $aK - bJ \neq 0$ one needs the log-product form to equal a nonzero integer multiple of $2\pi^2$. Since $i\pi = \log(-1)$, this is exactly the same wall over the prime set extended by the "prime" -1: the bilinear lattice gains the directions $\pi^2 = -(\log(-1))^2$ and $i\pi \log p$.

Status: [paper proof] — the independence criterion of Section 11, extended to this larger lattice, covers the multivalued reading as well, and it remains a consequence of Schanuel's conjecture. The four exponentials conjecture masters the multivalued reading wholesale, because the branch logarithms stay $\mathbb{Q}$-independent: it gives $x \in \mathbb{Q}$, and then the modulus argument forces $x \in \mathbb{Z}$ with all phases trivial. The predicted answer is unchanged; only the wall widens.

64.5 Complexifying the codomain: the Gaussian version is equivalent

Since extending the domain is free, the proper place to look for new content is the codomain.

Theorem 64.4 (Gaussian equivalence)**.** *The statement "$2^x, 3^x \in \mathbb{Z}[i] \Rightarrow x \in \mathbb{Z}$" is equivalent to Conjecture 1.1.*

Proof. One direction is trivial since $\mathbb{Z} \subset \mathbb{Z}[i]$. Conversely assume the conjecture and let $2^x, 3^x \in \mathbb{Z}[i]$ with $x = s + it$. Taking norms, $N(2^x) = 2^{2s}$ and $N(3^x) = 3^{2s}$ are positive integers, which is the real hypothesis at exponent $2s$, so $2s \in \mathbb{Z}$. Now 3 is inert in $\mathbb{Z}[i]$, so a Gaussian integer of norm 3^{2s} is a unit times 3^k with $9^k = 3^{2s}$, forcing $s = k \in \mathbb{Z}$: the inert prime upgrades the half-integer ambiguity to integrality. Then $2^{2s} = 4^s$ shows that 2^x is divisible only by the ramified prime $1 + i$, so $2^x = u(1+i)^{2s} = 2^s i^s u$. In both bases the values are 2^s and 3^s times fourth roots of unity, the only Gaussian integers of modulus one, so the argument conditions read $t \log 2,\ t \log 3 \in (\pi/2)\mathbb{Z}$, and irrationality of ϑ kills t as in Theorem 64.3. ☐

Status: [paper proof] — a self-contained equivalence whose only inputs are Gaussian factorization, irrationality of ϑ, and the real conjecture as a hypothesis. It is a low-cost Lean target on top of `ComplexSolutions`.

Two boundary markers make the picture sharp. Replacing $\mathbb{Z}[i]$ by $\mathbb{Q}(i)$ breaks the argument at exactly one point: unit-modulus elements of $\mathbb{Q}(i)$ need not be roots of unity, as $(3 + 4i)/5$ shows, so the imaginary slice reopens as a genuine unit-circle four-exponentials instance. It is integrality up to units, not rationality, that closes the argument direction. Replacing $\mathbb{Z}[i]$ by the integers of an imaginary quadratic field in which 2 or 3 *splits* — for instance $\mathbb{Q}(\sqrt{-2})$, where 3 splits and $1 + \sqrt{-2}$ has modulus $\sqrt{3}$ — destroys the root-of-unity conclusion, and eliminating t produces relations mixing $\vartheta\pi$ with logarithms of complex algebraic numbers: the product-of-logarithms wall again, with more rooms. These split-prime variants are recorded as open rather than pursued; they meet the same wall with worse constants.

64.6 Summary

Reading	Status
Exponent in $\mathbb{C}$, principal branch, values in $\mathbb{Z}$	reduces to the real integer formulation, **[kernel-verified Lean]**
Exponent in $\mathbb{C}$, principal branch, values in $\mathbb{Q}$	reduces to the real rational-output formulation at paper level; equivalence to integer AE still needs the denominator-support lemma of Proposition 64.2
Values in $\overline{\mathbb{Q}}$, exponent real	two-base: four exponentials; conclusion $\mathbb{Q}$, sharp
Values in $\overline{\mathbb{Q}}$, exponent in $\mathbb{C}$	splits into real and unit-circle slices
Three bases, values in $\overline{\mathbb{Q}}$, exponent in $\mathbb{C}$	solution set exactly $\mathbb{Q}$, **[classical/open background]**
Multivalued branches	wall widened by $\log(-1)$, **[paper proof]**
Values in $\mathbb{Z}[i]$	unconditionally equivalent, **[paper proof]**
Values in $\mathbb{Q}(i)$ or split-prime rings	open

In every reading the answer the conjecture predicts is the same.

65 Hausdorff moments and Müntz invisibility

The minimal transcendence and fractional-normalization inputs are already part of the accepted corpus. The new direction is the additive adjacency of the $\{2,3\}$-smooth numbers and the Hausdorff moment problem it creates.

65.1 The explicit moment measure

For $0<\alpha<1$ define

$$d_j = (j+2)^\alpha - (j+1)^\alpha \qquad (j \geq 0).$$

These are the moments of the finite positive measure

$$d\mu_\alpha(q) = \frac{\alpha}{\Gamma(1-\alpha)}(1-q)(-\log q)^{-\alpha-1}\,dq, \qquad d_j = \int_0^1 q^j\,d\mu_\alpha(q). \tag{502}$$

The same Mellin identity holds for every real exponent $z>-1$. It follows from the gamma integral for $t^{\alpha-1}$ and Tonelli after $q=e^{-s}$. The density is positive on $(0,1)$, integrable at both endpoints, and has full support. Consequently

$$(-1)^k\Delta^k d_j = \int_0^1 q^j(1-q)^k\,d\mu_\alpha(q) > 0, \tag{503}$$

every finite Hankel matrix (d_{i+j}) is positive definite, and strict Hölder gives

$$d_j^{k-i} < d_i^{k-j} d_k^{j-i} \qquad (0 \leq i < j < k).$$

These analytic identities are **[paper proof]** paper proofs using classical integration.

The only adjacent positive $\{2,3\}$-smooth pairs are

$$(1,2), \quad (2,3), \quad (3,4), \quad (8,9),$$

by coprimality and Catalan–Mihăilescu. Thus under a hypothetical fractional solution, with $u = 2^\alpha$ and $v = 3^\alpha$, the adjacent endpoints force

$$d_0 = u - 1, \qquad d_1 = v - u, \qquad d_2 = u^2 - v, \qquad d_7 = v^2 - u^3 \tag{504}$$

to be rational and positive. This is a list of the four moments whose rationality is *forced by adjacent smooth endpoints*; it does not exclude accidental rationality of other moments.

65.2 Exact arithmetic defects and finite rational simulation

Write $u = P/2^r$ and $v = Q/3^s$ in lowest terms. The first Hankel defect and three nonconsecutive log-convexity defects are

$$\mathcal{H} = d_0 d_2 - d_1^2, \quad \mathcal{F}_{017} = d_0^6 d_7 - d_1^7, \quad \mathcal{F}_{127} = d_1^5 d_7 - d_2^6, \quad \mathcal{F}_{027} = d_0^5 d_7^2 - d_2^7.$$

They are strictly positive, and their exact reduced denominators are respectively

$$2^{3r}3^{2s}, \qquad 2^{9r}3^{7s}, \qquad 2^{12r}3^{7s}, \qquad 2^{14r}3^{7s}. \tag{505}$$

Parity and reduction modulo 3 show that the displayed positive numerators are coprime to these denominators. The four common-denominator identities and their local-unit conclusions are **[kernel-verified Lean]** in `MomentArithmetic`; positivity comes from (502) and remains the paper layer.

Finite positivity data cannot distinguish the special measure. If rational $m_0 > m_1 > m_2 > 0$ satisfy $m_0 m_2 > m_1^2$, then

$$\left(m_0 - \frac{m_1^2}{m_2}\right)\delta_0 + \frac{m_1^2}{m_2}\delta_{m_2/m_1} \tag{506}$$

is a positive rational two-atom measure reproducing m_0, m_1, m_2. This identity is also **[kernel-verified Lean]** in `MomentArithmetic`. More generally, a finite family of rational polynomial observations lying in the interior of the closed moment cone admits a positive rational finite quadrature: one first passes to the closed cone, uses $\operatorname{int} C = \operatorname{int} \overline{C}$, and then perturbs a finite generating set to rational nodes. This is a **[paper proof]** finite-dimensional paper theorem.

65.3 All smooth linear observations remain invisible

Put

$$\Lambda = \{N - 1 : N = 2^a 3^b \geq 2\}, \qquad K_m(q) = \frac{1 - q^m}{1 - q}.$$

Unique factorization gives the convergent reciprocal sum

$$\sum_{m \in \Lambda} \frac{1}{m} \leq 4. \tag{507}$$

The full L^1 Müntz theorem on an interval away from zero, followed by Hahn–Banach, now gives an exact infinite-dimensional no-go. For every $0 < a < b < 1$ there is a nonzero bounded real g, supported in $[a, b]$, such that

$$\int_0^1 K_m(q) g(q)\, d\mu_\alpha(q) = 0 \qquad (m \in \Lambda). \tag{508}$$

For sufficiently small $\varepsilon > 0$, the distinct positive measures $(1 \pm \varepsilon g)\mu_\alpha$ therefore have every same $\{2, 3\}$-smooth linear observation, agree with μ_α near both endpoints, and change the jth ordinary moment by only $O(b^j)$. This is a **[paper proof]** theorem conditional only on the classical Müntz/Hahn–Banach inputs. It is a no-go for reconstruction among positive measures; the perturbations need not belong to the one-parameter family $\{\mu_\beta\}$ and therefore do not produce another exponent.

Status: The exact two-atom and denominator arithmetic is **[kernel-verified Lean]** in `MomentArithmetic`. The Hausdorff integral, finite moment-cone quadrature, and Müntz invisible perturbations are paper/classical arguments with the scopes stated above. They prove that positivity and all smooth linear observations are insufficient, not that the conjecture fails or holds.

66 Depth-two rank geometry and rational symmetrization

The restricted matrix-coefficient, formal prime-symbol, and external-rank reductions are already carried by `RestrictedMatrixCoefficient`, `FormalPrimeDeterminant`, and the accepted external-rank modules. The new layer is a quadratic-rank classification and a rational symmetric normal form for the multiplicatively dependent branch.

66.1 Baker excludes quadratic rank at most two

Let $\lambda_1, \dots, \lambda_t$ be $\mathbb{Q}$-independent logarithms of positive algebraic numbers. A nonzero rational quadratic form q of rank at most two cannot vanish at these logarithms. Over $\overline{\mathbb{Q}}$, such a form factors as $q = L_1 L_2$; vanishing would give a nontrivial algebraic linear relation among the λ_i, contrary to Baker's homogeneous linear-independence theorem. This is a classical Baker consequence; its application to the two-star tensor is new to this corpus.

In prime-exponent space put $a = \nu(M)$, $b = \nu(N)$ and

$$q_{a,b} = e_2 b - e_3 a \in \mathrm{Sym}^2(V).$$

Its period specialization is $\log 2 \log N - \log 3 \log M$. A nonintegral solution makes this zero but does not make the formal tensor zero, so Baker forces $\mathrm{rank}(q_{a,b}) \geq 3$.

66.2 Hyperbolic pullback and the rank-three conic

For independent e, f and arbitrary a, b, the universal tensor $eb - fa$ is the pullback of the hyperbolic form $z_1 z_2 - z_3 z_4$. If $r = \dim \mathrm{span}\{e, b, f, a\}$, its rank is four for $r = 4$ and at most two for $r \leq 2$. At $r = 3$, write the unique relation as

$$c_1 e + c_2 b + c_3 f + c_4 a = 0.$$

The restricted form has rank three exactly when

$$c_1 c_2 - c_3 c_4 \neq 0; \tag{509}$$

otherwise its rank is two. Combining this finite classification with Baker shows that a hypothetical counterexample has equal multiplicative and quadratic rank, necessarily three or four. In rank three the two external prime-exponent vectors lie on one rational ray, and the determinant relation becomes a nondegenerate split ternary conic. This is a compression of the dependent branch, not a nonvanishing theorem for its prime-log point.

The hyperbolic normal identity and exact conic determinant are **[kernel-verified Lean]** in `RankGeometry`. Baker's theorem and the passage from the actual two-star tensor to the rank classification are the paper/classical layer. An earlier draft wrote the rank-three and rank-four kernel strata as intersections equal to $\{0\}$; literally those intersections must be empty, because the zero tensor is not in either stratum. The unified formulation uses the corrected nonvanishing statement.

The same conic has a useful scalar factorization. In the rank-three branch write the external prime support on a primitive ray as

$$M = 2^{m_2}3^{m_3}C^r, \qquad A = 2^{a_2}3^{a_3}C^s, \qquad C > 1, \quad (C, 6) = 1,$$

where the displayed exponents are nonnegative and $r + s > 0$. Put $X = \log 2$, $Y = \log 3$, and $Z = \log C$. The determinant relation becomes

$$-a_2X^2 + (m_2 - a_3)XY + m_3Y^2 + (rY - sX)Z = 0. \tag{510}$$

If

$$D = a_2r^2 - (m_2 - a_3)rs - m_3s^2,$$

then direct elimination in (510) gives

$$\boxed{DXY = (rY - sX)(a_2rX + m_3sY + rsZ).} \tag{511}$$

The existing conic determinant is $2D$, and Baker's rank-at-most-two exclusion gives $D \neq 0$. When $r, s > 0$, the second factor on the right is positive, so

$$\operatorname{sgn} D = \operatorname{sgn}(rY - sX), \qquad |r\vartheta - s| \geq \frac{\vartheta}{x(r\vartheta + s)}. \tag{512}$$

The edge cases $r = 0$ or $s = 0$ follow directly from the same factorization. This is a quantitative rational-approximation constraint internal to the dependent branch, not a contradiction. The conic determinant is kernel-checked in `RankGeometry`; (511) is an elementary paper identity.

The same integer D is also the determinant of the natural Möbius parametrization; this identifies two descriptions that otherwise look unrelated. Put

$$U = sm_2 - ra_2, \qquad V = sm_3 - ra_3. \tag{513}$$

Eliminating Z from the two output logarithms gives

$$x = \frac{U + V\vartheta}{s - r\vartheta}, \qquad D = -rU - sV, \qquad \frac{d}{dt}\frac{U + Vt}{s - rt}\bigg|_{t=\vartheta} = -\frac{D}{(s - r\vartheta)^2}. \tag{514}$$

Thus the Möbius map is nonconstant exactly when the ternary conic has rank three. Elementary height bounds give

$$1 \leq |D| \leq \frac{3\log 2\log 3}{(\log 5)^2}\,x^3 = 0.8819474591\ldots\,x^3. \tag{515}$$

Here one uses $r \le x \log_5 2$, $s \le x \log_5 3$ and the corresponding structural-prime bounds. This is a finite-height certificate, not a contradiction.

There is a particularly clean classification when the cross primes are absent:

$$3 \nmid M, \qquad 2 \nmid A, \qquad \text{so} \qquad m_3 = a_2 = 0. \tag{516}$$

The primitive-depth alternative $m_2 = 0$ or $a_3 = 0$, together with $D \ne 0$, forces $r, s > 0$ and exactly one of

$$m_2 = 0, \quad a_3 = d > 0, \quad M = C^r, \quad A = 3^d C^s, \quad r\vartheta > s, \quad x = \frac{rd\vartheta}{r\vartheta - s}, \tag{517}$$

$$a_3 = 0, \quad m_2 = d > 0, \quad M = 2^d C^r, \quad A = C^s, \quad s > r\vartheta, \quad x = \frac{sd}{s - r\vartheta}. \tag{518}$$

Both are equivalently summarized by the localized radical equation

$$\boxed{C^{|r\vartheta - s|} = 3^d}, \qquad C > 1, \quad (C, 6) = 1, \quad r, s, d \ge 1. \tag{519}$$

It produces a second sharp Four Exponentials square with bases $2, C$ and exponents ϑ, η, where $\eta = \log_C 3 = |r\vartheta - s|/d$: the four values $2^\vartheta, 2^\eta, C^\vartheta, C^\eta$ are algebraic, while both base logarithms and both exponents are rationally independent. This is another exact 2×2 frontier, not an application of the available Six Exponentials theorem; compare Section 6.

More generally the entire rank-three branch reduces to one explicitly parameterized radical. Define

$$\rho_* = \frac{a_2 + (a_3 - m_2)\vartheta - m_3\vartheta^2}{r\vartheta - s}, \qquad x_* = m_2 + m_3\vartheta + r\rho_*. \tag{520}$$

For nonnegative integral data with $(r, s) \ne (0, 0)$, $r\vartheta \ne s$, and $\rho_*, x_* > 0$, there is an integer $C > 1$, coprime to 6, satisfying

$$2^{x_*} = 2^{m_2} 3^{m_3} C^r, \qquad 3^{x_*} = 2^{a_2} 3^{a_3} C^s$$

if and only if the single number

$$\boxed{C_* = 2^{\rho_*}} \tag{521}$$

is such an integer. In that case $C = C_*$ uniquely; when the data come from an existing rank-three solution, its exponent is x_*. This equivalence simply solves the logarithmic conic for $\log_2 C$ and isolates the remaining arithmetic obstruction without claiming to remove it.

Status: The conic determinant and its nonvanishing core are already **[kernel-verified Lean]** in `RankGeometry`. The Möbius identification, height estimate, no-cross classification, localized radical equation, and one-special-radical equivalence are **[paper proof]** paper refinements. They compress the rank-three branch and expose another Four Exponentials threshold; they do not prove that the special radical is nonintegral.

66.3 A rational symmetric normal form

Let $L \in M_2(\mathbb{R})$ be nonzero and singular, suppose its entries span a three-dimensional rational vector space, and let its unique rational coefficient relation have matrix

$$C = \begin{pmatrix} \gamma & \alpha \\ \delta & \beta \end{pmatrix}, \qquad \Delta = \gamma\beta - \delta\alpha \ne 0.$$

The explicit right multiplier

$$Q = \begin{pmatrix} -\delta & \gamma \\ -\beta & \alpha \end{pmatrix}, \qquad \det Q = \Delta, \tag{522}$$

makes LQ symmetric. If $L = r(1, x)$ has rank one, symmetry forces

$$LQ = \kappa r r^T.$$

For a rank-three counterexample, $r = (\log 2, \log 3)^T$; after changing the global sign, $\kappa > 0$, and the three entries

$$\kappa(\log 2)^2, \qquad \kappa \log 2 \log 3, \qquad \kappa(\log 3)^2$$

are independent logarithms of positive algebraic numbers in geometric progression. Thus there would be a positive algebraic $\eta > 1$ for which

$$\eta, \qquad \eta^{\vartheta}, \qquad \eta^{\vartheta^2} \quad \text{are algebraic and multiplicatively independent.} \tag{523}$$

Current transcendence theorems do not exclude this two-step orbit. The determinant, right-multiplication symmetry, invertible-multiplier construction, and symmetric outer-product normal form are **[kernel-verified Lean]** in `RankGeometry`; algebraicity and the application to the counterexample remain the **[paper proof]** paper layer.

Status: The low-rank exclusion is a classical Baker consequence, and the hyperbolic pullback and rational symmetrization are new exact reductions. The finite matrix identities are kernel-checked in `RankGeometry`. The remaining rank-three and rank-four depth-two period nonvanishing statements are open; neither the conic nor the symmetric orbit proves the conjecture.

67 Unimodular gaps, transported secants, and finite-order barriers

The elementary and Four/Six Exponentials reductions are already part of the accepted material. The new contribution here is a determinant-distance law for binomial gaps, an associated family of rational secant brackets, and two precise no-go theorems for finite order data and bounded entrywise jets.

67.1 Determinant distance and adjacent gap coprimality

Let $A, B \geq 2$ be coprime, let p, q, r, s be positive integers, and put $D = |ps - rq| > 0$. Then

$$\gcd(B^q - A^p, B^s - A^r) \mid \gcd(A^D - 1, B^D - 1). \tag{524}$$

Indeed a common divisor is prime to AB; eliminating first B and then A from the two power congruences gives $A^D \equiv B^D \equiv 1$. When $D = 1$ the converse congruence is also automatic, so the exact identity is

$$\gcd(B^q - A^p, B^s - A^r) = \gcd(A - 1, B - 1). \tag{525}$$

Without coprimality, the same statement holds primewise away from AB. Consecutive continued-fraction convergents to $\log_2 3$ have determinant distance one; hence their signed gaps $3^{q_j} - 2^{p_j}$

are pairwise coprime at adjacent indices. For a hypothetical output pair $(M, N) = (2^x, 3^x)$, the corresponding adjacent output-gap gcd is $\gcd(M-1, N-1)$ when M, N are coprime, with the same away-from-MN restriction otherwise. These are **[paper proof]** exact paper theorems. The generic commutative-group collision and both determinant orientations of the modular implication are **[kernel-verified Lean]** in `UnimodularGapCollision`; the integer gcd packaging remains the paper layer.

In the canonical odd-core branch the determinant-one ladder also forms an exact radix-shadow square. Write $M = 2^a w^d$ and $x = a + d\log_2 w$, take adjacent convergents p_n/q_n to $d\log_2 w$, put $r_n = aq_n + p_n$, and restrict to the subsequence $\epsilon_n = q_n d\log_2 w - p_n > 0$. Discarding finitely many terms, assume also $\epsilon_n < \log_3 2$. Then $0 < \mathcal{B}_n < 2^{p_n}$ and $0 < \mathcal{C}_n < 3^{r_n}$ for the synchronized gaps

$$\mathcal{B}_n = w^{dq_n} - 2^{p_n}, \qquad \mathcal{C}_n = A^{q_n} - 3^{r_n}$$

have the same logarithmic error and satisfy

$$\begin{aligned} M^{q_n} &= 2^{r_n} + 2^{aq_n}\mathcal{B}_n, & A^{q_n} &= 3^{r_n} + \mathcal{C}_n, \\ L(M^{q_n}) &= 3^{r_n} + 3^{aq_n}L(\mathcal{B}_n), & R(A^{q_n}) &= 2^{r_n} + R(\mathcal{C}_n). \end{aligned} \tag{526}$$

Consequently the digit defects from Section 70.4 obey

$$D_{q_n} = \mathcal{C}_n - 3^{aq_n}L(\mathcal{B}_n), \qquad E_{q_n} = R(\mathcal{C}_n) - 2^{aq_n}\mathcal{B}_n. \tag{527}$$

The square is exact but not contractive: $3^{aq_n}L(\mathcal{B}_n)/\mathcal{C}_n = O(\epsilon_n^{\log_2 3-1}) \to 0$ and $2^{aq_n}\mathcal{B}_n/R(\mathcal{C}_n) = O(\epsilon_n^{1-\log_3 2}) \to 0$. Thus $D_{q_n} \sim \mathcal{C}_n$ and $E_{q_n} \sim R(\mathcal{C}_n)$; digit transfer discards rather than amplifies the small relative gap.

67.2 Rational secants and separated denominators

For positive p, q with $3^q \neq 2^p$, define

$$Q_{p,q} = \frac{N^q - M^p}{3^q - 2^p}.$$

Under a hypothetical solution $x > 1$ this is positive rational and the fundamental theorem of calculus gives

$$Q_{p,q} = x\int_0^1 \left((1-t)2^p + t3^q\right)^{x-1} dt. \tag{528}$$

Its reduced denominator is exactly $|3^q - 2^p|/\gcd(|3^q - 2^p|, |N^q - M^p|)$. Thus adjacent convergents give coprime reduced denominators. After setting

$$T_j = \frac{2^{p_j}}{M^{p_j}}Q_{p_j,q_j}, \qquad \delta_j = q_j\log 3 - p_j\log 2,$$

one has

$$T_j = \Phi_x(\delta_j), \qquad \Phi_x(\delta) = \frac{e^{x\delta}-1}{e^{\delta}-1}, \qquad T_j - x = \frac{x(x-1)}{2}\delta_j + O_x(\delta_j^2).$$

The integral representation of Φ_x makes it strictly increasing, so the rational T_j alternate above and below x. Away from primes dividing M, consecutive reduced denominators remain coprime. This is exact arithmetic separation, not an irrationality measure: the error is only $O(1/q_{j+1})$ while the rational heights can be exponential in p_j.

Symmetrization gives arbitrary fixed contact order without fixing the height problem. For a positive 3-smooth rational $r \neq 1$, define

$$\mathcal{J}_x(r) = \frac{r^x - r^{-x}}{r - r^{-1}}. \tag{529}$$

Under a hypothetical solution $\mathcal{J}_x(r) \in \mathbb{Q}$, and $\mathcal{J}_x(e^h) = \sinh(xh)/\sinh h$ is an even power series $x + \sum_{j\geq 1} a_j(x)h^{2j}$. For $m \geq 1$ put

$$c_{m,k} = (-1)^{k-1}\frac{2(m!)^2}{(m-k)!(m+k)!}, \qquad 1 \leq k \leq m. \tag{530}$$

These are the Lagrange weights at zero for the nodes $1^2, \dots, m^2$; hence

$$\sum_{k=1}^{m} c_{m,k} = 1, \qquad \sum_{k=1}^{m} c_{m,k}k^{2j} = 0 \ (1 \leq j < m), \qquad \sum_{k=1}^{m} c_{m,k}k^{2m} = (-1)^{m-1}(m!)^2. \tag{531}$$

The rational extrapolant

$$R_{m,x}(r) = \sum_{k=1}^{m} c_{m,k}\mathcal{J}_x(r^k) \tag{532}$$

therefore satisfies

$$R_{m,x}(e^h) = x + (-1)^{m-1}(m!)^2 a_m(x)h^{2m} + O_{x,m}(h^{2m+2}). \tag{533}$$

Taking $r = 3^q/2^p$ along convergents to $\log_2 3$ produces explicit rational approximations to x with error $O_{x,m}(q^{-2m})$ for every fixed m. This does not approach an irrationality-measure contradiction: the unreduced denominators remain exponential in $p+q$, and increasing m increases their coefficient and product costs.

67.3 Finite signatures and the entrywise jet ceiling

No finite list of strict monomial comparisons identifies the special slope. For every finite $F \subset \mathbb{Z}^2 \setminus \{0\}$ there are infinitely many coprime multiplicatively independent pairs (M, N) such that

$$\operatorname{sgn}(a \log M + b \log N) = \operatorname{sgn}(a \log 2 + b \log 3) \qquad ((a, b) \in F).$$

Choose a large prime $M = P$ and an integer N nearest $P^{\log_2 3}$ but not divisible by P; the log slope lies in the same finite sign chamber. This rules out only finite strict order signatures, not congruence, valuation, or uniform infinite information.

There is a complementary determinant ceiling. For an $R \times R$ integer decoration matrix C, let $\Sigma(C)$ be the set of permutations whose Leibniz products are nonzero and assume $\Sigma(C) \neq \varnothing$. Let $\tau_p(C)$ be the minimum p-valuation over $\Sigma(C)$ and put $T(C) = \prod_p p^{\tau_p(C)}$. If every nonzero $|c_{ij}| \leq B$, then for every integral E,

$$T(C) \mid \det(E \circ C), \qquad \log T(C) \leq R \log B. \tag{534}$$

For Euler or Hasse multipliers of order at most J on exponent scale L, this standalone guaranteed divisor contributes at most $RJ \log L$. It need not be coprime to, or additive with, existing determinant divisors. Therefore only the entrywise/termwise multiplier contribution is $o(R^2)$ when $J \log L = o(R)$; global confluence, cancellation, or factorization remains outside the theorem. The gap, secant, signature, and decoration results are **[paper proof]** paper arguments and do not prove the conjecture.

Status: The determinant-distance collision core is [kernel-verified Lean] in `UnimodularGapCollision`; the continued-fraction gcd, radix gap square, transported and symmetric secants, fixed-order Richardson extrapolants, finite-signature, and entrywise jet packages are paper proofs. Existing transcendence reductions are cross-referenced, and no fixed-order jet statement is promoted to a claim about global determinant cancellation.

68 Exact orbit heights and Farey entropy

The compact-orbit, recurrence, contact-rigidity, and Four Exponentials inputs are established earlier. The new contribution is quantitative: it computes every reduced orbit height exactly, converts Farey counting into a simultaneous two-coordinate denominator lower bound, and compares Baker recurrence with a fixed analytic contact order.

68.1 Exact reduced orbit law

Condition on a nonintegral counterexample and write $x = B + \alpha$ with $0 < \alpha < 1$. In lowest terms put

$$2^\alpha = \frac{W}{2^r}, \qquad 3^\alpha = \frac{Z}{3^s}, \qquad W > 1 \text{ odd}, \quad 3 \nmid Z.$$

Thus $r + \alpha = \log_2 W$ and $s + \alpha = \log_3 Z$. For $k \geq 0$ let $(R_k, S_k) = (2^{\{k\alpha\}}, 3^{\{k\alpha\}})$, and for a reduced rational a/b set $H(a/b) = \max(|a|, b)$. For every $k \geq 1$ the exact reduced expressions are

$$R_k = \frac{W^k}{2^{\lfloor k \log_2 W \rfloor}}, \qquad S_k = \frac{Z^k}{3^{\lfloor k \log_3 Z \rfloor}}, \tag{535}$$

so

$$H(R_k) = W^k, \quad H(S_k) = Z^k, \qquad \frac{W^k}{2} < \operatorname{den}(R_k) < W^k, \quad \frac{Z^k}{3} < \operatorname{den}(S_k) < Z^k. \tag{536}$$

The floor identity behind this is $rk + \lfloor k\alpha \rfloor = \lfloor k(r + \alpha) \rfloor$; coprimality makes the displayed fractions reduced. Consequently, with $C_0 = \max(W, Z)$,

$$\#\{k \geq 0 : \max(H(R_k), H(S_k)) \leq H\} = 1 + \lfloor \log_{C_0} H \rfloor \qquad (H \geq 1). \tag{537}$$

The floor and finite simultaneous-denominator cores are [kernel-verified Lean] in `OrbitHeightEntropy`; the reduced-fraction and height specialization is the [paper proof] paper layer.

68.2 Worst-case Farey-to-orbit distortion

Fix a nondegenerate closed interval $I \subset (0, 1)$ and let $\mathcal{F}_I(Q)$ be its reduced rationals with denominator at most Q. The classical estimate

$$\#\mathcal{F}_I(Q) = \frac{3|I|}{\pi^2} Q^2 + O_I(Q \log Q)$$

combines with (536) as follows. If

$$\Phi : \mathbb{Q} \cap I \longrightarrow \{(R_k, S_k) : k \geq 0\}$$

is injective and $D_2(Q), D_3(Q)$ are the maximum first- and second-coordinate denominators over $\mathcal{F}_I(Q)$, then

$$\liminf_{Q\to\infty} \frac{\log D_2(Q)}{Q^2} \geq \frac{3|I|}{\pi^2}\log W, \qquad \liminf_{Q\to\infty} \frac{\log D_3(Q)}{Q^2} \geq \frac{3|I|}{\pi^2}\log Z. \tag{538}$$

Indeed N distinct nonnegative orbit indices include one index at least $N-1$; the *same* image therefore has denominators exceeding $W^{N-1}/2$ and $Z^{N-1}/3$. The argument also permits fibers of a fixed integer multiplicity $M \geq 1$, dividing the two constants by M. The finite injective engine is **[kernel-verified Lean]** in `OrbitHeightEntropy`; the Farey asymptotic and limiting constants are **[paper proof]** paper mathematics.

This is a maximum-envelope theorem: for each large cutoff it forces at least one badly distorted input, not every input. It excludes polynomially tame injective, or uniformly bounded-multiplicity, relabelings of the existing orbit. It does not exclude an arithmetic operation that creates new rational graph points. A denominator-tame thickening with polynomially many genuinely new points is therefore a research target, not a consequence of the entropy theorem.

68.3 Baker recurrence versus fixed contact

Let p_j/q_j run through the convergents below α, so $0 < \delta_j = q_j\alpha - p_j < q_j^{-1}$. A two-logarithm Baker estimate [3] gives constants $c, \kappa > 0$ with $\delta_j \geq c q_j^{-\kappa}$. If $0 \neq P \in \mathbb{Q}[X, Y]$, $P(1, 1) = 0$, and

$$\sigma(P) = \operatorname{ord}_{t=0} P(2^t, 3^t),$$

then eventual nonvanishing and Taylor comparison give

$$c_P q_j^{-\kappa\sigma(P)} \leq |P(R_{q_j}, S_{q_j})| \leq C_P q_j^{-\sigma(P)}. \tag{539}$$

The separate elementary height estimate $|F(R_k, S_k)| \geq e^{-C_F k}$ applies when $F \in \mathbb{Q}(X, Y)$ is regular near the compact arc, $k \geq 1$, and $F(R_k, S_k) \neq 0$; this corrects the source statement, which fails at $k = 0$ for constants of absolute value below one. The contact-multiplicity core is already stronger in `RationalContactRigidity`; the new paper contribution here is its two-sided composition with the Baker return scale.

There is also an exact primitive-height calculation. Write

$$M = 2^m M_0, \quad 2 \nmid M_0, \qquad A = 3^a A_0, \quad 3 \nmid A_0,$$

and let $r_j/s_j \to x$ be convergents with $\varepsilon_j = r_j - s_j x$. For

$$u_j = \frac{2^{r_j}}{M^{s_j}} = 2^{\varepsilon_j}, \qquad v_j = \frac{3^{r_j}}{A^{s_j}} = 3^{\varepsilon_j},$$

one has, for all sufficiently large j,

$$h(1 : u_j : v_j) = s_j \log \operatorname{lcm}(M_0, A_0) + \max\{0, \varepsilon_j \log 2, \varepsilon_j \log 3\}. \tag{540}$$

The coefficient $\log \operatorname{lcm}(M_0, A_0)$ is positive for a nonintegral candidate. Corvaja–Zannier's rational-function height theorem [7] then gives

$$h\left(\frac{u_j - 1}{v_j - 1}\right) = s_j \log \operatorname{lcm}(M_0, A_0) + o(s_j), \qquad \frac{u_j - 1}{v_j - 1} \longrightarrow \log_3 2. \tag{541}$$

The analytic error is only polynomial in s_j, so its exponent relative to the reduced height tends to zero. More generally, every fixed nonconstant rational observable outside the exceptional torus cosets has height $\gg s_j$. Fixed Richardson order or fixed rational contact therefore cannot repair the exponential height loss; a successful transfer must have growing complexity or output-specific cancellation. For the cross-multiplied residuals

$$X_j = A^{s_j}|2^{r_j} - M^{s_j}|, \qquad Y_j = M^{s_j}|3^{r_j} - A^{s_j}|,$$

the same calculation gives

$$\log\gcd(X_j, Y_j) = \big(m\log 2 + a\log 3 + \log\gcd(M_0, A_0)\big)s_j + o(s_j). \tag{542}$$

The part supported away from the fixed primes dividing $6MA$ is only $\exp(o(s_j))$ by Corvaja–Zannier. Hence every linear common-divisor contribution is structural, not a reservoir of new auxiliary primes.

More abstractly, if nonzero rational certificates satisfy

$$\operatorname{den}(\Xi_N) \le e^{A_N N}, \qquad |\Xi_N| \le C_N N^{-d_N},$$

with $A_N, d_N \ge 0$ and $C_N > 0$, the elementary denominator lower bound can contradict the analytic estimate only if

$$d_N \log N > A_N N + \log C_N. \tag{543}$$

Here “subexponential” means $\log C_N = o(N)$. This is a necessary scale benchmark, not a construction and not a proof of the conjecture.

Status: The exact floor identities and finite worst-case entropy mechanism are **[kernel-verified Lean]** in `OrbitHeightEntropy`. The exact rational heights, Farey limits, and Baker–contact comparison are **[paper proof]** paper results. The entropy theorem concerns relabelings of the existing cyclic orbit; a genuine arithmetic thickening and every closing jet construction remain open.

69 Projective Padé defect and rational-transfer no-go

This section tests the weight-two polylogarithmic criterion of Kawashima [65] against the logarithm determinant. Its tensor-rank reduction is subsumed by the stronger rank geometry in Section 66; the separate directed-rounding computation is not used here. The new contribution is an optimized, representation-independent height defect and the no-go theorem it forces for rational polylogarithm points.

69.1 The projective defect bound

For $m \ge 1$ put $D_m = (m+1)^2 - 1$ and

$$C_m = D_m \log 2 + 3\log(m+1) + 2 + 2D_m.$$

Let K be a number field, v a place, let $\alpha_1, \ldots, \alpha_m \in K^\times$ be pairwise distinct, and suppose $\beta \in K$ satisfies $|\beta|_v > H_v(\boldsymbol{\alpha}) := \max(1, \max_i |\alpha_i|_v)$. Set $z_i = \alpha_i/\beta$ and write

$$a_v(\mathbf{z}) = -\log\max_i |z_i|_v, \qquad \overline{h}(\mathbf{z}) = \frac{1}{m}\sum_{i=1}^{m} h(z_i).$$

Kawashima's weight-two defect $\mathcal{V}_v(\boldsymbol{\alpha}, \beta)$ obeys the exact projective upper bound

$$\mathcal{V}_v(\boldsymbol{\alpha}, \beta) \leq (D_m + 1)a_v(\mathbf{z}) - D_m\overline{h}(\mathbf{z}) - C_m. \tag{544}$$

The proof writes $t = h_v(\beta)$ and optimizes

$$(D_m + 1)t - (D_m + 1)\max(0, t - a_v) - D_m \max(t, \overline{h}) - C_m.$$

Since $a_v \leq \overline{h}$, the maximum is attained throughout $a_v \leq t \leq \overline{h}$ and equals the right side of (544). This piecewise optimization and its abstract height-budget wrapper are **[kernel-verified Lean]** in `ProjectiveDefect`; the number-field height specialization and Kawashima's theorem are respectively **[paper proof]** and **[classical/open background]**.

If the defect is positive, then necessarily

$$a_v > C_m, \qquad \frac{a_v}{\overline{h}} > \frac{D_m}{D_m + 1} = 1 - \frac{1}{(m+1)^2}, \qquad \overline{h} < \left(1 + \frac{1}{D_m}\right) a_v. \tag{545}$$

These implications are also **[kernel-verified Lean]** in `ProjectiveDefect`. Thus mere archimedean proximity to one is insufficient: the local depth must consume almost all of the global height. Direct points $1 - 1/q$, high-root splittings, and ordinary unit-fraction telescoping all fail this necessary condition. A common integral rescaling is strictly harmful, decreasing the affine defect by $2D_m \log t$.

69.2 No rational transfer of the prime direction

Let distinct $z_1, \dots, z_m \in \mathbb{Q} \cap (0, 1)$ admit a positive-defect specialization at the real place, and put $\ell_i = -\log(1 - z_i)$. Then

$$\log 2 \notin \operatorname{span}_{\mathbb{Q}}\{\ell_1, \dots, \ell_m\}. \tag{546}$$

The same proof applies to a fixed $q \in \mathbb{Q}_{>0} \setminus \{1\}$ with a changed constant. Indeed (545) makes every $\ell_i = O(e^{-a_v})$ while bounding all prime valuations of $(1 - z_i)^{-1}$ by $O(ma_v)$. If the fixed prime direction lay in their rational span, Cramer's rule and Hadamard's inequality would supply integer coefficients of size polynomial in a_v; exponentiation would then force the nonzero constant $\log 2$ below an exponentially decaying bound. The final elementary comparison is uniform in m (with the logarithm taken of the ratio of that bound to $\log 2$ and with the endpoint derivative checked explicitly). This is a **[paper proof]** paper theorem.

Consequently a positive-defect rational tuple cannot reproduce the counterexample's formal four-log tensor. The range of that tensor contains the prime directions by Section 66, whereas every tensor built from these rational logarithms has range inside their forbidden valuation span. This rules out the factor-by-factor rational Li_1 transfer; it does not rule out algebraic points, varying fields, or a genuinely mixed-denominator determinant construction.

For a fixed number field K, place v, finitely generated $\Gamma < K^\times$, and fixed cardinality m, the effective approximation theorem of Calegari–Dimitrov–Tang [66] further implies that only finitely many tuples $z_i = 1 - \gamma_i$, $\gamma_i \in \Gamma$, can have positive defect, with effective height bounds. The fixed-m qualification is essential; this is not uniform over constructions with $m \to \infty$.

Finally, let p_n/q_n be the continued-fraction convergents to $\log_2 3$ and choose $\varepsilon_n \in \{\pm 1\}$ so that the units $r_n = (2^{p_n} 3^{-q_n})^{\varepsilon_n} > 1$; then $r_n \to 1$ and

$$\frac{-\log(1 - r_n^{-1})}{h(1 - r_n^{-1})} \longrightarrow 0,$$

and the output units have the same low-efficiency behavior. The present Padé criterion requires efficiency tending to one. This identifies an arithmetic-efficiency gap, not a closing theorem. A proposed cross-row, mixed-denominator cancellation remains only a schematic research target because its exceptional "zero coefficient after rational changes" alternative has not yet been defined precisely.

Status: The finite projective optimization and efficiency implications are **[kernel-verified Lean]** in `ProjectiveDefect`; the number-field specialization, rational-transfer no-go, and fixed-cardinality group finiteness are **[paper proof]** paper results using the cited external theorems. No mixed-denominator construction or proof of the conjecture is claimed.

70 Carry languages and fixed-target entropy

The logarithmic cone, one-orbit equivalence, one/two/three-base threshold, 3-smooth lattice count, and residual compact orbit are already part of the accepted corpus. The distinct contribution here is additive: a hypothetical counterexample forces a distinguished-descendant carry language whose words give many equal-arity representations of one fixed target. Character balancing explains why these families lie in algebraic parts rather than contradicting functional transcendence.

70.1 Carry paths and finite alphabets

Let $U, V \geq 2$ be multiplicatively independent integers and write $e_{a,b} = U^a V^b$. A step decreasing the first exponent replaces one distinguished copy by U copies, freezes $U - 1$, and continues with the remaining copy; the second-coordinate step does the same with V. A monotone word with r first steps and s second steps therefore yields a multiset representation

$$U^r V^s = \sum_{j=1}^{K_{r,s}} e_{a_j,b_j}, \qquad K_{r,s} = 1 + r(U-1) + s(V-1). \tag{547}$$

The frozen terms telescope to the initial monomial. At every intermediate total degree the multiset records the unique visited vertex; multiplicative independence then reconstructs the word. Hence there are at least

$$\binom{r+s}{r} \tag{548}$$

distinct representations of the fixed target. Under a hypothetical counterexample this applies to $(U, V) = (M, N) = (2^x, 3^x)$ and every term is itself $(2^a 3^b)^x$.

The construction extends to a finite alphabet of distinct moves $v_i = (\alpha_i, \beta_i) \in \mathbb{N}_0^2 \setminus \{(0,0)\}$. Put

$$D_i = U^{\alpha_i} V^{\beta_i}, \qquad (r, s) = \sum_i n_i v_i, \qquad K = 1 + \sum_i n_i (D_i - 1).$$

Words with the same symbol counts then give at least

$$\binom{\sum_i n_i}{n_1, \ldots, n_m} \tag{549}$$

distinct K-term representations of $U^r V^s$. The exact telescoping sum, term count, strict suffix-degree descent, and reconstruction from the endpoint and ordered suffix list are **[kernel-verified Lean]** in `CarryPath`. Recovering the word from the unordered frozen multiset and the multinomial enumeration remain the **[paper proof]** paper layer.

The two primitive counterexample carries $(2t)^x = Mt^x$ and $(3t)^x = Nt^x$ are also exactly character-balanced. More generally every finite sequence of such rewrites groups into independently vanishing character classes. Thus their positive-dimensional families are compatible with, rather than forbidden by, the usual functional-transcendence decomposition.

70.2 Exact carry-language pressure

For a fixed finite alphabet $\mathcal{S}$ of at least two 3-smooth integers $d_i = 2^{\alpha_i}3^{\beta_i} > 1$, let $D_i = d_i^x$ under the hypothetical solution. Entropy maximization gives two exact *carry-language* rates. They are the unique positive roots

$$\sum_i d_i^{-\delta_{\mathcal{S}}} = 1, \qquad \sum_i e^{-\tau_{\mathcal{S}}(D_i-1)} = 1. \tag{550}$$

For symbol proportions π_i, relative entropy proves

$$\sup_\pi \frac{H(\pi)}{\sum_i \pi_i \log d_i} = \delta_{\mathcal{S}}, \qquad \sup_\pi \frac{H(\pi)}{\sum_i \pi_i (D_i - 1)} = \tau_{\mathcal{S}}.$$

The alphabet $\{2, 3, 6\}$ already gives ambient exponent one. Exhausting all nonunit 3-smooth letters gives

$$(1 - 2^{-\delta_*})(1 - 3^{-\delta_*}) = \frac{1}{2}, \qquad \delta_* = 1.4352790836694176\ldots. \tag{551}$$

This is an exact supremum over the finite-alphabet carry construction, not over all additive representations.

There is also an unrestricted word pressure. Let $\tau_x > 0$ be the unique solution of

$$\sum_{(a,b)\in\mathbb{N}_0^2\setminus\{(0,0)\}} e^{-\tau_x(M^aN^b-1)} = 1, \qquad g = \gcd(M-1, N-1). \tag{552}$$

For every sufficiently large positive multiple n of g, some endpoint (r_n, s_n) with $r_n + s_n \le n$ receives

$$\gg_x \frac{e^{\tau_x n}}{n^2} \tag{553}$$

distinct carry-language representations of the fixed target $M^{r_n}N^{s_n}$ as a sum of $n+1$ values a^x, all with $a \le 3^n$. The word generating series is

$$\frac{1}{1-\Phi_x(z)}, \qquad \Phi_x(z) = \sum_{(a,b)\neq(0,0)} z^{M^aN^b-1}.$$

After factoring the period g, its unique dominant positive pole gives $[z^n](1-\Phi_x)^{-1} \asymp_x e^{\tau_x n}$ along $g \mid n$; pigeonholing among $O(n^2)$ endpoints yields (553). The matching upper rate is only for fibers generated by this unrestricted carry language, not for the full additive representation fiber.

These are **[paper proof]** exact paper theorems. They do not conflict with Harrison's fixed-arity and fixed-target bounds [67]: here the arity grows linearly with the logarithm of the input cutoff. A closing theorem would need uniform fixed-target bounds through that growing-arity range, strong enough to beat $e^{\tau_x n}$; no such estimate is known. Compact-orbit and point-counting reformulations do not supply it.

70.3 Multiplicative floor-carry rectangles

There is a second, genuinely different carry language on the multiplicative orbit itself. For integers $a, b \geq 2$, put

$$T_a(t) = \{at\}, \qquad c_a(t) = \lfloor at \rfloor \qquad (0 \leq t < 1).$$

Then $T_a T_b = T_b T_a$ and the two carry decompositions of ab, t give

$$b\, c_a(t) + c_b(T_a t) = a\, c_b(t) + c_a(T_b t). \tag{554}$$

On an $R \times S$ rectangle, the potentials $P_{ij}(t) = \lfloor a^i b^j t \rfloor$ therefore give horizontal digits $h_{ij} = P_{i+1,j} - aP_{ij} \in \{0, \ldots, a-1\}$, vertical digits $v_{ij} = P_{i,j+1} - bP_{ij} \in \{0, \ldots, b-1\}$, and the local law

$$bh_{ij} + v_{i+1,j} = av_{ij} + h_{i,j+1}.$$

Conversely, after fixing $P_{00} = 0$, the bounded edge labels and these plaquette laws reconstruct the potentials path-independently. If $Q = a^R b^S$, the half-open cylinders $[k/Q, (k+1)/Q)$ give an exact bijection

$$\{0, \ldots, Q-1\} \longleftrightarrow \{\text{compatible } R \times S \text{ floor-carry rectangles}\}; \tag{555}$$

in the cylinder indexed by k one has $P_{ij} = \lfloor k/(a^{R-i} b^{S-j}) \rfloor$. Thus there are exactly $a^R b^S$ rectangles, an aggregate fraction $(ab)^{-RS}$ of all unrestricted edge labelings. The four digit bounds and the local identity (554) are **[kernel-verified Lean]** in `MixedRadixCarryFinite`; the finite reconstruction and count are a **[paper proof]** paper argument.

Closed inverse-limit cylinders require the usual radix-endpoint convention. A nonsmooth point has one field, while every rational in $(0, 1)$ whose reduced denominator divides some $a^R b^S$ has both the floor and left-limit fields. In particular $t = 1/2$ already has two distinct compatible $(2, 3)$ fields. No uniqueness assertion at those endpoints is used below.

For multiplicatively independent a, b, the Furstenberg–Boshernitzan density theorem implies that every compatible finite rectangle occurs arbitrarily far out in the orbit of every irrational t [10]. In particular, for $Q = 2^R 3^S$,

$$\{\lfloor 2^r 3^s t \rfloor \bmod Q : r \geq R, s \geq S\} = \mathbb{Z}/Q\mathbb{Z}. \tag{556}$$

This rules out arguments based only on a forbidden finite floor-carry word. Frequencies, first-occurrence scales, heights, and arithmetic labels are not universal and remain available.

For a hypothetical nonintegral exponent x, a smooth anchor $q = 2^r 3^s$ with $\nu = \lfloor qx \rfloor$ and $t = \{qx\}$ satisfies

$$\lfloor 2^i 3^j qx \rfloor = 2^i 3^j \nu + \lfloor 2^i 3^j t \rfloor.$$

An all-zero $R \times S$ block, equivalently $0 < t < 1/Q$, therefore synchronizes both denominator-depth residues modulo Q and gives paired leading gaps

$$M^q = 2^\nu + \Delta_2, \qquad 0 < \Delta_2 < 2^{\nu - G_2(Q)}, \qquad A^q = 3^\nu + \Delta_3, \quad 0 < \Delta_3 < 3^{\nu - G_3(Q)},$$

where $G_b(Q) = \max\{0, \lfloor -\log_b(b^{1/Q} - 1) \rfloor\} = \log_b Q + O_b(1)$. This is stronger than either one-base leading-gap statement because the smooth index and leading exponent are shared.

The quantitative wall can be stated by the one-sided hitting time

$$H_x(Q) = \min\{2^r 3^s : r, s \in \mathbb{N}_0,\ 0 < \{2^r 3^s x\} < 1/Q\}.$$

Baker's two-logarithm estimate gives $H_x(Q) \gg_x Q^{\eta_x}$ for a counterexample. Combining its polynomial Diophantine consequence with the effective coprime-base orbit theorem of Bourgain–Lindenstrauss–Michel–Venkatesh and its explicit refinement by Gayfulin–Moshchevitin gives, for every $\varepsilon > 0$,

$$Q^{\eta_x} \ll_x H_x(Q) \leq \operatorname{expexpexp}\big(C_{x,\varepsilon} Q^{8+\varepsilon}\big) \tag{557}$$

for large Q [3, 11, 12]. The parameter map uses the coprime bases $2, 3$ and the polynomial Diophantine estimate; it is not a uniform statement for arbitrary bases. Any bound $H_x(Q) = Q^{o(1)}$ along an unbounded sequence would contradict the Baker lower bound and settle the conjecture, but finite-language universality alone gives no such acceleration.

70.4 Base-change digit lifts and integral carry defects

Write $m = \sum \varepsilon_j 2^j$ and $n = \sum d_j 3^j$, and evaluate the same digit strings in the other base:

$$L(m) = \sum \varepsilon_j 3^j, \qquad R(n) = \sum d_j 2^j.$$

With $\theta = \log_2 3$ and $\rho = \theta^{-1}$, elementary carry normalization gives

$$L(m) \leq m^\theta, \qquad n^\rho \leq R(n), \qquad R(L(m)) = m, \qquad L(R(n)) \geq n. \tag{558}$$

Equality in the first two inequalities occurs only at zero and at a pure power of the source base. If $m = 2^B + r$ with $0 < r < 2^B$, then the sharper bound $m^\theta - L(m) \geq (3/2)^B r$ holds. The exact valuation transfer is

$$v_3(L(m)) = v_2(m), \qquad v_2(R(n)) \geq v_3(n),$$

with equality in the second formula when the least nonzero ternary digit is one. More sharply, for distinct m, n,

$$v_3(L(m) - L(n)) = v_2(m - n). \tag{559}$$

Thus L extends from $\mathbb{Z}_2$ to the closed $\{0, 1\}$-digit subset of $\mathbb{Z}_3$ as a homeomorphism preserving the first-differing-digit index. It is not an isometry for the standard p-adic metrics: the two distances are respectively 2^{-r} and 3^{-r}. The finite congruence form of (559) is **[kernel-verified Lean]** in `RadixShadow`.

All addition and multiplication formulas are instances of one normalization identity. If a finite coefficient string r is normalized in base b, and q records the elementary carry counts, then evaluation in another base c satisfies

$$\operatorname{ev}_c(\operatorname{norm}_b r) = \operatorname{ev}_c(r) + (c - b)\operatorname{ev}_c(q). \tag{560}$$

This gives the sign of every forward or reverse carry without conflating positional radix carries with the exponent-rewrite paths of Section 70.1.

Multiplication has a signed carry expansion:

$$L(ab) - L(a)L(b) = \sum_j \kappa_j(a, b)3^j \geq 0, \qquad R(a)R(b) - R(ab) = \sum_j \lambda_j(a, b)2^j \geq 0, \tag{561}$$

where the coefficients count elementary base-2 and base-3 carries. For a hypothetical nonintegral solution define

$$D_n = A^n - L(M^n), \qquad E_n = R(A^n) - M^n.$$

Both are positive integers; at genuine integral exponents they vanish, so nonintegrality is essential. They have the same exponential rates as the outputs, but their exact carry recurrences retain additional combinatorics:

$$L(M^{m+n}) - L(M^m)L(M^n) = A^m D_n + A^n D_m - D_m D_n - D_{m+n}, \tag{562}$$
$$R(A^m)R(A^n) - R(A^{m+n}) = M^m E_n + M^n E_m + E_m E_n - E_{m+n}. \tag{563}$$

When $t_n = \{nx\} < \rho$, the exact tail bounds further give $D_n/A^n \asymp t_n$ and $E_n/M^n \asymp t_n^\rho$ on shrinking targets. The signs, valuation transfer, and recurrences are exact paper-level constraints, but their natural heights grow with the outputs and currently yield no contradiction.

The forward defect has an exact mantissa gauge. For the terminating binary fraction $\xi_n = 2^{t_n} - 1$, define its Cantor reevaluation $\mathcal{C}(\sum_{j\ge1}\varepsilon_j 2^{-j}) = \sum_{j\ge1}\varepsilon_j 3^{-j}$. Then

$$\frac{L(M^n)}{A^n} = 3^{-t_n}(1+\mathcal{C}(\xi_n)), \qquad \frac{t_n}{3} \le \frac{D_n}{A^n} \le (\log 3)t_n. \tag{564}$$

This couples the already-known Sturmian phase to a positional carry observable without repeating its density or natural-boundary theory.

There is also an exact first collision. If $m = 2^a u$ with $u > 1$ odd, then

$$v_3(L(m^2) - L(m)^2) = 2a + v_2(u-1), \qquad \frac{L(m^2) - L(m)^2}{3^{2a+v_2(u-1)}} \equiv 1 \pmod 3. \tag{565}$$

Indeed the two lowest set bits of m create the first raw square coefficient 2, and no earlier carry is possible. Writing $M = 2^a U$ therefore gives

$$v_3(L(M^{2n}) - L(M^n)^2) = 2an + v_2(U^n - 1)$$

for the non-power case $U > 1$. This supplies a precise structural-prime divisor of the normalization carry, but no matching archimedean upper bound is known.

The individual increments of the two radix shadows are especially simple:

$$L(n) - L(n-1) = \frac{3^{v_2(n)}+1}{2}, \qquad R(n) - R(n-1) = 2 - 2^{v_3(n)}. \tag{566}$$

Hence

$$L(N) = \frac{N}{2} + \frac{1}{2}\sum_{n\le N} 3^{v_2(n)}, \qquad R(N) = 2N - \sum_{n\le N} 2^{v_3(n)}. \tag{567}$$

For the defects above, put $C_n = L(M^n) = A^n - D_n$ and $K_n = L(R(A^n)) - A^n$. Telescoping over the special endpoint intervals gives

$$E_n = 2D_n - \sum_{C_n < j \le A^n} 2^{v_3(j)},$$
$$D_n + K_n = \frac{1}{2}\left(E_n + \sum_{M^n < j \le M^n + E_n} 3^{v_2(j)}\right). \tag{568}$$

In particular $0 < E_n \le D_n$ and $E_n \le R(D_n) \le 2D_n^{\log_3 2}$. This converts the archimedean defect into an exact excess of valuation weights on consecutive intervals; average-order estimates alone do not control their arithmetic endpoints.

There is also a rigid most-significant-digit description. Write

$$M^n = \sum_j \varepsilon_{n,j} 2^j, \qquad A^n = \sum_j \delta_{n,j} 3^j, \qquad P_n(Z) = \sum_j (\delta_{n,j} - \varepsilon_{n,j}) Z^j. \tag{569}$$

Then

$$P_n(3) = D_n, \qquad P_n(2) = E_n. \tag{570}$$

Its leading coefficient is 1 or 2. More generally, if $P(Z) = \sum_{i=0}^{J} c_i Z^i$ has $c_i \in \{-1, 0, 1, 2\}$, positive leading coefficient, and $E = P(2) > 0$, then

$$\begin{aligned} c_J = 1 &\Longrightarrow E = 1 + \sum_{i<J} (c_i + 1) 2^i, \\ c_J = 2 &\Longrightarrow E = 2^J + 1 + \sum_{i<J} (c_i + 1) 2^i. \end{aligned} \tag{571}$$

Thus $E < 2^s$ with $s \le J$ forces $c_J = 1$ and $c_i = -1$ for $s \le i < J$. For P_n, every such position has the digit pair $(\delta_{n,i}, \varepsilon_{n,i}) = (0, 1)$. This finite shifted-evaluation borrow step is **[kernel-verified Lean]** in `BorrowBlockCore`. The ensuing paper argument converts an abnormally small E_n into

$$M^n \equiv -u \pmod{2^J}, \qquad A^n \equiv v \pmod{3^J}, \qquad 1 \le u \le 2^s, \qquad 0 \le v < 3^s. \tag{572}$$

The common leading word has an explicit length. Put $\ell_n = \lfloor nx \rfloor$ and $u_n = \{nx\}$. The leading binary and ternary positions of M^n and A^n are both ℓ_n. For $0 < u < \log_3 2$, compare the fractional digits of 2^u in base 2 and of 3^u in base 3. Their common prefix is exactly a leading 1 followed by zeros; at the first mismatch the ternary digit is larger. Indeed

$$(1 + 2^{-t})^{\log_2 3} > 1 + 2\,3^{-t} \qquad (t \ge 1), \tag{573}$$

so a binary 1 at the first nonzero fractional position forces the ternary digit to be 2. If

$$t(u) = \left\lceil \log_3 \frac{1}{3^u - 1} \right\rceil, \tag{574}$$

then P_n has degree $\ell_n - t(u_n)$ when $u_n < \log_3 2$, and degree ℓ_n otherwise. Two-logarithm bounds give $t(u_n) = O_M(\log n)$, hence

$$D_n \gg_{M,A} \frac{A^n}{n^C} \tag{575}$$

for an effective C. The direct digit defect is therefore exponentially large even when long borrowing makes the return defect small. Finally, normalizing the ternary digits of A^n in base 2 with carry bits c_j gives the exact mask

$$K_n = \sum_{j \ge 0} c_j 3^j. \tag{576}$$

These identities expose a rigid cross-base word, but its two evaluations remain at incompatible scales and provide no contraction.

Status: The bounded carry-word sum, length, suffix-degree, and ordered reconstruction identities are [kernel-verified Lean] in `CarryPath`; the mixed-radix local floor-carry identity is [kernel-verified Lean] in `MixedRadixCarryFinite`; the finite radix-shadow congruence and valuation-index bridge are [kernel-verified Lean] in `RadixShadow`; and finite borrow-block rigidity is [kernel-verified Lean] in `BorrowBlockCore`. Multiset injectivity, multinomial fibers, character balancing, the finite and unrestricted pressure laws, finite floor-carry reconstruction, smooth-language consequences, hitting corridor, normalization carries, mantissa gauge, square-collision formula, interval-valuation identities, prefix separation, and carry-mask formula are [paper proof] paper results. Every "exact rate" is explicitly scoped to the carry-language construction; no contradiction with fixed-arity theorems or proof of Alaoglu–Erdős is claimed.

71 Critical entire interpolation, canonical products, and exact closure barriers

This section studies the power function on the universal cover of $\mathbb{C}^\times$ and the arithmetic sampling set $\{2^a 3^b : a, b \geq 0\}$. Its Six Exponentials monodromy and algebraic-multiplier inputs are established in Sections 48.6 and 12. The first new core is an exact analytic growth threshold for the 3-smooth lattice and an infinite boundary-defect identity at that threshold.

71.1 Regularity ladder for a power branch

On the universal cover, a holomorphic function satisfying

$$F(w + \log 2) = 2^x F(w), \qquad F(w + \log 3) = 3^x F(w), \qquad F(0) = 1$$

is necessarily $F(w) = e^{xw}$: after division by e^{xw}, the two incommensurable real periods have a dense subgroup, so continuity makes the quotient constant. The corresponding branch z^x descends to a single-valued holomorphic function on $\mathbb{C}^\times$ exactly when $e^{2\pi i x} = 1$, hence exactly when $x \in \mathbb{Z}$. Throughout the algebraic, Puiseux, and D-finite discussion, z^x means a branch on a simply connected punctured neighborhood or on this universal cover, not an ordinary analytic germ at zero.

Finite ramification makes the monodromy a root of unity and forces $x \in \mathbb{Q}$, after which the elementary rational-exponent argument closes the conjecture. A further exact criterion is arithmetic D-finiteness. If a branch z^λ satisfies a nonzero linear differential equation with coefficients in $\overline{\mathbb{Q}}[z]$, of order r and leading coefficient $p_r \neq 0$, then clearing powers of z gives

$$Q(z, \lambda) = \sum_{j=0}^{r} p_j(z)(\lambda)_j z^{r-j} = 0. \tag{577}$$

The polynomial $Q(z, X)$ is nonzero because $[X^r]Q = p_r(z)$, so λ is algebraic. The coefficient identity and noncancellation are [kernel-verified Lean] in `DfiniteExponentCore`; the differential-equation specialization is the [paper proof] paper layer. This would force integrality for the conditional power branch, whose nonintegral exponent is already known transcendental. Allowing coefficients that depend on x makes differential algebraicity vacuous: $zy' - xy = 0$. Even without admitting x as a coefficient, every power branch satisfies the coefficient-free nonlinear identity

$$zyy'' - z(y')^2 + yy' = 0,$$

so unrestricted nonlinear differential algebraicity still does not isolate the arithmetic exponents.

Integer samples accumulating at a puncture are also insufficient. The entire product below makes $1 + E(1/z)$ holomorphic on $0 < |z| < \varepsilon$, essential at zero, yet equal to 1 at every $z = 2^{-a}3^{-b}$. Exact dilation information, not sampling alone, is indispensable.

71.2 The sharp cubic logarithmic type

Define the locally uniformly convergent entire product

$$E(z) = \prod_{a,b \geq 0} \left(1 - \frac{z}{2^a 3^b}\right). \tag{578}$$

Its zeros are simple and exactly the positive 3-smooth integers. Factorwise equality at $z = -R$ gives $M_E(R) = E(-R)$, and lattice integration yields the exact leading growth

$$\log M_E(R) = \frac{(\log R)^3}{6 \log 2 \log 3} + O((\log R)^2). \tag{579}$$

Indeed, for $T = \log R$,

$$\sum_{a,b \geq 0} (T - a \log 2 - b \log 3)_+ = \frac{T^3}{6 \log 2 \log 3} + O(T^2).$$

For an entire F, set

$$\tau_3(F) = \limsup_{R\to\infty} \frac{\log^+ M_F(R)}{(\log R)^3}, \qquad \tau_* := \frac{1}{6 \log 2 \log 3}.$$

Jensen's formula gives the matching lower bound

$$F \not\equiv 0, \quad F(2^a 3^b) = 0 \ (a, b \geq 0) \quad \Longrightarrow \quad \tau_3(F) \geq \tau_*. \tag{580}$$

The constant is sharp by (579). These self-contained statements are **[paper proof]** to this corpus; no claim of priority in the literature is made.

The threshold gives a sharp sufficient analytic interface. More generally, let $p, q > 1$ be multiplicatively independent, $P, Q \in \mathbb{C}^\times$, and suppose an entire H satisfies

$$H(p^a q^b) = P^a Q^b \qquad (a, b \geq 0).$$

If

$$\limsup_{R\to\infty} \frac{\log^+ M_H(R)}{(\log R)^3} < \frac{1}{6 \log p \log q}, \tag{581}$$

then $H(z) = z^n$, $P = p^n$, and $Q = q^n$ for some $n \in \mathbb{N}_0$. To prove this, the defect $H(pz) - PH(z)$ vanishes on the whole lattice and remains subcritical, so (580) annihilates it; coefficient comparison then leaves a single monomial. Thus a subcritical interpolant of the hypothetical data would prove the conjecture. At equality, $1 + E$ is transcendental entire and integer-valued on every node, while $z^n + cE(z)$ interpolates the same integral-exponent data as z^n. These examples show failure of uniqueness at critical type; they do not prove that every more structured critical-type criterion is incapable of forcing exponent integrality.

71.3 The critical boundary cocycle

The product has exact edge factorizations

$$E(2z) = E(z)B_2(z), \qquad E(3z) = E(z)B_3(z), \tag{582}$$

where

$$B_2(z) = \prod_{b\geq 0}\left(1 - \frac{2z}{3^b}\right), \qquad B_3(z) = \prod_{a\geq 0}\left(1 - \frac{3z}{2^a}\right).$$

Their second-order growth is

$$\log M_{B_2}(R) = \frac{(\log R)^2}{2\log 3} + O(\log R), \qquad \log M_{B_3}(R) = \frac{(\log R)^2}{2\log 2} + O(\log R). \tag{583}$$

For an entire interpolant $H(2^a3^b) = M^aA^b$, set $D_2(z) = H(2z) - MH(z)$ and $D_3(z) = H(3z) - AH(z)$. Their lattice zeros and the simplicity of E give unique entire quotients $D_2 = EU$ and $D_3 = EV$. Commutativity of the two dilations is exactly the cocycle

$$B_3(z)U(3z) - AU(z) = B_2(z)V(2z) - MV(z). \tag{584}$$

This is an infinite analytic analogue of `BoundaryDilationDefect`, not a replacement for its finite algebra.

If U and V are polynomials, the strict quadratic gap in (583), evaluated on the negative real axis, first forces $U = 0$ and then $V = 0$. Both dilation equations are exact, so coefficient comparison gives $H(z) = z^n$, $M = 2^n$, and $A = 3^n$. This polynomial-boundary-quotient closure and the cocycle are **[paper proof]** paper theorems. Critical type alone does not establish the polynomial quotient hypothesis; manufacturing that finite-dimensional boundary regularity is the new open target.

The one-axis connection formalism in Section 62.4 gives a useful gauge for the same cocycle without introducing another compatibility equation. Abbreviate $\mathcal{N}_2(z) = \mathcal{N}_{2,M}(z)$, $\mathcal{E}_2(z) = (z; 1/2)_\infty$, and $\varkappa_2 = \varkappa_2(M)$. Since an entire interpolant H agrees with $\mathcal{N}_2$ on the dyadic axis, there is an entire W with

$$H(z) = \mathcal{N}_2(z) + \mathcal{E}_2(z)W(z).$$

Its normalized dyadic defect is

$$K_2(z) := \frac{H(2z) - MH(z)}{\mathcal{E}_2(z)} = \varkappa_2 + (1 - 2z)W(2z) - MW(z). \tag{585}$$

Because $E(z) = \mathcal{E}_2(z)E(z/3)$ and $D_2 = EU$, one also has

$$K_2(z) = E(z/3)U(z). \tag{586}$$

Thus K_2 vanishes on the shifted smooth grid and

$$\limsup_{R\to\infty} \frac{\log^+ M_{K_2}(R)}{(\log R)^3} < \frac{1}{6\log 2\log 3} \quad \Longrightarrow \quad K_2 = 0 \quad \Longrightarrow \quad x \in \mathbb{N}_0, \tag{587}$$

with a symmetric triadic statement. The scalar $\varkappa_2$ cannot by itself provide the strict saving: at the origin the free value of W absorbs it, and $K_2(0) = (1 - M)H(0)$. Equation (586) also shows that this is exactly the existing boundary quotient in a one-axis-normalized gauge, not a second cocycle.

Status: The D-finite noncancellation core is **[kernel-verified Lean]** in `DfiniteExponentCore`. The universal-cover classification, sharp canonical-product/Jensen threshold, subcritical interpolation rigidity, edge growth, critical cocycle, and polynomial-quotient closure are **[paper proof]** paper results. Overlapping monodromy material is omitted, and neither critical-type sampling nor the cocycle currently supplies the missing interpolant regularity.

71.4 One-edge residual rigidity at quadratic type

The two-dimensional product has cubic logarithmic growth, but each exposed edge has a sharp quadratic scale. For an entire function F, put

$$\sigma_2(F) = \limsup_{R\to\infty} \frac{\log^+ M_F(R)}{(\log R)^2}, \qquad M_F(R) = \max_{|z|\le R} |F(z)|. \tag{588}$$

Jensen's formula gives the exact edge barrier

$$F \not\equiv 0, \quad F(q^n/c) = 0 \ (n \ge 0) \quad \Longrightarrow \quad \sigma_2(F) \ge \frac{1}{2\log q} \qquad (q > 1,\ c > 0). \tag{589}$$

Indeed, the zeros through radius R contribute $(\log R)^2/(2\log q) + O(\log R)$ to Jensen's sum. Equality is attained by $\prod_{n\ge 0}(1 - cz/q^n)$; this is the one-dimensional version of the canonical-product calculation above [20].

The cocycle admits two asymmetric residuals

$$R_2(z) = AU(z) + B_2(z)V(2z) - MV(z), \tag{590}$$
$$R_3(z) = MV(z) + B_3(z)U(3z) - AU(z). \tag{591}$$

Using (584) once gives the exact factorizations

$$\boxed{R_2(z) = B_3(z)U(3z), \qquad R_3(z) = B_2(z)V(2z).} \tag{592}$$

Thus R_2 vanishes at every $2^a/3$, and R_3 at every $3^b/2$. Consequently

$$\boxed{\sigma_2(R_2) < \frac{1}{2\log 2} \quad \text{or} \quad \sigma_2(R_3) < \frac{1}{2\log 3} \quad \Longrightarrow \quad x \in \mathbb{N}_0.} \tag{593}$$

For example, the first inequality and (589) force $R_2 = 0$; then (592) gives $U = 0$. Comparing coefficients in $H(2z) = MH(z)$ leaves one monomial, and evaluation at 3 identifies the same integral exponent. In particular a counterexample forces both residuals to be nonzero and to meet or exceed their respective critical edge scales. The deliberately stronger separated hypothesis

$$\sigma_2(U) < \frac{1}{2\log 2}, \qquad \sigma_2(V) < \frac{1}{2\log 2} - \frac{1}{2\log 3}$$

implies the first inequality in (593), but the residual itself is the more flexible target because it retains cancellation among its three terms.

There is a matching coefficient-space criterion. Euler's q-binomial identity gives

$$[z^j]B_2 = \frac{(-6)^j}{\prod_{k=1}^{j}(3^k - 1)}, \qquad [z^j]B_3 = \frac{(-6)^j}{\prod_{k=1}^{j}(2^k - 1)}. \tag{594}$$

If $r_n = [z^n]R_2$, then any estimate

$$|r_n| \le C \exp\left[-\left(\frac{\log 2}{2} + \varepsilon\right) n^2\right] \quad (n \gg 1) \tag{595}$$

makes $\sigma_2(R_2) < 1/(2\log 2)$ by completing the square in $\sum_n |r_n| R^n$, and therefore closes the conjecture.

Arithmetic supplies a second sharp coordinate. For $F(z) = \sum_{n\ge 0} a_n z^n \in \mathbb{Q}[[z]]$ entire, define

$$\delta_2(F) = \limsup_{n\to\infty} \frac{\log \operatorname{den}(a_n)}{n^2}. \tag{596}$$

A Cauchy estimate at $R = \exp(n/(2\sigma'))$ proves the elementary uncertainty theorem

$$\boxed{4\sigma_2(F)\delta_2(F) < 1 \quad \Longrightarrow \quad F \text{ is a polynomial.}} \tag{597}$$

Indeed, a nonzero rational coefficient is at least the reciprocal of its reduced denominator, while Cauchy's bound is $|a_n| \le \exp(-n^2/(4\sigma'))$; strict separation of the two quadratic constants forces all large coefficients to vanish. It follows that

$$R_2 \in \mathbb{Q}[[z]], \quad \sigma_2(R_2) \le \frac{1}{2\log 2}, \quad \delta_2(R_2) < \frac{\log 2}{2} \quad \Longrightarrow \quad x \in \mathbb{N}_0, \tag{598}$$

with the symmetric triadic statement as well. The constants are exact: the edge product B_3 has the two values $1/(2\log 2)$ and $(\log 2)/2$, so their uncertainty product is one. In checking its reduced denominators, primes dividing the base q are separated from primes dividing the fixed numerator parameter; LTE on the remaining progressions shows that only $O(n \log n)$ can cancel from the quadratic denominator logarithm.

This arithmetic interface is nonempty. Expanding the two-dimensional product formally gives

$$ne_n = -\sum_{k=1}^{n} \frac{e_{n-k}}{(1-2^{-k})(1-3^{-k})}, \qquad E(z) = \sum_{n\ge 0} e_n z^n, \tag{599}$$

so $E \in \mathbb{Q}[[z]]$. Moreover the smooth nodes can be enumerated as $\lambda_0 < \lambda_1 < \cdots$ with $\lambda_n/n \to \infty$. Successively correcting a rational interpolation polynomial by

$$d_n \left(\frac{z}{\lambda_n}\right)^{k_n} \frac{\prod_{j<n}(z-\lambda_j)}{\prod_{j<n}(\lambda_n - \lambda_j)}, \qquad k_n \ge n,$$

and choosing k_n to make the correction at most 2^{-n} on $|z| \le n$ constructs an entire rational-coefficient interpolant. The same construction works verbatim for any prescribed rational values at the nodes: the correction d_n remains rational, and $k_n \ge n$ ensures that each Taylor coefficient receives only finitely many corrections. Formal division by $E \in 1 + z\mathbb{Q}[[z]]$ then makes U, V, R_2, R_3 rational-coefficient entire functions. This construction gives no useful growth or denominator bound; rather, it isolates the exact missing simultaneous estimate.

Status: [paper proof] — the edge Jensen barrier, residual factorization and closure, coefficient uncertainty, rational-coefficient interpolation, and the sharp type–denominator gauge target are paper theorems. Jensen's formula and the elementary q-product identities are classical; no priority claim is made. The strict inequalities are essential, since the canonical edge product realizes equality in both coordinates.

The critical analysis can be sharpened further. The strongest positive result is an explicit entire interpolant at the exact critical type, together with a strict second-order criterion that would force it to be a monomial. The same audit closes several tempting shortcuts negatively: low Hermite jets, Lambert cocycles, reciprocal Hankel rank, fixed-prime denominator support, and pure-2/3 convergent denominators all stop at quantified barriers. None of the statements below is advertised as a proof of the Alaoglu–Erdős conjecture.

71.5 The canonical critical interpolant and the quadratic threshold

Put

$$\alpha = \log 2, \qquad \gamma = \log 3, \qquad \tau = \frac{1}{6\alpha\gamma}, \qquad c_E = \frac{1}{4\alpha} + \frac{1}{4\gamma}, \qquad \delta_{\text{quad}} = \frac{1}{4\alpha} - \frac{1}{4\gamma} > 0.$$

For the canonical product E from (578), irrational-rotation equidistribution sharpens the lattice count to

$$\#\{(a,b) \in \mathbb{N}_0^2 : a\alpha + b\gamma \le T\} = \frac{T^2}{2\alpha\gamma} + \left(\frac{1}{2\alpha} + \frac{1}{2\gamma}\right) T + o(T). \tag{600}$$

Integrating this formula and comparing $\log(1+e^u)$ with u_+ in unit-width boundary shells gives

$$\boxed{\log M_E(e^T) = \tau T^3 + c_E T^2 + o(T^2).} \tag{601}$$

The same Jensen argument as in Section 71.2 now says that every nonzero entire function vanishing on all $2^a 3^b$ has maximum-modulus logarithm at least the right side of (601), up to $o(T^2)$.

There is a canonical entire function attaining this refined scale for *every* nonintegral $x > 0$. On the slit plane use the principal branch and define

$$w_x(t) = \frac{t^x}{E(-t)}, \qquad J_x(z) = \int_0^\infty \frac{w_x(t)}{t+z}\, dt, \qquad H_x(z) = z^x + \frac{\sin \pi x}{\pi} E(z) J_x(z). \tag{602}$$

All moments of w_x are finite. Across the negative axis the Plemelj jump of J_x is $-2\pi i r^x / E(-r)$, exactly cancelling the jump of z^x; the remaining singularity at zero is removable. Hence H_x is entire and

$$H_x(2^a 3^b) = (2^a 3^b)^x \qquad (a, b \ge 0). \tag{603}$$

Moreover, polynomial interpolation on any nested finite exhaustion of the smooth nodes converges locally uniformly to H_x. To see convergence across the cut as well, order the nodes as $s_0, s_1, \ldots$, put $F_j = \{s_0, \ldots, s_j\}$, and write P_j for the interpolating polynomial. Its Newton increment is

$$P_j(z) - P_{j-1}(z) = [F_j](t^x) \prod_{i<j} (z - s_i).$$

After discarding finitely many terms, choose $j_0 > x$. The positivity formula below gives, for $j \ge j_0$ and uniformly on $|z| \le R$,

$$|P_j(z) - P_{j-1}(z)| \le \frac{C_{F_j} E(-R)}{s_j} \le \frac{C_{F_{j_0}} E(-R)}{s_j}.$$

Since $\sum_j s_j^{-1} = \sum_{a,b \ge 0} 2^{-a} 3^{-b} < \infty$, the Weierstrass test gives local uniform convergence to an entire function. Off the negative cut, the finite keyhole-contour remainder and dominated convergence identify that limit with H_x.

The Cauchy transform in (602) is bounded in the two half-planes, while on the negative ray

$$R\,\mathrm{PV}\int_0^\infty \frac{w_x(t)}{t-R}\,dt \longrightarrow -\mu_0, \qquad \mu_0 := \int_0^\infty w_x(t)\,dt > 0.$$

Thus

$$|H_x(-R)| \sim \frac{|\sin\pi x|\mu_0}{\pi R}E(-R), \qquad \log M_{H_x}(e^T) = \tau T^3 + c_E T^2 + o(T^2). \tag{604}$$

The positive Stieltjes moment is the first exact obstruction: it disappears precisely when $\sin\pi x = 0$. Cancelling finitely many moments only replaces $E(z)/z$ by $E(z)/z^k$ and does not change the quadratic coefficient.

For an entire interpolant H, set

$$c_2(H) := \limsup_{T\to\infty} \frac{\log M_H(e^T) - \tau T^3}{T^2}.$$

The defect $D_2(z) = H(2z) - MH(z)$ vanishes at all smooth nodes. Dilation raises the quadratic coefficient by $3\alpha\tau = 1/(2\gamma)$, whereas refined Jensen requires c_E if $D_2 \neq 0$. Therefore

$$\boxed{c_2(H) < \delta_{\mathrm{quad}} \quad\Longrightarrow\quad H(z) = z^n, \quad M = 2^n, \quad A = 3^n.} \tag{605}$$

This is strictly sharper than merely assuming that the boundary quotients are polynomials. Indeed, if $D_2 = EU$ and $D_3 = EV$, and

$$u = \limsup_{r\to\infty} \frac{T(r,U)}{(\log r)^2}, \qquad v = \limsup_{r\to\infty} \frac{T(r,V)}{(\log r)^2},$$

then the cocycle (584) and the global Nevanlinna inequalities give

$$U, V \neq (0,0) \quad\Longrightarrow\quad u + v \geq \delta_{\mathrm{quad}}. \tag{606}$$

There are no exceptional radii in this estimate. If $U = 0$, zeros of B_2 propagate under halving to zero and force $V = 0$. Thus $u + v < \delta_{\mathrm{quad}}$ is another exact sufficient condition. The explicit H_x proves that ordinary residue interpolation lands on the wrong side of the gap: $c_2(H_x) = c_E > \delta_{\mathrm{quad}}$.

71.6 A four-value positivity closure and the canonical obstruction

There is a remarkably short sufficient condition at the opposite analytic extreme. Let

$$H(z) = \sum_{n\geq 0} c_n z^n, \qquad c_n \geq 0, \qquad H(1) = 1.$$

Absolute convergence at 6 and Tonelli's theorem give the exact covariance identity

$$H(6) - H(2)H(3) = \frac{1}{2}\sum_{m,n\geq 0} c_m c_n (2^m - 2^n)(3^m - 3^n). \tag{607}$$

Every summand is nonnegative, and it is strictly positive for two distinct indices carrying positive mass. Consequently

$$c_n \geq 0, \quad H(1) = 1, \quad H(6) = H(2)H(3) \quad\Longrightarrow\quad H(z) = z^N \tag{608}$$

for one $N \in \mathbb{N}_0$. An absolutely monotone entire interpolant of the smooth table would therefore prove the conjecture from the four values at $1, 2, 3, 6$. The finite covariance and single-support core is **[kernel-verified Lean]** in `PositiveCoefficientClosure`; the passage to an infinite nonnegative series is the displayed paper argument.

This rigidity is quantitatively stable. If (p_n) is a probability distribution with $\mathcal{M}(t) = \sum_n p_n t^n$ and $\mathcal{M}(6) < \infty$, then

$$\Delta := \mathcal{M}(6) - \mathcal{M}(2)\mathcal{M}(3) \geq 1 - \sum_n p_n^2 \geq 1 - \sup_n p_n. \tag{609}$$

Indeed the kernel in (607) is at least 2 off the diagonal. Thus $\Delta \leq \varepsilon < 1$ forces one atom to carry mass at least $1 - \varepsilon$. If every p_n is an integral multiple of $1/D$, then $D^2\Delta$ is a nonnegative integer; either $\Delta = 0$ or $\Delta \geq D^{-2}$. A positive rational approximation can therefore close only if its covariance error beats the square of its weight mesh.

The three values at $1, 2, 3$ already suffice analytically. For a probability distribution with $\mathcal{M}(3) < \infty$, strict Jensen convexity gives

$$\sum_n p_n 3^n \geq \left(\sum_n p_n 2^n\right)^{\log_2 3}, \tag{610}$$

with equality only for a point mass. The four-value identity is more useful arithmetically because its defect is rational or integral.

Geometric curvature, sums of squares, and the positivity tax. There is a broader three-value closure that does not refer to Taylor coefficients.

Proposition 71.1 (Three-value geometric-curvature closure)**.** *Let H be real entire, positive on $[1, 3]$, and suppose*

$$H(1) = 1, \qquad H(2) = 2^x, \qquad H(3) = 3^x$$

for some $x \in \mathbb{R}$. If $u \mapsto \log H(e^u)$ is convex or concave on $[0, \log 3]$, then $x = N \in \mathbb{N}_0$ and $H(z) = z^N$.

Proof. Put $a = \log 2$, $b = \log 3$, and $\phi(u) = \log H(e^u) - xu$. Then $\phi(0) = \phi(a) = \phi(b) = 0$, with $0 < a < b$. Equality at the interior point of the endpoint chord forces a convex or concave function to be affine on the whole segment, so $\phi = 0$ on $[0, b]$. Hence $zH'(z) - xH(z)$ vanishes on $(1, 3)$ and therefore identically. Comparing coefficients in $H(z) = \sum_n c_n z^n$ gives $(n - x)c_n = 0$ for every n; since $H(1) = 1$, exactly one coefficient survives, at an index $N = x \in \mathbb{N}_0$, and it equals one. $\square$

Equivalently, monotonicity of the elasticity $tH'(t)/H(t)$ on $[1, 3]$ already closes the problem. The proposition includes positive Mellin mixtures, for which log convexity follows from Hölder's inequality. It also includes a positive H whose multiplicative Hankel kernel $K(s, t) = H(st)$ is positive semidefinite on $[1, \sqrt{3}]$: its 2×2 Gram inequalities give midpoint log convexity. The corresponding TP_2 condition gives convexity, and reverse TP_2 gives concavity. Likewise, complete monotonicity of $t \mapsto H(3e^{-t})$ makes its logarithm convex, while a Bernstein log-coordinate is concave.

Pointwise positivity itself is completely flexible.

Proposition 71.2 (Positive rational sum-of-squares interpolation)**.** *For every pair $M, A \in \mathbb{N}_{>0}$ there is an entire $H_+ \in \mathbb{Q}[[z]]$ such that*

$$H_+(t) \geq 1 \quad (t \in \mathbb{R}), \qquad H_+(2^a 3^b) = M^a A^b \quad (a, b \in \mathbb{N}_0).$$

Indeed H_+ can be the diagonal of a positive semidefinite kernel on $\mathbb{R}$ of rank at most five.

Proof. For each smooth node $s = 2^a 3^b$, Lagrange's four-square theorem supplies integers $r_{1,s}, \ldots, r_{4,s}$ with $M^a A^b - 1 = \sum_{j=1}^4 r_{j,s}^2$. Apply the arbitrary-rational-value interpolation construction above separately to the four data sequences, obtaining entire $F_j \in \mathbb{Q}[[z]]$ with $F_j(s) = r_{j,s}$. Then

$$H_+(z) = 1 + \sum_{j=1}^4 F_j(z)^2 \quad \text{and} \quad K_+(s,t) = 1 + \sum_{j=1}^4 F_j(s)F_j(t)$$

have the asserted properties. The construction supplies no useful growth or denominator bound. □

Global positivity nevertheless has an exact growth cost.

Proposition 71.3 (Elasticity oscillation and refined positivity tax)**.** *Let $x \notin \mathbb{N}_0$, and let H be an entire interpolant of $H(2^a 3^b) = (2^a 3^b)^x$ that is positive on $[1, \infty)$. If $\lambda_0 < \lambda_1 < \cdots$ are the smooth nodes and*

$$L_x H(z) := zH'(z) - xH(z),$$

then $L_x H$ has a zero in every interval $(\lambda_j, \lambda_{j+1})$, and

$$\log M_H(e^T) \geq \tau T^3 + c_E T^2 + o(T^2). \tag{611}$$

Proof. Writing $\chi(t) = tH'(t)/H(t)$, consecutive interpolation values give

$$\int_{\lambda_j}^{\lambda_{j+1}} (\chi(t) - x)\, \frac{dt}{t} = 0.$$

The integrand cannot vanish identically on one gap, since the identity theorem would then give $L_x H \equiv 0$ and Proposition 71.1 would force $x \in \mathbb{N}_0$. It therefore takes both signs and has a zero in the gap. Jensen's formula, the interlacing, and (600) give, after removing a possible zero at the origin,

$$\log M_{L_x H}(e^T) \geq \sum_{\lambda_{j+1} \leq e^T} \log \frac{e^T}{\lambda_{j+1}} - O_H(T) = \tau T^3 + c_E T^2 + o(T^2).$$

For $\delta = T^{-2}$, Cauchy's estimate on the concentric disks of radii e^T and $e^{T+\delta}$ gives

$$M_{L_x H}(e^T) \leq (2T^2 + |x|) M_H(e^{T+\delta})$$

for large T. Replacing $T + \delta$ by the outer logarithmic radius preserves both displayed leading terms and proves (611). □

Thus positivity, real zero-freeness, rational Taylor coefficients, and even a finite-rank kernel diagonal can all be imposed without resolving the conjecture. The rigid extra input is one-signed geometric curvature; absent that curvature, a positive representative must oscillate in elasticity at every node scale and pay at least the full refined critical growth.

There is an explicit linear separator for the discrete moment cone. If $k < x < k + 1$, put

$$\psi_k(t) = (2^t - 2^k)(3^t - 3^{k+1}). \tag{612}$$

Then $\psi_k(n) \geq 0$ for every integer $n \geq 0$, whereas

$$\psi_k(x) = -(M - 2^k)(3^{k+1} - A) \in \mathbb{Z}_{\leq -1}. \tag{613}$$

Consequently any normalized positive mixture of $(1, 2^n, 3^n, 6^n)$ stays at sup-norm distance at least $(3^{k+1} + 2^k + 1)^{-1}$ from the target four-vector. More generally, a finite signed representation $\sum_n c_n(1, 2^n, 3^n, 6^n) = (1, M, A, MA)$ must satisfy

$$\sum_{c_n<0} |c_n|\psi_k(n) \geq (M - 2^k)(3^{k+1} - A) \geq 1. \tag{614}$$

Thus every exact interpolant needs a quantifiable negative component; moving a very small negative coefficient to a very large exponent can still pay the separator, so ordinary negative mass alone has no degree-independent lower bound.

The canonical Stieltjes representative cannot satisfy this condition. Write $m = \lfloor x \rfloor$ and $H_x(z) = \sum_{k\geq 0} h_k z^k$. The sign in (630), together with the signs of the elementary symmetric coefficients in the Newton basis and the locally uniform interpolation limit, gives

$$\boxed{(-1)^{k-m-1} h_k > 0 \qquad (k \geq m+1).} \tag{615}$$

This obstruction survives every finite positive-scale diagonal filter. If $0 < s_1 < \cdots < s_J$ and $p_j \in \mathbb{R}[T]$, then for the Euler operator $\mathcal{D} = z\, d/dz$ the coefficient of z^k in

$$\mathcal{T} H_x(z) = \sum_{j=1}^{J} p_j(\mathcal{D}) H_x(s_j z) \quad \text{is} \quad h_k R(k), \qquad R(k) = \sum_{j=1}^{J} p_j(k) s_j^k. \tag{616}$$

If $\mathcal{T} \neq 0$, choose the largest scale with nonzero combined polynomial. After division by that scale to the kth power and by the leading power of k, $R(k)$ tends to the nonzero leading coefficient. Hence $R(k)$ has a fixed sign eventually, so the filtered coefficients still alternate strictly. Adding a polynomial changes only finitely many coefficients. Thus no nonzero finite linear combination of positive smooth dilations—even with Euler multipliers—turns the canonical representative into the nonnegative-coefficient interpolant required by (608). The constant-multiplier finite core, including arbitrary signed weights with a unique largest active scale, is **[kernel-verified Lean]** in `FiniteDilationAlternation`; the Euler extension is a paper argument. Infinite filters, arbitrary nonlinear operations, and arbitrary representatives $H_x + EG$ remain outside this linear result.

The most direct nonlinear node-preserving corrections also miss by an exact cubic margin. For every integer $r \geq 2$,

$$V_r(z) := H_x(z)^r - H_x(z^r) \tag{617}$$

vanishes at every smooth node. Moreover $V_r \equiv 0$ if and only if H_x is a monomial: factoring $H_x = z^d u$ at the origin and taking the local logarithm of $u/u(0)$ reduces the identity to $L(z^r) = rL(z)$, whose least nonzero Taylor term is impossible. Since $H_x(1) = 1$, the monomial is z^d. For $r, s \geq 2$, the composition identity

$$V_{rs} = V_r \sum_{j=0}^{s-1} H_x^{r(s-1-j)} H_x(z^r)^j + V_s(z^r)$$

also shows that the dyadic tower is generated recursively by V_2 and its substitutions; it supplies no independent compatibility conditions. For a nonzero finite combination $W = \sum_{r=2}^{R} c_r V_r$ with $c_R \neq 0$, maximum-modulus comparison gives

$$\log M_W(e^T) = \tau R^3 T^3 + c_E R^2 T^2 + o(T^2). \tag{618}$$

Indeed, $H_x(z^R)$ has that growth and exponentially dominates $H_x(z)^R$ and every lower index. Since $W = EG$ for an entire G,

$$\log M_G(e^T) \geq (R^3 - 1)\tau T^3 + (R^2 - 1)c_E T^2 + o(T^2).$$

Thus every nonzero finite correction of this form is strongly supercritical. The interpolant $H_x + W$ also cannot have nonnegative coefficients for nonintegral x, by the four-value closure theorem.

Infinite scalar combinations have an exact convergence threshold. Put $a_r = -\log|c_r|$, with $a_r = +\infty$ when $c_r = 0$. The maximum-modulus comparison

$$M_{H_x}(e^{rT}) - M_{H_x}(e^T)^r \leq M_{V_r}(e^T) \leq M_{H_x}(e^{rT}) + M_{H_x}(e^T)^r$$

gives, for every fixed $T > 0$, as $r \to \infty$,

$$\log M_{V_r}(e^T) = \tau r^3 T^3 + O_T(r^2).$$

Consequently

$$\sum_{r\geq 2} c_r V_r \text{ converges normally (absolutely locally uniformly)} \iff \frac{a_r}{r^3} \longrightarrow +\infty. \tag{619}$$

An infinite support therefore has an absolute maximum-modulus majorant of infinite cubic type. Cancellation in an arbitrary normally convergent signed series remains open, but one large class is rigid: if such a series has infinite support, all its indices have one fixed 2-adic valuation k, and all nonzero c_r lie on one complex ray, evaluation at $z = e^T \exp(i\pi/2^k)$ sends every z^r to $-e^{rT}$. The asymptotic $H_x(-t) \sim \mathfrak{c}_x E(-t)/t$, $\mathfrak{c}_x \neq 0$, puts all dominant terms in one shrinking cone and proves that the sum itself has infinite cubic type.

Nor does multiplicative renormalization help. For integers $q, s \geq 2$, the quotients

$$\mathfrak{A}_q(z) = \frac{H_x(z^q)}{H_x(z)^q}$$

obey

$$\mathfrak{A}_{qs}(z) = \mathfrak{A}_s(z^q)\mathfrak{A}_q(z)^s = \mathfrak{A}_q(z^s)\mathfrak{A}_s(z)^q. \tag{620}$$

With a local logarithm at the origin, their natural rooted tower telescopes:

$$\prod_{j=0}^{n-1} \mathfrak{A}_q(z^{q^j})^{q^{-j-1}} = \frac{H_x(z^{q^n})^{q^{-n}}}{H_x(z)} \longrightarrow \frac{1}{H_x(z)}.$$

Globally each unrooted factor $\mathfrak{A}_q$ equals one at every smooth node. On the other hand, the pointwise modulus ratio

$$\frac{|H_x(z^{q^n})|^{q^{-n}}}{|H_x(z)|}$$

has supremum on $|z| = e^T$, for every $T > 0$, whose logarithm grows at least like $\tau q^{2n} T^3$. Hence the local rooted telescope cannot extend to a globally normally convergent holomorphic product. The dyadic and triadic towers are the same coboundary rather than independent constraints.

Finite Prouhet cancellation also fails beyond the polynomial asymptotic. For distinct finite multisets $\mathcal{U}, \mathcal{V}$ of positive substitution indices in one 2-adic class, assume $\sum_{r\in\mathcal{U}} r = \sum_{r\in\mathcal{V}} r$ and set

$$\mathcal{W}_{\mathcal{U},\mathcal{V}}(z) = \prod_{r\in\mathcal{U}} H_x(z^r) - \prod_{r\in\mathcal{V}} H_x(z^r).$$

This function vanishes on the smooth nodes, but

$$\tau_3(\mathcal{W}_{\mathcal{U},\mathcal{V}}) = \tau \max\left\{\sum_{r\in\mathcal{U}} r^3, \sum_{r\in\mathcal{V}} r^3\right\}. \tag{621}$$

Indeed Mellin inversion and Baker separation [3] write $\log E(-e^T)$ as a cubic polynomial plus two absolutely convergent Fourier series with frequencies in $2\pi(\log 2)^{-1}\mathbb{Z}$ and $2\pi(\log 3)^{-1}\mathbb{Z}$, up to an exponentially small remainder. If the polynomial moments tie, the least index whose multiplicities differ contributes a nonzero frequency that no larger index can reproduce; the two frequency lattices cannot collide because $\log_2 3$ is irrational. Hence the product ratio stays away from one along an unbounded sequence. Even an eight-against-eight Thue–Morse pair matching power sums through degree three retains type 65408τ.

There is a complementary defect using only smooth dilations:

$$\mathcal{R}_x(z) = H_x(z)H_x(6z) - H_x(2z)H_x(3z) = E(z)\mathcal{G}_x(z). \tag{622}$$

The rank-one table law gives the node zeros and hence the entire quotient $\mathcal{G}_x$. The exact product identity

$$E(z)E(6z) = (1-6z)E(2z)E(3z)$$

and the negative-axis Stieltjes asymptotic give

$$\mathcal{R}_x(-R) \sim \frac{\mathfrak{c}_x^2}{R}E(-2R)E(-3R), \qquad \log|\mathcal{G}_x(-e^T)| = \tau T^3 + 3c_E T^2 + o(T^2). \tag{623}$$

The Stieltjes formula supplies the matching global upper bound, so $\tau_3(\mathcal{G}_x) = \tau$: the most elementary four-dilation determinant produces another exactly critical multiple of E, not a subcritical one. At every integral control $x \in \mathbb{N}_0$, all the nonlinear and dilation defects above vanish identically. General signed infinite cancellation and mixed 2-adic supports remain outside these results.

In particular infinitely many Maclaurin coefficients are negative. Similarly,

$$H_x(t) - t^x = \frac{\sin \pi x}{\pi}E(t)J_x(t), \qquad J_x(t) > 0,$$

alternates sign between consecutive smooth nodes. This is the canonical sign-level manifestation of the general curvature and elasticity obstruction in Propositions 71.1 and 71.3.

The miss is also quantitative at the boundary. For the canonical quotients $H_x(2z) - 2^x H_x(z) = E(z)U_x(z)$ and $H_x(3z) - 3^x H_x(z) = E(z)V_x(z)$, put

$$K_x = \frac{\sin \pi x}{\pi}\int_0^\infty \frac{t^x}{E(-t)}\,dt \neq 0.$$

The signed principal-value asymptotic yields

$$U_x(-R) \sim -\frac{K_x}{2R}B_2(-R), \qquad V_x(-R) \sim -\frac{K_x}{3R}B_3(-R).$$

The maximum-modulus/Poisson comparison, with the outer-to-inner radius ratio subsequently sent to infinity, therefore gives

$$u_x \geq \frac{1}{2\log 3}, \qquad v_x \geq \frac{1}{2\log 2}, \qquad u_x + v_x \geq 2c_E > \delta_{\text{quad}}. \tag{624}$$

These are canonical-only lower bounds, not an improvement of the arbitrary-interpolant cocycle gap. The surviving positive direction is correspondingly precise: construct a *different* arithmetic representative in the nonnegative-coefficient cone. Ordinary Stieltjes/Newton interpolation provably does not do so.

The most natural positive measure attached to E also fails in an exact way. Normalize

$$d\nu_x(t) = \frac{t^{x-1}}{I_x E(-t)}\,dt, \qquad I_x = \int_0^\infty \frac{t^{x-1}}{E(-t)}\,dt, \qquad \mathcal{A}_s(t) = \frac{E(-t)}{E(-t/s)}. \tag{625}$$

For every suitable f and $r, s > 0$, substitution gives the exact quasi-invariance and cocycle

$$\int f(t)\mathcal{A}_s(t)\,d\nu_x(t) = s^x \int f(st)\,d\nu_x(t), \qquad \mathcal{A}_{sr}(t) = \mathcal{A}_s(t)\mathcal{A}_r(t/s). \tag{626}$$

Thus for a smooth $s = 2^a 3^b$ the normalizing constant is exactly $s^x = M^a A^b$, but this is only the original table written as a Radon–Nikodym identity. In particular

$$\mathcal{A}_6(t) = \mathcal{A}_2(t)\mathcal{A}_3(t/2), \qquad \mathcal{A}_2(t)\mathcal{A}_3(t) = (1+t)\mathcal{A}_6(t). \tag{627}$$

If $m_1 = \int t\,d\nu_x(t) > 0$, then

$$\int \mathcal{A}_2\mathcal{A}_3\,d\nu_x = 6^x(1+6m_1) > 6^x, \qquad \int \mathcal{A}_2(t)\mathcal{A}_3(t/2)\,d\nu_x = 6^x. \tag{628}$$

The shifted argument absorbs the whole positive covariance gap, so (607) cannot be applied to these transfer weights.

The same conclusion survives at every Hankel order. If $H_d = (\int t^{i+j}\,d\nu_x)_{0\le i,j\le d}$ and $H_d[\mathcal{A}_s]$ denotes the matrix with the tilted moments, then

$$H_d[\mathcal{A}_s] = s^x \operatorname{diag}(s^i) H_d \operatorname{diag}(s^i), \qquad \det H_d[\mathcal{A}_s] = s^{x(d+1)+d(d+1)} \det H_d. \tag{629}$$

After probability normalization this is merely dilation congruence, not an arithmetic lattice floor. Finally, normalizing the positive Taylor coefficients of $\mathcal{A}_s$ against the moments of ν_x produces an entire probability generating function Φ_s, but it has infinitely many positive coefficients and hence a strict four-value covariance defect. As smooth $s \to \infty$, every fixed coefficient of Φ_s tends to zero while $\Phi_s(1) = 1$: all mass escapes to infinity, so there is no locally uniform positive-coefficient interpolant in this family. These identities hold for every real $x > 0$, including all integral controls; integrality changes only the displayed normalizing constants.

71.7 The positive Stieltjes denominator target

Fix a hypothetical nonintegral counterexample, and let $F \subset \{2^a 3^b : a, b \in \mathbb{N}_0\}$ contain $N+1$ distinct nodes with $N > x$. A keyhole contour gives the exact divided difference

$$[F](t^x) = (-1)^{N+1}\frac{\sin \pi x}{\pi}\int_0^\infty \frac{t^x\,dt}{\prod_{s\in F}(t+s)}. \tag{630}$$

Consequently, along every nested exhaustion of the smooth semigroup,

$$C_F := \left(\prod_{s\in F} s\right) |[F](t^x)| \searrow C_x := \frac{|\sin \pi x|}{\pi}\int_0^\infty \frac{t^x}{E(-t)}\,dt > 0. \tag{631}$$

Under a counterexample all node values are integers, so $[F](t^x)$ is rational. If $\mathfrak{d}_F$ is its reduced positive denominator, nonvanishing and (631) imply

$$\boxed{\liminf_{F\nearrow\{2^a3^b\}} \frac{\mathfrak{d}_F}{\prod_{s\in F} s} \geq \frac{1}{C_x} > 0.} \tag{632}$$

Hence a single semigroup exhaustion satisfying $\mathfrak{d}_F = o(\prod_{s\in F} s)$ would prove the conjecture. This is a finite arithmetic target with no unspecified analytic constant in its leading scale.

Successive Markov ratios. The real common-power relation gives a stronger output-sensitive statement, but in the opposite direction. Write

$$\mathcal{D}_F(z) = [F](t^z).$$

For $|F| = N+1$, $N > x$, and a new smooth node $s \notin F$, the ratio of two successive divided differences is

$$\Phi_F(s) := -\frac{\mathcal{D}_{F\cup\{s\}}(x)}{\mathcal{D}_F(x)} = \int_0^\infty \frac{d\mu_F(t)}{s+t} \in \mathbb{Q}_{>0}, \qquad d\mu_F(t) = \frac{t^x\,dt}{I_F\prod_{u\in F}(t+u)}, \tag{633}$$

where I_F normalizes the measure to mass one. If $m \geq 1$ and $2m < N - x$, its moments through order $2m$ are rational:

$$\mu_k := \int_0^\infty t^k\,d\mu_F(t) = (-1)^k\frac{\mathcal{D}_F(x+k)}{\mathcal{D}_F(x)} \in \mathbb{Q} \qquad (0 \leq k \leq 2m). \tag{634}$$

Let $Q_m \in \mathbb{Q}[t]$ be the monic degree-m orthogonal polynomial for μ_F, put $h_m = \int Q_m^2\,d\mu_F$, and define

$$P_{m-1}(s) = \int_0^\infty \frac{Q_m(-s) - Q_m(t)}{s+t}\,d\mu_F(t) \in \mathbb{Q}[s].$$

Orthogonality gives the exact Markov–Padé remainder

$$\boxed{\Phi_F(s) - \frac{P_{m-1}(s)}{Q_m(-s)} = \frac{1}{Q_m(-s)^2}\int_0^\infty \frac{Q_m(t)^2}{s+t}\,d\mu_F(t) > 0.} \tag{635}$$

This identity has an integral floor. For $u \in F$, put

$$D_{F,u} = \prod_{v\in F\setminus\{u\}}(u-v), \qquad L_F = \mathrm{lcm}_{u\in F}\,|D_{F,u}|.$$

Then

$$R_k = L_F\mathcal{D}_F(x+k) \in \mathbb{Z}, \qquad \epsilon = \mathrm{sgn}(R_0), \qquad a_k = \epsilon(-1)^k R_k \in \mathbb{Z}_{>0} \quad (0 \leq k < N-x),$$

and set

$$\Delta_j = \det(a_{r+s})_{0\leq r,s<j} \in \mathbb{Z}_{>0} \qquad (1 \leq j \leq m+1).$$

The Hankel system shows that $\Delta_m Q_m \in \mathbb{Z}[t]$ and that $a_0\Delta_m P_{m-1} \in \mathbb{Z}[t]$, while

$$h_m = \frac{\Delta_{m+1}}{a_0\Delta_m}.$$

Thus $C_m = a_0\Delta_m$ is a simultaneous clearing and $C_m h_m = \Delta_{m+1}$. If $\mathfrak{b}_{F,s}$ is the reduced denominator of $\Phi_F(s)$, rational separation in (635), together with the fact that every zero of Q_m is positive, yields

$$\boxed{\mathfrak{b}_{F,s} > \frac{s|Q_m(-s)|}{\Delta_{m+1}} > \frac{s^{m+1}}{\Delta_{m+1}}.} \tag{636}$$

If $\mathcal{D}_F(x) = A_F/\mathfrak{d}_F$ and $\mathcal{D}_{F\cup\{s\}}(x) = A_{F,s}/\mathfrak{d}_{F,s}$ are reduced, then $\mathfrak{b}_{F,s} \le |A_F|\mathfrak{d}_{F,s}$, so the same estimate divided by $|A_F|$ applies to $\mathfrak{d}_{F,s}$. The strongest permitted choice has $m < (N-x)/2$, whereas the elementary append denominator has degree $N+1$ in s. The result therefore exposes an exact half-order barrier rather than the desired upper bound.

The nonzero integer behind the floor is a useful sharper target. Write both fractions in lowest terms:

$$\Phi_F(s) = \frac{\mathfrak{a}_{F,s}}{\mathfrak{b}_{F,s}}, \qquad \frac{P_{m-1}(s)}{Q_m(-s)} = \frac{p_s}{q_s}, \qquad \mathfrak{b}_{F,s}, q_s > 0.$$

Then

$$\begin{aligned} \delta_{m,s} &= \mathfrak{a}_{F,s}q_s - \mathfrak{b}_{F,s}p_s \in \mathbb{Z}_{>0}, \\ \frac{\delta_{m,s}}{\mathfrak{b}_{F,s}q_s} &= \frac{1}{Q_m(-s)^2}\int_0^\infty \frac{Q_m(t)^2}{s+t}\, d\mu_F(t), \\ \delta_{m,s} &< \frac{\mathfrak{b}_{F,s}q_s h_m}{sQ_m(-s)^2}. \end{aligned} \tag{637}$$

Thus a source-sensitive divisor of $\delta_{m,s}$ exceeding an independent upper bound for the last quantity would give a contradiction. No present cluster or Smith argument supplies one. For distinct new nodes s, u, the ratios also satisfy the exact resolvent recursion

$$\Phi_{F\cup\{u\}}(s) = \frac{\Phi_F(u) - \Phi_F(s)}{(s-u)\Phi_F(u)}. \tag{638}$$

This is the cleanest finite identity through which such a divisor could propagate across Newton levels. For a finite calibration, not a global counterexample, take $x = 1/2$, $F = \{1, 4, 9, 16\}$, and $s = 36$. Then $\Phi_F(36) = 1/40$, the primitive approximant $1/41$, and $\delta_{1,36} = 1$.

Smith saturation of the Padé defect. Raw collision content disappears on passing to the primitive approximant. Put

$$\mathsf{H}_m = (a_{i+j})_{\substack{0\le i<m,\\ 0\le j\le m}}, \qquad c_j = (-1)^{m+j}\det \mathsf{H}_m^{\widehat{j}}.$$

Then

$$\sum_{j=0}^{m} c_j t^j = \Delta_m Q_m(t).$$

Write $\Phi_F(s) = A/B$ in lowest terms, set

$$z_0 = a_0 A, \qquad z_j = Ba_{j-1} - sz_{j-1} \quad (1 \le j \le m),$$

and append this row to H_m. The moment identity $z_j = Ba_0 \int t^j (s+t)^{-1}\, d\mu_F(t)$ gives

$$\begin{aligned} J_{m,s} &:= \det\begin{pmatrix} \mathsf{H}_m & & \\ z_0 & \dots & z_m \end{pmatrix} = Ba_0\Delta_m \left(Q_m(-s)\Phi_F(s) - P_{m-1}(s)\right), \\ \delta_{m,s} &= \frac{|J_{m,s}|}{g_{m,s}}, \end{aligned} \tag{639}$$

where

$$\widehat{p}_{m,s} = a_0\Delta_m(-1)^m P_{m-1}(s), \qquad \widehat{q}_{m,s} = a_0\Delta_m(-1)^m Q_m(-s), \qquad g_{m,s} = \gcd(\widehat{p}_{m,s}, \widehat{q}_{m,s}).$$

If

$$\mathfrak{s}_m = \gcd(c_0, \ldots, c_m),$$

then $\mathfrak{s}_m$ is the top Smith determinantal divisor of H_m, divides both safe Padé coefficients, and $J_{m,s} = \mathfrak{s}_m\langle z, c^{\mathrm{sat}}\rangle$. Thus the relevant local quantity is

$$v_p(J_{m,s}) - min\{v_p(\widehat{p}_{m,s}), v_p(\widehat{q}_{m,s})\},$$

not $v_p(\mathfrak{s}_m)$.

This distinction is sharp at arbitrary collision depth. If $a_k = \sum_\nu w_\nu u_\nu^k$ with integral w_ν, u_ν and $u_\nu \equiv u \pmod{p^h}$ for every ν, a unimodular Pascal translation gives

$$p^{hm(m-1)} \mid \mathfrak{s}_m. \tag{640}$$

Indeed, after translation the (i,j) entry is divisible by $p^{h(i+j)}$, and every maximal minor contains at least $hm(m-1)$ powers of p. Nevertheless, let

$$\widetilde{M} = 5043, \qquad x_0 = \log_2 \widetilde{M}, \qquad K_h = 4\cdot 5^{h-1}, \qquad q_h = 2^{K_h}, \qquad w_h = \widetilde{M}^{K_h},$$

and take the finite one-axis table

$$F_h = \{1, q_h, \ldots, q_h^{19}\}, \qquad s_h = q_h^{20}, \qquad m = 2.$$

Here $12 < x_0 < 13$ and $4 < 19 - x_0$; every source node, the appended node, and every output w_h^j is congruent to 1 modulo 5^h. This finite one-axis calibration neither assumes nor addresses the condition $3^{x_0} \in \mathbb{Z}$. The geometric divided-difference formula yields

$$\Phi_{F_h}(s_h) = \frac{q_h^{19} - w_h}{q_h^{19}(q_h^{20} - 1)}, \qquad \mu_k = q_h^{19k - k(k+1)/2} \prod_{r=1}^{k} \frac{q_h^r w_h - 1}{q_h^{19-r} - w_h}.$$

The congruence

$$\frac{a^{K_h} - 1}{5^h} \equiv \frac{a^4 - 1}{5} \pmod 5$$

at the relevant unit bases gives

$$(\mu_1, \mu_2, \mu_3) \equiv (2, 2, 1) \pmod 5.$$

Hence $Q_2(-s_h) \equiv 1$ and $P_1(s_h) \equiv 3 \pmod 5$. Moreover

$$v_5(q_h^{19} - w_h) = h, \qquad v_5(q_h^{20} - 1) = h + 1.$$

After both fractions are reduced, the primitive defect δ_{2,s_h} is therefore a 5-adic unit for every h, although (640) gives $5^{2h} \mid \mathfrak{s}_2$. Thus no divisor depending only on source/output collision depths, append congruence, or unsaturated Hankel Smith factors can control $\delta_{m,s}$. This is only a no-go for common-collision and unsaturated-Smith arguments; a surviving divisor must use two-generator exponent geometry and the saturated transverse residue.

There is no hidden real cancellation behind these inequalities. Hermite–Genocchi gives

$$L_F\mathcal{D}_F(z) = L_F(z)_N \int_{\Delta_N} \left(\sum_{u\in F} \lambda_u u\right)^{z-N} d\lambda. \tag{641}$$

Here $d\lambda$ is coordinate-simplex measure of total mass $1/N!$. The second factor is positive on the real axis and is a compactly supported bilateral Laplace transform whose Hankel kernel is strictly totally positive. Hence the restricted circuit has exactly the real zeros $0, 1, \ldots, N-1$, all simple. Likewise, if $0 \le i_1 < \cdots < i_r$, $0 \le j_1 < \cdots < j_r$, and $i_r + j_r < N - x$, then $\det(a_{i_p+j_q})_{p,q=1}^r$ is a positive integer. In particular

$$R_0R_2 - R_1^2 \ge 1 \qquad (x + 2 < N). \tag{642}$$

The safe Hankel clearing can itself be factored exactly. Order $F = \{s_1 < \cdots < s_n\}$, put $\Delta_F^{\mathrm{V}} = \prod_{i<j}(s_j - s_i)$, $y_i = s_i^x$, and let $\mathcal{B}_{F,r}^{\mathrm{dir}}$ have ith row

$$(1, s_i, \ldots, s_i^{n-r-1}, y_i, y_is_i, \ldots, y_is_i^{r-1}).$$

Cauchy–Binet and complementary Vandermonde factorization give

$$\boxed{\Delta_r = \epsilon^r(-1)^{\binom{r}{2}} \frac{L_F^r}{\Delta_F^{\mathrm{V}}} \det \mathcal{B}_{F,r}^{\mathrm{dir}}.} \tag{643}$$

for $1 \le r \le m+1$. At Padé degree m, the exact prefactor is $L_F^{m+1}/\Delta_F^{\mathrm{V}}$. The shifted exponent block is interlaced with the ordinary block, so the shape-free HCIZ dominance from the reciprocal construction below is unavailable. The identity isolates primitive Cramer or Smith content as the part not measured by the full determinant.

Node-only denominator theory cannot prove that saving. For the size triangle

$$\mathcal{J}_T = \{(a,b) \in \mathbb{N}_0^2 : a\alpha + b\gamma \le T\}, \qquad F_T = \{2^a3^b : (a,b) \in \mathcal{J}_T\}, \qquad P_T = \prod_{s\in F_T} s,$$

put

$$D_s = \prod_{t\in F_T\setminus\{s\}} (s - t), \qquad L_T = \mathrm{lcm}_{s\in F_T} |D_s|.$$

The symbolic top divided difference has denominator exactly L_T. Already the endpoint $s = 1$ gives, after removing the 2- and 3-parts,

$$\log(L_T)_{(6)} \ge \log P_T - O(T) \ge \frac{T^3}{3\log 2\log 3} - O(T^2). \tag{644}$$

At the structural primes, the exact endpoint valuation comparison gives

$$\log(L_T)_{\{2,3\}} = \log P_T - 2T + O(1). \tag{645}$$

Thus both the away-prime and smooth-prime capacities can individually pay the leading Stieltjes requirement. Any proof of the needed $o(P_T)$ upper bound must exploit primitive cancellation for the *one fixed* rank-one value table M^aA^b; no Vandermonde, p-ordering, or node-only Smith theorem can supply it.

There is a precise fixed-divisor ceiling on any argument using only the abstract rank-one law. For a finite $E \subset \mathbb{N}_0^2$, put $s_{(a,b)} = 2^a 3^b$ and define D_e, L_E as above. The top divided difference of the table $s_{(a,b)} \mapsto U^a V^b$ is

$$[E]y = \frac{R_E(U,V)}{L_E}, \qquad R_E(U,V) = \sum_{(a,b)\in E} \frac{L_E}{D_{(a,b)}} U^a V^b \in \mathbb{Z}[U,V]. \tag{646}$$

The coefficients of R_E are primitive. If $r_E = \max_E a$, $s_E = \max_E b$, and $\delta_E = \gcd\{R_E(u,v) : u, v \in \mathbb{Z}\}$, then the binomial-basis finite-difference formula gives

$$\boxed{\delta_E \mid r_E! s_E!.} \tag{647}$$

Indeed, δ_E divides every integral binomial-basis coefficient, while returning to the monomial basis costs at most $r_E! s_E!$; primitivity proves the claim. Conversely, for every modulus q, local pairs attaining the p-adic minima defining δ_E can be combined by the Chinese remainder theorem. Thus arbitrarily large positive U, V satisfy

$$\gcd(R_E(U,V), q) = \gcd(\delta_E, q).$$

If $n = |E| - 1$, the integral controls calibrate exactly: $R_E(2^k, 3^k) = 0$ for $k < n$, while $R_E(2^n, 3^n) = L_E$. For the size triangles, take $q = L_T$ and combine (644)–(645). Suitable rank-one integer specializations then have

$$\log \mathfrak{d}_T \geq 2 \log P_T - O(T \log T), \qquad \mathfrak{d}_T \geq P_T^{2-o(1)}. \tag{648}$$

The pair (U, V) here may vary with T and need not lie on a common real power curve. Accordingly no universal bound obtained from adjugate, Sylvester, subresultant, or rank-one congruence identities can force $\mathfrak{d}_T = o(P_T)$ for all rank-one specializations; this does not rule out exceptional root concentration for the one fixed hypothetical pair (M, A). Such concentration would have to account for logarithmic mass $\Theta(T^3)$, whereas universal fixed-divisor content contributes only $O(T \log T)$.

For one fixed hypothetical pair, ordinary gap synchronization is also too sparse. Let

$$\mathrm{Dir}_T = (\mathcal{J}_T - \mathcal{J}_T) \setminus \{(0,0)\}.$$

For $d = (r, s) \in \mathrm{Dir}_T$, write in lowest terms

$$2^r 3^s = \frac{P_d}{Q_d}, \qquad M^r A^s = \frac{R_d}{S_d}, \qquad g_d = \gcd(|P_d - Q_d|, |R_d - S_d|).$$

Then the uniform triangular estimate is

$$\boxed{\sum_{d \in \mathrm{Dir}_T} \log g_d = o(T^3).} \tag{649}$$

The point is that all directions lie in the one fixed finitely generated group $\Gamma = \langle (2, M), (3, A) \rangle \subset \mathbb{G}_m^2(\mathbb{Q})$. For $d \neq 0$, put $u_d = 2^r 3^s$ and $v_d = M^r A^s = u_d^x$. Since $u_d \neq 1$ and x is irrational, u_d, v_d are multiplicatively independent. Corollary 1 of the Corvaja–Zannier S-unit gcd theorem [7] says that, for every $\varepsilon > 0$, all but finitely many multiplicatively independent $(u, v) \in \Gamma$ have generalized gcd at most $\varepsilon \max\{h(u), h(v)\}$. The quantity $\log g_d$ is its finite-place part, so the same bound holds for g_d. There are $O(T^2)$ directions and every height is $O_x(T)$; summing and then sending ε to zero proves (649). In particular the endpoint sum

$$\sum_{(a,b)\in\mathcal{J}_T\setminus\{(0,0)\}} \log\gcd(2^a3^b-1, M^aA^b-1) = o(T^3).$$

This has a precise local interpretation. Put $y_e = M^aA^b$ for $e=(a,b)$ and $\lambda_p = \max_e v_p(D_e)$. For $p\nmid 6MA$, choose a maximizing node i_p and define

$$\kappa_p = \sum_{j\neq i_p} \min\{v_p(s_{i_p}-s_j), v_p(y_{i_p}-y_j)\}. \tag{650}$$

Distinct j give distinct directions, so (649) implies

$$\sum_{p\nmid 6MA} \kappa_p \log p = o(T^3). \tag{651}$$

Define the collision graph modulo p to have the nodes as vertices and an edge $\{i,j\}$ exactly when $s_i \equiv s_j \pmod p$. If this graph has exactly one edge $\{i,j\}$ and $h_p = v_p(s_i-s_j)$, the two active barycentric coefficients are opposite modulo p^{h_p} and one has the exact isolated-collision formula

$$\min\{h_p, v_p(R_{\mathcal{J}_T}(M,A))\} = \min\{h_p, v_p(y_i-y_j)\}. \tag{652}$$

Thus every cancellation at an isolated collision prime is already charged to the subcubic budget. The exact mechanism for multiple collisions can also be exposed. For a general finite exponent set E, retain D_e, L_E, R_E from (646); for an arbitrary output vector $y=(y_e)_{e\in E}$ also write $R_E(y) = \sum_e (L_E/D_e)y_e$, and define κ_p by (650) with $\mathcal{J}_T$ replaced by E. Fix a prime p and put

$$d_e = v_p(D_e), \qquad \lambda = \max_e d_e, \qquad \omega_p = p^{-\lambda}L_E \in \mathbb{Z}_{(p)}^\times, \qquad u_e = p^{-d_e}D_e \in \mathbb{Z}_{(p)}^\times.$$

Then, exactly over $\mathbb{Q}_p$,

$$R_E(U,V) = \omega_p \sum_{e=(a,b)\in E} p^{\lambda-d_e}u_e^{-1}U^aV^b, \qquad \overline{R}_E = \overline{\omega}_p \sum_{d_e=\lambda} \overline{u}_e^{-1}U^aV^b. \tag{653}$$

There is no hidden sign: it is contained in u_e.

The coefficients in (653) are read directly from the collision tree. Let

$$C_k(e) = \{f\in E : s_f \equiv s_e \pmod{p^k}\}, \qquad C_0(e) = E.$$

Then

$$d_e = \sum_{k\geq 1}(|C_k(e)|-1). \tag{654}$$

For a depth-k cluster C, choose a common residue ρ_C modulo p^k and label each child B by $\xi_B = (s_b-\rho_C)/p^k \bmod p$, $b\in B$. The labels are defined up to common translation. If $B_k(e)$ is the child containing e, then

$$\overline{u}_e = \prod_{k\geq 0} \prod_{\substack{B \text{ child of } C_k(e) \\ B\neq B_k(e)}} (\xi_{B_k(e)}-\xi_B)^{|B|}. \tag{655}$$

Each other node contributes its valuation to (654) and its angular difference at the unique level where its branch separates from e, which proves both formulas.

There is a useful transfer theorem for the simplest geometry. Assume no pair that is congruent modulo p remains congruent modulo p^2. For each residue class C, let $m_C = |C|$, let r_C be its residue, choose an integer lift r_C^*, and write $t_e = (s_e - r_C^*)/p \bmod p$ for $e \in C$. If $m = \max_C m_C$, then $\lambda = m - 1$ and

$$\overline{\omega}_p^{-1}\overline{R}_E(y) = \sum_{m_C=m} \left(\prod_{C' \neq C} (r_C - r_{C'})^{-m_{C'}} \right) \sum_{e \in C} \frac{\overline{y}_e}{\prod_{f \in C \setminus \{e\}} (t_e - t_f)}. \tag{656}$$

These summands are precisely the top confluent coordinates of the largest local Smith blocks; the p-primary Vandermonde factors within a class of size m_C are $1, p, \ldots, p^{m_C - 1}$.

Proposition 71.4 (Unique affine cluster transfer)**.** *Assume $p \nmid 6MA$, the preceding simple-cluster hypothesis, and a unique largest class $C = \{e_0 + kd : 0 \leq k < m\}$, where $2 \leq m \leq p$ and $d = (r, s) \in \mathbb{Z}^2$. Then the collective excess at p is zero:*

$$(\min\{\lambda, v_p(R_E(M, A))\} - \kappa_p)_+ = 0$$

for every maximizing node used to define κ_p.

Proof. Put $q = 2^r 3^s$ and $\varrho = M^r A^s$. The source collision gives $q \equiv 1 \pmod{p}$ and, by simplicity, $c = (q-1)/p \in \mathbb{Z}_p^\times$. With $s_k = s_0 q^k$ and $y_k = y_0 \varrho^k$, the last sum in (656) is

$$\frac{\overline{y}_0 (\overline{\varrho} - 1)^{m-1}}{(\overline{s}_0 \overline{c})^{m-1} (m-1)!},$$

using the elementary $(m-1)$st divided difference of T^k; the factorial is a unit because $m \leq p$. Since the largest class is unique, $p \mid R_E(M, A)$ exactly when $\varrho \equiv 1 \pmod{p}$. In that case all $m - 1 = \lambda$ source collisions from a maximizing node transfer to the output, so $\kappa_p = \lambda$; otherwise the capped cancellation is zero. □

The hypotheses are sharp. The first complete size triangle,

$$\mathcal{J}_{\log 8} = \{(0,0), (1,0), (0,1), (2,0), (1,1), (3,0)\},$$

has $F = \{1, 2, 3, 4, 6, 8\}$ and

$$R_F(U, V) = U^3 + 35U^2 + 35U - 7UV - 56V - 8. \tag{657}$$

At $p = 5$, $\lambda = 1$ and

$$\overline{R}_F = (U - 2)(U^2 + 2U + 4 + 3V).$$

For the fixed specialization $(M, A) = (7, 6)$, $R_F(7, 6) = 1665$ has 5-adic valuation one, while the two collision pairs $\{1, 6\}$ and $\{3, 8\}$ have output gaps 41 and 337. Hence $\kappa_5 = 0$: even a full triangle has collective excess one.

The excess is unbounded on arbitrary smooth supports while keeping both the prime and the rank-one pair fixed. For $h \geq 1$, put $n = 2 \cdot 5^{h-1}$ and

$$E_h = \{(0,0), (0,n), (n,n), (n,2n)\}, \qquad \{s_e : e \in E_h\} = \{1, b, q, bq\}, \quad b = 3^n, \quad q = 6^n.$$

LTE gives two collision pairs $\{1, q\}$ and $\{b, bq\}$, each of depth h, so $\lambda_5 = h$. For $(M, A) = (7, 6)$ the output vector is congruent modulo 5^h to

$$(1, -b, -q, bq).$$

For arbitrary distinct $1, b, q, bq$, these four values lie on the quadratic

$$\frac{2X^2-(b+1)(q+1)X+2bq}{(b-1)(q-1)}.$$

Their third divided difference is therefore zero, and the integral cleared numerator gives $5^h \mid R_{E_h}(7,6)$. On both colliding pairs the output ratio is -1 modulo 5, so $\kappa_5 = 0$ and the collective excess equals h.

Accordingly no bound from the collision tree, local Smith form, support size, rank-one law, or a fixed multiplicatively independent output pair alone can improve the trivial λ_p ceiling for collective cancellation. For the full triangles, call a prime geometrically bad if it has a deeper collision, tied largest classes, or a non-affine largest class. The preceding proposition reduces the remaining capacity exactly to

$$\sum_{p\nmid 6MA} \min\{\lambda_p, v_p(R_{\mathcal{J}_T}(M,A))\}\log p \le o(T^3) + \sum_{p \text{ geometrically bad}} \lambda_p \log p. \tag{658}$$

No smallness of the last sum is asserted. The four-node family shows that any such estimate must use the downward-closed geometry of $\mathcal{J}_T$ or the real common-power relation, not smoothness or cluster data alone.

Two global calibrations further delimit the remaining capacity. For every fixed $p \ge 5$, elementary order lifting gives $\lambda_p = O_p(T^2)$, so no fixed set of nonstructural primes carries cubic mass. Also $R_{\mathcal{J}_T}$ is primitive of degree $O(T)$, whence

$$\#\{(u,v) \bmod p^e : R_{\mathcal{J}_T}(u,v) \equiv 0 \pmod{p^e}\} \le O(T)p^{2e-1};$$

this is only a density bound and cannot exclude one fixed exceptional pair. At every fixed integral control $x \in \mathbb{N}_0$, by contrast, the endpoint gcd sum has the full cubic scale and $R_{\mathcal{J}_T}(2^x, 3^x) = 0$ once the table is large enough. Thus (649) shows that, among nonstructural primes, any cancellation beyond the subcubic pairwise ratio-gap budget must be collective. The CRT specializations above evade it because their group changes with T.

The raw bad-cluster capacity in (658) cannot itself be made small. For $d = (r,s) \in \mathbb{Z}^2$, set

$$H_\triangle(d) = \max\{r_+ \log 2 + s_+ \log 3, r_- \log 2 + s_- \log 3\}.$$

Here $r_\pm = \max\{\pm r, 0\}$ and $s_\pm = \max\{\pm s, 0\}$.

Proposition 71.5 (Full capacity of the bad triangle geometry)**.** *For the complete size triangle, the primes that are not geometrically bad contribute only*

$$\sum_{\substack{p\ge 5\\ p \text{ not geometrically bad}}} \lambda_p \log p = O(T^2 \log(2+T)).$$

Consequently, for every fixed hypothetical pair (M,A),

$$\sum_{\substack{p\nmid 6MA\\ p \text{ geometrically bad}}} \lambda_p \log p \ge \frac{T^3}{3\log 2 \log 3} - O_{M,A}(T^2\log(2+T)). \tag{659}$$

Proof. A prime counted in the first sum has a unique simple affine maximal class $C = \{e_0 + kd : 0 \le k \le \lambda_p\}$. Each of $C-(1,0)$, $C-(0,1)$, and $C+(1,0)$, whenever it stays inside $\mathcal{J}_T$, is a congruent subset contained in a distinct residue class of size at least $|C|$. Uniqueness therefore forces the two coordinate minima of C to be zero and its top edge to lie within $\log 2$ of the boundary. For $v = \lambda_p d$, its endpoints are v_- and v_+ in the same coordinatewise convention, so

$$T - \log 2 < H_\triangle(v) \le T.$$

For fixed λ, this constant-width lattice shell contains $O(T/\lambda + 1)$ possible steps d; summing over $\lambda = O(T)$ gives $O(T \log(2+T))$ pairs (λ, d). Every prime assigned to one pair divides

$$2^{r_+}3^{s_+} - 2^{r_-}3^{s_-},$$

and hence the sum of $\lambda \log p$ over that pair is at most $\lambda H_\triangle(d) \le T$. This proves the first estimate. Now (644) gives the full nonstructural capacity, and the finitely many primes dividing MA contribute only $O_{M,A}(T^2)$ by the fixed-prime estimate above. Subtracting proves (659). ☐

Thus downward closure does not make the error term in (658) lower order: almost all source capacity is forced into its bad side. There is nevertheless one exact refinement. Under the simple-cluster hypothesis, if all maximal classes have the same affine step $d = (r, s)$ up to orientation, then their confluent coordinates share the factor

$$\overline{R}_E(U, V) = \left(U^{r_+}V^{s_+} - U^{r_-}V^{s_-}\right)^{m-1} Q_{p,E}(U, V) \tag{660}$$

over $\mathbf{F}_p$, where m is their common maximal size. This is the ordinary $(m-1)$st finite difference along each class. The common binomial is charged to the pairwise budget; the genuinely unresolved quantities are the vanishing and higher p-adic lifts of the output-sensitive transverse factor $Q_{p,E}(M, A)$, together with nonparallel and deeper clusters.

Finally, $E \in \mathbb{Q}[[z]]$: its reciprocal-zero power sums are

$$\sum_{a,b \ge 0} 2^{-ak}3^{-bk} = \frac{1}{(1-2^{-k})(1-3^{-k})} \in \mathbb{Q},$$

and Newton identities make every Taylor coefficient rational. Therefore rational Taylor coefficients and any finite collection of rational Hermite normalizations cannot select the critical interpolant: adding $qz^L P(z)E(z)$ preserves all node values and any prescribed finite jet while retaining critical growth.

71.8 The reciprocal Stieltjes moment lattice

The same transform has an output-sensitive dual. Assume a hypothetical nonintegral counterexample and put $\eta = 1/x \in (0, 1)$. Choose distinct exponent pairs $e_i = (a_i, b_i) \in \mathbb{N}_0^2$ so that

$$u_i = M^{a_i} A^{b_i} \quad (0 \le i \le n)$$

are increasing, and put $v_i = 2^{a_i}3^{b_i} = u_i^\eta$. For $P(T) = \prod_{i=0}^n (T - u_i)$ and $n \ge 2d+1$, define

$$h_r = \frac{\sin \pi\eta}{\pi} \int_0^\infty \frac{s^{\eta+r}}{\prod_{i=0}^n (s+u_i)}\, ds \qquad (0 \le r \le 2d). \tag{661}$$

Residues give the exact rational formula

$$h_r = (-1)^{n+r+1} \sum_{i=0}^{n} \frac{u_i^r v_i}{P'(u_i)} > 0. \tag{662}$$

Let $\Delta = \prod_{i<j}(u_j - u_i)$ and $k_r = \Delta h_r$. Since $\Delta/P'(u_i)$ is, up to sign, the product of the gaps not incident to u_i, every k_r is a positive integer. Consequently

$$K_d = (k_{i+j})_{0\le i,j\le d} \quad \text{is integral and positive definite,} \qquad \det K_d \ge 1. \tag{663}$$

After normalizing the measure in (661) by h_0, every nonzero $Q \in \mathbb{Z}[s]$ of degree at most d satisfies

$$\int_0^\infty Q(s)^2\, d\mu(s) = \frac{\text{a positive integer}}{k_0} \ge \frac{1}{k_0}, \qquad \det\left(\frac{h_{i+j}}{h_0}\right)_{0\le i,j\le d} \ge k_0^{-(d+1)}. \tag{664}$$

The determinant has substantially more integral content than the unit floor shows. Put $N = n+1$, $m = d+1$, and form the $N \times N$ alternant whose ith row is

$$\mathcal{B}_{i,\bullet} = (1, u_i, \ldots, u_i^{N-m-1}, v_i, u_i v_i, \ldots, u_i^{m-1} v_i). \tag{665}$$

Cauchy–Binet followed by complementary Vandermonde factorization gives the exact identity

$$\boxed{\det K_d = (-1)^{mN+\binom{m}{2}} \Delta^{m-1} \det \mathcal{B}.} \tag{666}$$

Since positivity fixes the sign of the right side and $\det \mathcal{B}$ is a nonzero integer,

$$\det K_d \ge \Delta^d, \qquad \det\left(\frac{h_{i+j}}{h_0}\right)_{0\le i,j\le d} \ge \frac{1}{\Delta h_0^{d+1}} = \frac{\Delta^d}{k_0^{d+1}}. \tag{667}$$

There is an entrywise refinement as well. If

$$L_F = \operatorname{lcm}_i |P'(u_i)|, \qquad G_F = \gcd_i \left|\frac{\Delta}{P'(u_i)}\right| = \frac{\Delta}{L_F},$$

then G_F divides every k_r. The normalized quadratic floor in (664) can therefore be strengthened from $1/k_0$ to $G_F/k_0 = 1/(L_F h_0)$.

In fact no choice of nodes or matrix size can make the residual alternant small. Let $q_0 < \cdots < q_{N-1}$ be the increasing rearrangement of

$$\{xk : 0 \le k < N-m\} \cup \{1 + xk : 0 \le k < m\}, \qquad G(N+1) = \prod_{r=1}^{N-1} r!.$$

For every $x \ge 2$, $1 \le m < N$, and distinct integers $1 \le v_0 < \cdots < v_{N-1}$, the HCIZ identity gives the shape-free estimate

$$\boxed{|\det \mathcal{B}| \ge \frac{\prod_{i<j}(q_j - q_i)}{G(N+1)} \prod_{i<j}(v_j - v_i) \ge \prod_{i<j}(v_j - v_i) \ge G(N+1).} \tag{668}$$

Indeed, the sorted exponent list begins with the pairs $kx, kx+1$ and then continues with the unused tail of the longer block. Thus $q_j - j$ is nonnegative and nondecreasing, so $\prod_{i<j}(q_j - q_i) \ge G(N+1)$. Put $a_i = \log v_i$. In the HCIZ integral the diagonal matrices $\operatorname{diag}(a_i)$ and $\operatorname{diag}(q_j - j)$ are positive

semidefinite; hence every integrand for the q-spectrum is at least the corresponding integrand for $0, 1, \ldots, N-1$. Dividing the two HCIZ identities proves the first inequality. The last follows from $v_j - v_i \geq j - i$. This proof uses real exponents, so it applies directly to a hypothetical nonintegral $x > 12$; it also covers $m > N/2$, where the Stieltjes integrals themselves cease to converge.

There is a matching free-energy law for arbitrary node geometries in the convergent range. Let $t_{i,T} = \log v_{i,T}$ with $0 \leq t_{1,T} < \cdots < t_{N,T} \leq CT$, let $N = N_T \to \infty$ and $m = m_T$ satisfy $2m \leq N$ for every T and $m/N \to \varrho \in (0, 1/2]$, and suppose the empirical quantiles $\mathcal{Q}_T(s) = T^{-1} t_{\lceil Ns \rceil, T}$, $0 < s \leq 1$, converge in $L^1(0,1)$ to a nondecreasing $\mathcal{Q} \geq 0$. With $\log^- y = \max\{0, -\log y\}$, assume also $\log(NT) = o(T)$ and the microscopic separation condition

$$\sum_{i<j} \log^- |t_{j,T} - t_{i,T}| = o(TN^2).$$

Writing

$$\mathsf{M}_{\mathcal{Q}} = \int_0^1 \mathcal{Q}(s)\, ds, \qquad \mathsf{A}_{\mathcal{Q}} = \int_0^1 s\mathcal{Q}(s)\, ds,$$

Andréief's identity and the HCIZ variational formula yield

$$\begin{aligned} \frac{\log \Delta_T}{xTN^2} &\longrightarrow \mathsf{A}_{\mathcal{Q}}, & \frac{\log h_{0,T}}{xTN} &\longrightarrow -\mathsf{M}_{\mathcal{Q}}, \\ \frac{\log \det G_d}{xTN^2} &\longrightarrow \frac{1}{2}\int_0^{2\varrho} (2\varrho - s)\mathcal{Q}(s)\, ds, & \frac{\log|\det \mathcal{B}|}{xTN^2} &\longrightarrow \Phi_\varrho(\mathcal{Q}), \end{aligned} \tag{669}$$

where

$$\Phi_\varrho(\mathcal{Q}) = \frac{1}{2}\int_0^{2\varrho} s\mathcal{Q}(s)\, ds + \int_{2\varrho}^1 (s - \varrho)\mathcal{Q}(s)\, ds. \tag{670}$$

The particle optimizer has quantile $Z(r) = \mathcal{Q}(2r)$ for $0 \leq r \leq \varrho$. Moreover

$$\Phi_\varrho(\mathcal{Q}) \geq \left(\frac{1}{2} - \varrho + \varrho^2\right) \mathsf{M}_{\mathcal{Q}} \geq \frac{1}{4}\mathsf{M}_{\mathcal{Q}}; \tag{671}$$

it decreases with ϱ and equals $\mathsf{A}_{\mathcal{Q}}/2$ at $\varrho = 1/2$. Hence every noncollapsed limiting shape has a strictly positive residual exponent. A collapsed scale cannot help either, by the exact superfactorial bound (668).

For calibration, take the complete input triangle

$$\mathcal{S}_T = \{v = 2^a 3^b : \log v \leq T\}, \qquad u = v^x, \qquad \mathcal{L} = \log 2 \log 3.$$

Here $N \sim T^2/(2\mathcal{L})$ and $\mathcal{Q}(s) = \sqrt{s}$, so (669) becomes

$$\begin{aligned} \log \Delta_T &= \frac{x}{10\mathcal{L}^2} T^5 + o(T^5), & \log h_{0,T} &= -\frac{x}{3\mathcal{L}} T^3 + o(T^3), \\ \log \det G_d &= \frac{2\sqrt{2}}{15} \frac{x\varrho^{5/2}}{\mathcal{L}^2} T^5 + o(T^5), \end{aligned} \tag{672}$$

and

$$\log|\det \mathcal{B}| = \frac{xT^5}{\mathcal{L}^2}\left(\frac{1}{10} - \frac{\varrho}{6} + \frac{2\sqrt{2}}{15}\varrho^{5/2}\right) + o(T^5). \tag{673}$$

The coefficient decreases to $1/20$ at $\varrho = 1/2$. At the exact endpoint $N = 2m$, $x = 2$ makes $\mathcal{B}$ the ordinary input Vandermonde, while $x = 3$ gives that Vandermonde times a positive Schur factor.

Thus the triangle calculation is the sharp $x = 2$ calibration of the universal no-go, not a peculiarity of that exhaustion.

The finite cofactor, integral-Hankel, quadratic-lattice transport, node–LCM content divisibility, and common-content amplification are **[kernel-verified Lean]** in `ReciprocalMomentLatticeFinite`; the gcd characterization, alternant identity, Stieltjes representation, positivity, shape-free HCIZ comparison, and asymptotic audit are paper results.

71.9 Cyclotomic defect cascades and familywise private support

The exact logarithmic determinant acquires a positive multiscale decomposition after the power values are shifted by ± 1. Put $X = \log 2$, $Y = \log 3$, and define, for $n \geq 1$,

$$\begin{aligned} \Delta_n^+ &= Y \log(M^n + 1) - X \log(A^n + 1), \\ \Delta_n^- &= Y \log(M^n - 1) - X \log(A^n - 1), \qquad E_n = -\Delta_n^-. \end{aligned} \tag{674}$$

Writing $u = M^{-n}$ and using $A^{-n} = u^\vartheta$ gives

$$\begin{aligned} \frac{\Delta_n^+}{X} &= \vartheta \log(1 + u) - \log(1 + u^\vartheta), \\ \frac{\Delta_n^-}{X} &= \vartheta \log(1 - u) - \log(1 - u^\vartheta). \end{aligned} \tag{675}$$

Strict convexity shows $\Delta_n^- < 0 < \Delta_n^+$. Moreover

$$\frac{d}{du} \Big(-\vartheta \log(1 - u) + \log(1 - u^\vartheta) \Big) = \frac{\vartheta(1 - u^{\vartheta - 1})}{(1 - u)(1 - u^\vartheta)} > 0,$$

so E_n strictly decreases to zero. Taylor expansion and $(T^n - 1)(T^n + 1) = T^{2n} - 1$ give

$$\begin{aligned} E_n &= Y M^{-n} - X A^{-n} + O(M^{-2n}), & \Delta_n^+ &= Y M^{-n} - X A^{-n} + O(M^{-2n}), \\ \Delta_n^+ &= E_n - E_{2n}, & E_n &= \sum_{j \geq 0} \Delta_{2^j n}^+. \end{aligned} \tag{676}$$

The constants depend only on the fixed pair. A bound without a hidden constant is

$$Y M^{-n} \frac{1 - (M/A)^n}{(1 + M^{-n})(1 + A^{-n})} \leq \Delta_n^+ < Y M^{-n}. \tag{677}$$

For $Q_{k,n}(T) = (T^{kn} - 1)/(T^n - 1)$ normalize its k monomials by

$$p_\ell = \frac{M^{\ell n}}{Q_{k,n}(M)} \qquad (0 \leq \ell < k).$$

Additivity of the logarithmic determinant gives the quotient law

$$Y \log Q_{k,n}(M) - X \log Q_{k,n}(A) = E_n - E_{kn} = X(\vartheta - 1) H_\vartheta(p) > 0, \tag{678}$$

where $H_\vartheta(p) = (1 - \vartheta)^{-1} \log \sum p_\ell^\vartheta$. For every prime p, the prime-power factors partition the same mass:

$$\begin{aligned} &\mathcal{C}_{p^j,n} := Y \log \Phi_{p^j}(M^n) - X \log \Phi_{p^j}(A^n) = E_{p^{j-1}n} - E_{p^j n} > 0, \\ &E_n = \sum_{j \geq 1} \mathcal{C}_{p^j,n}, \qquad \prod_{j \geq 1} \frac{\Phi_{p^j}(M^n)^\vartheta}{\Phi_{p^j}(A^n)} = \frac{1 - A^{-n}}{(1 - M^{-n})^\vartheta}. \end{aligned} \tag{679}$$

The Rényi notation records a deterministic strict ℓ^1–ℓ^ϑ inequality, not a probabilistic input.

The dyadic factors have additional arithmetic structure. For $T > 1$ and $j \ge 1$, put $W_j(T) = T^{2^j} + 1$. If $i \ne j$, then

$$\gcd(W_i(T), W_j(T)) = \begin{cases} 1, & T \text{ even}, \\ 2, & T \text{ odd}. \end{cases} \tag{680}$$

Each $W_j(T)$ has an odd prime divisor q_j, and every such divisor satisfies

$$\operatorname{ord}_{q_j}(T) = 2^{j+1}, \qquad 2^{j+1} \mid q_j - 1. \tag{681}$$

Thus q_j is private to its level, $\{W_j(T) : j \ge 1\}$ is multiplicatively independent, and

$$\sum_{j \le J} \log q_j \ge \frac{\log 2}{2} J^2 + O(J). \tag{682}$$

The order, congruence, and level-separation core is **[kernel-verified Lean]** in `DyadicPrivateSupport`; existence, the gcd law, and the finite valuation argument are the elementary paper layer.

Apply this separately to $U_j = M^{2^j} + 1$ and $V_j = A^{2^j} + 1$. With

$$L_j = \begin{pmatrix} X & \log U_j \\ Y & \log V_j \end{pmatrix}, \qquad \Gamma_j = -\det L_j = \Delta^+_{2^j},$$

the second columns acquire one-sided private odd support while

$$\Gamma_j \sim Y M^{-2^j}, \qquad \frac{-\log \Gamma_j}{\log V_j} \longrightarrow \frac{X}{Y}, \qquad \frac{U_j^\vartheta}{V_j} - 1 \sim \vartheta M^{-2^j}. \tag{683}$$

The Euclidean distance from $(\log U_j, \log V_j)$ to $\mathbb{R}(X, Y)$ is exactly $\Gamma_j/\sqrt{X^2 + Y^2}$. Yet the absolute power error grows like $\vartheta M^{(\vartheta-1)2^j}$, and every integral pair $(2^m, 3^m)$ has the same normalized critical slope. Privacy is only within each ladder: no disjointness between the U- and V-supports follows.

This extremality is also a contact ceiling for fixed transfers. Let P be a fixed monic polynomial of degree $d \ge 1$, suppose its coefficients in degrees $d-1, \ldots, d-r+1$ vanish, and let the coefficient in degree $d - r$ be the fixed nonzero c. Then

$$Y \log P(M^n) - X \log P(A^n) = cYM^{-rn} - cXA^{-rn} + O(M^{-(r+1)n}), \tag{684}$$

and

$$\frac{-\log |Y \log P(M^n) - X \log P(A^n)|}{\log P(A^n)} \longrightarrow \frac{r}{d\vartheta} \le \frac{1}{\vartheta}. \tag{685}$$

Equality forces $P(T) = T^d + c$. For a fixed cyclotomic polynomial the slope is $1/(\vartheta\varphi(\operatorname{rad} m))$; among prime-power ladders it is $1/(\vartheta(p-1))$, so only the dyadic ladder is critical. This theorem is deliberately fixed-polynomial and gives no uniform bound for degrees or coefficients varying with n.

Reciprocity gives a complementary sharp ceiling. Let $P, Q \in \mathbb{Z}[T]$ be distinct monic reciprocal polynomials of the same degree D, both with constant term 1, and suppose

$$\frac{P(T)}{Q(T)} = 1 + cT^{-K} + O(T^{-K-1}), \qquad c \ne 0.$$

Then

$$2K \le D, \tag{686}$$

with equality exactly when D is even and $P-Q=cT^{D/2}$. Indeed $P-Q$ is reciprocal, so its smallest and largest nonzero exponents are paired about $D/2$. Under a hypothetical counterexample, the integer

$$P(A^s)Q(M^s)-Q(A^s)P(M^s)$$

therefore has logarithmic size

$$(D(1+\vartheta)-K)s\log M+O(1)\geq D\left(\vartheta+\frac{1}{2}\right)s\log M+O(1). \tag{687}$$

Thus every fixed reciprocal or cyclotomic filter produces a rapidly growing nonzero integer. The finite support and equality classification in (686) are **[kernel-verified Lean]** in `ReciprocalHalfContact`; the orbit asymptotic is a paper argument.

These formulas isolate a sharp open interface. For an arbitrary fixed integer pair define $\lambda_j=-\log_{V_j}|\det L_j|$ only on the eventually nonzero tail. A theorem saying that simultaneous power-primitivity and external support force $\liminf\lambda_j<X/Y$ would prove the conjecture. Integral controls show that no one-sided private-support argument can establish it: cross-ladder synchronization is the missing datum.

71.10 Fresh-prime Newton synchronization

The primitive-cancellation target in (632) has an exact local form. Let $s_0,\dots,s_j$ be distinct positive integer nodes, let $y_i\in\mathbb{Z}$, and let P_{j-1} interpolate the first j pairs. With

$$W_{j-1}(X)=\prod_{i<j}(X-s_i),$$

the Newton append coefficient is

$$c_j=[s_0,\dots,s_j]y=\frac{y_j-P_{j-1}(s_j)}{W_{j-1}(s_j)}. \tag{688}$$

Suppose a prime p divides no old difference s_t-s_r but divides $W_{j-1}(s_j)$. There is a unique $i_p<j$ with $s_j\equiv s_{i_p}\pmod p$. Put $e_p=v_p(W_{j-1}(s_j))$. Lagrange interpolation is integral at p, and hence

$$y_j\not\equiv y_{i_p}\pmod p\quad\Longrightarrow\quad v_p(c_j)=-e_p. \tag{689}$$

Thus the entire fresh p-power survives in the reduced denominator. Conversely, any cancellation of its first power forces $y_j\equiv y_{i_p}\pmod p$; cancellation of higher powers requires genuinely higher congruences.

For the smooth table $s_i=2^{a_i}3^{b_i}$ and $y_i=M^{a_i}A^{b_i}$, and for $p\nmid 6MA$, a canceled fresh pole transfers the one relation

$$2^{\Delta a}3^{\Delta b}\equiv 1,\qquad M^{\Delta a}A^{\Delta b}\equiv 1\pmod p. \tag{690}$$

One relation is not a local common exponent. The exact finite-group criterion is

$$\ker\bigl((u,v)\mapsto 2^u3^v \bmod p\bigr)\subseteq\ker\bigl((u,v)\mapsto M^uA^v \bmod p\bigr)\quad\Longleftrightarrow\quad\begin{cases}M\equiv 2^{k_p}\pmod p,\\ A\equiv 3^{k_p}\pmod p\end{cases} \tag{691}$$

for some k_p modulo the order of $\langle 2,3\rangle\subset\mathbb{F}_p^\times$. Indeed, full relation transfer defines an endomorphism of that cyclic subgroup. A single small top denominator supplies neither the kernel inclusion nor an almost-all-prime family of such exponents.

Two exact calibrations prevent false positives. First, if an exponent pattern Ω has $N+1$ nodes and top coefficient $q_\Omega = a/d$ in lowest terms, translating it by (u,v), with $c = 2^u 3^v$ and $R = M^u A^v$, gives

$$q_{\Omega+(u,v)} = Rc^{-N} q_\Omega, \qquad \frac{d_{\Omega+(u,v)}}{P_{\Omega+(u,v)}} = \frac{d}{cP_\Omega \gcd(aR, dc^N)}. \tag{692}$$

The apparent saving for a drifting block is automatic for every value table. Such blocks are not nested; adjoining them restores the omitted old–new cross-resultants.

Second, fix a smooth ray and put $q = 2^u 3^v$, $r = M^u A^v$. Its order-n coefficient is

$$\frac{\prod_{i=0}^{n-1}(r - q^i)}{q^{n(n-1)/2}\prod_{k=1}^{n}(q^k - 1)}. \tag{693}$$

Let ℓ be an odd prime, let d_ℓ be the reduced denominator in (693), and remove the ℓ-primary part by writing

$$\Phi_\ell(q)^{[\ell]} = \frac{\Phi_\ell(q)}{\ell^{v_\ell(\Phi_\ell(q))}}.$$

Prime-power bookkeeping at the first collision gives the exact divisor

$$\frac{\Phi_\ell(q)^{[\ell]}}{\gcd(\Phi_\ell(q)^{[\ell]}, r^\ell - 1)} \mid d_\ell. \tag{694}$$

For $p \mid \Phi_\ell(q)$, $p \neq \ell$, one has $\mathrm{ord}_p(q) = \ell$; any canceled p-power in the unique numerator factor also divides $r^\ell - 1$. Under a hypothetical nonintegral solution, q and r are multiplicatively independent, because a rational exponent compatible with both original bases is integral. The BCZ gcd estimate therefore implies

$$\log d_\ell \geq \ell \log q - o(\ell). \tag{695}$$

This is an uncanceled fresh denominator *layer*; it need not be supported only on value-mismatch primes, since a synchronized prime can cancel one power while higher powers survive. Fixed rays therefore make the desired denominator upper bound worse, not better. The one-step append/evaluation and barycentric scaling identities are **[kernel-verified Lean]** in `NewtonCollisionCore`. The localization, valuation, synchronization, translation, and ray-divisor statements above are **[paper proof]** paper arguments; BCZ is the cited **[classical/open background]** input.

Fresh layers can in fact be made asymptotically as large as the whole node product, but this strengthens the obstruction rather than overcoming it. If a fixed pair $M, A > 1$ is not $(2^k, 3^k)$, at least one of

$$(B, C) = (2, M), \qquad (3, A), \qquad (6, MA)$$

is multiplicatively independent. Given a finite smooth-node set F containing 1, choose a sufficiently large prime n so that $\Phi_n(B)$ is coprime to the old Vandermonde. At the new node $B^n \mapsto C^n$ every prime factor of $\Phi_n(B)$ gives a unique collision with 1, and the fresh-pole calculation yields

$$R_n := \frac{\Phi_n(B)}{\gcd(\Phi_n(B), C^n - 1)} \mid \mathfrak{d}_{F\cup\{B^n\}}, \qquad \log R_n = n \log B - o(n). \tag{696}$$

The asymptotic is BCZ. Recursively interleaving the next missing smooth node with a prime spike whose height dominates all preceding nodes produces one nested exhaustion and a subsequence along which

$$\frac{\log \mathfrak{d}_F}{\log \prod_{s\in F} s} \geq 1 - o(1). \tag{697}$$

This construction works for every fixed off-curve integer pair and uses no real common exponent. More importantly, it has the wrong polarity for (632): both results are lower bounds for the reduced denominator. A larger denominator merely permits a larger reduced numerator, whereas closure requires an upper bound $\mathfrak{d}_F = o(\prod F)$. Likewise, even after combining a fixed-level Kummer density with the generic $O(\sqrt{p})$ short-relation estimate, the guaranteed contribution in a $K \times K$ table reaches only $p \ll K^2$, hence has quadratic squarefree logarithmic mass, while $\log \prod F_K = \Theta(K^3)$; residue-symbol depth is not a substitute for gap valuation. The missing statement is therefore an output-specific upper bound, or a theorem correlating prime divisors of the sparse S-unit gaps with the detecting Frobenius classes. Chebotarev alone supplies neither.

71.11 Support rigidity and Kummer defect primes

The finite determinant has a primitive integral model. For $u \in \mathbb{Q}_{>0}^{\times}$ put

$$L_u(X) = \sum_p v_p(u) X_p, \qquad Q_{M,A}(X) = L_A(X)X_2 - L_M(X)X_3.$$

In the simultaneous normalization

$$M = 2^a w^d, \qquad A = 3^b v^e,$$

where w is odd, $3 \nmid v$, and w, v have primitive valuation vectors, one has

$$\mathrm{cont}(Q_{M,A}) = \gcd(d, e, |a-b|). \tag{698}$$

Least-output primitivity makes this content 1. Thus $Q_{M,A}$ remains nonzero modulo every prime, while $Q_{M,A} = 0$ over $\mathbb{Z}$ exactly for $(M, A) = (2^n, 3^n)$. The coefficient-content identity, its least-output specialization, and nonzero reduction modulo every nontrivial modulus are **[kernel-verified Lean]** in `TwoStarContent`; the formal zero criterion is **[kernel-verified Lean]** in `FormalPrimeDeterminant`.

The local kernel criterion has a sharp classical global counterpart. A theorem of Corrales–Rodríguez and Schoof [14] states that if $u, v \in \mathbb{Q}^{\times}$ and, for all but finitely many primes and every integer n,

$$u^n \equiv 1 \pmod{p} \implies v^n \equiv 1 \pmod{p},$$

then $v = u^k$ for some $k \in \mathbb{Z}$. Consequently, for positive integers M, A, the following conditions are equivalent:

(i) $M = 2^k$ and $A = 3^k$ for some $k \in \mathbb{N}_0$;

(ii) relation-kernel inclusion holds for almost every good prime;

(iii) a local common exponent k_p exists for almost every good prime;

(iv) for almost every prime,

$$\mathrm{ord}_p(M) \mid \mathrm{ord}_p(2), \qquad \mathrm{ord}_p(A) \mid \mathrm{ord}_p(3), \qquad \mathrm{ord}_p(MA) \mid \mathrm{ord}_p(6). \tag{699}$$

The last implication applies the support theorem to the three pairs and then uses unique factorization. For a fixed candidate even one of the three almost-everywhere divisibilities forces $x \in \mathbb{Z}$. This is an exact local–global reformulation, not a proof that a counterexample supplies the local hypothesis.

The four Kummer coordinates give an exact finite density law. For every sufficiently large odd prime ℓ outside the saturation set, put $K_\ell = \mathbb{Q}(\zeta_\ell)$ and let $V_\ell \subset K_\ell^\times / K_\ell^{\times \ell}$ be the space spanned by $2, 3, M, A$. Its dimension is the ordinary multiplicative rank, and its dual character space $W_\ell = V_\ell^\vee$ embeds in $\mathbb{F}_\ell^4$ by evaluation on those four generators. For an unramified prime ideal $\mathfrak{p}$ write

$$(u_1, u_2, v_1, v_2) = (\kappa_\mathfrak{p}(2), \kappa_\mathfrak{p}(3), \kappa_\mathfrak{p}(M), \kappa_\mathfrak{p}(A)). \tag{700}$$

Exact local compatibility means that $(v_1, v_2) = t(u_1, u_2)$ for some $t \in \mathbb{F}_\ell$; it is stronger than the determinant equation $u_1 v_2 - u_2 v_1 = 0$ when the source pair vanishes. In the full space there are exactly

$$1 + \ell(\ell^2 - 1) = \ell^3 - \ell + 1 \tag{701}$$

compatible vectors. Chebotarev therefore gives the relative prime-ideal densities

$$\begin{aligned} \mathrm{dens}_{\mathrm{rank},4}(\text{compatible}) &= \frac{1}{\ell} - \frac{1}{\ell^3} + \frac{1}{\ell^4}, \\ \mathrm{dens}_{\mathrm{rank},3}(\text{compatible}) &= \frac{1}{\ell} - \frac{1}{\ell^2} + \frac{1}{\ell^3}. \end{aligned} \tag{702}$$

The second formula assumes that the rank-three Kummer hyperplane remains transverse modulo ℓ, which again excludes only finitely many primes. For every fixed t, compatibility has density exactly ℓ^{-2} in either rank. Thus a nonintegral candidate is locally incompatible on density $1 - 1/\ell + O(\ell^{-2})$, even though its real logarithm matrix has rank one. The exact existential-scalar count, including the zero-source case, is **[kernel-verified Lean]** in `KummerCompatibilityFinite`; the Kummer realization and Chebotarev passage are classical paper inputs.

Compatibility at higher prime-power depth has an exact count. Over $\mathbb{Z}/\ell^e\mathbb{Z}$, the number of compatible four-coordinate vectors in full rank is

$$N_4(\ell, e) = 1 + \ell(\ell^2 - 1)\frac{\ell^{3e} - 1}{\ell^3 - 1}, \qquad \mathrm{prop}_4 = \frac{N_4(\ell, e)}{\ell^{4e}}, \tag{703}$$

whereas a transverse rank-three hyperplane contains

$$N_3(\ell, e) = 1 + \ell\frac{\ell^{2e} - 1}{\ell + 1}, \qquad \mathrm{prop}_3 = \frac{N_3(\ell, e)}{\ell^{3e}}. \tag{704}$$

At $e = 1$ these recover (702); as $e \to \infty$ the local proportions decay like ℓ^{-e}. Here transverse means that, for the primitive rank-three relation $c = (c_1, c_2, c_3, c_4)$, the determinant $c_1 c_4 - c_2 c_3$ is an ℓ-adic unit. After Kummer realization these proportions are relative densities inside the cyclotomic split layer; the corresponding absolute rational-prime densities have the additional factor $1/\varphi(\ell^e)$. The full-rank compatibility count and the unit-transverse post-scalar normal-form count are **[kernel-verified Lean]** in `PrimePowerKummerCoherence`. Identifying the latter normal form with compatible vectors in the Kummer hyperplane, Kummer realization at depth ℓ^e, and its Chebotarev interpretation remain paper-level or classical inputs. At depth one the rank-three relation also determines every zero/nonzero status pattern. For its coefficient vector c, put $J_\ell = \{i : c_i \not\equiv 0 \pmod{\ell}\}$. If s coordinates in J_ℓ and h coordinates outside J_ℓ are prescribed nonzero, the count is

$$\frac{(\ell - 1)^s + (\ell - 1)(-1)^s}{\ell}\,(\ell - 1)^h;$$

in particular, a pattern with exactly one active nonzero coordinate is impossible. Full rank allows every pattern. This gives a finite detector for the support of the unique rank-three relation, but not an archimedean-to-local transfer theorem.

The determinant condition is weaker than compatibility at a zero source pair, but its nonvanishing has a complementary density law. Primitivity in (698) and the finite quadratic-form bound give, for every odd ℓ, relative detecting density at least $(1-1/\ell)^2$ among primes splitting in the cyclotomic layer, hence absolute density at least $(\ell-1)/\ell^2$. Characteristic two gives the absolute bound $1/4$. Outside the finite saturation and rank-discriminant set, the exact relative/absolute densities are

$$\begin{array}{c|cc} & \text{relative} & \text{absolute} \\ \hline \text{rank three} & 1-\ell^{-1} & \ell^{-1} \\ \text{rank four} & 1-\ell^{-1}-(\ell-1)\ell^{-3} & \ell^{-1}-\ell^{-3}. \end{array} \tag{705}$$

In particular the cubic universal bounds are $4/9$ relative and $2/9$ absolute. These are determinant-transversality densities, not replacements for the stronger common-scalar counts above. Chebotarev and Kummer theory are the classical inputs; the coefficient primitivity is the arithmetic hinge. At the fixed quadratic and cubic levels the radical extensions have degrees at most 16 and 162, with logarithmic discriminant $O(\log(6MA))$. Effective Chebotarev therefore gives a detecting prime $\ll \log^2(6MA)$ under GRH for the relevant Dedekind zeta function, and a fixed power of $6MA$ unconditionally. These bounds are not uniform in a growing torsion level.

The same coordinates localize common near-power divisors. If at such a prime

$$M^q \equiv 2^p, \qquad A^q \equiv 3^{p'} \pmod{\mathfrak{p}}, \tag{706}$$

then

$$q(u_1v_2-u_2v_1) = (p'-p)u_1u_2 \quad \text{in } \mathbb{F}_\ell. \tag{707}$$

In particular, when $p=p'$ and $\ell \nmid q$, a nonzero determinant excludes simultaneous division of the two gaps. The finite determinant identity and same-residue obstruction are **[kernel-verified Lean]** in `MixedGapKummerFinite`. Thus, for

$$D_2(r,s) = M^s - 2^r, \qquad D_3(r,s) = A^s - 3^r \quad (r \ge 0, s > 0),$$

every eligible good common prime divisor at a level $\ell \nmid s$ lies on the determinant-zero locus at that level. If s is odd this excludes a set of rational primes of density at least $1/4$; if $3 \nmid s$ the cubic exclusion has density at least $2/9$. The same module verifies the abstract retention bound

$$n_2 = 0 \text{ or } n_3 = 0 \quad \Longrightarrow \quad \min(d_2,d_3) \le \max(d_2-n_2, d_3-n_3),$$

where d_2, d_3 are denominator valuations and n_2, n_3 the corresponding numerator valuations. Hence, for some fixed ℓ, common gaps along every denominator q prime to ℓ avoid a positive-density Chebotarev set. This does not upper-bound the gcd: one large integer can be supported entirely on the complementary prime set. It does turn the missing step into a precise Frobenius-distribution problem for the prime divisors of the near-power gaps.

The direct ordinary K_2 encoding reaches a precise sign barrier. In Milnor $K_2(\mathbb{Q})$ [69, 70] put

$$\Xi(M,A) = \{M,3\}\{A,2\}. \tag{708}$$

It is trivial on every standard pair, and its tame symbols are

$$\partial_p \Xi = 2^{-v_p(A)}3^{-v_p(M)} \pmod p \quad (p \ne 2,3), \qquad \partial_3 \Xi = (-1)^{v_3(M)+v_3(A)}\overline{M/3^{v_3(M)}}. \tag{709}$$

The standard-vanishing exterior class has the opposite sign from the symmetric logarithmic determinant; retaining the symmetric sign makes the exterior class nonzero even on the integer ray. Thus the ordinary Steinberg symbol does not encode the required quadratic period. This rules out only the direct construction, not symmetric refinements or higher cohomological operations.

Archimedean closeness does not reduce this local entropy. With $k_j = \lfloor jx \rfloor$, $\theta_j = \{jx\}$, the compact rational orbit

$$U_j = M^j 2^{-k_j} = 2^{\theta_j}, \qquad V_j = A^j 3^{-k_j} = 3^{\theta_j}$$

satisfies, for every interval $I \subset [0,1)$ and every residues $r, s \pmod{\ell}$,

$$\mathrm{dens}\{j : \theta_j \in I,\ j \equiv r,\ k_j \equiv s \pmod{\ell}\} = \frac{|I|}{\ell^2}. \tag{710}$$

This is elementary equidistribution of an irrational rotation after restricting j to one residue class. At $\mathfrak{p}$, the Kummer symbols of (U_j, V_j) are the image of (j, k_j) under

$$\begin{pmatrix} \kappa(M) & -\kappa(2) \\ \kappa(A) & -\kappa(3) \end{pmatrix}. \tag{711}$$

Conditional on any real interval I, they are uniform on this matrix image. A nonzero Kummer determinant therefore gives all ℓ^2 symbol pairs even on an arbitrarily short real arc. The real orbit is dense and nearly one-dimensional, while its generic finite residue shadow remains fully two-dimensional.

Failure occurs on a quantitatively structured set. For multiplicatively independent $B, C > 1$, choose two valuation coordinates with nonzero 2×2 minor and an odd prime ℓ not dividing that minor. Then B, C are independent modulo ℓth powers, also over $\mathbb{Q}(\zeta_\ell)$. In

$$L = \mathbb{Q}(\zeta_\ell, B^{1/\ell}, C^{1/\ell})$$

the set of unramified primes satisfying

$$p \equiv 1 \pmod{\ell}, \qquad B \in (\mathbb{F}_p^\times)^\ell, \qquad C \notin (\mathbb{F}_p^\times)^\ell \tag{712}$$

has absolute natural density $1/\ell^2$ by Chebotarev. Every such prime obeys

$$v_\ell(\mathrm{ord}_p(B)) \le v_\ell(p-1) - 1 < v_\ell(\mathrm{ord}_p(C)),$$

so the order loss is at least one. For this fixed extension,

$$\sum_{\substack{p \le X \\ p \text{ satisfies } (712)}} \log p = \frac{X}{\ell^2} + o(X). \tag{713}$$

The witness ℓ, field, and error term depend on B, C; no uniformity in the unknown candidate is claimed.

These Frobenius defects enter interpolation denominators directly. If an integral table (x_i, y_i) is interpolated by $Q \in \mathbb{Q}[X]$ and $\delta(Q)$ is least positive with $\delta(Q)Q \in \mathbb{Z}[X]$, then

$$v_p(\delta(Q)) \ge \max\{0, v_p(x_i - x_j) - v_p(y_i - y_j)\} \qquad (i \ne j). \tag{714}$$

Indeed $x_i - x_j$ divides $\delta(Q)(y_i - y_j)$. On the ray $B^j \mapsto C^j$, $0 \le j \le N$, this yields

$$\mathrm{lcm}_{1 \le n \le N} \frac{B^n - 1}{\gcd(B^n - 1, C^n - 1)} \mid \delta_N^{\mathrm{ray}}(B, C). \tag{715}$$

For the Kummer witness, the squarefree product

$$R_N(B, C; \ell) = \prod_{\substack{p \le \ell N+1 \\ p \text{ satisfies } (712)}} p \quad \text{satisfies} \quad R_N \mid \delta_N^{\mathrm{ray}}, \qquad \log R_N = \frac{N}{\ell} + o(N). \tag{716}$$

It is supported on one Frobenius class, but remains too small by itself to close the archimedean estimate.

For a good prime define the defect radius

$$\rho_p(M, A) = \min_{\substack{2^r 3^s \equiv 1 \ (p) \\ M^r A^s \not\equiv 1 \ (p)}} \max(|r|, |s|),$$

with value ∞ if no relation defect exists. In the square-grid notation of Section 57.2, $\rho_p(M, A) \le K-1$ forces $p \mid \delta_K$. Reducing relation exponents modulo $p-1$ gives

$$\text{failure of a local common exponent at } p \quad \Longrightarrow \quad p \mid \delta_{p-1}. \tag{717}$$

The index is $p-1$ because the live grid uses exponents $0, \ldots, K-1$. Showing that many Kummer defects have ρ_p much smaller than p is a concrete remaining target.

An additive reformulation has a sharp ceiling. The semigroup homomorphism

$$\Phi(2^a 3^b) = M^a A^b$$

would force an integer exponent if it preserved all congruences away from the primes dividing $6MA$, by the local common-exponent criterion and the support theorem above. The primitive binary identities among 3-smooth integers correspond, up to exchange, to the four adjacent pairs listed in Section 65.1; all other binary identities are 3-smooth scalings. Their defects are $M-2$, $A-M-1$, M^2-A-1, and A^2-M^3-1. More generally,

$$\big(P(M, A) : P \in \mathbb{Z}[X, Y],\ P(2, 3) = 0\big) = \gcd(M-2, A-3)\mathbb{Z}. \tag{718}$$

At time n the corresponding ideal is generated by

$$C_n = \gcd(M^n - 2^n, A^n - 3^n), \qquad \log C_n = o(n) \tag{719}$$

for a nonintegral candidate, by the S-unit gcd theorem. Thus primitive binary relations and same-time polynomial defects cannot supply the exponential congruence support required by the local criterion; this does not exclude more elaborate, time-dependent additive data.

The support classification is the cited **[classical/open background]** input. The finite exact-compatibility counts are **[kernel-verified Lean]** in `KummerCompatibilityFinite`, and their prime-power extensions are **[kernel-verified Lean]** in `PrimePowerKummerCoherence`. The mixed-gap determinant and same-residue obstruction are **[kernel-verified Lean]** in `MixedGapKummerFinite`. Chebotarev densities, mantissa entropy, collision valuation, ray divisor, finite-box bridge, tame-symbol calculation, and additive ceiling are **[paper proof]** paper deductions. They strengthen the denominator lower bound without supplying the missing upper bound.

71.12 Directional gap quotients and exact capture cost

The diagonal ray estimate has a homogeneous fixed-direction form. Fix $(u, v) \in \mathbb{Z}^2 \setminus \{(0,0)\}$ and orient this direction so that

$$r = 2^u 3^v = P/Q > 1, \qquad s = M^u A^v = R/S = r^x > 1$$

are in lowest terms. The homogeneous Corvaja–Zannier gcd theorem gives, when r and s are multiplicatively independent,

$$\log \gcd(P^n - Q^n, R^n - S^n) = o(n). \tag{720}$$

Consequently the reduced directional quotient

$$G_n(r, s) = \frac{R^n - S^n}{P^n - Q^n} \tag{721}$$

has

$$\log \operatorname{den} G_n = n \log P + o(n), \qquad \log |\operatorname{num} G_n| = n \log R + o(n). \tag{722}$$

If $x = k \in \mathbb{N}$, by contrast, $s = r^k$ and G_n is an integer. Dependence of r, s would make x rational and therefore integral by the elementary rational-exponent reduction. Thus every fixed direction has an exact zero-versus-exponential denominator dichotomy. The fixed-group refinement in (649) sums the corresponding gcd bounds over the triangular direction set; it still supplies denominator mass, not the denominator upper bound required for closure.

There is also an exact group-theoretic measure of how far a source gap is from being captured by a target gap. Remove from $U_n = P^n - Q^n$ every prime power supported on RS, calling the remaining modulus $\widetilde{U}_n$, and let

$$t_n = \operatorname{ord}_{\widetilde{U}_n}(RS^{-1})$$

in the unit group. The least positive multiplier h for which $\widetilde{U}_n \mid R^{hn} - S^{hn}$ is

$$\boxed{\kappa(n) = \frac{t_n}{\gcd(t_n, n)}.} \tag{723}$$

Equivalently,

$$\widetilde{U}_n \mid R^{hn} - S^{hn} \iff \kappa(n) \mid h. \tag{724}$$

The finite-group identity behind (724) is **[kernel-verified Lean]** in `CaptureMultiplier`. For a fixed direction, $\kappa(n) = 1$ in the integral case, whereas in the nonintegral case $\kappa(n) \to \infty$: a bounded subsequence would fix some h infinitely often. By LTE, the part removed from $P^n - Q^n$ at the fixed primes dividing RS has logarithm $O(\log n)$, so $\widetilde{U}_n$ retains the full exponential scale. Its divisibility into the fixed target gap would then contradict the homogeneous Corvaja–Zannier theorem applied, after fixing h on that bounded subsequence, to the fixed pair (r, s^h). This qualitative phase transition is external-derived and again nonuniform in the direction.

A recent preprint supplies a complementary one-base signature. For integers $a, b > 1$, the sequence $\gcd(a^n - 1, b^n - 1)$ satisfies an integer-coefficient constant-recursion relation, even eventually, exactly when a, b are multiplicatively dependent [13]. Applied to $(2, M)$, dependence is equivalent to $x \in \mathbb{N}$; the identity

$$\gcd(2^n - 1, M^n - 1) = \gcd(2^n - 1, R_0^n - 1) \quad \text{when } M = 2^a R_0, R_0 \text{ odd}$$

shows that the irrelevant common prime-power content disappears. This is an exact conditional characterization, not a published proof of the conjecture: no recurrence for the gcd sequence follows from simultaneous integrality alone. Together, the gap quotient, capture cost, and recurrence criterion isolate the missing input as a synchronization theorem across directions rather than another fixed-ray estimate.

71.13 Higher Newton–Kummer cocycles and an exponential small algebraic integer

The order-one radical secant from the finite-place analysis extends to a full Newton hierarchy. For $N, m \geq 1$, take the positive roots

$$P = 2^{1/N}, \qquad Q = 3^{1/N}, \qquad U = M^{1/N} = P^x, \qquad V = A^{1/N} = Q^x,$$

and put

$$F_m(U,P) = \prod_{j=0}^{m-1} (U - P^j), \qquad D_m(P) = F_m(P^m, P).$$

The cross-base cocycle

$$\boxed{\Xi_{N,m} = F_m(U,P)D_m(Q) - F_m(V,Q)D_m(P)} \tag{725}$$

is an algebraic integer. If

$$C_{L,m,N}(y) = \prod_{j=0}^{m-1} \frac{e^{yL/N} - e^{jL/N}}{e^{mL/N} - e^{jL/N}},$$

then in the distinguished real embedding

$$\Xi_{N,m} = D_m(P)D_m(Q)\big(C_{\log 2,m,N}(x) - C_{\log 3,m,N}(x)\big). \tag{726}$$

For $m = \lfloor x \rfloor$ and nonintegral $x > m$, each factor $(q^{x-j} - 1)/(q^{m-j} - 1)$ is strictly increasing for $q > 1$; hence $\Xi_{N,m} < 0$ for every N. For an integral exponent $x = m$, the two normalized products are both one and $\Xi_{N,m} = 0$. Thus the adaptive cocycle distinguishes the two cases exactly, although integrality alone does not yet force its vanishing.

For fixed m, direct expansion at $N = \infty$ gives

$$\begin{aligned}\Xi_{N,m} = \frac{m\,m!}{2}\,& x(x-1)\cdots(x-m)(\log 2 \log 3)^m (\log 2 - \log 3) N^{-(2m+1)} \\ &+ O_{x,m}(N^{-(2m+2)}).\end{aligned} \tag{727}$$

Order $m = 1$ recovers the earlier cubic Kummer determinant. More importantly, the original table permits the order to grow. Write $x = m_0 + \delta$ with $0 < \delta < 1$. For a prime $\ell > m_0 + 1$, shift the solution by

$$t_\ell = \ell - 1 - m_0, \qquad (M_{t_\ell}, A_{t_\ell}) = (2^{t_\ell}M, 3^{t_\ell}A), \qquad m = \ell - 1, \qquad N = \ell,$$

and denote the resulting cocycle by $\Xi^{(t_\ell)}_{\ell,\ell-1}$. With

$$J(a) = a + \frac{\mathrm{Li}_2(e^{-a}) - \pi^2/6}{a},$$

Euler–Maclaurin for the endpoint products and the gamma-product asymptotic for the normalized quotient yield

$$\boxed{\lim_{\ell\to\infty}\frac{1}{\ell}\log\left|\Xi^{(t_\ell)}_{\ell,\ell-1}\right| = J(\log 2)+J(\log 3) = -0.9053250970927727\ldots.} \tag{728}$$

This is the first exponentially small algebraic integer supplied by the fixed-table Kummer construction. Genuine integer exponents instead make every adaptive shifted cocycle exactly zero.

The natural algebraic operations nevertheless lose the cancellation. Put

$$\rho = \operatorname{rank}_{\mathbb{Z}}\langle 2,3,M,A\rangle \in \{3,4\}.$$

For primes outside the fixed saturation index, the pure-radical field

$$K_\ell = \mathbb{Q}(2^{1/\ell},3^{1/\ell},M^{1/\ell},A^{1/\ell})$$

has degree ℓ^ρ. Every conjugate of the shifted cocycle is $\exp(O(\ell))$, so the elementary norm bound reaches only

$$\left|\Xi^{(t_\ell)}_{\ell,\ell-1}\right| \ge \exp(-O(\ell^{\rho+1})), \tag{729}$$

far below the linear exponential scale in (728).

Ordinary traces are worse. Let $F=\mathbb{Q}(\zeta_\ell,P,Q)$, $S=m(m-1)/2$, and take $m<\ell$. For every sufficiently large good prime, rank four gives

$$\operatorname{Tr}_{F(U,V)/F}(\Xi_{\ell,m}) = \ell^2(-1)^m\big(P^S D_m(Q)-Q^S D_m(P)\big). \tag{730}$$

In the non-axis rank-three branch write $U=cW^r$, $V=dW^s$ over F; for good $\ell\nmid rs$,

$$\operatorname{Tr}_{F(W)/F}(\Xi_{\ell,m}) = \ell(-1)^m\big(P^S D_m(Q)-Q^S D_m(P)\big). \tag{731}$$

Character orthogonality has erased M,A,x completely. At $m=\ell-1$ the dominant base-only term grows like $\exp(0.281244\ldots\ell)$, so trace destroys precisely the distinguished cancellation. This is not an accident of the invariant character: writing $F_m(T,P)=\sum_{k=0}^m c_k(P)T^k$, every rank-four character projection of Ξ is either the base-only constant term, one term $D_m(Q)c_k(P)U^k$, or one term $-D_m(P)c_k(Q)V^k$. All mixed nonconstant characters vanish. Linear Fourier projection therefore either erases both outputs or retains only one axis; the rank-three exception is exactly a support collision forced by its unique relation.

There is one useful rank calibration. If $\gcd(6,MA)=1$, then the base-prime and output-prime valuation lattices split. Rank three would force $M=W^r$, $A=W^s$, hence $\log_2 3 = s/r\in\mathbb{Q}$, impossible. Every hypothetical cross-coprime counterexample is therefore in the harder rank-four Kummer branch. The natural common ideal can be computed exactly, and it is still too small. Let $\ell>3$ be prime, $1\le m<\ell$, put $P=2^{1/\ell}$, $Q=3^{1/\ell}$, $L=\mathbb{Q}(P,Q)$, and write

$$S=\frac{m(m-1)}{2},\qquad H_a(m)=\prod_{r=1}^m(a^r-1),\qquad D_m(P)=P^S\prod_{r=1}^m(P^r-1)$$

with the analogous formula for $D_m(Q)$. For the ideal

$$\mathfrak{I}_{\ell,m} = (D_m(P))+(D_m(Q))\subset\mathcal{O}_L, \tag{732}$$

set $a_p = v_p(H_2(m))$ and $b_p = v_p(H_3(m))$. Then

$$\boxed{N_{L/\mathbb{Q}}\mathfrak{I}_{\ell,m} = 6^S \ell^{e_{\ell,m}} \prod_{p\nmid 6\ell} p^{\min(a_p,b_p)}, \qquad 0 \le e_{\ell,m} \le \ell \min(a_\ell, b_\ell).} \tag{733}$$

At a good prime, a prime ideal dividing $D_m(P)$ makes $P^r = 1$ for some $r < \ell$. Bézout between r and ℓ then puts P in the prime field and determines it uniquely from 2; the same holds for Q. Hence exactly one common degree-one prime occurs, with valuation $\min(a_p, b_p)$. The primes 2 and 3 contribute 2^S and 3^S; the displayed bound at ℓ follows from the two principal norms and LTE. In particular

$$\log N_{L/\mathbb{Q}}\mathfrak{I}_{\ell,m} = O(m^2 + m\ell) = O(\ell^2). \tag{734}$$

For fixed m and all sufficiently large ℓ, the exact norm is simply $6^S \gcd(H_2(m), H_3(m))$.

This settles the partial-norm scale. The cross-resultants give

$$|N_{L/\mathbb{Q}}D_m(P)| = (2^S H_2(m))^\ell, \qquad |N_{L/\mathbb{Q}}D_m(Q)| = (3^S H_3(m))^\ell.$$

In Kummer rank ρ, extension to K_ℓ raises (733) to the power $\ell^{\rho-2}$. At $m = \ell - 1$ its logarithm is therefore only $O(\ell^\rho)$, one full factor of ℓ below the $O(\ell^{\rho+1})$ conjugate cost in (729). In the cyclotomic Kummer closure, taking a norm over any subgroup of the output-twist group multiplies the ideal power and divides the remaining field degree in inverse proportions, so the absolute deficit is unchanged. Projective normalization does not repair it either:

$$(D_m(P)) \cap (D_m(Q)) = (D_m(P)D_m(Q))\mathfrak{I}_{\ell,m}^{-1}, \qquad \log N_{L/\mathbb{Q}}\big((D_m(P)) \cap (D_m(Q))\big) = \ell^3 \log 6 + O(\ell^2) \tag{735}$$

when $m = \ell - 1$. The normalized cross-base difference in (726) is only polynomial at this shifted order; the exponential smallness comes from the distinguished values of $D_m(P)D_m(Q)$, exactly where the projective denominator is largest. Thus ordinary traces, character projections, ideal gcds, and subgroup output partial norms do not improve these bounds at their exact scales. A signed group-ring operation faces a sharper integrality obstruction. Continue in the good rank-four branch over $F = \mathbb{Q}(\zeta_\ell, P, Q)$ and write

$$\mathcal{A}_a = D_m(Q)F_m(\zeta_\ell^a U, P), \qquad \mathcal{B}_b = D_m(P)F_m(\zeta_\ell^b V, Q), \qquad \mathcal{X}_{a,b} = \mathcal{A}_a - \mathcal{B}_b.$$

The output-twist group $G \simeq \mathbf{F}_\ell^2$ translates the two indices. Its smallest mixed additive projection vanishes identically,

$$\mathcal{X}_{0,0} - \mathcal{X}_{1,0} - \mathcal{X}_{0,1} + \mathcal{X}_{1,1} = 0,$$

whereas the corresponding multiplicative operation, wherever defined, is the genuine cross-ratio

$$\mathcal{C} = \frac{\mathcal{X}_{0,0}\mathcal{X}_{1,1}}{\mathcal{X}_{1,0}\mathcal{X}_{0,1}}, \qquad N_{F(U,V)/F}(\mathcal{C}) = 1, \qquad \mathcal{C} - 1 = \frac{(\mathcal{A}_0 - \mathcal{A}_1)(\mathcal{B}_0 - \mathcal{B}_1)}{\mathcal{X}_{1,0}\mathcal{X}_{0,1}}. \tag{736}$$

Thus an integral $\mathcal{C}$ would necessarily be a unit; otherwise it pays exactly the displayed denominator. Clearing that denominator leaves a product of one-axis differences and loses the distinguished cross-base subtraction. After specializing the formal cross-ratio to an integral rank-collapsed calibration with $\mathcal{X}_{0,0} = 0$ and nonzero denominator, $\mathcal{C} - 1 = -1$, not a small residual.

More generally, for $\Theta = \sum_{g\in G} n_g g \in \mathbb{Z}[G]$ and $\mathcal{X} = \mathcal{X}_{0,0}$,

$$v_\mathfrak{p}(\mathcal{X}^\Theta) = \sum_g n_g v_{g^{-1}\mathfrak{p}}(\mathcal{X}), \qquad N_{F(U,V)/F}(\mathcal{X}^\Theta) = N_{F(U,V)/F}(\mathcal{X})^{\operatorname{aug}\Theta}. \tag{737}$$

These are the exact divisor-convolution and augmentation tests. Zero augmentation plus integrality forces a unit. If a prime divisor of $\mathcal{X}$ has a free private G-orbit, the first formula forces every $n_g \geq 0$; zero augmentation then forces $\Theta = 0$, while positive augmentation retains ordinary norm mass. Establishing such a private prime for the primitive part of $\mathcal{X}$ would require a new cross-conjugate primitive-divisor theorem.

The missing statement has an exact resultant form. Let $\mathfrak{J} = \mathfrak{J}_{\ell,m}\mathcal{O}_F$, $\mathfrak{a} = (\mathcal{X}_{0,0})(\mathfrak{J}\mathcal{O}_{F(U,V)})^{-1}$, and

$$\Phi_A(T) = \prod_{a\in\mathbf{F}_\ell}(T - \mathcal{A}_a), \qquad \Phi_B(T) = \prod_{b\in\mathbf{F}_\ell}(T - \mathcal{B}_b).$$

Then

$$R_{\ell,m} := \operatorname{Res}(\Phi_A, \Phi_B) = N_{F(U,V)/F}(\mathcal{X}_{0,0}), \qquad N_{F(U,V)/F}\mathfrak{a} = (R_{\ell,m})\mathfrak{J}^{-\ell^2}. \tag{738}$$

Choose a nonzero submaximal minor $S_{\ell,m}$ of the Sylvester matrix. If a prime dividing the primitive ideal on the right avoids ramification, $\mathfrak{J}$, the two discriminants, and $S_{\ell,m}$, then the reduced orbit polynomials have a separable gcd of degree one. The unique common pair is Frobenius-fixed, so the prime splits in both output directions and gives a free private divisor of one conjugate difference. Thus absence of a private prime confines the radical of the primitive resultant to this explicit exceptional product. This criterion does not prove existence: the available bounds for the logarithms of the absolute norms of both the resultant and the discriminant–minor product are $O_x(\ell^5)$ for fixed m and $O_x(\ell^6)$ for $m = \ell - 1$. A squarefree divisor outside this potentially same-scale exceptional factor is precisely the missing arithmetic input.

At $m = 1$ that missing input has a particularly explicit form. Put $d_2 = P - 1$, $d_3 = Q - 1$, $c = P - Q$, $\mu = d_3^\ell M$, and $\nu = d_2^\ell A$. Shifting the exponent from x to $x + t$ replaces the resultant by

$$R_t = \mathcal{C}_\ell(2^t\mu, 3^t\nu; c), \qquad \mathcal{C}_\ell(X, Y; c) = \operatorname{Res}_z(z^\ell - X, (z + c)^\ell - Y).$$

The curve $\mathcal{C}_\ell = 0$ is the irreducible rational image $z \mapsto (z^\ell, (z + c)^\ell)$. Away from a fixed finite set $\mathcal{S}_\ell$ of primes, its only torus singularities are the finitely many ordinary double points

$$(X, Y) = c^\ell\left(\left(\frac{\zeta_\ell^b - 1}{\zeta_\ell^a - \zeta_\ell^b}\right)^\ell, \left(\frac{\zeta_\ell^a - 1}{\zeta_\ell^a - \zeta_\ell^b}\right)^\ell\right), \qquad a, b \in \mathbf{F}_\ell^\times,\ a \neq b.$$

Corvaja–Zannier [7] bounds the radical supported at this finite node locus by $\exp(o(t))$. The complementary radical has a standard conditional lower bound, rather than requiring a separate truncated-radical hypothesis. Take $K = F = \mathbb{Q}(\zeta_\ell, P, Q)$, which contains the coefficients and all nodes, put $D = [K : \mathbb{Q}]$, and enlarge $\mathcal{S}_\ell$ to contain every prime above $6\ell c\mu\nu$ together with the fixed coefficient and bad-reduction primes. Let h_K denote the absolute logarithmic projective height normalized by D^{-1}, and write

$$\operatorname{rad}_K(\alpha) = \frac{1}{D}\sum_{\mathfrak{q}\mid(\alpha)}\log N\mathfrak{q} \qquad (0 \neq \alpha \in \mathcal{O}_K),$$

while $\operatorname{rad}_K(a : b : r)$ is D^{-1} times the sum of $\log N\mathfrak{q}$ over primes at which $v_\mathfrak{q}(a), v_\mathfrak{q}(b), v_\mathfrak{q}(r)$ are not all equal. For odd ℓ set $W = X - Y + c^\ell$. There is a polynomial $G_\ell \in \mathbb{Z}[X, Y, Z]$, homogeneous of degree $\ell - 3$, such that

$$\boxed{\mathcal{C}_\ell(X, Y; c) = W^\ell + c^\ell XYG_\ell(X, Y, c^\ell).} \tag{739}$$

Indeed, $\mathcal{C}_\ell$ and W^ℓ agree when $X = 0$, $Y = 0$, or $c = 0$. Moreover $\mathcal{C}_\ell$ is invariant under $c \mapsto \zeta_\ell c$ and is weighted homogeneous of weight ℓ^2 when X, Y, c have weights $\ell, \ell, 1$. Their difference is

therefore divisible in $\mathbb{Z}[X,Y,c]$ by XYc^ℓ, and the quotient has the asserted integral homogeneity. Irreducibility of $\mathcal{C}_\ell$ gives
$$\gcd\big(X-Y+Z, G_\ell(X,Y,Z)\big)=1.$$
Homogeneity and $c\neq 0$ preserve this coprimality after the specialization $Z=c^\ell$; this is the coprime pair used below.

Put $W_t = 2^t\mu - 3^t\nu + c^\ell$ and $G_t = G_\ell(2^t\mu, 3^t\nu, c^\ell)$. Outside $\mathcal{S}_\ell$, the common ideal of W_t^ℓ and $c^\ell 2^t 3^t \mu\nu G_t$ has local order

$$\min\{\ell v_{\mathfrak{q}}(W_t), v_{\mathfrak{q}}(G_t)\} \le \ell \min\{v_{\mathfrak{q}}(W_t), v_{\mathfrak{q}}(G_t)\},$$

so Corvaja–Zannier gives logarithmic norm $o(t)$. The same estimate at the finitely many primes in $\mathcal{S}_\ell$ follows by applying the fixed-S theorem of Bugeaud–Evertse [8] to the simple nondegenerate integer recurrence $N_{K/\mathbb{Q}}(W_t)$. Its support contains the root 1, so no rational prime divides all of its recurrence coefficients, and its fixed-S part is $\exp(o(t))$. With projective radicals normalized by D^{-1} as above, this gives

$$\begin{aligned} h_K(W_t^\ell : c^\ell 2^t 3^t \mu\nu G_t : R_t) &= \ell t\log 3 + o(t),\\ \mathrm{rad}_K(W_t) &\le t\log 3 + O(1),\\ \mathrm{rad}_K(G_t) &\le (\ell-3)t\log 3 + O(1), \end{aligned} \tag{740}$$

while $2^t3^t\mu\nu c$ has fixed radical. The number-field *abc* conjecture over this fixed K, in the form

$$h_K(a:b:a+b) \le (1+\varepsilon)\,\mathrm{rad}_K(a:b:a+b) + O_{K,\varepsilon}(1),$$

therefore gives
$$\mathrm{rad}_K(R_t) \ge \left(\frac{\ell}{1+\varepsilon} - (\ell-2)\right) t\log 3 - o(t). \tag{741}$$
Consequently, conditionally on number-field *abc*, for every $\delta>0$,

$$\mathrm{rad}_K(R_t) \ge (2\log 3 - \delta)t$$

for all sufficiently large t. The node-supported radical is $o(t)$, so a divisor remains at a smooth point. Its unique normalization preimage is Frobenius-fixed and, in the good full rank-four branch, gives a free private prime for one conjugate difference. Thus no nontrivial integral augmentation-zero operation exists for any sufficiently large shifted $m=1$ instance at that fixed ℓ.

For $\ell=3$ the splitting is the concrete identity

$$\mathcal{C}_3(X,Y;c) = (X-Y+c^3)^3 + 27c^3XY.$$

Away from the structural primes, its unique torus singularity is the ordinary node $(c^3,-c^3)$. This checks the sign, the coefficient $G_3=27$, and the absence of moving common content in the second summand. Unconditionally, $N_{K/\mathbb{Q}}(R_t)$ is a simple nondegenerate integer recurrence with a unique dominant root, but Stewart's general squarefree-factor estimate gives only

$$\log \mathrm{rad}_{\mathbb{Q}} |N_{K/\mathbb{Q}}(R_t)| \gg_{K,\ell,\mu,\nu,c} \frac{\log t\,\log\log t}{\log\log\log t} \qquad [9],$$

which is $o(t)$ and does not dominate the $o(t)$ node bound. Classical Zsigmondy theorems do not apply to this nonbinary, nondivisibility recurrence.

The conclusion uses the full output-twist group. Shifting multiplies U, V by $P^t, Q^t \in F$, so it leaves the Kummer extension and its saturation unchanged. For an integral control $x = k$, however, $U = P^k$ and $V = Q^k$ already lie in F; the output extension collapses, the formal resultant is not an ℓ^2-degree Galois norm, and a smooth divisor supplies no free output orbit. The same caveat applies at a bad rank-collapsed prime ℓ. The fixed-ℓ conditional result is not uniform in ℓ and reaches neither the original unshifted instance nor the adaptive $m = \ell - 1$ cocycle.

These two failures are structural. First, the coefficient supplied by (741) is

$$c_\ell(\varepsilon) = \frac{\ell}{1+\varepsilon} - (\ell - 2) = 2 - \frac{\ell\varepsilon}{1+\varepsilon}.$$

It is positive only for $\varepsilon < 2/(\ell-2)$; obtaining $c_\ell(\varepsilon) \geq 2-\delta$ for $0 < \delta < 2$ requires $\varepsilon \leq \delta/(\ell-\delta)$. Thus even a field-uniform *abc* estimate with one fixed ε cannot be diagonalized. At the adaptive shift $t_\ell = \ell - 1 - \lfloor x \rfloor$, one would need $\varepsilon_\ell = O(\ell^{-1})$ and every fixed-field *abc*, gcd, and fixed-prime-part error to be $o(\ell)$. The Bugeaud–Evertse threshold used above is not uniform in ℓ, while

$$h(\mu), h(\nu) = \Theta_x(\ell), \qquad \log\|G_\ell\|_\infty = O(\ell^2),$$

so the fixed-ℓ asymptotic has no such diagonal error term.

Second, order one does not divide the higher Newton hierarchy. On its normalized common-zero curve write

$$U = 1 + \sigma(P-1), \qquad V = 1 + \sigma(Q-1), \qquad \mathcal{T}_{P,m}(\sigma) = \frac{F_m(1+\sigma(P-1), P)}{D_m(P)}.$$

For every $m \geq 2$, the difference $\mathcal{T}_{P,m}(\sigma) - \mathcal{T}_{Q,m}(\sigma)$ is a nonzero polynomial. Already

$$\mathcal{T}_{P,2}(\sigma) - \mathcal{T}_{Q,2}(\sigma) = \sigma(\sigma-1)\left(\frac{1}{P(P+1)} - \frac{1}{Q(Q+1)}\right), \tag{742}$$

and in general its leading coefficient is

$$\frac{P^{-m(m-1)/2}}{\prod_{r=1}^{m}(1+P+\cdots+P^{r-1})} - \frac{Q^{-m(m-1)/2}}{\prod_{r=1}^{m}(1+Q+\cdots+Q^{r-1})} \neq 0$$

for $1 < P < Q$. A private divisor of an order-one difference therefore has no systematic reason to divide the order-$(\ell-1)$ cocycle. There is no polynomial-factor descent in the shift parameter either: for distinct r, t the irreducible polynomials

$$\mathcal{C}_\ell(2^r X, 3^r Y; c) \quad \text{and} \quad \mathcal{C}_\ell(2^t X, 3^t Y; c)$$

are coprime. Their common constant term fixes any associate scalar, and their X^ℓ coefficients then force $r = t$.

Finally, the free-orbit conclusion exists only in rank four; rank three remains an unresolved branch under the hypothetical counterexample. At adaptive order in rank four, the absolute resultant and its discriminant–minor exceptional factor both have logarithmic size $O_x(\ell^6)$, whereas the known common ideal contributes only $O_x(\ell^5)$. The diagonalized order-one *abc* lower bound has unnormalized scale $\Theta_x(\ell^4)$ and, by (742), concerns a different resultant. The exact missing input is therefore a moving-family truncated primitive-divisor theorem for the order-$(\ell-1)$ Kummer resultants, together with a separate rank-three analysis; standard fixed-field *abc* does not provide either one.

The extracted common ideal $\mathfrak{I}_{\ell,m}$ cancels from every augmentation-zero ratio. Multiplication by cyclotomic units does not change valuations, and a Stickelberger class-group annihilator can principalize the resulting fractional ideal without removing its negative valuations. Consequently a successful algebraic operation must create exceptional cross-conjugate ideal overlap; changing the group-ring element alone cannot preserve the signed cancellation as an algebraic integer.

The product/cocycle identities and integral-exponent calibrations are **[kernel-verified Lean]** in `RankOneNewtonKummer`. The finite signed-orbit augmentation identity, mixed additive cancellation, and two-by-two minor factorization are **[kernel-verified Lean]** in `SignedOrbitNormFinite`; the divisor, integrality, private-prime, asymptotic, Kummer-degree, norm, ideal, and trace analyses are paper mathematics. The resultant splitting (739) and its local consequences are unconditional paper results; the linear radical and resulting private prime use the number-field *abc* conjecture explicitly.

71.14 Adjacent-smooth cyclic resultants and finite-logarithm descent

The four adjacent smooth pairs identified in Section 65.1 support a Kummer construction whose distinguished embedding has unit rather than exponential scale. Let $N/q < x$ be a reduced lower convergent, put $\varepsilon = qx - N > 0$ and $s = qx/N = 1 + \varepsilon/N$, and for $a \in \{1, 2, 3, 8\}$ set

$$P_a = a^{qx}, \qquad Q_a = (a+1)^{qx}, \qquad u_a = P_a^{1/N} = a^s, \qquad v_a = Q_a^{1/N} = (a+1)^s.$$

All P_a, Q_a are integers under a hypothetical solution.

For arbitrary positive integers $P < Q$, let $\alpha = (Q/P)^{1/N}$, $\zeta_N = e^{2\pi i/N}$, and

$$Z_j = P(\zeta_N^j \alpha - 1)^N \qquad (0 \le j < N).$$

The distinguished value Z_0 lies in the pure-radical field $\mathbb{Q}(\alpha)$ of degree at most N. The other Z_j generally do not; they lie in $\mathbb{Q}(\zeta_N, \alpha)$, whose Kummer splitting extension over $\mathbb{Q}(\zeta_N)$ is cyclic of degree dividing N. Nevertheless the complete multiset is Galois-stable, and

$$\Phi_{N;P,Q}(T) := \prod_{j=0}^{N-1} (T - Z_j) = P^{-N} \operatorname{Res}_X(PX^N - Q, T - P(X-1)^N) \in \mathbb{Z}[T]. \tag{743}$$

This is a degree-N characteristic/resultant polynomial, not necessarily the minimal polynomial of an individual Z_j. Its power sums are

$$\sum_{j=0}^{N-1} Z_j^m = N \sum_{r=0}^{m} \binom{Nm}{Nr} (-1)^{N(m-r)} P^{m-r} Q^r. \tag{744}$$

Define the integer

$$\mathcal{R}_N(P, Q) = \prod_{j=0}^{N-1} (Z_j - 1) = (-1)^N \Phi_{N;P,Q}(1). \tag{745}$$

For the adjacent data this integer is positive: the distinguished factor is positive, nonreal factors pair with their conjugates, and $|\zeta_N^j v_a - u_a| \ge v_a - u_a > 1$ excludes a zero factor.

Let

$$c_a = (a+1)\log(a+1) - a\log a.$$

The distinguished defect has the scale-free expansion

$$(v_a - u_a)^N - 1 = c_a\varepsilon + \frac{c_a^2}{2}\varepsilon^2 + O_a(\varepsilon^2/N + |\varepsilon|^3). \tag{746}$$

For the remaining embeddings, exact factorization and a Gaussian boundary layer give

$$\begin{aligned}
\mathcal{R}_N(P_a, Q_a) &= (Q_a - P_a)^N \prod_{j=0}^{N-1} |1 - (\zeta_N^j v_a - u_a)^{-N}|,\\
\log \mathcal{R}_N(P_a, Q_a) &= N\log(Q_a - P_a) + \log(c_a\varepsilon) - \gamma_a\sqrt{N} + o(\sqrt{N}),\\
\gamma_a &= \frac{\zeta(3/2)}{\sqrt{2\pi a(a+1)}}.
\end{aligned} \tag{747}$$

The Riemann-sum proof is uniform on $\delta\sqrt{N} \le j \le L\sqrt{N}$; the singular initial window is dominated first, then $\delta \downarrow 0$, and the Gaussian tail is removed after $L \uparrow \infty$. Uniform convergence all the way down to $j = 1$ is neither needed nor true in general.

Write $\mathcal{R}_a = \mathcal{R}_N(P_a, Q_a)$. Cancellation of both N^2 height components and of $\log\varepsilon$ has, up to sign, the unique primitive weight vector

$$(w_1, w_2, w_3, w_8) = (2, -2, -1, 1).$$

Thus

$$\Theta_{N,q} = \frac{\mathcal{R}_1^2\mathcal{R}_8}{\mathcal{R}_3\mathcal{R}_2^2} = C_0 \exp(-\kappa\sqrt{N} + o(\sqrt{N})), \tag{748}$$

where

$$\begin{aligned}
C_0 &= \frac{c_1^2 c_8}{c_3 c_2^2} = 0.7356241567479139\ldots,\\
\kappa &= 2\gamma_1 + \gamma_8 - \gamma_3 - 2\gamma_2 = 0.44490230205040846\ldots > 0.
\end{aligned} \tag{749}$$

If $\Theta_{N,q} = U_{N,q}/V_{N,q}$ is reduced, positivity of the numerator forces

$$\liminf \frac{\log V_{N,q}}{\sqrt{N}} \ge \kappa. \tag{750}$$

Consequently the conjecture would follow from any fixed $\eta > 0$ and infinitely many lower convergents for which

$$\frac{\mathcal{R}_3\mathcal{R}_2^2}{\gcd(\mathcal{R}_1^2\mathcal{R}_8, \mathcal{R}_3\mathcal{R}_2^2)} \le e^{(\kappa-\eta)\sqrt{N}}.$$

This makes the required cross-specialization gcd cancellation exact; it does not prove it.

The same four pairs have an exact tangent lattice. For $U = 2^s$, $V = 3^s$, put

$$e_1 = U - 2, \quad e_2 = V - U - 1, \quad e_3 = U^2 - V - 1, \quad e_8 = V^2 - U^3 - 1.$$

Then

$$\begin{aligned}
-3e_1 + e_2 + e_3 &= e_1^2,\\
6e_1 - 6e_2 + e_8 &= -e_1^3 - 5e_1^2 + 2e_1e_2 + e_2^2,
\end{aligned} \tag{751}$$

and all integer relations among (c_1, c_2, c_3, c_8) form

$$\mathbb{Z}(-3,1,1,0) \oplus \mathbb{Z}(6,-6,0,1). \tag{752}$$

Every nonzero vector has support at least three, and those directions span the rank-two Kummer lattice. This exact support statement alone does not prove that cancellation cannot occur in a degree-N subfield; that stronger claim would require a separate Kummer-character argument.

At a prime degree ℓ, Frobenius supplies a conditional finite-place shadow. In $\mathbb{F}_\ell[P, Q, T]$,

$$\Phi_{\ell;P,Q}(T) \equiv (T - Q + P)^\ell. \tag{753}$$

If $\ell \nmid P$ and $Q - P - 1 = \ell C$, put

$$H_\ell(X) = \frac{(X-1)^\ell - X^\ell + 1}{\ell} \in \mathbb{Z}[X].$$

Then $\ell^\ell \mid \mathcal{R}_\ell(P, Q)$ and, for $r = Q/P$ modulo ℓ,

$$\frac{\mathcal{R}_\ell(P,Q)}{\ell^\ell} \equiv PH_\ell(r) + C, \qquad H_\ell(X) \equiv \sum_{k=1}^{\ell-1} \frac{X^k}{k} \pmod{\ell}. \tag{754}$$

Now impose the additional simultaneous resonance $M^q \equiv 2$ and $A^q \equiv 3 \pmod{\ell}$, and write $M^q = 2 + \ell\xi$, $A^q = 3 + \ell\upsilon$ modulo ℓ^2. The four normalized first quotients $W_a = \mathcal{R}_a/\ell^\ell \bmod \ell$ are

$$\begin{aligned}
W_1 &= -2q_\ell(2) + \xi,\\
W_2 &= 2q_\ell(2) - 3q_\ell(3) + \upsilon - \xi,\\
W_3 &= -8q_\ell(2) + 3q_\ell(3) + 4\xi - \upsilon,\\
W_8 &= 24q_\ell(2) - 18q_\ell(3) + 6\upsilon - 12\xi.
\end{aligned} \tag{755}$$

They obey the same two tangent relations as (752). If all four are nonzero and $t = W_2/W_1$, the balanced residual collapses to

$$\Theta_{\ell,q} \equiv \frac{6(t-1)}{t^2(3-t)} \pmod{\ell}. \tag{756}$$

The resonance is not known to occur infinitely often. The finite tangent identities, conditional balance and tangent-lattice parametrizations, two residue relations, and a conditional residual cross identity are **[kernel-verified Lean]** in `CatalanTangentCore`. The universal polynomial, boundary law, balanced quotient, and conditional Frobenius formulas are **[paper proof]** paper results. They replace a generic Kummer-height loss by a sharply specified cross-resultant gcd and finite-logarithm problem, but do not yet close either one.

71.15 Common-zero Hermite jets stop at order two

The common-zero route admits a striking but non-closing normalization. For $a > 1$ put

$$F_a(z) = \frac{a^{-z/2}\sin(\pi a^z)}{\pi \log a}. \tag{757}$$

At a positive zero λ with $m = a^\lambda \in \mathbb{N}$ one has

$$F_a'(\lambda) = (-1)^m \sqrt{m} \in \overline{\mathbb{Q}}, \qquad F_a''(\lambda) = 0. \tag{758}$$

Hence at a common zero, with $u = 2^\lambda$ and $v = 3^\lambda$, the removable quotient $H = F_2/F_3$ satisfies

$$H(\lambda) = (-1)^{u-v}\sqrt{u/v} \in \overline{\mathbb{Q}}, \qquad H'(\lambda) = 0.$$

This algebraic first jet adds no new logarithmic direction because $H(\lambda)^2 = u/v$.

The next two normalized jets make the ceiling exact. With $L = \log a$,

$$r_3 := \frac{F_a'''(\lambda)}{F_a'(\lambda)} = L^2\left(\frac{1}{4} - \pi^2 m^2\right), \qquad r_4 := \frac{F_a''''(\lambda)}{F_a'(\lambda)} = -4\pi^2 L^3 m^2, \tag{759}$$

and therefore

$$L^3 - 4r_3 L + r_4 = 0. \tag{760}$$

The two higher jets cannot both be algebraic: the cubic would first make L algebraic, and then the formula for the nonzero r_4 would make π^2 algebraic, contrary to Lindemann–Weierstrass [3].

The extra multiplicity in the unit-shift defect is paid exactly by an explicit infinite-order multiplier. Writing $c_2(z) = \cos(\pi 2^z)$, $c_3(z) = \cos(\pi 3^z)$ and

$$B_1(z) = 3c_2(z) - 4c_3(z)^2 + 1,$$

one has at every common zero $\lambda = n + k\beta$ with $n, k \geq 1$

$$F_2(z+1)F_3(z) - \sqrt{2/3}\,F_2(z)F_3(z+1) = \frac{\sqrt{2}}{3}F_2(z)F_3(z)B_1(z). \tag{761}$$

Here B_1 has an exact double zero, so the cleared defect has order four, but B_1 is of infinite order. The quotient F_2/F_3 also has at least $3^R - 2^R + O(1)$ nonshared real poles up to real part R, and is not of bounded characteristic in the right half-plane. Thus the algebraic first jet does not feed Schneider–Lang, a finite-order argument, or a bounded-characteristic uniqueness theorem; third derivatives restore the original logarithmic obstruction.

71.16 Lambert universality and the reciprocal-Hankel universal-clearing deficit

Over a field, for parameters (s, r) satisfying $s^n \neq 1$ for every $n \geq 1$, define

$$h_n(s,r) = \frac{r^n - 1}{s^n - 1}, \qquad \mathcal{H}_{s,r}(X) = \sum_{n\geq 1} h_n(s,r)X^n,$$

and let $\Delta_c G(X) = G(cX) - G(X)$. Coefficientwise cancellation gives

$$\Delta_s \mathcal{H}_{s,r} = \frac{rX}{1-rX} - \frac{X}{1-X} =: R_r(X). \tag{762}$$

More importantly,

$$\boxed{\Delta_N \Delta_2 \mathcal{H}_{2,M} = \Delta_M \Delta_3 \mathcal{H}_{3,N}} \tag{763}$$

holds for arbitrary rational M, N; both sides have nth coefficient $(M^n - 1)(N^n - 1)$. Thus this visually cross-base identity contains no common-exponent input. The exact formal-series algebra, including the rational presentation and the fully generic four-parameter identity, is **[kernel-verified Lean]** in `LambertCocycleFinite`.

The leading Hankel determinants of the same sequence are also exact. Put

$$\mathcal{D}_n(s,r) = \det\left(\frac{r^{i+j+1}-1}{s^{i+j+1}-1}\right)_{0\le i,j<n}.$$

The Cauchy double alternant and the rank drops at $r = 1, s^{\pm 1}, \dots, s^{\pm(n-1)}$ give

$$\mathcal{D}_n(s,r) = (-1)^{\binom{n}{2}} s^{\binom{n}{3}} (r-1)^n \times \frac{\prod_{k=1}^{n-1}(s^k - r)^{n-k}(s^k r - 1)^{n-k}(s^k-1)^{2(n-k)}}{\prod_{i,j=0}^{n-1}(s^{i+j+1}-1)}. \tag{764}$$

For integer $s, r \ge 2$, the universal cyclotomic integerizer

$$Q_n(s) = \frac{\prod_{i,j<n}(s^{i+j+1}-1)}{\prod_{k=1}^{n-1}(s^k-1)^{2(n-k)}}$$

is an integer. Off the locus $r = s^k$ one has

$$\log Q_n(s) = \frac{2}{3}n^3 \log s + O_s(n), \tag{765}$$

$$\log|\mathcal{D}_n(s,r)| = -\frac{1}{6}n^3 \log s + \frac{1}{2}n^2 \log(r/s) + O_{s,r}(n). \tag{766}$$

Therefore the universally cleared nonzero integer $Q_n(s)\mathcal{D}_n(s,r)$ grows at rate $\frac{1}{2}n^3 \log s + O(n^2)$: this cyclotomic integerizer costs four times the cubic decay. When $r = s^k$, the determinant vanishes for $n > k$, recovering integral-exponent finite rank; off that locus its signs depend only on $\lfloor \log_s r \rfloor$. The detector is exact but this universal clearing has no subunit arithmetic regime. Specialization-dependent gcd cancellation can reduce the actual denominator and is not controlled by this calculation, so the statement is deliberately not a reduced-denominator lower bound.

There is an independent fixed-support obstruction. If $a, b > 1$ are multiplicatively independent integers, the reduced denominator B_n of $(b^n-1)/(a^n-1)$ satisfies

$$\log B_n = n \log a - o(n). \tag{767}$$

Indeed, the Bugeaud–Corvaja–Zannier gcd theorem [4] makes the cancelled part subexponential, whereas LTE makes the part supported on any fixed finite prime set only polynomial in n. Consequently the quotient lies outside every fixed S-integer ring for all sufficiently large n—there is not even an unbounded fixed-S subsequence. At a prime index n, every surviving denominator prime outside the fixed divisors of $a-1$ has exact multiplicative order n. Since the fixed part is only polynomial and the full denominator is exponential, fresh order-n primes occur for every sufficiently large prime n, and are distinct for distinct indices. Hence the literal Hadamard-quotient strategy requires a hypothesis that a conditional counterexample provably violates.

71.17 Pure denominators still miss the Subspace threshold

Write a hypothetical nonintegral exponent as $x = B + \eta$, where $B \in \mathbb{N}_0$ and $0 < \eta < 1$, and in lowest terms write

$$2^\eta = \frac{W}{2^a}, \qquad 3^\eta = \frac{Z}{3^b}, \qquad 2 \nmid W, \quad 3 \nmid Z.$$

For a convergent p/q to η, put

$$\varepsilon = q\eta - p, \qquad K = aq + p, \qquad L = bq + p.$$

For every sufficiently large convergent, $p, K, L \geq 1$; all statements below are restricted to that tail. Set

$$X = \frac{W^q}{2^K} = 2^\varepsilon, \qquad Y = \frac{Z^q}{3^L} = 3^\varepsilon.$$

The strongest joint pure-support fraction is already reduced:

$$X - Y = \frac{C_{p,q}}{D_{p,q}}, \qquad C_{p,q} = W^q 3^L - Z^q 2^K, \qquad D_{p,q} = 2^K 3^L, \qquad \gcd(C_{p,q}, 6) = 1. \tag{768}$$

The parity and mod-3 assertions prove the gcd statement, while

$$D_{p,q} = (WZ)^q 6^{-\varepsilon}, \qquad C_{p,q} = D_{p,q}(2^\varepsilon - 3^\varepsilon). \tag{769}$$

A two-logarithm Baker lower bound [3] and the convergent upper bound give constants $c_1, c_2, \kappa > 0$ such that

$$c_1 D_{p,q} q^{-\kappa} \leq |C_{p,q}| \leq c_2 D_{p,q} q^{-1}. \tag{770}$$

Thus $|C_{p,q}| = D_{p,q}^{1-o(1)}$. Ridout's theorem or Schmidt's Subspace theorem [15, 16] would require a fixed saving $D_{p,q}^{1-\rho}$ with $\rho > 0$; Baker proves that no such inequality can hold eventually. For every fixed nonzero $P \in \mathbb{Q}[X, Y]$, contact rigidity gives

$$\operatorname{ord}_{t=0} P(2^t, 3^t) \leq \deg_X P + \deg_Y P.$$

Indeed, the first nonzero homogeneous part of P at $(1, 1)$ evaluates at $(\log 2, \log 3)$; any further cancellation would make the transcendental ratio $\log_2 3$ algebraic. Thus fixed-degree contact supplies only a polynomial-in-q saving against the natural exponential common-denominator scale. This does not rule out q-dependent polynomials of controlled height; a uniform theorem for such a family would be genuinely new. The coprime 2/3 supports are exact, but the unrestricted primes of $C_{p,q}$ absorb the product formula.

Status: This critical interpolation and closure layer contains several bounded formal cores and exact paper-level advances. The universal Lambert identity is [kernel-verified Lean] in `LambertCocycleFinite`; because it holds for arbitrary rational M, N, it is a certified no-go. The finite positivity/covariance core, Newton append/collision algebra, and higher Newton–Kummer integral-exponent calibration are [kernel-verified Lean] in `PositiveCoefficientClosure`, `NewtonCollisionCore`, and `RankOneNewtonKummer`. Finite dilation-tail preservation and signed-orbit augmentation are [kernel-verified Lean] in `FiniteDilationAlternation` and `SignedOrbitNormFinite`. Private dyadic order, cleaned-gap capture, and adjacent-smooth tangent algebra are [kernel-verified Lean] in `DyadicPrivateSupport`, `CaptureMultiplier`, and `CatalanTangentCore`. Their infinite-series, valuation, asymptotic, Kummer-field, density, and resultant extensions remain explicitly [paper proof] paper arguments. Together with the refined Stieltjes/Jensen, Hermite, reciprocal-Hankel, BCZ, and pure-denominator results, they leave three narrower interfaces: a positive arithmetic interpolant lying in a rigid cone such as nonnegative coefficients or one-signed geometric curvature; an output-specific denominator upper bound on a balanced exhaustion, or a saturated two-generator divisor of the positive Markov–Padé defect integer, rather than more fresh divisors; or exceptional cross-conjugate ideal overlap that makes a signed Kummer operation integral. The natural Stieltjes transfer family, unsaturated Markov–Padé Hankel Smith content, finite positive-dilation filters, finite and one-ray infinite power-substitution corrections, four-dilation rank-one defect, sparse-spike aggregation, pairwise triangular gap synchronization, and collision-tree/Smith control without an output-sensitive estimate for the transverse bad-cluster factor and its p-adic lifts, reciprocal Hankel determinants, ordinary traces, character projections, common ideals, signed augmentation-zero ratios absent exceptional overlap, and subgroup output partial norms do not improve the relevant bounds at their exact scales. Fixed-level order-one defining polynomials also supply no factor descent in the shift or Newton order. None of the surviving bridges is supplied by the original hypotheses.

72 Prime-top cyclotomic isolation and growing linear stencils

Section 68 showed that a *fixed* Richardson order or fixed rational contact cannot repair the exponential height loss of the rational symmetric secants $J_k(r,s) = (s^k - s^{-k})/(r^k - r^{-k})$. This section removes the word "fixed." For a hypothetical counterexample the level-ℓ secant at a prime ℓ carries, in its *reduced* denominator, almost the entire homogeneous cyclotomic factor $\widehat{\Phi}_{2\ell}(P,Q) = (P^\ell + Q^\ell)/(P+Q)$, and no linear combination of lower levels with coefficients of small prime support can cancel it. Against a denominator of logarithmic size $(\ell - 1)\log P$ the real contact of an m-node maximally flat stencil supplies only $2m\log(1/|h|)$, and Baker's bound makes $\log(1/|h|) = O(\log\log P)$ along convergents. The approximation exponent of every prime-top maximal-contact integer stencil therefore tends to zero even when the order grows almost as fast as $1/|h|$. The finite algebra is [kernel-verified Lean] in `PrimeTopIsolation`; the uniform gcd input and the nearest-pole asymptotics are [paper proof] with a [classical/open background] citation.

72.1 Orbit-uniform generalized gcd

Let $x \notin \mathbb{Z}$ with $M = 2^x, A = 3^x \in \mathbb{Z}$. For $w = (a, b) \in \mathbb{Z}^2 \setminus \{0\}$ put $r_w = 2^a 3^b$ and $s_w = M^a A^b = r_w^x$, oriented so that $r_w > 1$, and write $r_w = P/Q$, $s_w = U/V$ in lowest terms. The decisive observation is that every pair (r_w^n, s_w^n) lies in the *one* finitely generated group

$$\Gamma_x = \langle (2, M), (3, A) \rangle \subset \mathbb{G}_m^2(\mathbb{Q}), \tag{771}$$

and r_w, s_w are multiplicatively independent for every $w \neq 0$ (a relation $r_w^c s_w^d = 1$ gives $c + dx = 0$, hence $x \in \mathbb{Q}$ and then $x \in \mathbb{Z}$). Corvaja–Zannier's theorem for a fixed finitely generated subgroup of $\mathbb{G}_m^2$ [7] has a finite exceptional set, which cannot follow an unbounded sequence of changing directions. Hence, uniformly in w,

$$\log \gcd(P^n - Q^n, U^n - V^n) = o_x(n \log P) \qquad (n \log P \to \infty). \tag{772}$$

The orbit-height calculation of Section 68 used the same theorem along one ray; (772) is the moving-direction form.

72.2 Order of every fresh prime

For an odd prime ℓ define $\widehat{\Phi}_{2\ell}(P, Q) = \sum_{i<\ell} P^i (-Q)^{\ell-1-i}$, so that $(P+Q)\widehat{\Phi}_{2\ell}(P, Q) = P^\ell + Q^\ell$ and $\log \widehat{\Phi}_{2\ell}(P, Q) = (\ell - 1) \log P + O(1)$.

Lemma 72.1 (Fresh primes)**.** *Let ℓ be an odd prime, P, Q coprime integers, and $p \neq \ell$ a prime dividing $\widehat{\Phi}_{2\ell}(P, Q)$. Then $p \nmid PQ(P+Q)$, the residue PQ^{-1} has exact order 2ℓ in $\mathbb{F}_p^\times$, $2\ell \mid p - 1$, and $p \nmid P^{2k} - Q^{2k}$ for $0 < k < \ell$. Moreover $v_\ell(\widehat{\Phi}_{2\ell}(P, Q)) \leq 1$.*

Proof. Coprimality excludes $p \mid P$ or $p \mid Q$ from $p \mid P^\ell + Q^\ell$. If $p \mid P + Q$ then $Q \equiv -P$ and $\widehat{\Phi}_{2\ell} \equiv \ell P^{\ell-1} \pmod p$, forcing $p = \ell$. Thus $y = PQ^{-1}$ satisfies $y^\ell = -1 \neq 1$ (here $p \neq 2$ because $P^\ell + Q^\ell$ is odd) and $y^2 \neq 1$ (as $y \neq \pm 1$), so $\operatorname{ord}(y) = 2\ell$ and $2\ell \mid p - 1$. If $p \mid P^{2k} - Q^{2k}$ then $2\ell \mid 2k$, impossible for $0 < k < \ell$. The ℓ-adic bound is lifting-the-exponent. □

All parts except the last are **[kernel-verified Lean]** in `PrimeTopIsolation` (the order, the divisibility $2\ell \mid p - 1$, and the lower-level integrality are separate theorems). Define the *fresh factor*

$$\mathfrak{F}_\ell = \Big(\frac{\widehat{\Phi}_{2\ell}(P, Q)}{\gcd(\widehat{\Phi}_{2\ell}(P, Q), U^{2\ell} - V^{2\ell})} \Big)_{(\ell)}, \tag{773}$$

the prime-to-ℓ part after removing target-gap cancellation. By (772) and the lemma,

$$\log \mathfrak{F}_\ell = (\ell - 1) \log P - o_x(\ell \log P) \qquad (\ell \log P \to \infty \text{ through odd primes}), \tag{774}$$

uniformly over varying directions.

72.3 Reduced-denominator survival and the coefficient tax

Let $p^d \parallel \mathfrak{F}_\ell$ with ℓ larger than every prime divisor of MA. Since $p > 2\ell$ and $p \nmid PQUV$, Lemma 72.1 gives $v_p(J_\ell) = -d$ and $v_p(J_k) \geq 0$ for $1 \leq k < \ell$. The ultrametric unique-minimum lemma (**[kernel-verified Lean]**, `padicValRat_sum_eq_top`) then yields:

Theorem 72.2 (Stencil survival)**.** *If $a_1, \dots, a_\ell \in \mathbb{Q}$ with $a_\ell \neq 0$ satisfy $v_p(a_\ell) = 0$ and $v_p(a_k) \geq 0$ at every prime $p \mid \mathfrak{F}_\ell$, then $\mathfrak{F}_\ell \mid \operatorname{den}(\sum_{k\leq \ell} a_k J_k)$, and hence $\log \operatorname{den} \geq (\ell-1)\log P - o_x(\ell \log P)$.*

Without any support hypothesis there is still a local tax. Put $C_p = \max_{a_k \neq 0} |v_p(a_k)|$ and $E_p = \max(0, -v_p(S))$ for $S = \sum a_k J_k$.

Proposition 72.3 (Coefficient-or-denominator tax)**.** $2(E_p + C_p) \geq d$ *for every* $p^d \parallel \mathfrak{F}_\ell$*. Consequently* $\log \operatorname{den}(S) + 2\sum_k h(a_k) \geq \frac{1}{2} \log \mathfrak{F}_\ell$*.*

Proof. If $2C_p \geq d$ there is nothing to prove. Otherwise the top summand has valuation $v_p(a_\ell) - d < -d/2$ while every other nonzero summand has valuation $\geq -C_p > -d/2$, so the unique-minimum lemma gives $v_p(S) = v_p(a_\ell) - d$ and $E_p \geq d - C_p$. □

The local inequality is [kernel-verified Lean] (`coefficient_tax`); the height form uses $\sum_p |v_p(a)| \log p \leq 2h(a)$. The factor $\frac{1}{2}$ is sharp for this dichotomy. The conclusion is not that adaptive coefficients are impossible, but that a proposed cancellation must visibly carry the moving prime mass it removes.

72.4 Maximal-contact stencils and nearest-pole asymptotics

Let $f_x(z) = \sinh(xz)/\sinh z$ and let $1 \leq n_1 < \cdots < n_m = N$ be integer nodes with $\lambda_i = n_i^2$. The weights $w_i = \prod_{j\neq i} \lambda_j/(\lambda_j - \lambda_i)$ are the Lagrange weights reproducing the value at 0 for polynomials in z^2 of degree $< m$, and $\mathcal{S}_{\mathbf{n},x}(e^h) = \sum_i w_i f_x(n_i h) = \sum_i w_i J_{n_i}(r,s)$ is rational for a 2–3 rational $r = e^h$. The moments satisfy $M_0 = 1$, $M_j = 0$ for $0 < j < m$, and $M_m = (-1)^{m-1} \prod \lambda_i$, with generating identity $\sum_j (M_j - \mathbf{1}_{j=0}) t^j = (-1)^{m-1} (\prod_i \lambda_i) t^m / \prod_i (1 - \lambda_i t)$. Every prime in a weight w_i is below $2N$ (only n_j, $n_j \pm n_i$ occur), so for prime $N = \ell$ larger than the prime support of MA, Theorem 72.2 applies with zero coefficients at unused levels:

$$\mathfrak{F}_N \mid \operatorname{den} \mathcal{S}_{\mathbf{n},x}(e^h), \qquad \log \operatorname{den} \mathcal{S}_{\mathbf{n},x}(e^h) \geq (N-1)\log P - o_x(N \log P). \tag{775}$$

For consecutive nodes the weights are $c_{m,k} = 2(-1)^{k-1}\binom{2m}{m+k}/\binom{2m}{m}$ and the stencil is a central difference: $\mathcal{R}_{m,x}(e^h) - x = (-1)^{m+1}\binom{2m}{m}^{-1} \delta_h^{2m} f_x(0)$, equivalently $\sum_{k\leq m} c_{m,k}\cosh(kt) = 1 - (-1)^m \binom{2m}{m}^{-1} (2\sinh(t/2))^{2m}$. On integer exponents this gives the exact finite sums $\mathcal{R}_{m,L}(e^h) - L = -2(-1)^m\binom{2m}{m}^{-1} \sum_j (2\sinh(jh))^{2m}$ over $j \leq (L-1)/2$ (odd L; half-integer shifts for even L), which separate the integral locus from the pole-dominated regime below.

For $x \notin \mathbb{Z}$ the nearest poles of f_x are $\pm i\pi$, with principal part $2\pi \sin(\pi x)/(z^2 + \pi^2)$; Cauchy estimates on $|z| \leq \rho < 2\pi$ give $b_j(x) = 2(-1)^j \sin(\pi x)\pi^{-2j-1} + O(\rho^{-2j})$ for the Taylor coefficients of f_x. Inserting this into the exact tail $\mathcal{S} - x = (-1)^{m-1}(\prod \lambda_i) h^{2m} \sum_r b_{m+r} h^{2r} h_r(\lambda)$ (complete homogeneous h_r) and summing the geometric series gives, whenever $m \to \infty$ and $N|h| \to 0$,

$$\mathcal{S}_{\mathbf{n},x}(e^h) - x = -\frac{2\sin \pi x}{\pi} \prod_{i=1}^{m} \frac{n_i^2 h^2}{\pi^2 + n_i^2 h^2} \left(1 + o_x(1)\right), \tag{776}$$

eventually nonzero with the fixed sign $-\operatorname{sgn}\sin \pi x$; the remainder is exponentially small in m because $h^2 \sum n_i^2 \leq mN^2h^2 = o(m)$.

72.5 The growing-order no-go

Let p_j/q_j be convergents to $\log 3/\log 2$, $h_j = q_j \log 3 - p_j \log 2$, oriented with $r_j = e^{|h_j|} = P_j/Q_j > 1$, so $\log P_j \asymp q_j$ and $|h_j| \ll q_j^{-1}$. A lower bound for linear forms in two logarithms [33, 57] gives $|h_j| \geq C_0 q_j^{-C_1}$, hence $\log(1/|h_j|) = O(\log q_j) = o(\log P_j)$.

Theorem 72.4 (Prime-top maximal-contact stencils are weak). *Let N_j be odd primes with $N_j|h_j| \to 0$, and let S_j be any maximal-contact integer stencil with $m_j \to \infty$ nodes and largest node N_j evaluated at $e^{|h_j|}$. Then*

$$\frac{-\log|S_j - x|}{\log \operatorname{den}(S_j)} \longrightarrow 0,$$

so $|S_j - x| > \operatorname{den}(S_j)^{-\eta}$ for every fixed $\eta > 0$ and all large j. In particular this holds for Richardson orders $\ell_j \asymp q_j^{1/2}$ (Bertrand).

Proof. By (775), $\log \operatorname{den}(S_j) \geq (1 - o(1))N_j \log P_j$. By (776), $-\log|S_j - x| \leq 2m_j \log(C/|h_j|) + o(m_j) \leq 2N_j \log(C/|h_j|) + o(N_j)$. Divide and use $\log(1/|h_j|) = o(\log P_j)$. □

Prime-top isolation contributes denominator mass $\asymp N \log P$ while real contact supplies at most $m \log(1/|h|)$ with $m \leq N$; the ratio is $O(\log q/q)$ no matter how the nodes move. The theorem does not cover nonlinear constructions, stencils whose largest index leaves the near-identity window $N|h| \to 0$, or coefficient systems not chosen by maximal polynomial contact; those remain genuine options, and Proposition 72.3 is the first test any of them must pass.

72.6 Covolume conservation for restricted denominators

A separate proposal forces approximation denominators to carry prescribed p-adic depth and then combines real and p-adic smallness. For $L = \prod_{p\in S} p^{e_p}$ the lattice $\{(q, q\beta - a) : q \in L\mathbb{Z},\ a \in \mathbb{Z}\}$ has covolume L, so Minkowski—or, more simply, Dirichlet's pigeonhole applied to $L\beta$—supplies $0 < q \leq nL$ with $L \mid q$ and $|q\beta - a| \leq 1/(n + 1) \leq L/q$, while every such q has $\prod_{p\in S} |q|_p \leq L^{-1}$. Therefore

$$|q\beta - a| \prod_{p\in S} |q|_p \leq \frac{1}{q}, \tag{777}$$

exactly the unrestricted Dirichlet exponent; and for each fixed L a badly approximable $\beta = \alpha/L$ shows $|q\beta - a| \geq cL/q$ on $L\mathbb{Z}$, so the scale L/Q is sharp. The one-prime statement is **[kernel-verified Lean]** (`covolume_conservation` in `RestrictedDenominatorDirichlet`); the several-prime version is the product of local factors. The law rules out only the direct strategy "impose $q \equiv 0 \pmod L$ and invoke Minkowski"; accidental extra divisibility, correlations with the numerator, or output-specific congruences are outside its scope.

72.7 What remains viable

The results narrow the search to three interfaces: a *nonlinear* construction that escapes prime-top isolation (any proposal should first be run through Lemma 72.1 and Proposition 72.3); an *output-specific* local-to-global cancellation involving several correlated cyclotomic factors rather than one pair (the orbit-uniform bound shows that direct cancellation between $\widehat{\Phi}_{2\ell}(P, Q)$ and $U^{2\ell} - V^{2\ell}$ has only $o(\ell \log P)$ mass); or a *tensor-period* statement controlling the quadratic relation

$\log M \log 3 - \log A \log 2 = 0$. Logarithms of algebraic numbers are periods of Kummer 1-motives, for which the period conjecture is known through the analytic subgroup theorem, but the quadratic relation lives in a tensor square outside that linear theory; the function-field analogue (algebraic independence for Drinfeld-module logarithms via Frobenius difference equations) illustrates the missing operator-level symmetry but is not applicable here. A concrete intermediate problem: for every rational expression of bounded degree in the prime-top values $J_1, \ldots, J_\ell$, either a positive proportion of $\log \widehat{\Phi}_{2\ell}(P, Q)$ survives in its reduced height or the expression factors through an explicit multiplicative dependence.

73 The sparse quadratic prime-logarithm class

The log-product criterion of Section 11 and the formal prime determinant of Section 71.11 treat the relation $\log m \log 3 - \log n \log 2 = 0$ as a linear form in products of prime logarithms. This section records the exact *local* structure of that form at an external prime, a q-adic realization that is unconditionally nonzero, the narrowest period-stability principle that would close the problem, and—new here—the calibration that this principle is *equivalent* to the conjecture. The canonical-generator, counting, and commensurability statements that accompany this viewpoint are already in Sections 4 and 4.5 and are not repeated.

73.1 The formal class and its external derivatives

Introduce formal variables L_p (p prime), $L(u) = \sum_p v_p(u) L_p$, and

$$\mathcal{F}_{m,n} = L(m)L_3 - L(n)L_2, \qquad \mathrm{ev}_\infty(L_p) = \log p. \tag{778}$$

A candidate has $\mathrm{ev}_\infty(\mathcal{F}_{m,n}) = 0$, and $\mathcal{F}_{m,n} = 0$ formally iff $m = 2^k, n = 3^k$ (compare the monomials L_qL_3, L_qL_2, L_3^2, L_2^2, L_2L_3). For an external prime $q \notin \{2, 3\}$,

$$\partial_q \mathcal{F}_{m,n} = a_q L_3 - b_q L_2, \qquad \Lambda_q := \mathrm{ev}_\infty(\partial_q \mathcal{F}_{m,n}) = \log \frac{3^{a_q}}{2^{b_q}}, \tag{779}$$

with $a_q = v_q(m)$, $b_q = v_q(n)$. By Section 8 every nonintegral candidate has an external prime in the support of mn, and for such q unique factorization gives $\Lambda_q \neq 0$, quantitatively

$$|\Lambda_q| \geq \frac{1}{\max(3^{a_q}, 2^{b_q})} \tag{780}$$

(from $|\log(A/B)| \geq |A - B|/\max(A, B)$). Both facts are **[kernel-verified Lean]** (`external_derivative_ne_zero`, `abs_external_derivative_ge` in `SparseCoaction`). Thus the hypothetical zero of $\mathcal{F}_{m,n}$ at the prime-logarithm point is *nonsingular*: its gradient is nonzero. Methods that detect only factorized or singular period relations cannot reach the conjecture. The derivatives at the structural primes are $\partial_2 \mathcal{F} = a_2 L_3 - b_2 L_2 - L(n)$ and $\partial_3 \mathcal{F} = a_3 L_3 + L(m) - b_3 L_2$, whose evaluations still involve x; only the external derivatives eliminate x and become logarithms of explicit rationals. Euler's identity $2\mathcal{F} = \sum_q L_q \partial_q \mathcal{F}$ exhibits the zero period as a weighted cancellation among first derivatives of which at least one is unconditionally nonzero.

Writing $r_{\mathrm{ext}} = \mathrm{rank}\{(a_q, b_q) : q \notin \{2, 3\}\} \in \{1, 2\}$, the rank-one case has $(a_q, b_q) = c_q(A, B)$ with coprime (A, B), so $m = 2^{a_2}3^{a_3}U^A$, $n = 2^{b_2}3^{b_3}U^B$ with $U = \prod q^{c_q}$ and the relation collapses to the three-logarithm form

$$a_3(\log 3)^2 - b_2(\log 2)^2 + (a_2 - b_3)\log 2 \log 3 + \log U\,(A \log 3 - B \log 2) = 0;$$

in rank two, two external primes give independent descendants. This refines the external rank classification of Section 13. As a symmetric operator on the prime-indexed vector space the form has rank at most four (image in $\mathrm{span}\{[m],[n],e_2,e_3\}$), independently of the number of prime divisors of mn: the prime-logarithm vector would be an isotropic, non-radical point of an "arrowhead" form of rank ≤ 4.

73.2 A nonarchimedean realization

For $q \ge 5$ let $\log_q$ be the Iwasawa logarithm on $\mathbb{Q}_q^\times$ ($\log_q q = 0$). On positive rationals its kernel is $q^{\mathbb{Z}}$ (the only positive rational root of unity is 1). Define $\rho_q([u] \odot [v]) = v_q(u)\log_q v + v_q(v)\log_q u$ on symmetric products of exponent vectors; then $\rho_q(\mathcal{F}_{m,n}) = \log_q(3^{a_q}/2^{b_q})$, which is nonzero because $3^{a_q}/2^{b_q}$ is a q-unit different from 1. A candidate would therefore have the profile $\mathrm{ev}_\infty(\mathcal{F}_{m,n}) = 0$, $\rho_q(\mathcal{F}_{m,n}) \neq 0$. The first nonzero q-adic digit is detected by a finite congruence: with $R = 3^{a_q}/2^{b_q}$ and $h_q(R) = v_q(R^{q-1} - 1) \ge 1$ one has $v_q(\log_q R) = h_q(R)$, so $R^{q-1} \not\equiv 1 \pmod{q^{h_q(R)+1}}$; for $h_q = 1$ this is the ordinary Fermat quotient (compare `FermatQuotientDeterminant`). No comparison theorem currently turns the archimedean/q-adic mismatch into a contradiction.

73.3 Sparse coaction separation is the conjecture

In the mixed-Tate category over $\mathbb{Z}[1/6mn]$ the decomposable weight-two class $\mathcal{K}_{m,n} = \ell^{\mathfrak{m}}(m)\ell^{\mathfrak{m}}(3) - \ell^{\mathfrak{m}}(n)\ell^{\mathfrak{m}}(2)$ has real period $\log m \log 3 - \log n \log 2$, and the prime-valuation functional applied to the de Rham factor of its $(1,1)$ coaction component gives $D_q\mathcal{K}_{m,n} = \ell^{\mathfrak{m}}(3^{a_q}/2^{b_q})$ with real period $\Lambda_q \neq 0$. So a candidate gives a *nonzero* motivic class with zero period and an explicit nonzero first descendant. The narrowest principle that would finish is:

Definition 73.1 (Sparse coaction separation)**.** For all positive integers u, v with $\log u \log 3 = \log v \log 2$ and every prime $q \notin \{2,3\}$: $v_q(u)\log 3 = v_q(v)\log 2$.

One must not infer $\mathrm{ev}_\infty(\partial_q F) = 0$ from $\mathrm{ev}_\infty(F) = 0$ for an arbitrary numerical relation; evaluation kernels are not stable under formal differentiation. The coaction argument is valid only once a numerical relation is known to come from a motivic relation, which is exactly the missing injectivity.

Theorem 73.2 (Calibration)**.** *Sparse coaction separation is* equivalent *to Conjecture 1.1.*

Proof. ($\Rightarrow$) Given a nonintegral candidate, pick an external support prime q; the principle forces $\Lambda_q = 0$, contradicting (780). ($\Leftarrow$) Given $u, v > 0$ with $\log u \log 3 = \log v \log 2$, put $t = \log u / \log 2$; then $2^t = u$ and $3^t = e^{t \log 3} = e^{\log v} = v$ are integers, so the conjecture gives $t \in \mathbb{Z}$, $t \ge 0$, hence $u = 2^t$, $v = 3^t$ and $v_q(u) = v_q(v) = 0$ for every external q. □

Both directions are **[kernel-verified Lean]** in `SparseCoaction`, as the single equivalence `sparseCoactionSeparation_iff_alaogluErdosConjecture`. The calibration matters: the principle was proposed as a "deliberately narrow target," but its surplus generality—arbitrary u, v rather than outputs of a single exponent—is illusory, exactly as for the prime-log-product criterion in Section 11.3. A proof of separation for decomposable weight-two Kummer classes with one fixed factor in $\{\ell 2, \ell 3\}$ would be a proof of the conjecture, not a reduction of it; conversely, nothing short of the conjecture proves the separation. The useful content is the precise *witness*: the required injectivity has a fully explicit, minimal-weight descendant, and any argument using only the numerical equality of products of logarithms, without a mechanism making the relation coaction-stable, silently assumes the conclusion.

73.4 Convergent gaps under *abc*

Write $m = 2^a M_0$ with M_0 odd and $n = 3^b N_0$ with $3 \nmid N_0$; for a nonintegral candidate neither M_0 nor N_0 is 1. Let p_j/q_j be convergents of $y = \log M_0/\log 2$ and $G_j = M_0^{q_j} - 2^{p_j}$; then $\gcd(G_j, 2M_0) = 1$. Similarly $H_j = N_0^{s_j} - 3^{r_j}$ from convergents of $\log N_0/\log 3$.

Proposition 73.3 (Radical-full gaps)**.** *Assume the abc conjecture. Then* $\log \operatorname{rad}|G_j|/\log|G_j| \to 1$ *and likewise for* H_j*; equivalently* $|G_j|/\operatorname{rad}|G_j| = |G_j|^{o(1)}$.

Proof. The three numbers $\min(M_0^{q_j}, 2^{p_j})$, $|G_j|$, $\max(\cdot)$ are pairwise coprime, so *abc* gives $\max(M_0^{q_j}, 2^{p_j}) \le K_\varepsilon \operatorname{rad}(2M_0G_j)^{1+\varepsilon}$, hence $\operatorname{rad}|G_j| \ge c_\varepsilon \max(\cdot)^{1/(1+\varepsilon)}$ while $\log|G_j| \le q_j \log M_0 + o(q_j) = \log\max(\cdot) + o(q_j)$. Let $\varepsilon \downarrow 0$. □

No lower bound for the continued-fraction error is used. The consequence for Frey-type curves $Y^2 = X(X - M_0^{q_j})(X - 2^{p_j})$ is that the repeated powers in the two main terms do *not* produce a bounded or slowly growing level: the gap injects exponentially many distinct primes into the discriminant. Baker-type bounds control the *size* of the gap (finite irrationality measure of y); *abc* controls its *radical*. The two are complementary, and a modular strategy must supply an independent reason for the gap primes to disappear from the residual conductor.

74 Mesoscopic factorial cocycles

This section develops an auxiliary construction not treated elsewhere in the report: signed products of the falling factorials $F_n = (A^n)_{M^n} = A^n!/(A^n - M^n)!$. Its archimedean side is exactly solvable, its arithmetic side is exactly hostile, and the two determine a precise linear-width escape target. The elementary identities are **[kernel-verified Lean]** in `FactorialCocycle`; the prime-window theorem is **[paper proof]** with **[classical/open background]** analytic inputs.

74.1 The one-divergent-cumulant regime

With $\alpha = \log M/\log A = \log 2/\log 3$, the inequalities $4 > 3$ and $8 < 9$ give $\frac{1}{2} < \alpha < \frac{2}{3}$ (**[kernel-verified Lean]**). Put $N_n = A^n$, $K_n = M^n = N_n^\alpha$, and

$$q = \frac{M}{A} = \left(\frac{2}{3}\right)^x, \qquad \lambda = \frac{M^2}{A} = \left(\frac{4}{3}\right)^x > 1, \qquad \eta = \frac{M^3}{A^2} = \left(\frac{8}{9}\right)^x < 1, \qquad \tau = \frac{M^4}{A^3} < q < \eta. \tag{781}$$

For K_n balls in N_n boxes the pair-collision scale $K_n^2/N_n = \lambda^n$ diverges while the triple-collision scale $K_n^3/N_n^2 = \eta^n$ vanishes; no information about the size of x is needed. The kernel bound $x \ge 12$ (`FiniteCheck4096`; now $x \ge 13$ by `FiniteCheck8192`) enters only through $q \le (2/3)^{12} = 0.0077\ldots$.

74.2 Exact expansion and fixed-shift classification

From $-\log(1-u) = u + u^2/2 + O(u^3/(1-u))$ and the first two power sums,

$$\log F_n = nM^n \log A - \tfrac{1}{2}\lambda^n + \tfrac{1}{2}q^n - \tfrac{1}{6}\eta^n + O(\tau^n). \tag{782}$$

For $P(T) = \sum_j c_j T^j \in \mathbb{Z}[T]$ let $\mathcal{F}_{P,n} = \prod_j F_{n+j}^{c_j}$, so $\log \mathcal{F}_{P,n} = P(E)(\log F)_n$ with E the forward shift. The shift identities

$$P(E)(\rho^n) = P(\rho)\rho^n, \qquad P(E)(n\rho^n) = \rho^n(nP(\rho) + \rho P'(\rho)) \tag{783}$$

([**kernel-verified Lean**], `shiftEval_geom`, `shiftEval_mul_geom`) give the master asymptotic

$$\log \mathcal{F}_{P,n} = \log A\, M^n(nP(M) + MP'(M)) - \tfrac{1}{2}P(\lambda)\lambda^n + \tfrac{1}{2}P(q)q^n - \tfrac{1}{6}P(\eta)\eta^n + O_P(\tau^n).$$

Theorem 74.1 (Near-unit classification)**.** *$\mathcal{F}_{P,n} \to 1$ if and only if $P(M) = P'(M) = P(M^2/A) = 0$, i.e. iff $P_* \mid P$ in $\mathbb{Z}[T]$ where, with $g = \gcd(A, M^2)$,*

$$P_*(T) = (T-M)^2\Big(\frac{A}{g}T - \frac{M^2}{g}\Big) = -\frac{M^4}{g} + \frac{M^2(A+2M)}{g}T - \frac{M^2+2AM}{g}T^2 + \frac{A}{g}T^3.$$

The minimal cocycle satisfies $\log \mathcal{F}_{P_*,n} = C_*\eta^n + O(q^n)$ with $C_* = \frac{M^2(A-M)}{6gA}\big(M - \frac{M^3}{A^2}\big)^2 > 0$, so $\mathcal{F}_{P_*,n} > 1$ for large n. Every nonzero annihilator has at least three coefficient sign variations (Descartes, three positive roots).

The three annihilations, the vanishing of all three growing modes under $P_*(E)$, and the coefficient expansion are [**kernel-verified Lean**] in `FactorialCocycle`; necessity uses the separation $M > \lambda > 1$ of the growing modes. When $P(M) = P'(M) = 0$ the visible powers $(A^n)^{M^n}$ cancel exactly (their exponent is $M^n(nP(M) + MP'(M)) = 0$), so $\prod_j R_{n+j}^{c_j} = \mathcal{F}_{P,n}$ for the no-collision probabilities $R_n = F_n/(A^n)^{M^n}$: the obstruction is the primitive prime content of the terminal intervals $(A^t - M^t, A^t]$ themselves. Cancelling further collision modes $\rho_{r,s} = M^s/A^r$ (only $r = 1, s = 2$ exceeds one; the subunit ladder begins $\eta, q, \tau, (M/A)^2, \dots$) makes $|\mathcal{F}_{P,n} - 1|$ smaller but never changes the sign-variation count or the fixed degree, and the next mode after η decays at the rate $x\log(3/2)$ that also governs the thickness threshold below—a critical coincidence rather than slack.

74.3 The prime-window denominator theorem

By Legendre, $V_p(t) := v_p(F_t) = \sum_{r\ge1}(\lfloor A^t/p^r\rfloor - \lfloor(A^t - M^t)/p^r\rfloor)$ counts multiples of p^r in the terminal interval, and $v_p(\mathcal{F}_{P,n}) = \sum_j c_j V_p(n+j)$, $\log \operatorname{den} \mathcal{F}_{P,n} = \sum_p \max(0, -v_p)\log p$. Two [**classical/open background**] inputs: Huxley's short-interval theorem $\vartheta(X) - \vartheta(X - X^\gamma) \sim X^\gamma$ for $\gamma > 7/12$ [5] (our exponent is $\alpha = 0.6309\ldots$), and Brun–Titchmarsh $\pi(y+h) - \pi(y) \le 2h/\log h$ [6].

Theorem 74.2 (Prime-window obstruction)**.** *Assume $x \ge 12$ and fix $\varepsilon > 0$. Let $P_n \in \mathbb{Z}[T]$ be nonzero polynomials with a negative coefficient, k_n the highest negative index, and suppose the span satisfies $d_n - k_n \le (\log_2 3 - 1 - \varepsilon)(n + k_n)$. Then*

$$\log \operatorname{den}(\mathcal{F}_{P_n,n}) \ge (-c_{n,k_n})(\Delta_\varepsilon + o(1))M^{n+k_n}, \qquad \Delta_\varepsilon = 1 - \frac{8q}{(2 - \log_2 3 + \varepsilon(1-\alpha))(1-q)} > 0.$$

In particular every fixed P with a negative coefficient—hence every fixed near-unit—has $\log \operatorname{den} \mathcal{F}_{P,n} \gg M^n$, while $|\mathcal{F}_{P_*,n} - 1| \sim C_*\eta^n$.

Proof sketch. At level $N = n + k$ put $X = A^N$, $H = M^N$. Each prime $p \in (X - H, X]$ occurs once in F_N and in no lower factor; it enters the denominator unless some higher positive factor $F_{N+\ell}$ captures a multiple $mp \in (Y_\ell - W_\ell, Y_\ell]$ with $Y_\ell = A^\ell X$, $W_\ell = M^\ell H$. For fixed $m \asymp A^\ell$ this confines p to an interval of length $\le 2Hq^\ell$, and there are at most $2 + 2A^\ell H/X$ multipliers. The span hypothesis makes $\log(2Hq^\ell)/\log X \ge 2 - \log_2 3 + \varepsilon(1-\alpha) =: \delta_\varepsilon > 0$, so Brun–Titchmarsh bounds the logarithmic prime mass of each capture interval by $(4/\delta_\varepsilon + o(1))Hq^\ell$; summing over multipliers and $\ell \le D$ gives covered mass $\le (8q/(\delta_\varepsilon(1-q)) + o(1))H$, the second geometric sum being negligible because $q^N M^D \le e^{-\varepsilon N \log M}$. For $x \ge 12$ the coefficient is $0.1497\ldots < 1$ already at $\varepsilon = 0$. Huxley gives total mass $(1 + o(1))H$, so uncovered primes have mass $\ge (\Delta_\varepsilon + o(1))H$, and each contributes $v_p \le c_{n,k} < 0$. □

The same argument applies to binomial cocycles $\mathcal{B}_{P,n} = \prod_j \binom{A^{n+j}}{M^{n+j}}^{c_j}$, whose near-unit classification adds a double root at 1 (Stirling's $\frac{1}{2}n \log M$ term): $P_{\text{bin}} = (T-1)^2 P_*$. Two universal thresholds appear: the multiplier m begins to drift at span $\ell/(n+k) \asymp 1 - \alpha = 0.369\ldots$, but covering survives until the higher interval is as long as the prime scale, at $\ell/(n+k) \asymp \log_2 3 - 1 = 0.585\ldots$. For a fixed polynomial the obstruction is visible in one contiguous slice without the $x \ge 12$ input: with $\rho = \max\{(M/A)^{j-k} : j > k, c_j > 0\}$, the primes in $(X - (1-\epsilon)H, X - (\rho+\epsilon)H)$ occur in no positive factor, giving the sharper $\log \operatorname{den} \ge (-c_k)(1 - \rho + o(1))M^{n+k}$.

74.4 The linear-width target and the structural-prime prefix

Research target 74.3 (Linear-width factorial feasibility)**.** Decide whether there are $P_n \in \mathbb{Z}[T]$ with $P_n(M) = P_n'(M) = P_n(M^2/A) = 0$, $v_p(\mathcal{F}_{P_n,n}) \ge 0$ for all p, $\mathcal{F}_{P_n,n} \to 1$, and $\deg P_n - k_n \ge (\log_2 3 - 1 - o(1))(n + k_n)$. A positive answer with an eventual strict inequality $\mathcal{F}_{P_n,n} \ne 1$ (e.g. a sign-controlled first surviving mode) proves the conjecture; a negative answer by a dual valuation certificate closes the family. *Update (session log): the constructive branch is empty — Corollary 88.2 shows conditions (2) and (3) already force $P_n = 0$ for any mesoscopic pair, so this target is resolved negatively; only integrality without convergence to 1 remains, and the LP computation of Section 88.10 indicates that it is empty too. Final update: the residue is also gone — the global terminal-prime contraction (Theorem 81.1) closes bare integrality for every nonzero P with $P(M) = 0$, uniformly in degree and span; see Section 81.*

For fixed (M, A, n, d) the conditions are three homogeneous linear equations plus the finite system $\sum_j c_j V_p(n+j) \ge 0$, $p \le A^{n+d}$, i.e. a rational polyhedral cone $\mathcal{C}_{n,d}$ intersected with a codimension-three subspace; the proof above is a dual certificate below the critical width, and the right search at the critical width is for sparse dual certificates supported on prime windows and transported multiples (LP with exact Farkas certificates, then ILP with $P_n = P_* S_n$, then prime-power constraints).

The factorial asymptotic uses only the ratio $\log 2/\log 3$ and cannot by itself distinguish a hypothetical primitive pair from the controls $(2^m, 3^m)$. The first primitive-sensitive ingredient is a structural prime $r \mid M$, $r \nmid A$ (or $s \mid A$, $s \nmid 3$), which a nonintegral solution must have. For such r with $r^e \parallel M$, the lowest en base-r digit positions create no carry when adding M^n and $A^n - M^n$, so Kummer's theorem gives

$$v_r \binom{A^n}{M^n} \le \lfloor n \log_r A \rfloor - en \tag{784}$$

(**[kernel-verified Lean]**, `padicValNat_choose_le_of_pow_dvd`). The high-prime obstruction has size M^n; the structural valuation is $O(n)$. A hybrid dual certificate combining the two weightings (high primes constrain signs and span; a structural prime constrains the same coefficients through long carry-free prefixes) is the first remaining direction that genuinely uses nonintegrality rather than the ratio alone.

75 Centered cyclotomic trace dynamics

Every rational approximation p/q to a hypothetical nonintegral β produces the *synchronized* rational near-units

$$z_2 = \frac{M^q}{2^p} = 2^\delta, \qquad z_3 = \frac{A^q}{3^p} = 3^\delta = z_2^\vartheta, \qquad \delta = q\beta - p, \tag{785}$$

one small real parameter encoded by two rationals whose denominators are supported at the structural primes. The symmetric secants of Section 68 put the unknown exponent in the dilation slot; here the dilation is an auxiliary *odd integer* d and the near-unit is the dynamical variable, so the maps can be iterated and every iterate reduced prime by prime. The finite core is **[kernel-verified Lean]** in `CenteredTrace`.

75.1 Traces, Bernoulli expansion, and exact defect

For odd $d = 2h+1$ define $T_d(z) = d^{-1}\sum_{j=-h}^{h} z^j = z^{-h}(z^d-1)/(d(z-1))$. It is reciprocal, $T_d(z) \geq 1$ for $z > 0$ with equality only at $z = 1$ (AM–GM), and $T_d(e^t) = \sinh(dt/2)/(d\sinh(t/2))$, with the Bernoulli–Jordan expansion

$$\log T_d(e^t) = \sum_{r\geq 1} \frac{B_{2r}(d^{2r}-1)}{2r(2r)!} t^{2r} = \frac{d^2-1}{24}t^2 - \frac{d^4-1}{2880}t^4 + \cdots, \tag{786}$$

obtained from $\log((e^u-1)/u) = u/2 + \sum B_{2r}u^{2r}/(2r(2r)!)$; the linear terms cancel, which is the analytic source of the quadratic contact. Arithmetically, $T_d(z) = d^{-1}\prod_{n|d,n>1} \mathcal{C}_n(z)$ with the centered factors $\mathcal{C}_n = z^{-\varphi(n)/2}\Phi_n$, and $\log \mathcal{C}_n(e^t) = \log \Phi_n(1) + \sum_r B_{2r}J_{2r}(n)t^{2r}/(2r(2r)!)$ with Jordan totients J_m; summing $\sum_{n|d} J_m(n) = d^m$ recovers (786). (Raw necklace coordinates have only first-order contact at 1; centering by reciprocity removes it.)

For coprime $a, b > 0$ and $S_d(a,b) = (a^d - b^d)/(a-b)$,

$$T_d(a/b) = \frac{S_d(a,b)}{d(ab)^h}, \qquad \gcd(S_d(a,b), ab) = 1, \qquad \gcd\big(S_d(a,b), d(ab)^h\big) = \gcd(S_d(a,b), d) \mid d, \tag{787}$$

so reduction of one trace value cancels only a divisor of the fixed integer d.

Theorem 75.1 (Exact defect factorization)**.** *$S_{2h+1}(a,b) - (2h+1)(ab)^h = \sum_{j=1}^{h}(ab)^{h-j}(a^j-b^j)^2 = (a-b)^2 P_d(a,b)$ with $P_d(a,b) = \sum_{j=1}^{h}(ab)^{h-j}\big((a^j-b^j)/(a-b)\big)^2 > 0$ symmetric of degree $d-3$; $P_3 = 1$, $P_5 = a^2+3ab+b^2$, $P_7 = a^4+3a^3b+6a^2b^2+3ab^3+b^4$. If $T_d(a/b) = a'/b'$ is reduced and $u = a-b$, then $a'-b' = u^2P_d(a,b)/\gcd(S_d(a,b),d)$.*

The sum-of-squares identity and the divisibility by $(a-b)^2$ are **[kernel-verified Lean]** in `CenteredTrace`, by induction on h. Since $S_d \geq d(ab)^h$, the reduced height satisfies $|h(T_d(z)) - (d-1)h(z)| \leq \log d$: local contact is always two, arithmetic degree is $d-1$. Hence for any composition $\Psi = T_{d_m} \circ \cdots \circ T_{d_1}$ the contact is 2^m while $\deg\Psi = \prod(d_i-1) \geq 2^m$, with equality exactly for pure cubic composition: the cubic trace is the unique *critical* member, every higher positive trace is subcritical before any cross-branch cancellation.

75.2 Exact cubic dynamics

$T_3(z) - 1 = (z-1)^2/(3z)$, an exact square. For reduced $z_k = a_k/b_k$ and $u_k = a_k - b_k$,

$$a_{k+1} = \frac{a_k^2 + a_k b_k + b_k^2}{g_k}, \quad b_{k+1} = \frac{3a_k b_k}{g_k}, \quad u_{k+1} = \frac{u_k^2}{g_k}, \quad g_k = 3^{\mathbf{1}_{3|u_k}}, \tag{788}$$

because $a^2 + ab + b^2 - 3ab = (a-b)^2$, $\gcd(a^2 + ab + b^2, ab) = 1$, and $3 \mid a^2 + ab + b^2 \iff 3 \mid a - b$ (all **[kernel-verified Lean]**). Therefore, for every prime $\ell \neq 3$, $v_\ell(u_k) = 2^k v_\ell(u_0)$ exactly (`padicValInt_sq_div_three`), and $v_3(u_k) = 1 + 2^k(v_3(u_0) - 1)$ when $3 \mid u_0$ (else 0). Two consequences:

Corollary 75.2 (No new cross-orbit support)**.** *For two reduced cubic orbits with defects u_k, v_k: $\operatorname{rad}_{\neq 3}\gcd(u_k, v_k) = \operatorname{rad}_{\neq 3}\gcd(u_0, v_0)$ and $v_\ell(\gcd(u_k, v_k)) = 2^k v_\ell(\gcd(u_0, v_0))$ for $\ell \neq 3$.*

Proposition 75.3 (Critical saturation)**.** *For $z_0 \in \mathbb{Q}_{>0} \setminus \{1\}$, $T_3^{\circ k}(z_0) \to 1$ with $\varepsilon_{k+1} = \varepsilon_k^2/(3(1+\varepsilon_k))$, while $H_k^2/3 \leq H_{k+1} \leq 3H_k^2$, so the canonical height $\widehat{h}_3(z_0) = \lim 2^{-k}\log H_k$ exists, is positive, and differs from $\log H_0$ by at most $\log 3$; and $\varepsilon_k \geq 1/b_k \geq 1/H_k$. Locally $T_3^{\circ k}(e^t) - 1 = 3^{1-2^k} t^{2^k}(1 + \alpha_k t^2 + O(t^4))$ with $\alpha_1 = 1/12$, $\alpha_k = -2^{k-3}/3$ for $k \geq 2$.*

Applied to the synchronized germ, $R_{k;p,q} = E_{3,k}/E_{2,k} \in \mathbb{Q}_{>0}$ with $E_{i,k} = T_3^{\circ k}(z_i) - 1$ satisfies $R_{k;p,q} = \vartheta^{2^k}(1 + \alpha_k((\log 3)^2 - (\log 2)^2)\delta^2 + O_k(\delta^4))$: the contact 2^k of each branch does *not* transfer to the ratio, whose relative error stays quadratic in δ, while its height is $O(2^k q)$. By Corollary 75.2, any exceptional cancellation in $R_{k;p,q}$ is already encoded in the initial gaps $M^q - 2^p$ and $A^q - 3^p$; iteration only raises their common valuations to powers. This closes, exactly rather than by a loose height estimate, every "iterate until the rational number is too close to one" argument based on the cubic trace.

75.3 Prime traces and the two fifth-trace channels

For an odd prime $p \geq 5$ one has $P_p(a,b) \equiv (a-b)^{p-3} \pmod p$ (Frobenius on $S_p \equiv (a-b)^{p-1}$), $\gcd(S_p, p) = p$ iff $p \mid a - b$, and in that case $v_p(P_p(a,b)) = 1$ exactly (evaluate on the diagonal: $P_p(b,b) = pb^{p-3}(p^2-1)/24$, and the first derivative vanishes modulo p). So the normalized factor $C_p = P_p/p^{\mathbf{1}_{p|a-b}}$ is a p-free integer and the reduced defect is $a' - b' = (a-b)^2 C_p(a,b)$: the fifth trace is the first level that can introduce new defect primes, and the last whose new factor is a single rational reciprocal quadratic.

Lemma 75.4 (Reciprocal-quadratic bridge)**.** *In any commutative ring, $cd(a^2 + \lambda ab + b^2) - ab(c^2 + \lambda cd + d^2) = (ad - bc)(ac - bd)$.*

Theorem 75.5 (Two-channel decomposition)**.** *Let $F(a,b) = a^2 + 3ab + b^2$, (a,b) and (c,d) coprime pairs, and $p \neq 5$ a prime with $p^e \mid F(a,b)$ and $p^e \mid F(c,d)$, $e \geq 1$. Then p^e divides exactly one of the channels $D_- = ad - bc$, $D_+ = ac - bd$, and p does not divide the other. Consequently, with $G = \gcd(C_1, C_2)$ of the normalized factors, $G = \gcd(G, D_-)\gcd(G, D_+)$ with coprime factors; unnormalized, $\gcd(F(a,b), F(c,d)) \mid \operatorname{lcm}(|D_-|, |D_+|)$.*

Proof. $p \nmid abcd$ since $F(a,b)$ is coprime to ab. The bridge gives $p^e \mid D_- D_+$. If p divided both channels then in $\mathbb{F}_p$ the ratios $r = a/b$, $s = c/d$ would satisfy $r = s$ and $rs = 1$, so $r^2 = 1$; $r = -1$ contradicts $r^2 + 3r + 1 = 0$, and $r = 1$ forces $p = 5$. Hence p divides exactly one channel and, by coprimality, p^e divides it. $\square$

Both are [kernel-verified Lean] (`bridge_identity`, `fifth_trace_two_channel`). For the synchronized germ with $z_2 = a/b$, $z_3 = c/d$ reduced, the channels are the explicit synchronized binomial gaps

$$D_- \propto 3^p M^q - 2^p A^q = 6^p(2^\delta - 3^\delta), \qquad D_+ \propto (MA)^q - 6^p = 6^p(6^\delta - 1), \tag{789}$$

the first comparing the two branches in the same direction, the second a branch with the reciprocal of the other. Every shared prime power of the new normalized fifth-trace factors is carried, with full valuation, by exactly one of them. Exact examples: $F(35, 37) = 11 \cdot 19 \cdot 31$, $F(64, 65) = 11 \cdot 31 \cdot 61$, with $31 \mid D_- = -3 \cdot 31$ and $11 \mid D_+ = -3 \cdot 5 \cdot 11$.

The theorem says where shared content must go, not that any is forced: real synchronization does not transparently force modular synchronization. The two channels are, however, asymptotically disjoint, by an exact descent to the original gaps:

Proposition 75.6 (Channel descent)**.** *For all integers, $a(ad - bc) + b(ac - bd) = d(a^2 - b^2)$ and $d(ac - bd) - c(ad - bc) = b(c^2 - d^2)$. Hence any common divisor of the two channels divides $d(a^2 - b^2)$ and $b(c^2 - d^2)$, and a prime $\ell \nmid 6$ dividing both channels of (789) divides the doubled gaps $M^{2q} - 4^p = (M^q - 2^p)(M^q + 2^p)$ and $A^{2q} - 9^p = (A^q - 3^p)(A^q + 3^p)$.*

[kernel-verified Lean] (`dvd_of_dvd_both_channels`, `prime_dvd_doubled_gaps_of_dvd_both_channels`). The pair $(M^{2q}/4^p, A^{2q}/9^p)$ lies in the fixed group $\langle (M, A), (2, 3) \rangle$ with multiplicatively independent coordinates, so the fixed-group Corvaja–Zannier bound of Section 72.1 gives $\log \gcd(D_-, D_+) = o(q)$ for gcd taken prime to 6: shared channel support is, up to sign, shared support of the original synchronized gaps, and it is asymptotically negligible. Signed trace filters $\prod T_{d_i}^{c_i}$ can raise contact to $2R$ exactly when $\sum_i c_i(d_i^{2r} - 1) = 0$ for $r < R$, which needs $m \geq R$ factors (Vandermonde), and their cyclotomic degree budget $\sum_n |e_n|\varphi(n)$ grows much faster than the contact (orders 4 and 6 already cost 10 and 42)—the trace-language form of the finite-order height obstruction of Section 17.4.

Research target 75.7 (Trace interfaces)**.** (i) Force shared normalized fifth-trace support along an infinite sequence of approximants beyond what the two channels can carry; (ii) show that primes generated by the normalized reciprocal quadratic have a local or Kummer signature incompatible with both $3^p M^q - 2^p A^q$ and $(MA)^q - 6^p$ outside a negligible set; (iii) for $p \geq 7$ compute the bivariate common-root resultants of P_p (the seventh-trace quartic factors over a quadratic field) and classify the extension-field channels by Frobenius type.

76 Exponent-lattice moment masks

The last framework keeps the scalar viewpoint but changes variables: instead of the smooth nodes it uses *near-units attached to exponent-lattice relations*. For $\nu = (a, b) \in \mathbb{Z}^2$ put $\rho_\nu = 2^a 3^b = e^{\delta_\nu}$, $\delta_\nu = a \log 2 + b \log 3$, and $\sigma_\nu = M^a A^b = \rho_\nu^x$; for a convergent p/q of ϑ, $\nu = (-p, q)$ gives the output near-unit $z_\nu = A^q/M^p = e^{x\delta_\nu}$ (note the contrast with (785), which approximates β rather than ϑ). Integer polynomials in such near-units are "masks." The finite core is [kernel-verified Lean] in `ExponentLatticeMask`; the barriers are [paper proof] with classical inputs.

76.1 Exact height and the two-direction uncertainty

With $L_{M,A}(\nu) = \sum_{p|MA} |av_p(M) + bv_p(A)| \log p$, a norm on $\mathbb{R}^2$ (the valuation vectors of M, A are independent because M, A are multiplicatively independent), the reduced $\sigma_\nu = R_\nu/S_\nu$ satisfies

$$\log R_\nu + \log S_\nu = L_{M,A}(\nu), \qquad \log R_\nu - \log S_\nu = x\delta_\nu, \tag{790}$$

so both sides are exponentially large in $|\nu|$ even when $x\delta_\nu$ is tiny. For two vectors with determinant $D = ad - bc \neq 0$ and $H = \|\nu\|_1 + \|\omega\|_1$, $d\delta_\nu - b\delta_\omega = D \log 2$ and $a\delta_\omega - c\delta_\nu = D \log 3$, so $\max(|\delta_\nu|, |\delta_\omega|) \geq |D| \log 3/H$; adjacent convergents ($D = \pm 1$) are the optimally conditioned basis, and even they cannot make both errors smaller than the reciprocal exponent scale.

76.2 Toric collapse and the gap-power theorem

For a list $\nu_1, \ldots, \nu_k$, the kernel of $T_i \mapsto U^{a_i} V^{b_i}$ on Laurent polynomials is exactly the lattice ideal $I_\nu = \langle T^\alpha - T^\beta : \sum \alpha_i \nu_i = \sum \beta_i \nu_i \rangle$ (group terms by fibres), and every element of I_ν vanishes at *both* (ρ_{ν_i}) and (σ_{ν_i}). Modulo I_ν every mask is a Laurent polynomial in two adjacent near-units: adding convergent variables adds no formal degrees of freedom, and the continued-fraction recurrence itself gives $T_{n+1} - T_n^{a_{n+1}} T_{n-1} \in I_\nu$, so recurrence-only masks are identically zero, not small. Any claimed high-contact mask must be reduced modulo I_ν before its contact is credited. (The numerical kernel at $(2, 3)$ is larger—$3 - 2 - 1 = 0$—but such additive identities do not transfer to arbitrary base pairs and are the hard, candidate-specific part.)

For a mask $F \in \mathbb{Z}[U, V]$ the contact order at $(1, 1)$ is the largest r with $F \in (U - 1, V - 1)^r$, equivalently the vanishing of the two-dimensional Prouhet moments $\sum_\ell c_\ell a_\ell^i b_\ell^j$ for $i + j < r$; it is at most the total bidegree and at most (#monomials) $- 1$ (a generic linear functional reduces to a univariate exponential sum with distinct frequencies).

Theorem 76.1 (One-direction gap power)**.** *If $P \in \mathbb{Z}[T]$ has a zero of multiplicity $\geq r$ at 1 and $d \geq \deg P$, then for all integers R, $S \neq 0$ the cleared value $S^d P(R/S) = \sum_i P_i R^i S^{d-i}$ is divisible by $(R - S)^r$; if nonzero it is at least $|R - S|^r$ in absolute value.*

[kernel-verified Lean] (`gap_pow_dvd_clearedEval`, `abs_clearedEval_ge`): write $P = (T - 1)^r Q$ and clear. For the Thue–Morse mask $P_r = \prod_{j<r}(1 - T^{2^j})$ the cleared value is exactly $\prod_{j<r}(S^{2^j} - R^{2^j})$. This closes at once every one-dimensional Prouhet partition, finite-difference mask, and sign pattern: high contact on a single near-unit *factorizes* into a power of the output gap. Bivariately, contact r places $S_1^d S_2^e F(z_1, z_2)$ in the ideal $(G_1, G_2)^r$ of the two gaps, with the primewise Newton floor $\tau_p(F) = \min_{\gamma_{ij} \neq 0}(v_p(\gamma_{ij}) + ig_{1,p} + jg_{2,p} + (d - i)s_{1,p} + (e - j)s_{2,p})$ as guaranteed divisor and the scalar product-formula criterion $\prod_p p^{\tau_p} \leq |J_F| \leq \sum |\gamma_{ij}||G_1|^i |G_2|^j S_1^{d-i} S_2^{e-j}$.

76.3 Stable structural support and the shared-pole Newton law

For $\ell \mid MA$ write $m_\ell = v_\ell(M)$, $n_\ell = v_\ell(A)$ and $u_{\ell,k} = v_\ell(z_k) = q_k n_\ell - p_k m_\ell = q_k(n_\ell - \vartheta m_\ell) + m_\ell(q_k \vartheta - p_k)$. The coefficient $n_\ell - \vartheta m_\ell$ is nonzero ($\vartheta \notin \mathbb{Q}$), so for large k the sign of $u_{\ell,k}$ is fixed: the reduced numerators and denominators of the adjacent near-units are supported on the *same* fixed sets $\mathcal{P}_+ = \{\ell : n_\ell > \vartheta m_\ell\}$ and $\mathcal{P}_- = \{\ell : n_\ell < \vartheta m_\ell\}$, both nonempty because $\sum_\ell (n_\ell - \vartheta m_\ell) \log \ell = \log A - \vartheta \log M = 0$, with every depth linear in q_k. Sufficiently deep adjacent pairs therefore have *no private denominator prime*; the near-cancellation $z_k \to 1$ is a balance between two fixed, linearly growing valuation profiles, not a matter of small valuations. Adjacent depths are coupled exactly ([kernel-verified Lean], `transport_q`, `transport_p`):

$$q_{k+1}u_{\ell,k} - q_k u_{\ell,k+1} = \varepsilon_k m_\ell, \qquad p_{k+1}u_{\ell,k} - p_k u_{\ell,k+1} = \varepsilon_k n_\ell, \qquad \varepsilon_k = p_{k+1}q_k - p_k q_{k+1} = \pm 1, \tag{791}$$

i.e. $z_k^{q_{k+1}}/z_{k+1}^{q_k} = M^{\varepsilon_k}$ and $z_k^{p_{k+1}}/z_{k+1}^{p_k} = A^{\varepsilon_k}$.

At a private prime ($p \mid S_1$, $p \nmid S_2$) the leading U-face $f_d(V)$ of $F = \sum f_i(V)U^i$ controls the first pole layer: if $v_p(S_2^e f_d(z_2)) < s = v_p(S_1)$ then $v_p(J_F) = v_p(S_2^e f_d(z_2))$ and the reduced denominator keeps $p^{ds - v_p(A_d)}$; a unit or monomial top face keeps the full pole, and cancelling the first layer at all private primes costs $\log P_1 \le eh(z_2) + \log \|f_d\|_1$. Cyclotomic faces $A_{\mathcal{M}}(V) = \prod_{m\in\mathcal{M}} \Phi_m(V)$ turn this into a finite order-cover condition $\mathrm{ord}_p(z_2) \in \mathcal{M}$ (and fixed-order factors supply only logarithmic depth). Stable support shows this private-prime model is a finite diagnostic only. The asymptotic law is the shared one: for $\ell \in \mathcal{P}_-$ and $F = \sum c_{ij}U^iV^j$,

$$v_\ell(J_F) \ge \lambda_\ell := \min_{c_{ij}\neq 0}\big(v_\ell(c_{ij}) + (d-i)s_{\ell,1} + (e-j)s_{\ell,2}\big), \tag{792}$$

with equality when the minimum is unique; when it ties, $v_\ell(J_F) > \lambda_\ell$ iff the explicit initial form $\sum_{\text{face}} \overline{c_{ij}\ell^{-v_\ell(c_{ij})}}\,\overline{R_1}^i\,\overline{R_2}^j\,\overline{S_1'}^{d-i}\,\overline{S_2'}^{e-j}$ vanishes in $\mathbb{F}_\ell$, and deeper cancellation repeats this layer by layer. A unit top-right corner keeps the whole pole $\ell^{ds_{\ell,1}+es_{\ell,2}}$; the positive mask $(U-1)^{2r} + (V-1)^{2r}$ keeps $\ell^{2r\max(s_{\ell,1},s_{\ell,2})}$ unless the depths tie. A nonzero binomial mask cannot vanish at both $(1,1)$ and (z_1,z_2) (adjacent relation vectors are a basis, so z_1, z_2 generate the rank-two group of M, A; [kernel-verified Lean] in the form `det_eq_zero_of_orthogonal`, `not_dvd_of_dvd_reduced_gap`).

76.4 Barriers and the reduced-denominator endgame

Along adjacent convergents, with $\Delta_n = \max(|x\delta_n|, |x\delta_{n+1}|)$ and the effective two-logarithm bound $\log(1/\Delta_n) = O_x(\log q_n)$:

Theorem 76.2 (Three barriers)**.** *(i) Full clearing cannot become subunit: for nonzero F_n of bidegree (d_n, e_n) and contact r_n, $S_n^{d_n}S_{n+1}^{e_n}\|\widetilde{F}_n\|_1(2\Delta_n)^{r_n} > 1$ for all large n, uniformly in F_n, because $\log S \gg q(d+e)$ while $r \le d+e$ and $\log(1/\Delta) = O(\log q)$. (ii) Structural primes are never shared by the reduced gaps of independent directions, and $\gcd(G_\nu, G_\omega) \mid \gcd(M^D-1, A^D-1)$ with $D = |\det(\nu,\omega)|$ (the gap-determinant transport of Section 71.10); for adjacent directions the shared gap is a fixed divisor of $\gcd(M-1, A-1)$, and for nonadjacent ones it is $e^{o(H)}$ by Bugeaud–Corvaja–Zannier [4], so contact-supplied shared divisors are exponentially too small. (iii) If the exact scalar criterion $r\log(1/(2\Delta)) > \log b_F + \log\|F(1+X, 1+Y)\|_1$ (with b_F the* reduced *denominator) succeeds infinitely often with subheight coefficients, then $\log b_{F_n} = O_x((d_n+e_n)\log q_n)$: the compression index $\log(B_F/b_F)/\log B_F$ must tend to one, and since $b_{F_n} = \prod_{\ell\in\mathcal{P}_-}\ell^{\rho_{\ell,n}}$ with $\rho_{\ell,n} = (d_n s_{\ell,n} + e_n s_{\ell,n+1} - v_\ell(J_{F_n}))_+$, the mask must capture a $1-o(1)$ fraction of the pole depth at* every *stable denominator prime.*

Research target 76.3 (Shared structural-face capture)**.** Construct non-toric masks F_k on adjacent pairs with subheight shifted coefficients, certified nonzero, whose successive initial forms (792) vanish deeply enough that $\sum_{\ell\in\mathcal{P}_-}\rho_{\ell,k}\log\ell = O_x((d_k+e_k)\log q_k)$; or prove the obstruction $\log b_F \ge c_x(d\log S_1 + e\log S_2) - O_x(r\log H)$ for every such mask, which would eliminate the architecture. The supports should respect the transport relations (791); the search is an integer program over shared Newton faces, not over arbitrary high-contact coefficients.

77 Rank-two rigidity: the pair-lattice defect and rational Hankel rays

This section integrates a further external report. Its core — the canonical primitive counterexample, the free monoid $\mathcal{E}_{\text{int}}(2,3) = \mathbb{N}_0 + \mathbb{N}_0\beta$ of common integral exponents, the group $\mathcal{E}_{\text{rat}}(2,3) = \mathbb{Z} + \mathbb{Z}\beta$ of common rational exponents, the uniqueness and leastness of β, and the reciprocal duality $\mathcal{E}_{\text{int}}(2^\beta, 3^\beta) = \mathbb{N}_0 + \mathbb{N}_0\beta^{-1}$ — is the exact conditional structure already established in Sections 4 and 5 and kernel-verified in `PrimitiveGenerator`, `RationalPowerIndex`, and `LeastSolution`; none of it is repeated here. What is new is a compact *pair-lattice* packaging of the descent invariant, its local Kummer form, and an exact classification of rational Hankel (three-term exponentiation) branches inside the conditional orbit. The finite algebra of the latter is **[kernel-verified Lean]** in `RationalHankelTriple`.

77.1 The saturation defect of a pair

For a counterexample pair $(A, B) = (2^x, 3^x)$ write $a_p = v_p(A)$, $b_p = v_p(B)$ and define the *saturation defect*

$$\delta_{\text{sat}}(A, B) = \gcd(\{a_p : p \neq 2\},\ \{b_p : p \neq 3\},\ b_3 - a_2). \tag{793}$$

The valuation lattice spanned by the axis vector of $(2, 3)$ and the vector of (A, B) has Smith normal form $\operatorname{diag}(1, \delta_{\text{sat}})$, so δ_{sat} is exactly the index of the saturation; and $d \mid \delta_{\text{sat}}$ holds iff there are a residue r (unique mod d) and integers C, D with

$$A = 2^r C^d, \qquad B = 3^r D^d, \tag{794}$$

in which case $y = (x - r)/d$ is again a solution with outputs (C, D). Extracting the full defect at once and then removing the axis content $\min(v_2(A), v_3(B))$ lands on the canonical primitive pair in a single step — no iterated descent is needed. This is the pair-lattice form of the rational-power index of Section 4 and of the ambient saturation index d of Section 10; the three notions agree on a counterexample orbit. For a descendant $\alpha = n + k\beta$ the invariants are exact: axis content n, defect k, localized pair (A_0^k, B_0^k) — so the coordinate k is *intrinsic* (the Smith index of the pair lattice of α), and a candidate pair with defect $k > 1$ is never primitive. The change of basis from $(1, \beta)$ to $(1, \alpha)$ lifts to the determinant lift $G \oplus \det G$ with Smith form $\operatorname{diag}(1, k, k)$: the rank-three bookkeeping carries no third invariant.

77.2 Local Kummer form and the defect-forcing criterion

The defect localizes prime by prime: $\ell^e \mid \delta_{\text{sat}}(A, B)$ iff the two valuation columns satisfy $g_1 \equiv r g_0 \pmod{\ell^e}$ for some residue r, iff

$$[A] = [2]^r, \qquad [B] = [3]^r \qquad \text{in } \mathbb{Q}^\times_{>0}/\mathbb{Q}^{\times \ell^e}_{>0}$$

for one common r. Hence for the *canonical* pair the two valuation columns are linearly independent over $\mathbb{F}_\ell$ for *every* prime ℓ: at least one of

$$\ell \nmid a_p \text{ (some } p \neq 2), \qquad \ell \nmid b_p \text{ (some } p \neq 3), \qquad a_2 \not\equiv b_3 \pmod{\ell}$$

holds for each ℓ.

Proposition 77.1 (Defect-forcing criterion)**.** *If every counterexample pair necessarily satisfies $\delta_{\mathrm{sat}}(A, B) > 1$ — equivalently, if some prime ℓ always carries a matched Kummer alignment $[A] = [2]^r$, $[B] = [3]^r$ modulo ℓ-th powers — then the conjecture follows.*

This is a clean endpoint for an "archimedean-to-Kummer bridge": the real relation would have to force one congruence of valuation columns at one prime. It is the mod-ℓ counterpart of the local common-exponent criterion of Section 71.11 (via Corrales-Rodríguez–Schoof, a common exponent at almost every prime characterizes the integral case), and the same caution applies: nothing currently produces the bridge, and the control principle of Section 88.6 is satisfied — the criterion's hypothesis genuinely fails for the controls, so it is not control-blind. A cheap but telling special case of the mod-2 statement: if $A \equiv B$ (mod squares) in the precise matched sense above, descent halves the exponent.

77.3 Rational Hankel triples and the square valuation pattern

A three-term exponentiation orbit U, $V = U^{\xi}$, $W = U^{\xi^2}$ is encoded by the logarithmic Hankel identity $(\log V)^2 = \log U \log W$. Two multiplicatively independent initial values with a *common* slope activate the six exponentials theorem: if both triples (U_i, V_i, W_i), $i = 1, 2$, satisfy the Hankel identity with the same $\xi = \log V_i / \log U_i$, all six entries are algebraic, and U_1, U_2 are multiplicatively independent, then $\xi \in \mathbb{Q}$. (For algebraic a, b this is the general-bases form of the escape criterion of Section 8: simultaneous algebraicity of $a, a^{\xi}, a^{\xi^2}, b, b^{\xi}, b^{\xi^2}$ forces $\xi \in \mathbb{Q}$ via Gelfond–Schneider plus six exponentials.) The rational-slope case then has an exact integral classification:

Theorem 77.2 (Square valuation pattern)**.** *Let U, V, W be positive integers with $U^r = V^s$ and $V^r = W^s$ for coprime $r, s \geq 1$ (the monomial recurrences of the slope $\xi = r/s$). Then there is a unique positive integer T with*

$$U = T^{s^2}, \qquad V = T^{rs}, \qquad W = T^{r^2};$$

equivalently $(v_p(U), v_p(V), v_p(W)) = d_p\,(s^2, rs, r^2)$ for every prime p. Moreover the positive solutions of $m_1^2 = m_0 m_2$ are exactly $(m_0, m_1, m_2) = d\,(u^2, uv, v^2)$ with u, v coprime.

Both statements are **[kernel-verified Lean]** (`exists_common_base_of_recurrences` — three applications of the coprime-exponent common-base lemma, no factorization bookkeeping — and `sq_eq_mul_parametrization`). The exponent vector (s^2, rs, r^2) is the primitive integer kernel of the two recurrence rows $(r, -s, 0)$, $(0, r, -s)$, whose 2×2 minors $r^2, -rs, s^2$ have gcd 1: the square pattern is lattice saturation in disguise.

Inside a conditional orbit the classification is global. For $t_i = n_i + k_i\beta \in \mathcal{E}_{\mathrm{rat}}(2, 3)$, $t_i \neq 0$, transcendence of β turns the numerical relation $t_1^2 = t_0 t_2$ into the coefficient identities $n_1^2 = n_0 n_2$, $2n_1k_1 = n_0k_2 + n_2k_0$, $k_1^2 = k_0k_2$, and these force all cross determinants $n_ik_j - n_jk_i$ to vanish: the three exponents lie on one rational ray $t_i = m_i\tau$ with $m_1^2 = m_0m_2$, hence, for positive integral exponents,

$$(t_0, t_1, t_2) = d\,(s^2, rs, r^2)\,\tau,$$

and the localized solution pairs are $(\mathbf{T}^{s^2}, \mathbf{T}^{rs}, \mathbf{T}^{r^2})$ with $\mathbf{T} = (A_0, B_0)^d$ after the axis factor is removed. The determinant identity

$$(n_0k_1 - n_1k_0)(n_2k_1 - n_1k_2) = 0$$

and the full ray conclusion for nonzero forms, together with the transcendence bridge, are **[kernel-verified Lean]** (`crossDet_mul_crossDet_eq_zero`, `hankel_ray`, `coeff_identities_of_transcendental`).

Corollary 77.3 (Quadratic closure inside the orbit is rational)**.** *Let $t \neq 0$ lie in $\mathcal{E}_{\mathrm{rat}}(2,3)$ and $\theta \in \mathcal{E}_{\mathrm{rat}}(2^t, 3^t)$. If also $\theta^2 \in \mathcal{E}_{\mathrm{rat}}(2^t, 3^t)$, then $\theta \in \mathbb{Q}$.*

Proof. With $u = t\theta$, $v = t\theta^2$ in $\mathbb{Z} + \mathbb{Z}\beta$ one has $u^2 = tv$; the ray classification puts t, u, v on one rational ray, so $\theta = u/t \in \mathbb{Q}$. □

This makes exact the reason fractional-linear manipulations cannot manufacture the missing β^2 column: the orbit ratios $(m+\ell\beta)/(n+k\beta)$ with nonproportional coefficient vectors are transcendental (Gelfond–Schneider), each changed base pair $(2^t, 3^t)$ again has the two-dimensional exponent group $(\mathbb{Z} + \mathbb{Z}\beta)/t$ (Section 12 in spectrum form), and any *quadratic* closure collapses to a rational slope. The composable family of these base changes is a groupoid model of every transformation obtainable without new arithmetic information — all of it strictly inside the six-exponentials ceiling of rank two.

77.4 Near-axis remarks and cross-references

For the canonical pair, $\min(a_2, b_3) = 0$ prevents simultaneous cancellation at the axis: at least one of $A^q/2^p$, $B^q/3^p$ is already in lowest terms with denominator exactly 2^p resp. 3^p, giving the elementary bound $|q\beta - p| \geq 1/(2^{p+1} \log 2)$ (resp. base 3) — far weaker than the height laws of Sections 68 and 76.1 but assumption-free. The report's "synchronized difference pairs" $\Delta_2 = 2^p - A^q$, $\Delta_3 = 3^p - B^q$ with $(\Delta_3/3^p)/(\Delta_2/2^p) \to \log 3/\log 2$ are precisely the synchronized germ gaps of Section 75; the two-channel theorem and the channel-descent proposition (Theorem 75.5, Proposition 75.6) are the sharpest known statements about coupling their prime divisors, and the reconnaissance of Section 88.8 found no further exploitable coupling. The report's research programs map onto standing targets: its Kummer-alignment program is Criterion 77.1; its nonlinear-orbit and functional-source programs are the escape criterion (Section 8) and Section 72.7; its synchronized-primitive-divisor program is Target 75.7; its monoid-leakage program packages the negative-coefficient contradictions already implicit in Section 4.

Assessment. The merged material consolidates the conditional structure in a tighter normal form and contributes two exact classifications (the local Kummer form of the defect with its every-ℓ independence statement, and the Hankel-ray/square-pattern theorems, now kernel-checked). It does not move the four-exponentials barrier; its value is that the defect-forcing criterion is a genuinely non-control-blind endpoint, and that quadratic closure inside the orbit is now excluded exactly rather than heuristically.

78 Bhargava factorials of the two–three semigroup

The corpus has repeatedly normalized interpolation determinants by ad hoc Smith factors and noted (Section 71.6 environs) that Bhargava orderings remained an undeveloped interface. This section integrates a report that develops it completely for the input semigroup

$$\mathcal{B} = \{2^a 3^b : a, b \in \mathbb{N}_0\},$$

computing the *exact* generalized factorial $k!_{\mathcal{B}}$ — hence the optimal universal denominator of integer-valued interpolation on the smooth nodes — and reformulating the conjecture as a subunit problem in the canonical lattice $\mathrm{Int}(\mathcal{B},\mathbb{Z}) = \{P \in \mathbb{Q}[T] : P(\mathcal{B}) \subset \mathbb{Z}\}$. Status: the local formulas, the cutoff, the normalized determinant, the defect theorems, and the localized no-go are **[paper proof]** paper proofs (with Bhargava's choice-invariance theorem [**classical/open background**]); the tables are **[exact computation]** with source; two finite cores added here are **[kernel-verified Lean]** (`SmoothSemigroupCore`).

78.1 The exact local factorial

For a prime p let $\nu_{\mathcal{B},p}(k)$ be Bhargava's minimal valuation of a k-th difference product, so that $k!_{\mathcal{B}} = \prod_p p^{\nu_{\mathcal{B},p}(k)}$ and a regular basis $F_0, F_1, \ldots$ of $\mathrm{Int}(\mathcal{B},\mathbb{Z})$ has leading coefficients $1/k!_{\mathcal{B}}$. Passing to p-adic closures and using that every closed subgroup of $\mathbb{Z}_p^\times$ (p odd) is procyclic, one gets, with $h_p(e) = |\langle 2,3\rangle \bmod p^e| = \mathrm{lcm}(\mathrm{ord}_{p^e} 2, \mathrm{ord}_{p^e} 3)$:

$$\nu_{\mathcal{B},p}(k) = \sum_{e\geq 1} \mathcal{B}igl\lfloor \frac{k}{h_p(e)} \mathcal{B}igr\rfloor \quad (p \geq 5), \qquad \nu_{\mathcal{B},3}(k) = v_3(k!), \tag{795}$$

the latter because 2 is a primitive root modulo 3^e, so $\mathcal{B}$ is dense in $\mathbb{Z}_3$ and the prime 3 gives *no gain* over $\mathbb{Z}$. At $p = 2$ the closure splits self-similarly, $K = \overline{\langle 3\rangle} \sqcup 2K$, giving an exact two-branch merge recursion for $a_k = \nu_{\mathcal{B},2}(k)$ (odd-branch costs $u_s = s + \lfloor s/2\rfloor + v_2(s!)$, even-branch costs $r + a_r$; the multiset of values is the ordered merge of the two streams) with first values $0,0,1,1,3,4,4,5,7,9,9,10,\ldots$ (independently reverified here) and, by a renewal argument for self-similar merges,

$$\lim_k \frac{\nu_{\mathcal{B},2}(k)}{k} = \frac{\sqrt{11}-1}{2} = 1.15831\ldots \tag{796}$$

(the equation $1/c = 1/(1+c) + 2/5$ from the odd-branch slope $5/2$). Every unramified prime carries strict excess over the ordinary factorial: $\alpha_p = \sum_e 1/h_p(e) \geq p/(p-1)^2$. A one-line pigeonhole *cutoff* makes the factorial finitely computable: for $p \geq 5$ and $R = \lfloor\sqrt{k}\rfloor$, if $h_p(e) \leq k$ then two of the $(R+1)^2$ box elements 2^a3^b ($a, b \leq R$) collide modulo p^e, so p^e divides a nonzero difference below 6^R:

$$h_p(e) \leq k \implies p^e < 6^{\lfloor\sqrt{k}\rfloor}. \tag{797}$$

The pigeonhole core of (797) is **[kernel-verified Lean]** (`modulus_lt_of_card_image_lt`). The resulting exact values (e.g. $k!_{\mathcal{B}}/k! = 1,1,1,1,5,2,7,2,5,20,22,92,10465,\ldots$; first proper-orbit prime 23 at $k = 11$ via $h_{23}(1) = 11$; exceptional orbit primes like 431 entering at degree 43) come with independent consistency checks. The same calculus applies verbatim to any two-generator semigroup — in particular to the *output* semigroup $\{M^aN^b\}$ of a counterexample, whose ramified branches at primes dividing MN obey analogous rooted merge systems with closed renewal slopes.

78.2 A counterexample as an isomorphism of factorial semigroups

Under a hypothetical solution the power map $q \mapsto q^x$ is an order-preserving monoid isomorphism from $\mathcal{B}$ onto the output semigroup, and both carry now-computable factorial spectra $(k!_{\mathcal{B}})$ and $(k!_{\mathcal{C}})$. The isomorphism controls all archimedean ratios exactly, while the two local spectra are a priori unrelated — Bhargava factorials are invariants of the embedded sets, not of the abstract monoid. *Factorial-spectrum rigidity* — find a property of order-preserving power isomorphisms forcing a quantitative relation between the two spectra — is a genuinely new formulation of the missing arithmetic–archimedean bridge, and it passes the control test of Section 88.6: for integer x the two spectra coincide up to the explicit change of generators.

78.3 The optimally normalized determinant and the factorial defect

For nodes $E = (q_0, \ldots, q_{d+1}) \subset \mathcal{B}$ and a regular basis, the determinant with columns $F_0, \ldots, F_d, t^x$ is, under a counterexample, a *nonzero integer* (each $F_j(q_i) \in \mathbb{Z}$, each $q_i^x \in \mathbb{Z}$; nonvanishing because $1, t, \ldots, t^d, t^x$ is a Chebyshev system on $\mathbb{R}_{>0}$), equal to the monomial determinant divided by $\prod_{j \le d} j!_{\mathcal{B}}$. Every m-node Vandermonde is divisible by $\prod_{j<m} j!_{\mathcal{B}}$, so each node set has an integer *factorial defect* $Q_{\mathcal{B}}(E) = |V(E)|/\prod_{j<m} j!_{\mathcal{B}}$, and the master archimedean inequality for a counterexample reads

$$Q_{\mathcal{B}}(E)\,(d+1)!_{\mathcal{B}}\,\mathcal{B}igl|\binom{x}{d+1}\mathcal{B}igr|\,\xi^{x-d-1} \;\ge\; 1 \qquad (\xi \in (\min E, \max E)). \tag{798}$$

The defect minima through eight nodes are exactly

$$1,\ 1,\ 1,\ 1,\ 1,\ 7,\ 7,\ 396,$$

by exhaustive exact certificates. The first failure is structural:

Theorem 78.1 (First factorial defect). *Every six-node subset of $\mathcal{B}$ has $Q_{\mathcal{B}}(E) \ge 7$, with equality for $E_6 = \{1, 2, 3, 4, 6, 8\}$. Hence $\mathcal{B}$ admits no simultaneous p-ordering prefix of length six, and the regular basis is not a Newton basis from degree* 6 *on (the first non-Newton basis element is explicit, glued by CRT from local orderings).*

The proof reduces, via a translation-invariant span bound, to 792 gap patterns, of which only $(0, 1, 2, 4, 5, 6)$ could beat 7; that pattern would need two triples of consecutive 3-smooth integers four apart, and *the only consecutive triples in $\mathcal{B}$ are* $\{1, 2, 3\}$ *and* $\{2, 3, 4\}$ — this last, decisive fact is **[kernel-verified Lean]** (`consecutive_threeSmooth_triples`). The verification $V(E_6) = 7{\cdot}345600$ and the denominator $\prod_{j \le 5} j!_{\mathcal{B}} = 345600$ are exact arithmetic.

The lattice reformulation is a closest-vector statement (*Bhargava locking*): for a counterexample, every $P \in \mathrm{Int}_{\le d}(\mathcal{B}, \mathbb{Z})$ misses the integer vector $(q_i^x)_i$ by at least 1 in ℓ^∞ on any $d+2$ nodes; the primitive left-kernel vector of the evaluation matrix is a dual certificate. Four finite criteria result (normalized determinant below $\prod j!_{\mathcal{B}}$; low-defect curvature (798) reversed; regular-basis CVP; paired input–output determinants for t^x and $t^{1/x}$), each unconditional as an implication.

78.4 Localization and the initial-form frontier

Over $R_6 = \mathbb{Z}[1/6]$ the full group $\{2^a 3^b : a, b \in \mathbb{Z}\}$ is dense in $\mathbb{R}_{>0}$ and its localized factorial keeps exactly the unramified part of (795). For a counterexample the localized determinant is a nonzero element of $\mathbb{Z}[1/(6MN)]$: every possible denominator is supported on the finitely many structural primes $S = \{p : p \mid 6MN\}$, so the conjecture follows from a single family of node sets with

$$|\Delta^\times|_\infty \cdot \mathfrak{d}_S(\Delta^\times) < 1.$$

But the natural construction fails for the familiar reason, here in sharp form: for a one-ratio geometric cluster $1, u, \ldots, u^{d+1}$ with $u = 2^a 3^{-b} \to 1$, the archimedean determinant decays only polynomially in b (like $|\log u|^{\binom{d+2}{2}}$), while termwise clearing at the structural primes costs exponentially in b. This is the Bhargava-normalized incarnation of the height mismatch of Sections 68 and 76.4, with all universal denominators already removed; what survives is precisely a cancellation target: at each $p \in S$, write $\Delta^\times = p^{m_p}(I_p + pJ_p)$ with m_p the minimal term valuation — the *initial-form bridge* asks for families where the archimedean logarithm is $-A_d + o(A_d)$ while the initial forms I_p vanish to weighted depth exceeding the termwise budget by $A_d/\log p$. This is the regular-basis twin of the shared-pole Newton endgame (Target 76.3 and (792)); the two formulations should be attacked together, and the LP/Farkas methodology of Section 88.10 applies to both.

78.5 Subquadratic growth and the two-prime synchronization gap

A second, independently written report derives the same local calculus — identical formulas at every prime, the identical two-adic value stream, and the identical slope $(\sqrt{11}-1)/2$ — which functions as an external cross-check of Section 78.1. It adds three results that complete the picture.

First, the lifting depths are a gcd sequence: for $p \geq 5$ with first orbit size $d_p = h_p(1)$, the saturation depth satisfies $c_p = v_p(G_{d_p})$ where $G_d = \gcd(2^d - 1, 3^d - 1)$, so Bugeaud–Corvaja–Zannier [4] bounds the whole factorial:

$$\log(k!_{\mathcal{B}}) \leq O(k) + 2k \sum_{d \leq k} \frac{\log G_d}{d} = o(k^2). \tag{799}$$

The intrinsic local denominator of the smooth semigroup is therefore *subquadratic*: small enough that exponential-height nodes would beat it.

Second, the *conditional scalar bridge*: if there were simultaneous p-ordering prefixes $A_j \subset \mathcal{B}$ of lengths $n_j \to \infty$ whose least node satisfies $\log \min A_j \geq cn_j$, then the integer $n_j!_{\mathcal{B}}\,[A_j]\,t^x$ (integral by the unit-triangular Newton basis) would have absolute value $n_j!_{\mathcal{B}}\,|\binom{x}{n_j}|\,\xi^{x-n_j} \to 0$ by (799) — proving the conjecture.

Third, and decisively, *no such family exists*, for a reason more structural than the six-node defect: the two structural primes impose incompatible equilibrium populations. An asymptotically optimal dyadic ordering must have odd-node proportion $q_2 = (\sqrt{11}-1)/5 = 0.4633\ldots$ (from the self-similar merge slopes), while an asymptotically optimal triadic ordering must have 3-unit proportion $2/3$; on $\mathcal{B} \setminus \{1\}$ the odd nodes and the 3-unit nodes are *disjoint* ($2^a 3^b$ is odd iff $a = 0$, a 3-unit iff $b = 0$), and

$$q_2 + \tfrac{2}{3} = 1 + \tfrac{3\sqrt{11}-8}{15} > 1.$$

Quantitatively, with pair energies $\mathcal{E}_p(A) = \frac{2}{N^2}\sum_{s<t} v_p(s-t)$, every N-node set $A \subset \mathcal{B} \setminus \{1\}$ pays

$$\log 2\,(\mathcal{E}_2(A) - \mathcal{E}_2^{\min}) + \log 3\,(\mathcal{E}_3(A) - \mathcal{E}_3^{\min}) \;\geq\; \kappa + o(1), \qquad \kappa = \eta^2 \frac{ab}{a+b} = 0.0236\ldots, \tag{800}$$

($\eta = q_2 - \frac{1}{3}$, $a = \frac{\sqrt{11}+6}{2}\log 2$, $b = \frac{9}{4}\log 3$): at least $\frac{1}{2}\kappa N^2$ extra logarithmic Vandermonde mass beyond the separated local minima. This explains the defects $7, 7, 396$ of Section 78.3 asymptotically and closes the scalar-bridge route unconditionally — another entry for the closed-routes ledger of Section 88, with the same conclusion as always: the surviving frontier is output-sensitive cancellation, not sharper counting. A near-geometric chain $s_j = 2^{ju}3^{(n-j)v}$ makes the same point in exact form: its 2- and 3-adic interpolation tax is $\frac{n(n-1)}{2}(u\log 2 + v\log 3)$ against node heights $\asymp nu\log 2$, cancelling all but one power of the archimedean decay — consistent with the one-ratio cluster no-go of Section 78.4.

Assessment. The Bhargava calculus settles, exactly and permanently, the universal local-divisibility side of every interpolation attack on the smooth nodes: no future determinant experiment needs to guess its denominator normalization again, node sets can be scored by their exact defect $Q_{\mathcal{B}}$, and automatic output-prime mass is separated from genuine signed cancellation. It does not move the barrier — its own no-gos show the localized cluster route fails by the standard exponential-vs-polynomial margin, and the simultaneous-ordering route fails by the two-prime equilibrium clash (800) — but it replaces "try a larger determinant" by two canonical open interfaces: factorial-spectrum rigidity for the input–output isomorphism, and structural-prime initial-form cancellation in the optimally normalized lattice.

79 Additive isolation and the weighted smooth-support series

A further external report develops three themes. Its first — an "algebraic-orbit filtration" $\mathrm{Orb}_r(x)$ of bases whose first r iterates under $\alpha \mapsto \alpha^x$ stay algebraic, with ranks $2, \le 1, 0$ at $r = 1, 2, 3$, the rank identity $\operatorname{rank}\langle 2, 3, M, A\rangle = 4 - \operatorname{rank}\mathrm{Orb}_2(x) \in \{3, 4\}$, and the Möbius form of the exponent in the dependent branch — is precisely the kernel-verified spectrum/kernel/tower chain of Sections 12, 14, and the depth-three tower theorem (`TowerRigidity`), together with the rank-geometry Möbius parametrization and its height bound in Section 66; it is not repeated. The two remaining themes are new to this report and are recorded here.

79.1 Additive isolation

The spectrum theorem makes u^x algebraic exactly for 3-smooth positive rationals u. Combining it with the classification of *consecutive* smooth integers gives an exact additive statement:

Proposition 79.1 (Consecutive smooth pairs; additive isolation)**.** *The positive integers u with u and $u+1$ both 3-smooth are exactly $u \in \{1, 2, 3, 8\}$ (pairs $(1, 2), (2, 3), (3, 4), (8, 9)$). Consequently, under a hypothetical counterexample, $(u+1)^x$ is algebraic for a smooth u exactly when $u \in \{1, 2, 3, 8\}$, and transcendental otherwise; likewise $(u-1)^x$ is algebraic exactly for $u \in \{2, 3, 4, 9\}$. The longest run of consecutive smooth integers is $1, 2, 3, 4$.*

The external report proves the pair classification by invoking Mihăilescu's theorem; that is far more than needed. The equations $2^a = 3^b+1$ and $3^d = 2^a+1$ are elementary (reduce modulo 3 resp. 8, force an even exponent, and factor y^2-1 into two powers of the other base differing by 2), and the whole classification is **[kernel-verified Lean]** in `SmoothSemigroupCore` (`pow_two_eq_pow_three_add_one`, `pow_three_eq_pow_two_add_one`, `consecutive_threeSmooth_pairs`), alongside the triple classification used by the six-node defect. Additive isolation sharpens the "one more base" picture of Section 8: a counterexample is multiplicatively closed but additively barren — any derivation of algebraicity of 5^x, or of the successor of any nonexceptional smooth integer, is already a complete proof, not an auxiliary step.

79.2 The weighted smooth-support series

The report encodes a hypothetical solution in the weighted series

$$H_{M,A}(z) = \sum_{a,b\ge 0} M^a A^b \, z^{2^a 3^b} \in \mathbb{Z}[[z]],$$

whose coefficient at a smooth exponent s is exactly s^x. It satisfies an exact mixed 2–3 Mahler identity (inclusion–exclusion over the two dilations $z \mapsto z^2$, $z \mapsto z^3$), extending the finite mixed-Mahler algebra of `MixedMahlerFinite`. Two quantitative results are proved about the automatic side: the number of binary and ternary Cartier states of the unweighted support series grows exponentially (so the support is far from 2- or 3-automatic — quantifying the carry-language analysis of Section 70), and the reductions of $H_{M,A}$ modulo a prime ℓ are classified: they are algebraic over $\mathbb{F}_\ell(z)$ exactly according to the ℓ-divisibility geometry of M and A. The diagnosis matches the corpus: finite characteristic sees the divisibility geometry of the outputs, while the conjecture is an archimedean equality of two pole abscissae; the missing ingredient is a bridge between the two layers, the same missing coupling as everywhere else in this report. **[paper proof]** throughout this subsection; the elementary saturation and normalization layers of the external report duplicate kernel-verified corpus modules and were not re-integrated.

80 The external-prime Newton wall and nonarchimedean ray rigidity

A further external report opens the first genuinely p-adic *analytic* layer of the corpus: it works at an external prime of a hypothetical counterexample, where the sampled input geometry acquires an accumulation point that the real picture lacks. Its elementary preliminaries (positivity, irrationality, and transcendence of a nonintegral x; multiplicative independence of $2, M$ and of $3, A$; existence of an external prime $p \geq 5$ dividing MA) are kernel-verified corpus facts (Sections 8 and 12) and were not re-integrated. Everything below consumes the external prime itself, so the control principle of Section 88.6 is passed *by construction*: for an integer control $(2^d, 3^d)$ no prime $p \geq 5$ divides MA and the package cannot even be formed.

80.1 Oriented data and nonarchimedean ray rigidity

Orient the data at an external prime $p \geq 5$: if $p \mid M$ set $(Q, R; \alpha, \lambda) = (M, A; 2, 3)$, otherwise $(A, M; 3, 2)$; in both cases

$$m = v_p(Q) > 0, \qquad a = v_p(R) \geq 0, \qquad F(Q^k) = R^k, \qquad F(\alpha Q^k) = \lambda R^k$$

are the two rows of sampling data that any local interpolation F of the conditional grid would have to satisfy near the p-adic accumulation point $Q^k \to 0$. Because $p \geq 5$, the numbers $2, 3, \alpha, \lambda, \alpha - 1$ are all p-adic units. The report proves:

Theorem 80.1 (Geometric-ray rigidity and the edge law)**.** *Let K be a complete nonarchimedean field of characteristic zero and $0 < |t| < 1$.*

1. *If an analytic germ F at 0 satisfies $F(t^k) = s^k$ for all large k, then $s = t^d$ and $F(z) = z^d$ for a unique $d \in \mathbb{N}_0$. A meromorphic germ forces $s = t^d$ with $d \in \mathbb{Z}$; a germ on a finite ramified cover $z = w^e$ forces $s^e = t^d$.*
2. *If F is analytic on a disk D with $qD \subseteq D$ and $F(qt^k) = \mu F(t^k)$ for infinitely many k, then the Taylor support of F lies in the spectrum $\{n : q^n = \mu\}$; if the spectrum is empty, then $F \equiv 0$.*

Consequently no nonzero analytic, meromorphic, or finite-Puiseux germ at 0 can carry even the single row $F(Q^k) = R^k$ (the forced relation $R = Q^d$ contradicts the kernel-verified multiplicative *independence of the outputs), and none can carry both rows: the edge pairs* $(\alpha, \lambda) = (2, 3)$ *and* $(3, 2)$ *have empty spectrum.*

The two arithmetic pivots are kernel-checked in the new module `ExternalPrimeNewtonWall`. First, *unit-convergence rigidity*: in any normed division ring, a unit-norm element whose power sequence converges equals 1 (`eq_one_of_tendsto_pow`); this is the step that turns "the powers of the unit R/Q^d converge to the nonzero leading coefficient" into the exact equality $R = Q^d$. Second, the *formal edge law*: for formal power series over any commutative ring the dilation-eigenvalue identity $F(qz) = \mu F(z)$ forces $(q^n - \mu) \cdot c_n = 0$ for every n (`edge_spectrum`); over a field the support lies in the spectrum (`eq_zero_of_edge_law`); and for the Alaoglu–Erdős edge pairs the spectrum is empty over $\mathbb{Q}$ — $2^n \neq 3$ and $3^n \neq 2$ — so the only formal solution is zero (`no_formal_edge_realization`). What remains of the analytic statements beyond these cores is the classical nonarchimedean identity theorem for accumulating zeros (Strassmann), **[classical/open background]**.

80.2 The quantitative wall: exact valuations and last-term dominance

Rigidity says that no exact local model exists; the report's central computation measures *how* every polynomial approximation fails. Let

$$c_n(Q, R) = \frac{\prod_{j<n}(R - Q^j)}{Q^{\binom{n}{2}} \prod_{j=1}^{n}(Q^j - 1)}$$

be the q-Newton coefficients of the native row, so that $P_N(z) = \sum_{n \le N} c_n \prod_{j<n}(z - Q^j)$ is the unique interpolant of degree at most N through $(Q^k, R^k)_{0 \le k \le N}$. With the threshold $q_* = \lfloor (a-1)/m \rfloor$ and a single tie excess ε (nonzero only when $m \mid a$, where $\varepsilon = v_p\Big(R - Q^{a/m}\Big) - a$), the report computes the valuation of *every* coefficient exactly: $v_p(c_n) = v_p(R-1) - m\binom{n}{2}$ when $a = 0$, and

$$v_p(c_n) = m\binom{q_*+1}{2} + a(n - 1 - q_*) + \varepsilon - m\binom{n}{2} = -\frac{m}{2}\,n^2 + \Big(a + \frac{m}{2}\Big)n + O(1) \qquad (n \ge q_* + 2)$$

when $a > 0$. Three consequences form the wall:

1. *Universal quadratic denominator escape.* The least D_n with $D_n c_n \in \mathbb{Z}_{(p)}$ has $v_p(D_n) = \frac{m}{2}n^2 - O(n)$; and by a Gauss-norm division argument this is not a Newton-basis artifact: *every* rational polynomial of arbitrary degree agreeing with the native values at $Q^0, \ldots, Q^N$ has a coefficient of p-adic denominator exponent at least $-v_p(c_N) = \frac{m}{2}N^2 - O(N)$.
2. *Off-ray explosion with last-term dominance.* At an adjacent-row point $z = uQ^k$ ($u \in \mathbb{Z}_p^\times$, $u \neq 1$, fixed k) the Newton terms have exact eventual ratio valuation $v_p(T_{n+1}/T_n) = a + mk - mn < 0$, so the term valuations strictly decrease and the partial sum inherits the valuation of its *last* term — the same ultrametric unique-minimum mechanism as the coefficient tax of `PrimeTopIsolation` — giving $v_p\Big(P_N(uQ^k)\Big) = -\frac{m}{2}N^2 + (a + m(k + \frac{1}{2}))N + O(1) \to -\infty$.
3. *Adjacent-row repulsion.* Hence, for all large N, $v_p\Big(P_N(\alpha Q^k) - \lambda R^k\Big) = v_p\Big(P_N(\alpha Q^k)\Big)$: the native interpolants are not merely inaccurate at the second row, they are quadratically p-adically *repelled* from the prescribed integers.

An exact rational audit (`fractions.Fraction`, no floating point; script with pinned SHA-256 digest reproduced in the source report) confirms the formulas in three exemplar regimes, including the delicate valuation-tie case. **[paper proof]** for the exact ledger; the valuation layer is Lean-ready q-binomial bookkeeping and is recorded as a pending target in Section B.

Why the wall is not yet a contradiction. The wall shows that the canonical archimedean completion of the native data has no p-adic convergence off the ray. It does not show that the complex value at an off-ray rational point equals a p-adic sum of the same terms: a series of rationals may converge in two completions to different limits, and at the adjacent row the series does not truncate, so equality is not place-independent. The product formula does not close the gap either: a nonzero rational residual may carry a p-denominator of size $p^{\Theta(N^2)}$ while being only geometrically small archimedeanly, so the elementary denominator lower bound $\exp(-\Theta(N^2))$ sits far below the archimedean error $\exp(-\Theta(N))$. Quadratic escape therefore *defeats* naive unit-locking rather than helping it — the shortcut "the entire function has rational data, so evaluate it p-adically" is closed, and any repair needs a separate overconvergence or algebraicity theorem. This is the p-adic face of the coupling problem diagnosed throughout this report.

80.3 Tate edge operator, finite-strip defects, and overconvergence

On polynomials the edge law is diagonal and quantitative. For $\mathcal{L}_{\alpha,\lambda}P(z) = P(\alpha z) - \lambda P(z)$ the monomial eigenvalues are $\alpha^n - \lambda$, so on degrees $\leq d$ the operator has the finite spectral gap

$$\sigma_{p,d}(\alpha, \lambda) = \min_{0 \leq j \leq d} |\alpha^j - \lambda|_p > 0,$$

positive precisely because $2^j \neq 3$ and $3^j \neq 2$ — the same finite fact as the empty edge spectrum, kernel-checked in `ExternalPrimeNewtonWall`. The report proves the resulting weighted-Gauss-norm inequality $\|\mathcal{L}_{\alpha,\lambda}P\|_\rho \geq \sigma_{p,d}(\alpha,\lambda)\|P\|_\rho$, extends it to Tate algebras on dilation-stable disks when the resonance statistic $\mu_{p,d} = \max_{j \leq d} v_p(\alpha^j - \lambda)$ stays bounded, and derives a *finite-strip defect theorem*: if a polynomial of degree $\leq d$ satisfies the second-row law on a strip $K \leq k \leq N$ and one native value, then in the forced factorization $\mathcal{L}_{\alpha,\lambda}P = \Phi_{K,N}S$ the quotient obeys $\|S\|_\rho \geq \sigma_{p,d}(\alpha,\lambda)\,|R|_p^K\,\rho^{-(N-K+1)}$. For the canonical two-row interpolants the defect quotient thus grows at least geometrically at nonresonant external primes: there is *no tame defect quotient*. A diagonal compactness argument upgrades this to: no family F_N analytic on one common disk of radius $\rho > 1$ can match even the one-row prefixes with bounded norms (*no uniformly overconvergent interpolation family*). The report closes this layer with two explicitly open interfaces — a removable-singularity (Robba-to-Tate) bridge and a coefficient-growth criterion — either of which would convert the wall into a contradiction; both are recorded as targets, **[proposed target]**.

80.4 The finite-rank two-dilation target

A genuine interpolation of the conditional grid would satisfy two dilation laws at once,

$$F(2z) = 3F(z), \qquad F(Mz) = AF(z),$$

with multiplicatively independent dilation parameters. The formal obstruction is the edge law again: a Laurent series solving the first equation alone is already zero ($(2^n - 3)c_n = 0$), while on the universal cover of $\mathbb{C}^\times$ the nonzero solution z^x exists — the gap is local regularity and monodromy, not consistency. Rigidity theorems for consistent pairs of independent q-difference systems [34, 35, 36] reduce sufficiently regular finite-rank systems to rational or constant normal forms, which the finite-Puiseux rigidity of Theorem 80.1 would then exclude; but the integer table is a discrete semigroup of values, not a difference module over $\overline{\mathbb{Q}}(z)$, and the geometric Newton completion satisfies only a one-axis *inhomogeneous* law with a q-product defect. The report therefore formulates the precise missing object as a target: a finite vector Y of arithmetic interpolation functions with rational transition matrices $B_2(z), B_M(z)$ obeying the cocycle consistency $B_2(Mz)B_M(z) = B_M(2z)B_2(z)$, one coordinate carrying the exact conditional grid, and regular-singular or overconvergent behavior at an external prime. **[proposed target]**; the current defect family contains nonrational infinite q-products and is not known to close in finite rank.

80.5 Assessment

The external-prime report adds the corpus's first quantitative nonarchimedean analytic layer, and it is control-clean by construction. Its permanent content is threefold: the ray-rigidity/edge-law pair, whose finite cores are now kernel-checked (`ExternalPrimeNewtonWall`) and whose analytic completions are classical; the exact valuation ledger, which shows that every polynomial or overconvergent model of the conditional data pays a quadratic p-adic price ($\frac{m}{2}N^2 - O(N)$) precisely where the archimedean models converge geometrically; and the sharpened statement of the obstruction — sampling is not regularity, and the two completions are decoupled at exactly the quadratic-versus-geometric scale. It does not advance the four-exponentials barrier: the wall excludes local models but produces no equality between completions, and its author is explicit that transferring the complex completion into a p-adic germ without a new arithmetic compatibility theorem would be circular. What it contributes to the frontier of Section 89 is a pair of well-posed interfaces (overconvergence bridge, finite-rank edge cocycle) that are *output-sensitive* in the required sense: both consume the external prime, and both would, if closed, terminate the conjecture through the kernel-verified rigidity layer rather than through counting.

81 The global terminal-prime contraction: the factorial route is closed

A further external report resolves the central open question of the factorial-cocycle program (Section 74) — and it resolves it *negatively, in full generality*. Where the session log had emptied the constructive branch (Corollary 88.2) and the LP layer for the control (Section 88.10), the new report closes the entire family in one stroke: no nonzero integer polynomial with $P(M) = 0$ ever makes the terminal factorial cocycle integral, uniformly in degree, span, height, and sign pattern.

81.1 The contraction theorem

Recall the objects of Section 74: $F_t = (A^t)_{\underline{M^t}}$ is the terminal falling factorial, $\mathcal{F}_{P,n} = \prod_j F_{n+j}^{c_j}$ for $P = \sum_j c_j T^j \in \mathbb{Z}[T]$, and $I_t = (A^t - M^t, A^t]$ is the terminal prime window, whose relative length exponent $\log 2/\log 3 = 0.6309\ldots > 7/12$ makes Huxley's theorem applicable: I_t contains $(1 + o(1))M^t$ of logarithmic prime mass. A prime $p \in I_t$ has $V_p(t) := v_p(F_t) = 1$ and $V_p(s) = 0$ for $s < t$ (both kernel-verified, `TerminalFactorialIndependence`), while at higher levels Legendre's interval formula gives the uniform transport bound

$$V_p(t+\ell) \le 2q^t M^\ell + 2\Big(1 + \frac{\ell}{t}\Big), \qquad q = \frac{M}{A} < 1. \tag{801}$$

Theorem 81.1 (Global terminal-prime contraction)**.** *Let (M, A) arise from a hypothetical counterexample (so $M \geq 2^{14}$ by the kernel-verified region of Section 3.2). There is $n_0(M, A)$ such that for every $n \geq n_0$ and every nonzero $P \in \mathbb{Z}[T]$ with $P(M) = 0$,*

$$\mathcal{F}_{P,n} \notin \mathbb{Z};$$

indeed for some negative coefficient index k every prime of I_{n+k} divides the reduced denominator, so $\log \operatorname{den}(\mathcal{F}_{P,n}) \geq (1 - o(1))M^{n+k}$.

Proof. Write $c_j^{\pm}$ for the positive and negative parts and $S_{\pm} = \sum_j c_j^{\pm} M^j$; the single archimedean input $P(M) = 0$ is exactly the mass balance $S_+ = S_- > 0$. Suppose $\mathcal{F}_{P,n} \in \mathbb{Z}$. For each k with $c_k < 0$ pick $p \in I_{n+k}$ (Huxley). Since $V_p(n + k) = 1$, lower levels vanish, and other negative coefficients only help, integrality forces

$$c_k^- \leq \sum_{j>k} c_j^+ V_p(n + j) \leq 2q^{n+k} \sum_{j>k} c_j^+ M^{j-k} + 2 \sum_{j>k} c_j^+ \Big(1 + \frac{j - k}{n + k}\Big)$$

by (801). Multiply by M^k and sum over all negative k. The transported-multiple kernel telescopes against the geometric series $\sum_k q^{n+k}$, and the prime-power allowance kernel against $\sum_{\ell \geq 1} M^{-\ell}$ and $\sum_{\ell \geq 1} \ell M^{-\ell}$:

$$S_- \ \leq\ \Big(\frac{2q^n}{1 - q} + \frac{2}{M - 1} + \frac{2M}{n(M - 1)^2}\Big) S_+.$$

With $S_+ = S_-$ this is $1 \leq \kappa_n$, and $\kappa_n < 0.008$ for $M \geq 2^{14}$, $q \leq (2/3)^{14}$ — a contradiction. No factor of the degree or of the number of negative coefficients ever appears: that is the decisive feature of the M-weighting. For the quantitative form, if the displayed inequality failed for every negative k the same summation would reproduce $1 \leq \kappa_n$; hence it fails for some k, and the transport bound is uniform over $p \in I_{n+k}$. □

Corollary 81.2 (Negative resolution of factorial feasibility)**.** *Target 74.3 is closed unconditionally: for all large n the conjunction $P_n \neq 0$, $P_n(M) = 0$, $v_p(\mathcal{F}_{P_n,n}) \geq 0$ for every prime p is impossible — the span condition, the near-unit condition, the further annihilations $P_n'(M) = P_n(M^2/A) = 0$, and any additional exact mode cancellations are all superfluous. The binomial-cocycle and no-collision variants close identically.*

The report also proves the sharp boundary statement *before* the mass balance is used: if the span satisfies $D_n \leq (\log_2 3 - 1)\, t_n - h_n$ with $h_n \to \infty$ ($t_n = n + k_n$), then the denominator retains $(-c_{n,k_n})(\Delta_0 - o(1))M^{t_n}$ of logarithmic prime mass with $\Delta_0 = 1 - 8q/(\delta_0(1 - q)) \geq 0.9337$, $\delta_0 = 2 - \log_2 3$ (Brun–Titchmarsh at relative thickness δ_0; Huxley for the total) — so an integral cocycle would need $D_n \geq (\log_2 3 - 1)\, t_n - O(1)$, and canonical Hermite stencils at first appear to live in exactly that bounded strip. A complete-homogeneous remainder identity ($T^N - H_{\mathcal{R}}(T^N) = D_{\mathcal{R}}(T) h_{N-d}(T, \mathcal{R})$, all tail coefficients positive) then shows the stencils have a unique negative coefficient of the wrong multiplicity, and the contraction supersedes the whole phase diagram anyway. The mesoscopic inequality $\frac{1}{2} < \log 2/ \log 3 < \frac{2}{3}$ (kernel-verified, `FactorialCocycle`) silently underlies both windows.

81.2 Kernel verification of the contraction core

The new module `CriticalWindowContraction` kernel-checks the complete finite skeleton of Theorem 81.1 in division-free form:

- `mass_balance`, `sum_cpos_pos`: the sign-mass identity $S_+ = S_- > 0$ from $\sum_j c_j M^j = 0$, $P \neq 0$;
- `geom_mul_le_one`, `geom_pow_mul_le`, `weighted_geom_pow_mul_le`: the two summable kernels $\sum_k u^k$ and $\sum_\ell \ell M^{-\ell}$ as exact polynomial inequalities;
- `sum_Ioc_swap`: the exchange of window and coefficient summations;
- `contraction_mul`: the abstract contraction — for any coefficient vector that is M-balanced, nonzero, and obeys the transport inequalities with parameters $(\varepsilon, u, \beta, \gamma)$, necessarily $(1-u)(M-1)(M-2) \leq \varepsilon(M-1)(M-2) + \beta(1-u)(M-2) + \gamma(1-u)(M-1)$;
- `kappa_mul_lt` and `no_balanced_transport`: with the Legendre parameters $\varepsilon = 2q^n$, $\beta = \gamma = 2$ the opposite strict inequality holds whenever $M \geq 2^{14}$ and $q \leq (2/3)^{14}$, so *no such coefficient vector exists*;
- `exponent_gap_dvd`: the rational-root exponent-gap divisibility $v^{N-b} \mid a$ used by the report's Hermite-stencil diagnostics.

The regime constants are exactly the corpus's kernel-verified zero-free region: a counterexample has $x \geq 14$, hence $M = 2^x \geq 2^{14}$ and $q = (2/3)^x \leq (2/3)^{14}$ (`FiniteCheck16384`). What remains outside the kernel is the arithmetic instantiation of the transport inequalities at window primes: $V_p(t) = 1$ and vanishing below the window are already kernel-verified (`TerminalFactorialIndependence`); the Legendre-interval upper bound (801) is elementary and Lean-ready; Huxley's prime-window theorem stays [**classical/open background**].

81.3 What survives: inflation-scale hybrids and two-dimensional transport

The contraction has four essential hypotheses: a one-parameter shift family, unit window multiplicity, geometric transport, and the same-base balance $P(M) = 0$. The report formulates the surviving interfaces as precise targets. First, a *two-dimensional coefficient transport*: archimedean cancellations $\sum_{ij} c_{ij} M^i A^j = 0$ with a two-parameter terminal product, where the transport kernel becomes a genuinely two-dimensional operator whose $M^i A^j$-weighted norm may or may not contract — either outcome is informative (a first finite exploration and a reduction to a top-cell pivot: Section 88.11). Second, a *coefficient-inflated Kummer dual*: the contraction shows that any near-feasible vector must carry exponentially inflated positive tails, which changes the comparison scale for the structural-prime route — a carry-free Kummer valuation of size $O(n)$ ((784), kernel-verified) multiplied by inflated coefficients is no longer negligible against a terminal window, so a hybrid dual combining terminal-window rows with one structural row deserves a fresh Farkas search *without* a span bound. Third, the report independently adopts the control principle of Section 88.6 as a design rule and confirms that the contraction itself is satisfied by the controls $(2^m, 3^m)$: it closes a method, and a proof of the conjecture must consume the external prime, the least-generator decomposition, or spectrum saturation.

Assessment. This merge upgrades the corpus in the strongest sense available for a negative result: a named open target (Target 74.3) becomes a theorem with a kernel-checked finite core, and two session-log partial closures (Corollary 88.2, the LP experiment) are subsumed and explained — the critical constant $\log_2 3 - 1$ was never the true obstacle; the M-weighted mass balance was. The conceptual residue is permanently useful: any auxiliary-integer construction whose archimedean cancellation shares a base with its coefficient weighting is subject to the same contraction, so future designs must either decouple the bases (two-dimensional transport) or exploit coefficient inflation at a structural prime. The conjecture itself is untouched — the closure holds for controls too — but the search region left to the factorial idea is now exactly two well-posed interfaces, both primitive-sensitive only through the standard three channels.

82 Compact charts, the exponent-two ceiling, and the relation-counting lock

A further external report recasts the conjecture as a rational-point counting problem on a *fixed compact curve*, and audits the strongest counting technology available against it. Its foundational layer — the canonical least generator β, the conditional monoid $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$, and the shell counts $\#(\mathcal{S} \cap [N, N+1)) = 1 + \lfloor (N+1)/\beta \rfloor$ — is the kernel-verified corpus monoid (Section 4.3) and is inherited, not re-proved. What is new is the geometry built on top of it and two sharp negative ledgers.

82.1 Two rational charts and the counting dichotomy

The *reciprocal chart* sends $t \mapsto (2^{-t}, 3^{-t})$, compactifying the power curve $y = x^\vartheta$ into the unit square with a cusp at the origin; solutions land exactly on the points with unit numerators. The *logistic (Farey) chart* sends $t \mapsto (\frac{m}{m+1}, \frac{n}{n+1})$ with $(m,n) = (2^t, 3^t)$; solutions land exactly on the unit-gap points, and the two charts are rationally conjugate. Both closures are C^1 but not C^2 at the cusp, with curvature blowing up like $s^{\vartheta-2}$ — the entire difficulty of the counting problem concentrates at one non-smooth boundary point. On either chart the conjecture becomes a *dichotomy on a single fixed compact curve*: marked-point counts of height $\le T$ grow like $T/\log 3$ under the conjecture and like $T^2/(2\beta(\log 3)^2)$ under failure, so

$$\text{Conjecture 1.1} \iff N_{\text{marked}}(T) = o(T^2) \iff Z_{\text{unit}}(s) = o(s^{-2}) \ (s \downarrow 0) \iff \#\mathcal{S}_N = o(N),$$

where Z_{unit} is the height zeta function of the marked points and $\mathcal{S}_N$ the shell occupancy. The report assembles an eight-item certificate package of such equivalences (marked counts on either chart, zeta pole order, shell occupancy, a one-sided S-integral identity, absence of a second diagonal contraction, two-sided foreign valuations in a rank-two group), together with strictly stronger sufficient statements: any $o(T^2)$ bound for *all* rational points on the compact cusp, or $o(N)$ points per shell, proves the conjecture, while six exponentials gives the unconditional ceiling $O(T^2)$ — the rank of the full rational-point group Γ_ϑ of $y = x^\vartheta$ is at most two, and the count is $O(T^r)$ for rank r. Subquadratic all-rational counting is therefore *exactly* the four-exponentials rank drop in counting clothes: it is equivalent to $\Gamma_\vartheta = \langle(2,3)\rangle$, the specialized four-exponentials statement of Section 6. The cleanest member of the package is the *one-point S-integral target*: the conjecture is equivalent to

$$\Gamma_\vartheta \cap \left(\mathbb{Z}[1/2]_{>0} \times \mathbb{Z}[1/3]_{>0}\right) = \langle(2,3)\rangle,$$

so it suffices to exclude a *single* noncyclic rational point whose denominators are concentrated on the structural primes. All of this consumes the canonical monoid (channel (ii) of the control principle, Section 88.6); the equivalences cannot even be stated for the integer controls.

82.2 The integer-area certificate and the marked three-point ledger

On the Farey chart the marked points have unit-gap coordinates, and the oriented area of any three of them is a rational number with an *integer numerator* over the six unit denominators: $\Delta = D/\prod_i(m_i+1)(n_i+1)$ with $D \in \mathbb{Z}$, so three genuinely non-collinear marked points obey $|\Delta| \geq \prod_i(m_i+1)^{-1}(n_i+1)^{-1}$, against the exact geometric value $\Delta = \frac{1}{2}F''(\xi)\prod_{i<j}(X_j - X_i)$ and the spacing identity $X_j - X_i = (m_j - m_i)/((m_i+1)(m_j+1))$. The arithmetic ledger is kernel-checked in the new module `CompactChartArea` (`area_eq_div`, `abs_area_ge`, `coord_diff`); the curvature side is classical. The certificate is a local exclusion tool adapted to the compactification — near the cusp the curvature blow-up compensates part of the small spacing, so it does not close anything by itself, but it iterates in higher-order determinants without undoing the chart, and a two-logarithm bound gives polynomial shell separation (e^{-N^ρ}-balls contain at most one marked point).

82.3 The alternant deficit and the relation-counting lock

Two negative ledgers quantify why the two most plausible counting technologies fall short. First, for plain two-variable exponential alternants $\det(m_i^{p_j}n_i^{q_j})$ on D marked points of one shell, the report proves a strict sign-regularity theorem (no such determinant vanishes at distinct curve points), an exact unit-numerator lower bound, and a Harish-Chandra–Itzykson–Zuber upper bound, whose combination forces the *shell ledger*: a monomial determinant can exclude D shell points only if some extra arithmetic divisibility compensates a loss of size $N\mathfrak{d}(\Lambda)$, and the minimal monomial deficit obeys

$$\delta_D = \frac{2^{3/2}}{3}\sqrt{ab}\,D^{3/2} + O(D).$$

Every plain monomial-determinant route therefore pays an unavoidable $ND^{3/2}$; the required compensation is exactly the output-sensitive divisibility that the corpus identifies as the frontier everywhere else (Sections 76, 78.4). Second, the report audits the 2026 relation-counting theorem (small-ball covering for algebraic points near trajectories of polynomial vector fields, admissible codimension $k < \sqrt{n} - 1$): a planar coefficient obstruction shows no nonzero polynomial vector field over a number field is tangent to either chart (so the plane is unusable), while the natural parameter and jet lifts to $n = 3$ represent the arithmetic event by codimension-two varieties, and

$$n - 2 < \sqrt{n} - 1 \quad \Longleftrightarrow \quad n = 2$$

— the *coefficient–codimension lock*, kernel-checked in squared form as `codimension_lock`. An invariant-prime trichotomy closes the redundant-coordinate escapes, and even a repaired covering theorem would need its polynomial exponent pushed below one before the shell-separation lemma could convert coverings into cardinality bounds. The route through relation counting is thus locked for structural reasons, not technical ones; the lock identifies precisely which two improvements (admissible codimension, covering exponent) would change the situation.

Assessment. The compact-chart frame adds no new unconditional theorem about the conjecture — its dichotomies consume the kernel-verified monoid and reprove the $O(T^2)$ ceiling from six exponentials — but it is the corpus's cleanest common coordinate system for the counting family of approaches, and its two ledgers are permanent: the $D^{3/2}$ alternant deficit quantifies the determinant wall on shells, and the codimension lock retires the hope that the 2026 relation-counting theorem applies off the shelf. The surviving interfaces it leaves are the one-point S-integral target (the sharpest known restatement of the four-exponentials special case) and determinant-with-divisibility hybrids, which reconnect to the shared-face mask target. **[paper proof]** for the charts, dichotomies, ledgers, and lock analysis; monoid, counting, and rank inputs **[kernel-verified Lean]**/**[classical/open background]** as cited.

83 Synchronized gaps: global gcd rigidity against local congruence cost

A further external report studies the synchronized gaps $D_2(r,s) = M^s - 2^r$, $D_3(r,s) = A^s - 3^r$ of a hypothetical counterexample, with $\delta_{r,s} = s\beta - r$ and the exact identities $D_2 = 2^r(2^\delta - 1)$, $D_3 = 3^r(3^\delta - 1)$: along $r/s \to \beta$ both gaps are simultaneously small relative to their ambient powers, so their common divisors are the natural place to look for a contradiction. The report proves matching global and local theorems whose *gap* is the finding.

83.1 Global rigidity and the closed approximation route

The global theorem is the *full synchronized-gap rigidity*: under a counterexample,

$$\log\gcd\big(M^s - 2^r,\ A^s - 3^r\big) = o(r+s),$$

by a Bugeaud–Corvaja–Zannier-type subexponential bound away from the structural primes [4] plus an $O((\log(r+s))^2)$ bound at 2 and 3; the two-logarithm input $|\delta_{r,s}| \geq c\max(r,s)^{-C}$ [33] is exactly where irrationality of β enters, so the statement is control-sensitive (for an integer control the gcd explodes on the diagonal $r = ms$). Two consequences are permanent. First, with $M_0 = M/2^{v_2(M)}$, $A_0 = A/3^{v_3(A)}$ (both > 1 precisely because $\beta \notin \mathbb{Z}$), the synchronized quotient $R_{r,s} = (3^\delta - 1)/(2^\delta - 1) \in \mathbb{Q}$ converges to $\log_2 3$ but has exact height $\log\mathrm{height}(R_{r,s}) = s\log(M_0A_0) + o(s)$: the gcd is too small to cancel the primitive exponential mass, so

$$\frac{-\log|R_{r,s} - \log_2 3|}{\log\mathrm{height}(R_{r,s})} \longrightarrow 0.$$

The *rational-approximation route is quantitatively closed*: these approximants have logarithmic quality relative to height, and no improvement of the irrationality measure of $\log_2 3$ can rescue them — a closed door, recorded in the ledger next to the factorial closure of Section 81.

83.2 The local congruence lattices and the square-root tax

The local theorem is an exact Smith-index law for the principal-unit congruence lattices $\Lambda_p(k) = \{(r,s) : M^s \equiv 2^r,\ A^s \equiv 3^r \pmod{p^{k+1}} \text{ on units}\}$: with Smith invariants $p^{\alpha_p} \mid p^{\gamma_p}$ of the local logarithm matrix B_p,

$$[\mathbb{Z}^2 : \Lambda_p(k)] = p^{(k-\alpha_p)_+ + (k-\gamma_p)_+},$$

stably $p^{2k-\tau_p}$; the mod-p shadow is the Fermat-quotient determinant $\det B_p \equiv q_p(M)q_p(3) - q_p(A)q_p(2) \pmod p$, whose kernel-verified form the corpus already holds (`FermatQuotientDeterminant`, Section 54.2); the finite-place product law multiplies the indices, giving $I = G^2/E$ where G is the enforced principal-unit precision and $E = \prod p^{\tau_p}$ the aggregate determinant credit. Against this, a *constrained Dirichlet theorem* (Minkowski applied to the sheared lattice, Lean-ready) finds $(r,s) \in \Lambda$ with $0 < s \leq H$ and $|s\beta - r| \leq (1+\eta)I/H$: any symmetric convex-body use of finite-index information conserves the product $H \cdot |s\beta - r| \asymp I$. Generic local precision therefore costs a *square*

($I = G^2$): the corpus's one-prime covolume conservation law (`RestrictedDenominatorDirichlet`) is exactly the $|S| = 1$ case of this tax.

83.3 The mismatch, the phase diagram, and the wedge

The synthesis is a precise mismatch: the global theorem forbids forced moduli $N = \exp(cH)$ at exponent height H, while generic local lifting produces only N polynomial in H because the congruence lattice has index $\approx N^2$. Between polynomial and exponential lies the entire gap; a proof must change the local cost *exponent*, not constants. The report's phase diagram makes the necessary structural shift explicit — essentially rank-one behavior at many places simultaneously ($I \approx G$), plus cross-prime coherence of the short directions — and organizes the missing principle adelically: the counterexample is the vanishing of the archimedean wedge $(\log 2, \log 3) \wedge (\log M, \log A)$, the local data are the finite-place wedges $-p^2 \det B_p$, and there is no product formula tying them together because each is bilinear in logarithms rather than a valuation of one algebraic number. Its five targets (adelic defect transfer; projective coherence of rank-one places; a second-order Fermat law; an auxiliary wedge with a product formula; the gcd contrapositive) are all forms of one question — transfer the archimedean rank loss to aggregate finite-place rank loss — which is the corpus's coupling problem (Sections 79.2, 80.2) in adelic form. *Assessment*: nothing moves the four-exponentials barrier; the durable additions are the closed approximation route, the exact square-cost benchmark that any lattice-based strategy must beat, and the cleanest adelic statement so far of what a decisive principle must do. **[paper proof]** throughout; BCZ/Baker inputs **[classical/open background]**; mod-p shadow and one-prime covolume law already **[kernel-verified Lean]** in the corpus.

84 Near-curve certificates: orbit pressure against fewnomial budgets

A further external report builds a computational front end for the conjecture from three elementary components, then audits the 2026 near-curve algorithm against it.

84.1 Pressure, locking, and budgets

Orbit pressure: a counterexample x places at least $\lceil (L+1)/x \rceil$ integer points of $Y = X^\theta$ ($\theta = \log_2 3$) in every dyadic block $2^L \le X < 2^{L+1}$ — linearly many per block, by the conditional monoid. *Subunit locking*: if $P \in \mathbb{Z}[X, Y]$ satisfies $\sup_{z \in [1,2]} |P(2^L z, 3^L z^\theta)| < 1$, every integer curve point of the block is a zero of P (an integer of modulus < 1 vanishes). *Fewnomial budget*: for irrational θ the monomials $X^i Y^j$ restrict to the curve as $X^{i+j\theta}$ with pairwise distinct exponents, a Chebyshev system, so a nonzero P with support size $T(P)$ has at most $T(P) - 1$ zeros on the curve. Combining the three: every global subunit certificate for block L has $T(P_L) \ge \lceil (L+1)/x \rceil + 1$, hence degree $d_L \gg \sqrt{L/x}$, and every certified cover of capacity Cap_L has $\mathrm{Cap}_L \ge \lceil (L+1)/x \rceil$. Contrapositively — and this is the report's central sufficient criterion — exhibiting certified covers with $\mathrm{Cap}_L = o(L)$ along an unbounded block sequence *proves* the conjecture. Two conditioning barriers calibrate the search: no fixed support is subunit for arbitrarily large L (finite support libraries become extinct effectively), and an unconditional near-critical degree barrier $L \le Cd^2 \log(d + 1)$ shows certificates cannot be cheap — the hunt lives in the narrow corridor $d \asymp \sqrt{L}$ up to logarithms. The report specializes the Brisebarre–Hanrot near-curve algorithm [38] into a proof-oriented certificate: a single verified short-vector inequality for the rounded Chebyshev–Bernstein matrix implies the uniform subunit bound, for dense or arbitrary sparse supports, giving a machine-checkable route (Priority A of its program) whose success criterion is a proven sufficient condition. The dense asymptotics do not yet meet the capacity criterion; the open question is whether near-resonant sparse supports do.

84.2 The toric no-go and the Carlitz model

Why is a certificate search needed at all, rather than an operator argument? The report isolates the obstruction: the diagonal scaling $(X, Y) \mapsto (2X, 3Y)$ admits only *monomial* polynomial semi-invariants — $P(2X, 3Y) = cP(X, Y)$ forces a single monomial, since the eigenvalues $2^i 3^j$ are pairwise distinct. The coefficientwise kernel core is verified in the new module `ToricMonomialRigidity` (`toric_eigen_support_unique`, on top of the kernel-verified injectivity of $(i, j) \mapsto 2^i 3^j$): geometric scaling never creates a coupled finite-rank system, so any relation-lifting theorem must introduce a semilinear or inhomogeneous action — the same conclusion as the finite-rank cocycle target of Section 80.4, reached operator-theoretically. As a model for what the missing rigidity would look like, the report assembles the Carlitz-module picture: in positive characteristic the full relation ideal of Carlitz logarithms is generated by k-linear relations [41], the two-direction multiplier rigidity ($c\lambda_1, c\lambda_2$ both Carlitz logarithms forces $c \in k$) is a theorem, and Estienne's 2026 Mahler-method proof [39] shows the rigidity can be reached through functional equations alone; the exact characteristic-zero analogue (its "quadratic target") would settle the conjecture. The differential lift of fixed-endpoint logarithm germs has the right relation ideal but specializes incorrectly — a defect the report exhibits already at integral controls, and its route triage includes an exact mod-7 counterexample to same-prime congruence transfer at the control $x = 2$, applying the control principle of Section 88.6 as a falsification tool. Harrison's bounds on additive relations in irrational powers [67] bound the additive energy of the smooth grid and the exponential boundary fiber, quantifying additive isolation (Section 79.1) at grid scale. *Assessment*: the durable additions are the certified-search interface with its proven sufficiency threshold $\mathrm{Cap}_L = o(L)$ — the corpus's first falsifiable-by-computation route — and the toric no-go that explains, in operator language, why every naive lift has failed. **[paper proof]** throughout; near-curve algorithm, Carlitz rigidity, and additive-energy inputs **[classical/open background]**; toric core **[kernel-verified Lean]**.

85 Chebyshev trap lattices and the unconditional block bound

A further external report develops the near-curve certificate frame of Section 84 into a quantitative theory and extracts from it the corpus's first *unconditional* counting theorem about the solution set itself.

85.1 Rank–codimension traps and the Chebyshev lattice

Strict total positivity of exponential systems ($\det(e^{\lambda_j t_i}) > 0$ for increasing nodes and frequencies) upgrades the fewnomial budget of Section 84.1 from one polynomial to a family: r linearly independent integer polynomials supported on a common q-element support, all of absolute value < 1 on a curve segment, leave at most $q - r$ integer points on that segment (*rank–codimension trap*); certified covers of $[0, 1]$ with total defect $o(H)$ for infinitely many H would prove the conjecture, matching the capacity criterion of Section 84.1. Short trap polynomials are produced from a *Chebyshev lattice*:

coefficient rows from Bernstein-ellipse expansions of $z \mapsto P(2^{H+z}, 3^{H+z})$ on the block, diagonal tail rows to control truncation, covolume bounded through Cauchy–Binet by

$$\log \Delta \le (H + R(\rho)) \sum_{e \in E} \lambda_e - \frac{q(q-1)}{2} \log \rho + O(q \log q)$$

uniformly in the support (diagonal tail rows never weaken the Chebyshev decay), and Minkowski producing r independent short vectors at fixed r. The optimal supports are weighted simplices (smallest frequency sums), and optimizing the ellipse against the simplex yields the report's headline.

85.2 The unconditional block bound

Theorem 85.1 (Unconditional integer-point bound in a logarithmic unit block)**.** *With* $I(H) := \#\{x \in [H, H+1) : 2^x \in \mathbb{Z},\ 3^x \in \mathbb{Z}\}$,

$$\limsup_{H\to\infty} I(H)\, \frac{(\log H)^2}{H^2} \le \frac{32 \log 2 \log 3}{9} = 2.70755559\ldots,$$

via supports of size $(32 \log 2 \log 3/9 + \varepsilon)H^2/(\log H)^2$ *(outer frequency scale* $L = (8ab/3)H/\log H$*, ellipse* $\rho = 3L/(2ab)$*,* $a = \log 2$*,* $b = \log 3$*); the same argument gives* $32 \log u \log v/9$ *for any multiplicatively independent bases* $u, v > 1$*.*

The inputs are only multiplicative independence, Bernstein–Chebyshev decay, Cauchy–Binet, Minkowski, and total positivity — no conditional monoid. This is a *pointwise* per-block bound, complementing the cumulative $O(T^2)$ ceiling of six exponentials (Section 82.1), which controls only block averages; under a counterexample the true count is $I(H) = H/\beta + O(1)$, so the theorem is a $H/(\log H)^2$-factor short of a contradiction — the first unconditional statement in the corpus to come within a quantified polynomial factor of the conjecture. **[paper proof]**; the constant checks numerically and the mechanism is elementary throughout.

85.3 The two ceilings: rank tax and the conservation law

The report then audits, with unusual honesty, why its own mechanism stops there. The macroscopic-window optimization has exact constant $\kappa_* = \frac{64 \log 2 \log 3}{9} \cosh(t_0)/t_0^2 = 5.08726658\ldots$ with $t_0 \tanh t_0 = 2$, and a *closing* trap must certify a short-vector rank fraction $R_H/q > 0.98596$ — a threshold computed directly from the corpus's kernel-verified zero-free region $\beta \ge 14$ (Section 3.2; the software exclusion $\beta > 27$ raises it to 0.99272). Against this stand two proved ceilings. The *rank-tax inequality* shows the covolume-plus-Minkowski mechanism carries a $(r-1)q \log \rho$ penalty, so it cannot certify any rank fraction beyond 1/2 — volume is not spectrum, and near-full short rank is a shape statement requiring genuine successive-minima or transference input. And a *conservation law*: trading support size against rank fraction leaves the leading block constant invariant at $32ab/9$ — within the volume method, fixed-rank and fractional-rank optimization are exactly equally strong. The surviving routes are therefore structural: parametric geometry of numbers for the trap lattices (switch-density as a finite combinatorial target), the compound-matrix spectrum of the H-independent Bessel core ($A = D_{\text{row}} K_N(E) D_H$ with $K_{n,e} = I_n(\lambda_e/2)$, all H-dependence diagonal), and arithmetic rows tied to a primitive pair — the only channel that could be control-sensitive.

85.4 Dilation rigidity and the certificate program

Two supporting layers round out the framework. First, *consecutive-dilation rigidity*: a q-monomial polynomial vanishing at q consecutive diagonal dilations $(2^n X, 3^n Y)$ of one positive point is identically zero, so no fixed sparse trap retires a persistent normalized orbit point — traps must move with the block. The finite core is kernel-checked in the new module `ConsecutiveDilationRigidity` (`eq_zero_of_consecutive_vanishing`, Vandermonde elimination of the largest node; wrapper `consecutive_dilation_rigidity`), the q-shift upgrade of the one-shift toric rigidity of `ToricMonomialRigidity`; the report's classification of finite-dimensional dilation-invariant subspaces (all spanned by monomials) is its Lagrange-interpolation companion. A Hermite-multiplicity variant of the traps is audited and blocked by a row-reset obstruction: derivative rows restart the Chebyshev decay and are not free. Second, the report specifies *proof-producing certificates*: directed-rounding interval enclosures of the Chebyshev rows, an explicit finite verifier (Phase I/II of its Lean plan is exact algebra plus a certificate checker, matching this corpus's decide-table philosophy), and a discovery algorithm that beats point scanning. *Assessment*: this merge contributes the corpus's first unconditional quantitative theorem about $\mathcal{S}$ itself, two honest ceilings that calibrate every future volume-based attempt, and a concrete verifier architecture; the conjecture-shaped residue is exactly the 98.6%-versus-50% rank gap, to be attacked by lattice shape (not volume) or by primitive-sensitive arithmetic rows. **[paper proof]** throughout; dilation core **[kernel-verified Lean]**; constants **[exact computation]** (re-verified numerically during integration).

86 The saturated divisor lattice and the two-star control detector

A further external report recasts the conditional framework in the divisor-lattice language of the free abelian group on the primes. Most of its unconditional layer is an independent re-derivation of kernel-verified corpus results and is deduplicated by citation: the six-exponentials rank cap and the free-semigroup dichotomy $\mathcal{S} = \mathbb{N}_0$ or $\mathbb{N}_0 + \mathbb{N}_0\xi$ are the canonical monoid (Theorem 4.3, **[kernel-verified Lean]**); the saturation index and explicit root descent are the corpus's radical normalization (Section 5); the rank window $\rho \in \{3, 4\}$ with its Möbius normal form is Section 66; the polylogarithmic counting gap and the $o((\log H)^2)$ sufficiency target are the height-scale form of the compact-chart certificate package (Section 82.1); and mod-ℓ Kummer primitivity for every ℓ is the mod-ℓ independence of the pair lattice (Section 77). The independent derivations agree with the corpus at every point of contact — a useful external audit. Two layers are genuinely new and are integrated.

86.1 The two-star tensor as an exact control detector

The report packages a conditional pair into the symmetric tensor $\Sigma(a, b) = e_2 \circledast b - e_3 \circledast a \in \mathrm{Sym}^2(V)$, where a, b are the divisor vectors of (A, B) and $\circledast$ is the symmetrized product, and proves that *formal zeroes are exactly the controls*: $\Sigma(a, b) = 0$ iff $(A, B) = (2^m, 3^m)$. A hypothetical counterexample is equivalent to a *primitive tensor*: $\Sigma \neq 0$, formal period $\mathrm{per}_2 \Sigma = 0$, coefficient content one, $\min(a_2, b_3) = 0$, and a positive external coefficient — and any such tensor reconstructs the solution through $\xi = \ell(a)/\log 2$. The content layer of this object is the corpus's kernel-verified two-star quadratic (`TwoStarContent`: content equals the simultaneous coordinate gcd, content one at the least solution); the control characterization is now also kernel-checked, in the new module

TwoStarControlCharacterization (twoStarCoeff_eq_zero_iff: the tensor vanishes identically iff a is supported at 2, b at 3, and $a_2 = b_3$; twoStarCoeff_control). This gives the control principle of Section 88.6 its cleanest algebraic form to date: the nonvanishing tensor Σ *is* the difference between a candidate and the control locus, so any argument that consumes $\Sigma \neq 0$ is control-clean by construction.

86.2 The Kummer period orbit and the design checklist

At an external prime q in the support, the independent Kummer translation moves a primitive relation: the first orbit derivative of the two-star period is the nonzero logarithmic form $\Omega(b_q \log 2 - a_q \log 3)$, so the primitive tensor is *not* orbit-invariant — precisely the asymmetry a proof should consume. The report's conditional closure states that a positive answer to its Kummer-orbit problem for the two-star quadratics $F = X_2L_B - X_3L_A$ proves the conjecture (**[proposed target]** target), and it distills a determinant design checklist — gauge invariance, root-descent invariance, control annihilation (a factor of Σ), external-prime sensitivity, and grids rather than lines — that matches the session-log control tests item for item. *Assessment*: a consolidation report whose independent lattice derivations cross-check the corpus, plus one durable new object. The two-star control detector does not move the four-exponentials barrier, but it upgrades the control principle from a methodological test to kernel-checked algebra, and the Kummer-orbit problem is a well-posed primitive-sensitive target sitting exactly on the frontier channel (i) (external prime). **[paper proof]** for the tensor normal form and orbit obstruction; detector core **[kernel-verified Lean]**.

87 Frontier campaign: parallel attack on the five surviving targets

The frontier left by the eleven integrated reports consisted of five targets: the two-parameter transport interface (Section 81.3), certified near-curve covers (Section 84.1), the Kummer period orbit (Section 86.2), the adelic wedge transfer (Section 83.3), and the search for a new mechanism. Each was attacked independently and each surviving claim was then subjected to adversarial refereeing (a referee instructed to refute, to re-run every computation in exact arithmetic, and to apply the control test). Four of the five produced results; the near-curve lattice computation did not complete. What follows records only what survived refereeing or what was independently re-verified here.

87.1 The two-parameter interface is closed

Theorem 88.3 was obtained independently, in the equivalent and more suggestive form "the archimedean balance says exactly that positive and negative slice-tail mass agree, $P = N$", which makes visible *why* no primes, no size hypothesis and no integrality are needed to empty the top-heavy class. That same identity also supplies the loop-closer the Φ-starvation bound of Section 88.11 was missing: substituting $P = N$ into the starvation estimate yields $N \leq 16Z(m+j^*)(\log A)N/A$ on the admissible grid, hence $c \equiv 0$ whenever $A > 16Z(m+j^*)\log A$ — comfortably true in the candidate regime, where $A \geq 3^{14}$ against a threshold of order 10^4. Subject to a prime in every annulus and strict mesoscopic admissibility, the general (not merely top-heavy) two-parameter cocycle is therefore also forced to vanish. Exhaustive exact search agrees: in two independent full-integrality laboratories — $(M, A) = (5, 11)$ testing all 14,773 primes below 161,051, and $(M, A) = (7, 19)$ testing all 181,449 primes below 2,476,099 — no nonzero balanced pattern is integral among 3,251,860 exactly enumerated candidates. **[paper proof]**, with the top-heavy core **[kernel-verified Lean]** and the searches **[exact computation]**.

The one non-elementary hypothesis is the prime supply, and its failure mode is now exactly located: for cells with $\theta_{ij} = (n+i)\log M/((m+j)\log A) < 7/12$ the annulus is shorter than $N_j^{7/12}$ and Huxley supplies nothing (the exponent improves to 0.525 with Baker–Harman–Pintz and to 1/2 under RH). Every such annulus in both laboratories does contain many primes, so the obstruction is a gap in available prime-gap technology rather than a real arithmetic phenomenon — and, being ratio-only, it applies verbatim to the controls. *Control status.* The whole package holds at $(M, A) = (2^m, 3^m)$: it closes the two-parameter *method* and leaves the conjecture where it was. The first of the two surviving interfaces of Section 81.3 is thus retired.

87.2 The adelic wedge route is closed, quantitatively

Three facts, each re-verified here in exact integer arithmetic, settle the adelic programme of Section 83.3.

Control-direction blindness. The Fermat-quotient wedge is invariant under multiplication by a control: $\Delta_p(2^a b, 3^a c) = \Delta_p(b, c)$ for every a and every pair prime to p (verified with zero mismatches over four thousand exact instances, and kernel-checked in the new module `AdelicWedgeDecoupling`, whose `fermatQuotientDeterminant_control_shift` and `fermatQuotientDeterminant_eq_zero_iff_line` also identify rank drop at p with (q_pM, q_pA) lying on the line spanned by (q_p2, q_p3)). The finite wedge therefore sees the pair only modulo the control orbit, and every value in $\mathbb{F}_p$ is realised inside every control orbit.

The density is off by three orders of magnitude. Writing $I = G^\kappa$ for the exponent-lattice index, the tax exponent is exactly $\kappa = 2 - (\sum_p \tau_p \log p)/(\sum_p k_p \log p)$; reaching the $\kappa \leq 1$ needed to beat the square-root tax requires rank drop at some 90% of the primes below the cutoff. Measured drop density for candidate-shaped pairs matches the exact heuristic $\sum_p (1 - 1/p)^2/p$: an independent run here found 452 drops in 359,068 exact (p, pair) tests against a heuristic expectation of 443.2, a rate of 0.126%. The controls, by contrast, drop at 100% of primes (6000/6000 tested) — indeed $\det B_p = 0$ exactly in $\mathbb{Q}_p$ there. So the statistic separates controls from candidates sharply while falling short of the requirement by a factor of order 10^3.

No product formula, and a hardness transfer. A bilinear-in-logarithms local pairing reproducing D_p modulo p is a norm-residue symbol over $\mathbb{Q}(\zeta_p)$, whose reciprocity has identically trivial archimedean terms because that field is totally complex for $p \geq 3$; so no reciprocity law can tie the finite wedges to the vanishing archimedean wedge. Moreover the finite wedge data are *decoupled* from archimedean alignment: for any finite S, any depth, and any $\varepsilon > 0$ there are pairs with $|\log 2 \log A - \log 3 \log M| < \varepsilon$ and arbitrarily prescribed wedges on S. Any unconditional theorem bounding the rank-drop count of candidate-shaped pairs below Y would imply the existence of non-Wieferich primes below Y — a hardness transfer that explains why the route resists. **[paper proof]** throughout; decoupling core **[kernel-verified Lean]**. The surviving suggestion is to abandon the unit-level wedge for the valuation identity $\sum_p (v_p(A) \log 2 - v_p(M) \log 3) \log p = 0$, which *is* control-sensitive because a counterexample has external prime support and a control has none.

87.3 The Kummer orbit hypothesis is equivalent to the conjecture

The proposed conditional closure of Section 86.2 splits into two halves, and both are now understood. The Kummer action on the divisor lattice is the *unipotent* period translation $x \mapsto x + \Omega n$, not a diagonal dilation; consequently the relevant rigidity is polynomial interpolation of a quadratic, not the Vandermonde elimination of `ConsecutiveDilationRigidity`, and the sharp threshold is a single orbit point per external prime. On the external sublattice the quadratic term drops out identically, so the orbit period is the *linear* form $\Omega \sum_q n_q \Lambda_q$ with $\Lambda_q = a_q \log 3 - b_q \log 2$. The first half of the closure — that the period cannot vanish along the whole orbit at an external prime — is then unconditional with a two-line proof from unique factorization, and is already the corpus's `external_derivative_ne_zero`; it carries no new arithmetic. The second half is an **[proposed target]** obstruction of the sharpest kind: the hypothesis "the period vanishes at the one-step orbit points for every external prime in the support" is, term by term, sparse coaction separation (Definition 73.1), which Theorem 73.2 shows is *equivalent* to Conjecture 1.1. The orbit route is therefore not a weaker foothold but a restatement, and the underlying deficit is one of *weight*: the counterexample relation is a $\mathbb{Q}$-linear relation among the weight-two products $\{\log p \log q\}$, whereas Baker–Wüstholz and Matveev are weight-one theorems. A related structural fact closes a natural hope: the two-star tensor is *monoid blind*, since $\Sigma(je_2 + ka, je_3 + kb) = k\,\Sigma(a, b)$ places every element of $\mathcal{S}$ on the same ray, so the tensor cannot consume channel (ii) of the control principle.

87.4 What the trap architecture can and cannot do

The block-bound machinery of Section 85 was rebuilt from scratch and its published constants independently reproduced to full printed precision ($32 \log 2 \log 3/9 = 2.70755559$, $\kappa_* = 5.08726658$ with $t_0 \tanh t_0 = 2$, and the corank thresholds 0.98595934 and 0.99271966). Three exact conclusions follow. The optimal support is the greedy set $\{2^i 3^j \le X\}$ with mean frequency $\frac{2}{3}\sqrt{2abq} - \log\sqrt{6} + o(1)$, so $32ab/9$ is the exact optimum of the support-shape problem and cannot be improved by reshaping. A certified short-vector rank fraction f yields *exactly* the implication $\beta \ge 1/((1-f)\kappa_*)$ and nothing stronger: the architecture is a zero-free-region generator, and the published thresholds are precisely this dictionary. And the one control-clean input available to a trap — the external prime, which makes p^{ek} divide every solution value and so relaxes the archimedean target — is worth only $\Theta(\log H/H)$ in the required support size (4.9×10^{-3} at $H = 10^3$, 8.9×10^{-5} at $H = 10^5$), and is strictly dominated, by a factor 1.43 to 1.52, by the control-*blind* lattice-discreteness term $-\log\sqrt{6}$. Raising the monomial floor to amplify the p-adic gain is a strict net loss. Finally, the unit block is the *worst* scale at which to state the bound: the deficiency against the conditional count is $H/(\log H)^2$ at $\delta = 1$ but only the constant $\kappa_*\beta$ at every δ of order H. **[paper proof]**; constants **[exact computation]**.

87.5 The spectrum question is a difference-module finiteness question

The sharp form of what remains — is $V_\tau = V_0$ for every finite τ? — has structure that was not visible when it was posed. The monoid is closed under both of its generators,

$$\mathcal{S} + 1 \subseteq \mathcal{S}, \qquad \mathcal{S} + \beta \subseteq \mathcal{S} \qquad (s = n + k\beta \mapsto (n{+}1) + k\beta,\ n + (k{+}1)\beta),$$

so the two difference operators

$$(\Delta_1 g)(z) = g(z+1) - g(z), \qquad (\Delta_\beta g)(z) = g(z+\beta) - g(z)$$

carry integer-valued functions on $\mathcal{S}$ to integer-valued functions on $\mathcal{S}$. They also preserve finite quadratic type, since $M(g(\cdot+h),r) \le M(g,r+h)$ leaves $\lim r^{-2}\log M$ unchanged. Hence *each* V_τ *is a module over the commutative ring* $\mathbb{Z}[\Delta_1,\Delta_\beta]$.

On the monomials $e_{ij}(z) = 2^{iz}3^{jz}$ both operators are diagonal, and — this is the point — with *integer* eigenvalues:

$$\Delta_1 e_{ij} = (2^i 3^j - 1)\, e_{ij}, \qquad \Delta_\beta e_{ij} = (M^i A^j - 1)\, e_{ij},$$

the second being an integer precisely because $M = 2^\beta$ and $A = 3^\beta$ are integers under the counterexample hypothesis. So $V_0 \supseteq \mathbb{Z}[2^z, 3^z]$ is spanned by joint eigenvectors with integer eigenvalue pairs, and the two operators separate them: the pair $(2^i3^j, M^iA^j)$ determines (i,j), by the kernel-verified injectivity of $(i,j) \mapsto 2^i3^j$ (`SmoothSemigroupCore`).

This turns the spectrum question into a finiteness question. If for some $g \in V_\tau$ the $\mathbb{Z}[\Delta_1,\Delta_\beta]$-orbit of g is finitely generated, then g satisfies a linear difference equation with integer coefficients in the two commuting shifts; its entire solutions are combinations of exponentials e_{ij}, forcing $g \in V_0$ and $\tau_2(g) = 0$. Consequently:

Research target 87.1 (Difference-module finiteness)**.** Decide whether every g that is entire, integer-valued on $\mathcal{S}$, and of finite quadratic type generates a finitely generated $\mathbb{Z}[\Delta_1,\Delta_\beta]$-module. A positive answer gives $V_\tau = V_0$ for all finite τ, hence a type spectrum $\{0\}\cup\{\infty\}$, and closes Target 87.6 *negatively.* A negative answer produces a candidate for the construction that would settle the conjecture.

Two remarks make this the natural next step rather than a reformulation. It is control-sensitive in the right way: at an integer control $\mathcal{S} = \mathbb{N}_0$ is closed under $+1$ alone, the second operator degenerates to $\Delta_\beta = \Delta_1^{(\text{iterated})}$ composition and the two-variable structure — the whole content — disappears. And it is of exactly the shape the literature survey singled out: a finite-rank statement for a system of difference equations in two independent shifts, which is the discrete counterpart of the arithmetic-holonomy programme of Section 87.7. The corpus's two surviving routes therefore meet here: the transcendence route needs independence of weight-two products, the interpolation route needs finiteness of a two-shift difference module, and both are finite-rank statements about the same pair $(2,3)$ acting on the same monoid. **[paper proof]**; target **[proposed target]**.

87.6 Where the threshold comes from: forced vanishing on an initial segment

Attacking Target 87.1 directly with the classical Pólya–Gel'fond descent turns out to be informative in a way that neither confirms nor refutes it, and it explains *why* a threshold of the shape $1/(C\beta^2ab)$ appears at all.

Because $\mathbb{N}_0 \subseteq \mathcal{S}$, the ordinary difference table $\Delta^k g(0) = \sum_j (-1)^{k-j}\binom{k}{j} g(j)$ consists of *integers*. Estimating it by Cauchy's formula on $|z| = R$ with $R = 2k$ — which is legitimate, the poles sitting at $0,1,\ldots,k$ — gives

$$|\Delta^k g(0)| \;\le\; M(g,R)\,\frac{k!\,R}{(R-k)^{k+1}} \;=\; 2\,M(g,2k)\,\frac{k!}{k^k}, \qquad \text{so} \qquad \log|\Delta^k g(0)| \le 4\tau k^2 - k + O(\log k).$$

The right-hand side is negative — indeed below -1 — for every $k < 1/(4\tau)$, and an integer of absolute value less than one is zero. Newton's expansion on $\mathbb{N}_0$, in which $\binom{n}{k} = 0$ for $k > n$, then propagates this to the values themselves.

Status: Retraction — the initial-segment proposition is false

An earlier version of this subsection asserted: *if g is entire, integer-valued on $\mathbb{N}_0 \subseteq \mathcal{S}$, with $\tau_2(g) \le \tau$, then $\Delta^k g(0) = 0$ for every $k < 1/(4\tau)$ and hence $g(n) = 0$ for $n \le K = \lfloor 1/(4\tau) \rfloor$.* That is *false*, and $g(z) = 2^z$ is a counterexample: it is integer-valued on all of $\mathcal{S}$ (since $2^{n+k\beta} = 2^n M^k$), it has $\tau_2 = 0 \le \tau$ for every $\tau > 0$, and yet $\Delta^k g(0) = (2-1)^k = 1$ for every k, with $g(n) = 2^n \ne 0$.

Two independent errors produced it. First, $\tau_2(g) \le \tau$ is a *limsup* statement: it gives $\log M(g,r) \le (\tau + \varepsilon) r^2$ only for $r \ge r_0(g,\varepsilon)$, not for all r. The Cauchy estimate below takes the contour radius $R = 2k$, so it is available only for $k \ge r_0/2$, whereas the vanishing range claimed was $k < 1/(4\tau)$; for 2^z with $\tau = 10^{-6}$ these are $k \ge 346{,}574$ against $k < 250{,}000$ — disjoint, so the estimate was never valid where it was invoked. Second, $k = 0$ is never constrained ($|g(0)| \le M(g,0)$ is vacuous), so even granting the bound, Newton's expansion yields $g(n) = g(0)$ on the segment — the function is *constant* there, not zero.

What survives is the estimate itself, and a lesson about why this problem is hard.

Proposition 87.2 (Window of vanishing differences)**.** *Let g be entire and integer-valued on $\mathbb{N}_0 \subseteq \mathcal{S}$, and suppose $\log M(g,2k) \le \tau(2k)^2$. Then*

$$\log\left|\Delta^k g(0)\right| \;\le\; 4\tau k^2 - k + O(\log k),$$

so $\Delta^k g(0) = 0$ whenever $k < 1/(4\tau)$ and k is large enough for the $O(\log k)$ to be dominated. Under the limsup hypothesis $\tau_2(g) \le \tau$ this applies exactly on the window $r_0/2 \le k < 1/(4\tau)$, which is empty unless $r_0 < 1/(2\tau)$.

A window of vanishing differences that does not begin at $k = 0$ says nothing about the values, so no conclusion about g follows; and requiring the bound at *all* radii, $M(g,r) \le e^{\tau r^2}$ for every r, defines a class so small that it excludes 2^z and indeed every function of positive exponential type, since $e^{cr} > e^{\tau r^2}$ for $r < c/\tau$. Either way the descent gives nothing.

The lesson is worth more than the failed proposition, and it is precisely the methodological point the literature survey makes about this problem. At order one Pólya's theorem works because $M(g,r) = o(2^r)$ constrains *every* radius, so the difference table is controlled from $k = 0$ upward and integrality can be cashed in immediately. At order two the natural hypothesis is a limsup, which says nothing at small radii, exactly where the difference table lives; and the constant term is free. This is why a Pólya-style descent cannot substitute for Pila's theorem, why that theorem needs interpolation determinants over the full two-dimensional monoid rather than a one-variable table, and why the constant in it is hard-won rather than elementary. It also retrospectively explains the shape of Section 87.8: the determinant derivation there succeeds precisely because it uses *many* points of $\mathcal{S}$ at a single large radius, never needing control at small ones. **[paper proof]** for the estimate; the retracted conclusion is withdrawn.

87.7 The terminal obstruction inside the Calegari–Dimitrov–Tang family

The survey's strongest methodological observation is that Calegari–Dimitrov–Tang is the only existing technology obtaining arithmetic control over a *product of two* logarithms, which is precisely the weight at which this problem sits (Section 87.3). It leaves open the question of whether the

Alaoglu–Erdős lattice can be encoded in that family at all. It can, exactly, and the encoding is finite.

Write

$$L_n := \log\Big(1+\frac{1}{n}\Big) = \log\frac{n+1}{n}, \qquad \text{so that} \qquad \log p = \sum_{n=1}^{p-1} L_n$$

for every prime p — every logarithm of an integer is a finite sum of logarithms of consecutive-integer ratios, which are exactly the generators appearing in the Calegari–Dimitrov–Tang theorems. Expanding the counterexample relation $\log 2\cdot\log A - \log 3\cdot\log M = 0$ in these generators turns the terminal obstruction into a $\mathbb{Z}$-linear relation among the weight-two products L_iL_j.

Status: Erratum — the generators L_n are dependent

An earlier version of this subsection asserted the encoding in the products L_iL_j and concluded that $\mathbb{Q}$-linear independence of the occurring products would exclude counterexamples. That is *false, not open*, for every $q \ge 5$: the generators satisfy

$$L_1 = L_2 + L_3, \qquad \text{i.e.} \qquad \log 2 = \log\tfrac{3}{2} + \log\tfrac{4}{3},$$

so $L_1^2 - L_1L_2 - L_1L_3 = 0$ is a nonzero $\mathbb{Q}$-relation lying inside the $q=5$ support. The relation space of $L_1,\dots,L_{q-1}$ has dimension $0,1,2,5,6$ at $q=3,5,7,11,13$ (computed exactly by reducing each L_n in the prime basis). The telescoping identity $\log p = \sum_{n<p} L_n$ is correct, and the encoding *into* $\mathbb{Z}[\{L_iL_j\}]$ is correct; what fails is the passage from "the relation holds" to "its coefficients vanish". The corrected statement below works in the prime basis, which *is* independent by unique factorization, and the $q=3$ quadratic form and its conclusion are unaffected.

Proposition 87.3 (Finite weight-two encoding)**.** *Let q be a prime and suppose MA is q-smooth. Expanding $\log M = \sum_p m_p \log p$ and $\log A = \sum_p a_p \log p$ over the primes $p \le q$, the relation $\log 2\cdot\log A = \log 3\cdot\log M$ reads*

$$\sum_{p\le q} a_p\,(\log 2\cdot\log p) \;-\; \sum_{p\le q} m_p\,(\log 3\cdot\log p) \;=\; 0,$$

a $\mathbb{Z}$-linear relation among the $2\pi(q)-1$ distinct weight-two products $\{\log 2\cdot\log p\}\cup\{\log 3\cdot\log p\}$ (the product $\log 2\log 3$ occurring in both families). If those products are $\mathbb{Q}$-linearly independent then $a_p = 0$ for $p\ne 3$, $m_p = 0$ for $p \ne 2$, and $a_3 = m_2$, i.e. $M = 2^t$, $A = 3^t$: the integer controls. Consequently such independence excludes every counterexample with q-smooth outputs.

The count is $2\pi(q)-1 = 5,7,9,11$ at $q = 5,7,11,13$. It is worth noting that these are exactly the dimensions computed for the image subspace C below — the earlier table's numbers were right, for this reason, even though the basis in which they were derived was not.

There is a better observation to make. The coefficient array of the corrected relation is *literally* the two-star tensor of Section 86.1: the coefficient of the unordered product $\{x,y\}$ in $\sum_p a_p(\log 2\log p) - \sum_p m_p(\log 3\log p)$ is

$$[x{=}2]\,a_y + [y{=}2]\,a_x - [x{=}3]\,m_y - [y{=}3]\,m_x,$$

which is exactly $\Sigma(m,a) = e_2 \circledast a - e_3 \circledast m$ evaluated at $\{x,y\}$ (verified with zero mismatches over three thousand random coefficient pairs and all prime pairs). Consequently the finite core of Proposition 87.3 requires no new formalization: it *is* the kernel-verified `TwoStarControl.twoStarCoeff_eq_zero_iff`, which states precisely that the array vanishes iff m is supported at 2, a at 3, and $m_2 = a_3$ — the controls.

This unifies two strands developed independently in this report. The two-star detector arrived from the saturated divisor lattice (Section 86) as an exact *control detector*: $\Sigma \neq 0$ is equivalent to primitivity, and is kernel-verified. The weight-two encoding arrives from the transcendence side as the terminal obstruction. They are the same object viewed twice, and putting them together isolates the missing ingredient exactly:

a counterexample makes $\Sigma(m, a) \neq 0$ (kernel-verified), yet the corresponding $\mathbb{Q}$-linear combination of weight-two products vanishes; so what is missing is precisely enough independence among $\{\log 2 \log p\} \cup \{\log 3 \log p\}$ to see a nonzero tensor.

The algebra is finished and formalized; the entire remaining content of this route is the transcendence input. **[exact computation]** for the identification; core **[kernel-verified Lean]**.

The coefficient system was computed symbolically and solved exactly in the prime basis. For $q = 3$ the relation is the quadratic form

$$(c + d - a - b)\, L_1^2 + (d - a - 2b)\, L_1 L_2 - b\, L_2^2 = 0, \qquad M = 2^a 3^b, \quad A = 2^c 3^d,$$

in $L_1 = \log 2$ and $L_2 = \log \frac{3}{2}$, whose vanishing gives $b = c = 0$ and $a = d$: the 3-smooth case therefore follows from $\mathbb{Q}$-linear independence of just the three products $(\log 2)^2$, $\log 2 \log \frac{3}{2}$, $(\log \frac{3}{2})^2$. (That case is already known by transcendence of ϑ, so this is a calibration rather than a new result.) The first genuinely new instance is $q = 5$, since a counterexample must have an external prime ≥ 5: there the relation involves the 5 products $\log 2 \log 2, \log 2 \log 3, \log 2 \log 5, \log 3 \log 3, \log 3 \log 5$, and independence forces $M = 2^t$, $A = 3^t$. **[exact computation]** (exact symbolic solve); the encoding itself **[paper proof]**.

Three consequences are worth stating. First, the obstruction is not merely *analogous* to the Calegari–Dimitrov–Tang setting; it lives in literally the same family of logarithms, with the number of generators bounded by the largest prime dividing MA. Second, the required independence statement is *finite* for each q — at most $\binom{q}{2}$ products, and in fact only the $O(q)$ of them that actually occur — so this is a sequence of well-posed finite problems rather than a single infinite one. Third, the system is heavily overdetermined (equations grow like q, monomials like q^2), which is why its solution set collapses so cleanly onto the controls.

The independence hypothesis can moreover be weakened substantially, which matters because full $\mathbb{Q}$-linear independence of a triangular array of weight-two products is far beyond current technology. The relation's coefficient vector depends linearly on the exponent vector $(m_p, a_p)_{p \leq q}$; write C for the image of that linear map inside the space spanned by the occurring products. Then what is actually needed is not that the products have *no* relations, but only that

$$R \cap C = 0,$$

where R is their true relation space. Computing the map exactly in the prime basis gives a uniform picture: its kernel is *exactly* the control line $\{m_2 = a_3 = t\}$ in every case — so the encoding is faithful, its only degeneracy being the controls themselves — and its image has dimension $2\pi(q) - 1$, which is the full span of the occurring products:

q	$2\pi(q)-1$ products	$\dim C$	kernel	consequence
5	5	5	control line	independence suffices
7	7	7	control line	independence suffices
11	9	9	control line	independence suffices

Since C is the full span of the occurring products, the subspace weakening $R \cap C = 0$ coincides here with independence of those $2\pi(q)-1$ products; the weakening bites only if one enlarges the generating set beyond the prime basis. **[exact computation]** (`scripts/cdt_weight_two_encoding.py`).

What this does and does not deliver. It does not prove anything: weight-two independence is open even for a single product, which is exactly why Calegari–Dimitrov–Tang is notable. What it delivers is the missing bridge the survey asked for. The research question ceases to be "can the lattice be encoded in an arithmetic holonomic family?" and becomes the concrete one: *obtain $\mathbb{Q}$-linear independence of the $2\pi(q)-1$ products $\{\log 2 \log p\} \cup \{\log 3 \log p\}$, $p \le q$, for a single prime $q \ge 5$.* A calibration against the source is in order here. Calegari–Dimitrov–Tang's admissibility is a near-equality condition $|m/n - 1| < 10^{-6}$ on the parameters of $\log(1+1/m)\log(1+1/n)$, with the correct statistic $\min(|m|,|n|)$ rather than $\min(m,n)$ (negative parameters are allowed, since $\log(1+1/(-(k+1))) = -\log(1+1/k)$, so each product has several representations). Among the 3-smooth products relevant to the $q=3$ fragment — where m and $m+1$ must both be 3-smooth, so $m \in \{1,2,3,8\}$ by the kernel-verified pair classification of Section 79.1 — the smallest defect is $\frac{1}{4}$, attained by $\log\frac{3}{2}\cdot\log\frac{4}{3}$. That fragment is precisely the $q=3$ calibration, already known. Crucially, the route is *not* closed by admissibility: a counterexample's external prime is unbounded, so any counterexample carrying a prime $q > 10^6+1$ places genuinely admissible pairs $(i, i+1)$, $i \ge 10^6$, inside its own weight-two encoding. What is missing is not admissibility but the *joint* statement — their theorem gives independence within one product's four-element family, whereas one needs injectivity on the whole symmetric square. Any such extension excludes all counterexamples with q-smooth outputs, and does so by consuming the external prime — channel (i) of the control principle — since a control has no prime ≥ 5 in its support at all. **[proposed target]** target.

87.8 A self-contained threshold, independent of the quoted constant

The dependency flagged above — that Pila's constant 832 and his exact hypotheses are quoted rather than checked — can be removed, at the cost of a weaker constant, by running the interpolation-determinant argument directly against the corpus's own data. The point is not to beat the literature but to make the corpus's conclusions self-supporting.

Set $a = \log 2$, $b = \log 3$. Suppose g is entire, integer-valued on $\mathcal{S}$, with $\tau_2(g) \le \tau$, and consider the square matrix with rows indexed by L points of $\mathcal{S} \cap [0,T]$ and columns by the functions

$$\varphi_{i,j,k}(z) = 2^{iz}3^{jz}g(z)^k, \qquad ia + jb \le \Lambda, \quad 0 \le k \le d.$$

Every entry is an *integer*, because $2^s, 3^s, g(s) \in \mathbb{Z}$ for $s \in \mathcal{S}$; so if g is not algebraic of degree $\le d$ over the field generated by $2^z, 3^z$ the determinant Δ is a nonzero integer and $|\Delta| \ge 1$. Against this stands the Gel'fond-type analytic bound

$$|\Delta| \le L!\,(T/R)^{L(L-1)/2} \prod_{i,j,k} \max_{|z|\le R} |\varphi_{i,j,k}|,$$

in which $\log \max |\varphi_{i,j,k}| = R(ia+jb) + \tau R^2 k + o(\cdot)$. Two exact inputs of this report enter here: the monoid density $\#(\mathcal{S} \cap [0,T]) = T^2/(2\beta) + O(T)$, which bounds the number of available rows;

and the greedy-support mean frequency $\frac{2}{3}\Lambda$ of Section 87.4, which evaluates $\sum(ia+jb)$. Taking all available rows, $L = T^2/(2\beta)$ and $\Lambda^2(d+1) = T^2ab/\beta$, and writing $R = cT$, the contradiction $|\Delta| \geq 1$ versus the upper bound reduces after dividing by T^2 to

$$\frac{\log c}{4\beta} > \frac{2c}{3}\sqrt{\frac{ab}{\beta(d+1)}} + \frac{\tau c^2 d}{2}, \tag{802}$$

the remaining terms ($\log L!$ and the endpoint corrections) being $O(T^2 \log T)$ against the T^4 scale and hence negligible. Optimising (802) over the dilation c and the degree d gives a threshold of exactly the same shape as the quoted one:

$$\tau < \tau_{\mathrm{crit}}(\beta) = \frac{1}{C\,\beta^2 \log 2 \log 3}, \qquad C = 4570.$$

The constant C is stable to 0.063% over $\beta \in \{14, 20, 27, 50, 100\}$ — i.e. the β^{-2} scaling is exact, not fitted — and the resulting τ_{crit} is 1.4667×10^{-6} at $\beta = 14$ and 3.9420×10^{-7} at $\beta = 27$. **[exact computation]** (`scripts/own_type_threshold.py`).

Three consequences. First, $C = 4570$ against the quoted 832: our version is 5.49 times more conservative, exactly as expected from a crude determinant estimate that uses all rows and does not optimise the vanishing order — which is what makes it safe to rely on, since a smaller threshold yields a *weaker but self-contained* statement. Second, every statement in this section may therefore be re-read with 832 replaced by 4570 and becomes independent of the unaudited input: in particular Theorem 87.5 holds unconditionally on the literature in the form *no entire function integer-valued on $\mathcal{S}$ has* $0 < \tau_2 \leq 1/(4570\,\beta^2 \log 2 \log 3)$, and Target 87.6 correspondingly asks for a construction inside the smaller interval. Third, and most usefully, the *qualitative* conclusions do not depend on the constant at all: the excess factor of the vanishing branch is $C\beta\,ab/4$, which is 158β with the quoted constant and 870β with ours, but is *linear in β* either way. The finding that improving the zero-free region makes this gate harder is thus robust.

87.9 The conditional type gap, and a construction that would settle the conjecture

The measurement (803) kills the branch in which g is made independent of $2^z, 3^z$ by vanishing on the monoid. The remaining branch — *designed* transcendence — turns out to need no design at all, and the resulting statement is sharper than the original lead.

Proposition 87.4 (Algebraic dependence forces vanishing quadratic type)**.** *Let g be entire and algebraically dependent with 2^z and 3^z, i.e. $P(2^z, 3^z, g(z)) \equiv 0$ for some nonzero $P \in \mathbb{Z}[X,Y,Z]$. Then $\log M(g,r) = O(r \log r)$, and in particular $\tau_2(g) = 0$.*

Proof. Clearing the Z-degree, g satisfies $\sum_{k=0}^{d} a_k(z) g(z)^k = 0$ with $a_k \in \mathbb{Z}[2^z, 3^z]$ exponential polynomials and $a_d \not\equiv 0$. The elementary root bound for a monic-after-division polynomial gives, wherever $a_d(z) \neq 0$,

$$|g(z)| \leq \max\Big(1, \sum_{k<d} |a_k(z)/a_d(z)|\Big),$$

and this bound is exactly attained in the worst case (checked numerically on 2×10^4 random instances, worst ratio 1.0000). Each a_k has order one, so $\log|a_k(z)| = O(r)$ on $|z| = r$; and $\log^+|1/a_d|$ is bounded *in mean* by $O(r)$ through the Nevanlinna first main theorem applied to the order-one function a_d ([**classical/open background**]), which avoids any exceptional-set argument. Hence $\log M(g,r) = O(r)$ and in particular $\tau_2(g) = \lim r^{-2} \log M(g,r) = 0$. (An earlier version used the classical minimum-modulus theorem on circles avoiding an exceptional set and obtained the weaker $O(r \log r)$; the conclusion $\tau_2 = 0$ is the same, and the Nevanlinna route is both stronger and cleaner.) ☐

Combining Proposition 87.4 with Pila's theorem in the specialization $a = \log 2$, $b = \log 3$, $\alpha = 1$, $\gamma = \beta$ gives a conditional *gap*.

Theorem 87.5 (Conditional type gap)**.** *Assume Conjecture 1.1 fails, with canonical generator β. Then no entire function g that is integer-valued on $\mathcal{S}$ satisfies*

$$0 < \tau_2(g) \le \frac{1}{832\,\beta^2 \log 2 \log 3}.$$

Proof. If $\tau_2(g)$ were at most the threshold, Pila's theorem would make g algebraically dependent with 2^z and 3^z, whence $\tau_2(g) = 0$ by Proposition 87.4 — contradicting $\tau_2(g) > 0$. ☐

Research target 87.6 (Small-type integer-valued interpolant)**.** Exhibit an entire function g, integer-valued on $\mathcal{S}$, with $0 < \tau_2(g) \le 1/(832\beta^2 \log 2 \log 3)$, constructed from data available under a hypothetical counterexample. By Theorem 87.5 any such construction *proves* Conjecture 1.1.

Two features make this the most attractive of the surviving targets. First, it is *control-clean in the strong sense*: the quadratic density $\#(\mathcal{S} \cap [0,T]) = T^2/(2\beta) + O(T)$ is the exact signature of the least-generator monoid, so at an integer control $\mathcal{S} = \mathbb{N}_0$ has *linear* density and the entire quadratic-type analysis is vacuous. The route therefore consumes channel (ii) of the control principle (Section 88.6) — which most of the corpus's machinery does not. Second, it converts an open-ended search into a single explicit inequality. What is ruled out is only the *vanishing* construction, and by Proposition 87.7 that branch fails by an infinite margin rather than the finite one of (803); a construction whose small type comes from controlled interpolation rather than from zeros is untouched. The natural constructions therefore realise only the two extremes $\tau_2 = 0$ (any $\mathbb{Z}$-polynomial in $2^z, 3^z$, which is integer-valued on $\mathcal{S}$) and $\tau_2 = \infty$ (anything vanishing on $\mathcal{S}$), while Theorem 87.5 forbids the interval $(0, 1/(832\beta^2 \log 2 \log 3)]$. Whether the achievable spectrum meets $(0,\infty)$ at all is exactly the open question. [**paper proof**]; Pila's theorem and the minimum-modulus input [**classical/open background**]; target [**proposed target**].

87.10 Literature synthesis: five clusters, and one gate measured

A parallel survey of the published literature (retained in the repository as `alaoglu-erdos-literature-synthesis`) identifies five clusters bearing on the problem, and confirms that none contains an off-the-shelf theorem: every route meets the same terminal obstruction, a nontrivial integer relation among products of logarithms of primes. Its negative audit is worth recording, since each item has been proposed at some point in this corpus: a counterexample is necessarily irrational, so results about rational x add nothing; $\{(2^t, 3^t)\}$ is a transcendental arc, so Bombieri–Masser–Zannier, Maurin and Zilber–Pink do not apply merely because orbit points are S-integral; Baker theory controls linear forms with *algebraic* coefficients and not the terminal coefficients $m_p \log 3 - a_p \log 2$, which are themselves logarithmic; $(1-z)^{\vartheta}$ is not a G-function, since its coefficients are not algebraic, so arithmetic algebraization cannot be invoked without an arithmetic replacement; and generic rational-point counting tolerates far more points than the compact orbit produces.

The most valuable positive lead is the closest published program: Pila's theorems on entire functions sharing arguments of integrality apply to *exactly* the semigroup forced by a counterexample, since with $a = \log 2$, $b = \log 3$, $\alpha = 1$, $\gamma = \beta$ the four corner values are $2, M, 3, A$ and $X_{1,\beta} = \mathcal{S}$. In this specialization an entire g integer-valued on $\mathcal{S}$ with $\tau_2(g) := \limsup r^{-2} \log M(g, r) \le 1/(832\beta^2 \log 2 \log 3)$ is algebraically dependent on 2^z and 3^z, which turns the interpolation idea into a measured target. *Dependency to be audited.* Pila's precise hypotheses and the constant 832 are quoted from the survey and have not been checked against the source here; that they must contain more than is stated is visible from the control side, where $\mathcal{S} = \mathbb{N}_0$ and $g(z) = \sin(\pi z)e^{cz^2}$ is entire, integer-valued on $\mathcal{S}$, and has $\tau_2 = c$ arbitrarily small — so some hypothesis must exclude it. Note that this construction cannot cross to the conditional world: by Proposition 87.7 every function vanishing on an irrational-β monoid has $\tau_2 = \infty$. Every occurrence of the factor $208\beta \log 2 \log 3$ below is conditional on that unaudited input — and Section 87.8 removes the dependency altogether by deriving a threshold of the same shape from the corpus's own data. We settle that measurement. Any g forced to be independent of $2^z, 3^z$ by vanishing on the monoid pays, by Jensen's formula against the exact density $\#(\mathcal{S} \cap [0, T]) = T^2/(2\beta) + O(T)$,

$$\tau_2(g) \;\ge\; \sup_{c>0} \frac{2 \log c + 1}{4\beta c^2} = \frac{1}{4\beta} \qquad \text{(optimum at } c = 1),$$

so the gate fails by the factor

$$\frac{1/(4\beta)}{1/(832\beta^2 \log 2 \log 3)} = 208\,\beta \log 2 \log 3 = 158.392\ldots \times \beta, \tag{803}$$

that is by 2217.5 at the kernel-verified $\beta \ge 14$, by 4276.6 at the computational $\beta > 27$, and *growing linearly in* β — so every improvement of the zero-free region makes this particular gate harder, not easier.

In fact the vanishing branch fails by an infinite margin, which is worth recording because it is both sharper and simpler than the Jensen estimate. The monoid has counting function $T^2/(2\beta)+O(T)$ and *all its points are real and positive*, so $\sum_{s\in\mathcal{S}} s^{-3}$ converges while $\sum_{s\in\mathcal{S},\, s\le r} s^{-2}$ diverges, at the exact rate $\beta^{-1} \log r$: measured increments per doubling are 0.0719 against $1/\beta = 0.0714$, with $\sum s^{-3} = 1.20647$ at $\beta = 14 + \sqrt{2}/1000$, converged to six digits in T (`scripts/sol_canonical_type.py`); that constant is β-dependent — it is 1.20628 at $\beta = 14.3178$ and 1.20321 at $\beta = 27.1$ — and only its finiteness matters here. Hence the Hadamard genus equals the order, both being 2, with divergent coefficient sum, and the canonical product — which has minimal growth among all entire functions vanishing on $\mathcal{S}$ — satisfies $\log M(r) \asymp \beta^{-1} r^2 \log r$. Therefore:

Proposition 87.7 (Vanishing on the monoid forces infinite type)**.** *Every entire function vanishing on all of $\mathcal{S}$ has $\tau_2 = +\infty$.*

So no function that is independent of $2^z, 3^z$ by virtue of vanishing on the monoid comes within any finite factor of Pila's threshold, and (803) is merely the elementary shadow of this.

Proposition 87.7 has an immediate consequence which is of independent interest, because it says the conditional monoid is a *uniqueness set* for the whole finite-type class.

Corollary 87.8 (Monoid rigidity)**.** *Assume Conjecture 1.1 fails, with generator β. If g_1, g_2 are entire, both of finite quadratic type, and $g_1 = g_2$ on $\mathcal{S}$, then $g_1 = g_2$ identically.*

Proof. $g_1 - g_2$ vanishes on $\mathcal{S}$ and satisfies $\tau_2(g_1 - g_2) \le \max(\tau_2(g_1), \tau_2(g_2)) < \infty$, since $M(g_1 - g_2, r) \le 2 \max(M(g_1, r), M(g_2, r))$. By Proposition 87.7 a *nonzero* entire function vanishing on $\mathcal{S}$ has infinite type, so $g_1 - g_2 = 0$. □

Writing V_τ for the set of entire functions that are integer-valued on $\mathcal{S}$ and have $\tau_2 \le \tau$, the situation is therefore this: V_0 contains every $\mathbb{Z}$-polynomial in 2^z and 3^z; the type gap (Theorem 87.5) says $V_{1/(832\beta^2 \log 2 \log 3)} = V_0$; and evaluation on $\mathcal{S}$ is *injective* on $\bigcup_{\tau<\infty} V_\tau$ by Corollary 87.8, so each such function is pinned by its integer values alone. Whether $V_\tau = V_0$ for *every* finite τ — equivalently, whether the achievable type spectrum is exactly $\{0\} \cup \{\infty\}$ — is the sharp form of the open question, and a positive answer would close Target 87.6 negatively rather than leaving it open. **[paper proof]**. **[paper proof]**; the divergence rate **[exact computation]**. **[paper proof]**. *Erratum on the numerical check.* The ratios $1.219, 1.109, 1.054$ quoted earlier were computed at $\beta = 14$ — an *integer*, hence a control, where $\mathcal{S}$ collapses to $\mathbb{N}_0$ and the quadratic density fails — and over the pair multiset rather than the point set. At an admissible irrational β the two coincide and the figures are correct: recomputed at $\beta = 14 + \sqrt{2}/1000$ the ratios are $1.2193, 1.1086, 1.0540, 1.0269$ at $T = 140, 280, 560, 1120$, and at $\beta = 14.3178$ they are $1.2244, 1.1110, 1.0552, 1.0275$ — converging to 1 from above, so the bound is valid and slightly conservative. At the control $\beta = 14$ itself the true ratios instead tend to zero $(0.390, 0.197, 0.099, 0.050)$, exactly as the degeneration of the density predicts. The remaining way to use Pila's theorem therefore requires small type produced by controlled interpolation rather than by zeros; transcendence itself is then automatic, and the precise statement is the conditional type gap of Theorem 87.5 with its construction target 87.6. The survey's other four clusters — arithmetic holonomy for products of logarithms (Calegari–Dimitrov–Tang, which notably does *not* cover $\log 2 \log 3$ and must not be quoted as if it did), multivariate and adelic generalized factorials, global determinant architecture (Heath-Brown, Salberger, Walsh), and conjugate control via Schinzel–Zassenhaus and transfinite diameter — are templates and mechanisms rather than plug-ins, and are recorded there with the concrete stop conditions the survey supplies.

Assessment of the campaign. Two of the five frontier targets are now closed as methods (two-parameter transport, adelic wedge), one is revealed to be a restatement of the conjecture rather than a foothold (Kummer orbit), one is exactly calibrated as a zero-free-region generator with a measured and unfavourable dictionary (trap architecture), and the literature's closest published program has had its single numerical gate measured and found to fail by a linearly growing factor. Nothing here moves the four-exponentials barrier. What the campaign buys is precision: the surviving frontier is now two routes with *proven sufficient conditions* — the certified sparse near-curve search ($\mathrm{Cap}_L = o(L)$, verification finite) and the small-type integer-valued interpolant of Target 87.6, which is moreover control-clean through the monoid's quadratic density — together with the two output-sensitive cancellation targets (Sections 76, 78.4) that have survived every campaign so far. Of these the interpolant target is the sharpest, being a single explicit inequality whose satisfaction would settle the conjecture outright. Alongside them sits the weight-two encoding of Proposition 87.3, which is the only target that attacks the terminal obstruction head-on rather than routing around it, and which reduces it — for each prime q separately — to a finite independence statement inside the one family where products of two logarithms have ever been controlled.

88 Direct attempts during the integration session

This section is a working log, kept deliberately plain: each entry records an attack tried while Sections 72–76 were being integrated, the exact point at which it stops, and—where the failure is quantitative—the constant by which it fails. Entries marked *dead end* should not be retried in the same form; entries marked *direction* are not exhausted.

88.1 Schneider's auxiliary function with integral values (dead end, quantified)

The classical route to four exponentials is an auxiliary function $F(z) = \sum_{a,b<L} c_{ab}\, 2^{az}3^{bz}$ vanishing on the N^2 semigroup points $z_{n,k} = n + k\beta$ $(n, k < N)$. For a counterexample every value $2^{az_{n,k}}3^{bz_{n,k}} = 2^{an}3^{bn}M^{ak}A^{bk}$ is a *rational integer*, so Siegel's lemma can be applied over $\mathbb{Z}$ rather than over a number field: the degree factor $[K : \mathbb{Q}]$ and the denominator of the algebraic values—the two quantities that make the general four exponentials argument fail—are simply absent. This is the most obvious place where integrality could pay, so the constants were recomputed.

With $L = cN$ the L^2 unknowns face N^2 linear conditions whose coefficients have logarithmic size at most $LN(1 + x)\log 6$, so Siegel gives integer coefficients with $\log\|c\| \le \frac{N^2}{L^2-N^2}\, LN(1+x)\log 6 = \frac{c}{c^2-1}(1+x)\log 6 \cdot N^2$. On the circle $|z| = R = (1+\beta)N\, e^s$ one has $\log|F|_R \le cN \cdot R\log 6 + \log\|c\|$, and Schwarz's lemma with N^2 zeros inside radius $(1 + \beta)N$ gives, at a new semigroup point z' of modulus $\le (1 + \beta)N$,

$$\log|F(z')| \le \Big[c(1+\beta)e^s \log 6 + \frac{c(1+x)\log 6}{c^2-1} - s\Big]N^2.$$

A contradiction with $F(z') \in \mathbb{Z} \setminus \{0\}$ needs the bracket to be negative. Since $c(1 + \beta)e^s \log 6 \ge s$ for every $s \ge 0$ and $c \ge 1$, $\beta > 1$, the bracket is positive for every choice of c, s; moreover the second term grows with x, so the kernel-verified bound $x \ge 13$ makes the deficit *larger*. Integrality removes a bounded factor from an argument that fails by an unbounded one. *Dead end.* The same bookkeeping, with the zero lemma replaced by interpolation determinants, is the min-plus ceiling of Section 20.

88.2 Structural-prime relaxation of the archimedean target (dead end, quantified)

Every counterexample has a prime $r \nmid 6$ with $r^e \parallel M$ for some $e \ge 1$ (Section 8). If the auxiliary function above is restricted to monomials with $a \ge 1$, every value $F(z_{n,k})$ is divisible by r^{ek}, so a nonzero value has $|F(z_{n,k})| \ge r^{ek}$ and the archimedean target relaxes from $|F| < 1$ to $|F| < r^{ek}$. This is a genuine second source of size, but it is *linear* in N $(k < N)$, while the deficit in Section 88.1 is quadratic. The same holds for any fixed finite set of structural primes and for the 3-side prime $s \mid A$. *Dead end* in this form; the only way a structural prime could matter is through a mechanism whose gain is itself quadratic in the interpolation parameters, which is the hybrid Target 74.3 restated.

88.3 Column divisibility in interpolation determinants (dead end, quantified)

The structural-prime gain of Section 88.2 is additive over points, so it is natural to put it into an interpolation determinant, where the arithmetic side is a *sum* over columns. Take $L = N^2$ functions $2^{az}3^{bz}$ $(1 \le a \le N,\ 0 \le b < N)$ and the L points $z_{n,k}$, $0 \le n, k < N$. The determinant $\Delta = \det\big(2^{az_{n,k}}3^{bz_{n,k}}\big)$ is a rational integer; the column of $z_{n,k}$ is divisible by r^{ek}, hence

$$v_r(\Delta) \;\ge\; e\sum_{n,k<N} k \;=\; \tfrac{e}{2}N^3(1 + o(1)),$$

a *cubic* lower bound for the arithmetic size of a nonzero Δ. The archimedean side (Laurent's lemma, with the points in a disc of radius $\rho = (1 + \beta)N$ and the functions bounded on $|z| = R = e\rho$ by

6^{NR}) gives $\log|\Delta| \le -\frac{1}{2}L^2 + L\cdot NR\log 6 = \big(e(1+\beta)\log 6 - \frac{1}{2}\big)N^4$. The right side is positive, i.e. the archimedean bound is already useless before the cubic r-adic term is subtracted; and even if the archimedean constant could be made negative, a cubic gain cannot compensate a quartic deficit. Summing valuations over columns turns a linear gain per point into a cubic total, one degree short of the quartic scale of the determinant. *Dead end.* A quadratic gain *per point* would be needed (Section 88.12, item 3).

88.4 Integer-valued exponential polynomials on the solution semigroup (dead end)

Every $G(z) = \sum c_{ab}2^{az}3^{bz}$ with $c_{ab} \in \mathbb{Z}$ is integer-valued on $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$, an entire function of exponential type $(a+b)\log 3$ taking integer values on a set with $\asymp T^2/(2\beta)$ points in $[0,T]$. Pólya's theorem (type $< \log 2$ on $\mathbb{N}$ forces a polynomial) does not apply because the type exceeds $\log 2$, and Gel'fond's two-dimensional theorem does not apply because $\mathcal{S}$ lies on a line: the N^2 points in a disc of radius N have mutual distances $n - n' + (k-k')\beta$, and Newton's divided differences at them have denominators in $\mathbb{Z}[\beta]$, for which no Liouville-type lower bound is available (β is transcendental with no known irrationality measure). Restricting to points with integer differences recovers only N points per line, i.e. the finite-difference obstruction of Section 17. *Dead end.*

88.5 The *abc* conjecture against convergent gaps (dead end)

For a convergent p/q of β, the triple $3^p + G = A^q$ with $G = A^q - 3^p$ has $|G| \asymp 3^p|q\beta - p|\log 3 \le 3^p/q$, so *abc* yields only $1 \ll (\mathrm{rad}(A)/q)^{1+\varepsilon}3^{p\varepsilon}$, which is never violated: a contradiction would need $|q\beta - p| \ll 3^{-p\varepsilon}$, exponentially small, whereas convergents are only polynomially good. This is the quantitative reason that Proposition 73.3 is a radical statement and not an exclusion. *Dead end.*

88.6 The control principle

Every hypothesis used so far that is stated in terms of the integers (M, A) and the ratio $\log 2/\log 3$ alone is also satisfied by the *controls* $(M, A) = (2^m, 3^m)$, $m \ge 13$, for which the conclusion $x \in \mathbb{Z}$ is true and no contradiction can exist. Hence any proof must use a hypothesis that fails for the controls. The corpus knows exactly three such unconditional facts: (i) an external prime $r \ge 5$ divides MA (Section 8); (ii) β is the least generator and $\mathcal{S} = \mathbb{N}_0 + \mathbb{N}_0\beta$ with unique coordinates (Section 4); (iii) the spectrum is saturated: $u^\beta \in \mathbb{Q}$ for a positive rational u iff u is 3-smooth (Section 12). Every framework in Sections 72–76 uses (ii) at most through the existence of many near-units and (i) not at all in its decisive estimate; this is why all of them are control-blind and all of them fail. A proposed argument should be checked against the controls *first*: if it does not visibly consume (i), (ii), or (iii), it cannot work.

88.7 Kernel frontier (progress)

The control principle has one positive consequence: the only statements in this report that are *not* control-blind and are fully proved are the finite exclusions, which use the actual values of 2^x and 3^x. The kernel-verified region was raised during this session from $2^x < 4096$ to $2^x < 8192$ and then to $2^x < 16384$ (Theorem 3.3); the hardest rows need semiconvergent enclosures rather than the next convergent. The 16384 table has 8191 rows, its hardest value $m = 9329$ uses the enclosure 5914137/3731405 (ten seconds in the kernel, a hundred in exact Python), and the whole table replays in under eight minutes. The remaining gap to the software frontier 2^{27} is 13 further doublings of mechanical work.

88.8 Channel reconnaissance on near-ϑ pairs (computed)

Section 75.3 left open whether the two fifth-trace channels carry a stable arithmetic signature that a Kummer or S-unit argument could grip. An exact computation (**[exact computation]**, `gmpy2` big integers) over the surrogate pairs (M, A) with $A = \lfloor M^{\vartheta} \rceil$, $M \in \{5, 7, 11, 13, 41\}$, and all convergents p/q of $\log_2 M$ with $q \le 4000$ gives: the gcd of the two normalized fifth-trace factors is 1 in every case except three, and in those three it is a single small prime (11, 19, 19) carried with full valuation by exactly one channel — twice the direct gap, once the reciprocal gap — as Theorem 75.5 mandates. No channel preference and no growth of shared content along q is visible. This is what one expects if nothing forces modular synchronization; as a target for "forced shared trace content" (Target 75.7(i)), the fifth trace looks barren, and effort should go to channel-*incompatibility* (Target 75.7(ii)) or higher traces instead.

88.9 The terminal factorials are multiplicatively independent (proved)

Analyzing the exact-unit case of the factorial target yields a clean unconditional statement. Let $1 < M < A$ with $\log M/\log A > 0.525$ — satisfied by every mesoscopic pair, where $\log M/\log A = \log 2/\log 3 = 0.6309\ldots$ — and $F_t = (A^t)!/(A^t - M^t)!$.

Proposition 88.1 (Multiplicative independence of the terminal factorials)**.** *There is $n_0 = n_0(M, A)$ such that the integers $F_{n_0}, F_{n_0+1}, F_{n_0+2}, \ldots$ are multiplicatively independent: $\prod_j F_{n+j}^{c_j} = 1$ with integer exponents forces every $c_j = 0$. Consequently $\mathcal{F}_{P,n} = 1$ is impossible for a nonzero $P \in \mathbb{Z}[T]$ once $n \ge n_0$.*

Proof. The interval $(A^t - M^t, A^t]$ has length X^{α} with $X = A^t$ and $\alpha > 0.525$, so for X large it contains a prime p_t (Baker–Harman–Pintz [32]); fix such a prime for each $t \ge n_0$. Suppose $\prod_j F_{n+j}^{c_j} = 1$ with not all $c_j = 0$, and let j^* be the *largest* index with $c_{j^*} \ne 0$. The prime p_{n+j^*} divides F_{n+j^*} exactly once, divides no F_{n+j} with $j < j^*$ (it exceeds A^{n+j}), and the levels above j^* carry zero exponent. Taking $v_{p_{n+j^*}}$ of the relation gives $c_{j^*} = 0$, a contradiction. □

The finite core is **[kernel-verified Lean]** in `TerminalFactorialIndependence`: the window valuation $v_p(F_t) = 1$ (`padicValNat_termFall_eq_one`, from pure inequalities $A^t - M^t < p \le A^t$, $A^t \le 2(A^t - M^t)$, $A^t < p^2$), the exclusion $p \nmid F_{t'}$ for $A^{t'} < p$, and the downward induction (`termFall_multiplicative_independent`), with the window primes supplied as an explicit hypothesis. The short-interval prime existence filling that hypothesis is **[classical/open background]** (Bertrand's postulate does not suffice: the window has sublinear length; Huxley's asymptotic used in Theorem 74.2 is stronger than needed, existence at exponent 0.525 is enough). The proposition does more than remove the exact-unit case: it *closes the constructive branch of Target 74.3 entirely.*

Corollary 88.2 (The constructive branch of the linear-width target is empty)**.** *Let (M, A) be any pair in the mesoscopic window (in particular the output pair of any hypothetical counterexample), and let $P_n \in \mathbb{Z}[T]$ be nonzero polynomials of arbitrary degrees with $v_p(\mathcal{F}_{P_n,n}) \ge 0$ for every prime p and $\mathcal{F}_{P_n,n} \to 1$. Then no such sequence exists: for all large n the two conditions force $P_n = 0$.*

Proof. Nonnegative valuations make $\mathcal{F}_{P_n,n}$ a positive integer; convergence to 1 puts it in $(0,2)$, hence equal to 1, for all large n; and Proposition 88.1 then forces every coefficient of P_n to vanish. ☐

No width hypothesis, coefficient bound, or archimedean sign analysis enters: items (2) and (3) of the target are jointly unsatisfiable with $P_n \neq 0$. The "linear-width escape" left open by Theorem 74.2 therefore does not exist, and the factorial-cocycle architecture of Section 74 is a complete no-go for proving the conjecture through an integer sequence converging to one. (The proof is embarrassingly short in hindsight; what was missing was the multiplicative independence of the F_t, which makes the endgame "integer $\to 1$ is eventually 1" terminal rather than a step.) The same argument closes the binomial variant of Section 74: a window prime has $v_p\binom{A^t}{M^t} = v_p(F_t) = 1$ (it exceeds M^t, so the lower factorial is prime to it), and the levels below are again clean, so the binomials $\binom{A^t}{M^t}$ are multiplicatively independent as well. What the argument does not exclude is a cocycle that is an integer *without* converging to 1 — there the LP computation below gives strong evidence of emptiness as well, now the only remaining question in this family.

88.10 The LP layer of the linear-width factorial target is empty for the control (computed)

Target 74.3 asked whether a factorial cocycle of *linear* width can be integral; the prime-window theorem closes widths below $\log_2 3 - 1 = 0.585\ldots$ and was expected to leave the wider regime open. The LP layer of the target was therefore computed exactly for the control pair $(M, A) = (2, 3)$ (**[exact computation]**, exact valuations, floating LP with margins far from zero). Writing $P = P_* S$ (which encodes the three archimedean annihilations identically), the question is whether the rational polyhedral cone $\{s : \sum_j c_j(s) V_p(n + j) \geq 0 \text{ for all } p \leq 3^{n+d}\}$ contains a nonzero vector. The result, over every affordable shape — $n \in \{2, 3, 4, 6\}$, all $4 \leq d \leq 12$ with $3^{n+d} \leq 3^{15}$, hence width ratios $d/(n + d)$ up to 0.857 — is that the cone is $\{0\}$: the optimal min-margin per unit coefficient is between -17 and -28 valuation units and shows no trend toward feasibility as the width ratio passes 0.585. In particular, for these (n, d) *no* real-coefficient multiple of P_* of any size makes every prime valuation of the cocycle nonnegative; a fortiori no integer one does.

The Farkas dual is more informative than the infeasibility. At $(n, d) = (2, 12)$ the certificate is carried by eight rows: the *dense* rows of the small primes $3, 11, 13$ — whose leading behaviour $V_p(t) \approx 2^t/(p - 1)$ contributes nothing against the family, because $\sum_j c_j 2^{n+j} = 2^n P(2) = 0$ kills the main term identically — and five *sparse* rows of medium "capture" primes (107, 263, 1129, 1583, 6427), each a single prime, which price the residual error terms of the dense rows. The mechanism is thus not the short-interval window of Theorem 74.2 (which needs n large) but a mixed small–medium-prime balance: the archimedean annihilations force the dense main terms to cancel, and what the dense rows leave behind is negative against the sparse captures.

Two caveats and one consequence. The computation is at small n and for the control pair only, and the LP is the relaxation of an integer question — but infeasibility of the relaxation is the strong direction. The consequence is a change of expectation: Target 74.3 should be expected to have a *negative* answer at every width, and the right theorem to attempt is an infeasibility certificate family $\{\mu_p^{(n,d)}\}$, built from the dense rows of a fixed small-prime set plus $O(1)$ capture primes, valid for all (n, d) and for every pair (M, A) in the mesoscopic window. Such a theorem would close the entire factorial-cocycle architecture unconditionally, converting Section 74 from "one escape width left open" to a complete no-go; the dual data above is the template for its proof.

88.11 The two-dimensional transport kernel: first finite exploration (computed)

Following the contraction-evasion target of Section 81.3, we computed both natural transport kernels for the two-parameter terminal products $G_{i,j} = (A^{m+j})_{\underline{M^{n+i}}}$, windows $I_{i,j} = (A^{m+j} - M^{n+i}, A^{m+j}]$, on truncated admissible grids (`scripts/two_dim_transport.py`; log-space, $x = 14$ scales, several base offsets). Two structural facts emerge. First, the *per-prime* sup kernel does not contract: M^iA^j-weighted column sums reach 10^{46}, and the escaping direction is exact — same-top pairs ($j = j'$) with a long-window source against a short-factor target ($i \gg i'$), whose weight ratio $M^{i-i'}$ is unbounded. The one-dimensional contraction of Section 81 survived because its single shift index totally orders windows below tails; a second parameter destroys that total order. Second, the escape is an artifact of per-prime worst-casing. A same-top shorter level captures only its sub-window, hence only the fraction $M^{i'-i} \ll 1$ of the window primes, and in the divisor-counting model (an integer of size $A^{m+j'}$ has at most $(m + j')/(m + j)$ prime factors of window size) triangularity is restored in both indices. On every grid tested the counting model then has *exactly one* cell whose window no single other level can cover: the top cell of the support (maximal j, then maximal i). For that cell the union bound is even stronger — all other levels together capture at most a $\sum_{i'<i} M^{i'-i} \le 1/(M - 1)$ fraction of its window — so a negative coefficient at the top cell of its own support makes the two-parameter cocycle nonintegral outright, while every other cell can in principle be covered by the top level. The two-dimensional route therefore reduces to a lexicographic pivot: any viable sign pattern must keep the top cell positive and route all rescues through the top levels, and the question is the multiplicity bookkeeping of those rescues against the two archimedean balances $\sum_{ij} c_{ij}M^iA^j = 0$ and its derivative. Neither a two-dimensional contraction theorem nor an escaping construction falls out cheaply; the pivot sharpens the open target to a concrete combinatorial question on sign patterns.

The pivot upgrades to an *exact partial rigidity theorem* on the top slice. Fix the maximal j^* of the support and list its cells $i_1 < \cdots < i_s = i^*$. The windows of one slice are nested, so the annulus $\mathcal{A}_k = (A^{m+j^*} - M^{n+i_k}, A^{m+j^*} - M^{n+i_{k-1}}]$ is met by the levels (i_l, j^*) with $l \ge k$ and by nothing else in the support; and for a prime $p \in \mathcal{A}_k$ each such level captures p *exactly once to first order* (the unique multiple in range is p itself, and $p^2 > A^{m+j^*}$), while all lower slices miss p and no higher slice exists. Hence

$$v_p(\mathcal{F}) = \sum_{l \ge k} c_{i_l,j^*} \qquad (p \in \mathcal{A}_k)$$

exactly, and integrality of the two-parameter cocycle forces every tail sum of the top slice to be nonnegative — with Huxley supplying primes in each annulus (the annulus has the same length exponent as its window) whenever the cells are mesoscopic. In particular a negative top-cell coefficient is impossible, and the top slice of any viable sign pattern is constrained by s exact linear inequalities before any transport estimate is invoked. A second exact experiment (`scripts/two_dim_signpatterns.py`) computes the integer nullspace of the three archimedean balances $\sum c_{ij}M^iA^j = \sum ic_{ij}M^iA^j = \sum jc_{ij}M^iA^j = 0$ on small admissible supports and filters the sign patterns through the tail-sum constraint, the coverage necessity, and the valuation-mass necessity ($c^- \cdot \#\text{primes} \le \sum c^+ \cdot \text{incidence}$). Two calibration lessons: instantiating the *control* pair degenerates the balance lattice (relations of the shape $(A - 3^{14})^2 = 0$) — the control principle bites the experiment itself, so generic perturbed pairs must be used, as the external-prime report's protocol prescribes; and the counting bound without multiplicities is far too weak (it passes embedded vertical one-parameter families that the mass constraint then kills by factors of 10^5). After all exact constraints, the surviving sign patterns collapse to a single shape: a negative cell rescued by a *same-top longer-window* positive with coefficient inflation ratio $\approx M^{\Delta i}$, sitting within $O(1)$ of the mass-balance boundary — the coefficient-inflated top-routing predicted by Section 81.3, now with its scale pinned. The two-dimensional escape corridor, if it exists at all, is exactly this razor-thin inflated channel.

The analysis closes further. For a negative tail $T_k^{(j)} < 0$ at *any* slice there is a trichotomy: either the annulus primes make the cocycle nonintegral outright; or some higher-slice level is a *dense capturer* ($M^{n+i'} \geq A^{m+j}$: every annulus prime is hit, with uniform count $\approx M^{n+i'}/A^{m+j}$); or the sparse capturers must cover the whole annulus, which the incidence bound blocks outside a thin near-dense boundary layer. The dense channel is then starved by the balance weight itself: writing $\Phi = c\, M^i A^j$ for the B_0 weight, the per-prime rescue bound $c^- A^{m+j} \leq 3c^+ M^{n+i'}$ converts through the exact exponent identity $M^{n+i'}(M^i A^j)A^{m+j'} = (M^{i'} A^{j'})M^{n+i} A^{m+j}$ into

$$\Phi^- \, A^{m+j'} \leq 3\, \Phi^+ \, M^{n+i}, \qquad \text{while admissibility gives} \qquad M^{n+i}\, A \leq A^{m+j'} :$$

every cross-slice rescue costs at least one factor A of Φ-mass, the summed source kernel stays geometric, and hence the total Φ-mass of negative slice tails is $O(1/A)$ of the positive mass. The finite core is kernel-checked in the new module `CrossSliceContraction` (`phi_transport_identity`, `phi_transport`, `phi_contraction_factor`), and a targeted exact experiment (`scripts/two_dim_dense_check.py`) confirms the starvation: on grids containing dense (capturer, slice) pairs, *zero* of 27,000 balance-nullspace directions survive the dense-rescue mass test. Conclusion for the two-dimensional interface of Section 81.3: there is no cheap escape (all rescues are Φ-starved or prime-starved) and no blanket contraction theorem either — *top-heavy* patterns, with every slice tail essentially nonnegative, satisfy the entire window layer, so the window layer alone would leave the two-parameter route open at the sub-window primes of top-heavy patterns.

That apparent residue is illusory, and the window layer is not in fact the last word: once the archimedean balance is imposed as well, the top-heavy class is *empty*, so the sub-window primes and the structural primes are never reached.

Theorem 88.3 (Top-heavy collapse)**.** *Let $c = (c_{ij})$ be integer coefficients on an admissible two-parameter grid, with strictly increasing window weights inside each slice, satisfying the balance $\sum_{i,j} c_{ij} M^i A^j = 0$. If* every *slice tail sum is nonnegative ($T_k^{(j)} = \sum_{l \geq k} c_{i_l,j} \geq 0$ for all j and all k), then $c \equiv 0$.*

Proof. Fix a slice j and write $w_l = M^{n+i_l}$ for its window weights, which are positive and strictly increasing. Abel summation gives the exact decomposition

$$S_j := \sum_l c_{i_l,j} w_l = T_0^{(j)} w_0 + \sum_{k \geq 1} T_k^{(j)} (w_k - w_{k-1}), \tag{804}$$

in which every term is a product of a nonnegative tail with a nonnegative weight increment; hence $S_j \geq 0$ for every slice. The balance reads $\sum_j A^j S_j = 0$ with $A^j > 0$, so a sum of nonnegative terms vanishes and therefore $S_j = 0$ for *every* j. Returning to (804) with $S_j = 0$: a nonnegative decomposition summing to zero is termwise zero, so $T_0^{(j)} w_0 = 0$ and $T_k^{(j)}(w_k - w_{k-1}) = 0$ for all k; since $w_0 > 0$ and the increments are strictly positive, all tails of the slice vanish. Finally $c_{i_l,j} = T_l^{(j)} - T_{l+1}^{(j)} = 0$. ☐

Theorem 88.3 has a quantitative refinement that explains *why* the collapse is forced, and locates where any surviving pattern must put its mass. If a slice has all tails nonnegative, then a negative coefficient at index l satisfies $c_l^- \le \sum_{l'>l} c_{l'}^+$ (immediately from $T_l \ge 0$), while the window weights obey $w_{l'} \ge M w_l$ for $l' > l$; hence

$$c_l^- w_l \;\le\; \frac{1}{M}\sum_{l'>l} c_{l'}^+ w_{l'}, \qquad \text{so} \qquad N_j \le \frac{s_j}{M}\,U_j, \qquad S_j \;\ge\; \Big(1-\frac{s_j}{M}\Big)U_j,$$

where U_j, N_j are the positive and negative window-weighted masses of the slice and s_j its number of cells. Since $M \ge 2^{14}$, a top-heavy slice has mass essentially equal to its *entire* positive mass: cancellation inside a top-heavy slice is impossible up to a factor $1 - s_j/M$. Feeding this into the balance shows that a surviving pattern must concentrate its weighted mass at *low* slices, since the top slice then obeys $U_J A^J \le \big(s\,(m+J)\log A/(A-1)\big)\sum_j U_j A^j$ with a prefactor of order 10^{-5}.

Both nontrivial steps are kernel-verified in the new module `AbelSliceMass`: the Abel identity and the resulting slice-mass positivity (`abel_identity`, `slice_mass_nonneg`), and the sharp collapse from a vanishing mass (`slice_vanishes_of_mass_zero`, with `tail_sub_tail` and `telescope`), together with the quantitative weight-gap bounds (`neg_le_pos_above`, `neg_weighted_le`). An exact experiment (`scripts/two_dim_allslice.py`) confirms the theorem on truncated grids: imposing the all-slice tail condition on the exact balance nullspace leaves *zero* survivors out of 14,787 directions tested, whereas imposing it on the top slice alone leaves 68.

Theorem 88.3 closes the corridor that the sign-pattern search had isolated. Combining it with the dichotomy and the Φ-contraction gives the exact structure of what remains: *a nonzero balanced integral two-parameter cocycle must carry a negative tail at a non-top slice, sustained entirely by cross-slice rescues, each of which loses at least a factor A of Φ-mass.* The surviving question is therefore purely the multiplicity bookkeeping of those starved rescues — a single, sharply posed finite question, in place of the open-ended "design a two-dimensional construction" target of Section 81.3.

Two exact facts sharpen that residual bookkeeping and are worth recording, because they replace the crude Legendre allowance by something rigid.

Multiplicity one. A level (i', j') can contribute a prime power to an annulus prime $p \asymp A^{m+j}$ only if $p^2 \le A^{m+j'}$, i.e. only if $j' \ge m + 2j$. Throughout the mesoscopic range — and on every grid inspected (`scripts/annulus_starvation.py`) — every higher slice satisfies $j' < m + 2j$, so *the prime-power allowance vanishes identically*: each higher level either misses p or captures it exactly once. The transport bound's additive term $2(1 + \ell/t)$, which is what defeats naive counting in the one-parameter setting, is simply absent here.

Exact double counting. Consequently, for a negative tail at slice j, index k, every annulus prime must be covered by a set of higher levels whose positive coefficients already sum to $-T_k^{(j)}$; and a level (i', j') can cover at most $M^{n+i'}$ annulus primes, since an integer of size $A^{m+j'}$ has at most one prime factor of annulus size. Summing the covering requirement over all $P_k^{(j)}$ annulus primes gives the exact, coefficient-weighted necessity

$$-T_k^{(j)} \;\cdot\; P_k^{(j)} \;\le\; \sum_{j'>j}\sum_{i'} c_{i'j'}^+\, M^{n+i'}, \qquad P_k^{(j)} \sim \frac{M^{n+i_k} - M^{n+i_{k-1}}}{(m+j)\log A}, \tag{805}$$

which is the rigorous form of the valuation-mass filter used in the sign-pattern search. Coefficient-free counting alone does *not* close the lower slices — the capacity $\sum_{j'>j,i'} M^{n+i'}$ exceeds the annulus prime count by some fifteen orders of magnitude on the grids inspected, while it is $-\infty$ at the top slice, recovering Theorem 88.3 — so (805) must be run against the balances rather than alone. That is exactly the remaining finite question. (Control status: like the one-parameter global contraction, this argument holds verbatim for the integer controls, so it closes the two-parameter *method*; it is not a route to the conjecture.) A third exact layer is transversal to both: at the structural primes, Legendre gives exactly $v_2(\mathcal{G}) = \sum_{ij} c_{ij}\big(M^{n+i} + s_2(A^{m+j} - M^{n+i}) - s_2(A^{m+j})\big)$ (and its v_3 analogue), whose leading weight M^{n+i} is independent of the slice — a constraint neither the balances nor the window layer sees. Adding the exact $v_2, v_3 \geq 0$ filter to the sign-pattern search leaves the marginal same-top corridor unchanged (its dominant coefficients are positive, so σ-positivity is automatic there) while excluding the opposite corner, where large negative coefficients sit at long windows.

[exact computation] for the kernel computations. The tail-sum theorem's finite core is now kernel-verified in `TopSliceTailSum`: the exact window valuation $v_p((A)_{\underline{K}}) = 1$ for $A - K < p \leq A < 2p$ (the unique multiple among the factors is p itself), the exact below-window vanishing for $p \leq A - K$, and the exact tail-sum identity `top_slice_tail_sum` for the signed product over a nested slice — so the assertion "integrality forces every realizable top-slice tail sum to be nonnegative" rests on kernel-checked valuations, with only Huxley's prime supply in the annuli remaining classical.

88.12 Directions not yet exhausted

1. **Negative closure of the linear-width family.** Done, and subsequently completed in full generality: the constructive branch is empty for every mesoscopic pair (Corollary 88.2), the LP layer is empty for the control (Section 88.10), and the global terminal-prime contraction (Theorem 81.1, kernel core `CriticalWindowContraction`) closes bare integrality itself — the certificate theorem anticipated here exists and is uniform in (n, d).
2. **Shared-face masks on the actual stable support.** Target 76.3 is a finite integer program once (M, A) is fixed; run it on control pairs to learn the shape of near-optimal masks, then test whether that shape consumes (i) above.
3. **Quadratic-gain structural mechanisms.** Section 88.2 shows that a structural prime gives a linear gain per point. A mechanism that makes the r-adic divisibility grow like k^2 (for example, values at $z_{n,k}$ divisible by $r^{e\binom{k}{2}}$) would change the bookkeeping; r-adic Kummer congruences for $\binom{A^n}{M^n}$ (Section 74.4) are the one place such growth has been seen.
4. **Thirteen doublings.** Closing the gap between the kernel region and the directed-rounding scan is mechanical with the semiconvergent device; it converts **[software computation]** into **[kernel-verified Lean]** for every exponent bound used in the descent constants. (Deliberately not pursued further in this session: the marginal value of each doubling is small, and the work is pure arithmetic replay.)

89 What the integrated frameworks add to the frontier

Sections 72–76 came from five independent investigations. Read together, they make three points that the earlier material stated less sharply.

One mechanism, four disguises. Prime-top stencils, factorial cocycles, centered traces, and moment masks all manufacture a rational number from a counterexample, make it archimedeanly close to a target, and then lose because the *reduced* height grows linearly in the approximation index q while Baker-type bounds cap the archimedean gain at $O(\log q)$ per unit of contact. The new results identify, in each case, the exact finite place where the reduced denominator survives: a fresh prime of order 2ℓ (Lemma 72.1), an uncovered short-interval prime (Theorem 74.2), an unreduced cubic defect (Corollary 75.2), or a unique corner of a shared Newton face ((792)). Every one of these survivals is a finite statement now checked by the kernel. What is never available is a *second* source of size—the same diagnosis as Section 21.

Equivalences are not reductions. Theorem 73.2 shows that sparse coaction separation, like the log-product criterion restricted to the pairs that actually occur (Section 11.3) and the subquadratic counting bound (Section 4.5), is the conjecture in different clothing. Its value is the explicit witness $\Lambda_q = \log(3^{a_q}/2^{b_q})$ with the elementary bound (780), not a weakening of the goal.

Does this bring the proof closer? Honestly assessed: no theorem in Sections 72–76 narrows the *truth* of the conjecture; the open problem is unchanged and still sits exactly on the four-exponentials boundary of Section 6. What has changed is the map of the search space. Four constructions that looked like candidate proof engines are now closed exactly (growing-order linear stencils, fixed and subcritical-width factorial cocycles, cubic-trace iteration, one-direction Prouhet masks), one proposed intermediate target has been shown to be the conjecture itself (Theorem 73.2), and the finite algebra behind every closure is kernel-checked, so none of these routes will be re-opened by an oversight. The residual targets below are the honest statement of what a proof would still have to supply.

The one literal advance on the conjecture itself made during this integration is finite: the kernel-verified zero-free region moved from $2^x < 4096$ to $2^x < 16384$ (Theorem 3.3), so a counterexample now has $x \geq 14$ with kernel trust. The software scans already reach $x > 27$; the gap between the two is engineering, not mathematics, and the semiconvergent device makes each further doubling a routine table.

Where genuinely new input could enter. Three targets survive all five audits and are not yet known to be equivalent to the conjecture: shared structural-face capture (Target 76.3), a finite-prime statement about repeated cancellation on fixed Newton faces coupled by (791); and the nonlinear prime-top problem of Section 72.7. (The third such target at the time of integration, the linear-width factorial feasibility problem, was resolved negatively during the session — Corollary 88.2 — and then closed in full generality by the global terminal-prime contraction of Section 81.) Each has a concrete computational entry point and a precise kill criterion, and each would have to use the structural-prime asymmetry ((784), $\mathcal{P}_\pm$) that distinguishes a hypothetical primitive pair from the integer controls $(2^m, 3^m)$.

A Status matrix for major statements

The status tags have the following meanings. **[kernel-verified Lean]** marks a statement that the Lean kernel accepts at the current repository state, with no `sorry`, no extra axioms, and no `native_decide`. **[paper proof]** marks a conventional paper proof documented in the thematic section cited by its row but not yet formalized. **[Lean draft, not yet accepted]** marks a statement for which Lean source exists in the repository but has not yet been accepted by the kernel. **[software computation]** marks a finite software computation with its local trust boundary—for example

CPython/`mpmath` interval arithmetic or C/MPFR/GMP directed rounding—which is not a kernel certificate. [**classical/open background**] marks classical or open background. No entry below asserts the full Alaoglu–Erdős conjecture; it remains open.

No row of the table carries [**Lean draft, not yet accepted**]. Every theorem module catalogued in Section B is accepted by the kernel and placed on the aggregate import surface. The tag applies only to the separate, unassembled 16384 finite-certificate attempt described in Section 23, which is not a row of this major-statement table and supplies no theorem.

Statement	Status	Location
Classical frame		
$2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$	[**kernel-verified Lean**]	`IntegerExponent`
Gelfond–Schneider theorem	[**kernel-verified Lean**]	`GelfondSchneider`
Four exponentials conjecture (hypothesis only)	[**classical/open background**]	[28, 29]
Full Alaoglu–Erdős conjecture	Open	Section 5, Section 25
Transcendence and reduction		
Nonintegral solution irrational and transcendental	[**kernel-verified Lean**]	`TwoBaseIntegerExponent`, `Transcendence`
$\vartheta = \log_2 3$ transcendental	[**kernel-verified Lean**]	`Transcendence`
Four-exponentials conjecture $\Rightarrow$ Conjecture 1.1	[**kernel-verified Lean**]	`FourExponentialsReduction`
Prime-log-product independence $\Rightarrow$ Conjecture 1.1	[**kernel-verified Lean**]	Section 11; `LogProductIndependence`
Prime-log-product independence $\Rightarrow$ the rational-output form (24)	[**kernel-verified Lean**]	Section 11; `LogProductIndependence`
Two-prime case of the independence criterion, and its sharpness	[**kernel-verified Lean**]	Section 11.5
Zero-free regions and finite searches		
$2^x < 100$ and $2^x < 256 \Rightarrow x \in \mathbb{Z}$	[**kernel-verified Lean**]	`FiniteCheck`, `FiniteCheck256`
$2^x < 4096 \Rightarrow x \in \mathbb{Z}$ (so $x \geq 12$)	[**kernel-verified Lean**]	`FiniteCheck4096`
$2^x < 8192 \Rightarrow x \in \mathbb{Z}$ (so $x \geq 13$)	[**kernel-verified Lean**]	`FiniteCheck8192`
$2^x < 16384 \Rightarrow x \in \mathbb{Z}$ (so $x \geq 14$)	[**kernel-verified Lean**]	`FiniteCheck16384`
No nonintegral candidate with $2^x < 2^{20}$	[**software computation**]	Section 23, Appendix F
No nonintegral candidate with $2^x < 2^{27}$	[**software computation**]	Section 34
Claimed scan through 2^{29}; canonical logs and replay artifacts unavailable in the repository artifact tree	[**software computation**]	Section 48.7
Exact conditional structure		
Rank of candidate exponent vectors ≤ 2	[**kernel-verified Lean**]	`MultiplicativeRank`
Cross-valuation identity and conditional box growth	[**kernel-verified Lean**]	`CandidateStructure`
Common odd core $2^i w^j$ (existence)	[**kernel-verified Lean**]	`OddCore`
Counting $\mathrm{clog}_2 N \cdot \mathrm{clog}_3 N$; linear dyadic blocks	[**kernel-verified Lean**]	`OddCore`
Primitive (canonical) odd core; unique coordinates	[**kernel-verified Lean**]	`PrimitiveOddCore`
Unique least solution; affine primitive output pair	[**kernel-verified Lean**]	`LeastSolution`

Statement	Status	Location
Simultaneous primitive output normalization; gcd one; size dichotomy	[kernel-verified Lean]	`OutputNormalization`
Power-of-three denominator normalization	[kernel-verified Lean]	`ThreeDenominatorNormalization`
Rational-power index and exact solution monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$	[kernel-verified Lean]	`RationalPowerIndex`, `PrimitiveGenerator`
Canonical primitive generator; exact candidate monoid $\{2^n M^k\}$	[kernel-verified Lean]	`PrimitiveGenerator`
Denominator-controlled rational third output and determinant margin	[kernel-verified Lean]	`RationalThirdBase`
Unrestricted rational-third-output determinant grid	[kernel-verified Lean]	`RationalSixExponentials`, `RationalBaseThird`
3-smooth classification of further bases	[kernel-verified Lean]	`ThreeSmooth`
Both outputs 3-smooth $\Rightarrow x \in \mathbb{Z}$ (unconditional exclusion)	[kernel-verified Lean]	`SmoothOutputs`
Sharp form: a 3-smooth output forces a partner with a prime factor ≥ 5	[kernel-verified Lean]	`SmoothOutputs`
Primitive output cores are never exactly 3 and 2; exact core rationality classification	[kernel-verified Lean]	`CoreIndependence`
Algebraic auxiliary bases and output loci		
Six exponentials for three positive rational bases and rational outputs	[kernel-verified Lean]	`RationalSixExponentials`
Field-norm determinant bound over rings of integers	[kernel-verified Lean]	`AlgebraicIntegerDeterminant`
One number field and one natural denominator for a finite algebraic family	[kernel-verified Lean]	`AlgebraicOutputBridge`, `FiniteAlgebraicOutputBridge`
Six exponentials for three positive algebraic real bases and algebraic outputs	[kernel-verified Lean]	`AlgebraicSixExponentials`
Algebraic output at a natural third base	[kernel-verified Lean]	`AlgebraicThirdBase`
Exact rational-base spectrum: $q^x \in \mathbb{Q}$ iff $x \in \mathbb{Z}$ or q is a $2,3$-unit	[kernel-verified Lean]	`RationalBaseThird`
Exact algebraic-output form of the same classification	[kernel-verified Lean]	`AlgebraicRationalBase`
Divisible-hull criterion: an algebraic base with algebraic power has a positive $2,3$-unit power	[kernel-verified Lean]	`AlgebraicBaseUnitPower`
Pointwise algebraic natural-base classification and iterated-output transcendence	[kernel-verified Lean]	`AlgebraicConsequences`
Exact algebraic output monomial locus is a coordinate face or the origin	[kernel-verified Lean]	`AlgebraicOutputLocus`
Algebraic output ratios lie on one external p-adic line (lattice rank ≤ 1)	[kernel-verified Lean]	`AlgebraicOutputLocus`
Common algebraic-output exponent plane: $y \in \mathbb{Q} + \mathbb{Q}x$	[kernel-verified Lean]	`AlgebraicExponentLocus`
Exact rational-function locus $P/Q \in \mathbb{Q} + \mathbb{Q}x$ iff $P = (q + rX)Q$; canonical $\mathbb{N}_0$-cone comparison	[kernel-verified Lean]	`RationalFunctionAlgebraicLocus`
General two-base saturated spectrum $\Gamma(a, b) = \{a^r b^s : r, s \in \mathbb{Q}\}$	[paper proof]	Section 12

Statement	Status	Location
Strong spectrum: $x \log \alpha \notin \mathcal{L}$ for $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$	**[paper proof]**	Section 12
Every $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \Gamma$ is a one-output detector for Conjecture 1.1	**[paper proof]**	Section 12
One-base equivalent formulations		
Conjecture 1.1 $\iff$ every solution has $5^x \in \overline{\mathbb{Q}}$	**[kernel-verified Lean]**	`AlgebraicConsequences`
Conjecture 1.1 $\iff$ every solution has $2^{x^2}, 3^{x^2} \in \overline{\mathbb{Q}}$	**[kernel-verified Lean]**	`AlgebraicConsequences`
Conjecture 1.1 $\iff$ every solution has $6^{x^2} \in \overline{\mathbb{Q}}$	**[kernel-verified Lean]**	`AlgebraicConsequences`
Conjecture 1.1 $\iff$ the algebraic locus among $(2^i/3^j)^{x^2}$ has rank two	**[kernel-verified Lean]**	`AlgebraicOutputLocus`
Conjecture 1.1 $\iff$ the algebraic real-power output locus has rank three	**[kernel-verified Lean]**	`AlgebraicRankThreeEquivalence`
Conjecture 1.1 $\iff$ no squared-monomial algebraic coordinate face	**[kernel-verified Lean]**	`AlgebraicOutputLocus`
Conjecture 1.1 $\iff$ $2^y, 3^y \in \overline{\mathbb{Q}}$ at every natural power $y = x^n$	**[kernel-verified Lean]**	`AlgebraicExponentLocus`
Conjecture 1.1 $\iff$ both outputs are algebraic after evaluation of any fixed $P \in \mathbb{Q}[X]$ of degree at least two	**[kernel-verified Lean]**	`RationalFunctionAlgebraicLocus`
Canonical radical: valuation, degree, and hull		
Odd-prime valuation gcd; outer-power exponents of the canonical radical	**[kernel-verified Lean]**	`CanonicalRadicalValuation`
Exact rational outer-power criterion; the triple gcd is not replaceable by either pairwise gcd	**[kernel-verified Lean]**	`CanonicalRadicalSharpness`
Norm and trace of the pure-radical generator; the base-swapped relation blocks monomial injectivity	**[kernel-verified Lean]**	`CanonicalRadicalFieldNorm`
Algebraic integrality of the radical iff trivial reduced denominator	**[kernel-verified Lean]**	`CanonicalRadicalThirdOutput`
Algebraicity of the radical's β-th power iff $2, 3$-smooth numerator	**[kernel-verified Lean]**	`CanonicalRadicalAlgebraicOutput`
External numerator prime excludes the positive $2, 3$-unit divisible hull	**[kernel-verified Lean]**	`CanonicalRadicalDivisibleHull`
Mixed radical monomials retain an external prime; transcendental outputs	**[kernel-verified Lean]**	`CanonicalRadicalMixedOutput`
Exact rational-power indices and radical degrees of the simultaneous radicals; product degree de	**[kernel-verified Lean]**	`SimultaneousRadicalDegrees`
Localized-radical conjecture target itself	Open	Section 5
Valuation lattice, kernel, and towers		
Rational-output lattice Λ_x of the prime-valuation matrix V_x	**[paper proof]**	Section 13
Smith classification: $\Lambda_x/\mathbb{Z}^2 \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z}$ and the finite rational-output defect	**[paper proof]**	Section 13

Statement	Status	Location
Second-iterate kernel $K_x = \{2^r 3^s : r, s \in \mathbb{Q},\ M^r A^s \in \Gamma\}$; its coordinate dimension is at most one	[kernel-verified Lean]	`SecondIterateKernel`
Kernel invariance $K_x = K_y$ across nonintegral solutions	[kernel-verified Lean]	`SecondIterateKernel`
Möbius criterion: $K \neq \{0\}$ iff $x \in \mathrm{PGL}_2(\mathbb{Q}) \cdot \vartheta$, iff $1, \vartheta, x, x\vartheta$ are $\mathbb{Q}$-dependent	[kernel-verified Lean]	`KernelDichotomy`
Multiplicative-dependence form: $K \neq \{0\}$ iff $M, A, 2, 3$ are multiplicatively dependent	[kernel-verified Lean]	`KernelDichotomy`
Four-type classification of counterexamples by external valuation rank	[paper proof]	Section 14
No nontrivial two-step return inside Γ	[paper proof]	Section 15
Depth-three tower rigidity: some α^{x^j}, $j \leq 3$, is transcendental, with exact first depth	[paper proof]	Section 15
Universal three-level detector: Conjecture 1.1 $\iff$ $\alpha^x, \alpha^{x^2}, \alpha^{x^3} \in \overline{\mathbb{Q}}$ for one fixed $\alpha \neq 1$	[paper proof]	Section 15
Base-2 cubic-tower equivalence: Conjecture 1.1 $\iff$ $2^{x^2}, 2^{x^3} \in \overline{\mathbb{Q}}$	[kernel-verified Lean]	`TowerRigidity`
Base-2 depth-two dichotomy: $2^{x^2} \in \overline{\mathbb{Q}}$ iff 2^x is 3-smooth, and then $2^{x^2} \in \mathbb{N}$ while 2^{x^3} is transcendental	[kernel-verified Lean]	`TowerRigidity`
Exact transition $2^y, 2^{xy} \in \overline{\mathbb{Q}}$ iff $y \in \mathbb{Q} + \mathbb{Q}\vartheta$; at most one positive depth starts an adjacent algebraic pair; two fixed distinct depths give an AE equivalence	[kernel-verified Lean]	`AlgebraicTowerTransition`
Exact base-3 transition locus; no positive base-2/base-3 pair can coexist; across both principal towers at most one positive transition depth, with higher rational-unit directions excluded in the dependent-output branch	[kernel-verified Lean]	`CrossBaseTowerTransition`
Symmetric tower constants: at least one of $\log E_3, \log E_2$ lies outside $\mathcal{L}$	[paper proof]	Section 16
Exact conditional classification of $E_3 = 3^\vartheta$ and $E_2 = 2^{1/\vartheta}$	[kernel-verified Lean]	`SymmetricTowerConstants`
E_3 and E_2 are never both algebraic under a nonintegral solution	[kernel-verified Lean]	`SymmetricTowerConstants`
Counting, growth, and equidistribution		
Exact block counts and cumulative finite sum	[kernel-verified Lean]	`ExactCounting`
Cumulative $T^2/(2\beta) + O(T)$ asymptotic; subquadratic-growth equivalence	[kernel-verified Lean]	`QuadraticAsymptotic`
Vanishing nonzero block Fourier modes	[kernel-verified Lean]	`BlockWeyl`
Exact block averages for finite trigonometric polynomials	[kernel-verified Lean]	`BlockEquidistribution`

Statement	Status	Location
Full continuous-test/interval blockwise equidistribution	**[paper proof]**	Section 4
Density dichotomy, gap equivalences, infinitude	**[kernel-verified Lean]**	`FractionalPartDensity`
Nonintegral candidate refinement; density zero	**[kernel-verified Lean]**	`NonintegerDensity`
Shrinking-target Diophantine bounds	**[kernel-verified Lean]**	`ShrinkingTarget`
Convergent-denominator growth $q_{n+1} < 2m^{q_n}$	**[kernel-verified Lean]**	`DenominatorGrowth`
Difference and progression obstructions		
Affine descent, $r + 8k \le x$	**[kernel-verified Lean]**	`ArithmeticRigidity`
Three-term AP obstruction under $h^5 \le n$	**[kernel-verified Lean]**	`ZeroDensity`
Sharp three-term and four-term AP obstructions ($r = 2, 3$)	**[kernel-verified Lean]**	`SharperProgressions`, `ThirdDifference`
Uniform arbitrary-order AP obstruction ($r \ge 2$)	**[kernel-verified Lean]**	`HigherDifference`
Effective rational three-term criterion $h^{400} \le n^{83}$	**[kernel-verified Lean]**	`SharpProgression`
Radical and localized reformulations		
Localized-radical and symmetric reformulations	**[kernel-verified Lean]**	`RadicalReformulation`, Appendix C
Primitive localized-radical reduction	**[kernel-verified Lean]**	`PrimitiveRadical`
Exact radical degree and binomial irreducibility	**[kernel-verified Lean]**	`RadicalDegree`
Exact translated canonical-radical norm and two-logarithm factorization	**[paper proof]**	Theorem 5.3
Candidate-form localization equivalence	**[kernel-verified Lean]**	`Localization`
Structural, determinant, dynamical, and closure developments		
Restricted logarithm-matrix coefficient principle $\iff$ Conjecture 1.1; rational anisotropy of a counterexample	**[kernel-verified Lean]**	`RestrictedMatrixCoefficient`; Section 48.1
Sparse irrational-power-curve zero bound, nonzero integral evaluation determinant, and the exact $R + 1$-term positive-node interpolant	**[kernel-verified Lean]**	`SparsePowerCurve`; Section 48.3
Exact rational rectangular orbit-collision count, irrational grid cardinality, and the one-monomial-saving rational-orbit square-grid certificate	**[kernel-verified Lean]**	`RationalOrbitGrid`; Section 49.1
Formal prime-symbol determinant and its specialization; formal vanishing exactly in the integral branch; sparse Structural Rank equivalence	**[kernel-verified Lean]**	`FormalPrimeDeterminant`; Section 48.2
BK degree threshold $R - 1$ and the factor-four comparison on the $2N$ grid	**[paper proof]**	Section 48.4
Boundary-dilation defect factorization, unique degree-$K - 1$ residuals, and the degree-$2K - 2$ commutator	**[kernel-verified Lean]**	`BoundaryDilationDefect`; Section 48.5

Statement	Status	Location
Unbounded $o((\log H)^2)$ and synchronized compact $o(\log H)$ point-counting targets	[paper proof]	Section 48.6
Transcendence of $e^{2\pi i q\beta}$ for $q \in \mathbb{Q}^{\times}$, deduced from classical six exponentials	[paper proof]	Section 48.6; classical input [classical/open background]
Exponential-polynomial zero count; generalized Vandermonde nonvanishing at ordered nodes	[kernel-verified Lean]	`ExponentialPolynomialZeros`
Generalized Vandermonde positivity (ordered chamber)	[kernel-verified Lean]	`OrderedChamberPositivity`
Arbitrary-node determinant integrality and divided-difference repulsion bound	[kernel-verified Lean]	`ArbitraryNodeDeterminant`
Universal nonnegative-node superfactorial interpolation divisor, exact consecutive-node witness, and strengthened candidate repulsion/packing	[kernel-verified Lean]	`SuperfactorialDeterminant`; final repulsion endpoints retain the named mean-value hypothesis
Normalized geometric signed-progression determinant, primitive quotient coprimality, and exact denominator-tax decomposition with bounded remainder	[kernel-verified Lean]	`SignedProgressionDeterminant`; Section 49.5
Normalized Fejér kernel, exact finite weighted Fourier score, and exact integer-phase hit bound	[kernel-verified Lean]	`FejerKernelCertificate`; Section 50.2
Uniform metric near-integrality and Haar random-phase extinction	[paper proof]	Section 50.1; finite Fejér bridge separately kernel-checked
Exact denominator-exponent drift, common centered defect, and structural-prime p-adic escape	[kernel-verified Lean]	`AdelicDrift`; Section 51.2
Sturmian carry telescope, exact prefix mean/variance, joint-carry probability, and covariance kernel	[kernel-verified Lean]	`SturmianProbability`; Section 51.3
Complete synchronized gap products, orbit-polynomial discriminants, logarithmic energy, matched-gap surplus κ, and unmatched quotient	[paper proof]; one-gap and finite-product amplification [kernel-verified Lean]	`MatchedGapAmplification`; Section 27.5
Dense denominator clearing, finite-difference approximation floor, and all-degree normalized tax obstruction	[kernel-verified Lean]	`PolynomialApproximationTax`; Section 51.4
Rational homogeneous contact nonvanishing and first-order Padé transversality	[kernel-verified Lean]	`RationalContactRigidity`; Section 51.5
Carry/floor and denominator formal-series identities, including the exact common defect	[kernel-verified Lean]	`SturmianGeneratingSeries`; Section 52.2
De-integralized common-zero lattice and canonical divisor; compact $\Theta(\log H)$ height dichotomy	[paper proof]	Section 52.1, Section 52.3

Statement	Status	Location
Pólya–Carlson natural boundary for the carry, floor, and reduced-denominator series	[**classical/open background**]	Section 52.2; exact series algebra separately kernel-checked
Fixed 2×2 integral matrix: integral real power iff both original outputs are integral	[**kernel-verified Lean**]	`IntegralPowerMatrix`; Section 53.1
Affine-exponent and rational-approximation alternant factorizations	[**kernel-verified Lean**]	`CandidateAlternant`; Section 53.3
Cross-Loewner positive integer certificate and optimized two-coset one-half barrier	[**paper proof**]	Section 53.2, Section 53.3
Fermat-quotient exact identity, power law, first-log determinant interface, and vanishing on $(2^n, 3^n)$	[**kernel-verified Lean**]	`FermatQuotientDeterminant`; Section 54.2
Structural-prime Iwasawa transversality, exact adelic wedge, conditional cascade depth, and Kummer third-order diagnostic	[**paper proof**]	Section 54; classical Iwasawa kernel where stated
Exact consecutive-logarithmic-mean enclosure	[**kernel-verified Lean**]	`ConsecutiveLogMeanBounds`; Section 55.1
Prime-switch antichain, microscopic resonance window, and 2–3 convergent sieve	[**paper proof**]	Section 55; rational-output switch family only
Exact exclusion of simultaneous 2–3 switches through $b = 10000$	[**software computation**]	Section 55.3; Python big-integer/rational verifier
Exact two-coordinate grid displacement identities and rank-K boundary defects	[**kernel-verified Lean**]	`BoundaryDisplacement`; rank under $K > 0$, nonsingularity, and nonzero multipliers; Section 56.4
Exact rational bilinear minimum, projective arithmetic density, and integral Hadamard-power no-go	[**paper proof**]	Section 56.1, Section 56.2
Determinantal-divisor compound sandwich and triangular method ceiling	[**paper proof**]	Section 56.3; ceiling applies to the stated termwise/Frobenius bounds
Two-variable finite-difference divisor for every compound of the square-grid matrix	[**paper proof**]	Section 56.3; top gain remains lower order
Boundary transfer determinant and Kraus–Loewner positive integral/denominator tax	[**paper proof**]	Section 56.4, Section 56.5
Integral dilation-difference quotient and its mod-2 derivative reduction	[**kernel-verified Lean**]	`DilationDifferenceQuotient`; Section 57.3
Structural-prime denominator lattice, favorable-branch residual scaling, root depths, and odd-degree dilation-resultant valuations	[**paper proof**]	Section 57; only the two displayed residue branches

Statement	Status	Location
Prime-power cross-residual cancellation, exact node-divisor scaling, and conditional exact nonzero cross-resultant orders at 2 and 3	[kernel-verified Lean]	`CrossResidualScaling`, `ResultantScaling`, `NodeDivisorScaling`, `CrossResultantBridge`; primary interpolant scalings and nonzero residue reductions remain hypotheses
Opposite-residual coefficient laws and the full local cross-Sylvester Smith profile in the simultaneous favorable branch	[paper proof]	Section 57.4; branch is nonexhaustive and the residual-specific Smith reduction remains paper-level
Eventual full-content residual profile for $v_2(A), v_3(M) > 0$ and $n > \max(v_2(M), v_3(A))$	[paper proof]	Section 57.4; finite cost and assignment core [kernel-verified Lean] in `ContentNormalizedProfileFinite`
Convex matching lower bound and $p^{m\binom{m}{2}}$ divisibility for every canonical weighted lower-Toeplitz m-minor	[kernel-verified Lean]	`SmithProfileCore`
Uniformly negative compound endpoint-rate functional	[kernel-verified Lean]	`CompoundEndpointRate`; this certifies a no-closure bound, not the conjecture
Exact structural-prime order of every leading principal subresultant	[paper proof]	Section 57.4; simultaneous favorable coefficient profiles
Sharp direct/shallow displacement-cost inequality and equality classification	[kernel-verified Lean]	`DeepSubresultantCost`; combinatorial core of the principal-subresultant theorem
Exact $n = 1$ cancellation of the full interpolation denominator from the paired residual resultant	[kernel-verified Lean]	`DenominatorCancellation Counterexample`; synthetic $(M, A) = (9, 2)$ test data, not a common-exponent pair
Common-grid transformation no-go for simultaneous structural-prime axis support	[kernel-verified Lean]	`GridTransformNoGo`; nonsingular direction changes cannot enlarge the favorable branch
All-n $\Omega(n^4)$ denominator-cancellation tax for the synthetic pair $(9, 2)$	[paper proof]	Section 57.4; exact paper consequence of the denominator and cross-resultant formulas, not common-exponent data
Universal leading/constant Cramer factors of the paired residual resultant	[paper proof]	Section 57.4; finite $K = 2, 3, 4$ core factorizations are unbundled exact-CAS diagnostics only
Exact unordered-pair decoding from the ordinary and synchronized p-power sums	[kernel-verified Lean]	`SynchronizedSumDecoder`; Section 58.3
Rational synchronized-sum decoder degree bound $L \le 2d + 1$	[kernel-verified Lean]	`RationalSumDecoder`; Section 58.3
Fixed-rank sum entropy, algebraic decoder degree growth, and finite Fourier parity–residue separation	[paper proof]	Section 58; unit-equation input classical where stated
Generic algebraic-base spectrum for an arbitrary independent positive algebraic pair	[kernel-verified Lean]	`AlgebraicBaseSpectrum`; Section 12

Statement	Status	Location
Finite rectangular mixed-Mahler inclusion–exclusion identity	[kernel-verified Lean]	`MixedMahlerFinite`; Section 59.2; no infinite convergence claim
Entire gamma quotient, corrected mixed-Mahler double-pole resonance, and transfer-spectrum diagnosis	[paper proof]	Section 59; holonomic, Pólya–Carlson, Barnes, and q-digamma inputs classical where stated
Exact determinant-one Smith reduction of the geometric Vandermonde, q-Pascal identity, factor formula, divisibility order, and positivity	[kernel-verified Lean]	`GeometricVandermondeSmith`; Section 60.2
Cubic fixed-base/cyclotomic half-mass law and escape beyond every fixed prime set	[paper proof]	Section 60.2; exact factorization plus LTE
Finite row-content amplification for a short prime-power relation	[kernel-verified Lean]	`FinitePlaceShortRelationCore`; Section 60.3
Pair-specific row/column Smith tomography, discrete-log image formula, and finite orbit-energy certificate	[paper proof]	Section 60.3; no numerical scouting input
Exterior-minor probability identity, weighted Smith-tail inequalities, and multilevel modular divisor	[paper proof]	Section 60.1, Section 60.2
Rational cross-Loewner Andreief–Selberg determinant, quadratic singular hierarchy, and weighted compound certificate	[paper proof]	Section 60.4; reduced denominator or Smith gain remains open
Infinitely divisible 3-smooth Gram no-go and exact semigroup entropy saturation	[paper proof]	Section 60.5; no closing arithmetic inequality
Finite ordinary two-scale support inclusion–exclusion identity	[kernel-verified Lean]	`TwoScaleSupportFinite`; Section 61.3; no infinite convergence claim
Spectral-zeta/logarithmic-Weyl reformulations and complete rank-two heat expansion	[paper proof]	Section 61.1, Section 61.2; exact counting core already kernel-checked, Mellin–Baker expansion paper-level
Sharp fixed-window polynomial node-budget barrier and capacity constant	[paper proof]	Section 61.4; no claim about rational or growing-window approximants
Exact finite geometric Newton expansion, interpolation, omitted tail, and first extrapolation product	[kernel-verified Lean]	`GeometricNewtonInterpolation`; Section 62.3
Finite q-Newton endpoint dilation defect	[kernel-verified Lean]	`QNewtonEndpointDefect`; Section 62.3
Entire q-Newton completion and exact edge-product defect	[paper proof]	Section 62.3; local-uniform convergence and classical basic-hypergeometric identity
Mixed geometric-Newton words, complementary product, exact shifted-block valuation, and one-signed dyadic tail	[paper proof]	Section 62.3

Statement	Status	Location
One-axis connection growth, sign/phase/integer amplification, meromorphic quotient, Mittag–Leffler expansion, and negative-axis asymptotics	**[paper proof]**	Section 62.4; classical normally convergent q-products
Pole–monodromy exchange and the exact two-axis branch connection system	**[paper proof]**	Section 62.4; exact but nonclosing compatibility normal form
Finite off-axis transfer denominator, forcing convolution, and closed solution	**[kernel-verified Lean]**	`QNewtonOffAxisTransfer`; Section 62.4
LCM-cleared arbitrary-node coefficients, vanishing moments, and the four-node cubic functional	**[kernel-verified Lean]**	`ClearedDividedDifference`; Section 18.3
Arithmetic unit locking, optimal-block no-go, dyadic certificates, and cofactor minimax unit barrier	**[paper proof]**	Section 62; no improvement to the accepted finite scan
Rational scalar/matrix-entry first-jet recovery and base-two integral-exponent closure	**[kernel-verified Lean]**	`JordanFirstJet`; Section 63.3; matrix real-power formula remains paper-level
Exact rationality locus for semisimple first-order self-extensions	**[paper proof]**	Section 63.3; false positives are precisely commutator/split directions
Operator local-universality no-go and global/finite/affine/trace/Hankel closing criteria	**[paper proof]**	Section 63; criteria are not consequences of the two outputs
Exact rational moment-defect denominators and rational two-atom reproduction	**[kernel-verified Lean]**	`MomentArithmetic`; Section 65.2
Hausdorff moment representation, finite rational quadrature, and Müntz invisible positive perturbations	**[paper proof]**	Section 65; full Müntz and Hahn–Banach inputs classical
Hyperbolic rank geometry and rational symmetrization identities	**[kernel-verified Lean]**	`RankGeometry`; Section 66
Baker rank-two exclusion, counterexample rank-3/4 classification, and conditional algebraic three-term orbit	**[paper proof]**	Section 66; Baker input classical
Möbius–conic determinant identification, no-cross families, localized radical, and one-special-radical reduction	**[paper proof]**	Section 66.2
Generic determinant-distance group collision and modular unimodular wrappers	**[kernel-verified Lean]**	`UnimodularGapCollision`; Section 67.1
Integer gap gcds, transported rational secants, finite order-signature no-go, and entrywise jet ceiling	**[paper proof]**	Section 67; jet ceiling is termwise only
Symmetric rational secants and arbitrary fixed even-order Richardson contact	**[paper proof]**	Section 67.2; unreduced denominator height remains exponential
Exact rotation-denominator floor law and finite simultaneous worst-case entropy bound	**[kernel-verified Lean]**	`OrbitHeightEntropy`; Section 68.1, Section 68.2

Statement	Status	Location
Exact compact-orbit heights, Farey limiting distortion, and Baker–contact scale comparison	**[paper proof]**	Section 68; entropy concerns relabelings, not new graph points
Exact projective secant height and structural cross-residual gcd rate; fixed-observable height-exponent zero	**[paper proof]**	Section 68.3; Corvaja–Zannier input classical
Piecewise projective Padé-defect optimization and efficiency thresholds	**[kernel-verified Lean]**	`ProjectiveDefect`; Section 69.1
Number-field projective defect, rational-transfer no-go, and fixed-cardinality group finiteness	**[paper proof]**	Section 69; external Padé and approximation theorems cited there
Exact finite carry-word telescoping, arity, suffix descent, and ordered reconstruction	**[kernel-verified Lean]**	`CarryPath`; Section 70.1
Mixed-radix digit bounds and floor-carry plaquette identity	**[kernel-verified Lean]**	`MixedRadixCarryFinite`; Section 70.3
Binary-to-ternary radix-shadow congruence and exact valuation-index bridge	**[kernel-verified Lean]**	`RadixShadow`; Section 70.4
Finite shifted-evaluation borrow step, forced $(0, 1)$ digit pairs, and leading-2 exclusion	**[kernel-verified Lean]**	`BorrowBlockCore`; Section 70.4
Finite floor-carry reconstruction, smooth-language universality, hitting corridor, and normalization-defect, interval-valuation, prefix-separation, and carry-mask formulas	**[paper proof]**	Sections 70.3 and 70.4
Multiset carry paths, character balancing, multinomial fibers, and carry-language pressure	**[paper proof]**	Section 70; exact-rate claims are restricted to the stated carry-language construction
D-finite exponent-relation polynomial and top-coefficient noncancellation	**[kernel-verified Lean]**	`DfiniteExponentCore`; Section 71.1
Sharp 3-smooth canonical-product/Jensen threshold, subcritical interpolation rigidity, and critical defect cocycle	**[paper proof]**	Section 71; polynomial quotient regularity remains an explicit hypothesis
Sharp geometric-edge Jensen barrier, one-residual closure, rational coefficient uncertainty, and rational-coefficient interpolant existence	**[paper proof]**	Section 71.4
Universal finite Lambert cocycle and base-$(2, 3)$ mixed double-difference no-go	**[kernel-verified Lean]**	`LambertCocycleFinite`; Section 71.16
Finite positive-coefficient covariance identity and single-support closure	**[kernel-verified Lean]**	`PositiveCoefficientClosure`; Section 71.6
Finite constant-weight positive-dilation filters preserve strict alternating tails and cannot produce eventual coefficient nonnegativity	**[kernel-verified Lean]**	`FiniteDilationAlternation`; Section 71.6
Newton append/collision identities and barycentric translation covariance	**[kernel-verified Lean]**	`NewtonCollisionCore`; Section 71.10

Statement	Status	Location
Higher Newton–Kummer product identities and integral-exponent calibration	**[kernel-verified Lean]**	`RankOneNewtonKummer`; Section 71.13
Finite signed-orbit augmentation identity, mixed additive cancellation, and difference-grid minor factorization	**[kernel-verified Lean]**	`SignedOrbitNormFinite`; Section 71.13
Private odd-prime order and dyadic-level separation	**[kernel-verified Lean]**	`DyadicPrivateSupport`; Section 71.9
Finite cleaned-gap capture-multiplier criterion	**[kernel-verified Lean]**	`CaptureMultiplier`; Section 71.12
Adjacent-smooth tangent identities, conditional balance and tangent-lattice parametrizations, two residue relations, and a conditional residual cross identity	**[kernel-verified Lean]**	`CatalanTangentCore`; Section 71.14
Canonical critical Stieltjes interpolant, refined quadratic threshold, positive reduced denominator target, direct-node Markov–Padé repulsion, saturated-Smith identity, arbitrarily deep unit-defect calibration, and real-axis root factorization, universal rank-one fixed-divisor ceiling, and fixed-pair triangular synchronization ceiling; exact collision-tree initial form, simple affine-cluster transfer, sharp collective examples, and full bad-geometry source capacity	**[paper proof]** paper deductions; Corvaja–Zannier input **[classical/open background]**	Sections 71.5 and 71.7
One-Vandermonde reciprocal moments, integral positive-definite Hankel lattice, common-content floors, full alternant divisor, shape-free HCIZ dominance, and sharp T^5 scale audit	finite arithmetic and common-content core **[kernel-verified Lean]**; HCIZ/alternant/asymptotic layer **[paper proof]**	`ReciprocalMomentLatticeFinite`; Section 71.8
Infinite positive-series closure, concentration/mesh gap, Jensen and separator rigidity, three-value geometric-curvature closure, rational sum-of-squares positivity, refined positive-interpolant growth tax, canonical alternating tail, finite Euler–dilation, controlled infinite substitution, Prouhet-composition, and four-dilation node-defect no-gos, boundary-quotient lower bounds, and Stieltjes-transfer no-go	**[paper proof]**	Section 71.6
Cyclotomic defect partitions, private-support critical slope, and fixed-polynomial contact ceiling	**[paper proof]**	Section 71.9

Statement	Status	Location
Reciprocal-polynomial half-contact and exact equality classification; growing two-scale integer filter	finite support core [**kernel-verified Lean**]; orbit asymptotic [**paper proof**]	`ReciprocalHalfContact`; Section 71.9
Fresh-prime pole, local synchronization, BCZ ray divisor, and asymptotically maximal nested spike exhaustion	[**paper proof**]	Section 71.10; BCZ input classical
Exact finite Kummer compatibility and transverse-hyperplane counts	[**kernel-verified Lean**]	`KummerCompatibilityFinite`; Section 71.11
Exact full-rank prime-power compatibility count and unit-transverse rank-three normal-form count	[**kernel-verified Lean**]	`PrimePowerKummerCoherence`; Section 71.11
Primitive two-star coefficient content and nonzero reduction at every modulus	[**kernel-verified Lean**]	`TwoStarContent`; Section 71.11
Mixed-gap Kummer determinant, same-residue obstruction, and denominator-retention core	[**kernel-verified Lean**]	`MixedGapKummerFinite`; Section 71.11
Support classification, exact compatibility and determinant Chebotarev densities, mixed-gap sieve, tame-symbol sign barrier, additive ceiling, mantissa entropy, collision divisor, and finite-box detection	[**paper proof**]	Section 71.11; support, Kummer, and Chebotarev inputs classical
Directional quotient dichotomy, exact capture cost, and recurrence criterion	[**paper proof**]	Section 71.12; external gcd and recurrence inputs cited
Shifted Newton–Kummer exponential smallness, character-projection ceiling, exact common-ideal norm, one-factor partial-norm deficit, private-prime resultant criterion, Fermat-like resultant splitting, number-field-*abc* radical bridge, diagonal nonuniformity, and order/shift polynomial nondivisibility	[**paper proof**]	Section 71.13; radical conclusion conditional on *abc*
Adjacent-smooth cyclic resultant, boundary law, balanced gcd target, and conditional Frobenius descent	[**paper proof**]	Section 71.14
Half-density algebraic first jet and exact higher-jet/infinite-order ceiling	[**paper proof**]	Section 71.15
Reciprocal-Hankel product and cubic universal-integerizer deficit; BCZ fixed-support no-go	[**paper proof**]	Section 71.16; BCZ input classical
Exactly reduced pure-2/3 convergent fraction and the Baker–Ridout/Subspace barrier	[**paper proof**]	Section 71.17; Baker input classical
Arbitrary-node repulsion theorem and its packing corollary	[**kernel-verified Lean**]	`ArbitraryNodeRepulsion`, conditional on its named divided-difference mean-value hypothesis
Newton divided-difference factorization of the interpolation determinant	[**kernel-verified Lean**]	`ArbitraryNodeRepulsion`, unconditional

Statement	Status	Location
Semigroup generalized-Vandermonde hierarchy (integrality half)	[kernel-verified Lean]	`MixedExponentSemigroup`, `SemigroupDeterminant`
Falling-factorial coefficient bound; dyadic-strip divided-difference bound	[kernel-verified Lean]	`FallingFactorialBound`
Semigroup gap theorem in quantitative form	[kernel-verified Lean]	`SemigroupGapBound`
Leibniz determinant bound from separated row/column bounds	[kernel-verified Lean]	`SemigroupGapBound`, unconditional
Window-span counting bridge for mixed exponents	[kernel-verified Lean]	`SemigroupWeightBound`
Explicit dyadic $O((\log X)^2)$ bound from that hierarchy	[kernel-verified Lean]	`SemigroupGapBound`, conditional on the named hypothesis `NewtonFactorization`
One-logarithm deficit analysis	[paper proof]	Section 21
Min-plus ceiling: term-wise p-adic content never exceeds a correct archimedean bound	[kernel-verified Lean]	`MinPlusCeiling`
Exact termwise-divisor ceilings (4/5, 7/10, 3/10, $\pi/4$, 0.8710)	[paper proof]	Section 20
Exact cross-coprime correction $7/10 - \mathcal{E}_{\mathrm{val}}/20$ and explicit β^{-4} separation	[paper proof]; abstract finite inequalities [kernel-verified Lean]	`ValuationSlopeEnergy`; Section 30.9
A unique tropical minimizer forbids cancellation	[kernel-verified Lean]	`MinPlusCeiling`
Tropical quotient factorization, all-depth divisibility, and tied-fiber first-layer criterion (no cancellation lower bound)	[kernel-verified Lean]	`TropicalInitialForm`
Block-preserving tied sums factor into block determinants and consecutive-power Vandermondes; Beatty minimizers, floor gap, quotient divisibility, and exact expansion	[kernel-verified Lean]	`BlockFiberFactorization`, `TropicalBlockApplication`, `BeattyTropicalFiber`
Exact all-layer row-partition expansion and cutoff-K support truncation (no bound on surviving cancellation)	[kernel-verified Lean]	`RowBlockLaplaceExpansion`
Prefix-crossing lower bound and stars-and-bars count of row partitions below a Beatty excess cutoff (no valuation/cancellation bound)	[kernel-verified Lean]	`AssignmentExcessGeometry`
Exact Vandermonde collision energy, elementary power-difference valuation, and finite variable-block cube budget	[kernel-verified Lean]	`BlockVandermondeValuationBudget`
Whole-Vandermonde dyadic and triadic unit fixed divisors under monic column changes (no bound on the residual cascade)	[kernel-verified Lean]	`AllLayerUnitBasis`, `UnitOrderingGain`
Uniform scaled-block minor divisors and additive whole-determinant gain $p^{\tau+U} \mid \det$ (no residual-cancellation bound)	[kernel-verified Lean]	`RowBlockUnitDivisibility`

Statement	Status	Location
Column-permuted raw mixed-matrix bridge and additive dyadic/triadic divisors under explicit finite fiber certificates	**[kernel-verified Lean]**	`MixedDeterminantUnitBridge`
Genuine global first-N mixed prefix, automatic consecutive fibers, exact dyadic/triadic cutoff expansions, and structural-output divisors (no residual-cancellation upper bound)	**[kernel-verified Lean]**	`GlobalMixedPrefixCascade`; strictly decreasing depths, with A odd for the dyadic structural divisor and $3 \nmid M$ for the triadic one
Exact synthetic cascade computations, including counterexamples to first-layer equality and minimum-effective-layer equality	**[exact computation]**	Section 32.5
Compact synchronized-orbit reformulation	**[kernel-verified Lean]**	`CompactOrbit`
Block-growth reformulations (bounded, $o(T)$, $\liminf$, $\equiv 1$)	**[kernel-verified Lean]**	`BlockGrowthReformulations`
Rational-function rigidity, multiplication and quotient criteria, closure reformulations	**[paper proof]**	Section 7
Perfect-power content, matched affine decompositions, primitive-output statistics	**[paper proof]**	Section 9
Candidate gcd/lcm lattice; prime-support rigidity of the whole branch	**[kernel-verified Lean]**	`CandidateLattice`
Ambient root saturation index d	**[paper proof]**	Section 10
Operator formulation on the prime-logarithm space	**[kernel-verified Lean]**	`PrimeLogSpaceOperator`
Every complex solution is real (principal branch)	**[kernel-verified Lean]**	`ComplexSolutions`
Gaussian-integer version is equivalent to the conjecture	**[paper proof]**	Section 64
Multivalued branches: the log-product wall widened by $\log(-1)$	**[paper proof]**	Section 64
Exact cost of the rational antecedent (denominator-support lemma)	**[paper proof]**	Section 64
Prime-top isolation, sparse quadratic class, cocycles, traces, masks		
Fresh-prime order 2ℓ, $2\ell \mid p-1$, lower-level p-integrality; ultrametric unique-minimum lemma; coefficient-or-denominator tax	**[kernel-verified Lean]**	`PrimeTopIsolation`; Section 72
Orbit-uniform gcd, fresh-factor mass, reduced-denominator survival, nearest-pole product, growing-order prime-top stencil no-go	**[paper proof]**; Corvaja–Zannier and two-log inputs **[classical/open background]**	Section 72
Covolume conservation for restricted denominators (one prime)	**[kernel-verified Lean]**	`RestrictedDenominatorDirichlet`; Section 72.6
External-prime derivative $\Lambda_q \neq 0$ with $\lvert\Lambda_q\rvert \geq 1/\max(3^{a_q}, 2^{b_q})$	**[kernel-verified Lean]**	`SparseCoaction`; Section 73.1

Statement	Status	Location
Sparse coaction separation ⇔ Conjecture 1.1	**[kernel-verified Lean]**	`SparseCoaction`; Theorem 73.2
q-adic witness, rank-one three-logarithm normal form, arrowhead rank ≤ 4	**[paper proof]**	Section 73
Radical-full convergent gaps	**[paper proof]**, conditional on *abc*	Proposition 73.3
Mesoscopic window $\frac{1}{2} < \log 2/\log 3 < \frac{2}{3}$, mode inequalities, shift identities, annihilator roots and expansion, carry-free Kummer bound	**[kernel-verified Lean]**	`FactorialCocycle`; Section 74
Factorial near-unit classification, minimal cocycle defect, prime-window denominator theorem (uses $x \geq 12$)	**[paper proof]**; Huxley and Brun–Titchmarsh **[classical/open background]**	Theorems 74.1, 74.2
Centered-trace defect factorization, cubic reduction and valuation law, bridge identity, fifth-trace two-channel theorem	**[kernel-verified Lean]**	`CenteredTrace`; Section 75
Bernoulli–Jordan expansion, unique criticality, critical saturation, prime-trace normalization, signed-filter moment conditions	**[paper proof]**	Section 75
One-direction gap-power theorem, cleared-value floor, adjacent valuation transport, unshared structural primes	**[kernel-verified Lean]**	`ExponentLatticeMask`; Section 76
Toric kernel, contact ceilings, stable structural support, shared-pole Newton law, three barriers, reduced-denominator endgame	**[paper proof]**; BCZ and two-log inputs **[classical/open background]**	Section 76
Session log: integral Schneider constants, structural-prime relaxation, Pólya/Gel'fond view, *abc* against convergent gaps (all quantified dead ends); control principle	**[paper proof]**	Section 88
Terminal factorials multiplicatively independent (window primes as hypothesis; finite core kernel-checked)	**[kernel-verified Lean]** core; prime existence **[classical/open background]**	`TerminalFactorialIndependence`; Section 88.9
Constructive branch of the linear-width factorial target empty for every mesoscopic pair	**[paper proof]** (one-paragraph proof from the independence proposition)	Corollary 88.2
Pair-lattice defect: gcd formula, Smith form $\mathrm{diag}(1, \delta_{\mathrm{sat}})$, matched root extraction, local Kummer form, mod-ℓ independence for every ℓ, defect-forcing criterion	**[paper proof]** (agrees with the kernel-verified rational-power index on the orbit)	Section 77
Square valuation pattern, $m_1^2 = m_0 m_2$ parametrization, Hankel-ray classification with transcendence bridge; quadratic closure inside the orbit is rational	**[kernel-verified Lean]** core	`RationalHankelTriple`; Section 77.3

Statement	Status	Location
Bhargava factorial of $\langle 2,3\rangle_+$: exact local formulas, two-adic merge recursion and slope $(\sqrt{11}-1)/2$, square-box cutoff, complete factorization, excess slopes	**[paper proof]** (Bhargava invariance **[classical/open background]**); tables **[exact computation]**	Section 78.1
Pigeonhole cutoff core and consecutive three-smooth triples	**[kernel-verified Lean]**	`SmoothSemigroupCore`; Sections 78.1, 78.3
Bhargava-normalized determinant, factorial defect $Q_{\mathcal{B}}$, six-node defect ≥ 7, defect minima $1,1,1,1,1,7,7,396$, Bhargava locking, four criteria	**[paper proof]**; enumeration **[exact computation]**	Section 78.3
Localized finite-support criterion, one-ratio cluster no-go, initial-form bridge	**[paper proof]**	Section 78.4
Factorial-spectrum rigidity (input–output isomorphism)	**[proposed target]** target	Section 78.2
Subquadratic factorial growth via $c_p = v_p(\gcd(2^{d_p}-1, 3^{d_p}-1))$ and BCZ; conditional scalar bridge; no high simultaneous orderings ($q_2 + \frac{2}{3} > 1$); quadratic synchronization gap $\kappa = 0.0236\ldots$	**[paper proof]**; BCZ **[classical/open background]**	Section 78.5
Consecutive smooth pairs $(1,2),(2,3),(3,4),(8,9)$, Catalan-free, and additive isolation	pairs **[kernel-verified Lean]** (`SmoothSemigroupCore`); isolation **[paper proof]**	Section 79.1
Weighted smooth-support series: mixed Mahler identity, Cartier-state growth, mod-ℓ algebraicity classification	**[paper proof]**	Section 79.2
LP layer of the linear-width factorial target empty for the control; dense-plus-capture Farkas mechanism	**[exact computation]**	Section 88.10; `scripts/linear_width_lp.py`
External-prime Newton wall		
Unit-convergence rigidity in normed division rings; formal edge law, spectrum support, empty spectrum for the pairs $(2,3),(3,2)$, no formal edge realization	**[kernel-verified Lean]**	`ExternalPrimeNewtonWall`; Section 80.1
Geometric-ray, meromorphic, and finite-Puiseux rigidity; accumulating edge law (analytic completions via Strassmann)	**[paper proof]**; identity theorem **[classical/open background]**	Section 80.1
Exact Newton coefficient valuations (both branches, tie excess), universal Gauss-norm lower bound, quadratic denominator escape, last-term dominance, adjacent-row repulsion	**[paper proof]**; audit **[exact computation]**	Section 80.2
Tate spectral gap $\sigma_{p,d}$, finite-strip defect growth, no tame defect quotient, no uniformly overconvergent family	**[paper proof]**	Section 80.3

Statement	Status	Location
Arithmetic compatibility bridge (Robba-to-Tate, coefficient-growth criterion) and finite-rank edge cycle with consistency $B_2(Mz)B_M(z) = B_M(2z)B_2(z)$	**[proposed target]** targets	Sections 80.3, 80.4
Global terminal-prime contraction (critical window)		
Abstract M-weighted contraction (mass balance + transport $\Rightarrow \kappa \geq 1$), candidate-regime numerics $\kappa < 1$ at $M \geq 2^{14}$, $q \leq (2/3)^{14}$, impossibility of balanced transport vectors, exponent-gap divisibility	**[kernel-verified Lean]**	`CriticalWindowContraction`; Section 81.2
Global contraction theorem: $\mathcal{F}_{P,n} \notin \mathbb{Z}$ for every nonzero P with $P(M) = 0$; negative resolution of Target 74.3; binomial and no-collision closures	**[paper proof]**; window inputs Huxley/Legendre **[classical/open background]**	Theorem 81.1; Corollary 81.2
Additive critical-window sharpening $D_n \geq (\log_2 3 - 1)t_n - O(1)$ with density $\Delta_0 \geq 0.9337$; Hermite remainder identity and positive-tail no-go	**[paper proof]**; Brun–Titchmarsh/Huxley **[classical/open background]**	Section 81.1
Two-dimensional transport norm; coefficient-inflated Kummer hybrid dual	**[proposed target]** targets	Section 81.3
Compact charts and counting (reciprocal / Farey)		
Integer-area certificate on the Farey chart: integer numerator, area floor, spacing identity; coefficient–codimension lock $(n-1)^2 < n \iff n = 2$	**[kernel-verified Lean]**	`CompactChartArea`; Section 82.2
Chart dichotomies and certificate package (counts, zeta pole order, shells, S-integral one-point target, contraction semigroup); cusp regularity C^1 not C^2	**[paper proof]** (monoid and rank inputs **[kernel-verified Lean]**/**[classical/open background]**)	Section 82.1
Alternant shell ledger: sign-regularity, unit-numerator floor, HCIZ ceiling, optimal monomial deficit $\delta_D = \frac{2^{3/2}}{3}\sqrt{ab}\, D^{3/2} + O(D)$	**[paper proof]**	Section 82.3
Planar coefficient obstruction; parameter/jet lift equivalence; invariant-prime trichotomy; relation-counting lock analysis	**[paper proof]**; covering theorem **[classical/open background]**	Section 82.3
Synchronized gaps and near-curve certificates		
Full synchronized-gap rigidity $\log\gcd(M^s - 2^r, A^s - 3^r) = o(r+s)$; exact height of the synchronized quotient; zero power-of-height exponent (approximation route closed)	**[paper proof]**; BCZ and two-log inputs **[classical/open background]**	Section 83.1

Statement	Status	Location
Smith-index law for principal-unit congruence lattices; finite-place product law $I = G^2/E$; constrained Dirichlet and the square-root tax; adelic wedge program	**[paper proof]**; mod-p shadow and one-prime covolume **[kernel-verified Lean]**	Sections 83.2, 83.3
Orbit pressure per dyadic block; subunit locking; fewnomial budget $T-1$; capacity criterion $\mathrm{Cap}_L = o(L)$ suffices; degree barriers $L \le Cd^2 \log d$	**[paper proof]**; Chebyshev/Rolle **[classical/open background]**	Section 84.1
Toric monomial rigidity (semi-invariants are monomials)	**[kernel-verified Lean]**	`ToricMonomialRigidity`; Section 84.2
Carlitz-model rigidity and quadratic target; certified sparse near-curve search program	**[classical/open background]** inputs; program **[paper proof]**	Section 84.2
Two-dimensional transport kernels: sup-kernel escape direction, counting-model triangularity, top-cell pivot; exact top-slice tail-sum rigidity ($v_p(\mathcal{F})$ equals the slice tail sum on nested annuli; integrality forces all top-slice tail sums ≥ 0)	tail-sum core **[kernel-verified Lean]** (`TopSliceTailSum`); prime supply **[classical/open background]**; kernels **[exact computation]**	Section 88.11; `scripts/two_dim_transport.py`
All-slice tail dichotomy; Φ-contraction of cross-slice rescues (factor $\ge A$ per rescue); dense-rescue starvation (0/27,000 directions survive)	Φ-core **[kernel-verified Lean]** (`CrossSliceContraction`); **[exact computation]** experiment	Section 88.11; `scripts/two_dim_dense_check.py`
Top-heavy collapse: all-slice nonnegative tails + balance $\Rightarrow$ the two-parameter cocycle vanishes identically; remaining case is exactly cross-slice-rescued negative tails below the top slice	**[kernel-verified Lean]** core (`AbelSliceMass`); experiment **[exact computation]** (0/14,787)	Theorem 88.3; `scripts/two_dim_allslice.py`
Multiplicity one in the mesoscopic range (no prime-power allowance for $j' < m+2j$); exact double-counting necessity $-T_k^{(j)} P_k^{(j)} \le \sum_{j'>j} c^+ M^{n+i'}$; counting alone starves only the top slice	**[paper proof]**; grids **[exact computation]**	Section 88.11; `scripts/annulus_starvation.py`
Chebyshev trap lattices (block bound)		
Unconditional block bound $\limsup I(H)(\log H)^2/H^2 \le 32 \log 2 \log 3/9$; two-base version; rank–codimension traps; Cauchy–Binet covolume; frequency-simplex optimality	**[paper proof]**	Theorem 85.1; Section 85.1
Macroscopic constant $\kappa_* = 5.0873$ ($t_0 \tanh t_0 = 2$); corank frontier 98.596% (from kernel $\beta \ge 14$); rank-tax one-half ceiling; $32ab/9$ conservation law	**[paper proof]**; constants **[exact computation]**	Section 85.3
Consecutive-dilation rigidity (no fixed trap for a persistent orbit point)	**[kernel-verified Lean]**	`ConsecutiveDilationRigidity`; Section 85.4

Statement	Status	Location
Bessel factorization, compound-matrix bottleneck, proof-producing certificate verifier, parametric-geometry and arithmetic-row targets	[paper proof]/[proposed target]	Section 85.4
Saturated divisor lattice (two-star tensor)		
Two-star tensor vanishes exactly at the controls; symmetric coefficient identities	[kernel-verified Lean]	`TwoStarControlCharacterization`; Section 86.1
Frontier campaign		
Two-parameter interface closed: $P = N$ closes the Φ-starvation loop, $c \equiv 0$ once $A > 16Z(m + j^*)\log A$; sub-mesoscopic prime supply the only residue	[paper proof]; core [kernel-verified Lean]; searches [exact computation] (3.25M patterns)	Section 87.1
Adelic wedge closed: control-direction blindness, drop density 0.126% versus 90% required, norm-residue no-go, non-Wieferich hardness transfer	[paper proof]; decoupling [kernel-verified Lean] (`AdelicWedgeDecoupling`); statistics [exact computation]	Section 87.2
Kummer orbit hypothesis $\equiv$ sparse coaction separation $\equiv$ Conjecture 1.1; unipotent (interpolation, not Vandermonde); two-star tensor is monoid blind; weight-two versus weight-one deficit	[paper proof]/[proposed target]	Section 87.3
Trap dictionary $\beta \geq 1/((1-f)\kappa_*)$; greedy-support optimality with mean frequency $\frac{2}{3}\sqrt{2abq} - \log\sqrt{6}$; external prime worth $\Theta(\log H/H)$ and dominated by discreteness; unit block the worst scale	[paper proof]; constants [exact computation]	Section 87.4
Pila integrality gate measured: $\tau_2 \geq 1/(4\beta)$ by Jensen against the monoid density, failing the threshold by $208\beta \log 2 \log 3 \approx 158\beta$ (linear in β)	[paper proof]	Section 87.10, (803)
Self-contained threshold $\tau_{\mathrm{crit}} = 1/(4570\beta^2 \log 2 \log 3)$ from the interpolation determinant against the exact monoid density and greedy mean frequency; β^{-2} scaling exact to 0.063%; conclusions independent of the quoted constant	[paper proof]; optimisation [exact computation]	Section 87.8, (802)
Monoid rigidity: a finite-type entire function is determined by its values on $\mathcal{S}$; spectrum question $V_\tau = V_0$	[paper proof]/[proposed target]	Corollary 87.8
Weight-two encoding *is* the two-star tensor; its finite core is already kernel-verified	[kernel-verified Lean] core; identification [exact computation]	`TwoStarControlCharacterization`; Section 87.7

Statement	Status	Location
Weight-two collapse in divisor-vector form: vanishing coefficients produce a single t with $M = 2^t$, $A = 3^t$; the control pair satisfies the relation, so the vanishing locus is exactly the control line	**[kernel-verified Lean]**	`WeightTwoCollapse`; Section 87.7
V_τ is a $\mathbb{Z}[\Delta_1, \Delta_\beta]$-module; both shifts diagonal on monomials with integer eigenvalues $2^i 3^j - 1$, $M^i A^j - 1$; spectrum question $\equiv$ difference-module finiteness	**[paper proof]**; target **[proposed target]**	Section 87.5, Target 87.1
Window of vanishing differences $\log \lvert\Delta^k g(0)\rvert \le 4\tau k^2 - k + O(\log k)$; the initial-segment conclusion drawn from it is *retracted* (2^z is a counterexample: limsup type does not control small radii, and $k = 0$ is unconstrained)	estimate **[paper proof]**; conclusion withdrawn	Proposition 87.2
Terminal obstruction encoded in the CDT family: $\log p = \sum_{n<p} L_n$ turns the relation into a $\mathbb{Z}$-relation among $\{L_i L_j\}$; vanishing coefficients force the controls at $q = 3, 5, 7$; finite independence per q suffices, consuming the external prime	**[exact computation]** solves; encoding **[paper proof]**; target **[proposed target]**	Proposition 87.3
Algebraic dependence with $2^z, 3^z$ forces $\log M(g, r) = O(r \log r)$, hence $\tau_2 = 0$; conditional type gap (no integer-valued entire g on $\mathcal{S}$ with $0 < \tau_2 \le$ threshold); small-type interpolant target, control-clean via the monoid's quadratic density	**[paper proof]**; Pila and minimum modulus **[classical/open background]**; target **[proposed target]**	Proposition 87.4, Theorem 87.5, Target 87.6
Primitive tensor normal form (period zero, content one, external coefficient) $\Leftrightarrow$ counterexample; Kummer period-orbit obstruction and conditional closure; determinant design checklist	**[paper proof]**/**[proposed target]**	Sections 86.1, 86.2

B Formalization inventory

All modules live under `Analysis/ExponentialIdentities/Lean/ExponentialIdentities/`, in namespace `LeanProofs`. Every theorem module listed below is reachable from the aggregate import surface; `AuditNewModules` is built separately as an assumption-audit target. The repository tree builds with no errors. None of the theorem modules contains `sorry`, an extra axiom, or `native_decide`; certificate tables are replayed by the kernel.

B.1 Kernel-accepted modules

Module	Contents
`IntegerExponent` (+3 modules)	three-base theorem, interpolation determinants
`TwoBaseIntegerExponent`	conjecture statement, candidates, gaps, $2^x < 10$
`.../FiniteCheck`, `FiniteCheck256`	zero-free regions $2^x < 100$, $2^x < 256$
`.../FiniteCheck4096`	zero-free region $2^x < 4096$ (kernel `decide` table)
`.../FiniteCheck8192`	zero-free region $2^x < 8192$ (one unsplit `decide` table, nine tiers, one semiconvergent enclosure)
`.../FiniteCheck16384`	zero-free region $2^x < 16384$ (8191 rows, 39 tiers, exponents to $5.9 \cdot 10^6$)
`.../ZeroDensity`	AP obstruction, Roth-number counting, $o(N)$
`.../NonintegerDensity`	nonintegral candidate refinement, density zero
`.../MultiplicativeRank`	rank ≤ 2, ϑ irrational, $(c\log_2 N)^2$
`.../CandidateStructure`	cross-valuation identity, conditional box growth
`.../OddCore`	common odd core, $c\log_2 N\ c\log_3 N$, dyadic blocks
`.../PrimitiveOddCore`	gcd-normalized canonical core, unique coordinates
`.../LeastSolution`	unique positive least solution, exact affine primitivity
`.../ThreeDenominatorNormalization`	reduced denominator supported at 3
`.../RationalPowerIndex`	least rational-power index, exact solution monoid
`.../PrimitiveGenerator`	canonical primitive generator and unique coordinates
`.../ExactCounting`	exact unit/dyadic block counts and cumulative finite sum
`.../QuadraticAsymptotic`	sharp quadratic limit; subquadratic-growth equivalence
`.../BlockWeyl`	exact-block geometric sums and vanishing nonzero Fourier modes
`.../BlockEquidistribution`	finite trigonometric-polynomial block averages
`.../OutputNormalization`	simultaneous primitive outputs, gcd one, size dichotomy
`.../RationalThirdBase`	denominator divisibility and determinant margin
`.../RadicalReformulation`	localized-radical equivalences, base-swapped form
`.../PrimitiveRadical`	canonical primitive-power reduction of radical target
`.../RadicalDegree`	irreducible binomial, exact minimal polynomial and degree
`.../SharperProgressions`	sharp three-candidate curvature obstruction
`.../ThirdDifference`	exact third difference, four-candidate obstruction
`.../HigherDifference`	uniform arbitrary-order forward-difference obstruction
`.../PolynomialApproximationTax`	exact dense denominator clearing, integer resolution, finite-difference approximation floor, and the all-degree normalized tax obstruction
`.../SharpProgression`	effective rational three-term criterion $h^{400} \leq n^{83}$
`.../ArithmeticRigidity`	descent, $r + 8k \leq x$
`.../ThreeSmooth`	3-smooth classification of further bases
`.../ShrinkingTarget`	fractional-part shrinking targets, Diophantine lower bounds
`.../DenominatorGrowth`	convergent-denominator growth cap $q_{n+1} < 2m^{q_n}$
`.../Transcendence`	Gelfond–Schneider applications: solutions and $\log_2 3$
`.../FractionalPartDensity`	density dichotomy, gap equivalences, infinitude
`.../Localization`	candidate-form localization equivalence
`.../FourExponentialsReduction`	four exponentials $\Rightarrow$ the conjecture
`GelfondSchneider/...`	Gelfond–Schneider theorem (port of Karatarakis)
Rational and algebraic auxiliary bases	
`.../RationalSixExponentials`	six exponentials for three positive rational bases and rational outputs
`.../RationalBaseThird`	specialization to bases $2, 3$; exact $2, 3$-unit classification of positive rational third bases
`.../AlgebraicOutputBridge`	a number field, an embedding, and one natural denominator clearing an algebraic value to an algebraic integer
`.../FiniteAlgebraicOutputBridge`	one common number field and one denominator for a finite family of algebraic values
`.../AlgebraicIntegerDeterminant`	field-norm lower bound for a nonzero algebraic-integer determinant; house and factorial bounds on the n^{11} scale
`.../AlgebraicSixExponentials`	field-norm six-exponentials determinant for three positive algebraic real bases and three algebraic outputs

Module	Contents
`.../AlgebraicThirdBase`	three-base determinant for algebraic output at a natural third base, first over a number field and then by clearing denominators
`.../AlgebraicRationalBase`	algebraic outputs at positive rational bases; the algebraic locus is exactly the $2,3$-units unless the exponent is integral
`.../AlgebraicBaseUnitPower`	positive $2,3$-unit powers and the multiplicative rank of an algebraic auxiliary base
`.../AlgebraicBaseSpectrum`	arbitrary independent positive algebraic pair; exact positive-power/divisible-hull classification of positive algebraic bases with algebraic xth output
`.../AlgebraicConsequences`	pointwise algebraic natural-base classification; iterated-output transcendence; the 5^x, 6^{x^2} and both-iterate equivalences
`.../AlgebraicOutputLocus`	exact algebraic output monomial locus (coordinate face or origin); external p-adic line for output ratios; coordinate-face and rank-two equivalences
`.../AlgebraicRankThreeEquivalence`	rank three in the algebraic real-power output locus forces integrality; the resulting equivalence
`.../AlgebraicExponentLocus`	the common algebraic-output exponent plane $\mathbb{Q} + \mathbb{Q}x$, and its natural-power equivalence
`.../RationalFunctionAlgebraicLocus`	exact rational-function affine identity at a counterexample; nonaffine rational-function and degree-≥ 2 polynomial detectors; canonical rational affine plane versus nonnegative integral cone
Canonical radical arithmetic	
`.../CanonicalRadicalValuation`	odd-prime valuation gcd; divisors of the gcd are genuine outer-power exponents
`.../CanonicalRadicalSharpness`	exact criterion for a reduced $c/3^a$ to be a rational outer power; the triple gcd resists both pairwise relaxations
`.../CanonicalRadicalFieldNorm`	norm and trace of the pure-radical generator; the base-swapped relation blocks monomial injectivity
`.../CanonicalRadicalThirdOutput`	algebraic integrality of the normalized radical is equivalent to a trivial reduced denominator
`.../CanonicalRadicalAlgebraicOutput`	algebraicity of the radical's β-th power is equivalent to $2,3$-smoothness of the normalized numerator
`.../CanonicalRadicalDivisibleHull`	an external numerator prime excludes the whole positive $2,3$-unit divisible hull
`.../CanonicalRadicalMixedOutput`	genuinely mixed radical monomials keep an external prime, so all their outputs are transcendental
`.../SimultaneousRadicalDegrees`	the displayed exponents are the least rational-power indices; exact radical degrees and the product degree de
`.../CoreIndependence`	the two primitive output cores are never exactly 3 and 2; exact rationality classification of the simultaneous cores
Smoothness and log-products	
`.../SmoothOutputs`	both outputs 3-smooth forces an integral exponent; sharp form supplying a partner with a prime factor ≥ 5
`.../LogProductIndependence`	prime-log-product independence as a sufficient condition distinct from four exponentials, in both the integer-output and rational-output forms
Determinant hierarchy and reformulations	
`.../ArbitraryNodeDeterminant`	interpolation determinant at arbitrary ordered nodes; integrality and the divided-difference repulsion bound
`.../ClearedDividedDifference`	signed barycentric denominators, minimal LCM clearing, integer coefficients, vanishing low moments, real divided-difference bridge, and the exact four-node cubic functional

Module	Contents
`.../SuperfactorialDeterminant`	universal nonnegative-node superfactorial divisor and exact consecutive-node witness; strengthened candidate determinant, repulsion, and packing bounds
`.../SignedProgressionDeterminant`	geometric Vandermonde factorization; primitive invariant quotient and coprimality; exact finite logarithmic denominator-tax identity
`.../GeometricVandermondeSmith`	integral q-Pascal and Newton-basis matrices; determinant-one two-sided reduction to the exact positive Smith diagonal for $q \geq 2$
`.../FinitePlaceShortRelationCore`	determinant divisibility from row contents, cardinality amplification for p^t-divisible rows, and the eliminated-row endpoint
`.../GeometricNewtonInterpolation`	exact finite q-Newton power expansion; interpolation through the truncation degree, omitted-tail identity, and first extrapolation product over arbitrary integer parameters
`.../QNewtonEndpointDefect`	exact finite endpoint-dilation defect for the geometric Newton interpolant, with the coefficient and node-denominator recurrences used in its proof
`.../QNewtonOffAxisTransfer`	arbitrary-commutative-ring transfer-denominator successor, forcing convolution, transported off-axis recurrence, and exact finite sum solution; no infinite-product or analytic-existence claim
`.../ReciprocalHalfContact`	reciprocal coefficient-support symmetry, sharp $2K \leq D$ contact bound, and exact even/odd equality classifications
`.../ReciprocalMomentLatticeFinite`	one-Vandermonde cofactor clearing, integral Hankel moments, common-content determinant divisibility, and strengthened determinant and quadratic-form floors under an explicit paper-supplied positive-definiteness hypothesis
`.../FejerKernelCertificate`	normalized finite Fejér kernel; exact weighted Fourier score; exact phase and integer-phase hit-count certificates
`.../ExponentialPolynomialZeros`	zero-count theorem for real exponential polynomials and nonvanishing of the generalized Vandermonde determinant
`.../SparsePowerCurve`	sparse zero bound on an irrational power curve; nonzero integral evaluation determinant; exact-support positive-node interpolant
`.../RationalOrbitGrid`	exact rational rectangle collision count; irrational grid cardinality and sparse lower bound; canonical one-monomial-saving rational-orbit certificate
`.../RationalContactRigidity`	rational polynomial evaluation at a transcendental direction; homogeneous leading-contact nonvanishing and Padé transversality
`.../RestrictedMatrixCoefficient`	concrete logarithm matrix, rank-one coefficient factorization, rational anisotropy, and exact equivalence with the conjecture
`.../IntegralPowerMatrix`	explicit real powers of a fixed split integral matrix and their exact entry-integrality/conjecture equivalence
`.../JordanFirstJet`	rationality recovery from a scalar value and its nonzero first jet; rational matrix-entry extraction; base-two integral-exponent closure conditional on the formal first-jet matrix
`.../MomentArithmetic`	rational two-atom moment reproduction; four exact common-denominator identities and their structural-prime local-unit conclusions
`.../RankGeometry`	hyperbolic normal identity, ternary-conic determinant, explicit rational symmetrizer, and symmetric rank-one outer-product normal form
`.../UnimodularGapCollision`	determinant-distance power collisions in a commutative group and forward/reverse modular and unimodular congruence wrappers

Module	Contents
`.../OrbitHeightEntropy`	exact integral-slope and logarithmic floor identities; finite injective-index entropy forcing simultaneous two-coordinate denominator bounds
`.../ProjectiveDefect`	exact piecewise-linear optimization of the abstract weight-two defect and the resulting depth and arithmetic-efficiency thresholds
`.../CarryPath`	exact distinguished-descendant carry-word sum and length; strict suffix-degree descent and reconstruction from endpoint plus ordered suffix vertices
`.../MixedRadixCarryFinite`	four mixed-radix digit bounds and the exact local floor-carry plaquette identity
`.../RadixShadow`	injective binary-to-ternary digit reinterpretation, exact power-modulus congruence bridge, divisibility form, and valuation-index equality
`.../BorrowBlockCore`	finite shifted-evaluation coefficient vanishing, forced $(0, 1)$ digit pairs, and exclusion of the leading-2 branch under a subpower bound
`.../DfiniteExponentCore`	bivariate exponent-relation polynomial for a formal linear differential operator; exact top exponent coefficient and nonzero certificate
`.../LambertCocycleFinite`	exact formal Lambert coefficient clearing, rational cocycle presentation, and universal mixed double-difference identity for arbitrary parameters; the base-$(2, 3)$ specialization uses no common-exponent hypothesis
`.../PositiveCoefficientClosure`	finite covariance identity for the increasing base-two/base-three kernels; nonnegative rank-one coefficient vectors have exactly one positive unit mass
`.../FiniteDilationAlternation`	finite positive-scale coefficient multipliers; positive filters and signed filters with a unique largest active scale preserve strict alternating tails and cannot produce eventual nonnegativity
`.../NewtonCollisionCore`	one-step Newton append and evaluation preservation, residue-collision mismatch, and exact barycentric scaling under simultaneous node/value dilation
`.../MixedGapKummerFinite`	mixed-gap determinant identity, same-residue obstruction, and a finite valuation denominator-retention inequality
`.../KummerCompatibilityFinite`	exact finite-field common-scalar compatibility count and the transverse-hyperplane count, with the zero-source case included
`.../PrimePowerKummerCoherence`	exact full-rank compatibility count and unit-transverse post-scalar rank-three normal-form count over prime-power residue rings
`.../RankOneNewtonKummer`	higher geometric Newton product and cross-base cocycle; exact vanishing for every common natural exponent at or below the interpolation order
`.../SignedOrbitNormFinite`	full signed-orbit product as the ordinary orbit product to the augmentation; mixed additive cancellation and the factorization of a two-by-two difference-grid minor
`.../DyadicPrivateSupport`	exact odd-prime order, congruence, and level privacy for dyadic cyclotomic factors
`.../CaptureMultiplier`	exact finite-group divisibility criterion for the least cleaned-gap capture multiplier
`.../CatalanTangentCore`	adjacent-smooth tangent identities, conditional balance and tangent-lattice parametrizations, two residue relations, and a conditional residual cross identity
`.../MatchedGapAmplification`	monotonic power increments, the strict scale-free one-gap inequality, its logarithmic base-$(2, 3)$ form, and finite-product amplification

Module	Contents
`.../ValuationSlopeEnergy`	zero-safe finite probability energy and total variation; sharp two-sided energy bounds and the opposite-sign wedge lower certificate
`.../CandidateAlternant`	affine real-exponent Vandermonde factorization and the exact rational-approximation candidate alternant
`.../SynchronizedSumDecoder`	strict monotonicity on fixed-sum pairs and exact unordered-pair recovery from ordinary and synchronized real-power sums
`.../RationalSumDecoder`	exact $L \leq 2d+1$ bound for a degree-d rational decoder on the synchronized slice $(2^n+1, 3^n+1)$
`.../FormalPrimeDeterminant`	multivariate prime-symbol determinant, specialization at prime logarithms, exact formal-zero classification, and Structural Rank equivalence
`.../TwoStarContent`	exact coefficient content of the formal two-star determinant, least-output content one, and nonzero reduction modulo every nontrivial modulus
`.../PrimeTopIsolation`	homogeneous cyclotomic factor, fresh-prime order 2ℓ and $2\ell \mid p-1$, lower-level p-integrality, ultrametric unique-minimum lemma, and the coefficient-or-denominator tax
`.../RestrictedDenominatorDirichlet`	restricted-denominator Dirichlet and the one-prime covolume conservation law
`.../SparseCoaction`	external-prime derivative certificate with explicit lower bound; sparse coaction separation defined and proved equivalent to the conjecture
`.../FactorialCocycle`	mesoscopic window and mode inequalities, shift-evaluation identities, annihilator P_* (roots, expansion, kills growing modes), carry-free prefix and Kummer bound at a structural prime
`.../CenteredTrace`	homogeneous geometric sums, sum-of-squares defect factorization, cubic reduction (3 is the only cancellation), valuation doubling, reciprocal-quadratic bridge, fifth-trace two-channel theorem, synchronized channel identities
`.../ExponentLatticeMask`	cleared evaluation, one-direction gap-power theorem and cleared-value floor, adjacent valuation transport (additive and multiplicative), unshared structural primes, orthogonality determinant lemma
`.../TerminalFactorialIndependence`	window-prime valuation $v_p(F_t) = 1$ from pure inequalities, high-prime exclusion, finite-product valuation, and multiplicative independence of the terminal factorials under a window-prime hypothesis
`.../RationalHankelTriple`	square valuation pattern from coprime-slope recurrences, parametrization of $m_1^2 = m_0 m_2$, Hankel-ray cross-determinant identity and classification, transcendence bridge to coefficient identities
`.../SmoothSemigroupCore`	pigeonhole modulus cutoff for distinct residues, injectivity of $(a,b) \mapsto 2^a 3^b$, square-box cutoff, and the classification of consecutive three-smooth triples and pairs (Catalan-free)
`.../ExternalPrimeNewtonWall`	unit-convergence rigidity in normed division rings (convergent unit-norm powers force $u = 1$); formal edge law $(q^n - \mu)c_n = 0$, spectral support, empty spectrum of the edge pairs over $\mathbb{Q}$, and vanishing of every formal edge solution
`.../CriticalWindowContraction`	sign-mass identity from $P(M) = 0$; division-free geometric and weighted-geometric kernel bounds; window/coefficient summation exchange; abstract M-weighted contraction theorem; candidate-regime numerics ($M \geq 2^{14}$, $q \leq (2/3)^{14}$) and impossibility of balanced transport vectors; rational-root exponent-gap divisibility

Module	Contents
`.../CompactChartArea`	integer-numerator identity and reciprocal-product floor for the oriented area of three Farey-chart marked points; exact chart spacing; squared form of the relation-counting coefficient–codimension lock
`.../ToricMonomialRigidity`	toric eigenvalue equation determines the eigenvalue at every support index; diagonal $(2X, 3Y)$-semi-invariants are supported on one monomial
`.../ConsecutiveDilationRigidity`	Vandermonde elimination for exponential sums at consecutive shifts: strictly increasing positive nodes and q consecutive vanishing sums force all coefficients to zero; q-monomial dilation-rigidity wrapper on the two-power curve
`.../TwoStarControlCharacterization`	symmetric two-star coefficient array; the tensor vanishes identically iff the divisor vectors are those of an integer control $(2^m, 3^m)$; vanishing at every control pair
`.../TopSliceTailSum`	exact window valuation of a falling factorial at a prime above half the top ($v_p = 1$ in the window, 0 below, no error terms); exact tail-sum valuation of a signed nested-slice factorial product
`.../CrossSliceContraction`	exact transport-weight exponent identity; conversion of the per-prime rescue bound into balance weights $(\Phi^- A^{m+j'} \le 3\Phi^+ M^{n+i})$; admissibility contraction factor $M^{n+i}A \le A^{m+j'}$
`.../WeightTwoCollapse`	the weight-two coefficient array of the terminal relation is definitionally the two-star tensor; vanishing produces a single t with $m = t\delta_2$, $a = t\delta_3$; the control pair satisfies the relation
`.../AdelicWedgeDecoupling`	exact multiplicativity of the Fermat quotient; control-direction invariance of the Fermat-quotient wedge; rank drop at p characterized as collinearity of (q_pM, q_pA) with (q_p2, q_p3)
`.../AbelSliceMass`	Abel summation identity for slice tails, telescoping of weight increments, slice-mass positivity from nonnegative tails, the sharp collapse (a vanishing weighted mass with nonnegative tails and strictly increasing positive weights forces every coefficient to vanish), and the geometric weight-gap bounds crushing the negative mass of a top-heavy slice by a factor $1/M$
`.../FermatQuotientDeterminant`	exact Fermat quotient and power law modulo p; first local-log determinant interface and integral-power determinant vanishing
`.../ConsecutiveLogMeanBounds`	exact two-sided enclosure for the reciprocal consecutive logarithmic mean, with an explicit $1/(24H^2)$ remainder
`.../BoundaryDisplacement`	exact two-coordinate evaluation-matrix displacement identities, terminal-boundary projector ranks, and rank-K displacement consequences under explicit $K > 0$, nonsingularity, and nonzero-multiplier hypotheses
`.../BoundaryDilationDefect`	rectangular node products, dilation-defect divisibility, unique square-grid residuals, and the degree-$2K-2$ boundary commutator
`.../DilationDifferenceQuotient`	integral factorization of $P(cX) - P(X)$; dyadic $2X$ quotient and its formal-derivative reduction modulo two
`.../CrossResidualScaling`	generic prime-power cancellation through a unit polynomial; normalized opposite-residual transfer through a scaled defect identity
`.../ResultantScaling`	exact resultant change under $X \mapsto sX$; symmetric cross-normalization identities, nonzero residue quotients, and two-prime divisibility
`.../NodeDivisorScaling`	exact dyadic and triadic normalization of the two structural-prime boundary node divisors, with structural-prime-unit constant terms

Module	Contents
`.../CrossResultantBridge`	conditional composition of the primary and opposite residual scalings into one nonzero cross resultant with exact 2- and 3-orders
`.../DenominatorCancellation Counterexample`	exact smallest-grid certificate that the full interpolation denominator can disappear from the paired rational residual resultant
`.../GridTransformNoGo`	exact classification showing that one nonsingular common direction change cannot manufacture simultaneous dyadic and triadic axis separation
`.../SmithProfileCore`	convex injective-matching cost; strict equality case; universal $p^{m\binom{m}{2}}$ divisibility of weighted lower-Toeplitz minors
`.../ContentNormalizedProfileFinite`	endpoint-gap, quotient-cost, direct-floor, wrap-path, and compressed-permutation inequalities for the eventual primitive residual Smith profile
`.../DeepSubresultantCost`	sharp direct/shallow displacement-cost bound, strict case, and equality classification for the leading principal-subresultant matching
`.../CompoundEndpointRate`	kernel-checked negativity of the normalized structural compound gain after the mandatory base-2/base-3 endpoint costs
`.../OrderedChamberPositivity`	positivity of generalized real-power Vandermonde determinants and of the semigroup determinant on ordered chambers
`.../MixedExponentSemigroup`	pair-indexed exponent $a + b\vartheta$, injectivity, and integrality of $m^{a+b\vartheta}$
`.../MixedMahlerFinite`	exact finite rectangular inclusion–exclusion identity behind the mixed $(2, 3)$-Mahler equation; no convergence or natural-boundary assertion
`.../TwoScaleSupportFinite`	exact finite rectangular inclusion–exclusion identity behind the ordinary two-scale support equation; no convergence assertion
`.../SemigroupDeterminant`	integrality half of the semigroup generalized-Vandermonde hierarchy, with $\lvert\det\rvert \geq 1$
`.../FallingFactorialBound`	falling-factorial coefficient bound and dyadic-strip divided-difference bound
`.../SemigroupWeightBound`	mixed-exponent family construction and the window-summation counting bridge
`.../BlockGrowthReformulations`	boundedness, $o(T)$, $\liminf$, and $\equiv 1$ reformulations of the conjecture
`.../CompactOrbit`	compact synchronized-orbit reformulation with shared floor exponent, in both directions
`.../AdelicDrift`	exact integer denominator exponents, common centered defect, structural-prime normalized orbit norms, and p-adic escape
`.../SturmianProbability`	carry/floor telescope, exact prefix law, mean, variance, joint-carry intervals, probability, and covariance kernel
`.../SturmianGeneratingSeries`	exact carry/floor and denominator formal power-series identities, twice-cleared defect, and cross-base cancellation
`.../ArbitraryNodeRepulsion`	divided differences, the Newton factorization of the interpolation determinant, and the repulsion and packing bounds conditional on the named mean-value hypothesis
`.../SemigroupGapBound`	Leibniz determinant bound, semigroup span inequality, and the explicit dyadic $10^4(\log X)^2$ block bound conditional on the named Newton factorization
`.../TropicalInitialForm`	common-power factorization of a Leibniz determinant; all-depth quotient divisibility and the modulo-p tied-fiber criterion, without a cancellation lower bound
`.../BlockFiberFactorization`	block-preserving signed Leibniz sums as products of block determinants, including variable-size and consecutive-power Vandermonde forms

Module	Contents
`.../TropicalBlockApplication`	first-extra-prime divisibility iff the product of block determinants, or its consecutive-power Vandermonde specialization, vanishes modulo p; application hypotheses remain explicit
`.../BeattyTropicalFiber`	exact repeated-weight minimizers for rank-one assignment costs; Beatty floor gap; normalized quotient divisibility; exact block-determinant expansion
`.../RowBlockLaplaceExpansion`	exact grouping of all assignment layers by row partitions; normalized and cutoff-K expansions into signed products of square unit minors
`.../AssignmentExcessGeometry`	exact prefix-crossing energy; Beatty excess lower bound; stars-and-bars binomial count for row partitions below any fixed cutoff
`.../BlockVandermondeValuationBudget`	exact collision-depth formula for finite Vandermondes; elementary difference-of-powers valuation and variable-block cubic budgets
`.../AllLayerUnitBasis`	complete-determinant divisors from monic unitriangular or determinant-one column changes; strengthened positive dyadic-odd and triadic-unit p-ordering weights, without a residual-cascade bound
`.../UnitOrderingGain`	exact unit block-weight identities and finite square/cube aggregate bounds; no triangular-lattice asymptotic or residual-cascade estimate
`.../RowBlockUnitDivisibility`	row-scaled consecutive-power matrices; a common p-ordering divisor for every assignment-minor product; determinant-one fiberwise basis change; additive divisor $p^{\tau+U}$ for the full rank-one power determinant
`.../MixedDeterminantUnitBridge`	finite sorted-box mixed-exponent prefix; exact column-permuted identification with the rank-one scaled model; dyadic and triadic additive divisors under explicit block-fiber certificates
`.../GlobalMixedPrefixCascade`	global initial-segment certificate, automatic lexicographic/colexicographic fiber equivalences, exact truncated cascade identities, and structural-output determinant divisors under the stated decreasing-depth and residue hypotheses
Second iterates, towers, and the candidate lattice	
`.../CandidateLattice`	coordinatewise gcd and lcm for $2^s w^t$; unconditional closure of the candidate set under gcd and lcm; prime-support rigidity of the branch
`.../KernelDichotomy`	multiplicative dependence of $M, A, 2, 3$ is equivalent to a rational linear relation among $1, \vartheta, x, x\vartheta$ and to the Möbius orbit
`.../SecondIterateKernel`	the second-iterate kernel as a subgroup of $\mathbb{Z}^2$; no same-sign vectors, rank at most one, smooth coordinate cases, and solution-independence
`.../TowerRigidity`	base-two second iterate as a natural number; the depth-two algebraic dichotomy; the depth-three transcendence and the cubic-tower equivalence
`.../AlgebraicTowerTransition`	exact transition locus $2^y, 2^{xy} \in \overline{\mathbb{Q}} \iff y \in \mathbb{Q} + \mathbb{Q}\vartheta$ under a counterexample; global uniqueness of a positive transition depth, the all-depth three-level exclusion, and the fixed-distinct-depth AE equivalence
`.../CrossBaseTowerTransition`	exact scaled base-three transition locus; cross-base mutual exclusion, combined uniqueness across both principal towers, and higher smooth-direction exclusions in the dependent-output branch

Module	Contents
`.../SymmetricTowerConstants`	$E_3 = 3^{\vartheta}$ and $E_2 = 2^{1/\vartheta}$, their conditional rational powers, and their mutual exclusion
`.../PrimeLogSpaceOperator`	the ϑ-pullback of the prime-logarithm span is exactly $\mathbb{Q}\log 2$ if and only if the rational-output conjecture holds
`.../ComplexSolutions`	on the principal branch every complex solution is real, by incommensurability of the two argument gratings
`.../MinPlusCeiling`	the $\{2,3\}$-part of an integer divides it, hence term-wise p-adic content can never exceed a correct archimedean bound
`.../AuditNewModules`	`#print axioms` audit of the headline theorems listed above

B.2 Kernel acceptance and the finite-certificate boundary

The following determinant, semigroup, growth, orbit, and log-product modules are kernel-accepted and lie on the aggregate import surface:

`ArbitraryNodeDeterminant`, `ExponentialPolynomialZeros`, `MixedExponentSemigroup`, `SemigroupDeterminant`, `FallingFactorialBound`, `SemigroupWeightBound`, `BlockGrowthReformulations`, `CompactOrbit`, `LogProductIndependence`.

Statements that these nine modules actually contain are therefore **[kernel-verified Lean]**.

Separately, the attempted $2^x < 16384$ finite-certificate assembly described in Section 23 remains a genuine **[Lean draft, not yet accepted]**: its individual arithmetic rows replay, but its covering theorem does not compile end to end. It is not on the aggregate import surface and raises no kernel-verified bound. The accepted finite frontier remains $2^x < 4096$.

The candidate-lattice, tower, operator, repulsion, and gap layer comprises eight additional kernel-accepted modules: `CandidateLattice`, `KernelDichotomy`, `SecondIterateKernel`, `TowerRigidity`, `SymmetricTowerConstants`, `PrimeLogSpaceOperator`, `ArbitraryNodeRepulsion` and `SemigroupGapBound`. All lie on the aggregate import surface and are audited with `#print axioms`.

The algebraic-locus, determinant, orbit, decoder, spectral, and closure layer comprises the following kernel-accepted modules:

`AlgebraicExponentLocus`, `AlgebraicOutputLocus`,
`RationalFunctionAlgebraicLocus`, `AlgebraicTowerTransition`,
`CrossBaseTowerTransition`, `BlockVandermondeValuationBudget`, `AllLayerUnitBasis`,
`UnitOrderingGain`,
`CanonicalRadicalMixedOutput`, `SimultaneousRadicalDegrees`,
`TropicalInitialForm`, `BlockFiberFactorization`, `TropicalBlockApplication`,
`BeattyTropicalFiber`,
`RowBlockLaplaceExpansion`, `AssignmentExcessGeometry`, `RowBlockUnitDivisibility`,
`MixedDeterminantUnitBridge`,
`RestrictedMatrixCoefficient`, `FormalPrimeDeterminant`, `SparsePowerCurve`,
`BoundaryDilationDefect`, `SuperfactorialDeterminant`, `RationalOrbitGrid`,
`SignedProgressionDeterminant`, `FejerKernelCertificate`,
`AdelicDrift`, `SturmianProbability`, `SturmianGeneratingSeries`,
`PolynomialApproximationTax`,

`RationalContactRigidity`, `IntegralPowerMatrix`, `CandidateAlternant`, `FermatQuotientDeterminant`, `ConsecutiveLogMeanBounds`, `BoundaryDisplacement`, `GlobalMixedPrefixCascade`, `DilationDifferenceQuotient`, `SynchronizedSumDecoder`, `RationalSumDecoder`,
`AlgebraicBaseSpectrum`, `MixedMahlerFinite`, `GeometricVandermondeSmith`, `TwoScaleSupportFinite`, `GeometricNewtonInterpolation`, `QNewtonEndpointDefect`, `QNewtonOffAxisTransfer`, `ClearedDividedDifference`, `FinitePlaceShortRelationCore`, `JordanFirstJet`,
`MomentArithmetic`, `RankGeometry`, `UnimodularGapCollision`, `OrbitHeightEntropy`, `ProjectiveDefect`, `CarryPath`, `MixedRadixCarryFinite`, `RadixShadow`, `BorrowBlockCore`, `DfiniteExponentCore`,
`MixedGapKummerFinite`, `KummerCompatibilityFinite`, `CrossResidualScaling`, `ResultantScaling`, `NodeDivisorScaling`, `CrossResultantBridge`, `DenominatorCancellationCounterexample`, `GridTransformNoGo`, `SmithProfileCore`, `ContentNormalizedProfileFinite`, `DeepSubresultantCost`, `CompoundEndpointRate`, `LambertCocycleFinite`,
`PositiveCoefficientClosure`, `FiniteDilationAlternation`, `NewtonCollisionCore`, `RankOneNewtonKummer`, `SignedOrbitNormFinite`, `DyadicPrivateSupport`, `CaptureMultiplier`, `CatalanTangentCore`, `MatchedGapAmplification`, `ValuationSlopeEnergy`, `ReciprocalHalfContact`, `ReciprocalMomentLatticeFinite`, `TwoStarContent`, `PrimePowerKummerCoherence`,
`PrimeTopIsolation`, `RestrictedDenominatorDirichlet`, `SparseCoaction`, `FactorialCocycle`, `CenteredTrace`, `ExponentLatticeMask`.

Their headline APIs are on the aggregate surface and covered by the axiom-audit module.

Two trust cautions apply across these modules, and the status matrix reflects them. First, a module being accepted certifies exactly the theorems it states. The analytic halves of the repulsion and gap theorems are now formal, but each carries its missing analytic input as an explicit, named *hypothesis* — `RpowDividedDifferenceMeanValue` and `NewtonFactorization` — rather than as an axiom, so those two conclusions are conditional in a way the reader can see in the theorem statement; everything upstream and downstream of them is unconditional. `ExponentialPolynomialZeros` supplies the nonvanishing engine, while the separate accepted module `OrderedChamberPositivity` supplies the ordered-chamber sign proved on paper in Appendix D. Second, the interval scan through 2^{20} and the combined directed-rounding scan through 2^{27} are **[software computation]**, not kernel certificates; the kernel-verified exponent bound is still $x \geq 12$ from `FiniteCheck4096`.

A headline theorem receives **[kernel-verified Lean]** only after it is added to the aggregate import surface and checked with `#print axioms`. Numerical constants in the tables of Appendix E are explanatory only; the formal statements should use exact expressions or safe rational constants such as 1/24 and 10000.

C A symmetric formulation

Everything above has a base-swapped version. Put $\eta = \log_3 2 = 1/\vartheta$. Failure of the conjecture is also equivalent to the existence of an integer $v > 1$ with $3 \nmid v$ and

$$v^\eta \in \mathbb{Z}[1/2].$$

Status: [kernel-verified Lean] — `not_alaogluErdosConjecture_iff_exists_threeFree_localized_radical` and `alaogluErdosConjecture_iff_threeFreeLocalizedRadicalExclusion` in `RadicalReformulation`.

For the primitive pair (M, A), factoring A into a power of 3 times a 3-coprime part produces a symmetric primitive core and rational-power index; the two normalizations give the same least exponent β, and their compatibility condition is exactly $\log M \log 3 = \log A \log 2$. The symmetry is useful in formalization — some divisibility arguments are easier on one side — and confirms that the localized-radical reduction of Section 5 is not an artifact of privileging the base 2.

D Detailed proof of the generalized Vandermonde sign

This appendix expands the generalized Vandermonde positivity lemma of Section 18 to make its analytic input fully transparent.

Let $0 \le \lambda_0 < \cdots < \lambda_r$ and define $f_j(x) = x^{\lambda_j}$ on $(0, \infty)$. For $k \le r$,

$$f_j^{(k)}(x) = (\lambda_j)_{\underline{k}}\, x^{\lambda_j - k}.$$

The kth initial Wronskian is

$$\begin{aligned} W_k(x) &= \det\big(f_j^{(i)}(x)\big)_{0\le i,j\le k} \\ &= x^{\sum_{j=0}^{k} \lambda_j - k(k+1)/2} \det\big((\lambda_j)_{\underline{i}}\big)_{0\le i,j\le k}. \end{aligned}$$

The polynomial $(z)_{\underline{i}}$ is monic of degree i. Replacing the monomial basis $1, z, \ldots, z^k$ by the falling-factorial basis is unit lower-triangular. Therefore

$$\det\big((\lambda_j)_{\underline{i}}\big)_{i,j=0}^{k} = \prod_{0\le i<j\le k} (\lambda_j - \lambda_i) > 0.$$

Hence every initial Wronskian is positive.

One standard characterization of extended complete Chebyshev systems [19] now implies that for $x_0 < \cdots < x_r$,

$$\det\big(f_j(x_i)\big)_{i,j=0}^{r} > 0.$$

To avoid using that characterization as a black box, proceed inductively. Suppose a nontrivial combination

$$F(x) = \sum_{j=0}^{r} c_j x^{\lambda_j}$$

has $r+1$ distinct positive zeros. Put $x = e^t$ and factor the lowest exponential:

$$e^{-\lambda_0 t} F(e^t) = c_0 + \sum_{j=1}^{r} c_j e^{(\lambda_j - \lambda_0)t}.$$

Rolle's theorem gives at least r zeros of its derivative, an exponential polynomial with at most r terms. Induction gives a contradiction unless all coefficients vanish. Thus the evaluation determinant is nonzero. Its sign cannot change on the connected chamber $x_0 < \cdots < x_r$. Let the nodes coalesce as $x_i = x + i\varepsilon$ and divide by the positive ordinary Vandermonde. As $\varepsilon \downarrow 0$, the quotient tends to $W_r(x)/\prod_{k=0}^{r} k! > 0$. Hence the original determinant is positive everywhere in the ordered chamber.

Status: [paper proof] and [kernel-verified Lean] — paper proof in this report; kernel-accepted zero-count/nonvanishing in `ExponentialPolynomialZeros` and the positive ordered-chamber determinant in `OrderedChamberPositivity`.

E Numerical constants

For reference,

$$\vartheta = 1.58496250072115618145\ldots, \qquad \vartheta(\vartheta-1) = 0.92714362797110482756\ldots.$$

The three-point span constant is

$$\left(\frac{8}{\vartheta(\vartheta-1)}\right)^{1/3} = 2.051072395661874\ldots.$$

For the ordinary-power order r, the raw span constant and exponent are

$$C_r = \left(\frac{r!}{|\,(\vartheta)_{\underline{r}}\,|}\right)^{2/(r(r+1))}, \qquad \alpha_r = \frac{2(r-\vartheta)}{r(r+1)}.$$

The continuous critical point for α_r solves

$$-r^2 + 2\vartheta r + \vartheta = 0,$$

so

$$r_* = \vartheta + \sqrt{\vartheta^2 + \vartheta} = 3.6090841940\ldots,$$

confirming that $r = 4$ is the optimal integer order.

For the semigroup hierarchy of Section 19, the first exponents of $\Lambda_\vartheta = \{a + b\vartheta : a, b \in \mathbb{N}_0\}$ and their pair labels are:

Index	Pair (a,b)	Exponent $a+b\vartheta$
0	$(0,0)$	0
1	$(1,0)$	1
2	$(0,1)$	ϑ
3	$(2,0)$	2
4	$(1,1)$	$1+\vartheta$
5	$(3,0)$	3
6	$(0,2)$	2ϑ
7	$(2,1)$	$2+\vartheta$
8	$(4,0)$	4
9	$(1,2)$	$1+2\vartheta$
10	$(3,1)$	$3+\vartheta$
11	$(0,3)$	3ϑ
12	$(5,0)$	5

These decimal values are explanatory. They are not part of any machine-checked statement, and formal versions of the corresponding theorems should carry exact expressions instead.

F Interval computation: script and outputs

The script below verifies one range. Set `START` to 1 or 2^{19} and `LIMIT` to 2^{19} or 2^{20} to obtain the two halves reported afterwards.

```
import math, time
import mpmath as mp

START = 1
LIMIT = 1 << 19
DPS = 40

mp.iv.dps = DPS
ln2 = mp.iv.log(2)
ln3 = mp.iv.log(3)
theta_float = math.log2(3.0)

checked = 0
failures = []
min_lower = (float("inf"), None, None)
min_upper = (float("inf"), None, None)
start_time = time.time()

for m in range(START, LIMIT):
    if (m & (m - 1)) == 0:      # integral exponent: m is a power of two
        continue

    # Only proposes a neighboring integer.  The interval tests below certify it.
    n = math.floor(math.exp(theta_float * math.log(m)))

    lhs = mp.iv.log(m) * ln3
    lower_diff = lhs - mp.iv.log(n) * ln2
    upper_diff = mp.iv.log(n + 1) * ln2 - lhs

    lower_lb = float(lower_diff.a)
    upper_lb = float(upper_diff.a)
    if not (lower_lb > 0.0 and upper_lb > 0.0):
        failures.append((m, n, str(lower_diff), str(upper_diff)))
        break

    min_lower = min(min_lower, (lower_lb, m, n))
    min_upper = min(min_upper, (upper_lb, m, n))
    checked += 1

print("checked_nonpowers=", checked)
print("failures=", len(failures))
print("min_lower_log_margin=", min_lower)
print("min_upper_log_margin=", min_upper)
print("elapsed_seconds=", time.time() - start_time)
assert not failures
```

Listing 2: Interval verification

The two 40-digit runs produced:

```
LIMIT=524288
DPS=40
checked_nonpowers=524268
```

```
failures=0
min_lower_log_margin=(1.4147089634746718e-15, 189427, 231498127)
min_upper_log_margin=(3.0841010256409534e-15, 306982, 497586055)
elapsed_seconds=18.505
```

Listing 3: Lower half output.

```
LIMIT=1048576
DPS=40
checked_nonpowers=524287
failures=0
min_lower_log_margin=(9.490232037370244e-16, 958460, 3023921230)
min_upper_log_margin=(2.3019781003173366e-15, 556061, 1275861767)
elapsed_seconds=18.904
```

Listing 4: Upper half output.

Elapsed times are machine-dependent sanity checks; the counts, zero-failure results, extremal tuples, and displayed code are the reproducibility data.

Status: [software computation] — interval arithmetic in `mpmath` [30] at 40 digits. This is a software trust boundary, not a kernel certificate. The kernel-verified zero-free region remains $2^x < 4096$ in `FiniteCheck4096`.

G Suggested experimental determinant families

Before developing a full p-adic theory, several finite experiments can test whether the required amplification exists.

G.1 Rectangular row and column boxes

Choose rows (n, k) with $0 \le n < N_0$, $0 \le k < K_0$ and columns (a, b) with $0 \le a < A_0$, $0 \le b < B_0$. The entry is

$$(2^a 3^b)^n (M^a A^b)^k.$$

For complete rectangles, the matrix is a rearranged Kronecker product only when the row and column cardinalities match in a compatible way. Its determinant or maximal minors can be factored symbolically for small boxes. Record valuations at 2 and 3 as polynomials in $a_0 = v_2(M)$ and $b_0 = v_3(A)$.

G.2 Triangular boxes adapted to height

Use

$$n + k\beta \le T, \qquad a + b\vartheta \le L.$$

These are the natural row and column shapes from the exact counting theorem and the mixed-exponent semigroup. Search for square minors with minimal archimedean weight and maximal forced valuation. The linear boundary slopes are irrational, so floor-sum fluctuations may prevent cancellation patterns present in rectangles.

G.3 Finite-difference row operations

Order rows by n for fixed k, or by k for fixed n, and apply repeated difference operators. Differences introduce factors such as

$$2^a 3^b - 1, \qquad M^a A^b - 1.$$

These may have systematic 2- or 3-adic valuations for carefully selected column pairs. Lifting-the-exponent lemmas can then provide exact lower bounds. The risk is that the archimedean size of the same factors grows too quickly; experiments should track both sides.

G.4 Confluent columns

Instead of only distinct mixed monomials, consider derivatives with respect to an auxiliary continuous exponent parameter before specializing. Confluent generalized Vandermonde matrices introduce logarithmic factors, which are not integral, but common-denominator or linear-form constructions may restore arithmetic control. This is closer to Laurent's interpolation determinant method [25] and may reduce total exponent weight.

H Dependency map and Lean-oriented proof sketches

The conditional arithmetic criteria of this report were written with formalization in mind: each one is a short argument over an already kernel-verified base. This appendix records the shortest route from the existing corpus to each of them, and then gives skeleton statements for the four criteria whose formalization is least obvious.

Status: Design note. The snippets below are schematic ASCII transliterations and are not claimed to compile unchanged. The mathematics they target is **[paper proof]**; none of it has been accepted by the kernel, and nothing here may be read as a claim of verification.

H.1 Dependency map

Each row names one new result, the principal existing input it is built on, and the extra mathematics a formalization would have to supply. All of the new results are conditional: they describe the structure forced on $\mathcal{S}$ and $\mathcal{C}$ by a hypothetical counterexample to Conjecture 1.1, and they are vacuous if the conjecture holds.

New result	Principal existing input	Additional mathematics
rational-function rigidity	canonical primitive generator; transcendence of β	injectivity of polynomial evaluation at a transcendental element
product and power criteria	unique (n, k) coordinates	quadratic coefficient comparison
quotient criterion	unique (n, k) coordinates	denominator clearing and coefficient comparison
three-smooth output test	transcendence of ϑ	unique factorization in $\mathbb{N}$
product transcendence	three-smooth-base theorem	classical six exponentials
common power degree	unique (n, k) coordinates	positive root uniqueness; divisibility
primitive statistics	total lattice count	Möbius inversion
gcd and lcm lattice	exact candidate monoid	prime-valuation min and max
ambient root saturation	primitive valuation gcd of w	Bézout and linear cone arithmetic
optimized differences	arbitrary-order difference theorem	gamma identity and Stirling estimate

Every entry in the middle column is **[kernel-verified Lean]** in the corpus catalogued in Section 3. In the right-hand column, only the six exponentials theorem is external background (**[classical/open background]**); the remaining items are ordinary mathlib-level algebra and analysis, which is what makes these criteria cheap to formalize once the base is in place.

Transliteration conventions. The listings write Unicode in plain ASCII: `R`, `Q`, `Z` and `N` are the real, rational, integer and natural types; arrows are `->` and `<->`; membership is `in` and divisibility is `dvd`; anonymous constructors use angle brackets; the two logical connectives use their two-character ASCII spellings; and the focusing dot is written as a period. Coercions are written as explicit casts such as `Int.cast` rather than as the coercion arrow.

H.2 Multiplication skeleton

A direct statement can use the repository predicate `TwoBaseIntegralSolution`, which is the defining conjunction $2^x \in \mathbb{Z}$ and $3^x \in \mathbb{Z}$.

```
theorem product_solution_iff_integer_factor
    {x y : R}
    (hx : TwoBaseIntegralSolution x)
    (hy : TwoBaseIntegralSolution y) :
    TwoBaseIntegralSolution (x * y) <->
      x in Set.range (Int.cast : Z -> R) \/
      y in Set.range (Int.cast : Z -> R) := by
  classical
  constructor
  . intro hxy
    by_cases hAE : AlaogluErdosConjecture
    . left
      exact hAE x hx
    . obtain <w, d, a, c, beta, ..., hcoord, ...> :=
        exists_canonical_primitiveGenerator_of_not_alaogluErdosConjecture hAE
```

```
      obtain <nx, kx, hxcoord> := (hcoord x).mp hx
      obtain <ny, ky, hycoord> := (hcoord y).mp hy
      obtain <np, kp, hpcoord> := (hcoord (x * y)).mp hxy
      -- If kx and ky are both positive, expanding the product gives a
      -- nonzero quadratic polynomial over Q vanishing at beta, which
      -- contradicts transcendence of beta.  Hence kx = 0 or ky = 0.
      ...
  . rintro (hxint | hyint)
    . exact solution_nsmul_left hxint hy
    . exact solution_nsmul_right hyint hx
```

Listing 5: Product criterion: skeleton.

In the finished theorem the integer range can be narrowed to $\mathbb{N}_0$, because every solution is nonnegative by `nonneg_of_two_rpow_integer`. The algebraic step should be isolated, since three of the four criteria reduce to it. Note that the coefficients must be rational — with arbitrary real a, b, c the statement is false — and that is exactly how the applications supply them, as differences of the integer coordinates:

```
lemma transcendental_quadratic_coeff_eq_zero
    {beta : R} {a b c : Q}
    (hbeta : Transcendental Q beta)
    (h : (a : R) * beta ^ 2 + (b : R) * beta + (c : R) = 0) :
    a = 0 /\ b = 0 /\ c = 0 := ...
```

Listing 6: The shared coefficient lemma.

Mathlib's polynomial evaluation API is probably the cleanest route: build the polynomial $aX^2 + bX + c$, rewrite `h` as an evaluation, and apply the definition of transcendence to conclude that the polynomial is zero.

H.3 Output smoothness skeleton

Define the predicate

$$\mathrm{ThreeSmooth}(m) :\Longleftrightarrow \exists u, v \in \mathbb{N}_0,\ m = 2^u 3^v.$$

A practical implementation should avoid a partial "natural output" function and instead carry explicit witnesses $M, A \in \mathbb{N}$ for the two integer powers:

```
theorem integer_of_outputs_threeSmooth
    {x : R} {M A : N}
    (hM : (2 : R) ^ x = (M : R))
    (hA : (3 : R) ^ x = (A : R))
    (hMs : ThreeSmooth M) (hAs : ThreeSmooth A) :
    x in Set.range (Int.cast : Z -> R) := by
  obtain <a, b, rfl> := hMs
  obtain <c, d, rfl> := hAs
  -- Taking logarithms base two: x = a + b * theta and x * theta = c + d * theta.
  have hpoly :
      (b : R) * logThreeDivLogTwo ^ 2
        + ((a : R) - (d : R)) * logThreeDivLogTwo - (c : R) = 0 := by
    ...
  have htheta := transcendental_logThreeDivLogTwo
  -- The coefficients are integers, so all three vanish:
  -- b = 0, a = d, c = 0, whence 2 ^ x = 2 ^ a and x = a.
  ...
```

Listing 7: Output smoothness: skeleton.

The converse direction is immediate: for integral $x = a$ the outputs are 2^a and 3^a. The interesting content is that three-smoothness of *both* outputs is already enough to force integrality, with no appeal to the conditional structure theory.

H.4 Common-power skeleton

The formal theorem is most reusable when it is stated for explicit output witnesses and for a fixed canonical generator, so that the coordinate comparison is available as a hypothesis rather than reconstructed inside the proof.

```
theorem simultaneous_perfectPower_iff_divides_coordinates
    (hfail : not AlaogluErdosConjecture)
    {x beta : R} {n k r : N}
    (hr : 0 < r)
    (hbeta : CanonicalPrimitiveGenerator beta)
    (hxcoord : x = (n : R) + (k : R) * beta) :
    (Exists fun U : N => (U : R) ^ r = (2 : R) ^ x) /\
    (Exists fun V : N => (V : R) ^ r = (3 : R) ^ x) <->
      r dvd n /\ r dvd k := by
  -- Convert the two perfect powers into a solution at x / r,
  -- apply uniqueness of coordinates, and compare.
  ...
```

Listing 8: Common powers: skeleton.

The positive-root step can be discharged either by strict monotonicity of $t \mapsto t^r$ on the nonnegative reals or by injectivity of the logarithm; the first is usually shorter in mathlib.

H.5 Ambient saturation skeleton

The only genuinely arithmetic lemma in the saturation argument is the reconstruction of a root of a power of a primitive number:

```
lemma primitive_core_root
    {w u q r : N}
    (hw : oddPrimeValuationGCD w = 1)
    (hr : 0 < r)
    (hpow : u ^ r = w ^ q) :
    Exists fun t : N => q = r * t /\ u = w ^ t := by
  -- Compare every coordinate of the prime factorizations.
  -- A Bezout combination of the positive valuations of w gives r | q;
  -- factorization extensionality then gives u = w ^ t.
  ...
```

Listing 9: Primitive-core root lemma.

The proof in this report only needs the case $q = dk$, but the general lemma is worth stating once: the same reconstruction is used implicitly in `PrimitiveOddCore` and in `OutputNormalization`, and a shared version would remove that duplication.

H.6 Suggested theorem order for a pull request

A low-conflict sequence, each step reviewable on its own, is:

1. add a small `ThreeSmooth` API together with the output-smoothness theorem;
2. add coordinate comparison lemmas for affine, quadratic and quotient expressions;
3. prove the product, power and quotient criteria;
4. prove the common-power and affine-descent criteria;
5. add the candidate divisibility and support results;
6. add primitive-core root reconstruction and ambient saturation;
7. only then add the asymptotic and six-exponentials extensions.

This ordering keeps the first several commits elementary, and it defers everything that depends on external transcendence input to the end, where it can be reviewed against the status discipline of Appendix A rather than mixed into routine algebra.

I Directed-rounding verifier

Appendix F gives the interval-arithmetic script behind the 2^{20} verification. This appendix gives the second, independent verifier: a C program that uses MPFR directed rounding rather than an interval library, and that extends the scanned range sixty-fourfold, to 2^{26}.

Status: [software computation] — MPFR 4.2.2 at 192 bits with directed rounding modes. Like the 2^{20} scan this is a software trust boundary, not a kernel certificate. The kernel-verified zero-free region remains $2^x < 4096$ in `FiniteCheck4096`.

I.1 What the verifier certifies

For a non-power of two m, proving $m^\vartheta \notin \mathbb{N}$ amounts to exhibiting $n \in \mathbb{N}$ with $n < m^\vartheta < n+1$, and, as in Section 23, that pair of strict inequalities is certified in logarithmic form so that no exponentiation is ever performed. At precision $p = 192$ the program computes directed enclosures

$$L_m^- \leq \log m \leq L_m^+, \qquad L_2^- \leq \log 2 \leq L_2^+, \qquad L_3^- \leq \log 3 \leq L_3^+,$$

together with $L_n^+ \geq \log n$ and $L_{n+1}^- \leq \log(n+1)$, and then rounds *downward* in

$$\delta_- = L_m^- L_3^- - L_n^+ L_2^+, \qquad \delta_+ = L_{n+1}^- L_2^- - L_m^+ L_3^+.$$

Positivity of δ_- proves the lower inequality and positivity of δ_+ proves the upper one. Every input on the n side is below 2^{42} and is therefore exactly representable at 192 bits.

An ordinary `long double` evaluation proposes $n \approx \lfloor m^\vartheta \rfloor$ and the offsets $0, -1, 1, -2, 2, -3, 3$ are tried in that order. The proposal has no logical role: a proposal is accepted only when both directed lower bounds are strictly positive, and a failure to locate any admissible n is reported as a failure rather than silently reclassified. Locator error can therefore produce a false alarm but not a false proof. Powers of two are skipped rather than tested, since $(2^q)^{\vartheta} = 3^q$ is the known integral candidate; an exact integral value at a non-power of two would satisfy neither strict inequality for any n and would be reported.

I.2 Environment

The recorded run used MPFR 4.2.2 linked as `libmpfr.so.6`, GCC 14.2.0 with `-O3 -fopenmp`, GNU OpenMP with five worker threads, Linux on x86-64, and 192-bit precision with the directed modes `MPFR_RNDD` and `MPFR_RNDU`. The execution image contained the MPFR shared library but not its development header, so the source carries a guarded minimal declaration of the MPFR 4.x ABI; on a system with `mpfr.h` present the standard header branch is taken instead. That ABI detail is part of the software trust boundary and is disclosed here rather than hidden.

I.3 Source

The executed C99 source is stored at `source/mpfr_scan.c`. It uses MPFR directed rounding and OpenMP.

I.4 Build and invocation

On a system with the MPFR development headers installed, the whole build and a single full-range run are:

```
gcc -O3 -fopenmp source/mpfr_scan.c -lmpfr -lm -o mpfr_scan
OMP_NUM_THREADS=5 ./mpfr_scan 1 67108864 192
```

Listing 10: Build and invocation (POSIX shell).

The recorded audit instead used sixteen shorter invocations over half-open subranges, so that each chunk produces an independently inspectable log. The exit status is nonzero exactly when some input failed to be certified, which makes the chunked run scriptable without parsing the output.

J Chunk audit and reproducibility

Status: [software computation] — the numbers in this appendix are the output of the verifier of Appendix I. They are reproducible data at a software trust level, not kernel certificates; the kernel-verified exponent bound remains $x \geq 12$.

There are $2^{26} - 1 = 67{,}108{,}863$ positive integers below 2^{26}, of which exactly 26 are powers of two, namely 2^0 through 2^{25}. The verifier therefore tests $67{,}108{,}863 - 26 = 67{,}108{,}837$ inputs. The range was split at multiples of 2^{22} into one initial chunk $[1, 2^{22})$ and fifteen chunks of width 2^{22}. The canonical chunk log holds the full standard output of every invocation.

Table 8 reports, for each chunk, the exact scanned integer interval, the number of non-powers certified, and the smallest directed lower and upper logarithmic margins recorded in that chunk. Every chunk returned a failure count of zero.

Table 8: Directed-rounding audit through 2^{26}.

Chunk	Integer range	Checked	Smallest lower gap	Smallest upper gap
0	1–4,194,303	4,194,281	1.1539×10^{-17}	3.0076×10^{-17}
1	4,194,304–8,388,607	4,194,303	4.6319×10^{-20}	1.3515×10^{-19}
2	8,388,608–12,582,911	4,194,303	1.2811×10^{-18}	1.3515×10^{-19}
3	12,582,912–16,777,215	4,194,304	4.6319×10^{-20}	2.9082×10^{-19}
4	16,777,216–20,971,519	4,194,303	1.1898×10^{-18}	1.3515×10^{-19}
5	20,971,520–25,165,823	4,194,304	5.4212×10^{-19}	1.5111×10^{-19}
6	25,165,824–29,360,127	4,194,304	4.6319×10^{-20}	1.7669×10^{-19}
7	29,360,128–33,554,431	4,194,304	1.4439×10^{-19}	5.4132×10^{-21}
8	33,554,432–37,748,735	4,194,303	4.2839×10^{-19}	1.3515×10^{-19}
9	37,748,736–41,943,039	4,194,304	3.9292×10^{-19}	1.0460×10^{-19}
10	41,943,040–46,137,343	4,194,304	1.9177×10^{-19}	1.5111×10^{-19}
11	46,137,344–50,331,647	4,194,304	1.0117×10^{-19}	3.4145×10^{-19}
12	50,331,648–54,525,951	4,194,304	5.6927×10^{-20}	1.2322×10^{-19}
13	54,525,952–58,720,255	4,194,304	3.6447×10^{-20}	1.1345×10^{-19}
14	58,720,256–62,914,559	4,194,304	6.7779×10^{-20}	5.4132×10^{-21}
15	62,914,560–67,108,863	4,194,304	5.0923×10^{-20}	3.9197×10^{-20}
	total	**67,108,837**		

J.1 Extremal records

The global lower-gap record is the chunk 13 value 3.6447×10^{-20}, attained at

$$(m, n) = (58{,}570{,}438,\ 2{,}048{,}703{,}434{,}093),$$

where a separate 100-digit evaluation gives $m^{\vartheta} - n = 1.07725158750919017548\ldots \times 10^{-7}$. The global upper-gap record is the chunk 7 value 5.4132×10^{-21}, attained at

$$(m, n) = (29{,}411{,}704,\ 687{,}581{,}893{,}443),$$

where $n + 1 - m^{\vartheta} = 5.36974926710988874953\ldots \times 10^{-9}$. The same upper margin reappears in chunk 14 at the scaled pair (58,823,408, 2,062,745,680,331); this is not a coincidence but the identity $(2m)^{\vartheta} = 3\,m^{\vartheta}$, which multiplies the input by two and its ϑ-power by three while leaving the logarithmic margin unchanged.

Both extremal inequalities were reevaluated independently at 100 decimal digits. That secondary check is not an independent proof system — it shares the same floating-point philosophy — but it does guard against transcription and aggregation mistakes, and the certified margins sit many orders of magnitude above the nominal 192-bit rounding scale of the logarithmic operations.

The 2^{20} run of Appendix F and the present run are independent implementations, one using interval arithmetic in `mpmath` and the other MPFR directed rounding, and they agree on the overlapping range $1 \le m < 2^{20}$: no failures. Their recorded extremal metadata are not directly comparable, because the chunk boundaries differ.

J.2 Reproducibility checklist

To reproduce the finite claim independently:

1. obtain MPFR 4.2.x and GMP on a 64-bit platform;
2. save the source of Appendix I;
3. compile with optimization and OpenMP, or delete the OpenMP pragmas for a serial run;
4. run the sixteen half-open intervals of Table 8 at 192 bits;
5. verify that every invocation reports `failures=0`;
6. sum the reported `checked_nonpowers` and obtain 67,108,837;
7. compare the two global extremal records above;
8. optionally rerun at 256 or 320 bits, and on a second MPFR build;
9. for a theorem-grade result, translate the intervals into exact rational certificates and replay them in Lean rather than trusting the executable.

The finite statement is deterministic. Thread count changes execution time and the reduction order that produces the recorded minimum metadata, but it cannot change whether an individual directed inequality is certified. Step nine is the one that would change the status of the result rather than its range, and it is the reason the scan is presented here as data for certificate generation — in the sense of Section 25 — and not as an extension of the kernel-verified zero-free region.

K Classical transcendence inputs and their exact roles

Status: **[classical/open background]** — this appendix states external theorems, not new ones. Exactly one of them, Gelfond–Schneider, is formalized in this repository (Theorem 3.7); the remaining ones enter only through paper proofs, which are tagged **[paper proof]** where they appear. The corpus does not introduce any of them as an axiom: where a conjectural or unformalized statement is needed in Lean it is defined as a proposition and the implication is proved conditionally on it. Deleting six exponentials, five exponentials, Baker's theorems, or Roy's theorem from the report would change which paper results stand, and would change none of the kernel-verified results catalogued in Appendix A.

The purpose of this appendix is auditability. A reader who wants to know how much of the report is elementary consequence of the repository hypotheses, and how much rests on major transcendence theory, should be able to answer that question by looking in one place. For each classical input below we give the statement in the form actually used, the exact places where the report uses it, and the exact reason it stops short of Conjecture 1.1. The last reason is the same in every case, seen from a different angle: every input listed here constrains a *linear* configuration of logarithms or an *algebraic* exponent, while the hypothetical counterexample is a *bilinear* relation between logarithms with a transcendental exponent.

Gelfond–Schneider (Hilbert's seventh problem)

Statement. Let $\alpha, \beta \in \overline{\mathbb{Q}}$ with $\alpha \neq 0, 1$ and $\beta \notin \mathbb{Q}$. Then every value of α^β is transcendental [17, 24, 3]. **[classical/open background]**

Role here. This is the one classical transcendence theorem the corpus proves rather than cites; the Lean proof follows the classical exposition [17] and is recorded as Theorem 3.7. It is used twice. First, a nonintegral solution x is shown irrational by the elementary rational-exponent argument, and then must be transcendental because 2^x is an algebraic number; this is the algebraic–integral dichotomy of Corollary 3.8, and it is what makes the conditional primitive generator β and the odd-core exponent $\log_2 w$ transcendental throughout Section 4 and Section 5. Second, $\vartheta = \log_2 3$ is irrational, hence transcendental, because $2^\vartheta = 3$.

Why it does not settle Conjecture 1.1. The theorem speaks only about algebraic irrational exponents. Once the dichotomy has removed algebraic exponents, the remaining hypothetical exponent is transcendental, and Gelfond–Schneider has nothing further to say about 2^x for transcendental x. The theorem closes the easy half of the problem and defines the hard half.

Six exponentials

Statement. Let $u_1, u_2, u_3 \in \mathbb{C}$ be linearly independent over $\mathbb{Q}$ and let $v_1, v_2 \in \mathbb{C}$ be linearly independent over $\mathbb{Q}$. Then at least one of the six numbers

$$e^{u_i v_j} \qquad (1 \leq i \leq 3,\ 1 \leq j \leq 2)$$

is transcendental [25, 26, 27]. **[classical/open background]**

Role here. It is the analytic engine behind every exact spectrum statement in the report. The saturated algebraic-base spectrum — for multiplicatively independent $a, b \in \overline{\mathbb{Q}}_{>0}$ with $a^x, b^x \in \overline{\mathbb{Q}}$ and x irrational, $\alpha^x \in \overline{\mathbb{Q}}$ if and only if $\alpha \in \Gamma(a, b)$ — is the case with rows $\log a, \log b, \log \alpha$ and columns $1, x$. The conditional classification of the symmetric constant $E_3 = 3^\vartheta$ is the case with rows $1, x, \vartheta$ and columns $\log 2, \log 3$, and the classification of $E_2 = 2^\eta$ is its mirror image under $\vartheta \mapsto \eta = 1/\vartheta$. The ordinary form of the unconditional E_3/E_2 dichotomy uses the same configuration. The kernel-verified 3-smooth classification of auxiliary bases and the three-base theorem $2^x, 3^x, 5^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$ are the formalized integer-level shadows of this theorem; the Lean proofs proceed by interpolation determinants and do not cite it.

Why it does not settle Conjecture 1.1. Six exponentials needs three independent directions on one side of the configuration. The two-base problem supplies exactly two logarithmic directions, $\log 2$ and $\log 3$, and every auxiliary base built from 2 and 3 has logarithm in their rational span, so it creates no third column. The remaining 2×2 configuration of Section 6 is precisely the four exponentials conjecture, which is open.

Five exponentials

Statement. Let u_1, u_2 be linearly independent over $\mathbb{Q}$, let v_1, v_2 be linearly independent over $\mathbb{Q}$, and let $\gamma \in \overline{\mathbb{Q}}$ with $\gamma \neq 0$. Then at least one of the five numbers

$$e^{u_1 v_1}, \quad e^{u_1 v_2}, \quad e^{u_2 v_1}, \quad e^{u_2 v_2}, \quad e^{\gamma u_2 / u_1}$$

is transcendental [26, 23]. **[classical/open background]**

Role here. It supplies the two forbidden exponential rays of the hypothetical world: for a nonintegral solution x and any nonzero algebraic γ, both $e^{\gamma x}$ and $e^{\gamma\vartheta}$ are transcendental. For the first, apply the theorem to the independent pairs $(1, x)$ and $(\log 2, \log 3)$, whose four rectangular exponentials are the algebraic numbers $2, 3, M, A$, leaving the fifth transcendental; the second is the same argument with the roles of the two logarithms exchanged. Formalizing it would also upgrade the denominator-controlled auxiliary-base theorem in `RationalThirdBase` from a controlled denominator to an unrestricted rational output, which is why it appears as a research priority in Section 25.

Why it does not settle Conjecture 1.1. The theorem does not remove the fifth exponential, so it does not prove four exponentials. More concretely, the natural auxiliary quantities in this problem are $p^x = e^{x \log p}$ for a prime p, whose coefficient $\log p$ is transcendental rather than an admissible algebraic γ. The forbidden rays therefore never convert a prime divisor of the integer outputs into a contradiction, as Section 24 records.

Baker's theorem on linear forms in logarithms

Statement. Let $\gamma_1, \ldots, \gamma_n$ be nonzero algebraic numbers of degree at most d and height at most H, and let $\alpha_1, \ldots, \alpha_n$ be algebraic numbers of height at most B, not all zero. If

$$\Lambda = \alpha_1 \log \gamma_1 + \cdots + \alpha_n \log \gamma_n \neq 0,$$

then $|\Lambda| > B^{-C}$ for an effectively computable $C > 0$ depending only on n, d, H and the degree of the α_i [3, 25]. **[classical/open background]**

Role here. The effective form is used as a measuring stick rather than as a step in any proof. In the rational-approximation analysis of Section 24 it fixes the scale of what is available: Baker-type lower bounds are polynomial in the relevant heights, whereas the integer gaps produced by the hypotheses only assert $|\varepsilon| \gtrsim 2^{-p}$, exponentially weaker, so the two scales are consistent and no contradiction arises. It is also the theorem that effective irrationality measures for ϑ would refine.

Why it does not settle Conjecture 1.1. Baker's theory bounds forms whose coefficients are *algebraic*. The decisive relation under failure is $\log M \, \log 3 = \log A \, \log 2$, and its radical form is $d \log w \log 3 = (\log A - a \log 3) \log 2$: each side is a product of two logarithms, so the coefficient of every logarithm is itself a logarithm. Expanded over prime factorizations $m = \prod_p p^{m_p}$ and $A = \prod_p p^{a_p}$, failure is the nontrivial vanishing of

$$D = \sum_p \big(m_p \log 3 - a_p \log 2\big) \log p,$$

an integer-coefficient linear form in the *products* $\log p \cdot \log 2$ and $\log p \cdot \log 3$. No lower-bound technology exists for such forms, and even the $\mathbb{Q}$-linear independence of $\{\log p \cdot \log q\}$ is open; Section 11 turns exactly that independence into a sufficient condition.

Baker's homogeneous theorem

Statement. Let $\gamma_1, \ldots, \gamma_n$ be nonzero algebraic numbers. If $\log \gamma_1, \ldots, \log \gamma_n$ are linearly independent over $\mathbb{Q}$, then $1, \log \gamma_1, \dots, \log \gamma_n$ are linearly independent over $\overline{\mathbb{Q}}$. The homogeneous consequence used here is the shorter one: logarithms of algebraic numbers that are linearly independent over $\mathbb{Q}$ are linearly independent over $\overline{\mathbb{Q}}$ [3, 25]. **[classical/open background]**

Role here. It is the bridge from a rational-independence hypothesis to the $\overline{\mathbb{Q}}$-independence hypothesis that Roy's theorem requires. In the strong spectrum, the three logarithms $\log a, \log b, \log \alpha$ are first shown $\mathbb{Q}$-independent by an elementary saturation argument, and Baker's homogeneous theorem upgrades them to $\overline{\mathbb{Q}}$-independence; the same upgrade produces the logarithmic independence used in the strong forms of the E_3 and E_2 classifications. The ordinary spectrum does not need it, since rational independence already suffices for six exponentials.

Why it does not settle Conjecture 1.1. It is a statement about linear relations among logarithms. No independence statement about the family $\{\log p\}$, over $\mathbb{Q}$ or over $\overline{\mathbb{Q}}$, implies anything about the family of products $\{\log p \cdot \log q\}$, and it is the products that carry the obstruction.

Roy's strong and sharp six exponentials theorems

Statement. Let $\mathcal{L}$ be the $\overline{\mathbb{Q}}$-vector space spanned by 1 together with all logarithms of nonzero algebraic numbers. If u_1, u_2, u_3 are linearly independent over $\overline{\mathbb{Q}}$ and v_1, v_2 are linearly independent over $\overline{\mathbb{Q}}$, then at least one of the six products $u_i v_j$ lies outside $\mathcal{L}$. The sharp form replaces the independence hypotheses by a rank condition on a matrix whose entries are linear forms in logarithms of algebraic numbers [23, 26]. **[classical/open background]**

Role here. It produces the strengthened conclusions only. Concretely: the strong auxiliary-base spectrum, which asserts $x \log \alpha \notin \mathcal{L}$ for $\alpha \notin \Gamma(a, b)$ rather than merely $\alpha^x \notin \overline{\mathbb{Q}}$; the strong form of the symmetric-constant dichotomy, which places $\vartheta \log 3$ and $\eta \log 2$ outside $\mathcal{L}$; and the log-strong transcendence clauses in the second-iterate kernel and depth-three tower theorems.

Why it does not settle Conjecture 1.1. Every ordinary transcendence conclusion in the report survives the removal of this theorem; only the "lies outside $\mathcal{L}$" clauses disappear. Structurally it inherits the six exponentials dimension count and still demands three independent directions on one side, so it does not reach the 2×2 configuration where the conjecture actually lives.

Lindemann–Weierstrass

Statement. If $\alpha_1, \dots, \alpha_n$ are distinct algebraic numbers, then $e^{\alpha_1}, \dots, e^{\alpha_n}$ are linearly independent over $\overline{\mathbb{Q}}$. The special case used here is Hermite–Lindemann: for algebraic $\alpha \neq 0$ the value e^{α} is transcendental, equivalently every nonzero value of $\log \gamma$ is transcendental for algebraic $\gamma \neq 0, 1$ [3]. **[classical/open background]**

Role here. It is the standing background fact that each $\log p$, and in particular $\log 2$ and $\log 3$, is transcendental. That fact is what makes the coefficients in the log-product form D above transcendental rather than algebraic, and therefore what places the wall outside Baker's reach; it is also the precise content of the audit warning against treating $\log 2$ and $\log 3$ as algebraic coefficients (Remark L.3).

Why it does not settle Conjecture 1.1. Lindemann–Weierstrass constrains exponentials of *algebraic* arguments. Under failure the exponent x is transcendental and the arguments $x \log 2$ and $x \log 3$ are transcendental, so none of the four corner exponentials $2, 3, 2^x, 3^x$ is within its scope, and it gives no information whatever about products of two logarithms.

K.1 Logical dependency graph

The table records, for each family of results in the report, the classical input it actually consumes. The rows whose input column reports no classical theorem are elementary or kernel-verified, and depend on no transcendence theory beyond the Gelfond–Schneider dichotomy that the corpus already formalizes.

Table 9: Classical transcendence input consumed by each family of results.

Result	Required classical input
Nonintegral solution is transcendental	Rational-exponent lemma and Gelfond–Schneider; both kernel-verified
Reduction to real four exponentials	None beyond $\mathbb{Q}$-independence of $1, x$ and of $\log 2, \log 3$; kernel-verified
3-smooth auxiliary-base classification and the three-base theorem	None; interpolation determinants, kernel-verified
Exact algebraic-base spectrum for an independent pair	Six exponentials
Strong $\mathcal{L}$ spectrum	Exact row independence, Baker's homogeneous theorem, and Roy
Valuation lattice and Smith normal form of rational outputs	Exact spectrum and unique factorization
Second-iterate kernel: dimension and the four branch types	Exact spectrum and elementary prime valuations
Kernel invariance across all nonintegral solutions	Exact spectrum applied twice
Möbius orbit criterion	Kernel definition and real logarithms
Depth-three tower theorem	Exact spectrum and kernel dimension; Roy only for the strong form
Base-2 tower equivalence	Natural-base algebraicity classification and smooth branch algebra
Symmetric constants E_3 and E_2	Six exponentials; Baker's homogeneous theorem and Roy for the strong form
Two forbidden exponential rays $e^{\gamma x}$, $e^{\gamma \vartheta}$	Five exponentials
Transcendence of the coefficients $\log p$ in the log-product form	Lindemann–Weierstrass, in its Hermite–Lindemann case

L Common false shortcuts and audit warnings

Status: [**classical/open background**] — this appendix records negative facts and standard cautions. Nothing in it is a new result, and nothing in it changes the status of any statement elsewhere in the report.

The problem attracts several plausible but invalid one-line arguments. Recording them is part of the audit trail: each is a specific way in which a proposed proof can be circular, can silently assume the conclusion, or can mistake numerical evidence for exclusion. Future paper and formal work should be checked against this list before a short argument is believed.

Remark L.1 (Fractional-part error). Writing $x = n + \alpha$ with $n \in \mathbb{Z}$ and $0 < \alpha < 1$, and assuming $2^x \in \mathbb{Z}$, does *not* imply $2^\alpha \in \mathbb{Z}$. It implies only $2^\alpha \in \mathbb{Z}[1/2]$, and an irrational α is not an ordinary radical exponent $1/q$. The integrality of the output does not descend to the fractional part.

Remark L.2 (Rational-root circularity). From $M^\vartheta \in \mathbb{Z}$ one cannot conclude that ϑ is rational, nor that M is a perfect power. Gelfond–Schneider constrains algebraic irrational exponents, whereas ϑ is transcendental. An argument that quietly treats ϑ as a rational exponent has assumed the conclusion.

Remark L.3 (Baker is linear, not bilinear). The relation $\log M \, \log 3 = \log A \, \log 2$ is not a Baker linear form. Treating $\log 2$ and $\log 3$ as algebraic coefficients is invalid: by Hermite–Lindemann both are transcendental. The same objection applies to the radical form $d \log w \log 3 = (\log A - a \log 3) \log 2$ and to the prime-expanded log-product form of Section 11.

Remark L.4 (No modular real exponent). The expression $2^x \bmod p$ has no canonical meaning for a general real x. A p-adic exponential requires a p-adic exponent and a unit in its convergence domain; the real equality $2^x \in \mathbb{Z}$ supplies neither. Congruence arguments must therefore be stated for the integer outputs M and A, never for the real exponent.

Remark L.5 (Candidate quotients need not be candidates). If $m_1, m_2 \in \mathcal{C}$ and $m_1 \mid m_2$, it does not follow that $m_2/m_1 \in \mathcal{C}$, because the corresponding quotient of the outputs need not be integral. Multiplication is safe; division requires a separate valuation check. This is the reason $\mathcal{C}$ is treated as a multiplicative monoid rather than as a divisor-closed set.

Remark L.6 (Density alone is compatible). The orbit $\{kx\}$ is dense modulo one for irrational x, but the associated rational points have exponential height. A sequence of $O(\log H)$ points up to height H is compatible with generic transcendental-curve counting, so density by itself yields no contradiction without a denominator-sensitive refinement.

Remark L.7 (Numerical tolerance is not exclusion). Computing M^ϑ in floating point and comparing its distance to the nearest integer against a fixed epsilon cannot certify a zero-free region, however wide. Outward directed interval bounds or an exact certificate are required, which is why the finite scans in this report are tagged [**software computation**] and the kernel-verified exponent bound remains $x \geq 12$.

Remark L.8 (Algebraicity of a power is branch-sensitive). From $z^d \in \overline{\mathbb{Q}}$ with an integer $d > 0$ one may conclude $z \in \overline{\mathbb{Q}}$, but from $z^x \in \overline{\mathbb{Q}}$ with transcendental x no such algebraic closure argument is available. The finite-degree argument uses the polynomial $X^d - z^d$, which has no analogue when the exponent is not an integer.

Every warning above was re-derived independently during the preparation of the present unified report: each was proposed at some point as a short route to the conjecture, and each failed on exactly the ground stated here.

M Logarithmic-energy convergence for projected lattice triangles

We record the technical point needed in the triangular determinant limit. Let $\lambda_1, \lambda_2 > 0$ and assume their ratio has finite irrationality type: there are $c, \tau > 0$ such that

$$|r\lambda_1 + s\lambda_2| \geq c(1 + |r| + |s|)^{-\tau}$$

for all nonzero $(r, s) \in \mathbb{Z}^2$. Let $t_{1,N} < \cdots < t_{N,N}$ be the first N values of $a\lambda_1 + b\lambda_2$ with $a, b \in \mathbb{N}_0$, and set $\widetilde{t}_{i,N} = t_{i,N}/\sqrt{N}$. Then the empirical measures converge to

$$d\mu(t) = \frac{2t}{U^2}\,dt, \qquad U = \sqrt{2\lambda_1\lambda_2},$$

and

$$\frac{2}{N^2}\sum_{i<j} \log|\widetilde{t}_{j,N} - \widetilde{t}_{i,N}| \longrightarrow \iint \log|x-y|\,d\mu(x)d\mu(y). \tag{806}$$

The weak convergence follows from a Riemann-sum lattice count in the triangle $\lambda_1 a + \lambda_2 b \leq U\sqrt{N}$. For the singular kernel, truncate $\log|x-y|$ below at $-M$. Weak convergence handles the truncated kernel. Uniform integrability of the negative tail follows by counting difference vectors (r, s) in thin strips around $r\lambda_1 + s\lambda_2 = 0$; the finite-type lower bound prevents an individual difference from being smaller than a fixed power of N, while the strip count shows that the proportion below ε is $O(\varepsilon)$ up to a power-saving discrepancy term. Integrating the distribution function of $-\log|x-y|$ and then sending $M \to \infty$ proves (806).

For $(\lambda_1, \lambda_2) = (\log 2, \log 3)$, the required finite type follows from linear forms in logarithms. For $(1, \beta)$, it follows from $\beta = \log_2 A$ and the multiplicative independence of A and 2.

N Integral evaluations used in the determinant constants

N.1 The beta-two logarithmic energy

For the probability density $f(x) = 2x$ on $[0, 1]$,

$$\begin{aligned}\iint_0^1 4xy\log|x-y|\,dx\,dy &= 8\int_0^1\int_0^x xy\log(x-y)\,dy\,dx\\ &= 8\int_0^1 x^3\left(\frac{1}{2}\log x - \frac{3}{4}\right)dx\\ &= -\frac{7}{4}.\end{aligned}$$

Scaling by U adds $\log U$, proving Equation (248).

N.2 The simplex assignment constant

A coordinate of a point uniformly distributed on the standard two-simplex has quantile $q(s) = 1-\sqrt{1-s}$. Therefore

$$\begin{aligned} J &= \int_0^1 q(s)q(1-s)\,ds \\ &= \int_0^1 \left(1-\sqrt{s}-\sqrt{1-s}+\sqrt{s(1-s)}\right)ds \\ &= 1-\frac{4}{3}+\frac{\pi}{8} = \frac{\pi}{8}-\frac{1}{3}. \end{aligned}$$

Its comonotone second-moment constant is $K = \int_0^1 q(s)^2 ds = 1/6$.

N.3 The balanced-rectangle energy

Let $Z = (X+Y)/2$ with independent uniform $X, Y \in [0,1]$. The density of $S = X+Y$ is triangular, and the density of the difference of two independent copies of S is

$$h(t) = \begin{cases} \frac{2}{3} - t^2 + \frac{1}{2}t^3, & 0 \le t \le 1, \\ \frac{1}{6}(2-t)^3, & 1 \le t \le 2. \end{cases}$$

A direct integration gives

$$\mathbb{E}\log|S-S'| = \frac{4}{3}\log 2 - \frac{25}{12}.$$

Since $Z - Z' = (S-S')/2$,

$$\mathbb{E}\log|Z-Z'| = \frac{1}{3}\log 2 - \frac{25}{12}.$$

Also $\mathbb{E}Z^2 = 7/24$.

O General two-prime version

The structural and determinant arguments extend verbatim to distinct primes p, q. If a nonintegral β satisfies $p^\beta, q^\beta \in \mathbb{Z}$, choose the least such β. Six Exponentials gives the same affine-rank collapse, and valuation minimality yields the exact monoid $\mathbb{N}_0 + \mathbb{N}_0\beta$. The triangular determinant then gives

$$g\left(\sqrt{\beta \log p \log q}\right) \ge 0.$$

This form may be useful for comparing the strength of the method across base pairs and for testing variational improvements.

P Finite determinant certificates

The asymptotic argument can be converted into finite certificates. For chosen finite node sets, a certificate consists of:

1. the exact integer or rational matrix entries;
2. sorted rational intervals for u_i, v_i and their differences;
3. an upper bound from Equation (243);
4. for each relevant prime, dual potentials for the assignment problem in Lemma 39.3, proving the claimed minimum;
5. a product-formula comparison showing the upper bound is smaller than the certified divisor.

Such certificates are compact enough for independent checking and eventual Lean verification. They also avoid trusting floating-point determinant computations, which are extremely ill-conditioned in the near-singular regime.

Q Source of the finite verifier

The following is the complete source used to establish Theorem 47.1. Its only non-rigorous arithmetic—the binary64 call used to suggest a starting integer—cannot affect the result: the subsequent directed-interval sign search is monotone and independently certifies the final pair of inequalities.

```
#!/usr/bin/env python3
"""Rigorous finite verification for the Alaoglu--Erdos problem.

For every integer A in [2, LIMIT], this proves that A^(log_2(3))
is an integer only when A is a power of two.  All decisive comparisons use
Decimal directed rounding plus the positive atanh series for logarithms.
The ordinary binary64 approximation is used only to choose an initial nearby
integer; a monotone certified search repairs any inaccurate guess.
"""
from __future__ import annotations

from dataclasses import dataclass
from decimal import Decimal, Context, ROUND_FLOOR, ROUND_CEILING, localcontext
import json
import math
import sys
import time

LIMIT = int(sys.argv[1]) if len(sys.argv) > 1 else 100_000
PREC = 70
TERMS = 28

DOWN = Context(prec=PREC, rounding=ROUND_FLOOR)
UP = Context(prec=PREC, rounding=ROUND_CEILING)

@dataclass(frozen=True)
class Interval:
    lo: Decimal
    hi: Decimal


def _atanh_log_ratio(p: int, q: int, terms: int = TERMS) -> Interval:
    """Enclose 2*atanh(p/q), for 0 <= p/q <= 1/3."""
    assert 0 <= p <= q // 3 + (1 if 3 * p == q else 0)
    if p == 0:
        z = Decimal(0)
        return Interval(z, z)

    with localcontext(DOWN):
```

```
        z_lo = Decimal(p) / Decimal(q)
        z2_lo = z_lo * z_lo
        power = z_lo
        total = Decimal(0)
        for j in range(terms):
            total = total + power / Decimal(2 * j + 1)
            power = power * z2_lo
        lo = Decimal(2) * total

    with localcontext(UP):
        z_hi = Decimal(p) / Decimal(q)
        z2_hi = z_hi * z_hi
        power = z_hi
        total = Decimal(0)
        for j in range(terms):
            total = total + power / Decimal(2 * j + 1)
            power = power * z2_hi
        partial = Decimal(2) * total
        # power is now z^(2*terms+1).  Since all omitted terms are positive,
        # replace every odd denominator by the first one and sum geometrically.
        rem = Decimal(2) * power / (
            Decimal(2 * terms + 1) * (Decimal(1) - z2_hi)
        )
        hi = partial + rem
    return Interval(lo, hi)


LOG2 = _atanh_log_ratio(1, 3)


def log_integer(n: int, terms: int = TERMS) -> Interval:
    """Rigorous interval for log(n), n >= 1, by binary range reduction."""
    assert n >= 1
    if n == 1:
        z = Decimal(0)
        return Interval(z, z)
    k = n.bit_length() - 1
    two_k = 1 << k
    # n/2^k in [1,2), and z=(y-1)/(y+1) lies in [0,1/3).
    local = _atanh_log_ratio(n - two_k, n + two_k, terms)
    with localcontext(DOWN):
        lo = Decimal(k) * LOG2.lo + local.lo
    with localcontext(UP):
        hi = Decimal(k) * LOG2.hi + local.hi
    return Interval(lo, hi)


LOG3 = log_integer(3)


def determinant_log_interval(log_a: Interval, b: int) -> Interval:
    """Enclose F(A,b)=log(b)log(2)-log(A)log(3)."""
    log_b = log_integer(b)
    with localcontext(DOWN):
        lo = log_b.lo * LOG2.lo - log_a.hi * LOG3.hi
    with localcontext(UP):
        hi = log_b.hi * LOG2.hi - log_a.lo * LOG3.lo
    return Interval(lo, hi)


def certified_sign(log_a: Interval, b: int) -> tuple[int, Interval]:
    f = determinant_log_interval(log_a, b)
    if f.hi < 0:
        return -1, f
    if f.lo > 0:
        return +1, f
    # At this finite precision an interval meeting zero is not silently accepted.
    raise ArithmeticError(f"inconclusive interval at A-log={log_a}, b={b}: {f}")
```

```
def is_power_of_two(n: int) -> bool:
    return n > 0 and (n & (n - 1)) == 0


def main() -> None:
    start = time.monotonic()
    rho_approx = math.log2(3.0)
    min_abs_endpoint = Decimal('Infinity')
    max_repairs = 0
    checked = 0

    for a in range(2, LIMIT + 1):
        if is_power_of_two(a):
            continue
        checked += 1
        log_a = log_integer(a)
        guess = max(1, int(math.floor(math.exp(rho_approx * math.log(float(a))))))
        repairs = 0

        # Find consecutive integers b,b+1 on opposite sides of the unique real root.
        while True:
            s0, f0 = certified_sign(log_a, guess)
            s1, f1 = certified_sign(log_a, guess + 1)
            min_abs_endpoint = min(
                min_abs_endpoint,
                abs(f0.lo), abs(f0.hi), abs(f1.lo), abs(f1.hi)
            )
            if s0 < 0 < s1:
                break
            if s1 < 0:
                guess += 1
            elif s0 > 0:
                guess -= 1
                assert guess >= 1
            else:
                raise AssertionError((a, guess, s0, s1))
            repairs += 1
            if repairs > 100:
                raise AssertionError(f"unexpectedly bad initial approximation at A={a}")
        max_repairs = max(max_repairs, repairs)

    elapsed = time.monotonic() - start
    result = {
        "limit": LIMIT,
        "checked_nonpowers": checked,
        "precision_decimal_digits": PREC,
        "atanh_terms": TERMS,
        "max_initial_guess_repairs": max_repairs,
        "minimum_absolute_interval_endpoint": str(min_abs_endpoint),
        "elapsed_seconds": elapsed,
        "conclusion": (
            f"For every integer 2 <= A <= {LIMIT}, "
            "A^(log_2(3)) is integral only if A is a power of two."
        ),
    }
    print(json.dumps(result, indent=2, sort_keys=True))


if __name__ == '__main__':
    main()
```

R External mathematical context

This appendix records the external mathematical results that delimit the methods used above. Nothing below is used as a step in a proof unless it is cited at the point of use. Two tempting claims are corrected where they occur.

R.1 Effective measures for the two logarithms

An explicit linear-independence measure for the relevant triple is Qiang Wu, *On the linear independence measure of logarithms of rational numbers*, Math. Comp. **72** (2003), 901–911, which gives $\mu(1, \log 2, \log 3) \le 7.6155$, improving Rhin's 7.616. For the single logarithm, Wu and Wang, J. Number Theory **142** (2014), 264–273, give $\mu(\log 3) \le 5.1163051$, improving Salikhov's 5.125. For two-logarithm forms, Laurent–Mignotte–Nesterenko, J. Number Theory **55** (1995), 285–321, and Laurent, Acta Arith. **133** (2008), 325–348, give explicit $\log |b_1 \log 2 + b_2 \log 3| \ge -C(\log B)^2$ with C of the order of a few tens. On the non-archimedean side, Bugeaud–Laurent (1996) and Bugeaud (1999) are the two-logarithm workhorses, and K. C. Chim, J. Number Theory (2025), improves the $(\log B)^2$ dependence to $\log B$ for two p-adic logarithms.

Caution R.1 (Limitation of linear forms)**.** A tempting proposal is that Wu's measure might yield a *uniform* zero-free criterion replacing the per-value certificates of Section 23. That is wrong, and the error is instructive. Near-integrality of U^{ϑ} is not a rational approximation to ϑ; it is the vanishing of the bilinear form $\log 3 \log U - \log 2 \log V$, which is exactly the wall of Section 6. No linear-in-logarithms measure touches it. The finite verification stays per-value.

What the measure does give unconditionally is an effective irrationality exponent $\mu_{\mathrm{eff}}(\vartheta) \le 8.62$, hence a polynomial recurrence $q_{n+1} \ll q_n^{7.62}$ for the convergent denominators of ϑ. This supersedes the exponential estimate; the kernel-verified module `DenominatorGrowth` proves only the weaker exponential form.

R.2 Interpolation determinants with non-archimedean amplification

The closest existing realization of the amplification idea audited in Section 20 is Bombieri, *Effective Diophantine approximation on* $\mathbb{G}_m$, Ann. Scuola Norm. Sup. Pisa **20** (1993), 61–89, and Bombieri–Cohen, *ibid.* **24** (1997), 205–225, with the elementary account in LMS Lecture Notes **303** (2003), 41–62. These combine archimedean and non-archimedean interpolation determinants through the equivariant Thue–Siegel principle and provide a non-Baker route to effective results on the multiplicative group. Laurent's interpolation-determinant papers (Acta Arith. **66** (1994), 181–199, and **133** (2008)) supply the determinant method in the form this report already uses.

This is a technique needing adaptation, not a plug-in. Its relevance is sharpened by Corollary 20.2: any amplification must obtain its archimedean bound from genuine cancellation, and Bombieri–Cohen is where that is done systematically.

R.3 Small-value estimates and the four-exponentials frontier

Roy, *An arithmetic criterion for the values of the exponential function*, Acta Arith. **97** (2001), 183–194, constructs a small-value criterion on $\mathbb{G}_a \times \mathbb{G}_m$ and proves it *equivalent* to Schanuel's conjecture. His subsequent estimates — Acta Arith. **135** (2008), 357–393; Int. J. Number Theory **6** (2010), 919–956; Mathematika **59** (2013), 333–363 — and Nguyen–Roy, Int. J. Number Theory **12** (2016), 1273–1293, provide partial progress toward the four-exponentials and Schanuel circle. Roy's Strong Six Exponentials Theorem, J. Number Theory **41** (1992), 22–47, is the object whose formalization would most directly strengthen the algebraic-base spectrum of Section 12.

Butler, *Some cases of Wilkie's conjecture*, Bull. London Math. Soc. **44** (2012), 642–660, shows that certain cases of Wilkie's conjecture are equivalent to real forms of the three- and four-exponentials conjectures. That is a direct bridge between the counting route below and the transcendence core.

R.4 Counting rational points on the transcendental arc

A counterexample produces $\asymp T$ *rational* points

$$P_k = \left(\frac{M^k}{2^{r_k}},\ \frac{A^k}{3^{r_k}}\right) = (2^{\{k\beta\}}, 3^{\{k\beta\}})$$

of logarithmic height at most T, lying simultaneously in a *fixed* rank-two multiplicative subgroup $\Gamma \subset \mathbb{G}_m^2(\overline{\mathbb{Q}})$ and on the *fixed* transcendental Pfaffian arc $\mathcal{C} = \{(u, u^{\vartheta}) : 1 \le u < 2\}$. This unlikely-intersection formulation is a particularly promising framing, and it is not used elsewhere in this report.

The counting technology is Binyamini–Novikov, Ann. of Math. **186** (2017), 237–275, and Binyamini–Novikov–Zak, Ann. of Math. **199** (2024), 795–821, which prove Wilkie's conjecture for restricted elementary and Pfaffian structures with $(\log H)^{\kappa}$ counting; and, on the non-archimedean side, Binyamini–Cluckers–Novikov (Duke Math. J.), Cluckers–Comte–Loeser, Forum Math. Pi **3** (2015), and *Counting rational points on transcendental curves in valued fields*, arXiv:2506.19411 (2025). Boxall–Jones–Schmidt, J. Eur. Math. Soc. **24** (2022), 1567–1592, combine archimedean counting with p-adic methods.

The precise verdict is negative in a familiar way. Polylogarithmic counting gives $(\log H)^{\kappa}$ where a counterexample supplies $\asymp \log H$ points, so the framing reproduces the one-logarithm deficit of Section 21 in counting form: a third independent route arriving at the same shortfall. Moreover no cited theorem is sensitive to *which* primes divide the denominators, which is precisely what the compact-arc reformulation of Section 22 needs. The statement required — that a compact transcendental arc of nonzero curvature carries $o(H)$ points of height at most H whose first coordinate has 2-power denominator and second 3-power denominator — is not implied by any of them and would be a new theorem.

R.5 Multiplicative dependence, S-units, and rigidity

Bugeaud–Corvaja–Zannier, Math. Z. **243** (2003), 79–84, prove that for multiplicatively independent $a, b \ge 2$ and every $\varepsilon > 0$ one has $\log \gcd(a^n - 1, b^n - 1) \le \varepsilon n$ for large n, and that this is sharp. Together with the Corvaja–Zannier subspace-theorem machinery and Evertse–Győry's effective S-unit theory, this confirms that S-unit methods deliver finiteness, not nonexistence, and cannot supply the exponential-in-q divisibility the rational-approximation route would need.

For the $\times 2 \times 3$ side, Bourgain–Lindenstrauss–Michel–Venkatesh, Ergodic Theory Dynam. Systems **29** (2009), 1705–1722, give an effective Furstenberg-type density result, with a density gap of order

$(\log\log L)^{-\kappa}$. Shmerkin and Wu, Ann. of Math. **189** (2019), resolved Furstenberg's intersection conjecture. None of these can force visits to targets shrinking like M^{-k}; the target-width mismatch is fatal for a direct application, exactly as Section 22 anticipates. The effective Bourgain–Lindenstrauss–Michel–Venkatesh rate is the one usable quantitative handle and confirms, rather than defeats, the shrinking-target obstruction.

R.6 Kummer theory and the radical reformulation

Perucca–Sgobba–Tronto, Int. J. Number Theory **17** (2021), 1091–1110, and *Kummer theory for number fields via entanglement groups*, Manuscripta Math. **169** (2022), 251–270, compute degrees of Kummer extensions exactly through entanglement groups, and Chan–Pajaziti–Perissinotto–Perucca, arXiv:2508.19211 (2025), complete Kneser's theorem on additive relations among radicals. These make the algebraic half of the radical reformulation of Section 5 rigorous and formalizable. They do not see the special value ϑ: none of the cited Kummer results supplies the mixed archimedean–radical invariant that the reformulation needs.

R.7 Why the generic-power model theory excludes this exponent

Bays, Kirby and Wilkie, J. London Math. Soc. (2010), prove a Schanuel property for raising to an *exponentially transcendental* power: for such λ and multiplicatively independent $\bar{y}$ one has $\operatorname{td}(\bar{y}, \bar{y}^{\lambda}) \geq n$. The result does not apply here, and the reason is worth recording precisely, because it explains a pattern rather than merely reporting a failure.

The exponent ϑ is exponentially *algebraic*. The pair $(z, w) = (\log 2, \vartheta)$ is a nonsingular solution of the system

$$e^{z} - 2 = 0, \qquad e^{zw} - 3 = 0,$$

whose Jacobian determinant is $6 \log 2 \neq 0$. So ϑ lies in exactly the exceptional set that the generic-power theorems are stated to avoid. The same holds for every version of the problem in this report, since all of them involve the same exponent. Generic o-minimal and model-theoretic machinery should therefore be treated as a structural guide here, not as a black box expected to apply.

R.8 Prioritized directions, and what the ceiling changes

A useful ranking by proximity to the obstruction gives highest priority to the vanishing and p-adic lifts of the output-sensitive transverse factors $Q_{p,E}(M, A)$ in (660), and to a saturated two-generator divisor of the positive Markov–Padé defect integer in (637); either target couples the fixed common-power table to both Archimedean and p-adic information. Comparable priority goes to the rank-two multiplicative group combined with specialized o-minimal or algebraic covering and S-unit rigidity; to a Laurent-style interpolation-determinant redesign of the semigroup determinant of Section 19; and to tropical or p-adic determinant amplification with modern p-adic logarithmic-form control. High priority to the Corvaja–Zannier recurrence and power-sum machinery, once an integral quotient has been manufactured, and to the Mahler–Ridout–Corvaja–Zannier exponential-nearness criterion, which supplies a sharp conditional target: construct one non-Pisot algebraic power sequence exponentially close to the integers. Medium priority to the explicit logarithmic-form technology of Matveev, Laurent, Gouillon and Mignotte–Voutier, excellent for finite verification and auxiliary lemmas but unlikely to cross the bilinear barrier alone, and to integer-valued o-minimal function theory, methodologically close but not fitted to a sparse multiplicative grid. Low priority, as a direct attack, to Skolem–Mahler–Lech, generic-power model theory and general Zilber–Pink.

Most items in that ranking remain external inputs rather than ingredients of the arguments; they are listed here so that the gap is visible rather than implicit.

The tropical or p-adic determinant-amplification direction is excluded at the term-wise level by Corollary 20.2: automatic divisibility can never contradict a correct archimedean bound, whatever the primes, nodes or columns. What survives of that direction is the cancellation problem localized by Proposition 20.3, which is a different and harder target than the one the ranking had in view.

R.9 Consequences for abundant-number questions

Alaoglu and Erdős proved that the ratio of consecutive superabundant numbers is prime, and, using the three-prime case that Siegel had claimed, that the ratio of consecutive colossally abundant numbers is a prime or a semiprime. Erdős and Nicolas later showed that exactly one, two, or four integers can attain the maximum in the colossally abundant construction, so a positive answer to the rational-power form of the conjecture bounds this to at most two and makes the ratio of consecutive colossally abundant numbers prime. Lagarias, Amer. Math. Monthly **109** (2002), 534–543, records the Robin/Riemann reformulation of the surrounding circle.

R.10 The formalization landscape

Karatarakis and Wiedijk formalized the Gelfond–Schneider theorem in Lean 4, including a formalized Siegel lemma for algebraic integers and the auxiliary-function machinery; that development is the basis of the port used by this corpus and cited in Appendix K. Mathlib's Lindemann–Weierstrass development supplies analytic scaffolding only. The present repository formalizes neither Baker's theorem, the six exponentials theorem, nor linear forms in logarithms; those transcendence inputs remain behind explicit interfaces.

R.11 Remaining gaps

Three gaps remain. No lower-bound technology handles linear or bilinear forms whose *coefficients* are themselves logarithms, which is the wall of Section 6. No counting theorem is sensitive to the arithmetic of denominators, which is what the compact-arc reformulation needs. And no unconditional algebraic-independence result reaches products of two logarithms, which is what the criterion of Section 11 would need. The conjecture remains open, and each route above faces the same rank-two-to-rank-one collapse.

Status: [classical/open background] throughout, with the two corrections marked as such. This appendix supplies mathematical context; no statement here is silently used as a proof input.

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